



```
, 30, 33, 34, 42, 43, 45, 46, 47],
polyclinic: [0, 1, 2, 3, 4, 6, 8, 9, 13, 14, 15, 20, 34, 42, 43, 45, 46, 47],
phc: [0, 1, 2, 3, 4, 6, 14, 15, 33, 34],
specialty_center: [0, 1, 2, 3, 4, 6, 8, 14, 15, 20, 34, 43],
diagnostic_center: [3, 4, 14, 15],
rehabilitation_center: [0, 1, 2, 14, 15, 24],
dialysis_center: [0, 1, 2, 14, 15, 20],
dental_center: [0, 1, 2, 14, 15, 43],
mental_health_center: [0, 1, 2, 14, 15, 22],
home_healthcare_unit: [0, 1, 2, 14, 15, 33],
mobile_clinic: [0, 1, 2, 14, 15, 33],
virtual_clinic: [0, 1, 2, 14, 15, 33],
health_unit: [0, 1, 2, 3, 14, 15, 33],
emergency_center: [0, 1, 2, 3, 14, 15, 19]
};
```

```
const tr = (en, ar) => isArabic ? ar : en;
```

```
// Navigation items
```

```
const NAV_ITEMS = [
  { icon: '?', en: 'Dashboard', ar: '???? ??????' },
  { icon: '?', en: 'Reception', ar: '?????????' },
  { icon: '?', en: 'Appointments', ar: '?????????' },
  { icon: '????', en: 'Doctor Station', ar: '???? ??????' },
  { icon: '?', en: 'Laboratory', ar: '?????????' },
  { icon: '?', en: 'Radiology', ar: '?????????' },
  { icon: '?', en: 'Pharmacy', ar: '?????????' },
  { icon: '?', en: 'HR', ar: '???????? ??????????' },
  { icon: '?', en: 'Finance', ar: '?????????' },
  { icon: '??', en: 'Insurance', ar: '?????????' },
  { icon: '?', en: 'Inventory', ar: '?????????' },
  { icon: '????', en: 'Nursing', ar: '?????????' },
  { icon: '?', en: 'Waiting Queue', ar: '????? ??????????' },
  { icon: '?', en: 'Patient Accounts', ar: '?????? ??????????' },
  { icon: '?', en: 'Reports', ar: '?????????' },
  { icon: '??', en: 'Messaging', ar: '?????????' },
  { icon: '?', en: 'Catalog', ar: '?????????' },
  { icon: '?', en: 'Dept Requests', ar: '????? ??????????' },
  { icon: '?', en: 'Surgery & Pre-Op', ar: '????????? ??? ??????' },
  { icon: '?', en: 'Blood Bank', ar: '??? ?????' },
  { icon: '?', en: 'Consent Forms', ar: '?????????????' },
  { icon: '?', en: 'Emergency', ar: '?????????' },
  { icon: '??', en: 'Inpatient ADT', ar: '?????????' },
  { icon: '?', en: 'ICU', ar: '????????? ??????????' },
  { icon: '?', en: 'CSSD', ar: '????????? ??????????' },
  { icon: '??', en: 'Dietary', ar: '?????????' },
  { icon: '?', en: 'Infection Control', ar: '?????? ??????????' },
  { icon: '?', en: 'Quality', ar: '?????????' },
  { icon: '?', en: 'Maintenance', ar: '?????????' },
  { icon: '?', en: 'Transport', ar: '??? ??????????' },
  { icon: '?', en: 'Medical Records', ar: '????????? ??????????' },
  { icon: '?', en: 'Clinical Pharmacy', ar: '????????? ??????????' },
  { icon: '??', en: 'Rehabilitation', ar: '?????? ??????????' },
  { icon: '?', en: 'Patient Portal', ar: '????? ??????????' },
```

```

    { icon: '?', en: 'ZATCA E-Invoice', ar: '????? ?????????' },
    { icon: '?', en: 'Telemedicine', ar: '???? ?? ???' },
    { icon: '?', en: 'Pathology', ar: '??? ??????' },
    { icon: '?', en: 'Social Work', ar: '?????? ??????????' },
    { icon: '??', en: 'Mortuary', ar: '???? ??????' },
    { icon: '?', en: 'CME', ar: '??????? ?????' },
    { icon: '?', en: 'Cosmetic Surgery', ar: '????? ??????' },
    { icon: '?', en: 'OB/GYN', ar: '?????? ?????????' },
    { icon: '??', en: 'Settings', ar: '?????????' },
    { icon: '?', en: 'Dental', ar: '????????' },
    // 44: Specialties ? placeholder page (renderSpecialties); kept hidden from the sidebar on purpose.
    { icon: '?', en: 'Specialties', ar: '????????? ??????', hidden: true },
    // 45: OVR / Incident Reports ? patient-safety governance (renderOVR). Surfaced for safety/quality roles.
    { icon: '?', en: 'Incident Reports (OVR)', ar: '?????? ??????? (OVR)' },
    // 46: Audit Trail viewer ? compliance (renderAuditLog). Read-only; grant to auditor/compliance/admin only.
    { icon: '?', en: 'Audit Trail', ar: '??? ??????' },
    // 47: Specialties Panel ? E1 clinical specialties exposed to sidebar.
    { icon: '?', en: 'Specialties Panel', ar: '???? ?????????', hidden: false }
];

const NAV_ACCESS_RULES = {
  44: {
    permissionKeys: ['44', 'specialties', 'clinical.specialties.view'],
    roles: ['Admin', 'Doctor', 'OB/GYN', 'Neonatologist', 'Pathologist', 'Radiologist']
  },
  45: {
    permissionKeys: ['45', 'quality', 'quality.ovr.view', 'quality.ovr.create', 'quality.ovr.review'],
    roles: ['Admin', 'Quality Manager', 'Infection Control']
  },
  46: {
    permissionKeys: ['46', 'audit', 'audit.read', 'audit.export', 'audit.investigate'],
    roles: ['Admin', 'IT', 'Quality Manager']
  },
  47: {
    permissionKeys: ['47', 'specialties', 'clinical.specialties.view'],
    roles: ['Admin', 'Doctor', 'OB/GYN', 'Neonatologist', 'Pathologist', 'Radiologist']
  }
};

function normalizeAccessToken(value) {
  return String(value || '').trim().toLowerCase();
}

function getCurrentUserPermissions() {
  return currentUser?.permissions
    ? (Array.isArray(currentUser.permissions) ? currentUser.permissions : currentUser.permissions.split(',')).
      map(normalizeAccessToken).filter(Boolean)
    : [];
}

function getCurrentUserRole() {
  return String(currentUser?.role || '').trim();
}

```

```

function roleMatchesRule(rule, role) {
    const normalizedRole = normalizeAccessToken(role);
    return (rule.roles || []).map(normalizeAccessToken).includes(normalizedRole);
}

function permissionMatchesRule(rule, permissions) {
    const allowedKeys = (rule.permissionKeys || []).map(normalizeAccessToken);
    return permissions.some((permission) => allowedKeys.includes(permission));
}

function canAccessNavItem(index, options = {}) {
    const item = NAV_ITEMS[index];
    if (!item) return false;
    if (item.hidden && !options.allowHidden) return false;

    const allowed = FACILITY_ALLOWED[facilityType];
    const isHiddenDirectAccess = item.hidden && options.allowHidden;
    if (!isHiddenDirectAccess && !options.ignoreFacility && allowed && !allowed.includes(index)) return false;

    const role = getCurrentUserRole();
    const isAdmin = role === 'Admin';
    const permissions = getCurrentUserPermissions();
    const rule = NAV_ACCESS_RULES[index];

    if (rule) {
        return isAdmin || roleMatchesRule(rule, role) || permissionMatchesRule(rule, permissions);
    }

    return isAdmin || index === 0 || permissions.includes(String(index));
}

// ===== INIT =====
// ===== INIT =====
(async function init() {
    try {
        const data = await API.get('/api/auth/me');
        currentUser = data.user;
    } catch {
        currentUser = null;
    }

    // Language: set direction
    document.documentElement.dir = isArabic ? 'rtl' : 'ltr';
    document.documentElement.lang = isArabic ? 'ar' : 'en';
    const langBtn = document.getElementById('langToggleBtn');
    if (langBtn) langBtn.textContent = isArabic ? '? EN' : '? ????';
    updateShellLanguage();

    if (!currentUser) {
        // Render Public Homepage Mode
        document.getElementById('app').classList.add('public-mode');
        renderPublicHomepage();

        // Bind public language toggle

```

```

if (langBtn) {
  langBtn.onclick = () => {
    isArabic = !isArabic;
    localStorage.setItem('namaLang', isArabic ? 'ar' : 'en');
    document.documentElement.dir = isArabic ? 'rtl' : 'ltr';
    document.documentElement.lang = isArabic ? 'ar' : 'en';
    langBtn.textContent = isArabic ? '? EN' : '? ???';
    updateShellLanguage();
    renderPublicHomepage();
  };
}
return;
}

// Authenticated Mode
document.getElementById('app').classList.remove('public-mode');
document.getElementById('userName').textContent = currentUser.name;
document.getElementById('userRole').textContent = currentUser.role;
document.getElementById('userAvatar').textContent = currentUser.name.charAt(0);

// Tenant Context Indicator
const tenantText = `M-${currentUser.tenantId} || '1'`;
const userRoleEl = document.getElementById('userRole');
if (userRoleEl) {
  const oldTenant = document.getElementById('userTenant');
  if (oldTenant) oldTenant.remove();
  userRoleEl.insertAdjacentHTML('afterend', `
    <div class="user-tenant" id="userTenant" style="font-size:11px;color:rgba(255,255,255,0.6);margin-top:2p
x">
      ? ${tr('Tenant', '????????')}: ${tenantText}
    </div>
  `);
}

// Top header tenant context & Facility Switcher
const headerLeft = document.querySelector('.header-left');
if (headerLeft) {
  const oldHeaderTenant = document.getElementById('headerTenantContext');
  if (oldHeaderTenant) oldHeaderTenant.remove();
  headerLeft.insertAdjacentHTML('beforeend', `
    <span id="headerTenantContext" class="badge badge-tenant-context" style="margin-left: 16px; margin-right
: 16px; background: var(--accent-glow); color: var(--accent); padding: 4px 12px; border-radius: 20px; font-siz
e: 12px; font-weight: 600; border: 1px solid var(--accent)">
      ? ${tr('Tenant Context', '???? ?????????')}: ${tenantText}
    </span>
  `);
}

// Add Facility Switcher in Header
const headerRight = document.querySelector('.header-right');
if (headerRight) {
  const oldSwitcher = document.getElementById('facilitySwitcher');
  if (oldSwitcher) oldSwitcher.remove();
  headerRight.insertAdjacentHTML('beforeend', `

```

```

<select id="facilitySwitcher" class="theme-select" style="background: var(--accent-glow); color: var(--a
ccent); border: 1px solid var(--accent); margin-left: 12px; margin-right: 12px; padding: 4px 8px; border-radiu
s: 8px; font-size: 13px; font-weight: 600;">
  <option value="medical_city" ${facilityType === 'medical_city' ? 'selected' : ''}>? ${tr('Medical City
', '?????? ?????')}</option>
  <option value="general_hospital" ${facilityType === 'general_hospital' ? 'selected' : ''}>? ${tr('Gene
ral Hospital', '???????? ?????')}</option>
  <option value="specialized_hospital" ${facilityType === 'specialized_hospital' ? 'selected' : ''}>? ${
tr('Specialized Hospital', '???????? ?????')}</option>
  <option value="tertiary_hospital" ${facilityType === 'tertiary_hospital' ? 'selected' : ''}>? ${tr('Te
rtiary Hospital', '???????? ?????')}</option>
  <option value="polyclinic" ${facilityType === 'polyclinic' ? 'selected' : ''}>? ${tr('Polyclinic / Co
mplex', '???? ??????? / ???????')}</option>
  <option value="phc" ${facilityType === 'phc' ? 'selected' : ''}>? ${tr('Primary Care (PHC)', '??????
????')}</option>
  <option value="specialty_center" ${facilityType === 'specialty_center' ? 'selected' : ''}>? ${tr('Spec
ialty Center', '????? ?????')}</option>
  <option value="diagnostic_center" ${facilityType === 'diagnostic_center' ? 'selected' : ''}>? ${tr('Di
agnostic Center', '????? ?????')}</option>
  <option value="rehabilitation_center" ${facilityType === 'rehabilitation_center' ? 'selected' : ''}>?
  ${tr('Rehabilitation Center', '???? ??????')}</option>
  <option value="dialysis_center" ${facilityType === 'dialysis_center' ? 'selected' : ''}>? ${tr('Dialys
is Center', '???? ????')}</option>
  <option value="dental_center" ${facilityType === 'dental_center' ? 'selected' : ''}>? ${tr('Dental Cen
ter', '????')}</option>
  <option value="mental_health_center" ${facilityType === 'mental_health_center' ? 'selected' : ''}>? ${
tr('Mental Health Center', '???? ?????')}</option>
  <option value="home_healthcare_unit" ${facilityType === 'home_healthcare_unit' ? 'selected' : ''}>? ${
tr('Home Healthcare Unit', '???? ????')}</option>
  <option value="mobile_clinic" ${facilityType === 'mobile_clinic' ? 'selected' : ''}>? ${tr('Mobile Cli
nic', '?????')}</option>
  <option value="virtual_clinic" ${facilityType === 'virtual_clinic' ? 'selected' : ''}>? ${tr('Virtual
Clinic', '?????')}</option>
  <option value="health_unit" ${facilityType === 'health_unit' ? 'selected' : ''}>? ${tr('Health Unit',
'?????')}</option>
  <option value="emergency_center" ${facilityType === 'emergency_center' ? 'selected' : ''}>? ${tr('Emer
gency Center', '????')}</option>
</select>
`);

document.getElementById('facilitySwitcher').addEventListener('change', (e) => {
  facilityType = e.target.value;
  buildNav();
  navigateTo(0);
});
}

// Load saved theme + facility type
try {
  const s = await API.get('/api/settings');
  if (s.theme) { document.documentElement.setAttribute('data-theme', s.theme); document.getElementById('them
eSelect').value = s.theme; }
  if (s.facility_type) {
    facilityType = s.facility_type;

```

```

        const switcher = document.getElementById('facilitySwitcher');
        if (switcher) switcher.value = facilityType;
    }
} catch { }

buildNav();
setupEvents();
navigateTo(0);
})();

function buildNav() {
    const nav = document.getElementById('navList');
    const userPerms = currentUser?.permissions ? currentUser.permissions.split(',') : [];
    const isAdmin = currentUser?.role === 'Admin';

    const CLINICAL_ORDER = [
        0,    // Dashboard (???? ??????)
        1,    // Reception (?????????)
        2,    // Appointments (?????????)
        12,   // Waiting Queue (????? ????????)
        3,    // Doctor Station (???? ??????)
        11,   // Nursing (?????????)
        21,   // Emergency (?????????)
        22,   // Inpatient ADT (?????????)
        23,   // ICU (????????? ????????)
        18,   // Surgery & Pre-Op (????????? ??? ??????)
        41,   // OB/GYN (??????? ??????????)
        40,   // Cosmetic Surgery (?????? ??????????)
        4,    // Laboratory (?????????)
        5,    // Radiology (?????????)
        36,   // Pathology (??? ??????????)
        19,   // Blood Bank (??? ??????)
        32,   // Rehabilitation (?????? ??????????)
        6,    // Pharmacy (?????????)
        31,   // Clinical Pharmacy (????????? ??????????)
        34,   // ZATCA E-Invoice (????? ??????????)
        13,   // Patient Accounts (?????? ????????)
        9,    // Insurance (?????????)
        8,    // Finance (?????????)
        14,   // Reports (?????????)
        7,    // HR (????????? ??????????)
        10,   // Inventory (?????????)
        16,   // Catalog (?????????)
        17,   // Dept Requests (????? ??????????)
        24,   // CSSD (????????? ??????????)
        26,   // Infection Control (??????? ????????)
        27,   // Quality (?????????)
        25,   // Dietary (?????????)
        29,   // Transport (??? ????????)
        37,   // Social Work (??????? ??????????????)
        28,   // Maintenance (?????????)
        20,   // Consent Forms (?????????????)
        30,   // Medical Records (????????? ????????)
        33,   // Patient Portal (?????? ????????)
    ]

```

```

35, // Telemedicine (???? ?? ???)
15, // Messaging (???????)
39, // CME (?????? ?????)
38, // Mortuary (???? ??????)
43, // Dental (???????)
47, // Specialties Panel (???? ????????)
42 // Settings (?????????)
];
const allIndices = [...CLINICAL_ORDER];
for (let idx = 0; idx < NAV_ITEMS.length; idx++) {
  if (!allIndices.includes(idx)) {
    allIndices.push(idx);
  }
}

nav.innerHTML = allIndices.map(i => {
  const item = NAV_ITEMS[i];
  if (!item) return '';
  if (!canAccessNavItem(i)) return '';
  return `<div class="nav-item${i === currentPage ? ' active' : ''}" data-page="${i}">
    <span class="nav-icon">${item.icon}</span>
    <span class="nav-label">${tr(item.en, item.ar)}</span>
  </div>`;
}).join('');

nav.querySelectorAll('.nav-item').forEach(el => {
  el.addEventListener('click', () => navigateTo(parseInt(el.dataset.page)));
});
}

// Add notification bell to header
const headerR = document.querySelector('.header-right') || document.querySelector('.header');
if (headerR) {
  const bellSpan = document.createElement('span');
  bellSpan.id = 'notifBell';
  bellSpan.style.cssText = 'cursor:pointer;font-size:20px;position:relative;margin-left:12px;margin-right:12px';
  bellSpan.innerHTML = '?';
  headerR.prepend(bellSpan);
}

function setupEvents() {
  document.getElementById('logoutBtn').addEventListener('click', async () => {
    await API.post('/api/auth/logout');
    window.location.href = '/login.html';
  });
  document.getElementById('themeSelect').addEventListener('change', (e) => {
    document.documentElement.setAttribute('data-theme', e.target.value);
    API.put('/api/settings', { theme: e.target.value }).catch(() => {});
  });
  document.getElementById('langToggleBtn').addEventListener('click', () => {
    isArabic = !isArabic;
    localStorage.setItem('namaLang', isArabic ? 'ar' : 'en');
  });
}

```



```

document.documentElement.dir = isArabic ? 'rtl' : 'ltr';
document.documentElement.lang = isArabic ? 'ar' : 'en';
// Update toggle button text
document.getElementById('langToggleBtn').textContent = isArabic ? '? EN' : '? ????' ;
updateShellLanguage();
buildNav();
navigateTo(currentPage);
});
document.getElementById('menuToggle').addEventListener('click', () => {
  const open = document.getElementById('sidebar').classList.toggle('open');
  document.getElementById('sidebarOverlay').classList.toggle('show');
  // a11y: announce the sidebar open/closed state on the toggle button.
  document.getElementById('menuToggle').setAttribute('aria-expanded', open ? 'true' : 'false');
});
document.getElementById('sidebarOverlay').addEventListener('click', () => {
  document.getElementById('sidebar').classList.remove('open');
  document.getElementById('sidebarOverlay').classList.remove('show');
});
document.getElementById('globalSearch').addEventListener('keydown', async (e) => {
  if (e.key === 'Enter') {
    const txt = e.target.value.trim();
    if (!txt) return;
    const res = await API.get('/api/patients?search=' + encodeURIComponent(txt));
    let html = `<div id="searchResultPopup" class="card" style="position:absolute;top:60px;right:20px;width:
700px;z-index:1000;box-shadow:0 10px 30px rgba(0,0,0,0.5)">
      <div style="display:flex;justify-content:space-between;margin-bottom:10px">
        <strong>? ${tr('Search Results', '???? ????')} (${res.length})</strong>
        <button class="btn btn-danger btn-sm" onclick="document.getElementById('searchResultPopup').r
remove()">?</button>
      </div>
      <div style="max-height:400px;overflow-y:auto">
        ${makeTable(
          [tr('File#', '??? ?????'), tr('Name', '????'), tr('National ID', '??????'), tr('Phone', '??????'), tr
('Dept', '????')],
          res.map(p => ({ cells: [p.file_number, isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar),
p.national_id, p.phone, p.department] })))
        </div>
      </div>`;
    const old = document.getElementById('searchResultPopup');
    if (old) old.remove();
    const div = document.createElement('div');
    div.innerHTML = html;
    document.body.appendChild(div.firstElementChild);
  }
});
// Set initial search placeholder based on language
const searchBox = document.getElementById('globalSearch');
if (searchBox) searchBox.placeholder = isArabic ? '??? ??????? ??????? ??????? ??? ??????...' : 'Search by na
me, ID, phone, file number...';
}

// ===== UPDATE SHELL LANGUAGE =====
// Updates all static HTML elements that are hardcoded in index.html

```

```

function updateShellLanguage() {
  // Sidebar title & subtitle
  const sidebarTitle = document.querySelector('.sidebar-title');
  if (sidebarTitle) sidebarTitle.textContent = isArabic ? '???? ????' : 'jumanaMedical';
  const sidebarSubtitle = document.querySelector('.sidebar-subtitle');
  if (sidebarSubtitle) sidebarSubtitle.textContent = isArabic ? 'jumanaMedical' : 'Medical ERP';

  // Search placeholder
  const searchBox = document.getElementById('globalSearch');
  if (searchBox) searchBox.placeholder = isArabic ? '?? ?????? ?????? ??? ?????...' : 'Search by name, ID, phone, file number...';

  // Logout button tooltip
  const logoutBtn = document.getElementById('logoutBtn');
  if (logoutBtn) logoutBtn.title = isArabic ? '???? ?????' : 'Logout';

  // Theme select options
  const themeNames = [
    { ar: '? ??? ????', en: '? Dark Blue' },
    { ar: '? ??? ????', en: '? Dark Green' },
    { ar: '? ?????', en: '? Purple' },
    { ar: '? ???', en: '? Red' },
    { ar: '? ???', en: '? Gold' },
    { ar: '? ??? ?????', en: '? Light Classic' },
    { ar: '? ??? ????', en: '? Light Blue' },
    { ar: '? ??? ????', en: '? Light Green' }
  ];
  const themeSelect = document.getElementById('themeSelect');
  if (themeSelect) {
    themeSelect.title = isArabic ? '???? ?????' : 'Choose Theme';
    Array.from(themeSelect.options).forEach((opt, i) => {
      if (themeNames[i]) opt.textContent = isArabic ? themeNames[i].ar : themeNames[i].en;
    });
  }

  // Language toggle button tooltip
  const langBtn = document.getElementById('langToggleBtn');
  if (langBtn) langBtn.title = isArabic ? '???? ?????' : 'Change Language';

  // Page title
  const pageTitle = document.querySelector('title');
  if (pageTitle) pageTitle.textContent = isArabic ? '???? ????? - jumanaMedical ERP' : 'jumanaMedical ERP';
}

// ===== PUBLIC HOMEPAGE =====
function renderPublicHomepage() {
  const el = document.getElementById('pageContent');
  if (!el) return;

  // Static lists for filters
  const regions = [
    { id: 'all', en: 'All Regions', ar: '???? ??????' },
    { id: 'Riyadh', en: 'Riyadh Region', ar: '???? ??????' },
    { id: 'Makkah', en: 'Makkah Region', ar: '???? ??? ??????' }
  ]

```

```

];

const types = [
  { id: 'all', en: 'All Facility Types', ar: '???? ????? ??????' },
  { id: 'medical_city', en: 'Medical City', ar: '????? ????' },
  { id: 'general_hospital', en: 'General Hospital', ar: '?????? ??' },
  { id: 'specialized_hospital', en: 'Specialized Hospital', ar: '?????? ?????' },
  { id: 'tertiary_hospital', en: 'Tertiary Hospital', ar: '?????? ?????' },
  { id: 'polyclinic', en: 'Polyclinic / Complex', ar: '???? ?????? / ??????' },
  { id: 'phc', en: 'Primary Healthcare Center', ar: '???? ?????? ???? ?????' },
  { id: 'specialty_center', en: 'Specialty Center', ar: '???? ??????' },
  { id: 'diagnostic_center', en: 'Diagnostic Center', ar: '???? ??????' },
  { id: 'rehabilitation_center', en: 'Rehabilitation Center', ar: '???? ?????? ??' },
  { id: 'dialysis_center', en: 'Dialysis Center', ar: '???? ???? ?????' },
  { id: 'dental_center', en: 'Dental Center', ar: '???? ??????' },
  { id: 'mental_health_center', en: 'Mental Health Center', ar: '???? ??? ??????' },
  { id: 'home_healthcare_unit', en: 'Home Healthcare Unit', ar: '????? ???? ??????' },
  { id: 'mobile_clinic', en: 'Mobile Clinic', ar: '???????? ?????????' },
  { id: 'virtual_clinic', en: 'Virtual Clinic', ar: '???????? ??????????' },
  { id: 'health_unit', en: 'Health Unit', ar: '?????? ??????' },
  { id: 'emergency_center', en: 'Emergency Center', ar: '???? ?????? ??????' }
];

// Render HTML structure
el.innerHTML = `
  <!-- Public Header -->
  <header class="public-header flex justify-between items-center px-8 py-4 bg-surface-container-lowest border-b border-outline-variant/30 shadow-sm">
    <div class="flex items-center gap-3">
      <div class="w-10 h-10 bg-primary/10 text-primary rounded-xl flex items-center justify-center border border-primary/10">
        <span class="material-symbols-outlined text-2xl">health_and_safety</span>
      </div>
      <div>
        <h1 class="text-lg font-bold text-primary">${tr('NamaMedical Platform', '???? ??? ??????')}</h1>
        <p class="text-xs text-on-surface-variant">${tr('National Health Cluster Network', '?????? ??????? ? ??????? ??????')}</p>
      </div>
    </div>
    <div class="flex items-center gap-4">
      <a href="/login.html" class="px-5 py-2 bg-primary text-on-primary rounded-xl font-label-md shadow-md hover:opacity-90 transition-all">
        ? ${tr('Login to EMR', '????? ?????? ?????????')}
      </a>
    </div>
  </header>

  <!-- Hero Section -->
  <section class="hero-section text-center py-16 px-6 bg-gradient-to-br from-primary to-primary-container text-on-primary relative overflow-hidden">
    <div class="max-w-3xl mx-auto relative z-10">
      <span class="px-3 py-1 bg-white/10 rounded-full text-xs font-medium tracking-wider uppercase mb-4 inline-block">
        ?? ${tr('Saudi Vision 2030 Healthcare Transformation', '?????? ??? ???? - ??? ???? 2030')}
      </span>
    </div>
  </section>

```

```
')}}

</span>
<h2 class="text-4xl font-bold mb-4 tracking-tight">${tr('Your Portal to Integrated Healthcare', '????
? ?????? ?????? ?????????')}</h2>

<p class="text-lg text-white/80 mb-8">${tr('Connecting Medical Cities, General Hospitals, and Primary
Care Centers across the Kingdom.', '??? ???????? ??? ?????? ?????? ?????????? ?????? ?????? ??????
?????????')}</p>

<!-- Search & Filter Bar -->
<div class="glass-card p-4 rounded-2xl max-w-2xl mx-auto flex flex-col md:flex-row gap-3 shadow-lg bor
der border-white/10">
  <select id="regionFilter" class="form-input text-on-surface flex-1" style="color:var(--primary)">
    ${regions.map(r => `<option value="${r.id}">${tr(r.en, r.ar)}</option>`).join('')}
  </select>
  <select id="typeFilter" class="form-input text-on-surface flex-1" style="color:var(--primary)">
    ${types.map(t => `<option value="${t.id}">${tr(t.en, t.ar)}</option>`).join('')}
  </select>
  <button id="searchBtn" class="px-6 py-2.5 bg-secondary text-on-secondary rounded-xl font-label-md ho
ver:opacity-90 transition-all flex items-center justify-center gap-2">
    ? ${tr('Search', '???')}
  </button>
</div>
</div>
</section>

<!-- Facilities Grid -->
<section class="py-12 px-8 max-w-7xl mx-auto">
  <h3 class="text-2xl font-bold text-primary mb-6 flex items-center gap-2">
    ? ${tr('Healthcare Facilities & Branches', '?????? ???????? ?????? ??????')}
  </h3>
  <div id="facilitiesContainer" class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
    <!-- Dynamically filled -->
  </div>
</section>

<!-- Specialties Showcase -->
<section class="py-12 bg-surface-container-low px-8 border-t border-b border-outline-variant/20">
  <div class="max-w-7xl mx-auto">
    <h3 class="text-2xl font-bold text-primary mb-8 text-center">
      ? ${tr('Specialized Centers of Excellence', '???? ?????? ?????????? ??????')}
    </h3>
    <div class="grid grid-cols-2 md:grid-cols-4 lg:grid-cols-6 gap-6">
      <div class="card p-4 text-center hover:shadow-md transition-shadow">
        <div class="text-3xl mb-2">?</div>
        <h5 class="font-bold text-sm">${tr('Cardiology', '???? ??????')}</h5>
      </div>
      <div class="card p-4 text-center hover:shadow-md transition-shadow">
        <div class="text-3xl mb-2">?</div>
        <h5 class="font-bold text-sm">${tr('Neurology', '???? ??????')}</h5>
      </div>
      <div class="card p-4 text-center hover:shadow-md transition-shadow">
        <div class="text-3xl mb-2">?</div>
        <h5 class="font-bold text-sm">${tr('Pediatrics', '?? ?????? ??????')}</h5>
      </div>
    </div>
  </div>
</section>
```

```

<div class="card p-4 text-center hover:shadow-md transition-shadow">
  <div class="text-3xl mb-2">?</div>
  <h5 class="font-bold text-sm">${tr('OB/GYN', '????? ???????')}</h5>
</div>
<div class="card p-4 text-center hover:shadow-md transition-shadow">
  <div class="text-3xl mb-2">?</div>
  <h5 class="font-bold text-sm">${tr('Pathology', '????????? ???????')}</h5>
</div>
<div class="card p-4 text-center hover:shadow-md transition-shadow">
  <div class="text-3xl mb-2">?</div>
  <h5 class="font-bold text-sm">${tr('Emergency', '????? ???????')}</h5>
</div>
</div>
</div>
</section>

<!-- Emergency Quick Banner -->
<section class="py-8 bg-error/10 border-l-4 border-error px-8 text-error-container flex flex-col md:flex-row justify-between items-center max-w-7xl mx-auto my-12 rounded-xl">
  <div class="flex items-center gap-4 mb-4 md:mb-0">
    <span class="material-symbols-outlined text-4xl animate-pulse">emergency</span>
    <div>
      <h4 class="font-bold text-lg text-error">${tr('Urgent Emergency Care 24/7', '????? ??????? ??????? 24/7')}</h4>
      <p class="text-sm text-on-error-container">${tr('If you are facing a life-threatening emergency, call 997 immediately or proceed to the nearest ER.', '??? ??? ?????? ??? ???? ???? ??????? ??? ???? 997 ?????? ?? ????? ????? ??? ??????')}</p>
    </div>
  </div>
  <a href="tel:997" class="px-6 py-2.5 bg-error text-on-error rounded-xl font-label-md shadow-md hover:opacity-90 transition-all font-bold">
    ? ${tr('Call 997 Now', '???? ?????? 997 ?????')}
  </a>
</section>

<!-- Compliance Footer -->
<footer class="py-8 bg-surface-container-highest border-t border-outline-variant/30 text-center px-6">
  <div class="max-w-7xl mx-auto flex flex-col md:flex-row justify-between items-center gap-4 text-xs text-on-surface-variant">
    <p>© ${new Date().getFullYear()} ${tr('NamaMedical Hospital Platform. All Rights Reserved.', '???? ???? ??????? ???? ?????? ??????')}</p>
    <div class="flex gap-4">
      <span class="badge badge-success">?? CBAHI Accredited</span>
      <span class="badge badge-info">?? ZATCA Compliant</span>
      <span class="badge badge-secondary">? PDPL Secure</span>
    </div>
  </div>
</footer>
`;

// Filter and render facilities
const filterAndRender = () => {
  const region = document.getElementById('regionFilter').value;
  const type = document.getElementById('typeFilter').value;

```

```

const container = document.getElementById('facilitiesContainer');
if (!container) return;

let filtered = FACILITY_CATALOG.facilities;
if (region !== 'all') {
  const cityIds = FACILITY_CATALOG.medical_cities.filter(c => c.region === region).map(c => c.id);
  filtered = filtered.filter(f => cityIds.includes(f.city_id));
}
if (type !== 'all') {
  filtered = filtered.filter(f => f.type === type);
}

const typeIcons = {
  medical_city: '?',
  general_hospital: '?',
  specialized_hospital: '?',
  tertiary_hospital: '?',
  polyclinic: '??',
  phc: '?',
  specialty_center: '?',
  diagnostic_center: '?',
  rehabilitation_center: '??',
  dialysis_center: '?',
  dental_center: '?',
  mental_health_center: '?',
  home_healthcare_unit: '?',
  mobile_clinic: '?',
  virtual_clinic: '?',
  health_unit: '?',
  emergency_center: '?'
};

const typeLabels = {
  medical_city: { en: 'Medical City', ar: '????? ????' },
  general_hospital: { en: 'General Hospital', ar: '?????? ??' },
  specialized_hospital: { en: 'Specialized Hospital', ar: '?????? ?????' },
  tertiary_hospital: { en: 'Tertiary Hospital', ar: '?????? ?????' },
  polyclinic: { en: 'Polyclinic / Complex', ar: '???? ?????? / ??????' },
  phc: { en: 'Primary Healthcare Center', ar: '???? ?????? ???? ??????' },
  specialty_center: { en: 'Specialty Center', ar: '???? ??????' },
  diagnostic_center: { en: 'Diagnostic Center', ar: '???? ??????' },
  rehabilitation_center: { en: 'Rehabilitation Center', ar: '???? ?????? ??' },
  dialysis_center: { en: 'Dialysis Center', ar: '???? ???? ??????' },
  dental_center: { en: 'Dental Center', ar: '???? ??????' },
  mental_health_center: { en: 'Mental Health Center', ar: '???? ??? ??????' },
  home_healthcare_unit: { en: 'Home Healthcare Unit', ar: '????? ???? ??????' },
  mobile_clinic: { en: 'Mobile Clinic', ar: '???????? ??????????' },
  virtual_clinic: { en: 'Virtual Clinic', ar: '???????? ??????????' },
  health_unit: { en: 'Health Unit', ar: '?????? ??????' },
  emergency_center: { en: 'Emergency Center', ar: '???? ?????? ??????' }
};

container.innerHTML = filtered.map(f => {
  const label = typeLabels[f.type] || typeLabels.general_hospital;
  return `

```

```

<div class="glass-card-premium p-6 rounded-2xl flex flex-col gap-4 hover:-translate-y-1 transition-all
duration-300">
  <div class="flex justify-between items-start">
    <span class="text-3xl">${typeIcons[f.type] || '?'}</span>
    <span class="badge badge-success">${tr(f.status, f.status === 'Active' ? '???' : '?? ?')}</span>
  </div>
  <div>
    <h4 class="font-bold text-lg text-primary">${tr(f.name_en, f.name_ar)}</h4>
    <span class="inline-block mt-1 text-xs px-2 py-0.5 bg-secondary/10 text-secondary rounded-full font-medium">
      ${tr(label.en, label.ar)}
    </span>
    <p class="text-xs text-on-surface-variant mt-2 line-clamp-2">${tr(f.desc, f.desc)}</p>
  </div>
  <div class="flex justify-between items-center mt-auto pt-4 border-t border-outline-variant/30 text-xs text-on-surface-variant font-medium">
    <span>?? ${f.deptsCount} ${tr('Active Specialties', '???? ?')}</span>
    <button onclick="window.location.href='/login.html'" class="px-4 py-2 bg-primary/10 text-primary rounded-xl font-label-sm hover:bg-primary hover:text-on-primary transition-all">
      ${tr('Enter Portal', '???? ?')}
    </button>
  </div>
</div>
`;
}).join('');
};

// Bind filter events
document.getElementById('searchBtn').onclick = filterAndRender;
filterAndRender();
}

// ===== DEPARTMENT WORKSPACE CORE UI =====
function renderDepartmentWorkspace(el, page) {
  const item = NAV_ITEMS[page];
  if (!item) return;

  const isIcuOrSurgery = [18, 23].includes(page);
  const isHighRisk = isIcuOrSurgery ? 'danger' : 'warning';
  const riskLabel = isIcuOrSurgery ? tr('High Risk', '???? ?') : tr('Medium Risk', '???? ?');

  // Check facility-specific restrictions
  const isPolyclinic = facilityType === 'polyclinic';
  const isHealthUnit = facilityType === 'health_unit';

  if (isIcuOrSurgery && (isPolyclinic || isHealthUnit)) {
    el.innerHTML = `
      <div class="flex flex-col items-center justify-center p-xl">
        <div class="error-card-premium max-w-md w-full text-center">
          <span class="material-symbols-outlined error-card-icon text-4xl">block</span>
          <h3 class="text-lg font-bold mt-4">${tr('Access Denied', '?? ?')}</h3>
          <p class="text-sm mt-2">${tr('ICU and Surgical services are not available at this facility type.', '

```

```
????? ??????? ??????? ??????? ?? ?????? ?? ??? ?????? ?? ??????? ??????.'})</p>
    </div>
  </div>
  `;
  return;
}

// Filter allowed encounter types for the current facility
const allowedEncounters = window.getEncounterTypesForFacility(facilityType);

el.innerHTML = `
  <!-- Facility Context Header -->
  <div class="flex justify-between items-center mb-md p-4 bg-surface-container-low rounded-xl border border-
outline-variant/30">
    <div>
      <h3 class="text-sm font-bold text-primary">${tr('Facility Context', '???? ??????')}</h3>
      <p class="text-xs text-on-surface-variant">
        ? ${tr('Network: King Fahad Medical City', '?????: -')} |
        ? ${tr('Type: ', '?????: ')} ${escapeHTML(facilityType.replace('_', ' ').toUpperCase())}
      </p>
    </div>
    <span class="badge badge-tenant-context font-bold">
      ${item.icon} ${tr(item.en, item.ar)}
    </span>
  </div>

  <!-- Department Overview Grid -->
  <div class="grid grid-cols-1 md:grid-cols-3 gap-md mb-lg">
    <div class="glass-card-premium p-6 rounded-2xl">
      <h4 class="font-bold text-md text-primary mb-2">? ${tr('Overview', '???? ?????')}</h4>
      <p class="text-xs text-on-surface-variant">
        ${tr('Clinical services, diagnostics, and patient care management for the ', '??????? ???????? ?????
????? ?????? ?????? ?????? ')}
        <strong>${tr(item.en, item.ar)}</strong>.
      </p>
    </div>
    <div class="glass-card-premium p-6 rounded-2xl">
      <h4 class="font-bold text-md text-primary mb-2">? ${tr('Requirements', '????????? ?????????')}</h4>
      <p class="text-xs text-on-surface-variant">? ${tr('Appointments Required: Yes', '??? ?????? ??????: ???
')}</p>
      <p class="text-xs text-on-surface-variant">?? ${tr('Consent Forms Required: Yes', '????? ???????? ?????
: ???')}</p>
    </div>
    <div class="glass-card-premium p-6 rounded-2xl">
      <h4 class="font-bold text-md text-primary mb-2">?? ${tr('Compliance & Risk', '????????? ?????? ??????')}
    </h4>
      <span class="badge badge-${isHighRisk}">${riskLabel}</span>
      <p class="text-[11px] text-on-surface-variant mt-2">? ${tr('PDPL Protected & CBAHI Audited', '???? ???
??? ?????? ?????? ?????????? ?????????')}</p>
    </div>
  </div>

  <!-- Appointments & Patient Queue Board -->
  <div class="grid grid-cols-1 md:grid-cols-2 gap-md mb-lg">
```



```
<!-- Appointment Booking Preview -->
<div class="glass-card-premium p-6 rounded-2xl border border-primary/10">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Book Appointment (Simulation)', '??? ??? (??
???)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Simulate booking a new clinical appointment. Final booking is disabled in this phase.', '????
? ??? ??? ??? ????. ????? ?????? ??? ? ? ? ? ???????.')}
  </p>
  <div class="space-y-3 text-xs">
    <div class="form-group">
      <label class="font-bold">${tr('Appointment Type', '??? ?????')}</label>
      <select id="apptTypeSelect" class="form-input text-xs">
        <!-- Populated dynamically -->
      </select>
    </div>
    <div class="form-group">
      <label class="font-bold">${tr('Preferred Date', '????? ?????')}</label>
      <input type="date" class="form-input text-xs" value="2026-06-30" disabled>
    </div>
    <button class="btn btn-secondary text-xs w-full" disabled>${tr('Request Booking (Disabled)', '??? ??
??? (??? ?????')}</button>
  </div>
</div>

<!-- Patient Queue Board -->
<div class="glass-card-premium p-6 rounded-2xl border border-primary/10">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Patient Queue Board', '??? ????? ?????')}</h
4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Active waiting queue for the current department. Launch encounter to start consultation.', '??
??? ?????? ????? ????? ??????. ??? ?????? ????? ???????.')}
  </p>
  <div class="overflow-x-auto">
    <table class="data-table text-[11px]">
      <thead>
        <tr>
          <th>${tr('Queue No', '??? ?????')}</th>
          <th>${tr('Type', '????')}</th>
          <th>${tr('Status', '????')}</th>
          <th>${tr('Priority', '????')}</th>
          <th>${tr('Wait Time', '????')}</th>
        </tr>
      </thead>
      <tbody id="deptQueueBody">
        <!-- Populated dynamically -->
      </tbody>
    </table>
  </div>
</div>

<!-- Encounters & Clinical Visits Initiation -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Encounters & Clinical Visits', '????? ??????
???)')}
```

```
? ???????')}</h4>
    <p class="text-xs text-on-surface-variant mb-4">
        ${tr('Initiate a simulated clinical encounter for triage, assessment, and care planning.', '??? ?????
????? ?????? ?????? ?????? ?????? ?????? ?????? ??????')}
    </p>
    <div class="grid grid-cols-1 md:grid-cols-3 gap-md items-end">
        <div class="form-group">
            <label class="text-xs font-bold text-on-surface-variant">${tr('Select Patient', '???? ??????')}</lab
el>
            <select id="encounterPatientSelect" class="form-input text-xs">
                <option value="1">Faisal Al-Otaibi (MRN-882713)</option>
                <option value="2">Sarah Al-Ghamdi (MRN-449128)</option>
                <option value="3">Abdulrahman Al-Shahri (MRN-773129)</option>
            </select>
        </div>
        <div class="form-group">
            <label class="text-xs font-bold text-on-surface-variant">${tr('Encounter Type', '??? ??????')}</lab
el>
            <select id="encounterTypeSelect" class="form-input text-xs">
                ${allowedEncounters.map(enc => `<option value="${enc.id}">${tr(enc.name_en, enc.name_ar)}</option>
`).join('')}
            </select>
        </div>
        <button id="initiateEncounterBtn" class="btn btn-primary text-xs py-2">
            ? ${tr('Initiate Clinical Encounter', '??? ?????? ??????')}
        </button>
    </div>
</div>

<!-- Bento Grid of Workspace Placeholders -->
<div class="grid grid-cols-1 md:grid-cols-3 gap-md">
    <!-- Services Panel -->
    <div class="glass-card-premium p-6 rounded-2xl col-span-1 md:col-span-2">
        <h4 class="font-bold text-md text-primary mb-4">? ${tr('Available Services', '?????? ?????? ??????')}
    </h4>
        <div class="flex flex-wrap gap-2">
            <span class="badge badge-info">${tr('General Consultation', '?????? ??????')}</span>
            <span class="badge badge-info">${tr('Follow-up Visit', '????? ??????')}</span>
            <span class="badge badge-info">${tr('Diagnostic Assessment', '????? ??????')}</span>
            <span class="badge badge-info">${tr('Clinical Procedure', '????? ?????? ??????')}</span>
        </div>
    </div>

    <!-- Staff Panel -->
    <div class="glass-card-premium p-6 rounded-2xl">
        <h4 class="font-bold text-md text-primary mb-4">???? ${tr('On-Duty Staff', '?????? ?????? ??????')}</h4
>
        <div class="space-y-3">
            <div class="flex justify-between text-xs font-medium text-on-surface-variant">
                <span>Dr. Sarah Ahmed</span>
                <span class="badge badge-success">${tr('Available', '????')}</span>
            </div>
            <div class="flex justify-between text-xs font-medium text-on-surface-variant">
                <span>Nurse Khalid Ali</span>
```

```

        <span class="badge badge-success">${tr('On Duty', '????')}</span>
    </div>
</div>
</div>

<!-- Active Orders Placeholder -->
<div class="glass-card-premium p-6 rounded-2xl">
    <h4 class="font-bold text-md text-primary mb-4">? ${tr('Active Orders', '?????? ?????')}</h4>
    <div class="empty-state-card text-center p-4">
        <span class="material-symbols-outlined text-2xl text-on-surface-variant">receipt_long</span>
        <p class="text-xs text-on-surface-variant mt-2">${tr('No active orders for this department', '?? ???
? ????? ??? ???? ????')}</p>
    </div>
</div>

<!-- Recent Results Placeholder -->
<div class="glass-card-premium p-6 rounded-2xl">
    <h4 class="font-bold text-md text-primary mb-4">? ${tr('Recent Results', '?? ?????')}</h4>
    <div class="empty-state-card text-center p-4">
        <span class="material-symbols-outlined text-2xl text-on-surface-variant">analytics</span>
        <p class="text-xs text-on-surface-variant mt-2">${tr('No recent diagnostic results', '?? ??? ????
????? ????')}</p>
    </div>
</div>

<!-- Quality Indicators -->
<div class="glass-card-premium p-6 rounded-2xl">
    <h4 class="font-bold text-md text-primary mb-4">? ${tr('Quality Metrics', '????? ?????')}</h4>
    <div class="space-y-2">
        <div class="flex justify-between text-xs font-medium text-on-surface-variant">
            <span>${tr('Hand Hygiene Compliance', '?????? ?????')}</span>
            <span class="font-bold text-success">98%</span>
        </div>
        <div class="flex justify-between text-xs font-medium text-on-surface-variant">
            <span>${tr('CBAHI Standards Met', '????? ?????')}</span>
            <span class="font-bold text-success">100%</span>
        </div>
    </div>
</div>
</div>
`;

// Appointments & Patient Queue Logic
const apptTypeSelect = document.getElementById('apptTypeSelect');
const allowedApptTypes = window.getAppointmentTypesForFacility(facilityType);
apptTypeSelect.innerHTML = allowedApptTypes.map(t => `<option value="${t.id}">${tr(t.name_en, t.name_ar)}</o
ption>`).join('');

const queueBody = document.getElementById('deptQueueBody');
// Filter queue for current department page (or show all if empty for demo purposes, but let's filter correc
tly)
const filteredQueue = window.MOCK_QUEUE_DATA.filter(q => q.deptPage === page);

if (filteredQueue.length === 0) {

```

```

        queueBody.innerHTML = `<tr><td colspan="5" class="text-center p-4 text-on-surface-variant">${tr('No patients waiting', '?? ??? ???? ?? ??????? ??????')}</td></tr>`;
    } else {
        queueBody.innerHTML = filteredQueue.map(q => {
            const priorityInfo = window.getTriagePriorityBadges(q.priority);
            const apptTypeObj = window.APPOINTMENT_TYPES[q.type.toUpperCase()] || { name_en: q.type, name_ar: q.type };
            const statusObj = window.QUEUE_STATUSES[q.status.toUpperCase()] || { name_en: q.status, name_ar: q.status };
            return `
                <tr>
                    <td><strong>${q.queueNo}</strong></td>
                    <td>${tr(apptTypeObj.name_en, apptTypeObj.name_ar)}</td>
                    <td><span class="badge badge-warning">${tr(statusObj.name_en, statusObj.name_ar)}</span></td>
                    <td><span class="badge badge-${priorityInfo.class}">${tr(priorityInfo.label_en, priorityInfo.label_ar)}</span></td>
                    <td>${q.waitTimeMin} mins</td>
                </tr>
            `;
        }).join('');
    }

    // Bind Encounter Initiation Event
    document.getElementById('initiateEncounterBtn').onclick = () => {
        const patientId = document.getElementById('encounterPatientSelect').value;
        const encounterTypeId = document.getElementById('encounterTypeSelect').value;
        renderEncounterWorkspace(el, page, patientId, encounterTypeId);
    };
}

// ===== ENCOUNTER WORKSPACE CORE UI =====
function renderEncounterWorkspace(el, page, patientId, encounterTypeId) {
    const item = NAV_ITEMS[page];
    const journeySteps = window.getEncounterJourneySteps(encounterTypeId);

    // Mock patients list
    const patients = [
        '1': { name: 'Faisal Al-Otaibi', mrn: 'MRN-882713', gender: tr('Male', '???'), age: 45, blood: 'O+' },
        '2': { name: 'Sarah Al-Ghamdi', mrn: 'MRN-449128', gender: tr('Female', '????'), age: 32, blood: 'A+' },
        '3': { name: 'Abdulrahman Al-Shahri', mrn: 'MRN-773129', gender: tr('Male', '???'), age: 58, blood: 'B-' }
    ];
    const patient = patients[patientId] || patients['1'];

    el.innerHTML = `
        <!-- Encounter Header -->
        <div class="flex justify-between items-center mb-md p-4 bg-surface-container-low rounded-xl border border-outline-variant/30">
            <div>
                <h3 class="text-sm font-bold text-primary">? ${tr('Active Clinical Encounter', '?????? ?????? ??????')}</h3>
                <p class="text-xs text-on-surface-variant">
                    ? ${tr('Facility: ', '??????? : ')} ${escapeHTML(facilityType.toUpperCase())} |
                    ? ${tr('Department: ', '????? : ')} ${tr(item.en, item.ar)}
                </p>
            </div>
        </div>
    `;
}

```

```

</div>
<button id="closeEncounterBtn" class="btn btn-secondary text-xs py-1.5 px-4">
  ?? ${tr('Close Encounter', '????? ?????? ??????')}
</button>
</div>

<!-- Security & Compliance Warning Notice -->
<div class="p-4 bg-warning/10 border-l-4 border-warning text-warning-container rounded-xl mb-lg flex gap-3
items-start">
  <span class="material-symbols-outlined text-warning text-2xl">shield</span>
  <div>
    <h4 class="font-bold text-sm text-warning">${tr('Clinical Simulation & Privacy Notice', '?????? ??????
?????? ??????')}</h4>
    <p class="text-xs text-on-surface-variant">
      ${tr('This is a read-only simulated clinical environment. In accordance with Saudi PDPL & CBAHI regu
lations, no real patient health information (PHI) is stored or transmitted.', '??? ???? ?????? ?????? ?????? ??
??? ????. ?????? ?? ?????? ???? ?????? ?????? (PDPL) ?????? ?????? ?? ??? ??? ?? ??? ?? ?????? ?????? ?
????? ??????.')}
    </p>
  </div>
</div>

<!-- Patient Card -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Patient Demographics', '?????? ??????')}</h4>
  <div class="grid grid-cols-2 md:grid-cols-5 gap-md text-xs font-medium text-on-surface-variant">
    <div><strong>${tr('Name', '????')}</strong> ${escapeHTML(patient.name)}</div>
    <div><strong>MRN:</strong> ${escapeHTML(patient.mrn)}</div>
    <div><strong>${tr('Gender', '????')}</strong> ${escapeHTML(patient.gender)}</div>
    <div><strong>${tr('Age', '????')}</strong> ${patient.age}</div>
    <div><strong>${tr('Blood Group', '????? ?????')}</strong> ${patient.blood}</div>
  </div>
</div>

<!-- Journey Timeline -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg">
  <h4 class="font-bold text-md text-primary mb-6">? ${tr('Clinical Journey Timeline', '???? ?????? ??????
??')}</h4>
  <div class="flex flex-col md:flex-row gap-lg justify-between relative">
    ${journeySteps.map(step => `
      <div class="flex-1 flex flex-col items-center text-center relative z-10">
        <div class="w-8 h-8 rounded-full bg-primary text-on-primary flex items-center justify-center font-
bold mb-2 shadow-md">
          ${step.step}
        </div>
        <h5 class="text-xs font-bold text-primary">${tr(step.name_en, step.name_ar)}</h5>
        <span class="text-[10px] text-on-surface-variant mt-1">${tr('Completed', '????')}</span>
      </div>
    `).join('')}
  </div>
</div>

<!-- Bento Grid of Clinical Worksheets -->
<div class="grid grid-cols-1 md:grid-cols-2 gap-md mb-lg">

```

```
<!-- Nurse Assessment (Triage) -->
<div class="glass-card-premium p-6 rounded-2xl">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Nursing Vitals & Assessment', '?????? ??????
? ???????? ??????')}</h4>
  <div class="grid grid-cols-2 gap-md text-xs">
    <div class="form-group"><label>BP (Blood Pressure)</label><input class="form-input" value="120/80" d
isabled></div>
    <div class="form-group"><label>Heart Rate (bpm)</label><input class="form-input" value="72" disabled
></div>
    <div class="form-group"><label>Temp (°C)</label><input class="form-input" value="37.0" disabled></di
v>
    <div class="form-group"><label>Respiratory Rate</label><input class="form-input" value="16" disabled
></div>
  </div>
  <button class="btn btn-secondary text-xs mt-4 w-full" disabled>${tr('Save Draft (Simulated)', '??? ???
??? (?????)')}</button>
</div>

<!-- Physician Consultation (SOAP) -->
<div class="glass-card-premium p-6 rounded-2xl">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Physician Notes (SOAP)', '?????? ?????? ?????
????')}</h4>
  <div class="space-y-3 text-xs">
    <div class="form-group">
      <label class="font-bold">Subjective</label>
      <textarea class="form-input" rows="2" disabled>${tr('Patient complains of mild chest tightness.',
'????? ?????? ?? ??? ????? ?? ?????.')}</textarea>
    </div>
    <div class="form-group">
      <label class="font-bold">Objective</label>
      <textarea class="form-input" rows="2" disabled>${tr('Lungs clear, heart sounds normal.', '??????
????????? ?????? ?????? ??????')}</textarea>
    </div>
  </div>
</div>

<!-- Clinical Orders Creator (Draft Workspace) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Clinical Orders Creator (Simulated Draft)', '???
? ?????? ?????? (????? ??????)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Create draft clinical orders. In accordance with safety rules, high-risk orders show validation
requirements.', '????? ?????? ?????? ?????? ?????? ?? ?????? ??????? ?????? ??????? ?????? ???????.'
)}
  </p>

  <div class="grid grid-cols-1 md:grid-cols-3 gap-md items-end mb-lg">
    <div class="form-group">
      <label class="text-xs font-bold text-on-surface-variant">${tr('Order Type', '??? ?????')}</label>
      <select id="orderTypeSelect" class="form-input text-xs">
        <!-- Populated dynamically -->
      </select>
    </div>
  </div>
</div>
```

```
<div class="form-group">
  <label class="text-xs font-bold text-on-surface-variant">${tr('Select Item', '???? ?????')}</label>
  <select id="orderItemSelect" class="form-input text-xs">
    <!-- Populated dynamically -->
  </select>
</div>
<button id="addOrderDraftBtn" class="btn btn-primary text-xs py-2">
  ? ${tr('Add to Drafts', '????? ???????')}
</button>
</div>

<!-- Safety Warnings Box -->
<div id="orderSafetyWarnings" class="p-3 bg-error/10 border-l-4 border-error text-xs text-on-error-container rounded-xl mb-md hidden">
  <!-- Warnings populated dynamically -->
</div>

<!-- Draft Orders List -->
<h5 class="font-bold text-xs text-primary mb-2">${tr('Draft Orders List', '????? ??????? ???????')}</h5>
<div id="draftOrdersList" class="space-y-2 mb-md">
  <div class="empty-state-card text-center p-4">
    <p class="text-xs text-on-surface-variant">${tr('No draft orders added yet', '?? ???? ????? ?? ??????
?? ??????')}</p>
  </div>
</div>

<div class="flex justify-end gap-md pt-4 border-t border-outline-variant/30">
  <button class="btn btn-secondary text-xs" disabled>${tr('Save as Draft', '??? ??????')}</button>
  <button class="btn btn-primary text-xs" disabled>${tr('Finalize & Submit (Disabled)', '?????? ?????? (
??? ??????')}</button>
</div>
</div>

<!-- Results Inbox (Simulated Workspace) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Results Inbox (Simulated)', '????? ??????? (????
??)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Review diagnostic and clinical results for this encounter. Abnormal findings require clinical re
view.', '?????? ??????? ?????????? ?????????? ??? ????????. ??????? ?? ????????? ????? ??????? ??????????')}
  </p>

  <div class="flex gap-2 border-b border-outline-variant/30 pb-2 mb-4" id="resultsInboxTabs">
    <button class="px-4 py-1.5 text-xs font-bold text-primary border-b-2 border-primary" data-tab="lab">${
tr('Laboratory', '????????')}</button>
    <button class="px-4 py-1.5 text-xs font-bold text-on-surface-variant" data-tab="radiology">${tr('Radio
logy', '????????')}</button>
    <button class="px-4 py-1.5 text-xs font-bold text-on-surface-variant" data-tab="procedure">${tr('Proce
dures', '????????')}</button>
    <button class="px-4 py-1.5 text-xs font-bold text-on-surface-variant" data-tab="consultation">${tr('Co
nsultations', '????????')}</button>
  </div>

  <div id="resultsInboxContent" class="space-y-3">
```

```
<!-- Populated dynamically -->
</div>
</div>

<!-- Clinical Documentation Workspace (Simulated Draft) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">?? ${tr('Clinical Documentation (Draft Note)', '??????
?????? (????? ????????)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Complete the clinical documentation. No final signature is enabled in this phase.', '????? ?????
?? ??????? ????????. ??????? ??????? ?? ????? ?? ?? ???????.')}
  </p>

  <div class="space-y-4 text-xs">
    <div class="form-group">
      <label class="font-bold text-primary">Subjective (S)</label>
      <textarea class="form-input" rows="2" placeholder="${tr('Chief complaint, history of present illness
...', '????? ?????????? ??????? ?????? ??????..')}>" disabled></textarea>
    </div>
    <div class="form-group">
      <label class="font-bold text-primary">Objective (O)</label>
      <textarea class="form-input" rows="2" placeholder="${tr('Physical exam findings, vitals review...',
'????? ?????? ?????????? ?????? ?????????? ??????????..')}>" disabled></textarea>
    </div>
    <div class="form-group">
      <label class="font-bold text-primary">Assessment (A)</label>
      <textarea class="form-input" rows="2" placeholder="${tr('Clinical impression, differential diagnosis
...', '????????? ?????????? ??????? ??????????..')}>" disabled></textarea>
    </div>
    <div class="form-group">
      <label class="font-bold text-primary">Plan (P)</label>
      <textarea class="form-input" rows="2" placeholder="${tr('Treatment plan, referrals, follow-up schedu
ling...', '??? ??????? ?????????? ?????? ??????????..')}>" disabled></textarea>
    </div>
  </div>

  <div class="flex justify-end gap-md pt-6 border-t border-outline-variant/30 mt-4">
    <button class="btn btn-secondary text-xs" disabled>${tr('Save Progress', '??? ??????')}</button>
    <button class="btn btn-primary text-xs" disabled>${tr('Sign & Lock Document (Disabled)', '????????? ???
???? (??? ??????')}</button>
  </div>
</div>

<!-- Clinical Pharmacy & FEFO Inventory Preview (Simulated Workspace) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">? ${tr('Clinical Pharmacy & FEFO Inventory (Simulated)',
'????????? ?????????? ?????? FEFO (???????)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Review medication safety, allergy warnings, and FEFO (First Expired, First Out) batch sorting. D
ispensing and stock decrement are disabled.', '?????? ?????? ??????? ??????? ?????????? ?????? ???????
??? ?????? (FEFO). ?????? ?????? ?????? ?????? ?????????? ??? ??????..')}
  </p>

  <div class="grid grid-cols-1 md:grid-cols-3 gap-md items-end mb-lg">
```



```
<div class="form-group">
  <label class="text-xs font-bold text-on-surface-variant">${tr('Select Medication', '???? ?????')}</
label>
  <select id="pharmacyMedSelect" class="form-input text-xs">
    <option value="PARA500">Paracetamol 500mg</option>
    <option value="AM0500">Amoxicillin 500mg (Penicillin)</option>
    <option value="MORPHINE">Morphine (Controlled Substance)</option>
  </select>
</div>
<button id="viewFefoBtn" class="btn btn-secondary text-xs py-2">
  ? ${tr('View FEFO Batches', '??? ????? FEFO')}
</button>
<button id="previewDispenseBtn" class="btn btn-primary text-xs py-2">
  ? ${tr('Preview Dispense', '????? ?????')}
</button>
</div>

<!-- Safety & Controlled Alerts -->
<div id="pharmacySafetyAlerts" class="p-3 bg-error/10 border-l-4 border-error text-xs text-on-error-cont
ainer rounded-xl mb-md hidden">
  <!-- Warnings populated dynamically -->
</div>

<!-- FEFO Batches Table -->
<div id="fefoBatchesContainer" class="mb-md hidden">
  <h5 class="font-bold text-xs text-primary mb-2">${tr('FEFO Batch Expiry Queue', '????? ?????? ??????
(FEFO)')}</h5>
  <div class="overflow-x-auto">
    <table class="data-table text-xs">
      <thead>
        <tr>
          <th>${tr('Batch Code', '??? ?????')}</th>
          <th>${tr('Expiry Date', '????? ??????')}</th>
          <th>${tr('Stock Available', '??????? ??????')}</th>
          <th>${tr('FEFO Rank', '????? ??????')}</th>
          <th>${tr('Status', '?????')}</th>
        </tr>
      </thead>
      <tbody id="fefoBatchesBody">
        <!-- Populated dynamically -->
      </tbody>
    </table>
  </div>
</div>

<!-- Dispense Preview Card -->
<div id="dispensePreviewCard" class="p-4 bg-surface-container-low rounded-xl border border-outline-variant/10 hidden mb-md">
  <h5 class="font-bold text-xs text-primary mb-2">? ${tr('Dispense Preview', '????? ??? ?????')}</h5>
  <div class="grid grid-cols-2 gap-md text-xs text-on-surface-variant">
    <div><strong>${tr('Medication', '?????')}</strong> <span id="dispenseMedName">-</span></div>
    <div><strong>${tr('Selected Batch', '????? ??????')}</strong> <span id="dispenseBatchCode">-</span>
an></div>
    <div><strong>${tr('Status', '?????')}</strong> <span class="badge badge-warning">${tr('Ready for P
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review', '???? ??????')}</span></div>
    </div>
</div>

<div class="flex justify-end gap-md pt-4 border-t border-outline-variant/30">
    <button class="btn btn-secondary text-xs" disabled>${tr('Approve Review', '????? ??????')}</button>
    <button class="btn btn-primary text-xs" disabled>${tr('Finalize Dispense (Disabled)', '??? ????? (???
?????)')}</button>
</div>
</div>

<!-- Clinical Billing & Claims Preview (Simulated Workspace) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
    <h4 class="font-bold text-md text-primary mb-4">? ${tr('Clinical Billing & Claims Preview (Simulated)',
'??????? ???????? ???????? ?????????? (?????)')}</h4>
    <p class="text-xs text-on-surface-variant mb-4">
        ${tr('Review charge capture, insurance eligibility, and claim drafts. Invoicing, financial posting, an
d NPHIES submissions are disabled.', '????? ?????? ?????? ?????? ???????? ???????? ??????????. ???????? ??????????
??????? ???????? ?????? ?????? ????.')}</p>

    <div class="grid grid-cols-1 md:grid-cols-3 gap-md items-end mb-lg">
        <div class="form-group">
            <label class="text-xs font-bold text-on-surface-variant">${tr('Select Insurance Payer', '???? ??? ??
?????')}</label>
            <select id="billingPayerSelect" class="form-input text-xs">
                <!-- Populated dynamically -->
            </select>
        </div>
        <button id="viewEligibilityBtn" class="btn btn-secondary text-xs py-2">
            ? ${tr('Check Eligibility', '????? ?? ??????')}</button>
        <button id="previewClaimBtn" class="btn btn-primary text-xs py-2">
            ? ${tr('Preview Claim Draft', '????? ?????? ?????????')}</button>
    </div>

    <!-- Eligibility Details -->
    <div id="eligibilityResultBox" class="p-4 bg-surface-container-low rounded-xl border border-outline-vari
ant/10 hidden mb-md">
        <h5 class="font-bold text-xs text-primary mb-2">? ${tr('Eligibility Status', '???? ??????')}</h5>
        <div class="grid grid-cols-2 gap-md text-xs text-on-surface-variant">
            <div><strong>${tr('Payer', '??? ??????')}</strong> <span id="eligPayerName">-</span></div>
            <div><strong>${tr('Status', '?????')}</strong> <span class="badge badge-success">${tr('Active Cove
rage', '??????? ?????')}</span></div>
            <div><strong>${tr('Copayment', '???? ??????')}</strong> <span id="eligCopay">-</span></div>
            <div><strong>${tr('Max Limit', '???? ??????')}</strong> <span id="eligMaxLimit">-</span></div>
        </div>
    </div>

    <!-- Charge Capture Preview -->
    <h5 class="font-bold text-xs text-primary mb-2">${tr('Charge Capture Preview', '????? ?????? ?????????')}</h5>
    <div id="chargeCaptureList" class="space-y-2 mb-md">
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<!-- Populated dynamically -->
</div>

<!-- Claim Draft Preview -->
<div id="claimPreviewCard" class="p-4 bg-surface-container-low rounded-xl border border-outline-variant/
10 hidden mb-md">
  <h5 class="font-bold text-xs text-primary mb-2">? ${tr('Claim Draft Preview', '????? ?????????')}
</h5>
  <div class="grid grid-cols-2 gap-md text-xs text-on-surface-variant mb-3">
    <div><strong>${tr('Total Amount', '????? ?????????')}</strong> <span id="claimTotalAmount">-</span>
  </div>
    <div><strong>${tr('NPHIES Status', '???? ?????')}</strong> <span class="badge badge-warning">${tr('D
raft (Submission Disabled)', '????? (??????? ?????')}</span></div>
  </div>
  <div id="claimValidationBox" class="p-2 bg-warning/10 text-[11px] text-warning rounded border-l-2 bord
er-warning">
    <!-- Validation warnings -->
  </div>
</div>

<!-- NPHIES Readiness Checklist -->
<div class="mb-md p-4 bg-surface-container-low rounded-xl border border-outline-variant/10">
  <h5 class="font-bold text-xs text-primary mb-2">? ${tr('NPHIES Readiness Checklist', '????? ?????? ??
?')}</h5>
  <ul class="space-y-1 text-xs text-on-surface-variant">
    <li>?? ${tr('Facility NPHIES ID Configured', '???? ????? ?????? ?????')}</li>
    <li>?? ${tr('Valid Payer ID Selected', '???? ??? ?????? ?????? ?????')}</li>
    <li>? ${tr('Encounter Final Signature (Missing)', '??????? ?????? ?????? (?????')}</li>
    <li>? ${tr('Final Invoice Authorization (Blocked)', '????? ?????? ?????? (?????')}</li>
  </ul>
</div>

<div class="flex justify-end gap-md pt-4 border-t border-outline-variant/30">
  <button class="btn btn-secondary text-xs" disabled>${tr('Post to Ledger', '????? ?????')}</button>
  <button class="btn btn-primary text-xs" disabled>${tr('Finalize Invoice & Submit Claim (Disabled)', '?
???? ?????? ?????? ?????? (??? ?????')}</button>
</div>
</div>

<!-- Security & Audit Log Preview (Simulated Workspace) -->
<div class="glass-card-premium p-6 rounded-2xl mb-lg border border-primary/20">
  <h4 class="font-bold text-md text-primary mb-4">?? ${tr('Security & Audit Log Preview (Simulated)', '??
?? ?????? ??? ?????? (?????)')}</h4>
  <p class="text-xs text-on-surface-variant mb-4">
    ${tr('Select an enterprise role to preview RBAC (Role-Based Access Control) permissions and simulated
audit events for this encounter.', '???? ?????? ?????? ?????? ?????? ?????? ?????? ?????? ??? ?????? (RBAC) ?????
? ?????? ?????? ??????')}
  </p>
  <div class="grid grid-cols-1 md:grid-cols-2 gap-md items-end mb-lg">
    <div class="form-group">
      <label class="text-xs font-bold text-on-surface-variant">${tr('Select Active Role', '???? ?????? ???
?')}</label>
      <select id="securityRoleSelect" class="form-input text-xs">
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        <option value="PHYSICIAN">Physician (???? ?????)</option>
        <option value="PHARMACIST">Pharmacist (????? ?????)</option>
        <option value="BILLING_OFFICER">Billing Officer (???? ?????)</option>
        <option value="READ_ONLY_AUDITOR">Auditor (???? ?????)</option>
    </select>
</div>
<button id="triggerAuditLogBtn" class="btn btn-secondary text-xs py-2">
    ? ${tr('Generate Audit Log Entry', '????? ??? ???????')}
</button>
</div>

<!-- Audit Log Preview Output -->
<div id="auditLogOutputBox" class="p-4 bg-surface-container-low rounded-xl border border-outline-variant
/10 hidden mb-md">
    <h5 class="font-bold text-xs text-primary mb-2">? ${tr('Simulated Audit Event (JSON)', '??? ??????? ??
????? (JSON)')}</h5>
    <pre id="auditJsonOutput" class="text-[10px] font-mono text-on-surface-variant whitespace-pre-wrap bg-
surface-container/50 p-3 rounded-lg border border-outline-variant/20"></pre>
</div>
</div>
`;

// Bind Close Event
document.getElementById('closeEncounterBtn').onclick = () => {
    renderDepartmentWorkspace(el, page);
};

// Populate Order Types
const allowedOrderTypes = window.getOrderTypesForEncounter(facilityType, encounterTypeId);
const typeSelect = document.getElementById('orderTypeSelect');
typeSelect.innerHTML = allowedOrderTypes.map(t => `<option value="${t.id}">${tr(t.name_en, t.ar || t.name_ar
)}</option>`).join('');

const itemSelect = document.getElementById('orderItemSelect');
const warningsBox = document.getElementById('orderSafetyWarnings');
const draftList = document.getElementById('draftOrdersList');
const draftOrders = [];

const updateItemsAndWarnings = () => {
    const selectedType = typeSelect.value;
    const items = window.MOCK_ORDER_CATALOG[selectedType] || [];
    itemSelect.innerHTML = items.map(item => `<option value="${item.code}">${tr(item.name_en, item.name_ar)}</
option>`).join('');

    updateWarnings();
};

const updateWarnings = () => {
    const selectedType = typeSelect.value;
    const selectedItemCode = itemSelect.value;
    if (!selectedItemCode) {
        warningsBox.classList.add('hidden');
        return;
    }
}

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const warnings = window.getOrderSafetyWarnings(selectedType, selectedItemCode);
if (warnings.length > 0) {
  warningsBox.innerHTML = warnings.map(w => `<p>?? ${tr(w.text_en, w.text_ar)}</p>`).join('');
  warningsBox.classList.remove('hidden');
} else {
  warningsBox.classList.add('hidden');
}
};

typeSelect.onChange = updateItemsAndWarnings;
itemSelect.onChange = updateWarnings;

// Initial population
updateItemsAndWarnings();

// Add to Drafts Event
document.getElementById('addOrderDraftBtn').onclick = () => {
  const selectedType = typeSelect.value;
  const selectedItemCode = itemSelect.value;
  const catalog = window.MOCK_ORDER_CATALOG[selectedType] || [];
  const itemObj = catalog.find(x => x.code === selectedItemCode);

  if (!itemObj) return;

  // Check Guarding rules
  const hasPermission = window.canCreateOrderDraft(facilityType, page, selectedType);
  if (!hasPermission) {
    showToast(tr('Order type not allowed at this facility type', '??? ????? ?? ????? ?? ?? ?? ???????'), '
error');
    return;
  }

  draftOrders.push({ type: selectedType, code: itemObj.code, name_en: itemObj.name_en, name_ar: itemObj.name
_ar, risk: itemObj.risk });
  renderDraftList();
};

const renderDraftList = () => {
  if (draftOrders.length === 0) {
    draftList.innerHTML = `<div class="empty-state-card text-center p-4"><p class="text-xs text-on-surface-v
ariant">${tr('No draft orders added yet', '?? ??? ???? ?? ?????? ??????')}</p></div>`;
    return;
  }

  draftList.innerHTML = draftOrders.map((o, idx) => {
    const riskBadge = o.risk === 'high' ? `<span class="badge badge-danger">${tr('Consent Required', '????
????? ?????')}</span>` : '';
    return `
    <div class="flex justify-between items-center p-3 bg-surface-container-low rounded-xl border border-ou
tline-variant/10">
      <div>
        <p class="font-bold text-xs text-primary">${tr(o.name_en, o.name_ar)} (${o.code})</p>
        <span class="text-[10px] text-on-surface-variant font-medium uppercase">${o.type}</span>

```

```

    </div>
    <div class="flex gap-2">
      ${riskBadge}
      <span class="badge badge-warning">${tr('Draft Only', '????')}</span>
      <button class="text-error text-xs font-bold px-2 hover:bg-error/10 rounded" onclick="window._remov
eOrderDraft(${idx})">?</button>
    </div>
  </div>
  `;
  }).join('');
};

window._removeOrderDraft = (idx) => {
  draftOrders.splice(idx, 1);
  renderDraftList();
};

// Results Inbox Logic
const resultsContent = document.getElementById('resultsInboxContent');
const resultsTabs = document.getElementById('resultsInboxTabs');

const renderResultsTab = (tabId) => {
  // Update tab active styling
  resultsTabs.querySelectorAll('button').forEach(btn => {
    if (btn.dataset.tab === tabId) {
      btn.className = 'px-4 py-1.5 text-xs font-bold text-primary border-b-2 border-primary';
    } else {
      btn.className = 'px-4 py-1.5 text-xs font-bold text-on-surface-variant';
    }
  });
});

const results = window.MOCK_RESULTS[tabId] || [];
if (results.length === 0) {
  resultsContent.innerHTML = `
    <div class="empty-state-card text-center p-4">
      <p class="text-xs text-on-surface-variant">${tr('No results available for this category', '?? ??? ?
???? ???? ????? ?????')}</p>
    </div>
  `;
  return;
}

resultsContent.innerHTML = results.map(r => {
  const isAbnormal = r.abnormal;
  const abnormalBadge = isAbnormal ? `<span class="badge badge-danger">${tr('Abnormal Finding', '????? ?
????')}</span>` : '';
  const warnings = window.getAbnormalResultWarnings(tabId, r.code);
  const warningHtml = warnings.map(w => `
    <div class="p-2 bg-error/5 text-[11px] text-error rounded border-l-2 border-error mt-2">
      ?? ${tr(w.text_en, w.text_ar)}
    </div>
  `).join('');

  return `

```

```

<div class="p-4 bg-surface-container-low rounded-xl border border-outline-variant/10">
  <div class="flex justify-between items-start mb-2">
    <div>
      <h5 class="font-bold text-xs text-primary">${tr(r.name_en, r.name_ar)}</h5>
      <span class="text-[10px] text-on-surface-variant">${r.code}</span>
    </div>
    <div class="flex gap-2">
      ${abnormalBadge}
      <span class="badge badge-warning">${tr(r.status, r.status === 'Needs Review' ? '????? ??????' :
'??? ??????')}</span>
    </div>
  </div>
  <p class="text-xs text-on-surface font-medium">${escapeHTML(r.value)}</p>
  ${warningHtml}
  <div class="flex justify-end mt-2">
    <button class="btn btn-secondary text-[10px] py-1 px-3" disabled>${tr('Acknowledge Result', '?????
? ??????')}</button>
  </div>
</div>
`;
}).join('');
};

// Bind Tab Click Events
resultsTabs.querySelectorAll('button').forEach(btn => {
  btn.onclick = () => renderResultsTab(btn.dataset.tab);
});

// Initial tab render
renderResultsTab('lab');

// Clinical Pharmacy & FEFO Logic
const medSelect = document.getElementById('pharmacyMedSelect');
const viewFefoBtn = document.getElementById('viewFefoBtn');
const previewDispenseBtn = document.getElementById('previewDispenseBtn');
const pharmacyAlerts = document.getElementById('pharmacySafetyAlerts');
const fefoContainer = document.getElementById('fefoBatchesContainer');
const fefoBody = document.getElementById('fefoBatchesBody');
const dispenseCard = document.getElementById('dispensePreviewCard');
const dispenseMedName = document.getElementById('dispenseMedName');
const dispenseBatchCode = document.getElementById('dispenseBatchCode');

const updatePharmacyAlerts = () => {
  const medCode = medSelect.value;
  const safetyWarnings = window.getMedicationSafetyWarnings(medCode);
  const controlledWarnings = window.getControlledMedicationWarnings(medCode);
  const allWarnings = [...safetyWarnings, ...controlledWarnings];

  if (allWarnings.length > 0) {
    pharmacyAlerts.innerHTML = allWarnings.map(w => `<p?? ${tr(w.text_en, w.text_ar)}</p>`).join('');
    pharmacyAlerts.classList.remove('hidden');
  } else {
    pharmacyAlerts.classList.add('hidden');
  }
}

```

```

};

medSelect.onChange = () => {
  updatePharmacyAlerts();
  fefoContainer.classList.add('hidden');
  dispenseCard.classList.add('hidden');
};

viewFefoBtn.onclick = () => {
  const medCode = medSelect.value;
  const batches = window.getFefoBatchPreview(medCode);

  if (batches.length === 0) {
    fefoBody.innerHTML = `|<td colspan="5" class="text-center p-4">${tr('No batches available', '?? ????'
????? ??????')}</td></tr>`;
  } else {
    fefoBody.innerHTML = batches.map(b => `
      <tr>
        <td><strong>${b.batchCode}</strong></td>
        <td>${b.expiryDate}</td>
        <td>${b.quantity}</td>
        <td><span class="badge badge-info">Rank ${b.fefoRank}</span></td>
        <td><span class="badge badge-success">${tr('Available', '????')}</span></td>
      </tr>
    `).join('');
  }
  fefoContainer.classList.remove('hidden');
};

previewDispenseBtn.onclick = () => {
  const medCode = medSelect.value;
  const batches = window.getFefoBatchPreview(medCode);
  if (batches.length > 0) {
    dispenseMedName.textContent = medSelect.options[medSelect.selectedIndex].text;
    dispenseBatchCode.textContent = batches[0].batchCode; // Select earliest expiry batch (FEFO)
    dispenseCard.classList.remove('hidden');
  } else {
    showToast(tr('No batches available for dispense preview', '?? ????' ????? ?????? ?????? ??????'), 'error')
;
  }
};

// Initial alerts update
updatePharmacyAlerts();

// Clinical Billing & Claims Logic
const billingPayerSelect = document.getElementById('billingPayerSelect');
const viewEligibilityBtn = document.getElementById('viewEligibilityBtn');
const previewClaimBtn = document.getElementById('previewClaimBtn');
const eligibilityBox = document.getElementById('eligibilityResultBox');
const eligPayerName = document.getElementById('eligPayerName');
const eligCopay = document.getElementById('eligCopay');
const eligMaxLimit = document.getElementById('eligMaxLimit');

|  |

```



```

const chargeCaptureList = document.getElementById('chargeCaptureList');
const claimPreviewCard = document.getElementById('claimPreviewCard');
const claimTotalAmount = document.getElementById('claimTotalAmount');
const claimValidationBox = document.getElementById('claimValidationBox');

// Populate Payer Select
billingPayerSelect.innerHTML = window.MOCK_PAYERS.map(p => `<option value="${p.id}">${tr(p.name_en, p.name_ar)}</option>`).join('');

// Render Charge Capture Preview (initially empty or showing consultation)
const renderChargeCapture = () => {
  const charges = window.getChargeCapturePreview(encounterTypeId, []);
  if (charges.length === 0) {
    chargeCaptureList.innerHTML = `<div class="empty-state-card text-center p-4"><p class="text-xs text-on-surface-variant">${tr('No charges captured', '?? ??? ???? ???')}</p></div>`;
    return;
  }

  chargeCaptureList.innerHTML = charges.map(c => `
    <div class="flex justify-between items-center p-3 bg-surface-container-low rounded-xl border border-outline-variant/10">
      <div>
        <p class="font-bold text-xs text-primary">${tr(c.name_en, c.name_ar)}</p>
        <span class="text-[10px] text-on-surface-variant font-medium uppercase">${c.code}</span>
      </div>
      <span class="text-xs font-bold text-primary">${c.price} SAR</span>
    </div>
  `).join('');
};

renderChargeCapture();

viewEligibilityBtn.onclick = () => {
  const payerId = billingPayerSelect.value;
  const elig = window.getEligibilityPreview(payerId);
  eligPayerName.textContent = billingPayerSelect.options[billingPayerSelect.selectedIndex].text;
  eligCopay.textContent = `${elig.copayPercent}%`;
  eligMaxLimit.textContent = `${elig.maxLimit} SAR`;
  eligibilityBox.classList.remove('hidden');
  showToast(tr('Eligibility checked successfully (Simulated)', '?? ?????? ?? ?????? ????? (?????)'), 'success');
};

previewClaimBtn.onclick = () => {
  const charges = window.getChargeCapturePreview(encounterTypeId, []);
  const claim = window.getClaimDraftPreview(encounterTypeId, charges);
  claimTotalAmount.textContent = `${claim.totalAmount} SAR`;
  claimValidationBox.innerHTML = claim.warnings.map(w => `<p>?? ${w}</p>`).join('');
  claimPreviewCard.classList.remove('hidden');
  showToast(tr('Claim draft generated (Simulated)', '?? ?????? ?????? ?????? (?????)'), 'success');
};

// Security & Audit Logic
const securityRoleSelect = document.getElementById('securityRoleSelect');

```

```

const triggerAuditLogBtn = document.getElementById('triggerAuditLogBtn');
const auditOutputBox = document.getElementById('auditLogOutputBox');
const auditJsonOutput = document.getElementById('auditJsonOutput');

triggerAuditLogBtn.onclick = () => {
  const role = securityRoleSelect.value;
  const auditEvent = window.buildAuditPreviewEvent(role, 'FINAL_SIGNATURE', {
    facility: facilityType,
    department: page,
    encounter: encounterTypeId
  });
  auditJsonOutput.textContent = JSON.stringify(auditEvent, null, 2);
  auditOutputBox.classList.remove('hidden');
  showToast(tr('Simulated audit event logged (JSON)', '?? ????? ??? ?????? ?????? (JSON)'), 'success');
};
}

async function navigateTo(page) {
  if (!canAccessNavItem(page, { allowHidden: true })) {
    showToast(tr('Access Denied', '????? ?????'), 'error');
    return;
  }
  currentPage = page;
  document.querySelectorAll('.nav-item').forEach((el) => el.classList.toggle('active', parseInt(el.dataset.page) === page));
  const item = NAV_ITEMS[page];
  document.getElementById('headerTitle').textContent = tr(item.en, item.ar);
  // Close sidebar on mobile after navigation
  document.getElementById('sidebar').classList.remove('open');
  document.getElementById('sidebarOverlay').classList.remove('show');
  await loadPage(page);
}

function showToast(msg, type = 'success') {
  let t = document.querySelector('.toast');
  if (!t) { t = document.createElement('div'); t.className = 'toast'; document.body.appendChild(t); }
  t.className = `toast toast-${type} show`;
  t.innerHTML = `${type === 'success' ? '?' : '?'} ${escapeHTML(msg)}`;
  setTimeout(() => t.classList.remove('show'), 3000);
}

function makeTable(headers, rows, actions) {
  if (!rows.length) return `<div class="empty-state"><div class="empty-icon">?</div><p>${tr('No data found', '?? ????? ??????')}</p></div>`;
  let html = '<table class="data-table"><thead><tr>';
  headers.forEach(h => html += `<th>${h}</th>`);
  html += '</tr></thead><tbody>';
  rows.forEach(row => {
    html += '<tr>';
    row.cells.forEach(c => html += `<td>${cellHtml(c)}</td>`);
    if (actions) html += `<td>${actions(row)}</td>`;
    html += '</tr>';
  });
  html += '</tbody></table>';
}

```

```

    return html;
}

window.createTable = function(container, id, headers, rows, actions) {
    if (!container) return;
    let html = `<table class="data-table" id="${id}"><thead><tr>`;
    headers.forEach(h => html += `<th>${h}</th>`);
    html += `</tr></thead><tbody>`;
    if (!rows || !rows.length) {
        html += `<tr><td colspan="${headers.length}" class="text-center py-4 text-on-surface-variant">${tr('No data found', '?? ???? ?????')}</td></tr>`;
    } else {
        rows.forEach(row => {
            html += `<tr data-id="${escapeHTML(row.id || '')}" class="hover:bg-primary/5 transition-colors cursor-pointer">`;
            row.cells.forEach(c => html += `<td>${cellHtml(c)}</td>`);
            if (actions) html += `<td>${actions(row)}</td>`;
            html += `</tr>`;
        });
    }
    html += `</tbody></table>`;
    container.innerHTML = html;
};

function badge(text, type) { return rawHtml(`<span class="badge badge-${escapeHTML(type)}">${escapeHTML(text)}</span>`); }

function statusBadge(status) {
    const map = { Waiting: 'warning', 'With Doctor': 'success', Confirmed: 'success', Pending: 'warning', Approved: 'success', Rejected: 'danger', Active: 'success', 'On Leave': 'info', Cancelled: 'danger', Completed: 'success', Requested: 'info', Done: 'success', Available: 'success', Reserved: 'warning', Used: 'info', Expired: 'danger', Compatible: 'success', Incompatible: 'danger', Signed: 'success', Dispensed: 'success', Scheduled: 'info', 'In Progress': 'warning' };
    return badge(status, map[status] || 'info');
}

// ===== PRINT UTILITY =====
window.printDocument = function (title, content, options = {}) {
    const rtl = isArabic ? 'dir="rtl"' : '';
    const w = window.open('', '_blank', 'width=800,height=600');
    if (!w) { showToast(tr('Please allow pop-ups to print', '???? ?????? ???????? ???????? ????????'), 'error'); }
    return; }
    const showHeader = options.showHeader !== false;
    w.document.write(`<!DOCTYPE html><html ${rtl}><head><meta charset="utf-8"><title>${escapeHTML(title)}</title>
<style>
    *{margin:0;padding:0;box-sizing:border-box}
    body{font-family:'Segoe UI',Tahoma,sans-serif;padding:20px;color:#333;font-size:13px;direction:${isArabic ? 'rtl' : 'ltr'}}
    .header{text-align:center;border-bottom:3px double #1a5276;padding-bottom:12px;margin-bottom:16px}
    .header h1{font-size:22px;color:#1a5276;margin-bottom:4px}
    .header p{font-size:11px;color:#666}
    .info-grid{display:grid;grid-template-columns:1fr 1fr;gap:8px;margin-bottom:16px;font-size:12px}
    .info-grid div{padding:4px 8px;background:#f8f9fa;border-radius:4px}

```

```

.info-grid strong{color:#1a5276}
table{width:100%;border-collapse:collapse;margin:12px 0}
th{background:#1a5276;color:#fff;padding:8px 10px;text-align:${isArabic ? 'right' : 'left'};font-size:12
px}

td{padding:6px 10px;border-bottom:1px solid #ddd;font-size:12px}
tr:nth-child(even){background:#f8f9fa}
.total-row{font-weight:700;font-size:14px;background:#e8f4fd!important}
.footer{text-align:center;margin-top:24px;padding-top:12px;border-top:1px solid #ddd;font-size:10px;colo
r:#999}

.signature{display:flex;justify-content:space-between;margin-top:40px}
.signature div{text-align:center;min-width:150px;border-top:1px solid #333;padding-top:4px;font-size:11p
x}

@media print{body{padding:10px} .no-print{display:none!important}}
</style></head><body>
  ${showHeader ? '<div class="header"><h1>' + (options.companyName || '????? ???? ? jumanaMedical') + '</h1>
<p>' + (options.companyInfo || '????? ???? ???? | jumanaMedical Hospital') + '</p></div><h2 style="text-al
ign:center;color:#1a5276;margin-bottom:16px">' + escapeHTML(title) + '</h2>' : ''}
  ${content}
  <div class="footer">${tr('Printed on', '???? ????')}: ${new Date().toLocaleString('ar-SA')} | ${tr('juma
naMedical ERP', '???? ????')}</div>
  <button class="no-print" onclick="window.print()" style="position:fixed;top:10px;right:10px;padding:10px 2
4px;background:#1a5276;color:#fff;border:none;border-radius:8px;cursor:pointer;font-size:14px">?? ${tr('Print'
, '????')}</button>
</body></html>`);
w.document.close();
setTimeout(() => w.print(), 500);
};

```

```

window.printInvoice = async function (id) {
  try {
    const data = await API.get('/api/print/invoice/' + id);
    const inv = data.invoice;
    const content = `<div class="info-grid">
      <div><strong>${tr('Invoice #', '????? ?')}</strong> ${escapeHTML(inv.invoice_number || inv.id)}</div>
      <div><strong>${tr('Date', '?????')}</strong> ${escapeHTML(inv.created_at?.split('T')[0])}</div>
      <div><strong>${tr('Patient', '????')}</strong> ${escapeHTML(inv.patient_name)}</div>
      <div><strong>${tr('Payment', '????')}</strong> ${escapeHTML(inv.payment_method || '-')}</div>
    </div>
    <table><thead><tr><th>${tr('Description', '????')}</th><th>${tr('Amount', '????')}</th><th>${tr('VAT',
'????')}</th><th>${tr('Total', '?????')}</th></tr></thead>
    <tbody><tr><td>${escapeHTML(inv.description || inv.service_type)}</td><td>${escapeHTML(inv.amount)} SAR</t
d><td>${escapeHTML(inv.vat_amount || 0)} SAR</td><td>${escapeHTML(inv.total)} SAR</td></tr>
    <tr class="total-row"><td colspan="3">${tr('Grand Total', '????? ????')}</td><td>${escapeHTML(inv.total
)} SAR</td></tr></tbody></table>
    <div class="signature"><div>${tr('Cashier', '???? ?????')}</div><div>${tr('Patient Signature', '???? ?
????')}</div></div>`;
    printDocument(tr('Tax Invoice', '????? ?????'), content);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```

window.printLabReport = async function (id) {
  try {
    const data = await API.get('/api/print/lab-report/' + id);

```

```

const content = `<div class="info-grid">
  <div><strong>${tr('Patient', '?????')}</strong> ${escapeHTML(data.patient?.name_ar || data.patient?.na
me_en || '-')}</div>
  <div><strong>${tr('File #', '??? ?????')}</strong> ${escapeHTML(data.patient?.file_number || '-')}</div>
</div>
  <div><strong>${tr('Test', '????')}</strong> ${escapeHTML(data.order?.description)}</div>
  <div><strong>${tr('Date', '??????')}</strong> ${escapeHTML(data.order?.created_at?.split('T')[0])}</div>
</div>
<table><thead><tr><th>${tr('Test', '????')}</th><th>${tr('Result', '??????')}</th><th>${tr('Normal Range', '????? ?????')}</th><th>${tr('Status', '????')}</th></tr></thead>
<tbody>${(data.results || []).map(r => `<tr style="${r.is_abnormal ? 'color:#e74c3c;font-weight:700' : ''}"><td>${escapeHTML(r.test_name || '-')}</td><td>${escapeHTML(r.result_value || '-')}</td><td>${escapeHTML(r.normal_range || '-')}</td><td>${r.is_abnormal ? '?? ' + tr('Abnormal', '??? ?????') : '? ' + tr('Normal', '????')}</td></tr>`).join('')}</tbody></table>
<div class="signature"><div>${tr('Lab Technician', '??? ?????')}</div><div>${tr('Doctor', '????')}</div></div>`;

printDocument(tr('Lab Report', '???? ????'), content);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== EXPORT UTILITY =====
window.exportCSV = function (filename, headers, rows) {
  const BOM = '\uFEFF';
  const csv = BOM + headers.join(',') + '\n' + rows.map(r => r.map(c => `"${String(c || '').replace(/"/g, '"')}"`).join(',')).join('\n');
  const blob = new Blob([csv], { type: 'text/csv;charset=utf-8' });
  const a = document.createElement('a'); a.href = URL.createObjectURL(blob); a.download = filename + '.csv'; a.click();
  showToast(tr('Exported!', '?? ?????'));
};

window.exportTableCSV = function (filename) {
  const table = document.querySelector('#pageContent table');
  if (!table) return showToast(tr('No table found', '?? ???? ????'), 'error');
  const headers = [...table.querySelectorAll('th')].map(h => h.textContent);
  const rows = [...table.querySelectorAll('tbody tr')].map(r => [...r.querySelectorAll('td')].map(c => c.textContent.trim()));
  exportCSV(filename, headers, rows);
};

// ===== CONSENT FORMS =====
async function renderConsentForms(el) {
  const content = el;

  // 1. Load real templates, patients, and pending consents from APIs
  const [templates, patients, pendingConsents] = await Promise.all([
    API.get('/api/consent-forms/templates/list').catch(() => []),
    API.get('/api/patients').catch(() => []),
    API.get('/api/consent-forms').catch(() => []) // pending consents list
  ]);

  content.innerHTML = `
    <div class="page-title">? ${tr('Consent Forms & Clinical Signatures', '????? ?????????? ?????????? ??????')}

```

</div>

```
<div class="tab-bar mb-16">
  <button class="tab-btn active" onclick="window.switchConsentTab('create')">${tr('Create Consent', '????
???? ????')}</button>
  <button class="tab-btn" onclick="window.switchConsentTab('sign')">${tr('Pending Signatures', '?????? ?
?????')}</button>
</div>

<!-- Create Tab -->
<div id="consentTabCreate" class="consent-tab-content">
  <div class="card mb-16" style="padding:20px">
    <h4 style="margin:0 0 16px;color:var(--primary-color)">?? ${tr('New Consent Document Request', '??? ?
??? ????')}</h4>
    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px;margin-bott
om:16px">
      <div class="form-group">
        <label>${tr('Select Patient:', '???? ??????:')}</label>
        <select id="cfPatient" class="form-input">
          <option value="">${tr('-- Choose Patient --', '-- ???? ?????? --')}</option>
          ${patients.map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(isArabic ? p.name_a
r : p.name_en)}">${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`
).join('')}
        </select>
      </div>
      <div class="form-group">
        <label>${tr('Select Template:', '???? ?????? ??????:')}</label>
        <select id="cfTemplate" class="form-input" onchange="window.loadConsentTemplate()">
          <option value="">${tr('-- Choose Template --', '-- ???? ?????? --')}</option>
          ${templates.map(t => `<option value="${t.id || t.form_type}" data-title="${escapeHTML(t.title ||
t.en || '')}" data-title-ar="${escapeHTML(t.title_ar || t.ar || '')}" data-content="${escapeHTML(t.body_text
|| t.content || '')}">${escapeHTML(isArabic ? (t.title_ar || t.ar || t.title) : (t.title || t.en || t.title_ar
))}</option>`
).join('')}
        </select>
      </div>
      <div class="form-group">
        <label>${tr('Physician Name:', '??? ??????:')}</label>
        <input type="text" id="cfDoctor" class="form-input" value="${escapeHTML(currentUser?.display_name
|| '')}">
      </div>
    </div>

    <div class="form-group mb-16">
      <label>${tr('Consent Title:', '???? ??????:')}</label>
      <input type="text" id="cfTitle" class="form-input">
    </div>

    <div class="form-group mb-16">
      <label>${tr('Consent Content / Terms:', '?? ?????? ??????:')}</label>
      <textarea id="cfContent" class="form-input" style="height:150px"></textarea>
    </div>

    <button class="btn btn-primary" onclick="window.createConsentForm()">${tr('Generate Consent Form', '??
??? ???? ??????')}</button>
```

```

    </div>
</div>

<!-- Sign Tab -->
<div id="consentTabSign" class="consent-tab-content" style="display:none">
    <div class="card mb-16" style="padding:20px">
        <h4 style="margin:0 0 16px;color:var(--accent-color)">?? ${tr('Sign Pending Consents', '????? ???????
? ?????? ???????')}</h4>
        <div class="form-group mb-16">
            <label>${tr('Select Document:', '???? ??????? ??????? ???????')}</label>
            <select id="cfSignSelect" class="form-input" onchange="window.loadConsentForSign()">
                <option value="">${tr('-- Select Pending Document --', '-- ???? ??????? ??????? --')}</option>
                ${pendingConsents.filter(c => c.status !== 'Signed').map(c => `<option value="${safeId(c.id)}">${e
scapeHTML(c.form_title)} - [${escapeHTML(c.patient_name)}]</option>`).join('')}
            </select>
        </div>

        <div id="cfSignArea" style="display:none">
            <div id="cfSignContent" class="card mb-16" style="padding:16px;background:var(--bg-light);border-lef
t:4px solid var(--accent-color)"></div>
            <div class="form-group mb-16">
                <label>${tr('Witness Name (Optional):', '??? ?????? (???????)')}</label>
                <input type="text" id="cfWitness" class="form-input" placeholder="${tr('Enter witness name', '????
??? ??????')}">
            </div>
            <div class="form-group mb-16">
                <label>${tr('Signature Area (Draw below):', '????? ??????? (???? ?????? ????????)')}</label>
                <div style="border:1px dashed var(--border-color);border-radius:8px;padding:8px;background:#fff;ma
x-width:400px">
                    <canvas id="cfSigCanvas" width="380" height="150" style="cursor:crosshair;background:#fff;displa
y:block"></canvas>
                </div>
                <div style="margin-top:8px;display:flex;gap:8px">
                    <button class="btn btn-sm btn-secondary" onclick="window.clearSigCanvas()">${tr('Clear Signature
', '??? ???????')}</button>
                    <button class="btn btn-sm btn-primary" onclick="window.signConsentForm()">${tr('Submit Signed Co
nsent', '?????? ?????? ???????')}</button>
                </div>
            </div>
        </div>
    </div>
</div>
`;

// Dynamic tab switcher
window.switchConsentTab = (tab) => {
    document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
    document.getElementById('consentTabCreate').style.display = 'none';
    document.getElementById('consentTabSign').style.display = 'none';

    if (tab === 'create') {
        document.querySelector('.tab-btn:nth-child(1)').classList.add('active');
        document.getElementById('consentTabCreate').style.display = 'block';
    } else {

```

```

        document.querySelector('.tab-btn:nth-child(2)').classList.add('active');
        document.getElementById('consentTabSign').style.display = 'block';
    }
};

// Autoload template text details
window.loadConsentTemplate = () => {
    const sel = document.getElementById('cfTemplate');
    const opt = sel.options[sel.selectedIndex];
    if (opt && opt.value) {
        document.getElementById('cfTitle').value = isArabic ? (opt.dataset.titleAr || opt.dataset.title) : opt.d
ataset.title;
        document.getElementById('cfContent').value = opt.dataset.content || '';
    }
};

// Submit and create new form request
window.createConsentForm = async () => {
    const pSel = document.getElementById('cfPatient');
    const tSel = document.getElementById('cfTemplate');
    const opt = tSel.options[tSel.selectedIndex];
    if (!pSel.value) return showToast(tr('Please select a patient', '????? ?????? ??????'), 'error');
    try {
        await API.post('/api/consent-forms', {
            patient_id: pSel.value,
            patient_name: pSel.options[pSel.selectedIndex]?.dataset?.name || '',
            form_type: tSel.value || 'general',
            form_title: document.getElementById('cfTitle').value,
            form_title_ar: opt?.dataset?.titleAr || document.getElementById('cfTitle').value,
            content: document.getElementById('cfContent').value,
            doctor_name: document.getElementById('cfDoctor').value
        });
        showToast(tr('Consent request created successfully!', '?? ????? ??? ??????? ?????? ??????!'));
        await renderConsentForms(content);
        window.switchConsentTab('sign');
    } catch (e) { showToast(tr('Error creating consent form', '??? ?? ?????? ??? ???????'), 'error'); }
};

// Sign module for loading specific selected document
window.loadConsentForSign = async () => {
    const fid = document.getElementById('cfSignSelect').value;
    if (!fid) { document.getElementById('cfSignArea').style.display = 'none'; return; }
    document.getElementById('cfSignArea').style.display = 'block';
    try {
        const f = await API.get(`/api/consent-forms/${fid}`);
        document.getElementById('cfSignContent').innerHTML = `<h3>${escapeHTML(isArabic ? (f.form_title_ar || f.
form_title) : f.form_title)}</h3><p>${escapeHTML(f.content)}</p><p><strong>${tr('Patient', '??????')}</strong>
> ${escapeHTML(f.patient_name)}<br><strong>${tr('Doctor', '??????')}</strong> ${escapeHTML(f.doctor_name)}</p>
>`;
        // Initialize signature canvas drawing events
        setTimeout(() => {
            const canvas = document.getElementById('cfSigCanvas');
            if (!canvas) return;
            const ctx = canvas.getContext('2d');

```



```

        let drawing = false;
        canvas.onpointerdown = (e) => { drawing = true; ctx.beginPath(); ctx.moveTo(e.offsetX, e.offsetY); };
        canvas.onpointermove = (e) => { if (!drawing) return; ctx.lineTo(e.offsetX, e.offsetY); ctx.strokeStyle
e = '#000'; ctx.lineWidth = 2; ctx.stroke(); };
        canvas.onpointerup = () => drawing = false;
        canvas.onpointerout = () => drawing = false;
    }, 100);
} catch (e) { console.error(e); }
};

// Sign canvas clearing logic
window.clearSigCanvas = () => {
    const c = document.getElementById('cfSigCanvas');
    if (c) c.getContext('2d').clearRect(0, 0, c.width, c.height);
};

// Submit the signing with signature image details
window.signConsentForm = async () => {
    const fid = document.getElementById('cfSignSelect').value;
    if (!fid) return;
    const canvas = document.getElementById('cfSigCanvas');
    const sig = canvas ? canvas.toDataURL('image/png') : '';
    try {
        await API.put(`/api/consent-forms/${fid}/sign`, {
            patient_signature: sig,
            witness_name: document.getElementById('cfWitness').value
        });
        showToast(tr('Consent signed successfully!', '?? ????? ??????? ??????? ??????!'));
        await renderConsentForms(content);
    } catch (e) { showToast(tr('Error signing document', '??? ????? ???????'), 'error'); }
};
}

let _signCtx = null, _signDrawing = false;
function initSignaturePad() {
    const canvas = document.getElementById('signaturePad');
    if (!canvas) return;
    _signCtx = canvas.getContext('2d');
    _signCtx.strokeStyle = '#000';
    _signCtx.lineWidth = 2;
    _signCtx.lineCap = 'round';

    canvas.addEventListener('mousedown', e => { _signDrawing = true; _signCtx.beginPath(); _signCtx.moveTo(e.off
setX, e.offsetY); });
    canvas.addEventListener('mousemove', e => { if (_signDrawing) { _signCtx.lineTo(e.offsetX, e.offsetY); _sign
Ctx.stroke(); } });
    canvas.addEventListener('mouseup', () => _signDrawing = false);
    canvas.addEventListener('mouseleave', () => _signDrawing = false);
    // Touch support
    canvas.addEventListener('touchstart', e => { e.preventDefault(); _signDrawing = true; const r = canvas.getBo
undingClientRect(); _signCtx.beginPath(); _signCtx.moveTo(e.touches[0].clientX - r.left, e.touches[0].clientY
- r.top); });
    canvas.addEventListener('touchmove', e => { e.preventDefault(); if (_signDrawing) { const r = canvas.getBoun
dingClientRect(); _signCtx.lineTo(e.touches[0].clientX - r.left, e.touches[0].clientY - r.top); _signCtx.strok

```

```

e(); } });
    canvas.addEventListener('touchend', () => _signDrawing = false);
}

function clearSignature() {
    const canvas = document.getElementById('signaturePad');
    if (canvas && _signCtx) _signCtx.clearRect(0, 0, canvas.width, canvas.height);
}

async function loadConsentText() {
    const id = document.getElementById('consentTemplate')?.value;
    const area = document.getElementById('consentTextArea');
    if (!id) { area.style.display = 'none'; return; }
    try {
        const t = await API.get('/api/consent/templates/' + id);
        document.getElementById('consentTitle').textContent = t.title_ar;
        document.getElementById('consentBody').textContent = isArabic ? t.body_text_ar : t.body_text;
        area.style.display = 'block';
        // Show witness section if required
        document.getElementById('witnessSection').style.display = t.requires_witness ? 'block' : 'none';
        clearSignature();
        setTimeout(() => initSignaturePad(), 50);
    } catch (e) { showToast(tr('Error loading form', '??? ?? ????? ??????'), 'error'); }
}

window.loadConsentText = loadConsentText;
window.clearSignature = clearSignature;

async function submitConsent() {
    const patientId = document.getElementById('consentPatient')?.value;
    const templateId = document.getElementById('consentTemplate')?.value;
    const canvas = document.getElementById('signaturePad');
    if (!patientId) return showToast(tr('Select patient', '???? ??????'), 'error');
    if (!templateId) return showToast(tr('Select consent form', '???? ??????'), 'error');
    // Check if canvas has content
    const sigData = canvas.toDataURL('image/png');
    const emptyCanvas = document.createElement('canvas');
    emptyCanvas.width = canvas.width; emptyCanvas.height = canvas.height;
    if (sigData === emptyCanvas.toDataURL('image/png')) return showToast(tr('Please sign the form', '?????? ????'
    '?? ?? ??????'), 'error');

    const patientSelect = document.getElementById('consentPatient');
    const patientName = patientSelect.options[patientSelect.selectedIndex]?.text || '';

    try {
        await API.post('/api/consent/sign', {
            template_id: templateId,
            patient_id: patientId,
            patient_name: patientName.split(' ')[0],
            signature_data: sigData,
            witness_name: document.getElementById('witnessName')?.value || '',
            doctor_name: document.getElementById('consentDoctor')?.value || '',
            procedure_details: document.getElementById('consentProcedure')?.value || ''
        });
    }

```

```

    showToast(tr('Consent signed!', '?? ?????? ?? ????!!'));
    renderConsentForms(document.getElementById('pageContent'));
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
}

window.submitConsent = submitConsent;

window.viewSignedConsent = async function (id) {
  try {
    const consents = await API.get('/api/consent/recent');
    const c = consents.find(x => x.id === id);
    if (!c) return;
    const tpl = await API.get('/api/consent/templates/' + c.template_id);
    let html = '<div style="direction:rtl;text-align:right">' +
      '<h3 style="margin-bottom:12px;color:var(--primary)">' + (tpl.title_ar || c.title) + '</h3>' +
      '<div style="white-space:pre-wrap;line-height:2;font-size:14px;padding:12px;background:var(--hover);border-radius:8px;max-height:300px;overflow-y:auto;margin-bottom:16px">' + tpl.body_text_ar + '</div>' +
      '<div style="display:flex;gap:20px;flex-wrap:wrap;margin-bottom:12px">' +
      '<div><strong>' + tr('Patient', '??????') + ':</strong>' + escapeHTML(c.patient_name) + '</div>' +
      '<div><strong>' + tr('Doctor', '??????') + ':</strong>' + escapeHTML(c.doctor_name || c.created_by) + '
</div>' +
      '<div><strong>' + tr('Date', '??????') + ':</strong>' + new Date(c.signed_at || c.created_at).toLocaleString('ar-SA') + '</div>' +
      (c.witness_name ? '<div><strong>' + tr('Witness', '??????') + ':</strong>' + escapeHTML(c.witness_name)
+ '</div>' : '') +
      '</div>';
    if (c.signature_data) html += '<div style="margin-top:12px"><strong>' + tr('Signature', '??????') + ':</strong><br><img src="" + escapeHTML(c.signature_data) + '" style="max-width:300px;border:1px solid var(--border);border-radius:4px;margin-top:4px"></div>';
    html += '</div>';
    showModal(tr('Signed Consent', '?????? ????'), html);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.printSignedConsent = async function (id) {
  try {
    const consents = await API.get('/api/consent/recent');
    const c = consents.find(x => x.id === id);
    if (!c) return;
    const tpl = await API.get('/api/consent/templates/' + c.template_id);
    let html = '<div style="direction:rtl;text-align:right;font-family:Arial,sans-serif">' +
      '<div style="text-align:center;margin-bottom:20px"><h2>???? ???? ????</h2><h3 style="color:#1a56db">'
+ tpl.title_ar + '</h3></div>' +
      '<div style="white-space:pre-wrap;line-height:2.2;font-size:14px;margin-bottom:20px">' + tpl.body_text_
ar + '</div>' +
      (c.procedure_details ? '<div style="margin-bottom:16px;padding:8px;border:1px solid #ccc;border-radius:4px"><strong>????? ?????:</strong>' + escapeHTML(c.procedure_details) + '</div>' : '') +
      '<div style="margin-top:30px;display:flex;justify-content:space-between">' +
      '<div><strong>??? ?????:</strong>' + escapeHTML(c.patient_name) + '</div>' +
      '<div><strong>?????:</strong>' + new Date(c.signed_at || c.created_at).toLocaleDateString('ar-SA') +
'</div>' +
      '</div>' +
      '<div style="margin-top:10px"><strong>?????:</strong>' + escapeHTML(c.doctor_name || '-') + '</div>' +
      (c.witness_name ? '<div style="margin-top:10px"><strong>?????:</strong>' + escapeHTML(c.witness_name)
+ '</div>' : '') +

```

```

    '<div style="margin-top:20px"><strong>??????:</strong><br>' +
    (c.signature_data ? '' : '_____') + '</div>' +
    '</div>';
    printDocument(tmpl.title_ar, html, { showHeader: false });
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```
// ===== OB/GYN DEPARTMENT PAGE =====
```

```

async function renderOBGYN(el) {
  let stats = { activePregnancies: 0, highRisk: 0, dueThisWeek: 0, deliveredThisMonth: 0 };
  try { stats = await API.get('/api/obgyn/stats'); } catch (e) { }
  const patients = await API.get('/api/patients');
  let patOpts = patients.map(p => '<option value="' + p.id + '" data-name="' + (p.name_ar || p.name_en) + '">'
+ (p.name_ar || p.name_en) + ' (' + p.file_number + ')</option>').join('');

  el.innerHTML = '<div class="page-title">? ' + tr('OB/GYN Department', '??? ?????? ?????????') + '</div>' +
    '<div class="stats-grid">' +
    '<div class="stat-card" style="--stat-color:#ec4899"><div class="stat-label">' + tr('Active Pregnancies',
'??? ???') + '</div><div class="stat-value">' + stats.activePregnancies + '</div></div>' +
    '<div class="stat-card" style="--stat-color:#ef4444"><div class="stat-label">' + tr('High Risk', '???? ???
????') + '</div><div class="stat-value">' + stats.highRisk + '</div></div>' +
    '<div class="stat-card" style="--stat-color:#f59e0b"><div class="stat-label">' + tr('Due This Week', '????
? ??? ???????') + '</div><div class="stat-value">' + stats.dueThisWeek + '</div></div>' +
    '<div class="stat-card" style="--stat-color:#22c55e"><div class="stat-label">' + tr('Delivered This Month'
, '?????? ??? ?????') + '</div><div class="stat-value">' + stats.deliveredThisMonth + '</div></div>' +
    '</div>' +

    '<div class="card" style="margin-top:16px"><h3 style="margin-bottom:12px">? ' + tr('New Pregnancy Record',
'??? ??? ?????') + '</h3>' +
    '<div class="form-grid" style="gap:10px">' +
    '<div class="form-group"><label>' + tr('Patient', '??????') + '</label><select id="obPatient" class="form
-control"><option value="">' + tr('-- Select --', '-- ????? --') + '</option>' + patOpts + '</select></div>' +
    '<div class="form-group"><label>' + tr('LMP (Last Menstrual Period)', '??? ??? ?????') + '</label><input
type="date" id="obLMP" class="form-control"></div>' +
    '<div class="form-group"><label>G (Gravida)</label><input type="number" id="obGravida" class="form-control
" value="1" min="1"></div>' +
    '<div class="form-group"><label>P (Para)</label><input type="number" id="obPara" class="form-control" valu
e="0" min="0"></div>' +
    '<div class="form-group"><label>A (Abortions)</label><input type="number" id="obAbort" class="form-control
" value="0" min="0"></div>' +
    '<div class="form-group"><label>' + tr('Blood Group', '????? ?????') + '</label><select id="obBlood" class=
"form-control"><option>A</option><option>A-</option><option>B</option><option>B-</option><option>AB</option>
<option>AB-</option><option>0</option><option>0-</option></select></div>' +
    '<div class="form-group"><label>Rh</label><select id="obRh" class="form-control"><option>Positive</option>
<option>Negative</option></select></div>' +
    '<div class="form-group"><label>' + tr('Risk Level', '????? ???????') + '</label><select id="obRisk" class
="form-control"><option value="Low">' + tr('Low', '????') + '</option><option value="Medium">' + tr('Medium',
'????') + '</option><option value="High">' + tr('High', '????') + '</option></select></div>' +
    '<div class="form-group"><label>' + tr('Previous C-Sections', '?????? ?????') + '</label><input type="num
ber" id="obPrevCS" class="form-control" value="0" min="0"></div>' +
    '<div class="form-group"><label>' + tr('Chronic Conditions', '????? ?????') + '</label><input id="obChroni

```

```

c" class="form-control" placeholder="" + tr('DM, HTN, etc', '????? ???...') + "></div>" +
  '<div class="form-group"><label>' + tr('Allergies', '?????') + '</label><input id="obAllergy" class="form-control"></div>' +
  '<div class="form-group"><label>' + tr('Attending Doctor', '????? ??????') + '</label><input id="obDoctor" class="form-control" value="" + (currentUser?.display_name || '') + "></div>' +
  '</div>' +
  '<button class="btn btn-primary" onclick="createPregnancy()" style="margin-top:12px">' + tr('Create Record', '????? ?????') + '</button></div>' +

  '<div class="card" style="margin-top:16px"><h3 style="margin-bottom:12px">' + tr('Active Pregnancies', '??????? ??????') + '</h3><div id="obActiveList">' + tr('Loading...', '???? ????????...') + '</div></div>' +

  '<div class="card" style="margin-top:16px"><h3 style="margin-bottom:12px">' + tr('Lab Panels', '??? ???') + '</h3><div id="obLabPanels">' + tr('Loading...', '???? ????????...') + '</div></div>';

// Load active pregnancies
try {
  const preg = await API.get('/api/obgyn/pregnancies?status=Active');
  const list = document.getElementById('obActiveList');
  if (preg.length === 0) { list.innerHTML = '<p style="color:var(--text-muted)">' + tr('No active pregnancies', '?? ???? ?????? ???') + '</p>'; }
  else {
    let html = '<table class="data-table"><thead><tr><th>' + tr('Patient', '???????') + '</th><th>GPAL</th><th>' + tr('EDD', '????? ?????? ??????') + '</th><th>' + tr('Risk', '???????') + '</th><th>' + tr('Doctor', '???????') + '</th><th>' + tr('Actions', '???????') + '</th></tr></thead><tbody>';
    preg.forEach(p => {
      const riskColor = p.risk_level === 'High' ? '#ef4444' : p.risk_level === 'Medium' ? '#f59e0b' : '#22c55e';
      html += '<tr><td>' + escapeHTML(p.patient_name) + '</td><td>G' + p.gravida + 'P' + p.para + 'A' + p.abortions + 'L' + p.living_children + '</td><td>' + escapeHTML(p.edd || '-') + '</td><td><span style="color:' + riskColor + ';font-weight:700">' + escapeHTML(p.risk_level) + '</span></td><td>' + escapeHTML(p.attending_doctor || '-') + '</td><td><button class="btn btn-sm" onclick="showAntenatalForm(' + safeId(p.id) + ', ' + safeId(p.patient_id) + ')">' + tr('Antenatal', '?????') + '</button> <button class="btn btn-sm" onclick="showOBGYNUltrasoundForm(' + safeId(p.id) + ', ' + safeId(p.patient_id) + ')">' + tr('Ultrasound', '????') + '</button> <button class="btn btn-sm" onclick="showPartogramForm(' + safeId(p.id) + ')">' + tr('Partogram', '???? ???') + '</button> <button class="btn btn-sm btn-primary" onclick="showDeliveryForm(' + safeId(p.id) + ')">' + tr('Delivery', '???????') + '</button></td></tr>';
    });
    html += '</tbody></table>';
    list.innerHTML = html;
  }
} catch (e) { document.getElementById('obActiveList').innerHTML = '<p style="color:red">Error loading</p>'; }

// Load lab panels
try {
  const panels = await API.get('/api/obgyn/lab-panels');
  let ph = '<div style="display:grid;gap:8px">';
  panels.forEach(p => {
    ph += '<div style="padding:12px;border-radius:8px;background:var(--hover);border-right:3px solid #ec4899"><strong>' + escapeHTML(p.panel_name_ar) + '</strong> (' + escapeHTML(p.trimester) + ')<br><small style="color:var(--text-muted)">' + escapeHTML(p.tests) + '</small></div>';
  });
  ph += '</div>';
}

```

```

    document.getElementById('obLabPanels').innerHTML = ph;
  } catch (e) { }
}

window.createPregnancy = async () => {
  const patSel = document.getElementById('obPatient');
  const pid = patSel.value;
  if (!pid) return showToast(tr('Select patient', '????? ?????'), 'error');
  const lmp = document.getElementById('obLMP').value;
  if (!lmp) return showToast(tr('Enter LMP date', '????? ????? ??? ?????'), 'error');
  try {
    await API.post('/api/obgyn/pregnancies', {
      patient_id: pid,
      patient_name: patSel.options[patSel.selectedIndex]?.dataset?.name || '',
      lmp, gravida: parseInt(document.getElementById('obGravida').value) || 1,
      para: parseInt(document.getElementById('obPara').value) || 0,
      abortions: parseInt(document.getElementById('obAbort').value) || 0,
      blood_group: document.getElementById('obBlood').value,
      rh_factor: document.getElementById('obRh').value,
      risk_level: document.getElementById('obRisk').value,
      previous_cs: parseInt(document.getElementById('obPrevCS').value) || 0,
      chronic_conditions: document.getElementById('obChronic').value,
      allergies: document.getElementById('obAllergy').value,
      attending_doctor: document.getElementById('obDoctor').value
    });
    showToast(tr('Pregnancy record created!', '?? ?????? ??? ??????!'));
    navigateTo(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.showAntenatalForm = async (pregId, patientId) => {
  const visits = await API.get('/api/obgyn/antenatal/' + pregId);
  let vRows = visits.map(v => '<tr><td>' + v.visit_number + '</td><td>' + escapeHTML(v.gestational_age || '-') +
    '</td><td>' + escapeHTML(v.blood_pressure) + '</td><td>' + escapeHTML(v.fetal_heart_rate) + '</td><td>' + escapeHTML(v.weight) + 'kg</td><td>' + escapeHTML(v.risk_flags || '?') + '</td></tr>').join('');

  let html = '<h4 style="margin-bottom:8px">' + tr('Previous Visits', '????????? ????????') + '</h4>' +
    (visits.length ? '<table class="data-table" style="margin-bottom:16px"><thead><tr><th>#</th><th>GA</th><th><th>BP</th><th>FHR</th><th>Wt</th><th>Flags</th></tr></thead><tbody>' + vRows + '</tbody></table>' : '<p style="color:var(--text-muted);margin-bottom:16px">' + tr('No visits yet', '?? ??????') + '</p>') +
    '<h4 style="margin-bottom:8px">' + tr('New Visit', '????? ??????') + '</h4>' +
    '<div class="form-grid" style="gap:8px">' +
    '<div class="form-group"><label>GA (weeks)</label><input id="antGA" class="form-control" placeholder="e.g. 28+3"></div>' +
    '<div class="form-group"><label>Weight (kg)</label><input type="number" id="antWt" class="form-control" step="0.1"></div>' +
    '<div class="form-group"><label>BP</label><input id="antBP" class="form-control" placeholder="120/80"></div>' +
    '<div class="form-group"><label>Systolic</label><input type="number" id="antSys" class="form-control"></div>' +
    '<div class="form-group"><label>Diastolic</label><input type="number" id="antDia" class="form-control"></div>' +
    '<div class="form-group"><label>FHR</label><input type="number" id="antFHR" class="form-control" placeholder="110-160"></div>' +

```

```

    '<div class="form-group"><label>Fundal Height</label><input type="number" id="antFH" class="form-control"
step="0.5"></div>' +
    '<div class="form-group"><label>Hb</label><input type="number" id="antHb" class="form-control" step="0.1">
</div>' +
    '<div class="form-group"><label>Presentation</label><select id="antPres" class="form-control"><option>Ceph
alic</option><option>Breech</option><option>Transverse</option></select></div>' +
    '<div class="form-group"><label>Edema</label><select id="antEdema" class="form-control"><option>None</opti
on><option>Mild +</option><option>Moderate ++</option><option>Severe +++</option></select></div>' +
    '</div>' +
    '<div class="form-group" style="margin-top:8px"><label>Complaints</label><textarea id="antComp" class="for
m-control" rows="2"></textarea></div>' +
    '<div class="form-group"><label>Plan</label><textarea id="antPlan" class="form-control" rows="2"></textare
a></div>' +
    '<button class="btn btn-primary" onclick="saveAntenatal(' + safeId(pregId) + ', ' + safeId(patientId) + ')"
style="margin-top:8px">? ' + tr('Save Visit', '??? ?????') + '</button>';
    showModal(tr('Antenatal Visit', '????? ?????? ?????') + ' #' + pregId, html);
};

```

```

window.saveAntenatal = async (pregId, patientId) => {
    const bp = document.getElementById('antBP').value;
    try {
        await API.post('/api/obgyn/antenatal', {
            pregnancy_id: pregId, patient_id: patientId,
            gestational_age: document.getElementById('antGA').value,
            weight: parseFloat(document.getElementById('antWt').value) || 0,
            blood_pressure: bp,
            systolic: parseInt(document.getElementById('antSys').value) || (bp ? parseInt(bp.split('/')[0]) : 0),
            diastolic: parseInt(document.getElementById('antDia').value) || (bp ? parseInt(bp.split('/')[1]) : 0),
            fetal_heart_rate: parseInt(document.getElementById('antFHR').value) || 0,
            fundal_height: parseFloat(document.getElementById('antFH').value) || 0,
            hemoglobin: parseFloat(document.getElementById('antHb').value) || 0,
            fetal_presentation: document.getElementById('antPres').value,
            edema: document.getElementById('antEdema').value,
            complaints: document.getElementById('antComp').value,
            plan: document.getElementById('antPlan').value
        });
        showToast(tr('Visit saved!', '?? ??? ???????!'));
        document.querySelector('.modal-overlay)?.remove();
        navigateTo(currentPage);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```

// ===== E14: APGAR component selector (server computes the total; client only picks enums) =====
function apgarComponentSelect(prefix) {
    // Each component maps to a 0/1/2 enum; the SERVER recomputes the total (anti-spoof).
    const comp = (id, label, opts) => '<div class="form-group"><label>' + label + '</label><select id="' + id +
'" class="form-control">' +
        '<option value="">--</option>' + opts.map(o => '<option value="' + o[0] + '">' + o[1] + '</option>').join(
'') + '</select></div>';
    return comp(prefix + 'Appearance', 'Appearance', [['blue', 'Blue/Pale (0)'], ['acrocyanotic', 'Body pink, ex
tremities blue (1)'], ['pink', 'Completely pink (2)']]) +
        comp(prefix + 'Pulse', 'Pulse', [['absent', 'Absent (0)'], ['below_100', '<100 (1)'], ['above_100', '>100
(2)']]) +
        comp(prefix + 'Grimace', 'Grimace', [['no_response', 'No response (0)'], ['grimace', 'Grimace (1)'], ['cry

```

```

    ', 'Cry/Cough (2)']) +
    comp(prefix + 'Activity', 'Activity', [['limp', 'Limp (0)'], ['some_flexion', 'Some flexion (1)'], ['active', 'Active motion (2)']) +
    comp(prefix + 'Respiration', 'Respiration', [['absent', 'Absent (0)'], ['slow', 'Slow/Irregular (1)'], ['good', 'Good/Strong cry (2)']));
}

function readApgarComponents(prefix) {
  return {
    appearance: document.getElementById(prefix + 'Appearance').value,
    pulse: document.getElementById(prefix + 'Pulse').value,
    grimace: document.getElementById(prefix + 'Grimace').value,
    activity: document.getElementById(prefix + 'Activity').value,
    respiration: document.getElementById(prefix + 'Respiration').value
  };
}

window.showPartogramForm = async (pregId) => {
  let rows = '';
  try {
    const entries = await API.get('/api/obgyn/partogram/' + pregId);
    rows = entries.map(e => '<tr><td>' + escapeHTML(String(e.recorded_at || '').replace('T', ' ').slice(0, 16)) + '</td><td>' + e.cervical_dilation + 'cm</td><td>' + e.descent_station + '</td><td>' + e.contractions_per_10min + '/10</td><td>' + e.fetal_heart_rate_baseline + '</td><td>' + escapeHTML(e.alert_flags || '-') + '</td></tr>').join('');
  } catch (e) { }

  const html = (rows ? '<table class="data-table" style="margin-bottom:16px"><thead><tr><th>' + tr('Time', '????') + '</th><th>Dilation</th><th>Station</th><th>Contractions</th><th>FHR</th><th>Flags</th></tr></thead><tbody>' + rows + '</tbody></table>' : '<p style="color:var(--text-muted);margin-bottom:16px">' + tr('No partogram entries', '???? ????') + '</p>') +
    '<h4 style="margin-bottom:8px">' + tr('New Timepoint', '???? ????') + '</h4><div class="form-grid" style="gap:8px">' +
    '<div class="form-group"><label>Cervical Dilation (0-10cm)</label><input type="number" id="pgDil" class="form-control" min="0" max="10"></div>' +
    '<div class="form-group"><label>Effacement %</label><input type="number" id="pgEff" class="form-control"></div>' +
    '<div class="form-group"><label>Descent Station (-3..+5)</label><input type="number" id="pgStation" class="form-control"></div>' +
    '<div class="form-group"><label>Contractions /10min</label><input type="number" id="pgContr" class="form-control"></div>' +
    '<div class="form-group"><label>Contraction Duration (s)</label><input type="number" id="pgDur" class="form-control"></div>' +
    '<div class="form-group"><label>FHR Baseline</label><input type="number" id="pgFHR" class="form-control" placeholder="110-160"></div>' +
    '<div class="form-group"><label>Decelerations</label><select id="pgDecel" class="form-control"><option>None</option><option>Early</option><option>Variable</option><option>Late</option></select></div>' +
    '<div class="form-group"><label>Maternal BP</label><input id="pgBP" class="form-control" placeholder="120/80"></div>' +
    '</div><div class="form-group" style="margin-top:8px"><label>Notes</label><textarea id="pgNotes" class="form-control" rows="2"></textarea></div>' +
    '<button class="btn btn-primary" onclick="savePartogram(' + safeId(pregId) + ')" style="margin-top:8px">' + tr('Record', '????') + '</button>';
  showModal(tr('Partogram', '???? ????') + ' #' + safeId(pregId), html);
};

window.savePartogram = async (pregId) => {

```



```

try {
  await API.post('/api/obgyn/partogram', {
    pregnancy_id: pregId,
    cervical_dilation: parseInt(document.getElementById('pgDil').value) || 0,
    cervical_effacement: parseInt(document.getElementById('pgEff').value) || 0,
    descent_station: parseInt(document.getElementById('pgStation').value) || 0,
    contractions_per_10min: parseInt(document.getElementById('pgContr').value) || 0,
    contraction_duration: parseInt(document.getElementById('pgDur').value) || 0,
    fetal_heart_rate_baseline: parseInt(document.getElementById('pgFHR').value) || 0,
    decelerations: document.getElementById('pgDecel').value,
    maternal_bp: document.getElementById('pgBP').value,
    notes: document.getElementById('pgNotes').value
  });
  showToast(tr('Partogram recorded!', '?? ???????!'));
  document.querySelector('.modal-overlay')?.remove();
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.showDeliveryForm = async (pregId) => {
  const html = '<div class="form-grid" style="gap:8px">' +
    '<div class="form-group"><label>' + tr('Delivery Type', '??? ???????') + '</label><select id="delType" cla' +
    ss="form-control"><option value="NVD">Vaginal (NVD)</option><option value="Vaginal+Episiotomy">Vaginal + Episi' +
    otomy</option><option value="Cesarean">Cesarean</option><option value="VBAC">VBAC</option><option value="Vacuu' +
    m">Vacuum</option><option value="Forceps">Forceps</option></select></div>' +
    '<div class="form-group"><label>' + tr('Blood Loss (ml)', '????? ?????') + '</label><input type="number" id' +
    ="delBlood" class="form-control"></div>' +
    '<div class="form-group"><label>' + tr('Anesthesia', '????????') + '</label><select id="delAnes" class="for' +
    m-control"><option>None</option><option>Spinal</option><option>Epidural</option><option>General</option><optio' +
    n>Local</option></select></div>' +
    '<div class="form-group"><label>' + tr('Perineal Tear', '????') + '</label><select id="delTear" class="for' +
    m-control"><option>None</option><option>1st degree</option><option>2nd degree</option><option>3rd degree</opti' +
    on><option>4th degree</option></select></div>' +
    '<div class="form-group"><label>' + tr('Baby Sex', '??? ?????????') + '</label><select id="delSex" class="fo' +
    rm-control"><option>Male</option><option>Female</option></select></div>' +
    '<div class="form-group"><label>' + tr('Baby Weight (g)', '??? ?????????') + '</label><input type="number" i' +
    d="delWt" class="form-control"></div>' +
    '</div>' +
    '<h4 style="margin:12px 0 8px">APGAR @ 1 min <small style="color:var(--text-muted)">(' + tr('total compute' +
    d server-side', '?????? ????? ?? ???????') + ')</small></h4>' +
    '<div class="form-grid" style="gap:8px">' + apgarComponentSelect('d1') + '</div>' +
    '<h4 style="margin:12px 0 8px">APGAR @ 5 min</h4>' +
    '<div class="form-grid" style="gap:8px">' + apgarComponentSelect('d5') + '</div>' +
    '<div class="form-group" style="margin-top:8px"><label>' + tr('Complications', '????????') + '</label><text' +
    area id="delComp" class="form-control" rows="2"></textarea></div>' +
    '<button class="btn btn-primary" onclick="saveDelivery(' + safeId(pregId) + ')" style="margin-top:8px">? ' +
    tr('Record Delivery', '????? ?????????') + '</button>';
    showModal(tr('Delivery Record', '??? ?????????') + ' #' + safeId(pregId), html);
  };

window.saveDelivery = async (pregId) => {
  try {
    const r = await API.post('/api/obgyn/deliveries', {
      pregnancy_id: pregId,
      delivery_type: document.getElementById('delType').value,
      blood_loss_ml: parseInt(document.getElementById('delBlood').value) || 0,

```

```

        anesthesia_type: document.getElementById('delAnes').value,
        perineal_tear: document.getElementById('delTear').value,
        baby_gender: document.getElementById('delSex').value,
        baby_weight: parseInt(document.getElementById('delWt').value) || 0,
        apgar_1min_components: readApgarComponents('d1'),
        apgar_5min_components: readApgarComponents('d5'),
        complications: document.getElementById('delComp').value
    });
    showToast(tr('Delivery recorded! APGAR ', '?? ?????? ! ????? ') + (r.apgar_1min) + '/' + (r.apgar_5min));
    document.querySelector('.modal-overlay)?.remove();
    if (r && r.id) showNeonatalForm(r.id);
    else navigateTo(currentPage);
} catch (e) { showToast(tr('Error: ', '???? ') + (e.message || ''), 'error'); }
};

window.showNeonatalForm = async (deliveryId) => {
    const html = '<div class="form-grid" style="gap:8px">' +
        '<div class="form-group"><label>' + tr('Mother/Infant Band ID', '??? ???? ?????? ?????') + '</label><input' +
        id="neoBandId" class="form-control" placeholder="e.g. BND-99234"></div>' +
        '<div class="form-group"><label>' + tr('Birth Weight (g)', '??? ??????') + '</label><input type="number"' +
        id="neoWt" class="form-control"></div>' +
        '<div class="form-group"><label>' + tr('Length (cm)', '?????') + '</label><input type="number" id="neoLen"' +
        class="form-control" step="0.1"></div>' +
        '<div class="form-group"><label>' + tr('Head Circ (cm)', '???? ?????') + '</label><input type="number" id="neoHC"' +
        class="form-control" step="0.1"></div>' +
        '<div class="form-group"><label>' + tr('Blood Group', '????? ????') + '</label><input id="neoBG" class="fo' +
        rm-control"></div>' +
        '<div class="form-group"><label>' + tr('Feeding', '??????') + '</label><select id="neoFeed" class="form-c' +
        ontrol"><option>Breast</option><option>Bottle</option><option>Mixed</option></select></div>' +
        '<div class="form-group"><label>' + tr('Resuscitation', '????') + '</label><select id="neoResus" class="f' +
        orm-control"><option value="0">No</option><option value="1">Yes</option></select></div>' +
        '<div class="form-group"><label>' + tr('Discharge Destination', '???? ?????') + '</label><select id="neoD' +
        est" class="form-control"><option>Home</option><option>NICU</option><option>Transfer</option></select></div>' +
        +
        '<div class="form-group"><label>' + tr('Status', '????') + '</label><select id="neoStatus" class="form-c' +
        ontrol"><option>Healthy</option><option>Requiring Follow-up</option><option>Critical</option></select></div>' +
        +
        '</div>' +
        '<h4 style="margin:12px 0 8px">APGAR @ 1 min</h4><div class="form-grid" style="gap:8px">' + apgarComponent
        Select('n1') + '</div>' +
        '<h4 style="margin:12px 0 8px">APGAR @ 5 min</h4><div class="form-grid" style="gap:8px">' + apgarComponent
        Select('n5') + '</div>' +
        '<div class="form-group" style="margin-top:8px"><label>' + tr('Congenital Abnormalities', '????? ????') +
        + '</label><textarea id="neoAbn" class="form-control" rows="2"></textarea></div>' +
        '<button class="btn btn-primary" onclick="saveNeonatal(' + safeId(deliveryId) + ')" style="margin-top:8px"' +
        >? ' + tr('Save Neonatal Record', '??? ??? ??????') + '</button>';
        showModal(tr('Neonatal Record', '??? ??????') + ' (Delivery #' + safeId(deliveryId) + ')', html);
    };

    window.saveNeonatal = async (deliveryId) => {
        const bandId = document.getElementById('neoBandId')?.value.trim();
        if (!bandId) {
            showToast(tr('Mother/Infant Band ID is required for clinical safety match!', '??? ???? ?????? ?????? ??????
            ?????? ?????? ?????? ?????? !'), 'error');
            return;
        }
    }

```

```

}
try {
  const r = await API.post('/api/obgyn/neonatal', {
    delivery_id: deliveryId,
    birth_weight_grams: parseInt(document.getElementById('neoWt').value) || 0,
    length_cm: parseFloat(document.getElementById('neoLen').value) || 0,
    head_circumference_cm: parseFloat(document.getElementById('neoHC').value) || 0,
    blood_group: document.getElementById('neoBG').value,
    feeding_type: document.getElementById('neoFeed').value,
    resuscitation_needed: document.getElementById('neoResus').value === '1' ? 1 : 0,
    discharge_destination: document.getElementById('neoDest').value,
    discharge_status: document.getElementById('neoStatus').value,
    congenital_abnormalities: document.getElementById('neoAbn').value,
    apgar_1min_components: readApgarComponents('n1'),
    apgar_5min_components: readApgarComponents('n5'),
    follow_up_plan: 'Mother/Infant Band ID: ' + bandId
  });
  showToast(tr('Neonatal record saved! APGAR ', '?? ?????! ????? ') + (r.apgar_1min) + '/' + (r.apgar_5min))
;
  document.querySelector('.modal-overlay')?.remove();
  navigateTo(currentPage);
} catch (e) { showToast(tr('Error: ', '??? : ') + (e.message || ''), 'error'); }
};

// ===== DENTAL CLINIC WORKSPACE =====
let dentalTab = 'charting';
let dentalSelectedPatientId = null;
let dentalSelectedTooth = null;
let dentalSelectedSurface = 'Occlusal';

async function renderDental(el) {
  const [patients, dentalRecords, periodontalRecords, dentalImages] = await Promise.all([
    API.get('/api/patients').catch(() => []),
    dentalSelectedPatientId ? API.get('/api/dental/records/' + dentalSelectedPatientId).catch(() => []) : Promise.resolve([]),
    dentalSelectedPatientId ? API.get('/api/dental/periodontal/' + dentalSelectedPatientId).catch(() => []) : Promise.resolve([]),
    dentalSelectedPatientId ? API.get('/api/dental/images/' + dentalSelectedPatientId).catch(() => []) : Promise.resolve([])
  ]);

  if (!dentalSelectedPatientId && patients.length > 0) {
    dentalSelectedPatientId = patients[0].id;
  }

  const selectedPatient = patients.find(p => parseInt(p.id, 10) === parseInt(dentalSelectedPatientId, 10));

  let toothStatusMap = {};
  if (dentalRecords && dentalRecords.length) {
    const sorted = [...dentalRecords].sort((a, b) => new Date(a.visit_date) - new Date(b.visit_date));
    sorted.forEach(rec => {
      toothStatusMap[rec.tooth_number] = {
        condition: rec.condition,
        treatment: rec.treatment_done,

```

```

        surfaces: rec.affected_surfaces ? rec.affected_surfaces.split(',') : []
    };
    });
}
window.dentalToothStatusMap = toothStatusMap;

el.innerHTML = `
<div class="page-title">? ${tr('Dental / Dentistry Center', '???? ?? ??????')}</div>

<div class="card mb-16">
    <div style="display:flex;align-items:center;gap:12px;flex-wrap:wrap">
        <label style="font-weight:bold">${tr('Select Patient File:', '???? ??? ??????')}</label>
        <select id="dentalPatientSelect" class="form-input" style="max-width:300px" onchange="window.selectDentalPatient(this.value)">
            ${patients.map(p => `<option value="${safeId(p.id)}" ${parseInt(p.id, 10) === parseInt(dentalSelectedPatientId, 10) ? 'selected' : ''}>${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}
        </select>
        ${selectedPatient ? `<div class="badge badge-info">${tr('MRN: ', '??? ?????: ')}${selectedPatient.file_number}</div>` : ''}
    </div>
</div>

<div class="tab-bar">
    <button class="tab-btn ${dentalTab === 'charting' ? 'active' : ''}" onclick="dentalTab='charting';navigateTo(43)">? ${tr('Odontogram & Surfaces', '???? ?????? ??????')}</button>
    <button class="tab-btn ${dentalTab === 'periodontal' ? 'active' : ''}" onclick="dentalTab='periodontal';navigateTo(43)">? ${tr('Periodontal Charting', '??? ????? ??????')}</button>
    <button class="tab-btn ${dentalTab === 'images' ? 'active' : ''}" onclick="dentalTab='images';navigateTo(43)">?? ${tr('X-Rays & Imaging', '???? ????? ??????')}</button>
    <button class="tab-btn ${dentalTab === 'history' ? 'active' : ''}" onclick="dentalTab='history';navigateTo(43)">? ${tr('Treatment History', '??? ?????? ??????')}</button>
</div>

<div id="dentalTabContent"></div>
`;

const content = document.getElementById('dentalTabContent');

if (dentalTab === 'charting') {
    content.innerHTML = `
        <div style="display:grid;grid-template-columns:1.5fr 1fr;gap:20px;align-items:start">
            <div class="card">
                <h3 style="margin-bottom:12px;text-align:center">? ${tr('Odontogram (Adult / Deciduous)', '???? ???? ???? ??????')}</h3>

                <div style="margin-bottom:16px;text-align:center">
                    <label style="font-weight:bold;margin-right:8px">${tr('Dentition Mode:', '??? ??????')}</label>
                    <select id="dentalMode" class="form-input" style="width:auto;display:inline-block" onchange="window.toggleDentalMode(this.value)">
                        <option value="adult" selected>${tr('Adult Dentition (32 Teeth)', '????? ???????? (32 ??)}</option>
                        <option value="child">${tr('Deciduous Dentition (20 Teeth)', '??????? ???????? ???????? (20 ??)}</option>
                    </select>
                </div>
            </div>
        </div>
    `;
}

```

```
    </select>
  </div>
```

```
    <div id="odontogramWrapper" style="padding:10px;background:rgba(255,255,255,0.7);border-radius:12px;
border:1px solid var(--outline);min-height:220px;display:flex;flex-direction:column;gap:20px;align-items:cente
r">
```

```
      ${renderOdontogramSVG(toothStatusMap, 'adult')}
    </div>
```

```
    <div class="legend" style="display:flex;justify-content:center;gap:16px;margin-top:12px;flex-wrap:wr
ap;font-size:12px">
```

```
      <span style="display:flex;align-items:center;gap:6px"><span style="width:14px;height:14px;backgrou
nd:#ffffff;border:1px solid #999;border-radius:3px"></span> ${tr('Healthy', '????')}</span>
```

```
      <span style="display:flex;align-items:center;gap:6px"><span style="width:14px;height:14px;backgrou
nd:#fecaca;border-radius:3px"></span> ${tr('Caries (Decay)', '????')}</span>
```

```
      <span style="display:flex;align-items:center;gap:6px"><span style="width:14px;height:14px;backgrou
nd:#bdfdbfe;border-radius:3px"></span> ${tr('Restored (Filling)', '???')}</span>
```

```
      <span style="display:flex;align-items:center;gap:6px"><span style="width:14px;height:14px;backgrou
nd:#e9d5ff;border-radius:3px"></span> ${tr('Root Canal', '???? ???')}</span>
```

```
      <span style="display:flex;align-items:center;gap:6px"><span style="width:14px;height:14px;backgrou
nd:#e5e7eb;border-radius:3px"></span> ${tr('Extracted', '????? / ?????')}</span>
```

```
    </div>
```

```
  </div>
```

```
  <div class="card">
```

```
    <h3>? ${tr('5-Surface Charting Simulator', '????? ???? ?????? ??????')}</h3>
```

```
    <div style="margin:12px 0;padding:8px 12px;background:rgba(14,165,233,0.1);border-radius:8px;font-we
ight:bold;color:var(--primary);display:flex;justify-content:between;align-items:center">
```

```
      <span>${tr('Selected Tooth:', '???? ??????:')} <span id="dentalSelectedToothLabel">${dentalSelecte
dTooth ? dentalSelectedTooth : tr('Click a tooth', '???? ??? ?')}</span></span>
```

```
      <span>${tr('Selected Surface:', '????? ??????:')} <span id="dentalSelectedSurfaceLabel">${dentalSe
lectedSurface}</span></span>
```

```
    </div>
```

```
  <!-- 5-Surface Tooth Interactive Graphic -->
```

```
  <div style="display:flex;justify-content:center;margin:20px 0">
```

```
    <svg width="150" height="150" viewBox="0 0 100 100" style="border:1px dashed var(--outline);border
-radius:10px;background:#f8fafc">
```

```
      <!-- Buccal/Outer (Top) -->
```

```
      <polygon points="10,10 90,10 70,30 30,30" fill="${getSurfaceColor('Buccal')}}" stroke="var(--outl
ine)" stroke-width="1.5" style="cursor:pointer" onclick="window.selectDentalSurface('Buccal')" />
```

```
      <text x="50" y="22" font-size="6" text-anchor="middle" font-weight="bold" fill="#475569">Buccal
(B)</text>
```

```
      <!-- Distal (Right) -->
```

```
      <polygon points="90,10 90,90 70,70 70,30" fill="${getSurfaceColor('Distal')}}" stroke="var(--outl
ine)" stroke-width="1.5" style="cursor:pointer" onclick="window.selectDentalSurface('Distal')" />
```

```
      <text x="78" y="52" font-size="6" text-anchor="middle" font-weight="bold" fill="#475569">Distal
(D)</text>
```

```
      <!-- Lingual/Inner (Bottom) -->
```

```
      <polygon points="10,90 90,90 70,70 30,70" fill="${getSurfaceColor('Lingual')}}" stroke="var(--out
line)" stroke-width="1.5" style="cursor:pointer" onclick="window.selectDentalSurface('Lingual')" />
```

```
      <text x="50" y="82" font-size="6" text-anchor="middle" font-weight="bold" fill="#475569">Lingual
```

(L)</text>

```
<!-- Mesial (Left) -->
<polygon points="10,10 10,90 30,70 30,30" fill="${getSurfaceColor('Mesial')}}" stroke="var(--outline)" stroke-width="1.5" style="cursor:pointer" onclick="window.selectDentalSurface('Mesial')" />
<text x="22" y="52" font-size="6" text-anchor="middle" font-weight="bold" fill="#475569">Mesial
(M)</text>

<!-- Occlusal (Center) -->
<rect x="30" y="30" width="40" height="40" fill="${getSurfaceColor('Occlusal')}}" stroke="var(--outline)" stroke-width="1.5" style="cursor:pointer" onclick="window.selectDentalSurface('Occlusal')" />
<text x="50" y="52" font-size="6" text-anchor="middle" font-weight="bold" fill="#475569">Occlusal (O)</text>
</svg>
</div>
```

```
<div class="form-grid" style="grid-template-columns:1fr;gap:12px">
  <div class="form-group">
    <label>${tr('Condition / Diagnosis', '????? ????????? ????')}</label>
    <select id="dentalCondition" class="form-input" onchange="window.autoSuggestDentalTreatment(this.value)">
      <option value="Healthy">${tr('Healthy', '????')}</option>
      <option value="Caries">${tr('Caries / Decay', '???? ?????')}</option>
      <option value="Missing">${tr('Missing / Extracted', '????? / ?????')}</option>
      <option value="Restored">${tr('Restored / Filled', '????? ?????')}</option>
      <option value="Pulpal Pathology">${tr('Pulpal Pathology (Requires Root Canal)', '????? ??? (???? ???? ????)}')}</option>
      <option value="Fracture">${tr('Tooth Fracture', '??? ?? ?????')}</option>
    </select>
  </div>
  <div class="form-group">
    <label>${tr('Treatment Rendered', '??????? ??????? ?????')}</label>
    <select id="dentalTreatment" class="form-input">
      <option value="None">${tr('None / Observation', '?? ???? / ?????')}</option>
      <option value="Composite Filling">${tr('Composite Restorative Filling', '???? ??????? ???????')}</option>
      <option value="Root Canal Therapy">${tr('Root Canal Therapy (RCT)', '??? ????? ??? ?????')}</option>
      <option value="Extraction">${tr('Surgical / Simple Extraction', '??? ?? ???? / ?????')}</option>
      <option value="Scaling & Polishing">${tr('Dental Scaling & Polishing', '????? ????? ?????? ??? ?????')}</option>
      <option value="Dental Crown">${tr('Porcelain/Zirconia Crown', '????? ??? / ?????? ??')}</option>
      <option value="Fluoride Application">${tr('Topical Fluoride Application', '????? ?????????? ??? ?????')}</option>
    </select>
  </div>
  <button class="btn btn-primary" onclick="window.saveDentalRecord()" style="margin-top:16px;width:100%">? ${tr('Save Tooth & Surface Record', '??? ??? ???? ?????')}</button>
</div>
</div>
`;
```

```

} else if (dentalTab === 'periodontal') {
  content.innerHTML = `
    <div style="display:grid;grid-template-columns:1.2fr 1fr;gap:20px;align-items:start">
      <div class="card">
        <h3>? ${tr('Record Periodontal Pocket Exam', '????? ??? ???? ?????')}</h3>
        <div class="form-grid" style="grid-template-columns:1fr 1fr;gap:12px;margin-top:12px">
          <div class="form-group">
            <label>${tr('Select Tooth#:', '???? ??? ?????:')}</label>
            <input type="number" id="perioToothNumber" class="form-input" min="1" max="48" value="${dentalSe
lectedTooth || 11}" />
          </div>
          <div class="form-group">
            <label>${tr('Probing Depth (1-10 mm):', '??? ?????? (1-10 ??):')}</label>
            <input type="number" id="perioProbingDepth" class="form-input" min="1" max="10" value="3" />
          </div>
          <div class="form-group">
            <label>${tr('Gingival Recession (mm):', '????? ?????? (??):')}</label>
            <input type="number" id="perioGingivalRecession" class="form-input" min="0" max="10" value="0" /
>
          </div>
          <div class="form-group flex items-center" style="padding-top:24px">
            <input type="checkbox" id="perioBleeding" style="width:20px;height:20px;margin-right:8px" />
            <label for="perioBleeding" style="cursor:pointer;font-weight:bold;color:var(--error)">? ${tr('Bl
eeding on Probing', '???? ??? ?????')}</label>
          </div>
        </div>
        <button class="btn btn-primary" onclick="window.savePeriodontalRecord()" style="margin-top:16px;widt
h:100%">? ${tr('Save Periodontal Record', '??? ??? ?????')}</button>
      </div>

      <div class="card">
        <h3>? ${tr('Periodontal Exam History', '????? ????????? ???????? ?????')}</h3>
        ${periodontalRecords.length ? makeTable(
          [tr('Date', '????????'), tr('Tooth#', '????'), tr('Probing Depth', '??? ??????'), tr('Recession', '
????? ??????'), tr('Bleeding', '???????')],
          periodontalRecords.map(r => ({
            cells: [
              new Date(r.exam_date).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US'),
              r.tooth_number,
              r.probing_depth + ' mm',
              r.gingival_recession + ' mm',
              r.bleeding_on_probing ? 'Yes / ???' : 'No / ??'
            ]
          }))
        ) : `<div class="empty-state"><p>${tr('No previous periodontal exams recorded', '?? ???? ????? ???? ?
?? ?????')}</p></div>`}
      </div>
    </div>
  `;
} else if (dentalTab === 'images') {
  content.innerHTML = `
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:20px;align-items:start">
      <div class="card">
        <h3>?? ${tr('Link Dental X-Ray / Image', '??? ???? ?? ???? ?????')}</h3>

```

```

<div class="form-grid" style="grid-template-columns:1fr;gap:12px;margin-top:12px">
  <div class="form-group">
    <label>${tr('Select Tooth#:', '??? ????:')}</label>
    <input type="number" id="imageToothNumber" class="form-input" min="1" max="48" value="${dentalSe
lectedTooth || 11}" />
  </div>
  <div class="form-group">
    <label>${tr('Image Type:', '??? ??????:')}</label>
    <select id="imageTypeSelect" class="form-input">
      <option value="Periapical X-Ray">${tr('Periapical X-Ray', '???? ???? ?????')}</option>
      <option value="Bitewing X-Ray">${tr('Bitewing X-Ray', '???? ???? ?????')}</option>
      <option value="Panoramic OPG">${tr('Panoramic OPG', '???? ????????? OPG')}</option>
      <option value="Intraoral Photo">${tr('Intraoral Camera Photo', '???? ?????? ???? ?????')}</opti
on>
    </select>
  </div>
  <div class="form-group">
    <label>${tr('Select Image / Mock RVG Path:', '???? ?????? / ???? ???? RVG:')}</label>
    <select id="imagePathSelect" class="form-input">
      <option value="/images/mock_xray1.jpg">? ${tr('Healthy Molar X-Ray', '???? ?? ???? ?????')}</op
tion>
      <option value="/images/mock_xray2.jpg">? ${tr('Caries Root X-Ray', '???? ???? ??? ?')}</optio
n>
      <option value="/images/mock_intraoral.jpg">? ${tr('Intraoral Filling Photo', '???? ???? ??????
? ??????')}</option>
    </select>
  </div>
  <div>
    <button class="btn btn-primary" onclick="window.saveDentalImage()" style="margin-top:16px;width:100%
">? ${tr('Link Dental Image', '??? ???? ?????')}</button>
  </div>

  <div class="card">
    <h3>?? ${tr('Linked Dental Images Gallery', '???? ?????? ?????? ?????')}</h3>
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-top:12px">
      ${dentalImages.length ? dentalImages.map(img => `
        <div style="border:1px solid var(--outline);border-radius:8px;padding:8px;text-align:center;back
ground:#fff">
          <div style="font-weight:bold;margin-bottom:4px">Tooth# ${img.tooth_number} (${img.image_type})
</div>
          <div style="width:100%;height:100px;background:#e2e8f0;display:flex;align-items:center;justify
-content:center;border-radius:4px;font-size:12px;color:#64748b;font-weight:bold">
            ? [Dental RVG Image: ${img.image_path.split('/').pop()}]
          </div>
          <div style="font-size:10px;margin-top:4px;color:var(--text-secondary)">${new Date(img.created_
at).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US')}</div>
        </div>
        `).join('') : `<div class="empty-state" style="grid-column: span 2"><p>${tr('No dental images link
ed yet', '?? ???? ???? ???? ?????? ?????')}</p></div>`}
      </div>
    </div>
  </div>
  `;
} else {

```



```

content.innerHTML = `
  <div class="card">
    <h3>? ${tr('Dental Records History', '????? ??????? ??????? ???????')}</h3>
    ${dentalRecords.length ? makeTable(
      [tr('Date', '???????'), tr('Tooth#', '??? ???'), tr('Affected Surfaces', '?????? ?????????'), tr('Co
ndition', '??????? / ??????'), tr('Treatment Done', '??????? ???????')],
      dentalRecords.map(r => ({
        cells: [
          new Date(r.visit_date).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US') + ' ' + new Date(r.visit
_date).toLocaleTimeString(isArabic ? 'ar-SA' : 'en-US', {hour:'2-digit', minute:'2-digit'}),
          r.tooth_number,
          r.affected_surfaces || 'Occlusal',
          r.condition,
          r.treatment_done || '-'
        ]
      })))
    ) : `<div class="empty-state"><p>${tr('No previous dental procedures recorded for this patient', '?? ?
??? ?????? ?????? ?????? ?????? ?????? ??????')}</p></div>`}
  </div>
`;
}
}

window.selectDentalPatient = function(val) {
  dentalSelectedPatientId = val;
  dentalSelectedTooth = null;
  navigateTo(43);
};

window.toggleDentalMode = function(mode) {
  const odontogramWrapper = document.getElementById('odontogramWrapper');
  odontogramWrapper.innerHTML = renderOdontogramSVG(window.dentalToothStatusMap, mode);
};

window.selectTooth = function(num) {
  dentalSelectedTooth = num;
  const label = document.getElementById('dentalSelectedToothLabel');
  if (label) label.textContent = num;
  document.querySelectorAll('.tooth-btn').forEach(btn => {
    const isSelected = parseInt(btn.dataset.tooth, 10) === num;
    btn.style.border = isSelected ? '2px solid var(--primary)' : '2px solid var(--outline)';
    btn.style.boxShadow = isSelected ? '0 0 8px rgba(14, 165, 233, 0.4)' : 'none';
  });
};

window.selectDentalSurface = function(surface) {
  dentalSelectedSurface = surface;
  const label = document.getElementById('dentalSelectedSurfaceLabel');
  if (label) label.textContent = surface;
  navigateTo(43);
};

window.autoSuggestDentalTreatment = function(condition) {
  const txSelect = document.getElementById('dentalTreatment');

```

```

    if (!txSelect) return;
    if (condition === 'Caries') txSelect.value = 'Composite Filling';
    else if (condition === 'Pulpal Pathology') txSelect.value = 'Root Canal Therapy';
    else if (condition === 'Missing') txSelect.value = 'Extraction';
    else if (condition === 'Healthy') txSelect.value = 'None';
  };

window.saveDentalRecord = async function() {
  if (!dentalSelectedPatientId) {
    showToast(tr('Please select a patient first', '???? ?????? ??? ????'), 'error');
    return;
  }
  if (!dentalSelectedTooth) {
    showToast(tr('Please click a tooth on the diagram to select it', '???? ?????? ?? ?? ?????? ?????'), 'error');
  };
  return;
}
const condition = document.getElementById('dentalCondition').value;
const treatment = document.getElementById('dentalTreatment').value;
try {
  await API.post('/api/dental/records', {
    patient_id: dentalSelectedPatientId,
    tooth_number: dentalSelectedTooth,
    condition,
    treatment_done: treatment,
    affected_surfaces: dentalSelectedSurface
  });
  showToast(tr('Tooth and Surface record saved successfully', '?? ??? ??? ???? ?????? ?????'));
  navigateTo(43);
} catch (e) {
  showToast(tr('Error saving record', '??? ?? ?????'), 'error');
}
};

window.savePeriodontalRecord = async function() {
  const tooth = document.getElementById('perioToothNumber').value;
  const depth = document.getElementById('perioProbingDepth').value;
  const recession = document.getElementById('perioGingivalRecession').value;
  const bleeding = document.getElementById('perioBleeding').checked;
  try {
    await API.post('/api/dental/periodontal', {
      patient_id: dentalSelectedPatientId,
      tooth_number: tooth,
      probing_depth: depth,
      bleeding_on_probing: bleeding,
      gingival_recession: recession
    });
    showToast(tr('Periodontal record saved', '?? ??? ??? ?????? ?????'));
    navigateTo(43);
  } catch (e) {
    showToast(tr('Error saving periodontal record', '??? ?? ??? ??? ?????'), 'error');
  }
};

```

```

window.saveDentalImage = async function() {
  const tooth = document.getElementById('imageToothNumber').value;
  const type = document.getElementById('imageTypeSelect').value;
  const path = document.getElementById('imagePathSelect').value;
  try {
    await API.post('/api/dental/images', {
      patient_id: dentalSelectedPatientId,
      tooth_number: tooth,
      image_path: path,
      image_type: type
    });
    showToast(tr('Dental image linked successfully', '?? ??? ??? ???? ?? ??'));
    navigateTo(43);
  } catch (e) {
    showToast(tr('Error linking image', '??? ?? ?? ????'), 'error');
  }
};

```

```

function getSurfaceColor(surface) {
  if (!dentalSelectedTooth || !window.dentalToothStatusMap) return '#ffffff';
  const state = window.dentalToothStatusMap[dentalSelectedTooth];
  if (!state) return '#ffffff';
  const hasSurface = state.surfaces && state.surfaces.includes(surface);
  if (hasSurface || surface === 'Occlusal') {
    if (state.condition === 'Caries') return '#fecaca';
    else if (state.condition === 'Restored') return '#bfdbfe';
    else if (state.condition === 'Pulpal Pathology') return '#e9d5ff';
    else if (state.condition === 'Missing') return '#e5e7eb';
  }
  return '#ffffff';
}

```

```

function renderOdontogramSVG(toothStatusMap, mode = 'adult') {
  let upperTeeth, lowerTeeth;
  if (mode === 'adult') {
    upperTeeth = [18,17,16,15,14,13,12,11, 21,22,23,24,25,26,27,28];
    lowerTeeth = [48,47,46,45,44,43,42,41, 31,32,33,34,35,36,37,38];
  } else {
    upperTeeth = [55,54,53,52,51, 61,62,63,64,65];
    lowerTeeth = [85,84,83,82,81, 71,72,73,74,75];
  }
}

```

```

const makeRow = (list) => `
<div style="display:flex;gap:6px;justify-content:center;flex-wrap:wrap">
  ${list.map(t => {
    const state = toothStatusMap[t] || { condition: 'Healthy', treatment: '' };
    let bgColor = '#ffffff';
    if (state.condition === 'Caries') bgColor = '#fecaca';
    else if (state.condition === 'Restored') bgColor = '#bfdbfe';
    else if (state.condition === 'Pulpal Pathology') bgColor = '#e9d5ff';
    else if (state.condition === 'Missing') bgColor = '#e5e7eb';

    return `
      <button class="tooth-btn btn btn-outline"

```

```

        data-tooth="${t}"
        style="width:40px;height:55px;padding:2px;display:flex;flex-direction:column;align-items:center;justify-content:space-between;background:${bgColor};border:2px solid ${dentalSelectedTooth === t ? 'var(--primary)' : 'var(--outline)'};border-radius:8px"
        onclick="window.selectTooth(${t})">
        <span style="font-size:10px;font-weight:bold">${t}</span>
        <span style="font-size:16px">${state.condition === 'Missing' ? '?' : ''}</span>
        <span style="font-size:8px;opacity:0.8">${state.condition.substring(0,3)}</span>
    </button>
    `;
    }).join('')}
</div>
`;

return `
    <div style="width:100%">
        <div style="text-align:center;font-weight:bold;font-size:12px;margin-bottom:6px">${mode === 'adult' ? tr
('Upper Jaw / ???? ??????', '???? ??????') : tr('Primary Upper Jaw / ???? ?????? ??????', '???? ?????? ??????')}</div>
        ${makeRow(upperTeeth)}
        <div style="text-align:center;font-weight:bold;font-size:12px;margin:12px 0 6px">${mode === 'adult' ? tr
('Lower Jaw / ???? ??????', '???? ??????') : tr('Primary Lower Jaw / ???? ?????? ??????', '???? ?????? ??????')}</div>
        ${makeRow(lowerTeeth)}
    </div>
    `;
}

// ===== PAGE LOADER =====
async function loadPage(page) {
    const el = document.getElementById('pageContent');
    el.style.animation = 'none'; el.offsetHeight; el.style.animation = '';
    const pages = [renderDashboard, renderReception, renderAppointments, renderDoctor, renderLab, renderRadiology, renderPharmacy, renderHR, renderFinance, renderInsurance, renderInventory, renderNursing, renderWaitingQueue, renderPatientAccounts, renderReports, renderMessaging, renderCatalog, renderDeptRequests, renderSurgery, renderBloodBank, renderConsentForms, renderEmergency, renderInpatient, renderICU, renderCSSD, renderDietary, renderInfectionControl, renderQuality, renderMaintenance, renderTransport, renderMedicalRecords, renderClinicalPharmacy, renderRehabilitation, renderPatientPortal, renderZATCA, renderTelemedicine, renderPathology, renderSocialWork, renderMortuary, renderCME, renderCosmeticSurgery, renderOBGYN, renderSettings, renderDental, renderSpecialties, renderOVR, renderAuditLog, renderSpecialtiesPanel];
    if (pages[page]) await pages[page](el);
    else if (NAV_ITEMS[page]) renderDepartmentWorkspace(el, page);
    else el.innerHTML = `<div class="page-title">${NAV_ITEMS[page]?.icon} ${tr(NAV_ITEMS[page]?.en, NAV_ITEMS[page]?.ar)}</div><div class="card"><p>${tr('Coming soon...', '??????...')}</p></div>`;
}

// ===== DASHBOARD =====

// ===== MEDICAL REPORT / SICK LEAVE =====
window.showMedicalReportForm = (type) => {
    const patientId = document.getElementById('drPatient')?.value || window._selectedPatientId;

```

```

if (!patientId) return showToast(tr('Select patient first', '???? ???? ?????'), 'error');
const patientName = document.getElementById('drPatient')?.selectedOptions[0]?.text || window._selectedPatientName || '';

const typeLabels = {
  sick_leave: { en: 'Sick Leave', ar: '????? ?????' },
  medical_report: { en: 'Medical Report', ar: '????? ???' },
  fitness: { en: 'Fitness Certificate', ar: '????? ?????' },
};
const label = typeLabels[type] || typeLabels.medical_report;

const modal = document.createElement('div');
modal.className = 'modal-overlay';
modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
modal.innerHTML = '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:24px;width:550px;max-height:90vh;overflow-y:auto;direction:rtl">' +
  '<h3 style="margin:0 0 16px;color:var(--primary)">' + (isArabic ? label.ar : label.en) + '</h3>' +
  '<div class="form-group"><label>' + tr('Patient', '??????') + '</label><input class="form-input" value="' + patientName + '" readonly></div>' +
  '<div class="form-group"><label>' + tr('Diagnosis', '???????') + '</label><input class="form-input" id="mrDiagnosis" placeholder="' + tr('Diagnosis', '???????') + '"></div>' +
  '<div class="form-group"><label>' + tr('ICD Code', '??? ICD') + '</label><input class="form-input" id="mrICD" placeholder="e.g. J06.9"></div>' +
  (type === 'sick_leave' ?
    '<div style="display:flex;gap:12px">' +
    '<div class="form-group" style="flex:1"><label>' + tr('From', '??') + '</label><input type="date" class="form-input" id="mrFrom"></div>' +
    '<div class="form-group" style="flex:1"><label>' + tr('To', '???') + '</label><input type="date" class="form-input" id="mrTo"></div>' +
    '<div class="form-group" style="flex:1"><label>' + tr('Days', '????') + '</label><input type="number" class="form-input" id="mrDays" min="1"></div>' +
    '</div>' : '') +
  (type === 'fitness' ?
    '<div class="form-group"><label>' + tr('Fitness Status', '???? ???????') + '</label>' +
    '<select class="form-input" id="mrFitness"><option value="fit">' + tr('Fit', '????') + '</option><option value="unfit">' + tr('Unfit', '??? ?????') + '</option><option value="conditional">' + tr('Conditional', '?????') + '</option></select></div>' : '') +
    '<div class="form-group"><label>' + tr('Notes', '???????') + '</label><textarea class="form-input" id="mrNotes" rows="3"></textarea></div>' +
    '<div style="display:flex;gap:12px;margin-top:16px">' +
    '<button class="btn btn-primary" onclick="saveMedicalReport(\'' + type + '\', ' + safeId(patientId) + ', \'' + jsStr(patientName) + '\')" style="flex:1">' + tr('Save & Print', '??? ???????') + '</button>' +
    '<button class="btn btn-secondary" onclick="this.closest(\'.modal-overlay\').remove()" style="flex:1">' + tr('Cancel', '?????') + '</button>' +
    '</div></div>';
document.body.appendChild(modal);
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };
};

window.saveMedicalReport = async (type, patientId, patientName) => {
  try {
    const data = {
      patient_id: patientId,

```

```

    patient_name: patientName,
    report_type: type,
    diagnosis: document.getElementById('mrDiagnosis')?.value || '',
    icd_code: document.getElementById('mrICD')?.value || '',
    start_date: document.getElementById('mrFrom')?.value || null,
    end_date: document.getElementById('mrTo')?.value || null,
    duration_days: document.getElementById('mrDays')?.value || null,
    notes: document.getElementById('mrNotes')?.value || '',
    fitness_status: document.getElementById('mrFitness')?.value || null,
  };

  const result = await API.post('/api/medical-reports', data);
  document.querySelector('.modal-overlay')?.remove();
  showToast(tr('Report saved!', '?? ?? ??????!!'));

  // Print the report
  printMedicalReport(result, type);
} catch (e) { console.error('Medical report error:', e); showToast(e?.message || tr('Error saving report', '
??? ?? ?? ??????'), 'error'); }
};

window.printMedicalReport = (report, type) => {
  const typeLabels = { sick_leave: { ar: '????? ?????', en: 'Sick Leave Certificate' }, medical_report: { ar:
'????? ???', en: 'Medical Report' }, fitness: { ar: '????? ????? ?????', en: 'Fitness Certificate' } };
  const label = typeLabels[type] || typeLabels.medical_report;

  let html = '<div style="font-family:Arial;padding:40px;direction:rtl;text-align:right;line-height:2">';
  html += '<div style="text-align:center;border-bottom:2px solid #1a73e8;padding-bottom:16px;margin-bottom:24px">';
  html += '<h2 style="color:#1a73e8;margin:0">????? ????? - jumanaMedical</h2>';
  html += '<p style="margin:4px 0;color:#666">??????? ???????? ?????????</p>';
  html += '</div>';
  html += '<h3 style="text-align:center;background:#f0f6ff;padding:12px;border-radius:8px;margin:20px 0">' + label.ar + ' / ' + label.en + '</h3>';
  html += '<table style="width:100%;margin:16px 0;border-collapse:collapse">';
  html += '<tr><td style="padding:8px;font-weight:bold;width:30%">??? ??????:</td><td style="padding:8px">' + escapeHTML(report.report_number || '') + '</td></tr>';
  html += '<tr><td style="padding:8px;font-weight:bold">??? ??????:</td><td style="padding:8px">' + escapeHTML(report.patient_name || '') + '</td></tr>';
  html += '<tr><td style="padding:8px;font-weight:bold">?????????:</td><td style="padding:8px">' + escapeHTML(report.diagnosis || '') + '</td></tr>';
  if (report.icd_code) html += '<tr><td style="padding:8px;font-weight:bold">??? ICD:</td><td style="padding:8px">' + escapeHTML(report.icd_code) + '</td></tr>';
  if (type === 'sick_leave') {
    html += '<tr><td style="padding:8px;font-weight:bold">?? ??????:</td><td style="padding:8px">' + escapeHTML(report.start_date || '') + '</td></tr>';
    html += '<tr><td style="padding:8px;font-weight:bold">??? ??????:</td><td style="padding:8px">' + escapeHTML(report.end_date || '') + '</td></tr>';
    html += '<tr><td style="padding:8px;font-weight:bold">??? ??????:</td><td style="padding:8px">' + escapeHTML(report.duration_days || '') + ' ' + (isArabic ? '???' : 'days') + '</td></tr>';
  }
  if (type === 'fitness') {
    const statusAr = { fit: '???? ??????', unfit: '??? ?????', conditional: '???? ??????' };
    html += '<tr><td style="padding:8px;font-weight:bold">?????????:</td><td style="padding:8px;font-weight:bold;

```

```

color:' + (report.fitness_status === 'fit' ? 'green' : 'red') + '">' + (statusAr[report.fitness_status] || '')
+ '</td></tr>';
}
if (report.notes) html += '<tr><td style="padding:8px;font-weight:bold">?????</td><td style="padding:8px"
>' + escapeHTML(report.notes) + '</td></tr>';
html += '</table>';
html += '<div style="margin-top:40px;display:flex;justify-content:space-between">';
html += '<div style="text-align:center"><p>_____</p><p>???? ????</p><p style="font-weight:bold"
>' + escapeHTML(report.doctor || '') + '</p></div>';
html += '<div style="text-align:center"><p>_____</p><p>??? </p></div>';
html += '</div>';
html += '<p style="text-align:center;margin-top:24px;font-size:11px;color:#999">???? ???? : ' + new Date(
).toLocaleDateString('ar-SA') + ' | Report #' + escapeHTML(report.report_number || '') + '</p>';
html += '</div>';

printDocument(label.ar, html, { showHeader: false });
};

// ===== DRUG INTERACTION CHECK =====
// Returns { hasCritical, failed }. IMPORTANT-2/IMPORTANT-3: a CRITICAL interaction is a HARD-STOP
// signal for the caller (sendRx), and a network/CDS failure FAILS CLOSED (failed:true) ? never a
// silent "all clear". The advisory modal is still shown for non-critical interactions.
window.checkDrugInteractions = async (drugs) => {
  try {
    if (!drugs || drugs.length < 2) return { hasCritical: false, failed: false };
    const result = await API.post('/api/drug-interactions/check', { drugs });
    const hasCritical = !(result.interactions && result.interactions.some(i => String(i.severity).toLowerCase
() === 'critical'));
    if (result.interactions && result.interactions.length > 0) {
      let alertHtml = '<div style="background:#fff3f3;border:2px solid #ff4444;border-radius:12px;padding:16px
;direction:rtl">';
      alertHtml += '<h4 style="color:#cc0000;margin:0 0 12px">?? ' + tr('Drug Interactions Found!', '?? ?????
??? ?????? ??????!') + '</h4>';
      result.interactions.forEach(i => {
        const color = i.severity === 'critical' ? '#cc0000' : i.severity === 'high' ? '#ff6600' : '#ff9900';
        alertHtml += '<div style="margin:8px 0;padding:8px;background:#fff;border-right:4px solid ' + color +
';border-radius:4px">';
        alertHtml += '<strong>' + escapeHTML(i.drugs.join(' ? ')) + '</strong><br>';
        alertHtml += '<span style="color:' + color + '">[' + i.severity.toUpperCase() + '] ' + escapeHTML(isAr
abic ? i.message_ar : i.message_en) + '</span>';
        alertHtml += '</div>';
      });
      alertHtml += '</div>';

      const modal = document.createElement('div');
      modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.6);z-index:9999;display:flex;align
-items:center;justify-content:center';
      modal.innerHTML = '<div style="background:#fff;border-radius:16px;padding:24px;width:500px;max-height:80
vh;overflow-y:auto">' +
        alertHtml +
        '<div style="display:flex;gap:12px;margin-top:16px">' +
        '<button class="btn btn-danger" onclick="this.closest(\''.modal-overlay\')?.remove();this.parentElement
.parentElement.parentElement.remove()" style="flex:1;background:#cc0000;color:#fff">' + tr('Cancel Prescript
ion', '???? ?????') + '</button>' +

```

```

        '<button class="btn" onclick="this.parentElement.parentElement.parentElement.remove()" style="flex:1">
?? ' + tr('Continue Anyway', '????? ??? ??????') + '</button>' +
        '</div></div>';
        document.body.appendChild(modal);
    }
    return { hasCritical, failed: false };
} catch (e) {
    // IMPORTANT-2: FAIL-CLOSED ? a failed interaction check must NOT read as "no interactions".
    console.error('Interaction check failed:', e);
    return { hasCritical: false, failed: true };
}
};

// ===== ALLERGY CHECK =====
window.checkAllergyBeforePrescribe = async (patientId, drugs) => {
    try {
        if (!patientId || !drugs || drugs.length === 0) return true;
        const result = await API.post('/api/allergy-check', { patient_id: patientId, drugs });

        // Merge clinical safety check alerts (scanning nursing vitals and patient problem list)
        try {
            const safetyCheck = await API.post('/api/clinical/safety-check', { patient_id: patientId, drug_name: dru
gs[0] });
            if (safetyCheck && safetyCheck.alert) {
                if (!result.alerts) result.alerts = [];
                const alreadyAlerted = result.alerts.some(a => a.drug.toLowerCase() === drugs[0].toLowerCase());
                if (!alreadyAlerted) {
                    result.alerts.push({
                        drug: drugs[0],
                        message_en: safetyCheck.message || 'Potential allergy conflict detected.',
                        message_ar: '?? ??? ????? ?????? ????? ?? ????? ????????? ??????????: ' + (safetyCheck.details || '')
                    });
                    result.patient_allergies = (result.patient_allergies ? result.patient_allergies + ', ' : '') +
                        (safetyCheck.details || tr('Nursing vitals/problem list', '???????? ?????????/????? ??????'));
                }
            }
        } catch (e) {
            console.error('Clinical safety allergy check failed:', e);
        }

        if (result.alerts && result.alerts.length > 0) {
            let alertHtml = '<div style="background:#ffe0e0;border:3px solid #ff0000;border-radius:12px;padding:20px
;direction:rtl">';
            alertHtml += '<h3 style="color:#cc0000;margin:0 0 12px">? ' + tr('ALLERGY ALERT!', '????? ??????!!') + '<
/h3>';
            alertHtml += '<p style="margin:0 0 12px">' + tr('Patient allergies:', '????? ??????:') + ' <strong styl
e="color:#cc0000">' + escapeHTML(result.patient_allergies) + '</strong></p>';
            result.alerts.forEach(a => {
                alertHtml += '<div style="margin:8px 0;padding:10px;background:#fff;border-right:5px solid #ff0000;bor
der-radius:4px">';
                alertHtml += '<strong style="color:#cc0000">? ' + escapeHTML(a.drug) + '</strong><br>';
                alertHtml += '<span>' + escapeHTML(isArabic ? a.message_ar : a.message_en) + '</span>';
                alertHtml += '</div>';
            });
        }
    }
};

```



```

    alertHtml += '</div>';

    const modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(200,0,0,0.3);z-index:9999;display:flex;align-items:center;justify-content:center';
    modal.innerHTML = '<div style="background:#fff;border-radius:16px;padding:24px;width:500px">' +
        alertHtml +
        '<div style="margin-top:16px;text-align:center">' +
        '<button class="btn" onclick="this.parentElement.parentElement.parentElement.remove()" style="background:#cc0000;color:#fff;width:100%;padding:12px">? ' + tr('Understood - Review Prescription', '???? - ?????? ??' + '????') + '</button>' +
        '</div></div>';
    document.body.appendChild(modal);
    return false;
}
return true;
} catch (e) {
    // IMPORTANT-2: FAIL-CLOSED ? if the allergy check cannot complete (network/server failure) we
    // must NOT silently allow the prescription. Surface a warning and BLOCK; the prescriber can
    // still proceed only by supplying an explicit, audited override reason in sendRx.
    console.error('Allergy check failed:', e);
    try {
        showToast(tr('Allergy check failed ? fail-closed (blocked). Override requires a reason.',
            '???? ???? ?????? ? ?? ????? ??????????. ?????? ????? ??????.'), 'error');
    } catch (_) { /* toast best-effort */ }
    return false;
}
};

// =====
// E1 DOCTOR STATION ? Problem List + CPOE + SOAP notes (additive tabs).
// Rendered into #drE1Panel by loadPatientInfo() once a patient is selected.
// All sinks use escapeHTML(); ids coerced via safeId(); RTL via tr().
// =====
window.renderE1Panel = async (pid) => {
    const panel = document.getElementById('drE1Panel');
    if (!panel || !pid) return;
    panel.innerHTML = `
        <div class="card mt-16" style="border:2px solid var(--accent,#0ea5e9)">
            <div style="display:flex;gap:8px;flex-wrap:wrap;border-bottom:1px solid var(--border-color,#e5e7eb);padding-bottom:8px;margin-bottom:12px">
                <button class="btn btn-sm e1-tab" data-tab="problems" onclick="e1SwitchTab('problems',${safeId(pid)})"
                >? ${tr('Problem List', '???? ??????')}</button>
                <button class="btn btn-sm e1-tab" data-tab="cpoe" onclick="e1SwitchTab('cpoe',${safeId(pid)})" >? ${tr(
                'CPOE Orders', '???? CPOE')}</button>
                <button class="btn btn-sm e1-tab" data-tab="soap" onclick="e1SwitchTab('soap',${safeId(pid)})" >? ${tr(
                'SOAP Note', '????? SOAP')}</button>
                <button class="btn btn-sm e1-tab" data-tab="cardiology" onclick="e1SwitchTab('cardiology',${safeId(pid)})" >? ${tr('Cardiology', '?? ?????')}</button>
                <button class="btn btn-sm e1-tab" data-tab="gastro" onclick="e1SwitchTab('gastro',${safeId(pid)})" >? $
                ${tr('Gastroenterology', '????? ?????')}</button>
                <button class="btn btn-sm e1-tab" data-tab="endocrine" onclick="e1SwitchTab('endocrine',${safeId(pid)})" >? ${tr('Endocrinology', '????? ?????')}</button>
                <button class="btn btn-sm e1-tab" data-tab="nephrology" onclick="e1SwitchTab('nephrology',${safeId(pid)}

```

```

    }}">? ${tr('Nephrology', '????? ???????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="pulmonology" onclick="e1SwitchTab('pulmonology', ${safeId(p
id)})">? ${tr('Pulmonology', '??????? ???????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="rheumatology" onclick="e1SwitchTab('rheumatology', ${safeId
(pid)})">? ${tr('Rheumatology', '??????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="neurology" onclick="e1SwitchTab('neurology', ${safeId(pid)}
)">? ${tr('Neurology', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="ophthalmology" onclick="e1SwitchTab('ophthalmology', ${safe
Id(pid)})">? ${tr('Ophthalmology', '?? ?????? ??????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="ent" onclick="e1SwitchTab('ent', ${safeId(pid)})">? ${tr('E
NT', '????? ?????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="plastic_burns" onclick="e1SwitchTab('plastic_burns', ${safe
Id(pid)})">? ${tr('Plastic & Burns', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="intensive_care" onclick="e1SwitchTab('intensive_care', ${sa
feId(pid)})">? ${tr('Intensive Care', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="orthopedics" onclick="e1SwitchTab('orthopedics', ${safeId(p
id)})">? ${tr('Orthopedics', '????? ??????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="general_surgery" onclick="e1SwitchTab('general_surgery', ${
safeId(pid)})">? ${tr('General Surgery', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="urology" onclick="e1SwitchTab('urology', ${safeId(pid)})">?
${tr('Urology', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="cardiothoracic" onclick="e1SwitchTab('cardiothoracic', ${sa
feId(pid)})">? ${tr('Cardiothoracic', '????? ?????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="anesthesia_pain" onclick="e1SwitchTab('anesthesia_pain', ${
safeId(pid)})">? ${tr('Anesthesia & Pain', '????????? ?????? ??????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="pediatrics" onclick="e1SwitchTab('pediatrics', ${safeId(pid
)})">? ${tr('Pediatrics', '?? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="obgyn" onclick="e1SwitchTab('obgyn', ${safeId(pid)})">? ${t
r('OBGYN', '????????? ?????????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="psychiatry" onclick="e1SwitchTab('psychiatry', ${safeId(pid
)})">? ${tr('Psychiatry', '????? ??????')}</button>
        <button class="btn btn-sm e1-tab" data-tab="dermatology" onclick="e1SwitchTab('dermatology', ${safeId(p
id)})">? ${tr('Dermatology', '????????? ?????????')}</button>
    </div>
    <div id="e1TabBody"></div>
</div>`;
window.e1SwitchTab('problems', pid);
};

window.e1SwitchTab = async (tab, pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    document.querySelectorAll('.e1-tab').forEach(btn => {
        btn.classList.toggle('btn-primary', btn.dataset.tab === tab);
    });

    if (tab === 'problems') return window.e1RenderProblems(pid);
    if (tab === 'cpoe') return window.e1RenderCpoe(pid);
    if (tab === 'soap') return window.e1RenderSoap(pid);
    if (tab === 'cardiology') return window.e1RenderCardiology(pid);
    if (tab === 'gastro') return window.e1RenderGastro(pid);
    if (tab === 'endocrine') return window.e1RenderEndocrine(pid);
    if (tab === 'nephrology') return window.e1RenderNephrology(pid);

```

```

    if (tab === 'pulmonology') return window.e1RenderPulmonology(pid);
    if (tab === 'rheumatology') return window.e1RenderRheumatology(pid);
    if (tab === 'neurology') return window.e1RenderNeurology(pid);
    if (tab === 'ophthalmology') return window.e1RenderOphthalmology(pid);
    if (tab === 'ent') return window.e1RenderEnt(pid);
    if (tab === 'plastic_burns') return window.e1RenderPlasticBurns(pid);
    if (tab === 'intensive_care') return window.e1RenderIntensiveCare(pid);
    if (tab === 'orthopedics') return window.e1RenderOrthopedics(pid);
    if (tab === 'general_surgery') return window.e1RenderGeneralSurgery(pid);
    if (tab === 'urology') return window.e1RenderUrology(pid);
    if (tab === 'cardiothoracic') return window.e1RenderCardiothoracic(pid);
    if (tab === 'anesthesia_pain') return window.e1RenderAnesthesiaPain(pid);
    if (tab === 'pediatrics') return window.e1RenderPediatrics(pid);
    if (tab === 'obgyn') return window.e1RenderObgyn(pid);
    if (tab === 'psychiatry') return window.e1RenderPsychiatry(pid);
    if (tab === 'dermatology') return window.e1RenderDermatology(pid);
  };

// ----- Problem List -----
window.e1RenderProblems = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;
  let probs = [];
  try { probs = await API.get('/api/problems?patient_id=' + encodeURIComponent(pid)); } catch (e) { probs = []
  ; }
  const rows = (probs || []).map(p => `
    <div style="display:flex;gap:8px;align-items:center;padding:8px;margin:4px 0;border-radius:8px;background:
var(--hover,#f8f9fa);border-right:4px solid ${p.status === 'resolved' ? '#94a3b8' : '#16a34a'}">
      <div style="flex:1">
        <strong>${escapeHTML(p.description)}</strong>
        ${p.icd10 ? `<span class="badge badge-info" style="margin-right:6px">ICD: ${escapeHTML(p.icd10)}</span>
>` : ''}
        <span class="badge ${p.status === 'resolved' ? '' : 'badge-success'}">${escapeHTML(p.status)}</span>
      </div>
      ${p.status === 'active'
        ? `<button class="btn btn-sm" onclick="e1ResolveProblem(${safeId(p.id)},{safeId(pid)})">? ${tr('Resol
ve', '??')}</button>`
        : ''}
    </div>`).join('');
  body.innerHTML = `
    <div class="form-group mb-8"><label>${tr('Problem description', '??? ?????')}</label><input class="form-
input" id="e1ProbDesc"></div>
    <div class="flex gap-8 mb-8">
      <div class="form-group" style="flex:1"><label>${tr('Problem ICD-10', '??? ????? ?????')}</label><input
class="form-input" id="e1ProbIcd"></div>
      <div class="form-group" style="flex:1"><label>SNOMED</label><input class="form-input" id="e1ProbSnomed">
</div>
    </div>
    <button class="btn btn-primary btn-sm mb-12" onclick="e1AddProblem(${safeId(pid)})">? ${tr('Add Problem',
'????? ?????')}</button>
    <div>${rows || `<div style="color:var(--text-dim)">${tr('No problems recorded', '?? ???? ????? ?????')}</d
iv>`}</div>`;
  };
};

```

```

window.e1AddProblem = async (pid) => {
  const desc = document.getElementById('e1ProbDesc')?.value;
  if (!desc || !desc.trim()) { showToast(tr('Enter a description', '???? ?????'), 'error'); return; }
  try {
    await API.post('/api/problems', {
      patient_id: pid, description: desc,
      icd10: document.getElementById('e1ProbIcd')?.value || '',
      snomed: document.getElementById('e1ProbSnomed')?.value || ''
    });
    showToast(tr('Problem added', '??? ?????? ????????'));
    window.e1RenderProblems(pid);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```

window.e1ResolveProblem = async (id, pid) => {
  try {
    await API.patch('/api/problems/' + id, { status: 'resolved' });
    showToast(tr('Problem resolved', '?? ?? ????????'));
    window.e1RenderProblems(pid);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

// ----- CPOE (orders via the E-X unified orders table, CDS-gated server-side) -----

```

window.e1RenderCpoe = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  // Required order type metadata mapping
  const orderTypeMeta = {
    lab: { name: 'Lab', nameAr: '?????', icon: '?' },
    rad: { name: 'Radiology', nameAr: '????', icon: '?' },
    med: { name: 'Medication', nameAr: '????', icon: '?' },
    consult: { name: 'Consult', nameAr: '??????', icon: '?' }
  };

  // Required status ranking for UI ordering
  const statusRank = { active: 1, pending: 2, completed: 3, cancelled: 4 };

  let orders = [];
  try {
    orders = await API.get('/api/orders?patient_id=' + encodeURIComponent(pid));
  } catch (e) {
    orders = [];
  }

  // Sort orders by status priority
  const sortedOrders = (orders || []).sort((a, b) => {
    const rankA = statusRank[(a.status || '').toLowerCase()] || 99;
    const rankB = statusRank[(b.status || '').toLowerCase()] || 99;
    return rankA - rankB;
  });

  const rows = sortedOrders.map(o => {
    const meta = orderTypeMeta[o.type] || { name: o.type, nameAr: o.type, icon: '?' };

```

```

const summary = o.item_summary || tr('No details', '?? ??? ????');
const count = o.item_count || 0;
const badgeColor = o.status === 'active' ? '#10b981' : o.status === 'pending' ? '#f59e0b' : '#6b7280';
return `
  <div style="display:flex;justify-content:space-between;align-items:center;padding:12px;margin:8px 0;border
er-radius:8px;background:var(--hover,#f8f9fa);border-left:4px solid ${badgeColor}">
    <div>
      <span style="font-size:14px;margin-right:6px">${meta.icon}</span>
      <span class="badge" style="background:${badgeColor};color:#fff">${escapeHTML(o.status)}</span>
      <span style="font-size:12px;color:var(--text-dim);margin-left:6px">#${safeId(o.id)}</span>
      <div style="font-size:12px;margin-top:4px;font-weight:600">${escapeHTML(isArabic ? meta.nameAr : met
a.name)}</div>
      <div style="font-size:11px;color:var(--text-dim);margin-top:2px">${escapeHTML(summary)} (${count} ${
tr('items', '????')}</div>
    </div>
  </div>`;
}).join('');

body.innerHTML = `
<div class="flex gap-8 mb-8">
  <div class="form-group" style="flex:1"><label>${tr('Order type', '???')}</label>
    <select class="form-input" id="e10rdType">
      <option value="lab">${tr('Lab', '???')}</option>
      <option value="rad">${tr('Radiology', '???')}</option>
      <option value="med">${tr('Medication', '???')}</option>
      <option value="consult">${tr('Consult', '????')}</option>
    </select>
  </div>
  <div class="form-group" style="flex:2"><label>${tr('Item / catalog ref', '???')}</label><input class="
form-input" id="e10rdRef"></div>
</div>
<div class="flex gap-8 mb-8">
  <div class="form-group" style="flex:1"><label>${tr('Qty', '????')}</label><input class="form-input" id
="e10rdQty" value="1"></div>
  <div class="form-group" style="flex:1"><label>${tr('Dose (mg, for meds)', '???? (mg ???)')}</label>
><input class="form-input" id="e10rdDose"></div>
  <div class="form-group" style="flex:2"><label>${tr('Instructions', '????')}</label><input class="form
-input" id="e10rdInstr"></div>
</div>
  <button class="btn btn-primary btn-sm mb-12" onclick="e1SubmitOrder(${safeId(pid)})">? ${tr('Place Order (
CDS checked)', '???? ? (???? ? CDS)')}</button>
  <div>${rows || `<div style="color:var(--text-dim)">${tr('No orders yet', '?? ??? ????')}</div>`}</div>`;
};

// Submit a CPOE order. The server runs CDS; a 422 hard-stop returns alerts. We then capture an
// override reason and resubmit (audited server-side). Warnings do not 422.
window.e1SubmitOrder = async (pid, overrideReason) => {
  const type = document.getElementById('e10rdType')?.value;
  const ref = document.getElementById('e10rdRef')?.value;
  if (!ref || !ref.trim()) { showToast(tr('Enter an item', '????'), 'error'); return; }
  const payload = {
    patient_id: pid, type,
    dose: document.getElementById('e10rdDose')?.value || undefined,
    items: [{ catalog_ref: ref, qty: document.getElementById('e10rdQty')?.value || 1, instructions: document.g

```

```

etElementById('e10rdInstr')?.value || '' ]],
  override_reason: overrideReason || undefined
};
try {
  // NOTE: API.request resolves the JSON body even for non-2xx (it does not throw), so a CDS
  // 422 hard-stop arrives as a resolved payload with {blocked:true, alerts:[...]}.
  const r = await API.post('/api/cpoe/order', payload);
  if (r && (r.blocked || r.requires_override_reason)) {
    // CRITICAL CDS hard-stop: capture an override reason (audited server-side) and resubmit.
    const alerts = r.alerts || [];
    const txt = alerts.map(a => '[' + (a.severity || '').toUpperCase() + ']' + (isArabic ? (a.message_ar ||
a.message) : (a.message_en || a.message))).join('\n');
    const reason = window.prompt('? ' + tr('CRITICAL CDS alert ? hard stop. To override you MUST enter a cli
nical reason (audited):',
      '????? CDS ??? ? ????? ??????. ?????? ??? ????? ??? ????? (?????):') + '\n\n' + txt, '');
    if (reason && reason.trim()) { return window.e1SubmitOrder(pid, reason.trim()); }
    showToast(tr('Order blocked by CDS (no override reason)', '?? ??? ????? ?????? CDS (???? ??? ?????)'), '
error');
    return;
  }
  if (r && r.error) {
    // Try placing order using legacy workflow as fallback
    try {
      await API.post('/api/orders/legacy', payload);
      showToast(tr('Order placed using legacy workflow', '?? ????? ?????? ?????????? ??? ?????? ??????????'));
      window.e1RenderCpoe(pid);
      return;
    } catch (err) {
      showToast(tr('Error placing order', '??? ? ???? ?????'), 'error');
      return;
    }
  }
  if (r && r.cds_alerts && r.cds_alerts.length) {
    const warn = r.cds_alerts.map(a => '[' + (a.severity || '').toUpperCase() + ']' + (isArabic ? (a.messag
e_ar || a.message) : (a.message_en || a.message))).join('\n');
    showToast(tr('Order placed (with CDS advisories)', '?? ????? ?????? (?? ????????? CDS)'));
    if (warn) alert(warn);
  } else {
    showToast(tr('Order placed', '?? ????? ??????'));
  }
  window.e1RenderCpoe(pid);
} catch (e) {
  // Legacy workflow fallback on connection/missing endpoint error
  try {
    await API.post('/api/orders/legacy', payload);
    showToast(tr('Order placed using legacy workflow', '?? ????? ?????? ?????????? ??? ?????? ??????????'));
    window.e1RenderCpoe(pid);
  } catch (err) {
    showToast(tr('Error placing order', '??? ? ???? ?????'), 'error');
  }
}
};

// ----- SOAP note -----

```

```

window.e1RenderSoap = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;
  let notes = [];
  try { notes = await API.get('/api/clinical-notes?patient_id=' + encodeURIComponent(pid)); } catch (e) { notes = []; }
  const rows = (notes || []).map(n => `
    <div style="padding:8px;margin:4px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid ${n.emr_status === 'locked' ? '#16a34a' : '#f59e0b'}">
      <div class="flex gap-8" style="align-items:center"><strong>SOAP #${safeId(n.id)}</strong>
        <span class="badge ${n.emr_status === 'locked' ? 'badge-success' : 'badge-warning'}">${escapeHTML(n.emr_status)}</span>
        ${n.emr_status !== 'locked' ? `<button class="btn btn-sm" onclick="e1SignNote(${safeId(n.id)},{safeId(pid)})"> ${tr('Sign & Lock', '?????')}</button>` : ''}
      </div>
      ${n.assessment ? `<div style="font-size:12px;margin-top:4px"><strong>A:</strong> ${escapeHTML(n.assessment)}</div>` : ''}
    </div>`).join('');
  body.innerHTML = `
    <div class="form-group mb-8"><label>S ? ${tr('Subjective', '??????')}</label><textarea class="form-input form-textarea" id="e1S"></textarea></div>
    <div class="form-group mb-8"><label>O ? ${tr('Objective', '????')}</label><textarea class="form-input form-textarea" id="e1O"></textarea></div>
    <div class="form-group mb-8"><label>A ? ${tr('Assessment', '???????')}</label><textarea class="form-input form-textarea" id="e1A"></textarea></div>
    <div class="form-group mb-8"><label>P ? ${tr('Plan', '????')}</label><textarea class="form-input form-textarea" id="e1P"></textarea></div>
    <button class="btn btn-primary btn-sm mb-12" onclick="e1SaveSoap(${safeId(pid)})"> ${tr('Save SOAP Note', '??? ?????? SOAP')}</button>
    <div>${rows || `<div style="color:var(--text-dim)">${tr('No SOAP notes', '?? ???? ??????? SOAP')}</div>`}</div>`;
};

```

```

window.e1SaveSoap = async (pid) => {
  try {
    await API.post('/api/clinical-notes', {
      patient_id: pid,
      subjective: document.getElementById('e1S')?.value || '',
      objective: document.getElementById('e1O')?.value || '',
      assessment: document.getElementById('e1A')?.value || '',
      plan: document.getElementById('e1P')?.value || ''
    });
    showToast(tr('SOAP note saved', '?? ??? ?????? SOAP'));
    window.e1RenderSoap(pid);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```

window.e1SignNote = async (id, pid) => {
  try {
    await API.post('/api/clinical-notes/' + id + '/sign', {});
    showToast(tr('Note signed & locked', '?? ????? ???? ?????????'));
    window.e1RenderSoap(pid);
  } catch (e) { showToast(tr('Error signing', '??? ?? ???????'), 'error'); }
};

```

```
// ===== APPOINTMENT CHECK-IN (Receptionist) =====
window.checkInPatient = async (appointmentId) => {
  try {
    const result = await API.put('/api/appointments/' + appointmentId + '/checkin', {});
    showToast(tr('Patient checked in! Added to waiting queue.', '?? ????? ???? ?????! ??? ?????? ?????? ?????
???') + ' ?');
    // Refresh the page
    if (typeof renderAppointments === 'function') {
      const el = document.getElementById('mainContent');
      if (el) renderAppointments(el);
    }
  } catch (e) { showToast(tr('Check-in failed', '??? ?????? ??????'), 'error'); }
};

window.markNoShow = async (appointmentId) => {
  if (!confirm(tr('Mark this patient as No-Show?', '????? ?????? ??????'))) return;
  try {
    await API.put('/api/appointments/' + appointmentId + '/noshow', {});
    showToast(tr('Marked as No-Show', '?? ??????? ??????') + ' ??');
    if (typeof renderAppointments === 'function') {
      const el = document.getElementById('mainContent');
      if (el) renderAppointments(el);
    }
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== NEXT PATIENT (Doctor) =====
window.callNextPatient = async () => {
  try {
    const result = await API.get('/api/doctor/next-patient');
    if (!result.hasNext) {
      showToast(tr('No patients waiting', '?? ???? ???? ?????????') + ' ?', 'info');
      return;
    }

    // Show patient info modal
    const p = result.patient || {};
    const v = result.vitals || {};
    const q = result.queue || {};

    let modal = document.createElement('div');
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
    modal.innerHTML = '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:28px;width:600px;direction:rtl;max-height:90vh;overflow-y:auto">' +
      '<div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px">' +
      '<h3 style="margin:0;color:var(--primary)">? ' + tr('Next Patient', '????? ??????') + '</h3>' +
      '<span style="background:#e3f2fd;padding:4px 12px;border-radius:20px;font-size:14px">? ' + result.waitin
g_count + ' ' + tr('waiting', '????????') + '</span>' +
      '</div>' +
      '<div style="background:#f8f9fa;border-radius:12px;padding:16px;margin-bottom:16px">' +
      '<h4 style="margin:0 0 8px;font-size:18px">' + escapeHTML(p.name_ar || p.name_en || q.patient_name || ''

```



```

) + '</h4>' +
  '<div style="display:grid;grid-template-columns:1fr 1fr;gap:8px;font-size:14px">' +
  '<span>' + tr('MRN:', '?????:') + ' <strong>' + escapeHTML(p.mrn || p.file_number || '') + '</strong><
/span>' +
  '<span>' + tr('Age:', '?????:') + ' <strong>' + escapeHTML(p.age || '') + '</strong></span>' +
  '<span>' + tr('Phone:', '?????:') + ' <strong>' + escapeHTML(p.phone || '') + '</strong></span>' +
  '<span>' + tr('ID:', '?????:') + ' <strong>' + escapeHTML(p.national_id || '') + '</strong></span>' +
  (p.allergies ? '<span style="grid-column:1/-1;color:#cc0000;font-weight:bold">' + tr('Allergies:', '?
?????:') + ' ' + escapeHTML(p.allergies) + '</span>' : '') +
  (p.chronic_diseases ? '<span style="grid-column:1/-1;color:#e65100">' + tr('Chronic:', '???? ?????:')
+ ' ' + escapeHTML(p.chronic_diseases) + '</span>' : '') +
  '</div></div>' +
  (v.blood_pressure || v.temperature || v.pulse ?
    '<div style="background:#e8f5e9;border-radius:12px;padding:12px;margin-bottom:16px">' +
    '<h5 style="margin:0 0 8px">' + tr('Latest Vitals', '??????? ??????') + '</h5>' +
    '<div style="display:flex;gap:16px;flex-wrap:wrap;font-size:14px">' +
    (v.blood_pressure ? '<span>' + tr('BP: <strong>' + v.blood_pressure + '</strong></span>' : '') +
    (v.temperature ? '<span>' + tr('T: <strong>' + v.temperature + '°C</strong></span>' : '') +
    (v.pulse ? '<span>' + tr('P: <strong>' + v.pulse + '</strong></span>' : '') +
    (v.spo2 ? '<span>' + tr('SpO2: <strong>' + v.spo2 + '%</strong></span>' : '') +
    (v.weight ? '<span>' + tr('W: <strong>' + v.weight + 'kg</strong></span>' : '') +
    '</div></div>' : '') +
    '<div style="display:flex;gap:12px">' +
    '<button class="btn btn-primary" onclick="selectPatientFromQueue(' + (p.id || 'null') + ');this.closest(
\'div\').parentElement.remove()" style="flex:2;padding:12px">' + tr('Start Consultation', '??? ?????????') +
    '</button>' +
    '<button class="btn btn-secondary" onclick="this.closest(\'div\').parentElement.remove()" style="flex:1"
>' + tr('Skip', '????') + '</button>' +
    '</div></div>';
  document.body.appendChild(modal);
  modal.onclick = (e) => { if (e.target === modal) modal.remove(); };

} catch (e) { showToast(tr('Error loading next patient', '???'), 'error'); console.error(e); }
};

window.selectPatientFromQueue = (patientId) => {
  if (!patientId) return;
  const select = document.getElementById('drPatient');
  if (select) {
    select.value = patientId;
    select.dispatchEvent(new Event('change'));
    showToast(tr('Patient loaded!', '?? ????? ??????!'));
  }
};

// ===== TRIAGE (Nursing) =====
window.showTriageForm = (patientId, patientName) => {
  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:24px;width:450px;direction:rtl">' +
    '<h3 style="margin:0 0 16px;color:var(--primary)">' + tr('Triage Assessment', '???? ??????') + '</h3>'
  +

```

```
'<p style="margin:0 0 12px;font-weight:bold">' + (patientName || '') + '</p>' +  
'<div class="form-group"><label>' + tr('Triage Level', '???? ????') + '</label>' +  
'<select class="form-input" id="triageLevel">' +  
'<option value="5" style="background:#4caf50;color:#fff">5 - ' + tr('Non-Urgent', '?? ????') + '</option>  
' +  
'<option value="4" style="background:#2196f3;color:#fff">4 - ' + tr('Less Urgent', '?? ??????') + '</opt  
ion>' +  
'<option value="3" selected style="background:#ff9800;color:#fff">3 - ' + tr('Urgent', '?????') + '</opti  
on>' +  
'<option value="2" style="background:#ff5722;color:#fff">2 - ' + tr('Emergency', '?????') + '</option>' +  
'<option value="1" style="background:#d50000;color:#fff">1 - ' + tr('Resuscitation', '?????') + '</option>  
' +  
'</select></div>' +  
'<div class="form-group"><label>' + tr('Pain Score (0-10)', '???? ???? (0-10)') + '</label>' +  
'<div style="display:flex;align-items:center;gap:12px">' +  
'<input type="range" id="painScore" min="0" max="10" value="0" style="flex:1" oninput="document.getElement  
ById(\'\`painValue\`).textContent=this.value">' +  
'<span id="painValue" style="font-size:24px;font-weight:bold;width:30px;text-align:center">0</span>' +  
'</div>' +  
'<div style="display:flex;justify-content:space-between;font-size:11px;color:#999"><span>?' + tr('No Pain  
", '?? ??') + '</span><span>' + tr('Worst Pain', '??? ??') + '</span></div></div>' +  
'<div class="form-group"><label>' + tr('Chief Complaint', '????? ????????) + '</label>' +  
'<input class="form-input" id="chiefComplaint" placeholder="" + tr('Main reason for visit', '?? ?????? ?  
??????') + '"></div>' +  
'<button class="btn btn-primary" onclick="saveTriageData(' + patientId + ')" style="width:100%;padding:12p  
x">' + tr('Save Triage', '?? ???????') + '</button>' +  
'</div>';  
  
document.body.appendChild(modal);  
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };  
};  
  
window.saveTriageData = async (patientId) => {  
  try {  
    await API.post('/api/nursing/triage', {  
      patient_id: patientId,  
      triage_level: document.getElementById('triageLevel').value,  
      pain_score: document.getElementById('painScore').value,  
      chief_complaint: document.getElementById('chiefComplaint').value,  
    });  
    document.querySelector('[style*="position:fixed"][style*="z-index:9999"]')?.remove();  
    showToast(tr('Triage saved!', '?? ??? ??????!') + ' ');  
  } catch (e) { showToast(tr('Error', '???'), 'error'); }  
};  
  
// ===== DUPLICATE APPOINTMENT CHECK =====  
window.checkDuplicateAppointment = async (patientId, date, doctor) => {  
  try {  
    const result = await API.post('/api/appointments/check-duplicate', { patient_id: patientId, date, doctor })  
  );  
    if (result.duplicate) {  
      showToast(tr('Warning: Patient already has appointment with this doctor on this date!', '?????: ????? ?  
??? ???? ???? ? ? ? ? ? ? ? ? ? ? ? ? !') + ' ', 'warning');  
      return true;  
    }  
  }
```

```

    return false;
  } catch (e) { return false; }
};

window.loadMyQueue = async () => {
  try {
    const queue = await API.get('/api/doctor/my-queue');
    if (!queue || queue.length === 0) {
      showToast(tr('No patients in your queue', '?? ???? ???? ?? ??????'), 'info');
      return;
    }
    let html = '<div style="position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center">' +
      '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:24px;width:500px;direction:rtl;max-height:80vh;overflow-y:auto">' +
      '<h3 style="margin:0 0 16px">? ' + tr('My Queue', '??????') + ' (' + queue.length + ')</h3>';
    queue.forEach((q, i) => {
      const isActive = q.status === 'In Progress';
      html += '<div style="padding:12px;margin:8px 0;background:' + (isActive ? '#e8f5e9' : '#f5f5f5') + ';border-radius:8px;border-right:4px solid ' + (isActive ? '#4caf50' : '#ccc') + ';cursor:pointer" onclick="selectPatientFromQueue(' + safeId(q.patient_id) + ');this.closest(\'[style*=position]\').remove()">' +
        '<div style="display:flex;justify-content:space-between"><strong>' + (i + 1) + '. ' + escapeHTML(q.patient_name || '') + '</strong><span style="font-size:12px;color:#666">' + (q.check_in_time ? new Date(q.check_in_time).toLocaleTimeString('ar-SA', { hour: '2-digit', minute: '2-digit' }) : '') + '</span></div>' +
        '<span style="font-size:12px;color:' + (isActive ? '#2e7d32' : '#999') + '">' + (isActive ? '? ' + tr('In Progress', '????') : '? ' + tr('Waiting', '????????')) + '</span>' +
        '</div>';
    });
    html += '<button class="btn btn-secondary" onclick="this.closest(\'[style*=position]\').remove()" style="width:100%;margin-top:12px">' + tr('Close', '????') + '</button></div></div>';
    document.body.insertAdjacentHTML('beforeend', html);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```

// ===== CSV EXPORT UTILITY =====
window.exportToCSV = (data, filename) => {
  if (!data || data.length === 0) { showToast(tr('No data to export', '?? ???? ?????? ???????'), 'info'); return; }
  const headers = Object.keys(data[0]);
  const csvRows = [headers.join(',')];
  data.forEach(row => {
    csvRows.push(headers.map(h => {
      let val = row[h] !== null && row[h] !== undefined ? String(row[h]) : '';
      val = val.replace(/"/g, '""');
      if (val.includes(',') || val.includes('"') || val.includes('\n')) val = '"' + val + '"';
      return val;
    })).join(',');
  });
  const bom = '\uFEFF';
  const blob = new Blob([bom + csvRows.join('\n')], { type: 'text/csv;charset=utf-8;' });
}

```

```

const url = URL.createObjectURL(blob);
const a = document.createElement('a');
a.href = url; a.download = (filename || 'export') + '_' + new Date().toISOString().slice(0, 10) + '.csv';
document.body.appendChild(a); a.click(); document.body.removeChild(a);
URL.revokeObjectURL(url);
showToast(tr('Exported successfully!', '?? ?????? ?????!') + ' ?');
};

// ===== RECEIPT PRINT =====
window.printReceipt = (invoice) => {
  if (!invoice) return;
  let h = '<div style="font-family:Arial;width:300px;margin:0 auto;padding:20px;direction:rtl;text-align:right'
">';
  h += '<div style="text-align:center;border-bottom:2px dashed #333;padding-bottom:12px;margin-bottom:12px">';
  h += '<h3 style="margin:0">????? ??????</h3><p style="margin:2px 0;font-size:12px">jumanaMedical</p>';
  h += '<p style="margin:2px 0;font-size:11px;color:#666">???????? ??????????</p>';
  h += '</div>';
  h += '<p style="margin:4px 0;font-size:13px"><strong>' + tr('Receipt', '?????') + '</strong></p>';
  h += '<p style="margin:4px 0;font-size:12px">' + tr('Invoice #:', '??? ?????????:') + ' ' + (invoice.invoice_
number || invoice.id || '') + '</p>';
  h += '<p style="margin:4px 0;font-size:12px">' + tr('Patient:', '?????:') + ' ' + (invoice.patient_name ||
'') + '</p>';
  h += '<p style="margin:4px 0;font-size:12px">' + tr('Date:', '?????:') + ' ' + new Date(invoice.created_at
|| Date.now()).toLocaleDateString('ar-SA') + '</p>';
  h += '<div style="border-top:1px dashed #999;border-bottom:1px dashed #999;padding:8px 0;margin:8px 0">';
  h += '<p style="margin:4px 0;font-size:12px">' + tr('Service:', '?????:') + ' ' + (invoice.description || i
nvoice.service_type || '') + '</p>';
  h += '<p style="margin:4px 0;font-size:14px;font-weight:bold">' + tr('Total:', '?????:') + ' ' + parseFlea
t(invoice.total || 0).toFixed(2) + ' ' + tr('SAR', '????') + '</p>';
  if (invoice.vat_amount && parseFloat(invoice.vat_amount) > 0) {
    h += '<p style="margin:4px 0;font-size:11px;color:#666">' + tr('Includes VAT:', '???? ??????:') + ' ' + p
arseFloat(invoice.vat_amount).toFixed(2) + '</p>';
  }
  if (invoice.amount_paid) {
    h += '<p style="margin:4px 0;font-size:12px">' + tr('Paid:', '?????:') + ' ' + parseFloat(invoice.amount
_paid).toFixed(2) + '</p>';
  }
  if (invoice.balance_due && parseFloat(invoice.balance_due) > 0) {
    h += '<p style="margin:4px 0;font-size:12px;color:#cc0000">' + tr('Balance:', '?????:') + ' ' + parseFloa
t(invoice.balance_due).toFixed(2) + '</p>';
  }
  h += '<p style="margin:4px 0;font-size:12px">' + tr('Payment:', '????? ?????:') + ' ' + (invoice.payment_met
hod || 'Cash') + '</p>';
  h += '</div>';
  h += '<p style="text-align:center;font-size:10px;color:#999;margin-top:12px">' + tr('Thank you for your visi
t', '????? ?????????') + '</p>';
  h += '<p style="text-align:center;font-size:10px;color:#999">www.nama-medical.com</p>';
  h += '</div>';

  const w = window.open('', '_blank', 'width=350,height=500');
  w.document.write('<html><head><title>' + tr('Receipt', '?????') + '</title></head><body style="margin:0">' +
h + '<script>setTimeout(()=>{window.print();},300);</script></body></html>');
  w.document.close();
};

```

```
// ===== PASSWORD CHANGE =====
window.showChangePassword = () => {
  const modal = document.createElement('div');
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:28px;width:400px;direction:rtl">' +
    '<h3 style="margin:0 0 20px;color:var(--primary)">? ' + tr('Change Password', '???? ???? ?????') + '</h3>' +
    '<div class="form-group"><label>' + tr('Current Password', '???? ?????? ??????') + '</label><input type="password" class="form-input" id="cpCurrent"></div>' +
    '<div class="form-group"><label>' + tr('New Password', '???? ?????? ??????') + '</label><input type="password" class="form-input" id="cpNew" placeholder="">' + tr('Min 6 characters', '6 ???? ????') + '"></div>' +
    '<div class="form-group"><label>' + tr('Confirm New', '????? ??????') + '</label><input type="password" class="form-input" id="cpConfirm"></div>' +
    '<div style="display:flex;gap:12px;margin-top:16px">' +
    '<button class="btn btn-primary" onclick="submitChangePassword()" style="flex:1">? ' + tr('Save', '???') +
    '</button>' +
    '<button class="btn btn-secondary" onclick="this.closest(\'[style*=position]\').remove()" style="flex:1">' +
    tr('Cancel', '????') + '</button>' +
    '</div></div>';
  document.body.appendChild(modal);
  modal.onclick = (e) => { if (e.target === modal) modal.remove(); };
};

window.submitChangePassword = async () => {
  const current = document.getElementById('cpCurrent')?.value;
  const newPw = document.getElementById('cpNew')?.value;
  const confirm = document.getElementById('cpConfirm')?.value;
  if (!current || !newPw) return showToast(tr('Fill all fields', '???? ???? ?????'), 'error');
  if (newPw !== confirm) return showToast(tr('Passwords do not match', '????? ?????? ??? ??????'), 'error');
  if (newPw.length < 6) return showToast(tr('Min 6 characters', '6 ???? ????'), 'error');
  try {
    await API.put('/api/auth/change-password', { current_password: current, new_password: newPw });
    document.querySelector('[style*="position:fixed"][style*="z-index:9999"]')?.remove();
    showToast(tr('Password changed!', '?? ????? ???? ??????!') + ' ');
  } catch (e) {
    showToast(e.message || tr('Error', '???'), 'error');
  }
};

// ===== DATABASE BACKUP =====
window.startBackup = async () => {
  showToast(tr('Creating backup...', '???? ?????? ?????? ??????????...') + ' ');
  try {
    const response = await fetch('/api/admin/backup', { method: 'POST', headers: { 'Content-Type': 'application/json' } });
    if (!response.ok) throw new Error('Backup failed');
    const blob = await response.blob();
    const url = URL.createObjectURL(blob);
    const a = document.createElement('a');
    a.href = url; a.download = 'nama_backup_' + new Date().toISOString().slice(0, 10) + '.sql';
    document.body.appendChild(a); a.click(); document.body.removeChild(a);
  }
};
```

```

        showToast(tr('Backup downloaded!', '?? ????? ??????????') + ' ');
    } catch (e) { showToast(tr('Backup failed', '??? ????? ?????????'), 'error'); }
};

// ===== KEYBOARD SHORTCUTS =====
document.addEventListener('keydown', (e) => {
    // Don't trigger if user is typing in input/textarea
    if (e.target.tagName === 'INPUT' || e.target.tagName === 'TEXTAREA' || e.target.tagName === 'SELECT') {
        if (e.key === 'Escape') e.target.blur();
        return;
    }

    if (e.ctrlKey || e.metaKey) {
        switch (e.key.toLowerCase()) {
            case 'k': // Search
                e.preventDefault();
                const searchInput = document.querySelector('.search-box input') || document.querySelector('input[type=
"search"]') || document.querySelector('#globalSearch');
                if (searchInput) { searchInput.focus(); searchInput.select(); }
                break;
            case 'n': // New patient ? go to reception
                e.preventDefault();
                if (typeof navigateTo === 'function') navigateTo(1);
                break;
            case 'p': // Print current page
                e.preventDefault();
                window.print();
                break;
        }
    }

    // Escape ? close any modal
    if (e.key === 'Escape') {
        const modal = document.querySelector('[style*="position:fixed"][style*="z-index:9999"]');
        if (modal) modal.remove();
    }

    // Number keys 1-9 for quick nav (Alt+number)
    if (e.altKey && e.key >= '1' && e.key <= '9') {
        e.preventDefault();
        const pageIdx = parseInt(e.key) - 1;
        if (typeof navigateTo === 'function') navigateTo(pageIdx);
    }
});

window.toggleCalendarView = async () => {
    const existing = document.getElementById('calendarGrid');
    if (existing) { existing.remove(); document.getElementById('calToggleBtn').textContent = '? ' + tr('Calendar
View', '??? ?????'); return; }
    document.getElementById('calToggleBtn').textContent = '? ' + tr('List View', '??? ?????');

    try {

```

```

const appts = await API.get('/api/appointments');
const today = new Date();
const startOfWeek = new Date(today); startOfWeek.setDate(today.getDate() - today.getDay());

const days = [];
for (let i = 0; i < 7; i++) {
  const d = new Date(startOfWeek); d.setDate(startOfWeek.getDate() + i);
  days.push(d);
}
const dayNames = isArabic ? ['???', '?????', '??????', '??????', '????', '????', '???'] : ['Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat'];
const hours = Array.from({ length: 12 }, (_, i) => i + 8); // 8am to 7pm

let html = '<div id="calendarGrid" style="margin-top:12px;overflow-x:auto">';
html += '<table style="width:100%;border-collapse:collapse;font-size:13px;table-layout:fixed">';
html += '<thead><tr><th style="width:60px;padding:8px;background:#f0f6ff;border:1px solid #e0e0e0">' + tr(
'Time', '?????') + '</th>';
days.forEach((d, i) => {
  const isToday = d.toDateString() === today.toDateString();
  html += '<th style="padding:8px;background:' + (isToday ? '#1a73e8;color:#fff' : '#f0f6ff') + '";border:1
px solid #e0e0e0;font-weight:' + (isToday ? 'bold' : 'normal') + '">' + dayNames[i] + '<br><span style="font-s
ize:11px">' + d.toLocaleDateString('ar-SA', { day: 'numeric', month: 'short' }) + '</span></th>';
});
html += '</tr></thead><tbody>';

hours.forEach(h => {
  html += '<tr>';
  html += '<td style="padding:4px 8px;background:#f9f9f9;border:1px solid #e0e0e0;font-size:12px;text-align:center">' + (h < 10 ? '0' : '') + h + ':00</td>';
  days.forEach(d => {
    const dateStr = d.toISOString().slice(0, 10);
    const hourAppts = appts.filter(a => {
      const aDate = (a.appt_date || a.date || '').substring(0, 10);
      const aTime = a.appt_time || a.time || '';
      const aHour = parseInt(aTime.split(':')[0]);
      return aDate === dateStr && aHour === h;
    });
    html += '<td style="padding:2px;border:1px solid #e0e0e0;vertical-align:top;height:50px">';
    hourAppts.forEach(a => {
      const statusColor = a.status === 'Cancelled' ? '#ffcdd2' : a.status === 'Checked-In' ? '#c8e6c9' : a
.status === 'No-Show' ? '#ffe0b2' : '#e3f2fd';
      html += '<div style="background:' + statusColor + '";border-radius:4px;padding:2px 4px;margin:1px 0;font-size:11px;overflow:hidden;text-overflow:ellipsis;white-space:nowrap;cursor:pointer" title="' + escapeHTML(
(a.patient_name || '') + ' - ' + (a.doctor_name || a.doctor || '')) + '">' +
escapeHTML((a.patient_name || '').split(' ')[0]) + ' <span style="color:#666">' + escapeHTML((a.doctor_name || a.doctor || '').split(' ').slice(-1)[0]) + '</span></div>';
    });
    html += '</td>';
  });
  html += '</tr>';
});
html += '</tbody></table></div>';

const table = document.querySelector('.data-table') || document.querySelector('table');

```

```

    if (table) table.insertAdjacentHTML('beforebegin', html);
    else {
        const content = document.getElementById('mainContent');
        if (content) content.insertAdjacentHTML('beforeend', html);
    }
} catch (e) { console.error('Calendar error:', e); }
};

window.exportPatients = async () => {
    try { const data = await API.get('/api/patients'); exportToCSV(data, 'patients'); } catch (e) { showToast(tr
('Error', '???'), 'error'); }
};
window.exportInvoices = async () => {
    try { const data = await API.get('/api/invoices'); exportToCSV(data, 'invoices'); } catch (e) { showToast(tr
('Error', '???'), 'error'); }
};
window.exportAppointments = async () => {
    try { const data = await API.get('/api/appointments'); exportToCSV(data, 'appointments'); } catch (e) { show
Toast(tr('Error', '???'), 'error'); }
};

// ===== LANGUAGE TOGGLE =====
window.toggleLanguage = () => {
    isArabic = !isArabic;
    localStorage.setItem('namaLang', isArabic ? 'ar' : 'en');
    // Update document direction
    document.documentElement.dir = isArabic ? 'rtl' : 'ltr';
    document.documentElement.lang = isArabic ? 'ar' : 'en';
    document.body.style.direction = isArabic ? 'rtl' : 'ltr';
    document.body.style.textAlign = isArabic ? 'right' : 'left';
    // Rebuild sidebar navigation in new language
    if (typeof buildNav === 'function') buildNav();
    // Update header text if exists
    const headerTitle = document.querySelector('.header h1, .app-title, .logo-text');
    if (headerTitle) headerTitle.textContent = isArabic ? '????? ?????' : 'jumanaMedical';
    // Re-render current page content
    if (typeof navigateTo === 'function') navigateTo(currentPage);
    // Update the lang button text
    const langBtn = document.getElementById('langToggleBtn');
    if (langBtn) langBtn.textContent = isArabic ? '? EN' : '? ?????';
};

// ===== MOBILE LANGUAGE PROMPT =====
window.showMobileLangPrompt = () => {
    // Only show on mobile if no lang preference saved
    const isMobile = window.innerWidth <= 768 || /Android|iPhone|iPad|iPod/i.test(navigator.userAgent);
    if (!isMobile || localStorage.getItem('namaLang')) return;

    const prompt = document.createElement('div');
    prompt.id = 'mobileLangPrompt';
    prompt.style.cssText = 'position:fixed;bottom:0;left:0;right:0;background:linear-gradient(135deg,#1a73e8,#0d

```



```

47a1);color:#fff;padding:16px 20px;z-index:99999;display:flex;align-items:center;justify-content:center;gap:12
px;box-shadow:0 -4px 20px rgba(0,0,0,0.3);animation:slideUp 0.4s ease';
    prompt.innerHTML = '<span style="font-size:15px;font-weight:500">Choose Language / ??? ????</span>' +
        '<button onclick="setLang(\'en\')" style="padding:8px 20px;border:2px solid #fff;border-radius:8px;backgro
und:transparent;color:#fff;font-size:14px;font-weight:bold;cursor:pointer">?? English</button>' +
        '<button onclick="setLang(\'ar\')" style="padding:8px 20px;border:2px solid #fff;border-radius:8px;backgro
und:rgba(255,255,255,0.2);color:#fff;font-size:14px;font-weight:bold;cursor:pointer">?? ???</button>';
    document.body.appendChild(prompt);

    // Add slide-up animation
    const style = document.createElement('style');
    style.textContent = '@keyframes slideUp{from{transform:translateY(100%)}to{transform:translateY(0)}}';
    document.head.appendChild(style);
};

window.setLang = (lang) => {
    localStorage.setItem('namaLang', lang);
    isArabic = lang === 'ar';
    document.documentElement.dir = isArabic ? 'rtl' : 'ltr';
    document.documentElement.lang = isArabic ? 'ar' : 'en';
    document.body.style.direction = isArabic ? 'rtl' : 'ltr';
    document.body.style.textAlign = isArabic ? 'right' : 'left';
    const prompt = document.getElementById('mobileLangPrompt');
    if (prompt) prompt.remove();
    if (typeof buildNav === 'function') buildNav();
    if (typeof navigateTo === 'function') navigateTo(currentPage);
    const langBtn = document.getElementById('langToggleBtn');
    if (langBtn) langBtn.textContent = isArabic ? '? EN' : '? ???';
};

window.sendDirectRad = async () => {
    const patientId = document.getElementById('drPatient')?.value || window._selectedPatientId;
    if (!patientId) return showToast(tr('Select patient first', '???? ??? ????'), 'error');
    const patientName = document.getElementById('drPatient')?.selectedOptions[0]?.text || window._selectedPatien
tName || '';
    const examType = document.getElementById('radDirectType')?.value || '';
    const details = document.getElementById('radDirectDesc')?.value || '';
    const priority = document.getElementById('radDirectPriority')?.value || 'routine';
    try {
        await API.post('/api/radiology/orders', { patient_id: patientId, patient_name: patientName, exam_type: exam
mType, details: details, priority: priority, status: 'Pending' });
        showToast(tr('Radiology order sent!', '?? ????? ?? ????!!'));
        document.getElementById('radDirectType') && (document.getElementById('radDirectType').value = '');
        document.getElementById('radDirectDesc') && (document.getElementById('radDirectDesc').value = '');
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

async function renderDashboard(el) {
    // Render skeleton loading immediately
    el.innerHTML = `
        <!-- Skeleton Header -->
        <div class="flex justify-between items-end mb-lg animate-pulse">

```

```

<div class="w-1/2">
  <div class="skeleton-loader skeleton-bar title"></div>
  <div class="skeleton-loader skeleton-bar w-1/3 mt-2"></div>
</div>
<div class="flex gap-3">
  <div class="skeleton-loader h-10 w-28 rounded-xl"></div>
  <div class="skeleton-loader h-10 w-28 rounded-xl"></div>
</div>
</div>

<!-- Skeleton Cards -->
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-md mb-lg">
  <div class="glass-card-premium p-6 rounded-2xl flex flex-col gap-3">
    <div class="flex justify-between items-center"><div class="skeleton-loader skeleton-circle"></div><div
class="skeleton-loader h-6 w-12 rounded"></div></div>
    <div class="skeleton-loader skeleton-bar w-1/2"></div>
    <div class="skeleton-loader h-10 w-full rounded-lg"></div>
  </div>
  <div class="glass-card-premium p-6 rounded-2xl flex flex-col gap-3">
    <div class="flex justify-between items-center"><div class="skeleton-loader skeleton-circle"></div><div
class="skeleton-loader h-6 w-12 rounded"></div></div>
    <div class="skeleton-loader skeleton-bar w-1/2"></div>
    <div class="skeleton-loader h-10 w-full rounded-lg"></div>
  </div>
  <div class="glass-card-premium p-6 rounded-2xl flex flex-col gap-3">
    <div class="flex justify-between items-center"><div class="skeleton-loader skeleton-circle"></div><div
class="skeleton-loader h-6 w-12 rounded"></div></div>
    <div class="skeleton-loader skeleton-bar w-1/2"></div>
    <div class="skeleton-loader h-10 w-full rounded-lg"></div>
  </div>
  <div class="glass-card-premium p-6 rounded-2xl flex flex-col gap-3">
    <div class="flex justify-between items-center"><div class="skeleton-loader skeleton-circle"></div><div
class="skeleton-loader h-6 w-12 rounded"></div></div>
    <div class="skeleton-loader skeleton-bar w-1/2"></div>
    <div class="skeleton-loader h-10 w-full rounded-lg"></div>
  </div>
</div>

<!-- Skeleton Bento Grid -->
<div class="grid grid-cols-12 gap-gutter mb-lg">
  <div class="col-span-12 lg:col-span-8 glass-card-premium p-6 h-80 flex flex-col gap-4">
    <div class="skeleton-loader skeleton-bar title"></div>
    <div class="skeleton-loader flex-1 rounded-xl"></div>
  </div>
  <div class="col-span-12 lg:col-span-4 glass-card-premium p-6 h-80 flex flex-col gap-4">
    <div class="skeleton-loader skeleton-bar title"></div>
    <div class="space-y-3 flex-1">
      <div class="skeleton-loader h-12 rounded-xl"></div>
      <div class="skeleton-loader h-12 rounded-xl"></div>
      <div class="skeleton-loader h-12 rounded-xl"></div>
    </div>
  </div>
</div>
</div>
`;

```

```

let s, enhanced;
try {
  [s, enhanced] = await Promise.all([
    API.get('/api/dashboard/stats'),
    API.get('/api/dashboard/enhanced').catch(() => ({}))
  ]);
} catch (err) {
  el.innerHTML = `
    <div class="flex flex-col items-center justify-center p-xl">
      <div class="error-card-premium max-w-md w-full">
        <span class="material-symbols-outlined error-card-icon">error</span>
        <h3 class="text-lg font-bold">${tr('Failed to Load Dashboard', '??? ?? ????? ???? ?????')}</h3>
        <p class="text-sm">${tr('Connection error or session expired. Please verify connection and retry.',
'??? ??? ?? ?????? ?????? ?? ????? ??????. ??? ?????? ?????? ???????.')}</p>
        <button onclick="navigateTo(0)" class="mt-4 px-6 py-2.5 bg-primary text-on-primary rounded-xl font-l
abel-md shadow-md hover:opacity-90 transition-all">
          ${tr('Retry Connection', '???? ?????')}
        </button>
      </div>
    </div>
  `;
  return;
}

// Schedule Chart.js rendering
setTimeout(() => {
  renderDashboardCharts(el, enhanced);
  loadDashboardCharts();
}, 50);

const bedOccupancy = Math.min(100, Math.round((s.patients || 0) / 300 * 100));
const avgWaitTime = Math.max(5, Math.round((s.waiting || 0) * 1.5));
const emergencyCount = enhanced.pendingLab || Math.round((s.patients || 0) * 0.1);

let topDrHtml = '';
if (enhanced.topDoctors && enhanced.topDoctors.length) {
  topDrHtml = enhanced.topDoctors.map((d, index) => `
    <div class="flex items-center justify-between p-3 bg-surface-container-low rounded-xl border border-outl
ine-variant/10 hover:bg-surface-container-high transition-colors">
      <div class="flex items-center gap-3">
        <span class="w-8 h-8 rounded-full bg-primary/10 text-primary flex items-center justify-center font-b
old text-sm">${index + 1}</span>
        <div>
          <p class="font-bold text-primary text-sm">???? ${escapeHTML(d.display_name || tr('Unknown', '??? ?
????'))}</p>
          <p class="text-xs text-on-surface-variant font-medium">${escapeHTML(d.patients)} ${tr('patients',
'????')}</p>
        </div>
      </div>
      <span class="font-bold text-secondary text-sm">${Number(d.revenue).toLocaleString()} SAR</span>
    </div>
  `).join('');
} else {

```

```

topDrHtml = `
  <div class="empty-state-card w-full">
    <span class="material-symbols-outlined empty-state-icon">person_off</span>
    <p class="text-xs font-bold text-on-surface-variant">${tr('No Active Physicians Today', '?? ????'
? ?????? ??????? ?????')}</p>
  </div>
`;

}

let revTypeHtml = '';
if (enhanced.revenueByType && enhanced.revenueByType.length) {
  const typeIcons = { 'File Opening': '?', 'Lab Test': '?', 'Radiology': '?', 'Consultation': '?', 'Pharmacy': '?', 'Appointment': '?' };
  const maxRev = Math.max(...enhanced.revenueByType.map(r => Number(r.total)));
  const colors = ['bg-primary', 'bg-secondary', 'bg-tertiary-container', 'bg-error', 'bg-secondary-container', 'bg-outline'];
  revTypeHtml = enhanced.revenueByType.map((r, i) => {
    const pct = Math.round(Number(r.total) / maxRev * 100);
    return `
      <div class="space-y-1">
        <div class="flex justify-between text-xs font-medium text-on-surface-variant">
          <span>${typeIcons[r.service_type] || '?'} ${escapeHTML(r.service_type)} (${escapeHTML(r.cnt)})</span>
          <span class="font-bold text-primary">${Number(r.total).toLocaleString()} SAR</span>
        </div>
        <div class="w-full bg-surface-container-highest h-2 rounded-full overflow-hidden">
          <div class="${colors[i % colors.length]} h-full rounded-full transition-all duration-1000" style="width: ${pct}%"></div>
        </div>
      </div>
    `;
  }).join('');
} else {
  revTypeHtml = `
    <div class="empty-state-card w-full">
      <span class="material-symbols-outlined empty-state-icon">receipt_long</span>
      <p class="text-xs font-bold text-on-surface-variant">${tr('No Operational Revenue Recorded', '?? ????'
??????? ?????? ?????? ???')}</p>
    </div>
  `;
}

el.innerHTML = `
  <!-- Page Header -->
  <div class="flex justify-between items-end mb-lg">
    <div>
      <h2 class="font-headline-lg text-headline-lg text-primary">${tr('Operational Command Center', '???? ?'
????? ????????? - ??????? ?????')}</h2>
      <div class="flex items-center gap-3 mt-2">
        <span class="flex items-center gap-1.5 px-3 py-1 bg-secondary/10 text-secondary rounded-full text-sm font-medium">
          <span class="w-2 h-2 bg-secondary rounded-full animate-pulse"></span>
          ${tr('Live Now', '????? ???')}
        </span>

```

```

        <span class="text-on-surface-variant font-label-md flex items-center gap-1">
            <span class="material-symbols-outlined text-lg">calendar_today</span>
            ${new Date().toLocaleDateString(isArabic ? 'ar-SA' : 'en-US', { weekday: 'long', year: 'numeric',
month: 'long', day: 'numeric' })}
        </span>
    </div>
</div>
<div class="flex gap-3">
    <button onclick="exportTableCSV('dashboard_report')" class="flex items-center gap-2 px-5 py-2.5 bg-sur
face-container-lowest border border-outline-variant/50 rounded-xl text-primary font-label-md shadow-sm hover:b
g-surface-container-low transition-all">
        <span class="material-symbols-outlined">download</span>
        ${tr('Export Report', '????? ??????')}
    </button>
    <button onclick="window.print()" class="flex items-center gap-2 px-5 py-2.5 bg-primary text-on-primary
rounded-xl font-label-md shadow-md hover:opacity-90 transition-all">
        <span class="material-symbols-outlined">print</span>
        ${tr('Print Portal', '????? ??????')}
    </button>
</div>
</div>

<!-- KPI Cards Grid -->
<div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-md mb-lg">
    <!-- Card: Bed Occupancy -->
    <div class="glass-card-premium p-6 rounded-2xl flex flex-col transition-all duration-300">
        <div class="flex justify-between items-start mb-4">
            <div class="p-3 bg-primary/5 text-primary rounded-xl">
                <span class="material-symbols-outlined text-2xl">bed</span>
            </div>
            <span class="text-secondary text-sm font-bold flex items-center gap-0.5">
                <span class="material-symbols-outlined text-sm">trending_up</span>
                +2%
            </span>
        </div>
        <p class="text-on-surface-variant font-label-md mb-1">${tr('Bed Occupancy', '????? ??????')}</p>
        <div class="flex items-baseline gap-2">
            <h3 class="font-display-lg text-display-lg text-primary leading-none">${bedOccupancy}%</h3>
            <span class="text-on-surface-variant text-sm font-medium">/ 100%</span>
        </div>
        <div class="mt-4 w-full bg-surface-container rounded-full h-1.5 overflow-hidden">
            <div class="bg-secondary h-full rounded-full" style="width: ${bedOccupancy}%"></div>
        </div>
    </div>

    <!-- Card: Wait Time -->
    <div class="glass-card-premium p-6 rounded-2xl flex flex-col transition-all duration-300">
        <div class="flex justify-between items-start mb-4">
            <div class="p-3 bg-primary/5 text-primary rounded-xl">
                <span class="material-symbols-outlined text-2xl">timer</span>
            </div>
            <span class="text-secondary text-sm font-bold flex items-center gap-0.5">
                <span class="material-symbols-outlined text-sm">trending_down</span>
                -4m
            </span>
        </div>

```

```
        </span>
    </div>
    <p class="text-on-surface-variant font-label-md mb-1">${tr('Average Wait Time', '???? ? ? ??????')}</p>
</p>

    <div class="flex items-baseline gap-2">
        <h3 class="font-display-lg text-display-lg text-primary leading-none">${avgWaitTime}</h3>
        <span class="text-on-surface-variant text-sm font-medium">${tr('minutes', '????')}</span>
    </div>
    <p class="mt-4 text-xs text-on-surface-variant">${tr('25% better than last week', '???? ????? 25% ? ?
????? ?????')}</p>
</div>

<!-- Card: Compliance Index -->
<div class="glass-card-premium p-6 rounded-2xl flex flex-col border-r-4 border-r-secondary transition-all duration-300">
    <div class="flex justify-between items-start mb-4">
        <div class="p-3 bg-secondary/5 text-secondary rounded-xl">
            <span class="material-symbols-outlined text-2xl" style="font-variation-settings: 'FILL' 1;">verified</span>
        </div>
        <span class="px-2 py-0.5 bg-secondary/10 text-secondary rounded text-[10px] font-bold">${tr('Secure', '???')}</span>
    </div>
    <p class="text-on-surface-variant font-label-md mb-1">${tr('Compliance Index', '???? ?????')}</p>
    <div class="flex items-baseline gap-2">
        <h3 class="font-display-lg text-display-lg text-primary leading-none">98.5%</h3>
    </div>
    <p class="mt-4 text-xs text-on-surface-variant">${tr('Fully compliant with CHI & MOH standards', '????
?? ?????? ? ? ?????? ?????? ?????? ?????? ?????')}</p>
</div>

<!-- Card: Emergency Cases -->
<div class="glass-card-premium p-6 rounded-2xl flex flex-col border-r-4 border-r-error transition-all duration-300">
    <div class="flex justify-between items-start mb-4">
        <div class="p-3 bg-error/5 text-error rounded-xl">
            <span class="material-symbols-outlined text-2xl">emergency</span>
        </div>
        <span class="flex items-center h-2 w-2 bg-error rounded-full animate-ping"></span>
    </div>
    <p class="text-on-surface-variant font-label-md mb-1">${tr('Emergency Cases', '????? ?????? ??????')}</p>
    <div class="flex items-baseline gap-2">
        <h3 class="font-display-lg text-display-lg text-primary leading-none">${emergencyCount}</h3>
        <span class="text-on-surface-variant text-sm font-medium">${tr('active cases', '???? ?????')}</span>
    </div>
    <div class="mt-4 flex -space-x-2 space-x-reverse">
        <div class="w-6 h-6 rounded-full border-2 border-white bg-error/20 flex items-center justify-center text-[8px] font-bold text-error">R1</div>
        <div class="w-6 h-6 rounded-full border-2 border-white bg-error/20 flex items-center justify-center text-[8px] font-bold text-error">R2</div>
        <div class="w-6 h-6 rounded-full border-2 border-white bg-error/20 flex items-center justify-center text-[8px] font-bold text-error">R3</div>
        <div class="w-6 h-6 rounded-full border-2 border-white bg-surface-container-highest flex items-cente
```

```
r justify-center text-[8px] text-on-surface-variant font-bold">+15</div>
    </div>
  </div>
</div>

<!-- Bento Grid Row -->
<div class="grid grid-cols-12 gap-gutter mb-lg">
  <!-- Main Analytics: Revenue Trend & Weekly Patient Flow -->
  <div class="col-span-12 lg:col-span-8 glass-card-premium rounded-3xl p-6 flex flex-col justify-between">
    <div class="flex justify-between items-start mb-6">
      <div>
        <h4 class="font-title-lg text-title-lg text-primary">${tr('Weekly Operations & Revenue Trend', '??
? ???????? ???????? ?????????? ??????????')}</h4>
        <p class="text-on-surface-variant text-sm">${tr('Real-time revenue monitoring and patient flow ana
lyses', '?????? ?????????? ?????? ???? ?????? ?????????')}</p>
      </div>
    </div>
    <div class="grid grid-cols-1 md:grid-cols-2 gap-4 h-64">
      <div class="h-60 relative flex items-center justify-center">
        <canvas id="revenueChart"></canvas>
      </div>
      <div class="h-60 relative flex items-center justify-center">
        <canvas id="dashBar"></canvas>
      </div>
    </div>
  </div>
</div>

<!-- Compliance Alerts Sidebar -->
<div class="col-span-12 lg:col-span-4 glass-card-premium rounded-3xl p-6 flex flex-col justify-between">
  <div class="flex items-center gap-2 mb-4">
    <span class="material-symbols-outlined text-secondary text-2xl">gavel</span>
    <h4 class="font-title-lg text-title-lg text-primary">${tr('Compliance Alerts Feed', '???????? ????????
? ?????????')}</h4>
  </div>
  <div class="space-y-3 flex-1 overflow-y-auto max-h-60">
    <div class="p-3 bg-surface-container-low rounded-xl border-r-4 border-r-secondary relative overflow-
hidden group hover:bg-surface-container-high transition-colors">
      <div class="flex justify-between items-start mb-1">
        <span class="text-[9px] font-bold text-secondary tracking-wider">MOH COMPLIANCE</span>
        <span class="text-[9px] text-on-surface-variant">${tr('15 mins ago', '??? ?? ?????')}</span>
      </div>
      <p class="text-xs font-bold text-primary mb-0.5">${tr('Nursing Staff Licensing Status', '????? ???
?? ?????? ?????????')}</p>
      <p class="text-[11px] text-on-surface-variant leading-relaxed">${tr('Renew licenses for 5 emergenc
y room staff immediately.', '??? ?????? ?????? ?????? ????????? ?? ? ?????? ?? ??? ?????????')}</p>
    </div>
    <div class="p-3 bg-surface-container-low rounded-xl border-r-4 border-r-secondary relative overflow-
hidden group hover:bg-surface-container-high transition-colors">
      <div class="flex justify-between items-start mb-1">
        <span class="text-[9px] font-bold text-on-tertiary-container tracking-wider">CHI AUDIT</span>
        <span class="text-[9px] text-on-surface-variant">${tr('2 hours ago', '??? ?????')}</span>
      </div>
      <p class="text-xs font-bold text-primary mb-0.5">${tr('Insurance Claims Verification', '?????? ???
?? ?????????')}</p>
    </div>
  </div>
</div>
```

```

    <p class="text-[11px] text-on-surface-variant leading-relaxed">${tr('Verify October claims documents prior to CHI transmission.', '????? ????? ????????? ?????????? ??? ?????? ??? ??????? ????????.')}}</p>
  </div>
  <div class="p-3 bg-error/5 rounded-xl border-r-4 border-r-error relative overflow-hidden group hover:bg-error/10 transition-colors">
    <div class="flex justify-between items-start mb-1">
      <span class="text-[9px] font-bold text-error tracking-wider">CRITICAL WARN</span>
      <span class="text-[9px] text-on-surface-variant">${tr('4 hours ago', '??? ? ?????')}</span>
    </div>
    <p class="text-xs font-bold text-primary mb-0.5">${tr('Defibrillators Inspection Overdue', '??? ??? ??????? ?????????')}</p>
    <p class="text-[11px] text-on-surface-variant leading-relaxed">${tr('2 defibrillators on the 3rd floor require immediate testing.', '????? ????? ?????? ??????? ??????? ?? ?????? ??????.')}}</p>
  </div>
</div>
</div>
</div>

<!-- Vision 2030 Strategic Metrics Bento -->
<div class="glass-card-premium rounded-3xl p-6 mb-lg">
  <div class="flex justify-between items-start mb-4">
    <div>
      <h4 class="font-title-lg text-title-lg text-primary">${tr('Vision 2030 Strategic Alignment', '????? ?????? ??????? 2030')}</h4>
      <p class="text-on-surface-variant text-sm">${tr('National healthcare transformation indicators', '??? ?????? ??????????? ?????????? ???????')}</p>
    </div>
    <span class="bg-secondary/10 text-secondary px-3 py-1 rounded-full text-xs font-bold">${tr('Live Sync', '?????')}</span>
  </div>
  <div class="grid grid-cols-1 md:grid-cols-3 gap-md">
    <div class="space-y-2 p-3 bg-surface-container-low rounded-2xl border border-outline-variant/10">
      <p class="text-xs font-bold text-on-surface-variant uppercase tracking-wider">${tr('Digital Transformation', '????? ?????')}</p>
      <div class="flex items-baseline gap-1">
        <span class="font-display-lg text-3xl font-bold text-primary">94%</span>
        <span class="text-secondary font-bold text-sm">? 1.2%</span>
      </div>
      <div class="w-full bg-surface-container-highest h-2 rounded-full overflow-hidden">
        <div class="bg-secondary h-full rounded-full" style="width: 94%;"></div>
      </div>
    </div>
    <div class="space-y-2 p-3 bg-surface-container-low rounded-2xl border border-outline-variant/10">
      <p class="text-xs font-bold text-on-surface-variant uppercase tracking-wider">${tr('Spending Efficiency', '????? ?????')}</p>
      <div class="flex items-baseline gap-1">
        <span class="font-display-lg text-3xl font-bold text-primary">82%</span>
        <span class="text-secondary font-bold text-sm">? 0.5%</span>
      </div>
      <div class="w-full bg-surface-container-highest h-2 rounded-full overflow-hidden">
        <div class="bg-secondary h-full rounded-full" style="width: 82%;"></div>
      </div>
    </div>
  </div>
</div>
```



```
<p class="text-xs font-bold text-on-surface-variant uppercase tracking-wider">${tr('Patient Satisfac
tion', '??? ?????????')}</p>
<div class="flex items-baseline gap-1">
  <span class="font-display-lg text-3xl font-bold text-primary">89%</span>
  <span class="text-secondary font-bold text-sm">? 2.4%</span>
</div>
<div class="w-full bg-surface-container-highest h-2 rounded-full overflow-hidden">
  <div class="bg-secondary h-full rounded-full" style="width: 89%"></div>
</div>
</div>
</div>
</div>

<!-- Department Status Cards Grid -->
<div class="mb-lg">
  <h4 class="font-title-lg text-title-lg text-primary mb-6">${tr('Departmental Operational Performance', '
???? ??????? ?????????')}</h4>
  <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-md">
    <!-- Emergency -->
    <div class="glass-card-premium rounded-2xl p-5 border-t-4 border-t-error">
      <div class="flex justify-between items-start mb-3">
        <div class="w-10 h-10 bg-error/10 rounded-lg flex items-center justify-center text-error">
          <span class="material-symbols-outlined">emergency</span>
        </div>
        <div class="text-right">
          <p class="text-xs text-on-surface-variant font-medium">${tr('Status', '?????')}</p>
          <p class="text-sm font-bold text-error">${tr('Crowded', '??? ?????')}</p>
        </div>
      </div>
      <h5 class="font-bold text-primary text-sm mb-3">${tr('Emergency Room (ER)', '??? ?????')}</h5>
      <div class="space-y-2 text-xs">
        <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Active Patients', '?
????? ?????????')}</span><span class="font-bold text-primary">42</span></div>
        <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Wait Time', '???? ??
?????')}</span><span class="font-bold text-error">18m</span></div>
      </div>
    </div>
    <!-- Surgery -->
    <div class="glass-card-premium rounded-2xl p-5 border-t-4 border-t-secondary">
      <div class="flex justify-between items-start mb-3">
        <div class="w-10 h-10 bg-secondary/10 rounded-lg flex items-center justify-center text-secondary">
          <span class="material-symbols-outlined">medical_mask</span>
        </div>
        <div class="text-right">
          <p class="text-xs text-on-surface-variant font-medium">${tr('Status', '?????')}</p>
          <p class="text-sm font-bold text-secondary">${tr('Active', '???')}</p>
        </div>
      </div>
      <h5 class="font-bold text-primary text-sm mb-3">${tr('Surgery Suite', '??? ?????')}</h5>
      <div class="space-y-2 text-xs">
        <div class="flex justify-between"><span class="text-on-surface-variant">${tr('ORs In Use', '??? ??
?????')}</span><span class="font-bold text-primary">6 / 8</span></div>
        <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Staff Status', '????
?? ?????')}</span><span class="font-bold text-primary">${tr('Optimal', '????')}</span></div>
```

```

    </div>
</div>
<!-- ICU -->
<div class="glass-card-premium rounded-2xl p-5 border-t-4 border-t-primary">
  <div class="flex justify-between items-start mb-3">
    <div class="w-10 h-10 bg-primary/10 rounded-lg flex items-center justify-center text-primary">
      <span class="material-symbols-outlined">vital_signs</span>
    </div>
    <div class="text-right">
      <p class="text-xs text-on-surface-variant font-medium">${tr('Status', '??????')}</p>
      <p class="text-sm font-bold text-primary">${tr('Critical', '???')}</p>
    </div>
  </div>
  <h5 class="font-bold text-primary text-sm mb-3">${tr('Intensive Care (ICU)', '??????? ???????')}</h5>
>

  <div class="space-y-2 text-xs">
    <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Bed Occupancy', '???
? ???????')}</span><span class="font-bold text-primary">92%</span></div>
    <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Available Beds', '??
???? ???????')}</span><span class="font-bold text-error">2 beds</span></div>
  </div>
</div>
<!-- Radiology -->
<div class="glass-card-premium rounded-2xl p-5 border-t-4 border-t-outline">
  <div class="flex justify-between items-start mb-3">
    <div class="w-10 h-10 bg-surface-container-high rounded-lg flex items-center justify-center text-o
n-surface-variant">
      <span class="material-symbols-outlined">radiology</span>
    </div>
    <div class="text-right">
      <p class="text-xs text-on-surface-variant font-medium">${tr('Status', '??????')}</p>
      <p class="text-sm font-bold text-on-surface-variant">${tr('Stable', '????')}</p>
    </div>
  </div>
  <h5 class="font-bold text-primary text-sm mb-3">${tr('Imaging & Radiology', '?????? ?????????')}</h5>
  <div class="space-y-2 text-xs">
    <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Scans Today', '?????
??? ?????')}</span><span class="font-bold text-primary">85</span></div>
    <div class="flex justify-between"><span class="text-on-surface-variant">${tr('Reporting Time', '??
? ?????')}</span><span class="font-bold text-primary">45m</span></div>
  </div>
</div>
</div>
</div>

<!-- Row: Top Doctors & Revenue Breakdown & Charts -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-md mb-lg">
  <div class="glass-card-premium rounded-2xl p-6">
    <h4 class="font-title-lg text-title-lg text-primary mb-4 flex items-center gap-2">
      <span class="material-symbols-outlined text-secondary">star</span>
      ? ${tr('Top Doctors (This Month)', '???? ??????? (??? ?????')}
    </h4>
    <div class="space-y-2 max-h-60 overflow-y-auto">
      ${topDrHtml}

```

```
</div>
</div>
<div class="glass-card-premium rounded-2xl p-6">
  <h4 class="font-title-lg text-title-lg text-primary mb-4 flex items-center gap-2">
    <span class="material-symbols-outlined text-secondary">payments</span>
    ? ${tr('Revenue by Service Type', '????????? ??? ??? ?????')}
  </h4>
  <div class="space-y-4 max-h-60 overflow-y-auto">
    ${revTypeHtml}
  </div>
</div>
</div>
<div class="glass-card-premium rounded-2xl p-6 flex flex-col justify-between">
  <h4 class="font-title-lg text-title-lg text-primary mb-4 flex items-center gap-2">
    <span class="material-symbols-outlined text-secondary">pie_chart</span>
    ? ${tr('Department Share & Payment Methods', '???? ?????????? ??? ?????')}
  </h4>
  <div class="grid grid-cols-2 gap-2 h-44">
    <div class="relative"><canvas id="deptChart"></canvas></div>
    <div class="relative"><canvas id="paymentChart"></canvas></div>
  </div>
</div>
</div>

<!-- Quick Actions Panel -->
<div class="glass-card-premium rounded-2xl p-6">
  <h4 class="font-title-lg text-title-lg text-primary mb-4">${tr('Quick Administrative Actions', '???????
????? ?????')}</h4>
  <div class="grid grid-cols-2 md:grid-cols-6 gap-3">
    <button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
hadow-sm" onclick="navigateTo(1)">
      <span class="material-symbols-outlined text-lg">hotel</span>
      ${tr('Reception', '????????')}
    </button>
    <button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
hadow-sm" onclick="navigateTo(2)">
      <span class="material-symbols-outlined text-lg">calendar_month</span>
      ${tr('Appointments', '????????')}
    </button>
    <button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
hadow-sm" onclick="navigateTo(4)">
      <span class="material-symbols-outlined text-lg">biotech</span>
      ${tr('Lab', '????????')}
    </button>
    <button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
hadow-sm" onclick="navigateTo(6)">
      <span class="material-symbols-outlined text-lg">prescriptions</span>
      ${tr('Pharmacy', '????????')}
    </button>
    <button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
```

```

hadow-sm" onclick="navigateTo(14)">
    <span class="material-symbols-outlined text-lg">bar_chart</span>
    ${tr('Reports', '???????')}
</button>
<button class="flex items-center justify-center gap-2 p-3 bg-surface-container-low hover:bg-secondary
hover:text-white rounded-xl text-primary font-medium text-xs transition-all border border-outline-variant/20 s
hadow-sm" onclick="navigateTo(8)">
    <span class="material-symbols-outlined text-lg">account_balance_wallet</span>
    ${tr('Finance', '???????')}
</button>
</div>
</div>
`;
}

```

```

function renderDashboardCharts(el, enhanced) {
  try {
    const revData = enhanced.revenueByType || [];
    if (revData.length === 0 || typeof Chart === 'undefined') return;
    const labels = revData.map(r => r.service_type);
    const values = revData.map(r => parseFloat(r.total) || 0);
    const clr = ['#001629', '#006970', '#ac8c44', '#ba1a1a', '#22c55e', '#8b5cf6', '#ec4899', '#06b6d4'];

    const barCtx = document.getElementById('dashBar');
    if (barCtx) {
      // Clear any existing chart instances to avoid overlap
      const existingChart = Chart.getChart(barCtx);
      if (existingChart) existingChart.destroy();

      new Chart(barCtx, {
        type: 'bar',
        data: {
          labels,
          datasets: [{
            label: tr('Revenue', '???????'),
            data: values,
            backgroundColor: clr.slice(0, labels.length),
            borderRadius: 6
          }]
        },
        options: {
          responsive: true,
          maintainAspectRatio: false,
          plugins: { legend: { display: false } },
          scales: { y: { beginAtZero: true } }
        }
      });
    }
  } catch (e) { console.log('Chart error:', e); }
}

```

```

// === CRITICAL LAB VALUE DEFINITIONS ===
const CRITICAL_LAB_VALUES = {
  'Hemoglobin': { low: 7.0, high: 20.0, unit: 'g/dL' },

```

```

'Platelets': { low: 50, high: 1000, unit: 'x103/μL' },
'WBC': { low: 2.0, high: 30.0, unit: 'x103/μL' },
'Potassium': { low: 2.5, high: 6.5, unit: 'mEq/L' },
'Sodium': { low: 120, high: 160, unit: 'mEq/L' },
'Glucose': { low: 40, high: 500, unit: 'mg/dL' },
'Creatinine': { low: 0, high: 10.0, unit: 'mg/dL' },
'Troponin': { low: 0, high: 0.04, unit: 'ng/mL' },
'INR': { low: 0, high: 5.0, unit: '' },
'Lactate': { low: 0, high: 4.0, unit: 'mmol/L' }
};

window.checkCriticalLabValue = (testName, resultText) => {
  const numMatch = resultText.match(/[\\d.]+/);
  if (!numMatch) return null;
  const val = parseFloat(numMatch[0]);
  for (const [key, range] of Object.entries(CRITICAL_LAB_VALUES)) {
    if (testName.toLowerCase().includes(key.toLowerCase())) {
      if (val < range.low) return { test: key, value: val, status: 'CRITICALLY LOW', range };
      if (val > range.high) return { test: key, value: val, status: 'CRITICALLY HIGH', range };
    }
  }
  return null;
};

// === DRUG ALLERGY CHECK ===
window.checkDrugAllergy = async (patientId, drugName) => {
  try {
    const patients = await API.get('/api/patients');
    const patient = patients.find(p => p.id == patientId);
    if (!patient || !patient.allergies) return false;
    const allergies = patient.allergies.toLowerCase().split(/[,?;]+/).map(a => a.trim());
    const drug = drugName.toLowerCase();
    for (const allergy of allergies) {
      if (allergy && (drug.includes(allergy) || allergy.includes(drug))) {
        return allergy;
      }
    }
    return false;
  } catch { return false; }
};

// === PATIENT STATEMENT (Printable Account) ===
window.printPatientStatement = async (patientId) => {
  try {
    const account = await API.get('/api/patients/' + patientId + '/account');
    const p = account.patient;
    const invoices = account.invoices || [];
    let rows = invoices.map(inv =>
      '<tr><td>' + (inv.created_at ? new Date(inv.created_at).toLocaleDateString('ar-SA') : '-') +
      '</td><td>' + escapeHTML(inv.description || inv.service_type || '-') +
      '</td><td>' + escapeHTML(inv.total || 0) + ' SAR</td><td>' +
      (inv.paid ? '\\u2705 ' + tr('Paid', '?????') : '\\u26A0\\uFE0F ' + tr('Unpaid', '??? ?????')) + '</td></tr>'
    );
    const content = '<div style="text-align:center;margin-bottom:20px"><h2>\\u{1F3E5} ' + tr('jumanaMedical', '

```

```

????? ?????') + '</h2><div style="margin-bottom:8px"><button class="btn btn-sm" onclick="exportPatients()" style="background:#e0f7fa;color:#00838f">? ${tr("Export CSV","????? CSV")}</button></div><h3>' + tr('Patient Financial Statement', '??? ???? ??????') + '</h3></div>' +

  '<table style="width:100%;margin-bottom:15px"><tr><td><strong>' + tr('Name', '?????') + ':</strong>' + escapeHTML(p.name_ar || p.name_en) + '</td><td><strong>MRN:</strong>' + escapeHTML(p.mrn || p.file_number) + '</td></tr>' +

  '<tr><td><strong>' + tr('ID', '?????') + ':</strong>' + escapeHTML(p.national_id || '-') + '</td><td><strong>' + tr('Phone', '?????') + ':</strong>' + escapeHTML(p.phone || '-') + '</td></tr></table>' +

  '<table border="1" cellpadding="6" cellspacing="0" style="width:100%;border-collapse:collapse"><thead><tr style="background:#f0f0f0"><th>' + tr('Date', '?????') + '</th><th>' + tr('Description', '?????') + '</th><th>' + tr('Amount', '?????') + '</th><th>' + tr('Status', '?????') + '</th></tr></thead><tbody>' + rows + '</tbody></table>' +

  '<div style="margin-top:20px;padding:10px;background:#f9f9f9;border-radius:8px"><strong>' + tr('Total Billed', '????? ????') + ':</strong>' + escapeHTML(account.totalBilled || 0) + ' SAR | <strong>' + tr('Total Paid', '?????') + ':</strong>' + escapeHTML(account.totalPaid || 0) + ' SAR | <strong style="color:' + (account.balance > 0 ? 'red' : 'green') + '>' + tr('Balance', '?????') + ':</strong>' + escapeHTML(account.balance || 0) + ' SAR</div>';

  printDocument(tr('Patient Statement', '??? ???? ??????'), content);
} catch (e) { showToast(tr('Error loading statement', '??? ? ???? ????'), 'error'); }
};

// === DIAGNOSIS TEMPLATES ===
let _diagTemplatesCache = null;
window.loadDiagTemplates = async () => {
  try {
    const templates = await API.get('/api/diagnosis-templates');
    _diagTemplatesCache = templates;
    const sel = document.getElementById('drDiagTemplate');
    if (!sel) return;
    sel.innerHTML = '<option value="">' + tr('-- Select Template --', '-- ???? ???? --') + '</option>';
    for (const [specialty, items] of Object.entries(templates)) {
      const group = document.createElement('optgroup');
      group.label = specialty;
      items.forEach((t, idx) => {
        const opt = document.createElement('option');
        opt.value = specialty + '|' + idx;
        opt.textContent = (isArabic ? t.name_ar : t.name) + ' [' + t.icd + ']';
        group.appendChild(opt);
      });
      sel.appendChild(group);
    }
    showToast(tr('Templates loaded!', '?? ????? ??????'));
  } catch (e) { showToast(tr('Error loading templates', '??? ? ???? ??????'), 'error'); }
};

window.applyDiagTemplate = () => {
  const val = document.getElementById('drDiagTemplate')?.value;
  if (!val || !_diagTemplatesCache) return;
  const [specialty, idx] = val.split('|');
  const t = _diagTemplatesCache[specialty]?.[parseInt(idx)];
  if (!t) return;
  document.getElementById('drDiag').value = isArabic ? t.name_ar : t.name;
  document.getElementById('drSymp').value = t.symptoms || '';
  document.getElementById('drIcd').value = t.icd || '';
  document.getElementById('drNotes').value = t.treatment || '';

```

```

};

// === PHARMACY LOW STOCK ALERTS ===
window.showPharmacyStockAlerts = async () => {
  try {
    const lowStock = await API.get('/api/pharmacy/low-stock');
    if (lowStock.length === 0) { showToast(tr('All stock levels OK!', '???? ????????? ????? ?????!')); return; }

    let html = '<div style="max-height:400px;overflow-y:auto">';
    lowStock.forEach(d => {
      const pct = d.min_stock_level > 0 ? Math.round((d.stock_qty / d.min_stock_level) * 100) : 0;
      const color = d.stock_qty <= 0 ? '#dc2626' : d.stock_qty <= 5 ? '#f59e0b' : '#eab308';
      html += '<div style="display:flex;justify-content:space-between;align-items:center;padding:10px;margin:4px 0;border-radius:8px;border-right:4px solid ' + color + ';background:var(--hover)">' +
        '<div><strong>' + d.drug_name + '</strong>' + (d.category ? '<br><small>' + d.category + '</small>' :
        '') + '</div>' +
        '<div style="text-align:center"><span style="font-size:20px;font-weight:700;color:' + color + '">' + d
        .stock_qty + '</span><br><small>' + tr('of', '??') + ' ' + (d.min_stock_level || 10) + ' min</small></div></div>';
    });
    html += '</div>';
    showModal(tr('Low Stock Alerts', '??????? ???????? ????????') + ' (' + lowStock.length + ')', html);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// === P&L REPORT VIEWER ===
window.renderPnlReport = async (fromDate, toDate) => {
  try {
    let url = '/api/reports/pnl';
    if (fromDate && toDate) url += '?from=' + fromDate + '&to=' + toDate;
    const data = await API.get(url);
    const el = document.getElementById('pnlResult');
    if (!el) return;
    let typeRows = data.byType.map(t =>
      '<tr><td>' + (t.service_type || '-') + '</td><td>' + t.cnt + '</td><td style="font-weight:600">' + Number(
      t.total).toLocaleString() + ' SAR</td></tr>'
    ).join('');
    el.innerHTML = '<div class="stats-grid" style="margin-bottom:16px">' +
      '<div class="stat-card" style="--stat-color:#22c55e"><div class="stat-label">' + tr('Total Revenue', '??
      ??? ????') + '</div><div class="stat-value">' + Number(data.totalRevenue).toLocaleString() + '</div></div>' +
      ' +
      '<div class="stat-card" style="--stat-color:#3b82f6"><div class="stat-label">' + tr('Collected', '?????
      ') + '</div><div class="stat-value">' + Number(data.totalCollected).toLocaleString() + '</div></div>' +
      '<div class="stat-card" style="--stat-color:#f59e0b"><div class="stat-label">' + tr('Discounts', '?????
      ??') + '</div><div class="stat-value">' + Number(data.totalDiscounts).toLocaleString() + '</div></div>' +
      '<div class="stat-card" style="--stat-color:#ef4444"><div class="stat-label">' + tr('Uncollected', '???
      ???') + '</div><div class="stat-value">' + Number(data.totalUncollected).toLocaleString() + '</div></div>' +
      '<div class="stat-card" style="--stat-color:#64748b"><div class="stat-label">' + tr('Est. Costs', '????
      ? ?????') + '</div><div class="stat-value">' + Number(data.estimatedCosts).toLocaleString() + '</div></div>' +
      ' +
      '<div class="stat-card" style="--stat-color:' + (data.netProfit >= 0 ? '#10b981' : '#ef4444') + '"><div
      class="stat-label">' + tr('Net Profit', '???? ?????') + '</div><div class="stat-value">' + Number(data.netProf
      it).toLocaleString() + '</div></div>' +
      '</div><table class="data-table"><thead><tr><th>' + tr('Service', '????') + '</th><th>' + tr('Count',

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'?????') + '</th><th>' + tr('Revenue', '???????') + '</th></tr></thead><tbody>' + typeRows + '</tbody></table>
';
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// === SHOW MODAL HELPER ===
window.showModal = (title, content) => {
  let modal = document.getElementById('genericModal');
  if (!modal) {
    modal = document.createElement('div');
    modal.id = 'genericModal';
    modal.style.cssText = 'position:fixed;top:0;left:0;right:0;bottom:0;background:rgba(0,0,0,0.5);display:flex;align-items:center;justify-content:center;z-index:9999';
    modal.innerHTML = '<div style="background:var(--card-bg,#fff);border-radius:16px;padding:24px;max-width:600px;width:90%;max-height:80vh;overflow-y:auto;box-shadow:0 20px 60px rgba(0,0,0,0.3)"><div class="flex" style="justify-content:space-between;align-items:center;margin-bottom:16px"><h3 id="genericModalTitle" style="margin:0"></h3><button onclick="this.closest(\'#genericModal\').style.display=\'none\'" style="border:none;background:none;font-size:20px;cursor:pointer">\u2715</button></div><div id="genericModalBody"></div></div>';
    document.body.appendChild(modal);
  }
  document.getElementById('genericModalTitle').textContent = title;
  document.getElementById('genericModalBody').innerHTML = content;
  modal.style.display = 'flex';
};

async function renderReception(el) {
  // Show premium loading skeletons instantly
  el.innerHTML = `
<div class="page-title">? ${tr('Reception', '????????')}</div>
<div class="split-layout">
  <div class="card glass-card-premium">
    <div class="skeleton-bar title mb-16"></div>
    <div class="skeleton-bar mb-12"></div>
    <div class="skeleton-bar mb-12"></div>
    <div class="skeleton-bar mb-12"></div>
    <div class="skeleton-bar mb-12"></div>
  </div>
</div>
<div class="card glass-card-premium mt-16">
  <div class="skeleton-bar title mb-16"></div>
  <div class="skeleton-bar mb-12"></div>
  <div class="skeleton-bar mb-12"></div>
</div>
`;

  let patients = [];
  let doctors = [];
  let invoices = [];
  try {
    [patients, doctors, invoices] = await Promise.all([
      API.get('/api/patients'),
      API.get('/api/employees?role=Doctor'),
      API.get('/api/invoices').catch(() => [])
    ]);
  }
};

```



```

window.currentPatientsList = patients;
window.currentDoctorsList = doctors;
window.currentInvoicesList = invoices;
} catch (e) {
  el.innerHTML = `
    <div class="page-title">? ${tr('Reception', '????????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load reception data', '??? ?? ????? ?????? ?????????')}</h3>
      <p>${escapeHTML(e.message || e)}</p>
      <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
button>
    </div>
  `;
  return;
}

const depts = ['???????? ??????', '????????', '????????', '??????', '????????', '????? ??????', '??????', '????
???', '????????'];
const deptsEn = ['General Clinic', 'Internal Medicine', 'Pediatrics', 'Orthopedics', 'Dermatology', 'ENT', '
Ophthalmology', 'Dental', 'Emergency'];
const maxFile = patients.length ? Math.max(...patients.map(p => p.file_number || 1000)) + 1 : 1001;
el.innerHTML = `
  <div class="grid grid-cols-1 md:grid-cols-4 gap-6 mb-6">
    <div class="glass-card glass-card-premium p-4 rounded-xl border border-primary/20 flex items-center gap-
3">
      <span class="material-symbols-outlined text-3xl text-primary" style="font-size:32px">groups</span>
      <div>
        <div class="text-xs text-on-surface-variant font-medium">${tr('Total Patients', '?????? ?????????')}</
div>
        <div class="text-xl font-bold text-primary mt-1" id="kpiTotalPatients">${patients.length}</div>
      </div>
    </div>
    <div class="glass-card glass-card-premium p-4 rounded-xl border border-emerald-500/20 flex items-center
gap-3">
      <span class="material-symbols-outlined text-3xl text-emerald-500" style="font-size:32px">person_add</s
pan>
      <div>
        <div class="text-xs text-on-surface-variant font-medium">${tr("Today's Registered", '????? ?????')}<
/div>
        <div class="text-xl font-bold text-emerald-500 mt-1" id="kpiTodayPatients">
          ${patients.filter(p => p.created_at && new Date(p.created_at).toDateString() === new Date().toDate
String()).length}
        </div>
      </div>
    </div>
    <div class="glass-card glass-card-premium p-4 rounded-xl border border-amber-500/20 flex items-center ga
p-3">
      <span class="material-symbols-outlined text-3xl text-amber-500" style="font-size:32px">medical_service
s</span>
      <div>
        <div class="text-xs text-on-surface-variant font-medium">${tr('Active Insurance', '????? ?????????')}<
/div>
        <div class="text-xl font-bold text-amber-500 mt-1" id="kpiInsurancePatients">

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        ${patients.filter(p => p.insurance_company).length}
      </div>
    </div>
  </div>
<div class="glass-card glass-card-premium p-4 rounded-xl border border-error/20 flex items-center gap-3"
>
  <span class="material-symbols-outlined text-3xl text-error" style="font-size:32px">payments</span>
  <div>
    <div class="text-xs text-on-surface-variant font-medium">${tr('Pending Payments', '????? ?????? ????'
??')}</div>
    <div class="text-xl font-bold text-error mt-1" id="kpiPendingPayments">0</div>
  </div>
</div>

<div class="split-layout">
  <div class="card glass-card-premium">
    <!-- Stepper Wizard Header -->
    <div class="flex justify-between items-center mb-6 border-b pb-4 border-outline-variant/30" style="flex
x-wrap:wrap;gap:8px">
      <div class="flex items-center gap-2 step-indicator active" id="stepIndicator1">
        <span class="step-number-badge active" id="stepNum1">1</span>
        <span class="text-xs font-bold text-primary" id="stepLabel1">${tr('Identity Profile', '??? ??????'
)}}</span>
      </div>
      <div class="h-0.5 bg-outline-variant flex-1 mx-4" id="stepLine1" style="min-width:40px;background:va
r(--primary)"></div>
      <div class="flex items-center gap-2 step-indicator" id="stepIndicator2">
        <span class="step-number-badge pending" id="stepNum2">2</span>
        <span class="text-xs text-on-surface-variant font-medium" id="stepLabel2">${tr('Medical & Contact'
, '????? ?????? ?????????')}</span>
      </div>
      <div class="h-0.5 bg-outline-variant flex-1 mx-4" id="stepLine2" style="min-width:40px;background:va
r(--border-color,#e5e7eb)"></div>
      <div class="flex items-center gap-2 step-indicator" id="stepIndicator3">
        <span class="step-number-badge pending" id="stepNum3">3</span>
        <span class="text-xs text-on-surface-variant font-medium" id="stepLabel3">${tr('Insurance (NPHIES)
', '???????? (????)')}</span>
      </div>
    </div>

    <!-- EMPI Duplicate File Notification Card -->
    <div class="mb-4" id="empiDuplicateAlert" style="display:none"></div>

    <!-- Step 1: Personal Profile & Identity -->
    <div class="reception-step-content" id="receptionStep1">
      <h4 class="font-title-sm text-primary mb-4">? ${tr('Identity & Personal Information', '????????? ????'
??? ?????????')}</h4>

      <div class="reception-form-group mb-12">
        <label class="reception-form-label">? ${tr('File No.', '??? ?????')}</label>
        <div class="reception-input-wrapper">
          <span class="material-symbols-outlined reception-input-icon">tag</span>
          <input class="reception-form-input readonly" value="${maxFile}" readonly id="rFileNum">

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```
</div>
</div>
```

```
<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('Full Name (Arabic)', '????? ?????????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">person</span>
    <input class="reception-form-input" id="rNameAr" placeholder="${tr('Enter Arabic name', '???? ?
??? ?????????')}>
  </div>
</div>
```

```
<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('Full Name (English)', '????? ?????????????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">translate</span>
    <input class="reception-form-input" id="rNameEn" placeholder="${tr('Enter English name', '???? ?
????? ?????????????')}>
  </div>
</div>
```

```
<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('National ID / Iqama', '??? ?????? / ?????????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">badge</span>
    <input class="reception-form-input" id="rNatId" placeholder="1XXXXXXXXX / 2XXXXXXXXX">
  </div>
</div>
```

```
<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('Phone', '??????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">call</span>
    <input class="reception-form-input" id="rPhone" placeholder="05XXXXXXXX">
  </div>
</div>
```

```
<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('Nationality', '????????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">public</span>
    <select class="reception-form-input" id="rNationality">
      <option value="?????">? ${tr('Saudi', '????')}</option>
      <option value="?????">? ${tr('Yemeni', '????')}</option>
      <option value="????????">? ${tr('Emirati', '????????')}</option>
      <option value="?????">? ${tr('Kuwaiti', '????')}</option>
      <option value="?????">? ${tr('Bahraini', '????')}</option>
      <option value="?????">? ${tr('Qatari', '????')}</option>
      <option value="?????">? ${tr('Omani', '????')}</option>
      <option value="?????">? ${tr('Iraqi', '????')}</option>
      <option value="?????">? ${tr('Jordanian', '????')}</option>
      <option value="?????">? ${tr('Syrian', '????')}</option>
      <option value="????????">? ${tr('Lebanese', '????')}</option>
      <option value="????????">? ${tr('Palestinian', '????')}</option>
    </select>
  </div>
</div>
```

<option value="????">?? \${tr('Egyptian', '????')}</option>  
<option value="?????">?? \${tr('Sudanese', '?????')}</option>  
<option value="????">?? \${tr('Libyan', '????')}</option>  
<option value="????">?? \${tr('Tunisian', '????')}</option>  
<option value="?????">?? \${tr('Algerian', '?????')}</option>  
<option value="?????">?? \${tr('Moroccan', '?????')}</option>  
<option value="?????????">?? \${tr('Mauritanian', '????????')}</option>  
<option value="?????">?? \${tr('Somali', '?????')}</option>  
<option value="?????">?? \${tr('Djiboutian', '?????')}</option>  
<option value="??? ?????">?? \${tr('Comoran', '??? ?????')}</option>  
<option value="????">?? \${tr('Turkish', '????')}</option>  
<option value="?????">?? \${tr('Iranian', '?????')}</option>  
<option value="?????">?? \${tr('Afghan', '?????')}</option>  
<option value="?????????">?? \${tr('Pakistani', '????????')}</option>  
<option value="????">?? \${tr('Indian', '????')}</option>  
<option value="?????????">?? \${tr('Bangladeshi', '????????')}</option>  
<option value="?????????">?? \${tr('Sri Lankan', '????????')}</option>  
<option value="?????">?? \${tr('Nepali', '?????')}</option>  
<option value="?????">?? \${tr('Filipino', '?????')}</option>  
<option value="?????????">?? \${tr('Indonesian', '????????')}</option>  
<option value="?????">?? \${tr('Malaysian', '?????')}</option>  
<option value="?????????">?? \${tr('Thai', '????????')}</option>  
<option value="?????????">?? \${tr('Vietnamese', '????????')}</option>  
<option value="?????????">?? \${tr('Myanmar', '????????')}</option>  
<option value="????">?? \${tr('Chinese', '????')}</option>  
<option value="?????">?? \${tr('Japanese', '?????')}</option>  
<option value="????">?? \${tr('Korean', '????')}</option>  
<option value="?????">?? \${tr('American', '?????')}</option>  
<option value="????">?? \${tr('Canadian', '????')}</option>  
<option value="?????">?? \${tr('Mexican', '?????')}</option>  
<option value="?????????">?? \${tr('Brazilian', '????????')}</option>  
<option value="?????????">?? \${tr('Argentine', '????????')}</option>  
<option value="?????????">?? \${tr('Colombian', '????????')}</option>  
<option value="?????????">?? \${tr('British', '????????')}</option>  
<option value="?????">?? \${tr('French', '?????')}</option>  
<option value="?????">?? \${tr('German', '?????')}</option>  
<option value="?????">?? \${tr('Italian', '?????')}</option>  
<option value="?????">?? \${tr('Spanish', '?????')}</option>  
<option value="?????????">?? \${tr('Portuguese', '????????')}</option>  
<option value="?????">?? \${tr('Dutch', '?????')}</option>  
<option value="?????">?? \${tr('Belgian', '?????')}</option>  
<option value="?????">?? \${tr('Swiss', '?????')}</option>  
<option value="?????">?? \${tr('Austrian', '?????')}</option>  
<option value="?????">?? \${tr('Swedish', '?????')}</option>  
<option value="?????">?? \${tr('Norwegian', '?????')}</option>  
<option value="?????????">?? \${tr('Danish', '????????')}</option>  
<option value="?????">?? \${tr('Finnish', '?????')}</option>  
<option value="?????">?? \${tr('Polish', '?????')}</option>  
<option value="????">?? \${tr('Russian', '????')}</option>  
<option value="?????????">?? \${tr('Ukrainian', '????????')}</option>  
<option value="?????">?? \${tr('Romanian', '?????')}</option>  
<option value="?????">?? \${tr('Greek', '?????')}</option>  
<option value="?????????">?? \${tr('Australian', '????????')}</option>  
<option value="?????????">?? \${tr('New Zealander', '????????')}</option>

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<option value="???? ??????">?? ${tr('South African', '???? ??????')}</option>
<option value="???????">?? ${tr('Nigerian', '??????')}</option>
<option value="?????">?? ${tr('Kenyan', '????')}</option>
<option value="???????">?? ${tr('Ethiopian', '??????')}</option>
<option value="???????">?? ${tr('Ugandan', '??????')}</option>
<option value="?????????">?? ${tr('Tanzanian', '????????')}</option>
<option value="?????">?? ${tr('Ghanaian', '????')}</option>
<option value="???????">?? ${tr('Senegalese', '??????')}</option>
<option value="?????????">?? ${tr('Cameroonian', '????????')}</option>
<option value="?????">?? ${tr('Chadian', '????')}</option>
<option value="?????">?? ${tr('Malian', '????')}</option>
<option value="???????">?? ${tr('Eritrean', '??????')}</option>
<option value="?????????????">?? ${tr('Azerbaijani', '????????????')}</option>
<option value="???????">?? ${tr('Uzbek', '??????')}</option>
<option value="???????">?? ${tr('Kazakh', '??????')}</option>
<option value="?????????????">?? ${tr('Turkmen', '????????????')}</option>
<option value="???????">?? ${tr('Kyrgyz', '??????')}</option>
<option value="???????">?? ${tr('Tajik', '??????')}</option>
<option value="?????">?? ${tr('Other', '????')}</option>
</select>
</div>
</div>

```

```

<div class="reception-form-group mb-12">
  <label class="reception-form-label">?? ${tr('Gender', '????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">wc</span>
    <select class="reception-form-input" id="rGender">
      <option value="???">? ${tr('Male', '???')}</option>
      <option value="?????">? ${tr('Female', '????')}</option>
    </select>
  </div>
</div>

```

```

<div class="flex gap-16 mb-12" style="flex-wrap:wrap">
  <div class="reception-form-group" style="flex:3;min-width:220px">
    <label class="reception-form-label">? ${tr('DOB (Gregorian)', '????? ??????? (??????)')}</label>
    <div class="flex gap-4">
      <select class="reception-form-input" id="rGregDay" style="flex:0.8;padding:0 8px"><option value="">${tr('Day', '???')}</option>${Array.from({ length: 31 }, (_, i) => `<option value="${i + 1}">${i + 1}</option>`)}</select>
      <select class="reception-form-input" id="rGregMonth" style="flex:1.5;padding:0 8px"><option value="">${tr('Month', '???')}</option>
        <option value="1">${tr('January', '????')}</option><option value="2">${tr('February', '????
        ??')}</option><option value="3">${tr('March', '????')}</option>
        <option value="4">${tr('April', '????')}</option><option value="5">${tr('May', '????')}</op
        tion><option value="6">${tr('June', '????')}</option>
        <option value="7">${tr('July', '????')}</option><option value="8">${tr('August', '????')}<
        /option><option value="9">${tr('September', '????')}</option>
        <option value="10">${tr('October', '????')}</option><option value="11">${tr('November', '?
        ?????')}</option><option value="12">${tr('December', '????')}</option>
      </select>
      <select class="reception-form-input" id="rGregYear" style="flex:1;padding:0 8px"><option value
      ="">${tr('Year', '???')}</option>${Array.from({ length: 97 }, (_, i) => `<option value="${2026 - i}">${2026 -

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i}</option>`).join('')}</select>
    </div>
  </div>
  <div class="reception-form-group" style="flex:3;min-width:220px">
    <label class="reception-form-label">? ${tr('DOB (Hijri)', '???? ???? (???)')}</label>
    <div class="flex gap-4">
      <select class="reception-form-input" id="rHijriDay" style="flex:0.8;padding:0 8px"><option val
ue="">${tr('Day', '??')}</option>${Array.from({ length: 30 }, (_, i) => `<option value="${i + 1}">${i + 1}</o
ption>`).join('')}</select>
      <select class="reception-form-input" id="rHijriMonth" style="flex:1.5;padding:0 8px"><option v
alue="">${tr('Month', '??')}</option>
        <option value="1">????</option><option value="2">??</option><option value="3">??? ????</o
ption><option value="4">??? ????</option>
        <option value="5">??? ????</option><option value="6">??? ????</option><option value
="7">??</option><option value="8">???</option>
        <option value="9">???</option><option value="10">??</option><option value="11">?? ????
</option><option value="12">?? ????</option>
      </select>
      <select class="reception-form-input" id="rHijriYear" style="flex:1;padding:0 8px"><option valu
e="">${tr('Year', '??')}</option>${Array.from({ length: 101 }, (_, i) => `<option value="${1350 + i}">${1350
+ i}</option>`).join('')}</select>
    </div>
  </div>
  <div class="reception-form-group" style="flex:1;min-width:70px">
    <label class="reception-form-label">${tr('Age', '????')}</label>
    <input class="reception-form-input readonly" id="rAge" readonly style="padding:0 8px;text-align:
center">
  </div>
</div>
</div>

<!-- Step 2: Emergency Contact & Medical Baseline -->
<div class="reception-step-content" id="receptionStep2" style="display:none">
  <h4 class="font-title-sm text-primary mb-4">? ${tr('Medical Profile & Emergency Contacts', '???? ?
?? ???? ????')}</h4>

  <div class="reception-form-group mb-12">
    <label class="reception-form-label">? ${tr('Address', '????')}</label>
    <div class="reception-input-wrapper">
      <span class="material-symbols-outlined reception-input-icon">location_on</span>
      <input class="reception-form-input" id="rAddress" placeholder="${tr('City, Street...', '?????
?????...')}>
    </div>
  </div>

  <div class="flex gap-8 mb-12" style="flex-wrap:wrap">
    <div class="reception-form-group" style="flex:1;min-width:180px">
      <label class="reception-form-label">? ${tr('Emergency Contact Name', '?? ? ????')}</label>
      <div class="reception-input-wrapper">
        <span class="material-symbols-outlined reception-input-icon">contact_emergency</span>
        <input class="reception-form-input" id="rEmergencyName">
      </div>
    </div>
    <div class="reception-form-group" style="flex:1;min-width:140px">

```

```

<label class="reception-form-label">? ${tr('Emergency Phone', '???? ?????')}</label>
<div class="reception-input-wrapper">
  <span class="material-symbols-outlined reception-input-icon">phone</span>
  <input class="reception-form-input" id="rEmergencyPhone" type="tel">
</div>
</div>
</div>

<div class="flex gap-8 mb-12" style="flex-wrap:wrap">
  <div class="reception-form-group" style="flex:1;min-width:120px">
    <label class="reception-form-label">? ${tr('Blood Type', '???? ?????')}</label>
    <div class="reception-input-wrapper">
      <span class="material-symbols-outlined reception-input-icon">bloodtype</span>
      <select class="reception-form-input" id="rBloodType">
        <option value="">--</option>
        <option value="A+">A+</option><option value="A-">A-</option>
        <option value="B+">B+</option><option value="B-">B-</option>
        <option value="AB+">AB+</option><option value="AB-">AB-</option>
        <option value="O+">O+</option><option value="O-">O-</option>
      </select>
    </div>
  </div>
  <div class="reception-form-group" style="flex:2;min-width:200px">
    <label class="reception-form-label">?? ${tr('Allergies', '????????')}</label>
    <div class="reception-input-wrapper">
      <span class="material-symbols-outlined reception-input-icon">warning</span>
      <input class="reception-form-input" id="rAllergies" placeholder="${tr('Drug allergies, food al
ergies...', '?????? ?????? ?????...')}>
    </div>
  </div>
  <div class="reception-form-group" style="flex:2;min-width:200px">
    <label class="reception-form-label">? ${tr('Chronic Diseases', '???????? ?????')}</label>
    <div class="reception-input-wrapper">
      <span class="material-symbols-outlined reception-input-icon">clinical_notes</span>
      <input class="reception-form-input" id="rChronicDiseases" placeholder="${tr('Diabetes, Hyperte
nsion, Asthma...', '????? ???? ????...')}>
    </div>
  </div>
</div>
</div>

<!-- Step 3: Insurance & NPHIES Verification -->
<div class="reception-step-content" id="receptionStep3" style="display:none">
  <h4 class="font-title-sm text-primary mb-4">? ${tr('Insurance & NPHIES Verification', '???????? ?????
???? ??????? ?????')}</h4>

  <div class="reception-form-group mb-12">
    <label class="reception-form-label">? ${tr('Insurance Company', '???? ?????')}</label>
    <div class="reception-input-wrapper">
      <span class="material-symbols-outlined reception-input-icon">domain</span>
      <input class="reception-form-input" id="rInsuranceCompany" placeholder="${tr('e.g. Bupa, Tawuniy
a, MedGulf...', '????: ????? ?????????...')}>
    </div>
  </div>
</div>

```

```

<div class="reception-form-group mb-12">
  <label class="reception-form-label">? ${tr('Policy Number', '??? ??????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">key</span>
    <input class="reception-form-input" id="rInsurancePolicyNo">
  </div>
</div>

<div class="reception-form-group mb-16">
  <label class="reception-form-label">?? ${tr('Class', '????')}</label>
  <div class="reception-input-wrapper">
    <span class="material-symbols-outlined reception-input-icon">stars</span>
    <select class="reception-form-input" id="rInsuranceClass">
      <option value="">--</option>
      <option value="VIP">VIP</option>
      <option value="A">A (Gold)</option>
      <option value="B">B (Silver)</option>
      <option value="C">C (Bronze)</option>
    </select>
  </div>
</div>

<div class="bg-surface-container-low p-4 rounded-xl border border-outline-variant/30 mb-16">
  <button type="button" class="btn btn-secondary w-full" id="btnVerifyNphies" style="height:38px">?
  ${tr('Verify Coverage (NPHIES)', '???? ???? ?? ??????? ?????????')}</button>
  <div id="nphiesVerificationStatus" class="mt-3"></div>
</div>
</div>

<!-- Navigation Controls -->
<div class="flex justify-between items-center mt-6 pt-4 border-t border-outline-variant/30">
  <button type="button" class="btn btn-secondary" id="btnStepPrev" style="display:none;min-width:100px
">?? ${tr('Back', '????')}</button>
  <button type="button" class="btn btn-primary" id="btnStepNext" style="margin-right:auto; margin-left
:0;min-width:100px">
    ${tr('Next', '????')} ??
  </button>
  <button class="btn btn-primary" id="rSaveBtn" style="height:44px;font-size:15px;display:none;margin-
right:auto;margin-left:0;min-width:150px">
    ? ${tr('Save & Generate File', '??? ?????? ???')}
  </button>
</div>
</div>
</div>
<div id="rPatientBannerContainer"></div>
<div class="card glass-card-premium mt-16">
  <div class="card-title">? ${tr('Patient Queue', '????? ??????')}</div>
  <input class="search-filter" id="rSearch" placeholder="${tr('Search by name, ID, phone, file#...', '???
??????? ???????? ???????? ??? ??????...')}">
  <div id="rTable"></div>
</div>
<div class="card glass-card-premium mt-16" id="pendingPaymentCard">
  <div class="card-title">? ${tr('Pending Payment Orders (Lab / Radiology)', '????? ?????? ???????? ??????

```



```

(???? / ????)')}</div>
<div id="pendingPaymentTable"></div>
</div>
<div id="editPatientModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;background:
nd:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">
<div style="background:#fff;padding:30px;border-radius:16px;width:550px;max-width:90%;max-height:85vh;ov
erflow-y:auto;direction:rtl">
<h3 style="margin-bottom:16px">?? ${tr('Edit Patient', '???? ???? ????')}</h3>
<input type="hidden" id="editPID">
<div class="form-grid" style="gap:12px">
<div><label>${tr('Name (Arabic)', '???? ?????')}</label><input id="editPNameAr" class="form-cont
rol"></div>
<div><label>${tr('Name (English)', '???? ?????')}</label><input id="editPNameEn" class="form-
control"></div>
<div><label>${tr('National ID', '?? ?????')}</label><input id="editPNatId" class="form-control"></
div>
<div><label>${tr('Phone', '????')}</label><input id="editPPhone" class="form-control"></div>
<div><label>${tr('Nationality', '????')}</label><input id="editPNationality" class="form-control"
></div>
<div><label>${tr('DOB', '????')}</label><input id="editPDob" type="date" class="form-contro
l"></div>
</div>
<div style="display:flex;gap:10px;margin-top:16px">
<button class="btn btn-primary" onclick="saveEditPatient()" style="flex:1">? ${tr('Save', '???')}</b
utton>
<button class="btn" onclick="document.getElementById('editPatientModal').style.display='none'" style
="flex:1">? ${tr('Cancel', '????')}</button>
</div>
</div>
<div id="newInvoiceModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;backgroun
d:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">
<div style="background:#fff;padding:30px;border-radius:16px;width:450px;max-width:90%;direction:rtl">
<h3 style="margin-bottom:16px">? ${tr('New Service Invoice', '???? ????')}</h3>
<input type="hidden" id="invPID">
<input type="hidden" id="invPName">
<p id="invPLabel" style="font-weight:700;margin-bottom:12px"></p>
<div class="form-grid" style="gap:12px">
<div><label>${tr('Service Type', '?? ?????')}</label><select id="invServiceType" class="form-contr
ol" onchange="var v=this.value==='\u0643\u0634\u0641';document.getElementById('invDeptRow').style.display=v?'b
lock':'none';document.getElementById('invDoctorRow').style.display=v?'block':'none'">
<option value="???">? ${tr('Consultation', '???')}</option>
<option value="????">? ${tr('Laboratory', '????')}</option>
<option value="???">? ${tr('Radiology', '????')}</option>
<option value="????">? ${tr('Procedure', '????')}</option>
<option value="????">? ${tr('Medications', '????')}</option>
<option value="????">? ${tr('Surgery', '????')}</option>
<option value="????">??? ${tr('Nursing', '????')}</option>
<option value="???">? ${tr('Other', '???')}</option>
</select></div>
<div id="invDeptRow"><label>${tr('Department', '????')}</label><select id="invDept" class="form-con
trol">
${depts.map((d, i) => `<option value="${isArabic ? d : deptsEn[i]}">${isArabic ? d : deptsEn[i]}</
option>`).join('')}

```

```

        </select></div>
        <div id="invDoctorRow"><label>${tr('Doctor', '?????')}</label><select id="invDoctor" class="form-control">
            <option value="">${tr('Select Doctor', '??? ????')}</option>
            ${doctors || []}.map(d => `<option value="${escapeHTML(d.name)}">${escapeHTML(d.name)}</option>`)
            .join('')}
        </select></div>
        <div><label>${tr('Description', '????')}</label><input id="invDescription" class="form-control" placeholder="${tr('Service details', '????? ????')}"></div>
        <div><label>${tr('Amount (SAR)', '????? (?.)')}</label><input id="invAmount" type="number" step="0.01" class="form-control" placeholder="0.00"></div>
        <div class="flex gap-8" style="flex-wrap:wrap">
            <div style="flex:1"><label>?? ${tr('Discount (SAR)', '????? (?.)')}</label><input id="invDiscount" type="number" step="0.01" class="form-control" placeholder="0" value="0"></div>
            <div style="flex:2"><label>${tr('Discount Reason', '??? ????')}</label><input id="invDiscountReason" class="form-control" placeholder="${tr('e.g. Staff, Insurance, Coupon...', '????: ???? ???? ????...')}"></div>
        </div>
        <div><label>${tr('Payment Method', '????? ????')}</label><select id="invPayMethod" class="form-control">
            <option value="???">? ${tr('Cash', '???')}</option>
            <option value="?????">? ${tr('Card/POS', '????')}</option>
            <option value="????? ?????">? ${tr('Bank Transfer', '????? ????')}</option>
            <option value="?????">? ${tr('Tabby', '????')}</option>
            <option value="?????">? ${tr('Tamara', '????')}</option>
        </select></div>
    </div>
    <div style="display:flex;gap:10px;margin-top:16px">
        <button class="btn btn-primary" onclick="confirmNewInvoice()" style="flex:1">? ${tr('Create Invoice', '????? ????')}</button>
        <button class="btn" onclick="document.getElementById('newInvoiceModal').style.display='none'" style="flex:1">? ${tr('Cancel', '????')}</button>
    </div>
</div>
</div>

<!-- Quick Clinic Check-In Modal -->
<div id="quickCheckInModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;background:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">
    <div style="background:#fff;padding:30px;border-radius:16px;width:450px;max-width:90%;direction:rtl;box-shadow:0 10px 30px rgba(0,0,0,.2)">
        <h3 style="margin-bottom:16px;display:flex;align-items:center;gap:8px;color:var(--primary)">? ${tr('Quick Clinic Check-In', '????? ???? ???? ????')}</h3>
        <input type="hidden" id="qciPIId">
        <p id="qciPLabel" style="font-weight:700;margin-bottom:16px;background:rgba(var(--primary-rgb),0.05);padding:10px;border-radius:8px;border-right:4px solid var(--primary)"></p>

        <div class="reception-form-group mb-12">
            <label class="reception-form-label">? ${tr('Select Department', '????? / ?????')}</label>
            <div class="reception-input-wrapper">
                <span class="material-symbols-outlined reception-input-icon">medical_services</span>
                <select id="qciDept" class="reception-form-input" onchange="window.filterQuickDoctors(this.value)">
                    <option value="">-- ???? ???? --</option>

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        ${depts.map((d, i) => `<option value="${isArabic ? d : deptsEn[i]}">${isArabic ? d : deptsEn[i]}
</option>`).join('')
    </select>
</div>
</div>

<div class="reception-form-group mb-12">
    <label class="reception-form-label">? ${tr('Select Doctor', '????? ?????')}</label>
    <div class="reception-input-wrapper">
        <span class="material-symbols-outlined reception-input-icon">person</span>
        <select id="qciDoctor" class="reception-form-input">
            <option value="">${tr('Select Doctor', '???? ?????')}</option>
            ${((doctors || []).map(d => `<option value="${escapeHTML(d.name)}" data-dept="${escapeHTML(d.depa
rtment || ' ')}">${escapeHTML(d.name)}</option>`).join('')
        </select>
    </div>
</div>

<div class="reception-form-group mb-16">
    <label class="reception-form-label">? ${tr('Priority', '????? ?????')}</label>
    <div class="reception-input-wrapper">
        <span class="material-symbols-outlined reception-input-icon">priority_high</span>
        <select id="qciPriority" class="reception-form-input">
            <option value="????">? ${tr('Routine', '????')}</option>
            <option value="?????">? ${tr('Urgent', '?????')}</option>
            <option value="?????">? ${tr('Emergency', '?????')}</option>
        </select>
    </div>
</div>

<div style="display:flex;gap:10px;margin-top:20px">
    <button class="btn btn-primary" onclick="confirmQuickCheckIn()" style="flex:1;height:42px">? ${tr('C
onfirm Check-in', '????? ????? ?????')}</button>
    <button class="btn btn-secondary" onclick="document.getElementById('quickCheckInModal').style.displa
y='none'" style="flex:1;height:42px">? ${tr('Cancel', '?????')}</button>
</div>
</div>
</div>
`;

renderPatientTable(patients);
loadPendingPaymentOrders();

// Arabic to English transliteration (improved)
const commonNames = {
    '????': 'Mohammed', '????': 'Ahmed', '???': 'Ali', '???????': 'Abdullah', '??? ?????': 'Abdullah',
    '?????????': 'Abdulrahman', '??? ??????': 'Abdulrahman', '?????????': 'Abdulaziz', '??? ??????': 'Abdulazi
z',
    '?????????': 'Abdulmalik', '??? ??????': 'Abdulmalik', '?????????': 'Abdulrahim', '??? ??????': 'Abdulrahim'
,
    '???': 'Fahad', '????': 'Saad', '????': 'Khalid', '?????': 'Sultan', '????': 'Turki',
    '???': 'Saad', '????': 'Naif', '????': 'Bandar', '????': 'Faisal', '????': 'Salman',
    '????': 'Nasser', '????': 'Saleh', '????': 'Yousef', '?????????': 'Ibrahim', '?????????': 'Ibrahim',
    '???': 'Hassan', '????': 'Hussein', '???': 'Omar', '?????': 'Othman', '????': 'Talal',

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'????': 'Majed', '????': 'Waleed', '????': 'Mishal', '?????': 'Mansour', '????': 'Sarah',
'????': 'Noura', '?????': 'Fatimah', '?????': 'Aisha', '????': 'Mariam', '???': 'Hind',
'?????': 'Latifah', '?????': 'Munirah', '???': 'Haya', '?????': 'Lamia', '???': 'Reem',
'????': 'Dana', '???': 'Leen', '?????': 'Jawaher', '????': 'Badoor', '???????': 'Al-Anzi',
'???????': 'Al-Shammari', '???????': 'Al-Harbi', '?????????': 'Al-Qahtani', '?????????': 'Al-Ghamdi',
'?????????': 'Al-Dosari', '?????????': 'Al-Mutairi', '?????????': 'Al-Zahrani', '?????????': 'Al-Otaibi',
'?????????': 'Al-Subaie', '?????????': 'Al-Rashidi', '?????????': 'Al-Balawi', '?????????': 'Al-Juhani',
'?????????': 'Al-Saadi', '?????????': 'Al-Malki'
};

const arToEn = {
  '?': 'a', '?': 'a', '?': 'e', '?': 'aa', '?': 'b', '?': 't', '?': 'th',
  '?': 'j', '?': 'h', '?': 'kh', '?': 'd', '?': 'dh', '?': 'r', '?': 'z',
  '?': 's', '?': 'sh', '?': 's', '?': 'd', '?': 't', '?': 'dh',
  '?': 'a', '?': 'gh', '?': 'f', '?': 'q', '?': 'k', '?': 'l', '?': 'm',
  '?': 'n', '?': 'h', '?': 'w', '?': 'y', '?': 'ah', '?': 'a',
  '?': '"', '?': 'o', '?': 'e', '?': ' ', '?': 'a', '?': 'u', '?': 'i', '?': ' ', '?': ' ', '?': ' ', '?': ' '
};

document.getElementById('rNameAr').addEventListener('input', (e) => {
  const words = e.target.value.trim().split(/\s+/);
  const result = words.map(word => {
    // Check common names first
    if (commonNames[word]) return commonNames[word];
    // Handle ?? prefix
    let prefix = '';
    let w = word;
    if (w.startsWith('??') && w.length > 2) {
      prefix = 'Al-';
      w = w.substring(2);
    }
    let trans = '';
    for (let i = 0; i < w.length; i++) {
      const ch = w[i];
      if (arToEn[ch] !== undefined) {
        trans += arToEn[ch];
      } else if (ch.match(/[a-zA-Z0-9]/)) {
        trans += ch;
      }
    }
    if (trans.length > 0) {
      trans = trans.charAt(0).toUpperCase() + trans.slice(1);
    }
    return prefix + trans;
  }).filter(w => w.length > 0).join(' ');
  document.getElementById('rNameEn').value = result;
});

// Date conversion helpers
const gToH = (g) => {
  try { return new Intl.DateTimeFormat('ar-SA-u-ca-islamic', { year: 'numeric', month: '2-digit', day: '2-digit' }).format(g).replace(/??/g, ' ').trim(); } catch (e) { return ''; }
};

const hToG = (hY, hM, hD) => {
  // Proper Hijri to Gregorian conversion using tabular Islamic calendar
  const a = Math.floor((11 * hY + 3) / 30);

```

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const b = Math.floor(hY / 100);
const c = Math.floor(hY - 100 * b);
const d = Math.floor(b / 4);
const e1 = Math.floor((8 * (b + 1)) / 25) - 1;
// Calculate Julian Day Number from Hijri date
const jd = Math.floor(29.5001 * (hM - 1 + 12 * (hY - 1))) + hD + 1948439.5 - Math.floor((3 * (Math.floor((hY - 1) / 100) + 1)) / 4) + Math.floor((hY - 1) / 100) - Math.floor((hY - 1) / 400);
// Simpler and more reliable method: iterate from a known epoch
// Hijri epoch: July 16, 622 CE (Julian) = July 19, 622 CE (Gregorian)
const hijriEpoch = 1948439.5; // Julian Day for 1/1/1 Hijri
const monthDays = [30, 29, 30, 29, 30, 29, 30, 29, 30, 29, 30, 29];
// Leap year adds 1 day to month 12
const isLeapYear = (y) => (11 * y + 14) % 30 < 11;
let totalDays = 0;
for (let y = 1; y < hY; y++) {
    totalDays += isLeapYear(y) ? 355 : 354;
}
for (let m = 1; m < hM; m++) {
    totalDays += monthDays[m - 1];
}
if (hM === 12 && isLeapYear(hY)) totalDays += 1; // already counted
totalDays += hD - 1;
// Hijri epoch in JavaScript Date: July 19, 622 CE
const epochMs = new Date(622, 6, 19).getTime();
const gDate = new Date(epochMs + totalDays * 86400000);
// Fix JS Date quirk for years < 100
if (gDate.getFullYear() < 100) gDate.setFullYear(gDate.getFullYear());
const age = Math.abs(new Date(Date.now() - gDate.getTime()).getUTCFullYear() - 1970);
const y = gDate.getFullYear();
const m = String(gDate.getMonth() + 1).padStart(2, '0');
const day = String(gDate.getDate()).padStart(2, '0');
return { gDate: `${y}-${m}-${day}`, age };
};

// Gregorian dropdowns -> convert to Hijri
const gregChange = () => {
    const gY = parseInt(document.getElementById('rGregYear').value);
    const gM = parseInt(document.getElementById('rGregMonth').value);
    const gD = parseInt(document.getElementById('rGregDay').value);
    if (!gY || !gM || !gD) return;
    const dob = new Date(gY, gM - 1, gD);
    const diff = Date.now() - dob.getTime();
    document.getElementById('rAge').value = Math.abs(new Date(diff).getUTCFullYear() - 1970);
    const hStr = gToH(dob);
    if (hStr) {
        const parts = hStr.replace(/^[0-9]/g, '').split('/');
        if (parts.length === 3) {
            document.getElementById('rHijriDay').value = parseInt(parts[0]);
            document.getElementById('rHijriMonth').value = parseInt(parts[1]);
            document.getElementById('rHijriYear').value = parseInt(parts[2]);
        }
    }
};
document.getElementById('rGregYear').addEventListener('change', gregChange);

```

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document.getElementById('rGregMonth').addEventListener('change', gregChange);
document.getElementById('rGregDay').addEventListener('change', gregChange);

// Hijri dropdowns -> convert to Gregorian
const hijriChange = () => {
  const hY = parseInt(document.getElementById('rHijriYear').value);
  const hM = parseInt(document.getElementById('rHijriMonth').value);
  const hD = parseInt(document.getElementById('rHijriDay').value);
  if (!hY || !hM || !hD) return;
  const res = hToG(hY, hM, hD);
  // Populate Gregorian dropdowns
  const gd = new Date(res.gDate);
  document.getElementById('rGregDay').value = gd.getDate();
  document.getElementById('rGregMonth').value = gd.getMonth() + 1;
  document.getElementById('rGregYear').value = gd.getFullYear();
  document.getElementById('rAge').value = res.age;
};
document.getElementById('rHijriYear').addEventListener('change', hijriChange);
document.getElementById('rHijriMonth').addEventListener('change', hijriChange);
document.getElementById('rHijriDay').addEventListener('change', hijriChange);

// --- REDESIGN STEPPER WIZARD EVENT HANDLERS ---
let currentStep = 1;

const updateStepperUI = () => {
  // Hide all steps
  document.getElementById('receptionStep1').style.display = 'none';
  document.getElementById('receptionStep2').style.display = 'none';
  document.getElementById('receptionStep3').style.display = 'none';

  // Show current step
  document.getElementById(`receptionStep${currentStep}`).style.display = 'block';

  // Toggle navigation buttons
  document.getElementById('btnStepPrev').style.display = currentStep === 1 ? 'none' : 'block';
  document.getElementById('btnStepNext').style.display = currentStep === 3 ? 'none' : 'block';
  document.getElementById('rSaveBtn').style.display = currentStep === 3 ? 'block' : 'none';

  // Update Stepper Header indicators
  for (let s = 1; s <= 3; s++) {
    const indicator = document.getElementById(`stepIndicator${s}`);
    const line = document.getElementById(`stepLine${s}`);
    const numSpan = document.getElementById(`stepNum${s}`);
    const labelSpan = document.getElementById(`stepLabel${s}`);

    if (s === currentStep) {
      if (indicator) {
        indicator.className = 'flex items-center gap-2 step-indicator active';
      }
      if (numSpan) {
        numSpan.className = 'step-number-badge active';
        numSpan.innerHTML = s;
      }
      if (labelSpan) {

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        labelSpan.className = 'text-xs font-bold text-primary';
    }
} else {
    if (indicator) {
        indicator.className = s < currentStep ? 'flex items-center gap-2 step-indicator completed' : 'flex i
tems-center gap-2 step-indicator';
    }
    if (numSpan) {
        numSpan.className = s < currentStep ? 'step-number-badg completed' : 'step-number-badg pending';
        numSpan.innerHTML = s < currentStep ? '?' : s;
    }
    if (labelSpan) {
        labelSpan.className = 'text-xs text-on-surface-variant font-medium';
    }
}

if (s < 3) {
    const stepLine = document.getElementById(`stepLine${s}`);
    if (stepLine) {
        stepLine.style.background = s < currentStep ? '#10b981' : 'var(--border-color,#e5e7eb)';
    }
}
}
};

document.getElementById('btnStepNext').addEventListener('click', () => {
    if (currentStep === 1) {
        // Validate Step 1 before proceeding
        const nameAr = document.getElementById('rNameAr').value.trim();
        const nameEn = document.getElementById('rNameEn').value.trim();
        if (!nameAr && !nameEn) { showToast(tr('Enter patient name', '???? ??? ??????'), 'error'); return; }

        const natId = document.getElementById('rNatId').value.trim();
        if (!natId) { showToast(tr('National ID is required', '??? ?????? ??????? ??????'), 'error'); return; }
        if (!/^[12]\d{9}$/.test(natId)) {
            showToast(tr('National ID must be exactly 10 digits starting with 1 or 2', '??? ?? ?????? ??? ?????? ??
10 ?????? ?????? ?? 1 ?? 2'), 'error');
            return;
        }

        const phone = document.getElementById('rPhone').value.trim();
        if (!phone) { showToast(tr('Phone number is required', '??? ?????? ??????'), 'error'); return; }
        if (!/^(05|9665)\d{8}$/.test(phone)) {
            showToast(tr('Phone number must start with 05 or 9665 followed by 8 digits', '??? ?????? ??? ?? ????? ?
? 05 ?? 9665 ?????? ?? 8 ??????'), 'error');
            return;
        }
    }
}

if (currentStep < 3) {
    currentStep++;
    updateStepperUI();
}
});

```

```

document.getElementById('btnStepPrev').addEventListener('click', () => {
  if (currentStep > 1) {
    currentStep--;
    updateStepperUI();
  }
});

// --- EMPI DUPLICATE CHECK ---
const checkDuplicates = () => {
  const natIdVal = document.getElementById('rNatId').value.trim();
  const phoneVal = document.getElementById('rPhone').value.trim();
  const dupCard = document.getElementById('empiDuplicateAlert');
  if (!dupCard) return;

  let dupPatient = null;
  if (natIdVal && natIdVal.length >= 5) {
    dupPatient = patients.find(p => p.national_id === natIdVal);
  }
  if (!dupPatient && phoneVal && phoneVal.length >= 7) {
    dupPatient = patients.find(p => p.phone === phoneVal);
  }

  if (dupPatient) {
    dupCard.style.display = 'block';
    dupCard.innerHTML = `
      <div style="background:rgba(220,38,38,0.1); border:1px solid rgba(220,38,38,0.2); color:#ef4444; font-size:12px; padding:12px; border-radius:8px; display:flex; flex-direction:column; gap:4px">
        <div style="font-weight:700">?? ${tr('Duplicate File Warning (EMPI)', '?????: ??? ??? ??? ?????????')}</div>
        <div>${tr('A patient record already exists with this ID or phone number:', '???? ?????? ??? ????? ??? ? ????? ?? ??????:')}</div>
        <div style="font-family:monospace; margin-top:4px; font-weight:700">${escapeHTML(isArabic ? (dupPatient.name_ar || dupPatient.name_en) : (dupPatient.name_en || dupPatient.name_ar))} (${tr('MRN', '?????')}: ${escapeHTML(dupPatient.mrn || dupPatient.file_number)})</div>
      </div>
    `;
  } else {
    dupCard.style.display = 'none';
  }
};

document.getElementById('rNatId').addEventListener('input', checkDuplicates);
document.getElementById('rPhone').addEventListener('input', checkDuplicates);

// --- NPHIES MOCK ELIGIBILITY CHECK ---
const btnVerify = document.getElementById('btnVerifyNphies');
if (btnVerify) {
  btnVerify.addEventListener('click', async () => {
    const insCompany = document.getElementById('rInsuranceCompany').value.trim();
    const policyNo = document.getElementById('rInsurancePolicyNo').value.trim();
    const statusDiv = document.getElementById('nphiesVerificationStatus');
    if (!statusDiv) return;
  });
}

```



```

    if (!insCompany || !policyNo) {
        showToast(tr('Please enter Insurance Company and Policy Number', '???? ????? ??? ??? ?????? ??? ????
?????'), 'error');
        return;
    }

    btnVerify.disabled = true;
    btnVerify.innerHTML = `? ${tr('Verifying with NPHIES...', '???? ????????? ?? ?????...')}`;
    statusDiv.innerHTML = `<div style="color:var(--text-dim);font-family:monospace;font-size:12px">? Connect
ing to NPHIES Gateway...</div>`;

    // Simulate network request delay (1.2s)
    await new Promise(r => setTimeout(r, 1200));

    btnVerify.disabled = false;
    btnVerify.innerHTML = `? ${tr('Verify Coverage (NPHIES)', '???? ??? ? ?????? ?????????')}`;

    // Generate a mock eligibility response based on policy number
    const isEligible = !policyNo.toLowerCase().includes('reject') && !policyNo.includes('999');
    if (isEligible) {
        statusDiv.innerHTML = `
            <div style="background:rgba(16,185,129,0.1); border:1px solid rgba(16,185,129,0.2); color:#059669; font-size:12px; padding:12px; border-radius:8px; display:flex; flex-direction:column; gap:4px">
                <div style="font-weight:700">? ${tr('NPHIES Coverage Active', '?? ??????: ?????? ????????? ???')}</div>
                <div><strong>${tr('Class', '????')}</strong> ${escapeHTML(document.getElementById('rInsuranceClass').value || 'A')}</div>
                <div><strong>${tr('Copayment', '???? ?????')}</strong> 10% (Max 100 SAR)</div>
                <div><strong>${tr('Limit', '???? ?????')}</strong> 100,000 SAR</div>
            </div>
        `;
        showToast(tr('NPHIES Coverage Approved!', '?? ?????? ??? ???? ??????!'));
    } else {
        statusDiv.innerHTML = `
            <div style="background:rgba(220,38,38,0.1); border:1px solid rgba(220,38,38,0.2); color:#ef4444; font-size:12px; padding:12px; border-radius:8px; display:flex; flex-direction:column; gap:4px">
                <div style="font-weight:700">? ${tr('NPHIES Eligibility Rejected', '??? : ?????? ????????? ??? ????
????')}</div>
                <div>${tr('The policy number is invalid or has expired.', '??? ???????? ??? ???? ? ???? ??????
.')}</div>
            </div>
        `;
        showToast(tr('NPHIES Coverage Rejected', '???? ??? ?????? ?????????'), 'error');
    }
});

const userPerms = currentUser?.permissions ? currentUser.permissions.split(',') : [];
const isAdmin = currentUser?.role === 'Admin';
const canRegister = isAdmin || userPerms.includes('patients') || userPerms.includes('1');

const saveBtn = document.getElementById('rSaveBtn');
if (saveBtn && !canRegister) {
    saveBtn.disabled = true;
}

```

```

saveBtn.title = tr('You do not have permission to register patients', '??? ??? ???? ???? ????');
saveBtn.classList.add('btn-disabled');
saveBtn.style.opacity = '0.5';
saveBtn.style.cursor = 'not-allowed';
}

document.getElementById('rSaveBtn').addEventListener('click', async () => {
  if (!canRegister) {
    showToast(tr('Access Denied: You do not have permission to register patients', '?? ?? ?????: ?? ?? ????
???? ???? ????'), 'error');
    return;
  }
  const nameAr = document.getElementById('rNameAr').value.trim();
  const nameEn = document.getElementById('rNameEn').value.trim();
  if (!nameAr && !nameEn) { showToast(tr('Enter patient name', '???? ?? ????'), 'error'); return; }

  // Saudi Compliance Validations (SDAIA / PDPL & NDMO Data Integrity)
  const natId = document.getElementById('rNatId').value.trim();
  if (natId) {
    if (!/^[12]\d{9}$/.test(natId)) {
      showToast(tr('National ID must be exactly 10 digits starting with 1 or 2', '?? ? ???? ? ???? ?
10 ???? ???? ? 1 ? 2'), 'error');
      return;
    }
  } else {
    showToast(tr('National ID is required', '?? ???? ???? ????'), 'error');
    return;
  }

  const phone = document.getElementById('rPhone').value.trim();
  if (phone) {
    if (!/^(05|9665)\d{8}$/.test(phone)) {
      showToast(tr('Phone number must start with 05 or 9665 followed by 8 digits', '?? ????? ? ? ? ? ?
? 05 ? 9665 ????? ? 8 ????'), 'error');
      return;
    }
  } else {
    showToast(tr('Phone number is required', '?? ????? ????'), 'error');
    return;
  }

  try {
    await API.post('/api/patients', {
      name_ar: nameAr, name_en: nameEn,
      national_id: document.getElementById('rNatId').value,
      nationality: document.getElementById('rNationality').value,
      gender: document.getElementById('rGender').value,
      phone: document.getElementById('rPhone').value,
      blood_type: document.getElementById('rBloodType').value,
      allergies: document.getElementById('rAllergies').value,
      chronic_diseases: document.getElementById('rChronicDiseases').value,
      emergency_contact_name: document.getElementById('rEmergencyName').value,
      emergency_contact_phone: document.getElementById('rEmergencyPhone').value,
      address: document.getElementById('rAddress').value,
    });
  } catch (error) {
    console.error('API Error: ', error);
    showToast(tr('Registration failed. Please try again.', '??? ???? ????'), 'error');
  }
});

```

```

        insurance_company: document.getElementById('rInsuranceCompany').value,
        insurance_policy_number: document.getElementById('rInsurancePolicyNo').value,
        insurance_class: document.getElementById('rInsuranceClass').value,
        dob: (document.getElementById('rGregYear').value && document.getElementById('rGregMonth').value && document.getElementById('rGregDay').value) ? `${document.getElementById('rGregYear').value}-${String(document.getElementById('rGregMonth').value).padStart(2, '0')}-${String(document.getElementById('rGregDay').value).padStart(2, '0')}` : '',
        dob_hijri: (document.getElementById('rHijriYear').value && document.getElementById('rHijriMonth').value && document.getElementById('rHijriDay').value) ? `${document.getElementById('rHijriYear').value}/${String(document.getElementById('rHijriMonth').value).padStart(2, '0')}/${String(document.getElementById('rHijriDay').value).padStart(2, '0')}` : ''
    });
    showToast(tr('Patient saved!', '?? ??? ??????'));
    await navigateTo(1);
} catch (e) { showToast(tr('Error saving patient', '??? ?? ??? ??????'), 'error'); }
});

document.getElementById('rSearch').addEventListener('input', (e) => {
    const txt = e.target.value.toLowerCase();
    let visibleCount = 0;
    document.querySelectorAll('#rTable tbody tr').forEach(row => {
        const match = row.textContent.toLowerCase().includes(txt);
        row.style.display = match ? '' : 'none';
        if (match) visibleCount++;
    });

    let noMatchDiv = document.getElementById('rTableNoMatch');
    if (visibleCount === 0) {
        if (!noMatchDiv) {
            noMatchDiv = document.createElement('div');
            noMatchDiv.id = 'rTableNoMatch';
            noMatchDiv.className = 'empty-state-card';
            noMatchDiv.innerHTML = `<div class="empty-state-icon">?</div><div>${tr('No matching patients found', '
?? ??? ?????? ??? ????? ??????')}</div>`;
            document.getElementById('rTable').appendChild(noMatchDiv);
        }
        const table = document.querySelector('#rTable table');
        if (table) table.style.display = 'none';
    } else {
        if (noMatchDiv) noMatchDiv.remove();
        const table = document.querySelector('#rTable table');
        if (table) table.style.display = '';
    }
});
}

function renderPatientTable(patients) {
    if (!patients || patients.length === 0) {
        document.getElementById('rTable').innerHTML = `
        <div class="empty-state-card">
            <div class="empty-state-icon">?</div>
            <div>${tr('No patients registered yet', '?? ???? ???? ?????? ???')}</div>
        </div>
        `;
    }
}

```

```

    return;
  }
  const headers = [tr('MRN/File#', '??? ?????'), tr('Name', '?????'), tr('ID', '??????'), tr('Phone', '??????'),
  ), tr('Insurance', '????????'), tr('File Opening Fee', '???? ??? ?????'), tr('Date/Time', '????????/?????'), tr(
  'Status', '??????'), tr('Actions', '????????')];
  const rows = patients.map(p => ({
    cells: [
      p.mrn || p.file_number,
      rawHtml(`${p.gender === '???' ? '?' : '?'} ${escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en
      || p.name_ar))}${p.allergies ? ' <span style="color:#ef4444;font-weight:700" title="' + escapeHTML(p.allergie
      s) + '>??</span>' : ''}`),
      p.national_id,
      p.phone,
      p.insurance_company ? rawHtml(`<span style="font-size:11px">${escapeHTML(p.insurance_company)}${p.insura
      nce_class ? ' (' + escapeHTML(p.insurance_class) + ')' : ''}</span>`) : '-',
      rawHtml(`<span class="payment-badge paid" style="background:rgba(16,185,129,0.15);color:#10b981;border:1
      px solid rgba(16,185,129,0.3);padding:2px 8px;border-radius:4px;font-weight:bold;font-size:11px">${tr('Free',
      '????')} ?</span>`),
      p.created_at ? new Date(p.created_at).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' }
      ) : '-',
      statusBadge(p.status)
    ],
    id: p.id, raw: p
  }));
  document.getElementById('rTable').innerHTML = makeTable(headers, rows, (row) =>
  `<button class="btn btn-sm btn-info" onclick="selectPatient(${safeId(row.id)})" title="${tr('View Banner',
  '??? ??????')}">??</button> <button class="btn btn-sm" onclick="showQuickCheckInModal(${safeId(row.id)},'${js
  Str(row.raw.name_ar || row.raw.name_en || '')}'" title="${tr('Quick Clinic Check-in', '????? ???? ??????')}"
  style="background:rgba(var(--primary-rgb),0.1);color:var(--primary);margin:0 2px">??</button> <button class="b
  tn btn-sm" onclick="editPatient(${safeId(row.id)})" title="${tr('Edit', '????')}>??</button> <button class="
  btn btn-sm btn-success" onclick="showNewInvoiceModal(${safeId(row.id)},'${jsStr(row.raw.name_ar || row.raw.nam
  e_en || '')}'" title="${tr('Invoice', '?????')}">??</button> <button class="btn btn-danger btn-sm" onclick="d
  eletePatient(${safeId(row.id)})" title="${tr('Delete', '???')}">??</button>`
  );
}

window.selectPatient = async (id) => {
  try {
    const patients = await API.get('/api/patients');
    const p = patients.find(x => x.id === id);
    if (!p) return showToast(tr('Patient not found', '?????? ??? ?????'), 'error');

    // Audit Event log feedback to user
    showToast(tr('Selected patient: ', '?? ?????? ??????: ') + (isArabic ? (p.name_ar || p.name_en) : (p.name_
    en || p.name_ar)));

    // Render Patient Banner with minimized PHI (Masking National ID)
    const bannerEl = document.getElementById('rPatientBannerContainer');
    if (bannerEl) {
      const maskedNatId = p.national_id ? p.national_id.substring(0, 3) + '*****' + p.national_id.substring(8)
      : '-';
      const ageText = (p.dob) ? Math.abs(new Date(Date.now() - new Date(p.dob).getTime()).getUTCFullYear() - 1
      970) : '-';
      bannerEl.innerHTML = `

```

```

<div class="patient-banner glass-card-premium mb-16 p-16" style="border: 2px solid var(--accent); border-radius: 12px; background: var(--card-bg); display: flex; flex-direction: column; gap: 12px">
  <div style="display: flex; justify-content: space-between; align-items: center; border-bottom: 1px solid var(--border); padding-bottom: 8px">
    <h4 style="margin: 0; color: var(--accent); display: flex; align-items: center; gap: 8px">
      <span>${p.gender === '???' ? '?' : ''}</span>
      <span>${escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</span>
    </h4>
    <span class="badge" style="background: var(--accent-glow); color: var(--accent); border: 1px solid var(--accent)">
      MRN/File: ${escapeHTML(p.mrn || p.file_number)}
    </span>
  </div>
  <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(150px, 1fr)); gap: 12px; font-size: 13px">
    <div><strong>${tr('Age', '????')}</strong> ${escapeHTML(ageText)}</div>
    <div><strong>${tr('Gender', '????')}</strong> ${escapeHTML(p.gender)}</div>
    <div><strong>${tr('Blood Type', '???? ????')}</strong> ${p.blood_type ? `<span class="badge" style="background: #dc2626; color: #fff">${escapeHTML(p.blood_type)}</span>` : ''}</div>
    <div><strong>${tr('National ID (Min PHI)', '?? ???? (???)')}</strong> <span style="font-family: monospace">${escapeHTML(maskedNatId)}</span></div>
  </div>
  <div style="display: flex; gap: 16px; font-size: 13px; flex-wrap: wrap">
    <div style="flex: 1; min-width: 200px; background: rgba(220,38,38,0.05); padding: 8px; border-radius: 6px; border-right: 4px solid #ef4444">
      <strong style="color: #ef4444">?? ${tr('Allergies', '????????')}</strong>
      <span>${p.allergies ? escapeHTML(p.allergies) : tr('No known allergies', '?? ???? ?????? ??????')}</span>
    </div>
    <div style="flex: 1; min-width: 200px; background: rgba(13,148,136,0.05); padding: 8px; border-radius: 6px; border-right: 4px solid var(--accent)">
      <strong style="color: var(--accent)">? ${tr('Chronic Diseases', '?????? ?????')}</strong>
      <span>${p.chronic_diseases ? escapeHTML(p.chronic_diseases) : tr('None', '?? ?????')}</span>
    </div>
  </div>
  <div style="display: flex; justify-content: space-between; font-size: 11px; color: var(--text-dim)">
    <span>? ${tr('Facility Context', '???? ?????')}: M-${escapeHTML(p.tenant_id || currentUser.tenantId)}</span>
    <div style="display: flex; gap: 8px">
      <button class="btn btn-sm btn-info" onclick="printPatientWristband(${safeId(p.id)})" style="padding: 2px 8px">?? ${tr('Print Wristband', '???? ?????')}</button>
      <button class="btn btn-sm btn-secondary" onclick="document.getElementById('rPatientBannerContainer').innerHTML=''" style="padding: 2px 8px">${tr('Clear Banner', '???? ?????')}</button>
    </div>
  </div>
</div>
`;
}
} catch (e) {
  showToast(tr('Error loading patient banner', '?? ? ? ???? ?????? ??????'), 'error');
}
};

window.filterQuickDoctors = (dept) => {

```

```

const select = document.getElementById('qciDoctor');
if (!select) return;

const firstOption = select.options[0];
select.innerHTML = '';
select.appendChild(firstOption);

if (window.currentDoctorsList) {
  window.currentDoctorsList.forEach(d => {
    if (!dept || d.department === dept) {
      const opt = document.createElement('option');
      opt.value = d.name;
      opt.textContent = d.name;
      opt.setAttribute('data-dept', d.department || '');
      select.appendChild(opt);
    }
  });
}

};

window.showQuickCheckInModal = (patientId, patientName) => {
  document.getElementById('qciPid').value = patientId;
  document.getElementById('qciLabel').innerHTML = `? ??????: ${escapeHTML(patientName)}`;
  document.getElementById('qciDept').value = '';
  document.getElementById('qciDoctor').value = '';
  document.getElementById('qciPriority').value = '????';
  document.getElementById('quickCheckInModal').style.display = 'flex';
};

window.confirmQuickCheckIn = async () => {
  const patientId = document.getElementById('qciPid').value;
  const dept = document.getElementById('qciDept').value;
  const doctorName = document.getElementById('qciDoctor').value;
  const priority = document.getElementById('qciPriority').value;

  if (!dept) return showToast(tr('Please select clinic', '????? ?????? ??????'), 'error');

  try {
    await API.put(`/api/patients/${patientId}`, {
      department: dept,
      doctor: doctorName || null,
      status: 'Waiting'
    });

    await API.post('/api/appointments', {
      patient_id: patientId,
      doctor_name: doctorName || '???? ??????',
      department: dept,
      appointment_date: new Date().toISOString().split('T')[0],
      appointment_time: new Date().toString().split(' ')[0].substring(0, 5),
      status: 'Checked-in',
      notes: `????? ???? ???? ?? ????????? - ?????????: ${priority}`
    });
  }
};

```

```

    showToast(tr('Patient Checked-in successfully', '?? ????? ???? ?????? ??????? ??????'));
    document.getElementById('quickCheckInModal').style.display = 'none';
    navigateTo(1);
  } catch (e) {
    showToast(tr('Error during check-in', '??? ?????? ?????? ??????? ???????'), 'error');
  }
};

window.printPatientWristband = (patientId) => {
  const p = window.currentPatientsList ? window.currentPatientsList.find(x => x.id === patientId) : null;
  if (!p) return showToast(tr('Patient not found', '?????? ??? ??????'), 'error');

  let printArea = document.getElementById('wristbandPrintArea');
  if (!printArea) {
    printArea = document.createElement('div');
    printArea.id = 'wristbandPrintArea';
    printArea.style.display = 'none';
    document.body.appendChild(printArea);
  }

  printArea.innerHTML = `
    <!-- Left: Hospital Branding -->
    <div style="display: flex; flex-direction: column; align-items: center; justify-content: center; width: 14
0px; border-left: 1px solid #ddd; padding-left: 8px; text-align: center">
      <span style="font-size: 20px">?</span>
      <span style="font-size: 11px; font-weight: bold; color: #000">???? ?????? ??????</span>
      <span style="font-size: 9px; color: #555">Jumana Medical</span>
    </div>

    <!-- Middle: Patient Demographic Info -->
    <div style="flex: 1; display: flex; flex-direction: column; justify-content: center; padding: 0 10px; text
-align: right">
      <div style="font-size: 11px; font-weight: bold; margin-bottom: 2px">
        ?????: ${escapeHTML(p.name_ar || '')}
      </div>
      <div style="font-size: 10px; color: #333; margin-bottom: 2px">
        Name: ${escapeHTML(p.name_en || '')}
      </div>
      <div style="display: flex; gap: 15px; font-size: 9px; color: #444">
        <span><strong>????? (MRN):</strong> ${escapeHTML(p.mrn || p.file_number)}</span>
        <span><strong>?????:</strong> ${escapeHTML(p.gender || '-')}</span>
        <span><strong>???????</strong> ${p.dob ? new Date(p.dob).toLocaleDateString('en-US') : '-'}</span>
        <span><strong>???????</strong> ${escapeHTML(p.blood_type || '-')}</span>
      </div>
    </div>

    <!-- Right: Active Barcode and print timestamp -->
    <div style="display: flex; flex-direction: column; align-items: center; justify-content: center; width: 16
0px; padding-right: 8px; border-right: 1px solid #ddd">
      <svg id="wristbandBarcode" style="max-height: 40px; width: 140px"></svg>
      <span style="font-size: 8px; color: #666; margin-top: 2px">
        ${new Date().toLocaleString('ar-SA', {dateStyle:'short', timeStyle:'short'})}
      </span>
    </div>
  `

```

```

`;

setTimeout(() => {
  if (window.JsBarcode) {
    JsBarcode("#wristbandBarcode", p.mrn || p.file_number || '000000', {
      format: "CODE128",
      width: 1.2,
      height: 25,
      displayValue: true,
      fontSize: 8,
      margin: 0
    });
  }

  document.body.classList.add('printing-wristband');
  window.print();
  document.body.classList.remove('printing-wristband');
}, 100);
};

window.deletePatient = async (id) => {
  if (!confirm(tr('Delete this patient and all records?', '??? ??? ?????? ????? ?????????'))) return;
  try {
    await API.del(`/api/patients/${id}`);
    showToast(tr('Patient deleted', '?? ??? ??????'));
    await navigateTo(1);
  } catch (e) { showToast(tr('Error deleting', '??? ?? ??????'), 'error'); }
};

window.editPatient = async function (id) {
  try {
    const patients = await API.get('/api/patients');
    const p = patients.find(x => x.id === id);
    if (!p) return showToast(tr('Patient not found', '?????? ??? ??????'), 'error');
    document.getElementById('editPID').value = id;
    document.getElementById('editPNameAr').value = p.name_ar || '';
    document.getElementById('editPNameEn').value = p.name_en || '';
    document.getElementById('editPNatId').value = p.national_id || '';
    document.getElementById('editPPhone').value = p.phone || '';
    document.getElementById('editPNationality').value = p.nationality || '';
    document.getElementById('editPDob').value = p.dob || '';
    document.getElementById('editPatientModal').style.display = 'flex';
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.saveEditPatient = async function () {
  const id = document.getElementById('editPID').value;
  try {
    await API.put(`/api/patients/${id}`, {
      name_ar: document.getElementById('editPNameAr').value,
      name_en: document.getElementById('editPNameEn').value,
      national_id: document.getElementById('editPNatId').value,
      phone: document.getElementById('editPPhone').value,
      nationality: document.getElementById('editPNationality').value,
      dob: document.getElementById('editPDob').value
    });
  }
};

```



```

});
document.getElementById('editPatientModal').style.display = 'none';
showToast(tr('Patient updated!', '?? ????? ?????? ??????!'));
await navigateTo(1);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.showNewInvoiceModal = function (id, name) {
  document.getElementById('invPID').value = id;
  document.getElementById('invPName').value = name;
  document.getElementById('invPLabel').textContent = name;
  document.getElementById('invDescription').value = '';
  document.getElementById('invAmount').value = '';
  if (document.getElementById('invDiscount')) document.getElementById('invDiscount').value = '0';
  if (document.getElementById('invDiscountReason')) document.getElementById('invDiscountReason').value = '';
  document.getElementById('newInvoiceModal').style.display = 'flex';
};

window.confirmNewInvoice = async function () {
  const id = document.getElementById('invPID').value;
  const name = document.getElementById('invPName').value;
  const amount = parseFloat(document.getElementById('invAmount').value);
  const serviceType = document.getElementById('invServiceType').value;
  const discount = parseFloat(document.getElementById('invDiscount')?.value) || 0;
  const discountReason = document.getElementById('invDiscountReason')?.value || '';
  if (!amount || amount <= 0) return showToast(tr('Enter amount', '???? ??????'), 'error');
  if (discount > amount) return showToast(tr('Discount cannot exceed amount', '????? ?? ????? ?? ?????? ??????'), 'error');
  try {
    let desc = document.getElementById('invDescription').value;
    if (serviceType === '???') {
      const dept = document.getElementById('invDept').value;
      const doctor = document.getElementById('invDoctor').value;
      const parts = [dept, doctor, desc].filter(x => x);
      desc = parts.join(' - ');
      await API.put('/api/patients/' + id, { department: dept });
    }
    const finalAmount = amount - discount;
    await API.post('/api/invoices', {
      patient_id: id, patient_name: name,
      total: finalAmount,
      description: desc + (discount > 0 ? ' (??? : ' + discount + ' SAR' + (discountReason ? ' - ' + discountReason : '')) : '' : ''),
      service_type: serviceType,
      payment_method: document.getElementById('invPayMethod').value,
      discount: discount,
      discount_reason: discountReason
    });
    document.getElementById('newInvoiceModal').style.display = 'none';
    showToast(tr('Invoice created!', '?? ?????? ??????????!') + (discount > 0 ? ' (' + tr('Discount', '???') + ': ' + discount + ') ' : ''));
    await navigateTo(1);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== APPOINTMENTS =====

```

```

async function renderAppointments(el) {
  const userPerms = currentUser?.permissions ? currentUser.permissions.split(',') : [];
  const isAdmin = currentUser?.role === 'Admin';
  const canBook = isAdmin || userPerms.includes('appointments') || userPerms.includes('2');

  // Show premium skeletons instantly
  el.innerHTML = `
    <div class="page-title">? ${tr('Appointments', '???????')}</div>
    <div class="split-layout">
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
    </div>
  `;

  let appts = [];
  let emps = [];
  let patients = [];
  try {
    [appts, emps, patients] = await Promise.all([
      API.get('/api/appointments'),
      API.get('/api/employees?role=Doctor'),
      API.get('/api/patients')
    ]);
  } catch (e) {
    el.innerHTML = `
      <div class="page-title">? ${tr('Appointments', '???????')}</div>
      <div class="error-card-premium">
        <div class="error-card-icon">??</div>
        <h3>${tr('Failed to load appointments data', '??? ?? ????? ?????? ?????????')}</h3>
        <p>${escapeHTML(e.message || e)}</p>
        <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
    button>
      </div>
    `;
    return;
  }

  const appointmentsTableHtml = appts.length === 0 ? `
    <div class="empty-state-card">
      <div class="empty-state-icon">?</div>
      <div>${tr('No appointments booked yet', '?? ????? ?????? ?????? ??')}</div>
    </div>
  ` : makeTable(
    [tr('Patient', '?????'), tr('Doctor', '?????'), tr('Dept', '?????'), tr('Date', '???????'), tr('Time', '
    ?????'), tr('Status', '?????'), tr('Delete', '???')],
    appts.map(a => ({ cells: [a.patient_name, a.doctor_name, a.department, a.appt_date, a.appt_time, statusBad

```

```

ge(a.status)], id: a.id })),
  (row) => `<button class="btn btn-sm" onclick="checkInPatient(${safeId(row.id)})" title="${tr('Check-in', '
???? ????')}" style="background:#e8f5e9;color:#2e7d32;margin:0 2px">?</button><button class="btn btn-sm" oncl
ick="markNoShow(${safeId(row.id)})" title="${tr('No-Show', '????')}" style="background:#fff3e0;color:#e65100;
margin:0 2px">??</button><button class="btn btn-danger btn-sm" onclick="delAppt(${safeId(row.id)})" style="mar
gin:0 2px">?</button>`
);

window.currentCalYear = window.currentCalYear || new Date().getFullYear();
window.currentCalMonth = window.currentCalMonth || new Date().getMonth();
window.currentAppointmentsList = appts;

const todayDateStr = new Date().toISOString().split('T')[0];
const todayAppts = appts.filter(a => (a.appt_date || '').startsWith(todayDateStr));
const todayCount = todayAppts.length;

const checkedInToday = todayAppts.filter(a => a.status === 'Checked-in' || a.status === 'Checked-In' || a.st
atus === '????').length;
const attendanceRate = todayCount > 0 ? Math.round((checkedInToday / todayCount) * 100) : 0;

const noShowToday = todayAppts.filter(a => a.status === 'No-Show' || a.status === '????').length;
const noShowRate = todayCount > 0 ? Math.round((noShowToday / todayCount) * 100) : 0;

const activeDoctors = [...new Set(todayAppts.map(a => a.doctor_name))].length || emps.length || 1;
const maxSlots = activeDoctors * 12;
const slotUtilization = Math.min(100, Math.round((todayCount / maxSlots) * 100));

el.innerHTML = `
  <div class="page-title">? ${tr('Appointments', '????????')}</div>

  <!-- KPI Cards Dashboard -->
  <div class="appt-kpis-container">
    <div class="appt-kpi-card">
      <div class="appt-kpi-icon-wrapper" style="background: rgba(13,148,136,0.1); color: var(--accent)">?</d
iv>

      <div class="appt-kpi-info">
        <span class="appt-kpi-val">${todayCount}</span>
        <span class="appt-kpi-label">${tr("Today's Appointments", '????? ????')}</span>
      </div>
    </div>

    <div class="appt-kpi-card">
      <div class="appt-kpi-icon-wrapper" style="background: rgba(16,185,129,0.1); color: #10b981">?</div>
      <div class="appt-kpi-info">
        <span class="appt-kpi-val">${attendanceRate}%</span>
        <span class="appt-kpi-label">${tr('Attendance Rate', '???? ?????')}</span>
      </div>
    </div>

    <div class="appt-kpi-card">
      <div class="appt-kpi-icon-wrapper" style="background: rgba(245,158,11,0.1); color: #f59e0b">?</div>
      <div class="appt-kpi-info">
        <span class="appt-kpi-val">${noShowRate}%</span>
        <span class="appt-kpi-label">${tr('No-Show Rate', '???? ?????')}</span>

```

```

    </div>
</div>

<div class="appt-kpi-card">
    <div class="appt-kpi-icon-wrapper" style="background: rgba(99,102,241,0.1); color: #6366f1">?</div>
    <div class="appt-kpi-info">
        <span class="appt-kpi-val">${slotUtilization}%</span>
        <span class="appt-kpi-label">${tr('Slot Utilization', '????? ?????????')}</span>
    </div>
</div>
</div>

<div class="split-layout">
    <div class="card glass-card-premium">
        <div class="card-title">? ${tr('Book Appointment', '??? ?????')}</div>
        <div class="form-group mb-12"><label>${tr('Patient', '??????')}</label><select class="form-input" id="aPatient"><option value="">${tr('Select patient', '????? ?????')}</option>${patients.map(p => `<option value="${escapeHTML(p.name_en)}" data-pid="${safeId(p.id)}">${escapeHTML(isArabic ? p.name_ar : p.name_en)} (#${escapeHTML(p.file_number)})</option>`).join('')}</select></div>
        <div class="form-group mb-12"><label>${tr('Doctor', '??????')}</label><select class="form-input" id="aDoctor"><option value="">${tr('Select doctor', '????? ?????')}</option>${emps.map(d => `<option>${escapeHTML(d.name)}</option>`).join('')}</select></div>
        <div class="form-group mb-12"><label>${tr('Date', '????????')}</label><input class="form-input" type="date" id="aDate" value="${new Date().toISOString().split('T')[0]}"></div>
        <div class="form-group mb-12"><label>${tr('Time', '?????')}</label><input class="form-input" type="time" id="aTime" value="${new Date().getTimeString().slice(0, 5)}"></div>
        <div class="form-group mb-12"><label>${tr('Notes', '????????')}</label><input class="form-input" id="aNotes"></div>
        <div class="form-group mb-16"><label>${tr('Appointment Fee', '???? ??????')}</label><input class="form-input" id="aFee" type="number" value="0" placeholder="0.00"></div>
        <button id="aBookBtn" class="btn btn-primary w-full" onclick="bookAppt()" style="height:44px">? ${tr('Book', '???')}</button>
    </div>
    <div class="card glass-card-premium">
        <div style="display: flex; justify-content: space-between; align-items: center; margin-bottom: 16px; border-bottom: 1px solid var(--border); padding-bottom: 8px">
            <div class="card-title" style="margin:0">? ${tr('Appointments', '????????')}</div>
            <div style="display: flex; gap: 8px">
                <button class="btn btn-sm" id="btnListView" onclick="window.toggleApptView('list')" style="background:var(--accent);color:#fff;padding:4px 12px;font-size:12px">? ${tr('List', '????????')}</button>
                <button class="btn btn-sm btn-secondary" id="btnCalView" onclick="window.toggleApptView('calendar')" style="padding:4px 12px;font-size:12px">? ${tr('Calendar', '????????')}</button>
            </div>
        </div>
        <input class="search-filter" placeholder="${tr('Search...', '???...')}" oninput="filterTable(this,'aTable')">
        <div id="aTable">${appointmentsTableHtml}</div>
        <div id="aCalendarGrid" style="display: none"></div>
    </div>
</div>`;

const bookBtn = el.querySelector('#aBookBtn');
if (bookBtn && !canBook) {
    bookBtn.disabled = true;

```

```

    bookBtn.title = tr('You do not have permission to book appointments', '??? ??? ???? ???? ??????');
    bookBtn.classList.add('btn-disabled');
    bookBtn.style.opacity = '0.5';
    bookBtn.style.cursor = 'not-allowed';
  }
}
window.bookAppt = async () => {
  const userPerms = currentUser?.permissions ? currentUser.permissions.split(',') : [];
  const isAdmin = currentUser?.role === 'Admin';
  const canBook = isAdmin || userPerms.includes('appointments') || userPerms.includes('2');
  if (!canBook) {
    showToast(tr('Access Denied: You do not have permission to book appointments', '?? ??? ??????: ??? ??? ???? ???? ???? ??????'), 'error');
    return;
  }
  const pSelect = document.getElementById('aPatient');
  const pName = pSelect.value;
  const pId = pSelect.options[pSelect.selectedIndex]?.dataset?.pid || '';
  if (!pName) { showToast(tr('Select patient', '???? ???'), 'error'); return; }

  // Date validation (Scheduling blueprint compliance)
  const apptDateStr = document.getElementById('aDate').value;
  if (!apptDateStr) { showToast(tr('Select appointment date', '???? ????? ??????'), 'error'); return; }
  const todayStr = new Date().toISOString().split('T')[0];
  if (apptDateStr < todayStr) {
    showToast(tr('Appointment date cannot be in the past', '???? ????? ?? ??? ? ???? ? ? ??????'), 'error')
  }
  ;
  return;
}

// Conflict Double-Booking Detection
const doctorName = document.getElementById('aDoctor').value;
const apptTimeStr = document.getElementById('aTime').value;
if (window.currentAppointmentsList) {
  const conflict = window.currentAppointmentsList.find(a =>
    a.doctor_name === doctorName &&
    (a.appt_date || '').startsWith(apptDateStr) &&
    (a.appt_time || '').substring(0, 5) === apptTimeStr.substring(0, 5) &&
    a.status !== 'Cancelled'
  );
  if (conflict) {
    const confirmProceed = confirm(tr(
      `Warning: Dr. ${doctorName} already has an appointment booked at ${apptTimeStr} on ${apptDateStr}. Do
you want to confirm a double-booking?`,
      `?????: ?????? ${doctorName} ??? ???? ????? ?????? ?? ?????? ${apptTimeStr} ?????? ${apptDateStr}. ?
? ???? ? ? ????? ????? ??????`
    ));
    if (!confirmProceed) return;
  }
}

try {
  await API.post('/api/appointments', { patient_name: pName, patient_id: pId, doctor_name: doctorName, depart-
ment: '', appt_date: apptDateStr, appt_time: apptTimeStr, notes: document.getElementById('aNotes').value, fee

```

```

: parseFloat(document.getElementById('aFee').value) || 0 });
    showToast(tr('Appointment booked!', '?? ?? ??????!'));
    await navigateTo(2);
} catch (e) { showToast(tr('Error booking', '??? ?? ?????'), 'error'); }
};

```

```

window.toggleApptView = (view) => {
    const listBtn = document.getElementById('btnListView');
    const calBtn = document.getElementById('btnCalView');
    const tableEl = document.getElementById('aTable');
    const calEl = document.getElementById('aCalendarGrid');
    const filterInput = document.querySelector('.search-filter');

    if (view === 'list') {
        if (listBtn) { listBtn.style.background = 'var(--accent)'; listBtn.style.color = '#fff'; }
        if (calBtn) { calBtn.style.background = ''; calBtn.style.color = ''; }
        if (tableEl) tableEl.style.display = 'block';
        if (filterInput) filterInput.style.display = 'block';
        if (calEl) calEl.style.display = 'none';
    } else {
        if (calBtn) { calBtn.style.background = 'var(--accent)'; calBtn.style.color = '#fff'; }
        if (listBtn) { listBtn.style.background = ''; listBtn.style.color = ''; }
        if (tableEl) tableEl.style.display = 'none';
        if (filterInput) filterInput.style.display = 'none';
        if (calEl) {
            calEl.style.display = 'block';
            window.renderApptCalendar();
        }
    }
};

```

```

window.changeCalMonth = (direction) => {
    window.currentCalMonth += direction;
    if (window.currentCalMonth < 0) {
        window.currentCalMonth = 11;
        window.currentCalYear -= 1;
    } else if (window.currentCalMonth > 11) {
        window.currentCalMonth = 0;
        window.currentCalYear += 1;
    }
    window.renderApptCalendar();
};

```

```

window.renderApptCalendar = () => {
    const gridEl = document.getElementById('aCalendarGrid');
    if (!gridEl) return;

    const year = window.currentCalYear;
    const month = window.currentCalMonth;
    const appts = window.currentAppointmentsList || [];

    const monthNamesAr = ["?????", "???????", "?????", "?????", "?????", "???????", "???????", "???????", "???????", "?????
???", "???????", "???????"];
    const monthNamesEn = ["January", "February", "March", "April", "May", "June", "July", "August", "September",

```

```

"October", "November", "December"];
const monthName = isArabic ? monthNamesAr[month] : monthNamesEn[month];

const firstDayIndex = new Date(year, month, 1).getDay();
const totalDays = new Date(year, month + 1, 0).getDate();
const prevMonthTotalDays = new Date(year, month, 0).getDate();

let cellsHtml = '';

// Padding for previous month
for (let i = firstDayIndex - 1; i >= 0; i--) {
  const dNum = prevMonthTotalDays - i;
  cellsHtml += `
    <div class="appt-calendar-cell other-month">
      <div class="appt-calendar-cell-num">${dNum}</div>
      <div class="appt-calendar-strips"></div>
    </div>
  `;
}

// Current month cells
const todayStr = new Date().toISOString().split('T')[0];
for (let day = 1; day <= totalDays; day++) {
  const dayStr = `${year}-${String(month + 1).padStart(2, '0')}-${String(day).padStart(2, '0')}`;
  const isToday = dayStr === todayStr;
  const dayAppts = appts.filter(a => (a.appt_date || '').startsWith(dayStr) && a.status !== 'Cancelled');

  let stripsHtml = '';
  dayAppts.slice(0, 3).forEach(a => {
    let bg = '#e3f2fd', color = '#1565c0';
    if (a.status === 'Checked-in' || a.status === 'Checked-In' || a.status === '????') {
      bg = '#e8f5e9'; color = '#2e7d32';
    } else if (a.status === 'No-Show' || a.status === '?????') {
      bg = '#fff3e0'; color = '#e65100';
    }
    stripsHtml += `
      <span class="appt-mini-strip" style="background:${bg};color:${color}" title="${escapeHTML(a.patient_name || '')} - Dr. ${escapeHTML(a.doctor_name)}">
        ${escapeHTML(a.appt_time || '00:00')} - ${escapeHTML(a.patient_name || '')}
      </span>
    `;
  });

  if (dayAppts.length > 3) {
    stripsHtml += `<span style="font-size: 8px; color: var(--text-dim); text-align: center; display: block">
    ${dayAppts.length - 3} ${tr('more', '?????')}</span>`;
  }

  cellsHtml += `
    <div class="appt-calendar-cell ${isToday ? 'today' : ''}" onclick="window.selectCalDate('${dayStr}')">
      <div class="appt-calendar-cell-num">${day}</div>
      <div class="appt-calendar-strips">${stripsHtml}</div>
    </div>
  `;
}

```

```

}

gridEl.innerHTML = `
  <div class="appt-calendar-header" style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px;background:var(--hover);padding:8px 12px;border-radius:8px">
    <button class="btn btn-sm btn-secondary" onclick="window.changeCalMonth(-1)">?</button>
    <span style="font-weight:bold;font-size:14px;color:var(--text-main)">${monthName} ${year}</span>
    <button class="btn btn-sm btn-secondary" onclick="window.changeCalMonth(1)">?</button>
  </div>
  <div class="appt-calendar-grid" style="display:grid;grid-template-columns:repeat(7, 1fr);gap:6px">
    ${isArabic ? ['???', '?????', '??????', '??????', '????', '????', '???'] : ['Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat']}.map(d => `
      <div style="text-align:center;font-weight:bold;font-size:11px;padding:6px;background:var(--hover);border-radius:4px;color:var(--text-dim)">${d}</div>
    `).join('')}
    ${cellsHtml}
  </div>
`;
};

window.selectCalDate = (dateStr) => {
  const dateInput = document.getElementById('aDate');
  if (dateInput) {
    dateInput.value = dateStr;
    showToast(tr('Selected booking date: ', '?? ?????? ????? ?????: ') + dateStr);
  }
};

window.delAppt = async (id) => {
  if (!confirm(tr('Delete this appointment?', '??? ??? ??????'))) return;
  try { await API.del(`/api/appointments/${id}`); showToast(tr('Deleted', '?? ?????')); await navigateTo(2); }
  catch (e) { showToast(tr('Error deleting', '??? ?? ?????'), 'error'); }
};

window.filterTable = (input, tableId) => {
  const txt = input.value.toLowerCase();
  document.querySelectorAll(`#${tableId} tbody tr`).forEach(r => r.style.display = r.textContent.toLowerCase().includes(txt) ? '' : 'none');
};

// ===== EMR Lock/Signature UI (Phase A1) =====
function emrStatusBadge(s) {
  if (s === 'locked') return rawHtml(`<span class="badge" style="background:#c8e6c9;color:#1b5e20">? ${tr('Locked/Signed', '?????/????')}</span>`);
  if (s === 'signed') return rawHtml(`<span class="badge" style="background:#fff9c4;color:#827717">?? ${tr('Signed', '????')}</span>`);
  return rawHtml(`<span class="badge" style="background:#eeeeee;color:#555">? ${tr('Draft', '????')}</span>`);
};

function emrRecordActions(r) {
  if (r.emr_status === 'locked') return rawHtml(`<button class="btn btn-sm" onclick="amendMedicalRecord(${parseInt(r.id, 10)})" style="background:#fff3e0;color:#e65100">? ${tr('Amend', '????? ?????')}</button>`);
  return rawHtml(`<button class="btn btn-sm" onclick="signMedicalRecord(${parseInt(r.id, 10)})" style="background:#c8e6c9;color:#1b5e20">?? ${tr('Sign & Finalize', '????? ?????')}</button>`);
};

window.signMedicalRecord = async (id) => {

```



```

    if (!confirm(tr('Sign and lock this record? Editing will be disabled after locking.', '????? ???? ??? ??????
?????? ??????? ??? ??????.'))) return;
    try { await API.post('/api/medical-records/' + id + '/sign', {}); showToast(tr('Record signed & locked', '??
?????? ??????')); await navigateTo(3); }
    catch (e) { showToast(tr('Sign failed (record may already be locked)', '?? ??????? (?? ???? ?????? ??????)'
, 'error')); }
};

window.amendMedicalRecord = async (id) => {
    const reason = prompt(tr('Amendment reason (required) ? original stays locked:', '?? ??????? (?????) ? ???
? ???? ??????'));
    if (!reason || !reason.trim()) return;
    try { await API.post('/api/medical-records/' + id + '/amend', { reason: reason, new_values_summary: '' }); s
howToast(tr('Amendment recorded (audited)', '?? ?????? ??????? (?????)')); await navigateTo(3); }
    catch (e) { showToast(tr('Amendment failed (record must be locked)', '?? ??????? (?? ? ? ???? ?????? ??????)
'), 'error'); }
};

// ===== DOCTOR STATION =====
async function renderDoctor(el) {
    // Show premium loading skeletons instantly
    el.innerHTML = `
        <div class="page-title">???? ${tr('Doctor Station', '???? ??????')}</div>
        <div class="doctor-split-layout">
            <div>
                <div class="card glass-card-premium mb-16">
                    <div class="skeleton-bar title mb-16"></div>
                    <div class="skeleton-bar mb-12"></div>
                </div>
                <div class="card glass-card-premium mb-16">
                    <div class="skeleton-bar title mb-16"></div>
                    <div class="skeleton-bar mb-12"></div>
                </div>
            </div>
            <div class="card glass-card-premium">
                <div class="skeleton-bar title mb-16"></div>
                <div class="skeleton-bar mb-12"></div>
            </div>
        </div>
    `;

    let patients = [];
    let records = [];
    let drugs = [];
    let allServices = [];
    let currentUser = {};
    let opdQueue = [];
    try {
        currentUser = await API.get('/api/auth/me');
        const docName = (currentUser.user && (currentUser.user.display_name || currentUser.user.username)) || '';
        [patients, records, drugs, allServices, opdQueue] = await Promise.all([
            API.get('/api/patients'),
            API.get('/api/medical/records'),
            API.get('/api/pharmacy/drugs'),
            API.get('/api/medical/services'),

```

```

    API.get(`/api/opd/doctor/queue?doctor_name=${encodeURIComponent(docName)}`).catch(() => [])
  });
} catch (e) {
  el.innerHTML = `
    <div class="page-title">???? ${tr('Doctor Station', '???? ?????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load doctor station data', '??? ?? ????? ?????? ????? ?????')}</h3>
      <p>${escapeHTML(e.message || e)}</p>
      <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????')}</
button>
    </div>
  `;
  return;
}

const drSpecialty = (currentUser.user && currentUser.user.speciality) || '';
window.__currentDoctorSpecialty = drSpecialty;
const filteredServices = drSpecialty ? allServices.filter(s => s.specialty === drSpecialty) : allServices;
// Group services by category for display
const svcCategories = {};
filteredServices.forEach(s => { if (!svcCategories[s.category]) svcCategories[s.category] = []; svcCategories[s.category].push(s); });

el.innerHTML = `
  <div class="page-title">???? ${tr('Doctor Station', '???? ?????')}</div>
  <div class="doctor-workspace-grid">
    <div class="doctor-main-col">
      <div class="card glass-card-premium mb-16">
        <div class="card-title">? ${tr('OPD Appointments Queue', '????? ?????? ????????? ?????????')}</div>
        <div style="max-height:220px;overflow-y:auto;margin-bottom:12px">
          ${opdQueue.length === 0 ? `<div style="text-align:center;padding:12px;color:var(--text-muted)">${tr(
            'No appointments today', '?? ????? ?????? ?????')}</div>` : `
            <table class="table" style="width:100%;font-size:12px">
              <thead>
                <tr>
                  <th>${tr('Time', '????')}</th>
                  <th>${tr('Patient', '????')}</th>
                  <th>${tr('Status', '????')}</th>
                  <th>${tr('Vitals', '????????')}</th>
                  <th>${tr('Action', '????????')}</th>
                </tr>
              </thead>
              <tbody>
                ${opdQueue.map(a => {
                  const v = a.vitals;
                  const vitalsText = v ? `T:${v.temp}°C | BP:${v.bp} | P:${v.pulse}` : tr('No vitals', '?? ?
                  ??? ?????');
                  const badgeColor = a.status === 'Checked-In' ? '#f59e0b' : a.status === 'In-Consultation'
                  ? '#3b82f6' : '#10b981';
                  return `
                    <tr>
                      <td><strong>${escapeHTML(a.appt_time)}</strong></td>
                      <td>${escapeHTML(isArabic ? (a.name_ar || a.name_en) : (a.name_en || a.name_ar))}</td>

```

```

        <td><span class="badge" style="background:${badgeColor};color:#fff">${escapeHTML(a.status)}</span></td>

        <td style="font-size:11px;color:var(--text-muted)">${escapeHTML(vitalsText)}</td>
    <td>
        ${a.status === 'Checked-In' ? `
            <button class="btn btn-xs btn-primary" onclick="window.startOpdEncounter(${a.id},
            ${a.patient_id})">? ${tr('Start', '????')}</button>
            ` : `
            <button class="btn btn-xs btn-secondary" onclick="window.selectOpdPatient(${a.patient_id})">? ${tr('Open', '???')}</button>
            `}
    </td>
</tr>
</tbody>
</table>
`}
</div>
<hr style="border:0;border-top:1px solid var(--border);margin:12px 0">
<div class="card-title">? ${tr('All Patients Search', '????? ?? ??? ????')}</div>
<select class="form-input w-full" id="drPatient" onchange="loadPatientInfo()">
    <option value="">${tr('-- Select --', '-- ??? ???? --')}</option>
    ${patients.map(p => `<option value="${safeId(p.id)}">${escapeHTML(p.file_number)} - ${escapeHTML(i
sArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))} (${statusText(p.status)})</option>`).join('')}
</select>
<div id="drPatientInfo" class="mt-16"></div>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Diagnosis & Notes', '?????? ????')}</div>
    <div class="form-group mb-8"><label>? ${tr('Quick Diagnosis Template', '??? ????')}</label>
        <div class="flex gap-8">
            <select class="form-input" id="drDiagTemplate" style="flex:1" onchange="applyDiagTemplate()">
                <option value="">${tr('-- Select Template --', '-- ??? ???? --')}</option>
            </select>
            <button class="btn btn-sm" onclick="loadDiagTemplates()" style="white-space:nowrap">? ${tr('Load
            ', '????')}</button>
        </div>
    </div>
    <div class="form-group mb-12" style="position:relative"><label>${tr('Diagnosis', '????')}</label>
        <input class="form-input" id="drDiag" placeholder="${tr('Type diagnosis...', '??? ??????...')}"
        oninput="window.handleDiagAutocomplete(this)" autocomplete="off">
        <div id="drDiagSuggestions" style="position:absolute;top:100%;left:0;right:0;background:var(--bg-c
ard,#fff);border:1px solid var(--border-color,#e5e7eb);border-radius:8px;z-index:100;max-height:200px;overflow
-y:auto;display:none;box-shadow:0 4px 12px rgba(0,0,0,0.1)"></div>
    </div>
    <div class="form-group mb-12"><label>${tr('Symptoms', '????')}</label><input class="form-input" i
d="drSymp"></div>
    <div class="form-group mb-12"><label>${tr('ICD-10', '???')}</label><input class="form-input"
id="drIcd"></div>
    <div class="form-group mb-16"><label>${tr('Notes', '????')}</label><textarea class="form-input fo
rm-textarea" id="drNotes"></textarea></div>
    <!-- Specialty Template Container -->
    <div id="drSpecialtyTemplateContainer" class="mb-16"></div>

```

```
<button class="btn btn-primary w-full" onclick="saveMedRecord()" style="height:44px"> ${tr('Save Record', '??? ?????')}</button>
</div>
<div style="display:flex;gap:8px;margin-top:12px;flex-wrap:wrap">
  <button class="btn btn-sm" onclick="showMedicalReportForm('sick_leave')" style="flex:1;background:#fff3e0;border:1px solid #ff9800;color:#e65100;min-width:120px"> ${tr('Sick Leave', '????? ?????')}</button>
  <button class="btn btn-sm" onclick="showMedicalReportForm('medical_report')" style="flex:1;background:#e3f2fd;border:1px solid #1565c0;color:#1565c0;min-width:120px"> ${tr('Med Report', '????? ???')}</button>
  <button class="btn btn-sm" onclick="showMedicalReportForm('fitness')" style="flex:1;background:#e8f5e9;border:1px solid #2e7d32;color:#2e7d32;min-width:120px"> ${tr('Fitness', '????? ?????')}</button>
</div>
</div>
<div class="doctor-side-col">
  <div class="card glass-card-premium mb-16">
    <div class="card-title"> ${tr('Procedures / Services Performed', '????????? / ??????? ???????')} ${drSpecialty} <span class="badge badge-info" style="font-size:11px;margin-right:8px">${escapeHTML(drSpecialty)}</span> : ''</div>
    <div class="form-group mb-12"><label>${tr('Search Procedures', '???? ?? ?????')}</label>
      <input class="form-input" id="drSvcSearch" placeholder="${tr('Type to search...', '???? ??????...')}" autocomplete="off" oninput="filterDrServices()">
      <div id="drSvcDropdown" style="max-height:200px;overflow-y:auto;border:1px solid var(--border);border-radius:8px;display:none;margin-top:4px;background:var(--card)"></div>
    </div>
    <div id="drSvcTags" class="flex gap-8" style="flex-wrap:wrap;margin-bottom:12px"></div>
    <div style="font-size:12px;color:var(--text-dim);margin-bottom:8px">${tr('Available categories', '????????? ???????')}: <strong>${escapeHTML(Object.keys(svcCategories).join(', '))} || tr('All', '????')</strong>
  </div>
  <button class="btn btn-success w-full" onclick="billDrProcedures()" id="drBillBtn" style="height:40px;margin-top:8px"> ${tr('Bill Selected Procedures', '???? ?????????? ?????????')}</button>
  <input type="hidden" id="drSvcData" value='${escapeHTML(JSON.stringify(filteredServices))}'>
</div>
<div class="card glass-card-premium mb-16">
  <div class="card-title"> ${tr('Refer to Laboratory', '????? ???????')}</div>
  <div class="form-group mb-12"><label>${tr('Test Type', '??? ?????')}</label>
    <select class="form-input" id="drLabType">
      <optgroup label="${tr('Hematology', '????? ?????')}">
        <option>CBC (Complete Blood Count)</option>
        <option>ESR (Erythrocyte Sedimentation Rate)</option>
        <option>Coagulation Profile (PT, PTT, INR)</option>
        <option>Blood Film / Reticulocyte Count</option>
        <option>Hemoglobin Electrophoresis</option>
        <option>G6PD Deficiency Test</option>
        <option>Sickle Cell Test</option>
        <option>Bleeding Time / Clotting Time</option>
        <option>D-Dimer</option>
      </optgroup>
      <optgroup label="${tr('Biochemistry', '????????? ???????')}">
        <option>Comprehensive Metabolic Panel (CMP)</option>
        <option>Basic Metabolic Panel (BMP)</option>
        <option>Fasting Blood Sugar (FBS)</option>
        <option>Random Blood Sugar (RBS)</option>
        <option>Oral Glucose Tolerance Test (OGTT)</option>
        <option>HbA1c (Glycated Hemoglobin)</option>
      </optgroup>
    </select>
  </div>
</div>
```

<option>Lipid Profile (Total Cholesterol, HDL, LDL, Triglycerides)</option>  
 <option>Renal Profile (Urea, Creatinine, Electrolytes: Na, K, Cl)</option>  
 <option>Liver Function Test (LFT: ALT, AST, ALP, Total/Direct Bilirubin, Albumin, Total Protein)</option>  
 <option>Cardiac Enzymes (Troponin T/I, CK-MB, CK-Total, LDH)</option>  
 <option>Uric Acid</option>  
 <option>Calcium / Phosphorus / Magnesium</option>  
 <option>Iron Profile (Serum Iron, TIBC, Ferritin, Transferrin)</option>  
 <option>Vitamin D3 (25-OH Cholecalciferol)</option>  
 <option>Vitamin B12 / Folate</option>  
 <option>Amylase / Lipase</option>  
 <option>Serum Osmolality</option>

</optgroup>

<optgroup label="{tr('Hormones & Endocrinology', '???????? ??????')}">  
 <option>Thyroid Profile (TSH, Free T3, Free T4, Total T3, Total T4)</option>  
 <option>Fertility Hormones (FSH, LH, Prolactin, Testosterone (Free/Total), Estradiol E2, Progesterone)</option>  
 <option>Beta-hCG (Pregnancy Test - Blood Qualitative/Quantitative)</option>  
 <option>Cortisol (AM/PM)</option>  
 <option>Insulin (Fasting/Random)</option>  
 <option>Parathyroid Hormone (PTH)</option>  
 <option>Growth Hormone (GH)</option>  
 <option>ACTH</option>  
 <option>C-Peptide</option>  
 <option>Anti-Mullerian Hormone (AMH)</option>  
 <option>Aldosterone / Renin</option>  
 <option>DHEA-S (Dehydroepiandrosterone Sulfate)</option>  
 <option>17-OH Progesterone</option>  
 <option>Calcitonin</option>

</optgroup>

<optgroup label="{tr('Immunology & Serology', '???????? ?????????')}">  
 <option>CRP (C-Reactive Protein - Qualitative/Quantitative)</option>  
 <option>Rheumatoid Factor (RF)</option>  
 <option>Anti-CCP (Anti-Cyclic Citrullinated Peptide)</option>  
 <option>ANA (Anti-Nuclear Antibody) / Anti-dsDNA</option>  
 <option>ANCA (Anti-Neutrophil Cytoplasmic Antibody)</option>  
 <option>Anti-Scl-70 / Anti-Centromere</option>  
 <option>ASO Titer</option>  
 <option>Hepatitis Profile (HBsAg, HBsAb, HCV Ab, HAV IgM/IgG)</option>  
 <option>HIV 1 & 2 Abs/Ag</option>  
 <option>VDRL / RPR (Syphilis)</option>  
 <option>Widal Test (Typhoid)</option>  
 <option>Brucella (Abortus/Melitensis)</option>  
 <option>Dengue NS1 Ag / IgM / IgG</option>  
 <option>Toxoplasmosis (IgG/IgM)</option>  
 <option>Rubella (IgG/IgM)</option>  
 <option>Cytomegalovirus CMV (IgG/IgM)</option>  
 <option>Herpes Simplex Virus HSV 1/2 (IgG/IgM)</option>  
 <option>EBV (Epstein-Barr Virus)</option>  
 <option>Celiac Disease Panel (Anti-tTG, Anti-Endomysial)</option>  
 <option>Food Allergy Panel (IgE)</option>  
 <option>Inhalant Allergy Panel (IgE)</option>  
 <option>Flow Cytometry (Immunophenotyping / CD4 Count)</option>

</optgroup>

```

<optgroup label="$tr('Microbiology & Parasitology', '?????? ?????? ?????????')}">
  <option>Urine Analysis (Routine & Microscopic)</option>
  <option>Urine Culture & Sensitivity</option>
  <option>Stool Analysis (Routine & Microscopic)</option>
  <option>Stool Culture</option>
  <option>Stool Occult Blood</option>
  <option>H. Pylori (Ag in Stool / Ab in Blood)</option>
  <option>Throat Swab Culture</option>
  <option>Sputum Culture & AFB (Tuberculosis)</option>
  <option>Wound/Pus Swab Culture</option>
  <option>Blood Culture (Aerobic/Anaerobic)</option>
  <option>Ear/Eye/Nasal Swab Culture</option>
  <option>High Vaginal Swab (HVS) Culture</option>
  <option>Urethral Swab Culture</option>
  <option>Fungal Culture (Skin/Nail/Hair)</option>
  <option>Malaria Film</option>
  <option>QuantiFERON-TB Gold / TB Spot</option>
  <option>Chlamydia trachomatis (PCR/Ag)</option>
  <option>Neisseria Gonorrhoeae (PCR/Culture)</option>
  <option>CSF Analysis (Cell Count, Protein, Glucose)</option>
  <option>Synovial Fluid Analysis</option>
  <option>Semen Analysis (Spermogram)</option>
</optgroup>
<optgroup label="$tr('Tumor Markers', '?????? ??????')}">
  <option>PSA (Prostate Specific Antigen - Total/Free)</option>
  <option>CEA (Carcinoembryonic Antigen)</option>
  <option>CA 125 (Ovarian)</option>
  <option>CA 15-3 (Breast)</option>
  <option>CA 19-9 (Pancreatic/GI)</option>
  <option>AFP (Alpha-Fetoprotein)</option>
  <option>Beta-2 Microglobulin</option>
  <option>Thyroglobulin</option>
</optgroup>
<optgroup label="$tr('Molecular Diagnostics / PCR', '?????? ?????? / PCR')}">
  <option>COVID-19 PCR</option>
  <option>HCV RNA PCR (Quantitative)</option>
  <option>HBV DNA PCR (Quantitative)</option>
  <option>HIV RNA PCR (Quantitative)</option>
  <option>Respiratory Pathogen Panel (PCR)</option>
  <option>HPV DNA Typing</option>
</optgroup>
<optgroup label="$tr('Histopathology / Cytology', '??? ?????? ?????????')}">
  <option>Pap Smear</option>
  <option>Biopsy Specimen Examination</option>
  <option>FNAC (Fine Needle Aspiration Cytology)</option>
  <option>Fluid Cytology (Pleural, Ascitic, CSF)</option>
</optgroup>
<optgroup label="$tr('Blood Bank / Transfusion', '??? ???? / ??? ?????')}">
  <option>Blood Group (ABO) & Rh Typing</option>
  <option>Crossmatch (Major & Minor)</option>
  <option>Direct Coombs Test (DAT)</option>
  <option>Indirect Coombs Test (IAT)</option>
  <option>Antibody Screening Panel</option>
  <option>Cold Agglutinins</option>

```



```

        </select>
    </div>
    <div class="form-group mb-12"><label>${tr('Details', '????????')}</label><input class="form-input" id="drLabDesc"></div>
    <button class="btn btn-success w-full" onclick="sendToLab()">? ${tr('Send to Lab', '????? ??????')}
</button>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Refer to Radiology', '????? ??????')}</div>
    <div class="form-group mb-12"><label>${tr('Scan Type', '??? ??????')}</label>
    <select class="form-input" id="drRadType">
        <optgroup label="${tr('X-Ray', '?????? ??????')}">
            <option>X-Ray Chest (PA/LAT)</option>
            <option>X-Ray Abdomen (Erect/Supine)</option>
            <option>X-Ray KUB (Kidney, Ureter, Bladder)</option>
            <option>X-Ray Cervical Spine (AP/LAT/Open Mouth)</option>
            <option>X-Ray Thoracic Spine</option>
            <option>X-Ray Lumbar Spine (AP/LAT)</option>
            <option>X-Ray Pelvis (AP)</option>
            <option>X-Ray Skull / Facial Bones / PNS</option>
            <option>X-Ray Shoulder / Clavicle</option>
            <option>X-Ray Arm (Humerus/Radius/Ulna)</option>
            <option>X-Ray Hand / Wrist</option>
            <option>X-Ray Hip/Femur</option>
            <option>X-Ray Knee (AP/LAT/Skyline)</option>
            <option>X-Ray Ankle / Foot</option>
            <option>X-Ray Bone Age</option>
        </optgroup>
        <optgroup label="${tr('Ultrasound', '??????? ??? ?????? / ??????')}">
            <option>Ultrasound Abdomen (Whole)</option>
            <option>Ultrasound Pelvis (Transabdominal/Transvaginal)</option>
            <option>Ultrasound Abdomen & Pelvis</option>
            <option>Ultrasound KUB / Prostate</option>
            <option>Ultrasound Thyroid / Neck</option>
            <option>Ultrasound Breast</option>
            <option>Ultrasound Scrotum / Testicular</option>
            <option>Obstetric Ultrasound (1st Trimester/Viability)</option>
            <option>Obstetric Ultrasound (Anomaly Scan 2nd Trimester)</option>
            <option>Obstetric Ultrasound (Growth 3rd Trimester)</option>
            <option>Folliculometry (Ovulation Tracking)</option>
            <option>Ultrasound Soft Tissue / Swelling</option>
            <option>Doppler Ultrasound - Carotid</option>
            <option>Doppler Ultrasound - Lower Limb Venous (DVT)</option>
            <option>Doppler Ultrasound - Lower Limb Arterial</option>
            <option>Doppler Ultrasound - Renal Artery</option>
            <option>Doppler Ultrasound - Obstetrics / Umbilical Artery</option>
            <option>Echocardiogram (Echo - Heart)</option>
        </optgroup>
        <optgroup label="${tr('CT Scan', '?????? ??????')}">
            <option>CT Brain / Head (Without Contrast)</option>
            <option>CT Brain / Head (With Contrast)</option>
            <option>CT PNS (Paranasal Sinuses)</option>
            <option>CT Neck (With Contrast)</option>
            <option>CT Chest (HRCT) Without Contrast</option>
        </optgroup>
    </select>
</div>

```



<option>CT Chest / Lungs (With Contrast)</option>  
 <option>CT Abdomen & Pelvis (Without Contrast - Triphasic)</option>  
 <option>CT Abdomen & Pelvis (With Contrast)</option>  
 <option>CT KUB (Stone Protocol - Non Contrast)</option>  
 <option>CT Urography (With Contrast)</option>  
 <option>CT Cervical Spine</option>  
 <option>CT Lumbar Spine</option>  
 <option>CT Angiography - Pulmonary (CTPA)</option>  
 <option>CT Angiography - Brain</option>  
 <option>CT Angiography - Aorta / Lower Limbs</option>  
 <option>CT Virtual Colonoscopy</option>

</optgroup>

<optgroup label="{tr('MRI', '????? ?????????')}">  
 <option>MRI Brain (Without Contrast)</option>  
 <option>MRI Brain (With Contrast)</option>  
 <option>MRI Pituitary Fossa</option>  
 <option>MRI Cervical Spine</option>  
 <option>MRI Thoracic Spine</option>  
 <option>MRI Lumbar Spine</option>  
 <option>MRI Whole Spine</option>  
 <option>MRI Pelvis (Male/Female)</option>  
 <option>MRI Prostate (Multiparametric)</option>  
 <option>MRI Shoulder Joint</option>  
 <option>MRI Knee Joint</option>  
 <option>MRI Ankle / Wrist Joint</option>  
 <option>MRI Abdomen</option>  
 <option>MRCP (Magnetic Resonance Cholangiopancreatography)</option>  
 <option>MR Venography (MRV)</option>  
 <option>MRA (Magnetic Resonance Angiography) - Brain</option>

</optgroup>

<optgroup label="{tr('Specialized Imaging & Scans', '???? ????? ?????????')}">  
 <option>Mammogram (Bilateral/Unilateral)</option>  
 <option>DEXA Scan (Bone Density)</option>  
 <option>Fluoroscopy - Barium Swallow</option>  
 <option>Fluoroscopy - Barium Meal / Follow Through</option>  
 <option>Fluoroscopy - Barium Enema</option>  
 <option>Fluoroscopy - HSG (Hysterosalpingography)</option>  
 <option>Fluoroscopy - IVP (Intravenous Pyelogram)</option>  
 <option>Panoramic Dental X-Ray (OPG)</option>  
 <option>Cephalometric X-Ray</option>  
 <option>CBCT (Cone Beam CT for Dentistry)</option>  
 <option>PET Scan (Positron Emission Tomography)</option>

</optgroup>

<optgroup label="{tr('Cardiology & Neuro', '?? ?????? ?????? ???')}">  
 <option>ECG (Electrocardiogram)</option>  
 <option>Holter Monitor (24/48 Hours)</option>  
 <option>Ambulatory Blood Pressure Monitoring (ABPM)</option>  
 <option>Treadmill Stress Test (TMT)</option>  
 <option>EEG (Electroencephalogram)</option>  
 <option>EMG (Electromyography) / NCS</option>  
 <option>Spirometry / Lung Function Test</option>  
 <option>Upper GI Endoscopy (OGD)</option>  
 <option>Colonoscopy</option>

</optgroup>

```

        <optgroup label="${tr('Other', '????')}">
            <option>${tr('Other Scan (Specify in details)', '????? ??? (??? ?? ????????)')}</option>
        </optgroup>
    </select>
</div>
<div class="form-group mb-12"><label>${tr('Details', '????????')}</label><input class="form-input" id="drRadDesc"></div>
<button class="btn btn-success w-full" onclick="sendToRad()">? ${tr('Send to Radiology', '????? ????'
??')}</button>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Write Prescription', '????? ???')}</div>
    <div class="form-group mb-12"><label>${tr('Medication', '?????')}</label>
        <input list="drugsDataList" class="form-input" id="drRxDrug" placeholder="${tr('Type to search medication...', '???? ?? ??? ??????...')}" autocomplete="off">
        <datalist id="drugsDataList">
            ${drugs.map(d => `<option value="${escapeHTML(d.drug_name)}">`).join('')}
            <option value="${tr('Other', '????')}">
        </datalist>
    </div>
    <div class="flex gap-8 mb-12" style="flex-wrap:wrap">
        <div class="form-group" style="flex:1;min-width:140px"><label>${tr('Dosage', '?????')}</label><input class="form-input" id="drRxDose" placeholder="${tr('e.g. 500mg', '????? 500??')}"></div>
        <div class="form-group" style="flex:0.6;min-width:90px"><label>${tr('Qty/Day', '?????/?')}</label><input class="form-input" id="drRxQty" type="number" min="1" value="1"></div>
    </div>
    <div class="flex gap-8 mb-12" style="flex-wrap:wrap">
        <div class="form-group" style="flex:1;min-width:160px"><label>${tr('Frequency', '???????')}</label>
            <select class="form-input" id="drRxFreq"><option>x1 ${tr('daily', '?????')}</option><option>x2 ${tr('daily', '?????')}</option><option>x3 ${tr('daily', '?????')}</option><option>x4 ${tr('daily', '?????')}</option><option>${tr('Every 8 hours', '?? 8 ?????')}</option><option>${tr('Every 12 hours', '?? 12 ?????')}</option><option>${tr('As needed', '??? ?????')}</option><option>${tr('Before meals', '??? ?????')}</option><option>${tr('After meals', '??? ?????')}</option><option>${tr('Before sleep', '??? ?????')}</option></select>
        </div>
        <div class="form-group" style="flex:0.8;min-width:120px"><label>${tr('Duration', '?????')}</label><input class="form-input" id="drRxDur" placeholder="${tr('e.g. 7 days', '????? 7 ?????')}"></div>
    </div>
    <button class="btn btn-primary w-full" onclick="sendRx()">? ${tr('Issue Prescription ? Pharmacy', '?
???? ???? ? ???????')}</button>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Medical Certificate', '???????? ?????')}</div>
    <div class="form-group mb-12"><label>${tr('Certificate Type', '??? ???????')}</label>
        <select class="form-input" id="drCertType">
            <option value="sick_leave">? ${tr('Sick Leave', '????? ?????')}</option>
            <option value="medical_report">? ${tr('Medical Report', '????? ???')}</option>
            <option value="fitness">? ${tr('Fitness Certificate', '????? ?????')}</option>
        </select>
    </div>
    <div class="form-group mb-12"><label>${tr('Diagnosis/Reason', '????????/????')}</label><input class="form-input" id="drCertDiag"></div>
    <div class="flex gap-8 mb-12">
        <div class="form-group" style="flex:1"><label>${tr('From', '??')}</label><input class="form-input"

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```
type="date" id="drCertFrom" value="${new Date().toISOString().split('T')[0]}"></div>
    <div class="form-group" style="flex:1"><label>${tr('To', '???')}</label><input class="form-input"
type="date" id="drCertTo"></div>
    <div class="form-group" style="flex:0.5"><label>${tr('Days', '????')}</label><input class="form-in
put" type="number" id="drCertDays" value="1" min="1"></div>
</div>
    <div class="form-group mb-12"><label>${tr('Notes', '???????')}</label><input class="form-input" id="
drCertNotes"></div>
    <button class="btn btn-primary w-full" onclick="issueCertificate()">? ${tr('Issue Certificate', '???
?? ?????')}</button>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Referral to Department', '????? ???')}</div>
    <div class="form-group mb-12"><label>${tr('To Department', '??? ?????')}</label>
    <select class="form-input" id="drRefDept">
        <option>????????</option><option>????????</option><option>????????</option><option>????????</option>
        <option>????? ?????</option><option>????????</option><option>????????</option><option>??????? ?????
???</option>
        <option>????? ?????????</option><option>??????</option><option>????????? ?????????</option><option>?????
???</option><option>????????</option>
    </select>
</div>
    <div class="form-group mb-12"><label>${tr('Reason', '????')}</label><input class="form-input" id="d
rRefReason"></div>
    <div class="form-group mb-12"><label>${tr('Urgency', '????????')}</label>
    <select class="form-input" id="drRefUrg">
        <option value="Normal">? ${tr('Normal', '????')}</option>
        <option value="Urgent">? ${tr('Urgent', '????')}</option>
        <option value="Emergency">? ${tr('Emergency', '????')}</option>
    </select>
</div>
    <button class="btn btn-warning w-full" onclick="sendReferral()">? ${tr('Send Referral', '????? ?????
??')}</button>
</div>
<div class="card glass-card-premium mb-16">
    <div class="card-title">? ${tr('Schedule Follow-up', '????? ?????')}</div>
    <div class="flex gap-8 mb-12">
        <div class="form-group" style="flex:1"><label>${tr('Date', '???????')}</label><input class="form-i
nput" type="date" id="drFollowDate"></div>
        <div class="form-group" style="flex:1"><label>${tr('Time', '????')}</label><input class="form-inp
ut" type="time" id="drFollowTime" value="09:00"></div>
    </div>
    <div class="form-group mb-12"><label>${tr('Notes', '???????')}</label><input class="form-input" id="
drFollowNotes"></div>
    <button class="btn btn-info w-full" onclick="scheduleFollowup()">? ${tr('Book Follow-up', '??? ????
????')}</button>
</div>
</div>
<div class="card glass-card-premium mt-20">
    <div class="card-title">? ${tr('Medical Records', '??????? ?????')}</div>
    <input class="search-filter" placeholder="${tr('Search...', '???...')}" oninput="filterTable(this, 'drTab
le')">
    <div id="drTable">${makeTable([tr('Patient', '?????'), tr('Diagnosis', '???????'), tr('Symptoms', '????
```

```

    ???'), tr('Date/Time', '??????/?????'), tr('Status', '?????'), tr('Actions', '???????')], records.map(r => (
    { cells: [r.patient_name || '', r.diagnosis, r.symptoms, r.visit_date ? new Date(r.visit_date).toLocaleString(
    'ar-SA', { dateStyle: 'short', timeStyle: 'short' }) : '', emrStatusBadge(r.emr_status), emrRecordActions(r)]
    })))</div>
    </div>`;
}
function statusText(s) { return s === 'Waiting' ? tr('Waiting', '????????') : s === 'With Doctor' ? tr('With
Doctor', '?? ?????') : tr('Done', '????'); }
window.loadPatientInfo = async () => {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { document.getElementById('drPatientInfo').innerHTML = ''; return; }
    document.getElementById('drPatientInfo').innerHTML = `
        <div class="card glass-card-premium" style="margin-top:12px;padding:16px">
            <div class="skeleton-bar title mb-12"></div>
            <div class="skeleton-bar mb-8"></div>
            <div class="skeleton-bar mb-8"></div>
        </div>
    `;
    try {
        await API.put(`/api/patients/${pid}`, { status: 'With Doctor' });
        const p = (await API.get('/api/patients')).find(x => x.id == pid);
        const vitals = await API.get(`/api/nursing/vitals/${pid}`).catch(() => []);
        const account = await API.get(`/api/patients/${pid}/account`).catch(() => null);
        const v = vitals.length > 0 ? vitals[0] : null;
        let vitalsHtml = '';
        if (v) {
            vitalsHtml = `<div style="margin-top:12px;padding:12px;border:1px solid var(--border-color,#e5e7eb);bord
er-radius:10px;background:var(--card-bg,#fff)">
                <div style="font-weight:600;margin-bottom:8px;font-size:13px">?? ${tr('Vitals from Nursing', '????????
???????? ?? ?????')}</div> <span style="font-weight:400;font-size:11px;color:var(--text-dim)">${v.created_at ? new
Date(v.created_at).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' }) : ''}</span></div>
                <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:6px;font-size:12px">
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
? ${tr('BP', '????')}<br><strong>${escapeHTML(v.bp || '-')}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
?? ${tr('Temp', '????')}<br><strong>${v.temp ? escapeHTML(v.temp) + '°' : '-'}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
?? ${tr('Pulse', '????')}<br><strong>${escapeHTML(v.pulse || '-')}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
? ${tr('O2', '????')}<br><strong>${v.o2_sat ? escapeHTML(v.o2_sat) + '%' : '-'}</strong></div>
                </div>
                <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:6px;font-size:12px;margin-top:6px">
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
? ${tr('Weight', '????')}<br><strong>${v.weight ? escapeHTML(v.weight) + ' kg' : '-'}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
? ${tr('Height', '????')}<br><strong>${v.height ? escapeHTML(v.height) + ' cm' : '-'}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
? ${tr('Sugar', '????')}<br><strong>${escapeHTML(v.blood_sugar || '-')}</strong></div>
                    <div style="background:var(--bg-secondary,#f8f9fa);padding:6px;border-radius:6px;text-align:center">
?? ${tr('Resp', '????')}<br><strong>${escapeHTML(v.respiratory_rate || '-')}</strong></div>
                </div>
                ${v.allergies ? `<div style="margin-top:6px"><span class="badge badge-danger">?? ${tr('Allergies', '??
????')}</span></div>` : ''}
                ${v.chronic_diseases ? `<div style="margin-top:4px"><span class="badge badge-warning">? ${tr('Chronic'

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, '????? ?????')}: ${escapeHTML(v.chronic_diseases)}</span></div>` : ''}

    ${v.current_medications ? `<div style="margin-top:4px"><span class="badge badge-info">? ${tr('Medications', '?????')}: ${escapeHTML(v.current_medications)}</span></div>` : ''}

    </div>`;
}
// Build patient history timeline
let historyHtml = '';
if (account) {
    const events = [];
    (account.records || []).forEach(r => events.push({ type: 'record', icon: '?', color: '#6366f1', label: tr('Visit/Diagnosis', '?????/?????'), detail: `${r.diagnosis || '-'}${r.symptoms ? ' | ' + r.symptoms : ''}${r.doctor_name ? ' | ' + r.doctor_name : ''}`, date: r.visit_date || r.created_at }));
    (account.labOrders || []).forEach(o => events.push({ type: 'lab', icon: '?', color: '#f59e0b', label: tr('Lab', '?????'), detail: `${o.order_type} ${o.status === 'Done' ? '?' : '?'} ${o.results ? ' | ' + o.results.substring(0, 80) : ''}`, date: o.created_at }));
    (account.radOrders || []).forEach(o => events.push({ type: 'rad', icon: '?', color: '#0ea5e9', label: tr('Radiology', '?????'), detail: `${o.order_type} ${o.status === 'Done' ? '?' : '?'}`, date: o.created_at }));
    (account.prescriptions || []).forEach(rx => events.push({ type: 'rx', icon: '?', color: '#10b981', label: tr('Prescription', '?????'), detail: `${rx.drug_name || rx.medication || '-'} | ${rx.dosage || ''} ${rx.frequency || ''}`, date: rx.created_at }));
    (account.invoices || []).forEach(inv => events.push({ type: 'inv', icon: '?', color: '#8b5cf6', label: tr('Invoice', '?????'), detail: `${inv.description || inv.service_type || '-'} | ${inv.total || 0} ${tr('SAR', '?..?')} ${inv.paid ? '?' : '?'}`, date: inv.created_at }));
    events.sort((a, b) => new Date(b.date || 0) - new Date(a.date || 0));
    if (events.length > 0) {
        historyHtml = `<div style="margin-top:12px;padding:12px;border:1px solid var(--border-color,#e5e7eb);border-radius:10px;background:var(--card-bg,#fff);max-height:350px;overflow-y:auto">
            <div style="font-weight:600;margin-bottom:10px;font-size:14px">? ${tr('Patient Full History', '????? ?????? ??????')} (${$events.length})</div>
            ${events.map(e => `<div style="display:flex;gap:10px;padding:8px;margin:4px 0;border-radius:8px;border-right:4px solid ${e.color};background:var(--hover,#f8f9fa);font-size:12px;align-items:flex-start">
                <span style="font-size:18px;min-width:24px">${e.icon}</span>
                <div style="flex:1"><strong style="color:${e.color}">${e.label}</strong><div style="margin-top:2px;color:var(--text)">${escapeHTML(e.detail)}</div></div>
                <span style="color:var(--text-dim);font-size:11px;white-space:nowrap">${e.date ? new Date(e.date).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' }) : '-'}</span>
            </div>`).join('')}
        </div>`;
    } else {
        historyHtml = `
            <div class="empty-state-card mt-12">
                <div class="empty-state-icon">?</div>
                <div>${tr('No clinical encounter history found', '?? ???? ??? ?????? ?????')}</div>
            </div>
        `;
    }
}
const consentBadge = p.privacy_consent_signed
    ? `<span class="badge badge-success" style="background:#10b981;color:#fff">? ${tr('PDPL Consent Signed', '??????? ?????? ?????????')}</span>`
    : `<button class="badge" style="background:#ef4444;color:#fff;cursor:pointer;border:none;border-radius:4px;padding:4px 8px;" onclick="signPrivacyConsent(${p.id})">? ${tr('PDPL Consent Required', '????? ?????? ?????')}</button>`;

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```

document.getElementById('drPatientInfo').innerHTML = `<div class="flex gap-8 mt-16" style="flex-wrap:wrap;
align-items:center"><span class="badge badge-info">? ${escapeHTML(p.mrn || p.file_number)}</span><span class="
badge badge-warning">? ${tr('Age', '????')}: ${escapeHTML(p.age || '?')}</span>${p.blood_type ? `<span class=
"badge" style="background:#dc2626;color:#fff;font-weight:700">? ${escapeHTML(p.blood_type)}</span>` : ''}<span
class="badge badge-success">? ${escapeHTML(p.phone)}</span><span class="badge badge-purple">? ${escapeHTML(p.
national_id)}</span>${p.gender ? `<span class="badge" style="background:${p.gender === '???' ? '#3b82f6' : '#e
c4899'};color:#fff">${p.gender === '???' ? '?' : ''} ${escapeHTML(p.gender)}</span>` : ''}${p.insurance_compa
ny ? `<span class="badge" style="background:#0d9488;color:#fff">? ${escapeHTML(p.insurance_company)}${p.insura
nce_class ? ' (' + escapeHTML(p.insurance_class) + ')' : ''}</span>` : ''}${consentBadge}<span class="badge" s
tyle="background:#0ea5e9;color:#fff">? ${tr('Visit', '??????')}: ${new Date().toLocaleString('ar-SA', { dateS
tyle: 'short', timeStyle: 'short' })}</span><button class="btn btn-sm btn-primary" onclick="viewPatientResults
(${safeId(p.id)})">? ${tr('View Lab & Radiology Results', '?????? ???? ???? ???? ????')}</button><button c
lass="btn btn-sm" onclick="dischargePatient(${safeId(p.id)})" style="margin-right:auto;background:#dc3545;colo
r:#fff;font-weight:600">? ${tr('Patient Done', '???? ?')}</button></div>${p.allergies ? `<div style="margi
n-top:8px;padding:10px;background:#fef2f2;border:2px solid #ef4444;border-radius:8px;font-size:13px;font-weigh
t:600;color:#dc2626">?? <strong>${tr('ALLERGIES', '????')}</strong> ${escapeHTML(p.allergies)}</div>` : ''}
${p.chronic_diseases ? `<div style="margin-top:6px;padding:8px;background:#fefce8;border:1px solid #facc15;bor
der-radius:8px;font-size:12px;color:#854d0e">? <strong>${tr('Chronic Diseases', '???? ?')}</strong> ${es
capeHTML(p.chronic_diseases)}</div>` : ''}${vitalsHtml}${historyHtml}<div id="drE1Panel"></div><div id="drResu
ltsPanel"></div>`;

// E1: render Problem List / CPOE / SOAP tabs for the selected patient.
if (typeof window.renderE1Panel === 'function') { window.renderE1Panel(p.id); }

// Dynamic specialty clinical templates
if (typeof window.renderSpecialtyTemplate === 'function') { window.renderSpecialtyTemplate(window.__curren
tDoctorSpecialty); }

} catch (e) { }

};

window.dischargePatient = async (pid) => {
  try {
    await API.put(`/api/patients/${pid}`, { status: 'Done' });
    showToast(tr('Patient discharged!', '?? ???? ?????? ?'), 'success');
    document.getElementById('drPatientInfo').innerHTML = `<div class="badge badge-success" style="font-size:14
px;padding:12px 20px;margin-top:12px">? ${tr('Patient discharged successfully', '?? ???? ?????? ?????')}</div>
`;
    document.getElementById('drPatient').value = '';
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.signPrivacyConsent = async (pid) => {
  if (!confirm(tr('Are you sure you want to sign the Privacy Consent (PDPL) for this patient?', '?? ??? ?????
?? ????? ???? ???? ???? ???? ???? ???? ???? (PDPL) ??? ???? ????')) return;
  try {
    await API.post(`/api/patients/${pid}/consent`);
    showToast(tr('Privacy consent signed successfully!', '?? ????? ?????? ?????? ???? ?'), 'success');
    if (typeof window.selectDrPatient === 'function') { window.selectDrPatient(); }
  } catch (e) { showToast(tr('Failed to sign consent', '??? ? ???? ??????'), 'error'); }
};

window.drawLabTrend = function(canvasId, points) {
  const canvas = document.getElementById(canvasId);
  if (!canvas) return;
  const ctx = canvas.getContext('2d');
  if (!ctx) return;

  // Clear canvas
  ctx.clearRect(0, 0, canvas.width, canvas.height);

```

```

// Sort points by date ascending
points.sort((a, b) => a.date - b.date);

const width = canvas.width;
const height = canvas.height;
const paddingLeft = 50;
const paddingRight = 20;
const paddingTop = 25;
const paddingBottom = 30;

const chartWidth = width - paddingLeft - paddingRight;
const chartHeight = height - paddingTop - paddingBottom;

// Find min/max values
let minVal = Math.min(...points.map(p => p.value));
let maxVal = Math.max(...points.map(p => p.value));

// Include reference range in min/max calculation if present
const refLows = points.filter(p => p.ref_low !== null).map(p => p.ref_low);
const refHighs = points.filter(p => p.ref_high !== null).map(p => p.ref_high);
if (refLows.length) minVal = Math.min(minVal, ...refLows);
if (refHighs.length) maxVal = Math.max(maxVal, ...refHighs);

// Add margin to min/max
const valRange = maxVal - minVal;
const margin = valRange === 0 ? 1 : valRange * 0.2;
minVal -= margin;
maxVal += margin;

// Helper to map value to Y coordinate
const getY = (val) => {
  return paddingTop + chartHeight - ((val - minVal) / (maxVal - minVal)) * chartHeight;
};

// Helper to map index to X coordinate
const getX = (idx) => {
  if (points.length <= 1) return paddingLeft + chartWidth / 2;
  return paddingLeft + (idx / (points.length - 1)) * chartWidth;
};

// Draw Grid & Reference Bands (if reference range exists)
const firstRefLow = refLows[0] !== undefined ? refLows[0] : null;
const firstRefHigh = refHighs[0] !== undefined ? refHighs[0] : null;
if (firstRefLow !== null && firstRefHigh !== null) {
  const yLow = getY(firstRefLow);
  const yHigh = getY(firstRefHigh);
  ctx.fillStyle = 'rgba(74, 222, 128, 0.08)'; // Light green band for normal range
  ctx.fillRect(paddingLeft, Math.min(yLow, yHigh), chartWidth, Math.abs(yLow - yHigh));

  // Draw reference lines
  ctx.strokeStyle = 'rgba(74, 222, 128, 0.25)';
  ctx.lineWidth = 1;
  ctx.setLineDash([4, 4]);

```

```

    ctx.beginPath();
    ctx.moveTo(paddingLeft, yLow);
    ctx.lineTo(paddingLeft + chartWidth, yLow);
    ctx.moveTo(paddingLeft, yHigh);
    ctx.lineTo(paddingLeft + chartWidth, yHigh);
    ctx.stroke();
    ctx.setLineDash([]);
}

// Draw Grid Lines (Y-axis grid)
ctx.strokeStyle = 'rgba(0, 0, 0, 0.05)';
ctx.lineWidth = 1;
const gridSteps = 4;
for (let i = 0; i <= gridSteps; i++) {
    const val = minVal + (i / gridSteps) * (maxVal - minVal);
    const y = getY(val);
    ctx.beginPath();
    ctx.moveTo(paddingLeft, y);
    ctx.lineTo(paddingLeft + chartWidth, y);
    ctx.stroke();

    // Draw Y axis label
    ctx.fillStyle = '#888';
    ctx.font = '10px sans-serif';
    ctx.textAlign = 'right';
    ctx.textBaseline = 'middle';
    ctx.fillText(val.toFixed(1), paddingLeft - 8, y);
}

// Draw X axis labels (Dates)
points.forEach((p, idx) => {
    const x = getX(idx);
    const dateStr = p.date.toLocaleDateString('ar-SA', { month: 'short', day: 'numeric' });
    ctx.fillStyle = '#888';
    ctx.font = '10px sans-serif';
    ctx.textAlign = 'center';
    ctx.textBaseline = 'top';
    ctx.fillText(dateStr, x, paddingTop + chartHeight + 8);
});

// Draw Trend Line
if (points.length > 1) {
    ctx.strokeStyle = '#006970';
    ctx.lineWidth = 2.5;
    ctx.beginPath();
    ctx.moveTo(getX(0), getY(points[0].value));
    for (let i = 1; i < points.length; i++) {
        ctx.lineTo(getX(i), getY(points[i].value));
    }
    ctx.stroke();
}

// Draw Points
points.forEach((p, idx) => {

```



```

const x = getX(idx);
const y = getY(p.value);

// Color point based on reference range
let pointColor = '#006970';
if (p.ref_low !== null && p.value < p.ref_low) pointColor = '#ef4444'; // Low (red)
if (p.ref_high !== null && p.value > p.ref_high) pointColor = '#ef4444'; // High (red)

// Outer circle
ctx.fillStyle = pointColor;
ctx.beginPath();
ctx.arc(x, y, 5, 0, 2 * Math.PI);
ctx.fill();

// Inner white circle
ctx.fillStyle = '#fff';
ctx.beginPath();
ctx.arc(x, y, 2.5, 0, 2 * Math.PI);
ctx.fill();

// Draw Value text near point
ctx.fillStyle = '#333';
ctx.font = 'bold 10px sans-serif';
ctx.textAlign = 'center';
ctx.textBaseline = 'bottom';
ctx.fillText(`${p.value}`, x, y - 6);
});
};

window.startOpdEncounter = async (apptId, patientId) => {
  try {
    await API.post('/api/opd/encounter/start', { appointment_id: apptId });
    showToast(tr('Consultation started', '?? ?? ???? ????'));
    const drP = document.getElementById('drPatient');
    if (drP) {
      drP.value = patientId;
      // trigger change manually to load patient info
      drP.dispatchEvent(new Event('change'));
    }
  } catch(e) { showToast(tr('Error starting encounter', '??? ?? ?? ????'), 'error'); }
};

window.selectOpdPatient = (patientId) => {
  const drP = document.getElementById('drPatient');
  if (drP) {
    drP.value = patientId;
    drP.dispatchEvent(new Event('change'));
  }
};

window.viewPatientResults = async (pid) => {
  try {
    const data = await API.get(`/api/patients/${pid}/results`);
    const p = data.patient;

```

```

let html = `<div class="card mt-16" style="border:2px solid var(--accent)">
  <div class="card-title">? ${tr('Results for', '????')} ${escapeHTML(isArabic ? (p.name_ar || p.name
_en) : (p.name_en || p.name_ar))}</div>`;

// Lab Results
if (data.labOrders.length > 0) {
  html += `<div class="mb-16"><h4 style="color:var(--accent);margin:0 0 8px">? ${tr('Lab Results', '????
????')} (${data.labOrders.length})</h4>`;
  data.labOrders.forEach(o => {
    html += `<div style="padding:10px;margin:6px 0;background:var(--hover);border-radius:8px;border-right:
4px solid ${o.status === 'Done' ? '#4ade80' : '#f59e0b'}">
      <div class="flex gap-8" style="flex-wrap:wrap;align-items:center"><strong>${escapeHTML(o.ord
er_type)}</strong> ${statusBadge(o.status)} <span style="color:var(--text-dim);font-size:12px">${o.created_at
? new Date(o.created_at).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' }) : ''}</span></div>
      <div style="margin-top:8px;padding:8px;background:var(--bg);border-radius:6px
;font-size:13px;white-space:pre-wrap">${escapeHTML(o.results)}</div>` : `<div style="margin-top:4px;color:var(
--text-dim);font-size:12px">${tr('No results yet', '?? ??? ???? ????')}</div>`
    </div>`;
  });
  html += `</div>`;
} else {
  html += `<div class="mb-16" style="color:var(--text-dim)">? ${tr('No lab orders', '?? ??? ???? ????')}
</div>`;
}

// Lab Trends Section
const trends = {};
if (data.labResults && data.labResults.length > 0) {
  data.labResults.forEach(r => {
    const name = r.test_name;
    if (!trends[name]) trends[name] = [];
    trends[name].push({
      value: parseFloat(r.value),
      date: r.order_date ? new Date(r.order_date) : new Date(r.created_at),
      unit: r.unit,
      ref_low: r.ref_low !== null ? parseFloat(r.ref_low) : null,
      ref_high: r.ref_high !== null ? parseFloat(r.ref_high) : null
    });
  });
}

const trendTests = Object.keys(trends);
if (trendTests.length > 0) {
  html += `<div class="mb-16" style="border-top:1px dashed var(--border);padding-top:16px;">
    <h4 style="color:var(--accent);margin:0 0 12px">? ${tr('Lab Results Trend Charts', '?????? ?
?? ???? ????')}</h4>
    <div style="display:grid;grid-template-columns:repeat(auto-fit, minmax(320px, 1fr));gap:16px;"
  >`;

  trendTests.forEach((testName, idx) => {
    const pts = trends[testName];
    const canvasId = `trend_canvas_${pid}_${idx}`;
    const refStr = pts[0].ref_low !== null && pts[0].ref_high !== null ? ` (Ref: ${pts[0].ref_low}-${pts[0]`

```

```

].ref_high} ${pts[0].unit})` : '';
    html += `<div class="card glass-card-premium" style="padding:12px;margin:0;box-shadow:none;border:1px
solid var(--border);">
        <div style="font-weight:bold;margin-bottom:8px;color:var(--text);display:flex;justify-conten
t:space-between;align-items:center;font-size:13px;">
            <span>? ${escapeHTML(testName)}</span>
            <span style="font-size:11px;color:var(--text-dim);">${escapeHTML(refStr)}</span>
        </div>
        <canvas id="${canvasId}" width="360" height="180" style="width:100%;height:180px;background:
var(--bg);border-radius:6px;"></canvas>
    </div>`;
    });

    html += `</div></div>`;
}

// Radiology Results
if (data.radOrders.length > 0) {
    html += `<div class="mb-16" style="border-top:1px dashed var(--border);padding-top:16px;"><h4 style="col
or:var(--accent);margin:0 0 8px 0"> ${tr('Radiology Results', '???? ????')} (${data.radOrders.length})</h4>`
    ;
    data.radOrders.forEach(o => {
        html += `<div style="padding:10px;margin:6px 0;background:var(--hover);border-radius:8px;border-right:
4px solid ${o.status === 'Done' ? '#4ade80' : '#f59e0b'}">
            <div class="flex gap-8" style="flex-wrap:wrap;align-items:center"><strong>${escapeHTML(o.ord
er_type)}</strong> ${statusBadge(o.status)} <span style="color:var(--text-dim);font-size:12px">${o.created_at
? new Date(o.created_at).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' }) : ''}</span></div>
            <div style="margin-top:8px">${renderRadResults(o.results)}</div> : <div sty
le="margin-top:4px;color:var(--text-dim);font-size:12px">${tr('No results yet', '?? ???? ????')}</div>`
            </div>`;
        });
    html += `</div>`;
} else {
    html += `<div class="mb-16" style="color:var(--text-dim)">? ${tr('No radiology orders', '?? ???? ????')}
</div>`;
}
html += `</div>`;

document.getElementById('drResultsPanel').innerHTML = html;

// Draw all trends
if (trendTests.length > 0) {
    trendTests.forEach((testName, idx) => {
        const pts = trends[testName];
        const canvasId = `trend_canvas_${pid}_${idx}`;
        window.drawLabTrend(canvasId, pts);
    });
}
} catch (e) { showToast(tr('Error loading results', '??? ?? ????? ?????'), 'error'); }
};

window.saveMedRecord = async () => {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { showToast(tr('Select patient first', '???? ????? ?????'), 'error'); return; }
}

```

```

try {
  let notesExtra = '';
  const container = document.getElementById('drSpecialtyTemplateContainer');
  if (container) {
    const inputs = container.querySelectorAll('.dr-spec-input');
    if (inputs.length > 0) {
      notesExtra = '\n\n--- ' + tr('Specialty Documentation', '??????? ?????? ??????') + ' (' + tr(window.__currentDoctorSpecialty || 'General', window.__currentDoctorSpecialty || '???') + ') ---\n';
      inputs.forEach(input => {
        const label = input.dataset.label || input.name;
        const val = input.value;
        notesExtra += `${label}: ${val}\n`;
      });
    }
  }
}

await API.post('/api/medical/records', {
  patient_id: pid,
  diagnosis: document.getElementById('drDiag').value,
  symptoms: document.getElementById('drSymp').value,
  icd10_codes: document.getElementById('drIcd').value,
  notes: document.getElementById('drNotes').value + notesExtra
});
showToast(tr('Record saved!', '?? ??? ?????!'));
await navigateTo(3);
} catch (e) { showToast(tr('Error saving', '??? ?? ?????'), 'error'); }
};

window.sendToLab = async () => {
  const pid = document.getElementById('drPatient').value;
  if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
  try {
    await API.post('/api/lab/orders', { patient_id: pid, order_type: document.getElementById('drLabType').value, description: document.getElementById('drLabDesc').value });
    showToast(tr('Sent to Reception for payment ? then Lab', '?? ??????? ?????????? ?????? ? ?? ???????'), 'success');
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.sendToRad = async () => {
  const pid = document.getElementById('drPatient').value;
  if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
  try {
    await API.post('/api/radiology/orders', { patient_id: pid, order_type: document.getElementById('drRadType').value, description: document.getElementById('drRadDesc').value });
    showToast(tr('Sent to Reception for payment ? then Radiology', '?? ??????? ?????????? ?????? ? ?? ???????'), 'success');
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.sendRx = async () => {
  // IMPORTANT-1: clear any stale override reason from a PRIOR prescription BEFORE any checks, so a
  // previous override can never carry over to this (or the next) prescription. Also cleared in finally.
  window.__rxOverrideReason = null;
  try {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }

```

```

const drugName = document.getElementById('drRxDrug').value;
if (!drugName || !drugName.trim()) { showToast(tr('Enter medication name', '???? ??? ??????'), 'error'); r
return; }

// E1 CDS HARDENING (clinical safety): use the SERVER-BACKED checks instead of the weak
// client-only substring helper. Allergy AND critical drug-drug interactions are HARD-STOPs that
// require an explicit, audited override reason (matching the server-enforced 422). We pass the
// patient's CURRENT meds (existing pharmacy queue) so interactions are evaluated, not just the
// single new drug. The server independently re-derives the authoritative active-med list.
// 1) Allergy cross-check (server: /api/allergy-check). FAIL-CLOSED: returns false on alert OR error.
const allergyOk = await checkAllergyBeforePrescribe(pid, [drugName]);
if (!allergyOk) {
  // Hard-stop unless the prescriber explicitly captures an override reason.
  const reason = window.prompt(tr('CRITICAL allergy alert (or allergy check unavailable). To override you
MUST enter a clinical reason (audited):',
  '????? ?????? ??? (?? ?????? ??? ?????????). ????????? ??? ?????? ??? ?????? (??????? ?? ?????????):'), '');
  if (!reason || !reason.trim()) { showToast(tr('Prescription cancelled (allergy, no override reason)', '?
????? ?????? (??????? ??? ??? ??????)'), 'error'); return; }
  window.__rxOverrideReason = reason.trim();
}
// 2) Drug-drug interaction check (server: /api/drug-interactions/check) against current meds.
// IMPORTANT-3: a CRITICAL interaction is a HARD-STOP (require explicit override reason),
// consistent with the server 422. IMPORTANT-2: a failed check FAILS CLOSED (treat as hard-stop).
let ddResult = { hasCritical: false, failed: false };
try {
  const current = await API.get('/api/pharmacy/queue').catch(() => ({ __failed: true }));
  const failedFetch = current && current.__failed;
  const mine = Array.isArray(current) ? current.filter(q => String(q.patient_id) === String(pid)).map(q =>
q.medication_name).filter(Boolean) : [];
  ddResult = await checkDrugInteractions([drugName, ...mine]);
  if (failedFetch) ddResult = { hasCritical: ddResult.hasCritical, failed: true };
} catch (e) { ddResult = { hasCritical: false, failed: true }; }
if (ddResult.hasCritical || ddResult.failed) {
  // Only prompt if we don't already hold an override reason from the allergy hard-stop.
  if (!window.__rxOverrideReason) {
    const reason = window.prompt(ddResult.hasCritical
    ? tr('CRITICAL drug-drug interaction. To override you MUST enter a clinical reason (audited):',
    '????? ?????? ??? ?????????? ??? ?????? ??? ?????? (??????? ?? ?????????):')
    : tr('Interaction check unavailable ? fail-closed. To proceed you MUST enter a clinical reason (audi
ted):',
    '????? ??? ????????? ? ?? ?????? ??????????. ????????? ??? ?????? ??? ?????? (??????? ?? ?????????):'), '');
  );
  if (!reason || !reason.trim()) { showToast(tr('Prescription cancelled (interaction, no override reason
)', '??????? ?????? (??????? ??? ??? ??????)'), 'error'); return; }
  window.__rxOverrideReason = reason.trim();
}
}

try {
  const qty = document.getElementById('drRxQty')?.value || '1';
  await API.post('/api/prescriptions', {
    patient_id: pid, medication_name: drugName, dosage: document.getElementById('drRxDose').value,
    quantity_per_day: qty, frequency: document.getElementById('drRxFreq').value, duration: document.getEle
mentById('drRxDur').value,

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        override_reason: window.__rxOverrideReason || undefined
    });
    showToast(tr('Prescription sent to Pharmacy!', '?? ????? ?????? ???????!'));
} catch (e) { showToast(tr('Error', '???'), 'error'); }
} finally {
    // IMPORTANT-1: ALWAYS clear the override reason so a stale value cannot leak to the NEXT Rx,
    // even when the POST fails or an early return is taken.
    window.__rxOverrideReason = null;
}
};

window.issueCertificate = async () => {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
    const pSelect = document.getElementById('drPatient');
    const pName = pSelect.options[pSelect.selectedIndex]?.text?.split(' - ')[1]?.split(' ')[0] || '';
    try {
        await API.post('/api/medical/certificates', {
            patient_id: pid, patient_name: pName,
            cert_type: document.getElementById('drCertType').value,
            diagnosis: document.getElementById('drCertDiag').value,
            start_date: document.getElementById('drCertFrom').value,
            end_date: document.getElementById('drCertTo').value,
            days: parseInt(document.getElementById('drCertDays').value) || 1,
            notes: document.getElementById('drCertNotes').value
        });
        showToast(tr('Certificate issued!', '?? ?????? ???????!'));
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.sendReferral = async () => {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
    const pSelect = document.getElementById('drPatient');
    const pName = pSelect.options[pSelect.selectedIndex]?.text?.split(' - ')[1]?.split(' ')[0] || '';
    try {
        await API.post('/api/referrals', {
            patient_id: pid, patient_name: pName,
            to_department: document.getElementById('drRefDept').value,
            reason: document.getElementById('drRefReason').value,
            urgency: document.getElementById('drRefUrg').value
        });
        showToast(tr('Referral sent!', '?? ?????? ???????!'));
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.scheduleFollowup = async () => {
    const pid = document.getElementById('drPatient').value;
    if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
    const followDate = document.getElementById('drFollowDate').value;
    if (!followDate) { showToast(tr('Select date', '???? ???????'), 'error'); return; }
    const pSelect = document.getElementById('drPatient');
    const pName = pSelect.options[pSelect.selectedIndex]?.text?.split(' - ')[1]?.split(' ')[0] || '';
    try {
        await API.post('/api/appointments/followup', {
            patient_id: pid, patient_name: pName,
            appt_date: followDate,

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    appt_time: document.getElementById('drFollowTime').value,
    notes: document.getElementById('drFollowNotes').value
  });
  showToast(tr('Follow-up booked!', '?? ??? ??? ????'), 'error');
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

let selectedServices = [];
window.filterDrServices = () => {
  const q = document.getElementById('drSvcSearch').value.toLowerCase().trim();
  const dd = document.getElementById('drSvcDropdown');
  if (!q || q.length < 1) { dd.style.display = 'none'; return; }
  const svcs = JSON.parse(document.getElementById('drSvcData').value || '[]');
  const matches = svcs.filter(s => s.name_en.toLowerCase().includes(q) || s.name_ar.toLowerCase().includes(q) || s.category.toLowerCase().includes(q)).slice(0, 15);
  if (!matches.length) { dd.innerHTML = `<div style="padding:10px;color:var(--text-dim)">${tr('No results', '? ? ??? ????')}</div>`; dd.style.display = 'block'; return; }
  dd.innerHTML = matches.map(s => `<div onclick="addDrService(${safeId(s.id)},'${jsStr(s.name_en)}','${jsStr(s.name_ar)}',${safeId(s.price)} || 0,'${jsStr(s.category)}')" style="padding:8px 12px;cursor:pointer;border-bottom:1px solid var(--border);display:flex;justify-content:space-between;align-items:center" onmouseover="this.style.background='var(--hover)'" onmouseout="this.style.background=''">
    <span><strong>${escapeHTML(isArabic ? s.name_ar : s.name_en)}</strong> <small style="color:var(--text-dim)">${escapeHTML(s.category)}</small></span>
    <span style="color:var(--accent);font-weight:600">${escapeHTML(s.price)} ${tr('SAR', '??.?')}</span>
  </div>`).join('');
  dd.style.display = 'block';
};

window.addDrService = (id, nameEn, nameAr, price, cat) => {
  if (selectedServices.find(s => s.id === id)) return;
  selectedServices.push({ id, nameEn, nameAr, price, cat });
  document.getElementById('drSvcSearch').value = '';
  document.getElementById('drSvcDropdown').style.display = 'none';
  renderSvcTags();
};

window.removeDrService = (id) => {
  selectedServices = selectedServices.filter(s => s.id !== id);
  renderSvcTags();
};

function renderSvcTags() {
  const c = document.getElementById('drSvcTags');
  if (!selectedServices.length) { c.innerHTML = `<span style="color:var(--text-dim);font-size:13px">${tr('No procedures selected', '? ? ??? ????')}</span>`; return; }
  const total = selectedServices.reduce((s, x) => s + x.price, 0);
  c.innerHTML = selectedServices.map(s => `<span class="badge badge-info" style="font-size:12px;padding:6px 10px">${escapeHTML(isArabic ? s.nameAr : s.nameEn)} (${escapeHTML(s.price)} ${tr('SAR', '??.?')}) <span onclick="removeDrService(${safeId(s.id)})" style="cursor:pointer;margin-right:4px;font-weight:bold">\u2715</span></span>`).join('') +
    `<span class="badge badge-success" style="font-size:12px;padding:6px 10px;margin-right:auto">\ud83d\udcb0 ${tr('Total', '????')}: ${total} ${tr('SAR', '??.?')}</span>`;
}

window.billDrProcedures = async () => {
  const pid = document.getElementById('drPatient').value;
  if (!pid) { showToast(tr('Select patient first', '????'), 'error'); return; }
  if (!selectedServices.length) { showToast(tr('Select procedures first', '????'), 'error'); return; }
};

```

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try {
    const result = await API.post('/api/medical/bill-procedures', { patient_id: pid, services: selectedServices });
    showToast(`${tr('Billed successfully', '?? ????? ?????????')}: ${result.totalBilled} ${tr('SAR', '??.?')}`);
    selectedServices = [];
    renderSvcTags();
} catch (e) { showToast(tr('Error billing', '??? ?? ?????????'), 'error'); }
};

// ===== LAB =====
// ===== LAB NORMAL RANGES (Gender-Specific: m=male, f=female) =====
const LAB_NORMAL_RANGES = {
    'CBC (Complete Blood Count)': {
        m: 'WBC: 4.5-11.0 ×103/μL | RBC: 4.5-5.5 ×103/μL | Hgb: 13.5-17.5 g/dL | Hct: 38-50% | Platelets: 150-400 ×103/μL | MCV: 80-100 fL | MCH: 27-33 pg',
        f: 'WBC: 4.5-11.0 ×103/μL | RBC: 4.0-5.0 ×103/μL | Hgb: 12.0-16.0 g/dL | Hct: 36-44% | Platelets: 150-400 ×103/μL | MCV: 80-100 fL | MCH: 27-33 pg'
    },
    'ESR (Erythrocyte Sedimentation Rate)': { m: '0-15 mm/hr', f: '0-20 mm/hr' },
    'Coagulation Profile (PT, PTT, INR)': 'PT: 11-13.5 sec | INR: 0.8-1.2 | PTT (aPTT): 25-35 sec',
    'Blood Film / Reticulocyte Count': 'Reticulocyte: 0.5-2.5%',
    'Hemoglobin Electrophoresis': 'HbA: 95-98% | HbA2: 1.5-3.5% | HbF: <2%',
    'G6PD Deficiency Test': 'Normal: 4.6-13.5 U/g Hb',
    'Sickle Cell Test': 'Negative (Normal)',
    'Bleeding Time / Clotting Time': 'Bleeding Time: 2-7 min | Clotting Time: 4-10 min',
    'D-Dimer': '<0.5 μg/mL (or <500 ng/mL)',
    'Comprehensive Metabolic Panel (CMP)': 'Glucose: 70-100 mg/dL | BUN: 7-20 mg/dL | Creatinine: 0.6-1.2 mg/dL | Na: 136-145 mEq/L | K: 3.5-5.0 mEq/L | Cl: 98-106 mEq/L | CO2: 23-29 mEq/L | Ca: 8.5-10.5 mg/dL | Total Protein: 6.0-8.3 g/dL | Albumin: 3.5-5.5 g/dL | Bilirubin(T): 0.1-1.2 mg/dL | ALP: 44-147 IU/L | ALT: 7-56 IU/L | AST: 10-40 IU/L',
    'Basic Metabolic Panel (BMP)': 'Glucose: 70-100 mg/dL | BUN: 7-20 mg/dL | Creatinine: 0.6-1.2 mg/dL | Na: 136-145 | K: 3.5-5.0 | Cl: 98-106 | CO2: 23-29 mEq/L | Ca: 8.5-10.5 mg/dL',
    'Fasting Blood Sugar (FBS)': 'Normal: 70-100 mg/dL (3.9-5.6 mmol/L) | Pre-diabetes: 100-125 | Diabetes: ≥126',
    'Random Blood Sugar (RBS)': 'Normal: <140 mg/dL (7.8 mmol/L) | Diabetes: ≥200',
    'Oral Glucose Tolerance Test (OGTT)': 'Fasting: <100 | 1hr: <180 | 2hr: <140 mg/dL | Diabetes 2hr: ≥200',
    'HbA1c (Glycated Hemoglobin)': 'Normal: <5.7% | Pre-diabetes: 5.7-6.4% | Diabetes: ≥6.5%',
    'Lipid Profile (Total Cholesterol, HDL, LDL, Triglycerides)': { m: 'Total Cholesterol: <200 | LDL: <100 | HDL: >40 mg/dL | Triglycerides: <150 | VLDL: 5-40', f: 'Total Cholesterol: <200 | LDL: <100 | HDL: >50 mg/dL | Triglycerides: <150 | VLDL: 5-40' },
    'Renal Profile (Urea, Creatinine, Electrolytes: Na, K, Cl)': { m: 'BUN: 7-20 | Creatinine: 0.7-1.3 mg/dL | eGFR: >90 | Na: 136-145 | K: 3.5-5.0 | Cl: 98-106', f: 'BUN: 7-20 | Creatinine: 0.6-1.1 mg/dL | eGFR: >90 | Na: 136-145 | K: 3.5-5.0 | Cl: 98-106' },
    'Liver Function Test (LFT: ALT, AST, ALP, Total/Direct Bilirubin, Albumin, Total Protein)': { m: 'ALT: 7-56 | AST: 10-40 | ALP: 44-147 | GGT: 8-61 IU/L | Bilirubin(T): 0.1-1.2 | Direct: 0-0.3 | Albumin: 3.5-5.5 | Protein: 6.0-8.3', f: 'ALT: 7-45 | AST: 10-35 | ALP: 44-147 | GGT: 5-36 IU/L | Bilirubin(T): 0.1-1.2 | Direct: 0-0.3 | Albumin: 3.5-5.5 | Protein: 6.0-8.3' },
    'Cardiac Enzymes (Troponin T/I, CK-MB, CK-Total, LDH)': { m: 'Troponin I: <0.04 ng/mL | CK-Total: 39-308 IU/L | CK-MB: <25 | LDH: 140-280 | BNP: <100 pg/mL', f: 'Troponin I: <0.04 ng/mL | CK-Total: 26-192 IU/L | CK-MB: <25 | LDH: 140-280 | BNP: <100 pg/mL' },
    'Uric Acid': { m: '3.4-7.0 mg/dL', f: '2.4-6.0 mg/dL' },
    'Calcium / Phosphorus / Magnesium': 'Ca: 8.5-10.5 mg/dL | Ionized Ca: 4.5-5.6 mg/dL | Phosphorus: 2.5-4.5 mg/dL | Magnesium: 1.7-2.2 mg/dL',
    'Iron Profile (Serum Iron, TIBC, Ferritin, Transferrin)': { m: 'Serum Iron: 65-175 μg/dL | TIBC: 250-370 | F

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erritin: 12-300 ng/mL | Transferrin Sat: 20-50%', f: 'Serum Iron: 50-170 µg/dL | TIBC: 250-370 | Ferritin: 12-150 ng/mL | Transferrin Sat: 20-50%' },

'Vitamin D3 (25-OH Cholecalciferol)': 'Deficient: <20 ng/mL | Insufficient: 20-29 | Sufficient: 30-100 | Toxic: >100 ng/mL',

'Vitamin B12 / Folate': 'B12: 200-900 pg/mL | Folate: 2.7-17.0 ng/mL',

'Amylase / Lipase': 'Amylase: 28-100 U/L | Lipase: 0-160 U/L',

'Serum Osmolality': '275-295 mOsm/kg',

'Thyroid Profile (TSH, Free T3, Free T4, Total T3, Total T4)': 'TSH: 0.27-4.2 mIU/L | Free T4: 0.93-1.7 ng/dL | Free T3: 2.0-4.4 pg/mL | Total T4: 5.1-14.1 µg/dL | Total T3: 80-200 ng/dL',

'Fertility Hormones (FSH, LH, Prolactin, Testosterone (Free/Total), Estradiol E2, Progesterone)': { m: 'FSH: 1.5-12.4 | LH: 1.7-8.6 mIU/mL | Prolactin: 4-15 ng/mL | Testosterone: 270-1070 ng/dL | Free Testosterone: 8.7-25.1 pg/mL | Estradiol: 10-40 pg/mL', f: 'FSH(follicular): 3.5-12.5 | LH(follicular): 2.4-12.6 mIU/mL | Prolactin: 4-23 ng/mL | Testosterone: 15-70 ng/dL | Estradiol(follicular): 12.5-166 pg/mL | Progesterone(luteal): 1.8-24 ng/mL' },

'Beta-hCG (Pregnancy Test - Blood Qualitative/Quantitative)': { m: 'Normal: <2 mIU/mL', f: 'Non-pregnant: <5 mIU/mL | Pregnant: >25 mIU/mL' },

'Cortisol (AM/PM)': 'AM (6-8am): 6.2-19.4 µg/dL | PM (4pm): 2.3-11.9 µg/dL',

'Insulin (Fasting/Random)': 'Fasting: 2.6-24.9 µIU/mL',

'Parathyroid Hormone (PTH)': '15-65 pg/mL',

'Growth Hormone (GH)': { m: '0-5 ng/mL', f: '0-10 ng/mL' },

'ACTH': 'AM: 10-60 pg/mL',

'C-Peptide': '0.5-2.0 ng/mL (fasting)',

'Anti-Mullerian Hormone (AMH)': { m: '1.4-14.0 ng/mL', f: 'Reproductive: 1.0-10.0 ng/mL | Low reserve: <1.0 | High (PCOS): >10' },

'CRP (C-Reactive Protein - Qualitative/Quantitative)': 'Normal: <3 mg/L | Mild inflammation: 3-10 | Moderate: 10-100 | Severe: >100',

'Rheumatoid Factor (RF)': 'Normal: <14 IU/mL',

'ANA (Anti-Nuclear Antibody) / Anti-dsDNA': 'ANA: Negative (<1:40) | Anti-dsDNA: <30 IU/mL',

'ASO Titer': 'Adults: <200 IU/mL | Children: <100 IU/mL',

'Hepatitis Profile (HBsAg, HBsAb, HCV Ab, HAV IgM/IgG)': 'HBsAg: Negative | HBsAb: >10 mIU/mL (immune) | HCV Ab: Negative | HAV IgM: Negative',

'HIV 1 & 2 Abs/Ag': 'Negative (Non-reactive)',

'VDRL / RPR (Syphilis)': 'Non-reactive (Negative)',

'Widal Test (Typhoid)': 'O & H Titers: <1:80 (Normal)',

'Brucella (Abortus/Melitensis)': 'Titer: <1:80 (Negative)',

'Toxoplasmosis (IgG/IgM)': 'IgG: <1.0 IU/mL (Negative) | IgM: Negative',

'Rubella (IgG/IgM)': 'IgG: >10 IU/mL (Immune) | IgM: Negative',

'Cytomegalovirus CMV (IgG/IgM)': 'IgG: Negative (<6 AU/mL) | IgM: Negative',

'Herpes Simplex Virus HSV 1/2 (IgG/IgM)': 'IgG: <0.9 (Negative) | IgM: Negative',

'EBV (Epstein-Barr Virus)': 'VCA IgM: Negative | VCA IgG: Negative | EBNA: Negative',

'Celiac Disease Panel (Anti-tTG, Anti-Endomysial)': 'Anti-tTG IgA: <4 U/mL (Negative) | Anti-Endomysial: Negative',

'Food Allergy Panel (IgE)': 'Total IgE: <100 IU/mL (Adults) | Specific IgE: <0.35 kU/L per allergen',

'Inhalant Allergy Panel (IgE)': 'Total IgE: <100 IU/mL | Specific IgE: Class 0 (<0.35 kU/L)',

'Urine Analysis (Routine & Microscopic)': 'pH: 4.6-8.0 | Specific Gravity: 1.005-1.030 | Protein: Negative | Glucose: Negative | Blood: Negative | WBC: 0-5/HPF | RBC: 0-2/HPF | Bacteria: None',

'Urine Culture & Sensitivity': 'Negative: <10,000 CFU/mL | Positive: ?100,000 CFU/mL',

'Stool Analysis (Routine & Microscopic)': 'Color: Brown | Consistency: Formed | Occult Blood: Negative | WBC: None | RBC: None | Parasites: None',

'Stool Culture': 'No pathogenic organisms',

'Stool Occult Blood': 'Negative',

'H. Pylori (Ag in Stool / Ab in Blood)': 'Stool Ag: Negative | Serum Ab: Negative',

'Throat Swab Culture': 'Normal Flora | No Group A Strep',

'Sputum Culture & AFB (Tuberculosis)': 'Culture: Normal flora | AFB Smear: Negative',

'Wound/Pus Swab Culture': 'No pathogenic growth',  
'Blood Culture (Aerobic/Anaerobic)': 'No growth after 5 days',  
'PSA (Prostate Specific Antigen - Total/Free)': { m: 'Total PSA: <4.0 ng/mL | Free/Total ratio: >25%', f: 'N/A (??? ??????)' },  
'CEA (Carcinoembryonic Antigen)': 'Non-smoker: <2.5 ng/mL | Smoker: <5.0 ng/mL',  
'CA 125 (Ovarian)': '<35 U/mL',  
'CA 15-3 (Breast)': '<30 U/mL',  
'CA 19-9 (Pancreatic/GI)': '<37 U/mL',  
'AFP (Alpha-Fetoprotein)': '<10 ng/mL (Adults)',  
'Beta-2 Microglobulin': '0.8-2.2 mg/L',  
'Thyroglobulin': '1.5-38.5 ng/mL (pre-thyroidectomy)',  
'COVID-19 PCR': 'Negative (Not Detected)',  
'HCV RNA PCR (Quantitative)': 'Not Detected (<15 IU/mL)',  
'HBV DNA PCR (Quantitative)': 'Not Detected (<10 IU/mL)',  
'HIV RNA PCR (Quantitative)': 'Not Detected (<20 copies/mL)',  
'Respiratory Pathogen Panel (PCR)': 'Negative for all targets',  
'HPV DNA Typing': 'Negative (No high-risk HPV detected)',  
'Pap Smear': 'NILM (Negative for Intraepithelial Lesion or Malignancy)',  
'Malaria Film': 'No parasites seen',  
// Blood Bank  
'Blood Group (ABO) & Rh Typing': 'A/B/AB/O | Rh+ or Rh-',  
'Crossmatch (Major & Minor)': 'Compatible (No agglutination)',  
'Direct Coombs Test (DAT)': 'Negative',  
'Indirect Coombs Test (IAT)': 'Negative',  
'Antibody Screening Panel': 'Negative (No clinically significant antibodies)',  
'Cold Agglutinins': 'Titer: <1:64',  
// Blood Gas  
'Arterial Blood Gas (ABG)': 'pH: 7.35-7.45 | pCO2: 35-45 mmHg | pO2: 80-100 mmHg | HCO3: 22-26 mEq/L | BE: -2 to +2 | O2 Sat: 95-100%',  
'Venous Blood Gas (VBG)': 'pH: 7.31-7.41 | pCO2: 41-51 mmHg | HCO3: 22-26 mEq/L',  
'Lactate (Lactic Acid)': 'Venous: 0.5-2.2 mmol/L | Arterial: 0.5-1.6 mmol/L | Critical: >4 mmol/L',  
'Ionized Calcium': '4.5-5.6 mg/dL (1.12-1.40 mmol/L)',  
'Methemoglobin / Carboxyhemoglobin': 'MetHb: <1.5% | COHb: Non-smoker: <2%, Smoker: <10%',  
// TDM  
'Digoxin Level': 'Therapeutic: 0.8-2.0 ng/mL | Toxic: >2.0',  
'Phenytoin (Dilantin) Level': 'Therapeutic: 10-20 µg/mL | Toxic: >20',  
'Valproic Acid Level': 'Therapeutic: 50-100 µg/mL | Toxic: >100',  
'Carbamazepine Level': 'Therapeutic: 4-12 µg/mL | Toxic: >12',  
'Lithium Level': 'Therapeutic: 0.6-1.2 mEq/L | Toxic: >1.5',  
'Vancomycin Level (Trough/Peak)': 'Trough: 10-20 µg/mL | Peak: 20-40 µg/mL',  
'Gentamicin / Amikacin Level': 'Gentamicin - Trough: <2 | Peak: 5-10 µg/mL | Amikacin - Trough: <10 | Peak: 20-30',  
'Theophylline Level': 'Therapeutic: 10-20 µg/mL | Toxic: >20',  
'Methotrexate Level': '24hr: <10 µmol/L | 48hr: <1 | 72hr: <0.1',  
'Tacrolimus / Cyclosporine Level': 'Tacrolimus: 5-20 ng/mL | Cyclosporine: 100-300 ng/mL (varies by transplant)',  
// Special Chemistry  
'Protein Electrophoresis (SPEP)': 'Albumin: 3.5-5.5 g/dL | Alpha-1: 0.1-0.3 | Alpha-2: 0.6-1.0 | Beta: 0.7-1.2 | Gamma: 0.7-1.6 g/dL',  
'Immunoglobulins (IgA, IgG, IgM, IgE)': 'IgG: 700-1600 mg/dL | IgA: 70-400 | IgM: 40-230 | IgE: <100 IU/mL',  
'Complement C3 / C4': 'C3: 90-180 mg/dL | C4: 10-40 mg/dL',  
'Ammonia Level': '15-45 µg/dL (11-32 µmol/L)',  
'Homocysteine': '5-15 µmol/L | High risk: >15',  
'Ceruloplasmin / Copper': 'Ceruloplasmin: 20-35 mg/dL | Serum Copper: 70-155 µg/dL | Wilson Disease: Cerulop

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lasmin <20',
'Lactate Dehydrogenase (LDH)': '140-280 IU/L',
'Haptoglobin': { m: '30-200 mg/dL', f: '30-200 mg/dL' },
'Procalcitonin (PCT)': 'Normal: <0.1 ng/mL | Bacterial unlikely: <0.25 | Likely: 0.25-0.5 | Severe sepsis: >
2.0',
'BNP / NT-proBNP': 'BNP: <100 pg/mL (no HF) | NT-proBNP: <300 pg/mL (<50y), <900 (50-75y), <1800 (>75y)',
'Fibrinogen': '200-400 mg/dL',
'Anti-Xa (Heparin) Assay': 'LMWH prophylaxis: 0.2-0.5 IU/mL | Treatment: 0.5-1.0 | UFH: 0.3-0.7',
'Cystatin C': '0.6-1.0 mg/L',
'Microalbumin (Urine)': 'Normal: <30 mg/day | Microalbuminuria: 30-300 | Macroalbuminuria: >300',
'Serum Free Light Chains (Kappa/Lambda)': 'Kappa: 3.3-19.4 mg/L | Lambda: 5.7-26.3 | Ratio: 0.26-1.65',
'24hr Urine Protein / Creatinine Clearance': { m: 'Protein: <150 mg/24hr | Creatinine Clearance: 97-137 mL/m
in', f: 'Protein: <150 mg/24hr | Creatinine Clearance: 88-128 mL/min' },
// Immunology additions
'Anti-CCP (Anti-Cyclic Citrullinated Peptide)': 'Negative: <20 U/mL | Positive: ?20',
'ANCA (Anti-Neutrophil Cytoplasmic Antibody)': 'Negative (<1:20) | c-ANCA & p-ANCA',
'Anti-Scl-70 / Anti-Centromere': 'Negative (<1.0 U) | Anti-Scl-70: Scleroderma | Anti-Centromere: CREST',
'Dengue NS1 Ag / IgM / IgG': 'NS1 Ag: Negative | IgM: Negative | IgG: Negative',
'Flow Cytometry (Immunophenotyping / CD4 Count)': 'CD4: 500-1500 cells/μL | CD4/CD8 ratio: 1.0-3.0',
// Microbiology additions
'QuantiFERON-TB Gold / TB Spot': 'Negative (<0.35 IU/mL) | Borderline: 0.35-0.50 | Positive: ?0.35',
'Chlamydia trachomatis (PCR/Ag)': 'Not Detected (Negative)',
'Neisseria Gonorrhoeae (PCR/Culture)': 'Not Detected (Negative)',
'CSF Analysis (Cell Count, Protein, Glucose)': 'WBC: 0-5/μL | RBC: 0 | Protein: 15-45 mg/dL | Glucose: 40-70
mg/dL | Opening pressure: 6-20 cmH2O',
'Synovial Fluid Analysis': 'Color: Clear/Yellow | WBC: <200/μL | Crystals: None | Culture: No growth',
'Semen Analysis (Spermogram)': { m: 'Volume: 1.5-5 mL | Count: ?15 million/mL | Total: ?39 million | Motilit
y: ?40% | Morphology: ?4% normal | pH: 7.2-8.0', f: 'N/A' },
// Hormones additions
'Aldosterone / Renin': 'Aldosterone (upright): 7-30 ng/dL | Renin (upright): 0.5-4.0 ng/mL/hr | Ratio: <30',
'DHEA-S (Dehydroepiandrosterone Sulfate)': { m: '80-560 μg/dL (varies by age)', f: '35-430 μg/dL (varies by
age)' },
'Calcitonin': { m: '<8.4 pg/mL', f: '<5.0 pg/mL' },
'17-OH Progesterone': { m: '0.5-2.1 ng/mL', f: 'Follicular: 0.2-1.0 | Luteal: 1.0-4.0 ng/mL' },
// Toxicology & Trace Elements
'Myoglobin': { m: '28-72 ng/mL', f: '25-58 ng/mL' },
'Vitamin A (Retinol)': '30-65 μg/dL (1.05-2.27 μmol/L)',
'Zinc Level': '60-120 μg/dL (9.2-18.4 μmol/L)',
'Selenium Level': '70-150 μg/L',
'Lead Level (Blood)': 'Normal: <5 μg/dL | Action: ?5 | Toxic: >70',
'Mercury Level (Blood)': 'Normal: <10 μg/L | At risk: 10-50 | Toxic: >50',
'Urine Drug Screen (UDS)': 'Negative for all classes (Amphetamines, Barbiturates, Benzodiazepines, Cannabino
ids, Cocaine, Opiates)',
'Serum Ethanol (Alcohol) Level': 'Negative: 0 | Legal limit: <80 mg/dL | Lethal: >400 mg/dL',
'Acetaminophen (Paracetamol) Level': 'Therapeutic: 10-30 μg/mL | Toxic (4hr): >150 μg/mL',
'Salicylate (Aspirin) Level': 'Therapeutic: 15-30 mg/dL | Toxic: >30 | Lethal: >60'
};
window.getLabNormalRange = (testName, gender) => {
    let entry = LAB_NORMAL_RANGES[testName];
    if (!entry) { for (const key in LAB_NORMAL_RANGES) { if (testName.includes(key) || key.includes(testName)) {
entry = LAB_NORMAL_RANGES[key]; break; } } }
    if (!entry) return '';
    if (typeof entry === 'string') return entry;
    const g = (gender || '').trim();

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if (g === '???' || g === 'Male' || g === 'male' || g === 'M') return '? ' + (entry.m || '');
if (g === '????' || g === 'Female' || g === 'female' || g === 'F') return '? ' + (entry.f || '');
return '? ' + entry.m + '\n? ' + entry.f;

// Auto-load diagnosis templates on page render
setTimeout(() => { if (document.getElementById("drDiagTemplate")) loadDiagTemplates(); }, 500);
};

async function renderLab(el) {
  // Show skeletons instantly
  el.innerHTML = `
    <div class="page-title">? ${tr('Laboratory', '???????')}</div>
    <div class="stats-grid">
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
    </div>
    <div class="split-layout">
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-12"></div>
        <div class="skeleton-bar mb-8"></div>
      </div>
      <div class="card glass-card-premium" style="flex:2">
        <div class="skeleton-bar title mb-12"></div>
        <div class="skeleton-bar mb-8"></div>
      </div>
    </div>
  `;

  let orders = [];
  let patients = [];
  try {
    [orders, patients] = await Promise.all([
      API.get('/api/lab/orders'),
      API.get('/api/patients')
    ]);
  } catch (e) {
    el.innerHTML = `
      <div class="page-title">? ${tr('Laboratory', '???????')}</div>
      <div class="error-card-premium">
        <div class="error-card-icon">??</div>
        <h3>${tr('Failed to load laboratory data', '??? ?? ????? ?????? ??????')}</h3>
        <p>${escapeHTML(e.message || e)}</p>
        <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
    button>
      </div>
    `;
    return;
  }
  el.innerHTML = `<div class="page-title">? ${tr('Laboratory', '???????')}</div>
    <div class="stats-grid">
      <div class="stat-card" style="--stat-color:#f59e0b"><div class="stat-label">${tr('Pending', '????????')}
    </div><div class="stat-value">${orders.filter(o => o.status === 'Requested').length}</div></div>
      <div class="stat-card" style="--stat-color:#3b82f6"><div class="stat-label">${tr('In Progress', '??? ???

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??')}</div><div class="stat-value">${orders.filter(o => o.status === 'In Progress').length}</div></div>
  <div class="stat-card" style="--stat-color:#4ade80"><div class="stat-label">${tr('Completed', '????')}<
/div><div class="stat-value">${orders.filter(o => o.status === 'Done').length}</div></div>
</div>
<div class="split-layout">
  <div class="card glass-card-premium" style="flex:1">
    <div class="card-title">? ${tr('Direct Lab Order', '????? ??? ???')}</div>
    <div class="form-group mb-12"><label>${tr('Select Patient (Optional)', '???? ???? (???????)')}</label>
      <select class="form-input" id="labPatientId"><option value="">--</option>${patients.map(p => `<optio
n value="${safeId(p.id)}">${escapeHTML(p.name_ar || p.name_en)}</option>`).join('')}</select>
    </div>
    <div class="form-group mb-12"><label>${tr('Test Name', '??? ???????')}</label>
      <select class="form-input" id="labDirectType">
        <optgroup label="${tr('Hematology', '????? ?????')}">
          <option>CBC (Complete Blood Count)</option>
          <option>ESR (Erythrocyte Sedimentation Rate)</option>
          <option>Coagulation Profile (PT, PTT, INR)</option>
          <option>Blood Film / Reticulocyte Count</option>
          <option>Hemoglobin Electrophoresis</option>
          <option>G6PD Deficiency Test</option>
          <option>Sickle Cell Test</option>
          <option>Bleeding Time / Clotting Time</option>
          <option>D-Dimer</option>
        </optgroup>
        <optgroup label="${tr('Biochemistry', '????????? ???????')}">
          <option>Comprehensive Metabolic Panel (CMP)</option>
          <option>Basic Metabolic Panel (BMP)</option>
          <option>Fasting Blood Sugar (FBS)</option>
          <option>Random Blood Sugar (RBS)</option>
          <option>Oral Glucose Tolerance Test (OGTT)</option>
          <option>HbA1c (Glycated Hemoglobin)</option>
          <option>Lipid Profile (Total Cholesterol, HDL, LDL, Triglycerides)</option>
          <option>Renal Profile (Urea, Creatinine, Electrolytes: Na, K, Cl)</option>
          <option>Liver Function Test (LFT: ALT, AST, ALP, Total/Direct Bilirubin, Albumin, Total Protein)
</option>
          <option>Cardiac Enzymes (Troponin T/I, CK-MB, CK-Total, LDH)</option>
          <option>Uric Acid</option>
          <option>Calcium / Phosphorus / Magnesium</option>
          <option>Iron Profile (Serum Iron, TIBC, Ferritin, Transferrin)</option>
          <option>Vitamin D3 (25-OH Cholecalciferol)</option>
          <option>Vitamin B12 / Folate</option>
          <option>Amylase / Lipase</option>
          <option>Serum Osmolality</option>
        </optgroup>
        <optgroup label="${tr('Hormones & Endocrinology', '????????? ???????')}">
          <option>Thyroid Profile (TSH, Free T3, Free T4, Total T3, Total T4)</option>
          <option>Fertility Hormones (FSH, LH, Prolactin, Testosterone (Free/Total), Estradiol E2, Progesterone)</option>
          <option>Beta-hCG (Pregnancy Test - Blood Qualitative/Quantitative)</option>
          <option>Cortisol (AM/PM)</option>
          <option>Insulin (Fasting/Random)</option>
          <option>Parathyroid Hormone (PTH)</option>
          <option>Growth Hormone (GH)</option>
          <option>ACTH</option>
        </optgroup>
      </select>
    </div>
  </div>
</div>

```

☐ C-Peptide</option>  
☐ Anti-Mullerian Hormone (AMH)</option>  
☐ Aldosterone / Renin</option>  
☐ DHEA-S (Dehydroepiandrosterone Sulfate)</option>  
☐ 17-OH Progesterone</option>  
☐ Calcitonin</option>
 </optgroup>
 <optgroup label="\$tr('Immunology & Serology', '?????? ??????')">
 <option>CRP (C-Reactive Protein - Qualitative/Quantitative)</option>  
 <option>Rheumatoid Factor (RF)</option>  
 <option>Anti-CCP (Anti-Cyclic Citrullinated Peptide)</option>  
 <option>ANA (Anti-Nuclear Antibody) / Anti-dsDNA</option>  
 <option>ANCA (Anti-Neutrophil Cytoplasmic Antibody)</option>  
 <option>Anti-Scl-70 / Anti-Centromere</option>  
 <option>ASO Titer</option>  
 <option>Hepatitis Profile (HBsAg, HBsAb, HCV Ab, HAV IgM/IgG)</option>  
 <option>HIV 1 & 2 Abs/Ag</option>  
 <option>VDRL / RPR (Syphilis)</option>  
 <option>Widal Test (Typhoid)</option>  
 <option>Brucella (Abortus/Melitensis)</option>  
 <option>Dengue NS1 Ag / IgM / IgG</option>  
 <option>Toxoplasmosis (IgG/IgM)</option>  
 <option>Rubella (IgG/IgM)</option>  
 <option>Cytomegalovirus CMV (IgG/IgM)</option>  
 <option>Herpes Simplex Virus HSV 1/2 (IgG/IgM)</option>  
 <option>EBV (Epstein-Barr Virus)</option>  
 <option>Celiac Disease Panel (Anti-tTG, Anti-Endomysial)</option>  
 <option>Food Allergy Panel (IgE)</option>  
 <option>Inhalant Allergy Panel (IgE)</option>  
 <option>Flow Cytometry (Immunophenotyping / CD4 Count)</option>
 </optgroup>
 <optgroup label="\$tr('Microbiology & Parasitology', '?????? ?????? ?????????')">
 <option>Urine Analysis (Routine & Microscopic)</option>  
 <option>Urine Culture & Sensitivity</option>  
 <option>Stool Analysis (Routine & Microscopic)</option>  
 <option>Stool Culture</option>  
 <option>Stool Occult Blood</option>  
 <option>H. Pylori (Ag in Stool / Ab in Blood)</option>  
 <option>Throat Swab Culture</option>  
 <option>Sputum Culture & AFB (Tuberculosis)</option>  
 <option>Wound/Pus Swab Culture</option>  
 <option>Blood Culture (Aerobic/Anaerobic)</option>  
 <option>Ear/Eye/Nasal Swab Culture</option>  
 <option>High Vaginal Swab (HVS) Culture</option>  
 <option>Urethral Swab Culture</option>  
 <option>Fungal Culture (Skin/Nail/Hair)</option>  
 <option>Malaria Film</option>  
 <option>QuantiFERON-TB Gold / TB Spot</option>  
 <option>Chlamydia trachomatis (PCR/Ag)</option>  
 <option>Neisseria Gonorrhoeae (PCR/Culture)</option>  
 <option>CSF Analysis (Cell Count, Protein, Glucose)</option>  
 <option>Synovial Fluid Analysis</option>  
 <option>Semen Analysis (Spermogram)</option>
 </optgroup>

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<optgroup label="{tr('Tumor Markers', '????? ?????')}">
  <option>PSA (Prostate Specific Antigen - Total/Free)</option>
  <option>CEA (Carcinoembryonic Antigen)</option>
  <option>CA 125 (Ovarian)</option>
  <option>CA 15-3 (Breast)</option>
  <option>CA 19-9 (Pancreatic/GI)</option>
  <option>AFP (Alpha-Fetoprotein)</option>
  <option>Beta-2 Microglobulin</option>
  <option>Thyroglobulin</option>
</optgroup>
<optgroup label="{tr('Molecular Diagnostics / PCR', '?????? ?????? / PCR')}">
  <option>COVID-19 PCR</option>
  <option>HCV RNA PCR (Quantitative)</option>
  <option>HBV DNA PCR (Quantitative)</option>
  <option>HIV RNA PCR (Quantitative)</option>
  <option>Respiratory Pathogen Panel (PCR)</option>
  <option>HPV DNA Typing</option>
</optgroup>
<optgroup label="{tr('Histopathology / Cytology', '?? ?????? ???????')}">
  <option>Pap Smear</option>
  <option>Biopsy Specimen Examination</option>
  <option>FNAC (Fine Needle Aspiration Cytology)</option>
  <option>Fluid Cytology (Pleural, Ascitic, CSF)</option>
</optgroup>
<optgroup label="{tr('Blood Bank / Transfusion', '??? ??? / ??? ???')}">
  <option>Blood Group (ABO) & Rh Typing</option>
  <option>Crossmatch (Major & Minor)</option>
  <option>Direct Coombs Test (DAT)</option>
  <option>Indirect Coombs Test (IAT)</option>
  <option>Antibody Screening Panel</option>
  <option>Cold Agglutinins</option>
</optgroup>
<optgroup label="{tr('Blood Gas & Electrolytes', '???? ??? ???????')}">
  <option>Arterial Blood Gas (ABG)</option>
  <option>Venous Blood Gas (VBG)</option>
  <option>Lactate (Lactic Acid)</option>
  <option>Ionized Calcium</option>
  <option>Methemoglobin / Carboxyhemoglobin</option>
</optgroup>
<optgroup label="{tr('Therapeutic Drug Monitoring', '????? ??? ??????)}">
  <option>Digoxin Level</option>
  <option>Phenytoin (Dilantin) Level</option>
  <option>Valproic Acid Level</option>
  <option>Carbamazepine Level</option>
  <option>Lithium Level</option>
  <option>Vancomycin Level (Trough/Peak)</option>
  <option>Gentamicin / Amikacin Level</option>
  <option>Theophylline Level</option>
  <option>Methotrexate Level</option>
  <option>Tacrolimus / Cyclosporine Level</option>
</optgroup>
<optgroup label="{tr('Special Chemistry', '????? ?????')}">
  <option>Protein Electrophoresis (SPEP)</option>
  <option>Immunoglobulins (IgA, IgG, IgM, IgE)</option>

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        <option>Complement C3 / C4</option>
        <option>Ammonia Level</option>
        <option>Homocysteine</option>
        <option>Ceruloplasmin / Copper</option>
        <option>Lactate Dehydrogenase (LDH)</option>
        <option>Haptoglobin</option>
        <option>Procalcitonin (PCT)</option>
        <option>BNP / NT-proBNP</option>
        <option>Fibrinogen</option>
        <option>Anti-Xa (Heparin) Assay</option>
        <option>Cystatin C</option>
        <option>Microalbumin (Urine)</option>
        <option>24hr Urine Protein / Creatinine Clearance</option>
        <option>Serum Free Light Chains (Kappa/Lambda)</option>
    </optgroup>
    <optgroup label="$tr('Toxicology & Trace Elements', '????? ??????? ??????')">
        <option>Myoglobin</option>
        <option>Vitamin A (Retinol)</option>
        <option>Zinc Level</option>
        <option>Selenium Level</option>
        <option>Lead Level (Blood)</option>
        <option>Mercury Level (Blood)</option>
        <option>Urine Drug Screen (UDS)</option>
        <option>Serum Ethanol (Alcohol) Level</option>
        <option>Acetaminophen (Paracetamol) Level</option>
        <option>Salicylate (Aspirin) Level</option>
    </optgroup>
    <optgroup label="$tr('Other', '????')">
        <option>$tr('Other Specific Test (Specify in details)', '?? ? (?? ? ??????)')</option>
    </optgroup>
</select>
</div>
<div class="form-group mb-12"><label>$tr('Details', '????????')</label><input class="form-input" id=
"labDirectDesc"></div>
    <button class="btn btn-success w-full" onclick="sendDirectLab()">? $tr('Direct Lab Order', '?? ?????
')</button>
</div>
<div class="flex-column" style="flex:2">
    <div class="card glass-card-premium mb-16">
        <div class="card-title">? $tr('Barcode Scanner', '???? ???????')</div>
        <div class="flex gap-8"><input class="form-input" id="labBarcodeInput" placeholder="$tr('Scan barco
de or enter order ID...', '???? ??????? ?? ???? ???? ?????...')>? style="flex:3" onkeydown="if(event.key==='En
ter')scanLabBarcode()"><button class="btn btn-primary" onclick="scanLabBarcode()" style="flex:1">? $tr('Searc
h', '???')</button></div>
        <div id="labScanResult" class="mt-16"></div>
    </div>
    <div class="card glass-card-premium">
        <div class="card-title">? $tr('Lab Orders', '????? ???????')</div>
        <input class="search-filter" placeholder="$tr('Search...', '????...')>? oninput="filterTable(this, 'l
abT')">
        <div id="labT"><div class="table-wrapper"><table class="data-table"><thead><tr>
            <th>$tr('Barcode', '????????')</th><th>$tr('Patient', '??????')</th><th>$tr('Type', '?????')</th><th>$tr('Normal Range', '?????? ???????')</th><th>$tr('Status', '??????')</th><th>$tr('Date', '?????
??')</th><th>$tr('Report & Results', '???????? ?????????')</th><th>$tr('Actions', '????????')</th>

```



```

</tr></thead><tbody>
  ${orders.length === 0 ? `<tr><td colspan="8"><div class="empty-state-card"><div class="empty-state-i
con">?</div><div>${tr('No lab orders found', '?? ???? ????? ?????')}</div></div></tr>` : orders.map(o =>
{
  const pt = patients.find(p => p.id == o.patient_id);
  const nRange = getLabNormalRange(o.order_type, pt ? pt.gender : ''); return `<tr>
    <td><svg id="labBC${safeId(o.id)}" class="barcode-svg"></svg><br><button class="btn btn-sm btn-inf
o" onclick="printLabBarcode(${safeId(o.id)}, '${jsStr(o.patient_name || '')}', '${jsStr(o.order_type || '')}',
${safeId(o.patient_id)})" style="margin-top:4px;font-size:11px">?? ${tr('Print', '????')}</button></td>
    <td>${escapeHTML(o.patient_name || '')}</td><td>${escapeHTML(o.order_type)}</td>
    <td style="font-size:11px;max-width:200px;color:var(--text-dim);white-space:pre-wrap">${nRange ||
'-'}</td>
    <td>${statusBadge(o.status)}</td><td>${o.created_at ? new Date(o.created_at).toLocaleString('ar-SA
', { dateStyle: 'short', timeStyle: 'short' }) : ''}</td>
    <td>${o.status === 'Done' && o.results ? `<div style="max-width:200px;padding:6px 10px;background:
var(--hover);border-radius:6px;font-size:12px;white-space:pre-wrap">${escapeHTML(o.results)}</div>` : o.status
!== 'Requested' ? `<textarea class="form-input form-textarea" id="labRpt${safeId(o.id)}" rows="2" placeholder
="${tr('Write report...', '???? ????????.')}" style="min-height:60px;font-size:12px">${escapeHTML(o.results |
| '')}</textarea><button class="btn btn-sm btn-primary mt-8" onclick="saveLabReport(${safeId(o.id)})">? ${tr('
Save', '???')}</button>` : `<span style="color:var(--text-dim)">?</span>`</td>
    <td>${o.status !== 'Done' ? `<button class="btn btn-sm btn-success" onclick="updateLabStatus(${saf
eId(o.id)},'${o.status === 'Requested' ? 'In Progress' : 'Done'}')">? ${o.status === 'Requested' ? tr('Start',
'????') : tr('Complete', '????')}</button>` : `<span class="badge badge-success">?</span>`</td>
    </tr>`;
  }).join('')}
</tbody></table></div></div>
</div>
</div>
</div>
<div id="lisPanel" class="mt-16"></div>`;
setTimeout(() => { orders.forEach(o => { try { JsBarcode('#labBC' + o.id, 'LAB-' + o.id + '-' + (o.patient_n
ame || '').replace(/^[^a-zA-Z0-9]/g, '').substring(0, 8), { format: 'CODE128', width: 1.2, height: 35, fontSize
: 9, displayValue: true, margin: 2, textMargin: 1 }); } catch (e) { } }); }, 100);
try { renderLisPanel(); } catch (e) { console.log('LIS panel error:', e); }
}

// =====
// E3 LIS panel ? specimen lifecycle, structured results (verify/hold),
// critical call-back capture, QC entry. All output escaped via escapeHTML/safeId.
// Degrades silently if the E3 endpoints are not yet deployed (404/500 -> empty).
// =====
async function renderLisPanel() {
  const host = document.getElementById('lisPanel');
  if (!host) return;
  let samples = [], results = [], qc = [];
  try { samples = await API.get('/api/lab/samples'); } catch (e) { samples = null; }
  // API.request returns parsed JSON even on errors; if the endpoint is missing/blocked it
  // returns a non-array {error} object. Render nothing rather than break the legacy screen.
  if (samples === null || !Array.isArray(samples)) { host.innerHTML = ''; return; }
  try { results = await API.get('/api/lab/results'); } catch (e) { results = []; }
  if (!Array.isArray(results)) results = [];
  try { qc = await API.get('/api/lab/qc'); } catch (e) { qc = []; }
  if (!Array.isArray(qc)) qc = [];

```

```

const stateBadge = (st) => {
  const map = { Collected: '#64748b', Received: '#3b82f6', InProcess: '#f59e0b', Verified: '#8b5cf6', Report
ed: '#22c55e', Rejected: '#ef4444' };
  return `<span class="badge" style="background:${map[st] || '#64748b'}">${escapeHTML(st)}</span>`;
};
const resStatusBadge = (r) => {
  if (r.is_critical) return `<span class="badge" style="background:#dc2626">? ${tr('CRITICAL', '????')}</spa
n>`;
  const m = { verified: '#22c55e', held: '#f59e0b', pending: '#64748b' };
  return `<span class="badge" style="background:${m[r.status] || '#64748b'}">${escapeHTML(r.status || '')}</
span>`;
};

host.innerHTML = `
<div class="card glass-card-premium">
  <div class="card-title">? ${tr('LIS ? Sample Lifecycle', '???? ?????? ? ???? ???? ?????')}</div>
  <div class="flex gap-8 mb-12" style="flex-wrap:wrap">
    <input class="form-input" id="lisCollectOrder" placeholder="${tr('Lab order ID (optional)', '??? ??? ?
????? (???????)')}" style="flex:1;min-width:140px">
    <input class="form-input" id="lisCollectPatient" placeholder="${tr('Patient ID (optional)', '??? ?????
? (???????)')}" style="flex:1;min-width:140px">
    <button class="btn btn-success" onclick="lisCollectSample()">? ${tr('Collect Sample', '??? ?????')}</bu
tton>
  </div>
  <div class="table-wrapper"><table class="data-table"><thead><tr>
    <th>${tr('Barcode', '????????')}</th><th>${tr('Patient', '??????')}</th><th>${tr('State', '??????')}</
th><th>${tr('Actions', '????????')}</th>
  </tr></thead><tbody>
    ${samples.length === 0 ? `<tr><td colspan="4"><div class="empty-state-card"><div class="empty-state-icon
">?</div><div>${tr('No samples', '?? ???? ?????')}</div></div></td></tr>` : samples.map(sm => `<tr>
    <td style="font-family:monospace;font-size:12px">${escapeHTML(sm.barcode || '')}</td>
    <td>${escapeHTML(sm.patient_name || (sm.patient_id ? '#' + safeId(sm.patient_id) : '?'))}</td>
    <td>${stateBadge(sm.state)}</td>
    <td>
      ${sm.state === 'Collected' ? `<button class="btn btn-sm btn-primary" onclick="lisSampleAction(${safe
Id(sm.id)},'receive')">? ${tr('Receive', '??????')}</button>` : ''}
      ${sm.state === 'Received' ? `<button class="btn btn-sm btn-info" onclick="lisSampleAction(${safeId(s
m.id)},'process')">?? ${tr('Process', '??????')}</button>` : ''}
      ${['Collected', 'Received', 'InProcess'].includes(sm.state) ? `<button class="btn btn-sm btn-danger"
onclick="lisRejectSample(${safeId(sm.id)})">? ${tr('Reject', '???')}</button>` : ''}
      ${sm.state === 'InProcess' ? `<button class="btn btn-sm btn-success" onclick="lisOpenResultForm(${sa
feId(sm.id)})">? ${tr('Enter Result', '????? ?????')}</button>` : ''}
    </td>
  </tr>`).join('')}
  </tbody></table></div>
  <div id="lisResultForm" class="mt-12"></div>
</div>

<div class="card glass-card-premium mt-16">
  <div class="card-title">? ${tr('Results ? Verify / Hold / Report', '??????? ? ???? / ???? / ?????')}</d
iv>
  <div class="table-wrapper"><table class="data-table"><thead><tr>
    <th>${tr('Test', '????')}</th><th>${tr('Value', '??????')}</th><th>${tr('Flag', '????')}</th><th>${t
r('Status', '????')}</th><th>${tr('Hold Reasons', '????? ?????')}</th><th>${tr('Actions', '???????')}</th>

```

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</tr></thead><tbody>
  ${results.length === 0 ? `<tr><td colspan="6"><div class="empty-state-card"><div class="empty-state-icon
">?</div><div>${tr('No results', '?? ???? ?????')}</div></div></td></tr>` : results.map(r => `<tr>
  <td>${escapeHTML(r.test_name || '')}${r.loinc ? `<span style="color:var(--text-dim);font-size:11px">(${
escapeHTML(r.loinc)})</span>` : ''}</td>
  <td>${escapeHTML(String(r.value ?? ''))} ${escapeHTML(r.unit || '')}</td>
  <td>${escapeHTML(r.abnormal_flag || '?')}</td>
  <td>${resStatusBadge(r)}${r.reported ? `<span class="badge" style="background:#22c55e">? ${tr('Report
ed', '????')}</span>` : ''}</td>
  <td style="font-size:11px;color:var(--text-dim);max-width:180px">${escapeHTML(r.hold_reasons || '')}</
td>
  <td>
    ${r.status === 'held' ? `<button class="btn btn-sm btn-primary" onclick="lisVerifyResult(${safeId(r.
id)})">? ${tr('Verify', '??????')}</button>` : ''}
    ${r.is_critical ? `<button class="btn btn-sm btn-danger" onclick="lisCriticalCallback(${safeId(r.id)
})">? ${tr('Call-back', '????? ?????')}</button>` : ''}
    ${r.status === 'verified' && !r.reported ? `<button class="btn btn-sm btn-success" onclick="lisRepor
tResult(${safeId(r.id)}, ${r.is_critical ? 'true' : 'false'})">? ${tr('Report', '?????')}</button>` : ''}
  </td>
</tr>`).join('')}
</tbody></table></div>
</div>

<div class="card glass-card-premium mt-16">
  <div class="card-title">? ${tr('Quality Control (Levey-Jennings)', '??? ??????')}</div>
  <div class="flex gap-8 mb-12" style="flex-wrap:wrap">
    <input class="form-input" id="qcAnalyzer" placeholder="${tr('Analyzer', '??????')}" style="flex:1;min
-width:110px">
    <input class="form-input" id="qcAnalyte" placeholder="${tr('Analyte', '?????')}" style="flex:1;min-wi
dth:110px">
    <input class="form-input" id="qcLevel" placeholder="${tr('Level', '??????')}" style="width:90px">
    <input class="form-input" id="qcValue" type="number" step="any" placeholder="${tr('Value', '?????')}"
style="width:100px">
    <input class="form-input" id="qcTarget" type="number" step="any" placeholder="${tr('Target', '?????')}"
" style="width:100px">
    <input class="form-input" id="qcSd" type="number" step="any" placeholder="SD" style="width:80px">
    <button class="btn btn-primary" onclick="lisEnterQc()">? ${tr('Enter QC', '?????')}</button>
  </div>
  <div class="table-wrapper"><table class="data-table"><thead><tr>
    <th>${tr('Analyzer', '??????')}</th><th>${tr('Analyte', '?????')}</th><th>${tr('Level', '??????')}</
th><th>${tr('Value', '?????')}</th><th>Z</th><th>${tr('Rule', '??????')}</th>
  </tr></thead><tbody>
    ${qc.length === 0 ? `<tr><td colspan="6"><div class="empty-state-card"><div class="empty-state-icon">?</
div><div>${tr('No QC points', '?? ???? ?????')}</div></div></td></tr>` : qc.map(q => `<tr style="${q.breach ? '
background:rgba(239,68,68,.12)'" : ''}">
    <td>${escapeHTML(q.analyzer || '')}</td><td>${escapeHTML(q.analyte || '')}</td><td>${escapeHTML(q.leve
l || '')}</td>
    <td>${escapeHTML(String(q.value ?? ''))}</td><td>${q.z !== null && q.z !== undefined ? escapeHTML(Numb
er(q.z).toFixed(2)) : '?'}</td>
    <td>${q.breach ? `<span class="badge" style="background:#ef4444">?? ${escapeHTML(q.westgard_flag || 'b
reach')}</span>` : `<span class="badge" style="background:#22c55e">${escapeHTML(q.westgard_flag || 'in_control
')}</span>`}</td>
  </tr>`).join('')}
</tbody></table></div>

```

```

    </div>`;
}

// API.request returns parsed JSON even on non-2xx (it does not throw); treat {error} as failure.
function lisErr(out) { return out && typeof out === 'object' && out.error ? (out.code ? out.code + ': ' : '')
+ out.error : null; }
window.lisCollectSample = async () => {
  const order = document.getElementById('lisCollectOrder').value.trim();
  const patient = document.getElementById('lisCollectPatient').value.trim();
  try {
    const out = await API.post('/api/lab/samples', { lab_order_id: order || null, patient_id: patient || null
});
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    showToast(tr('Sample collected', '?? ??? ??????'));
    await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisSampleAction = async (id, action) => {
  try {
    const out = await API.put(`/api/lab/samples/${id}`, { action });
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    showToast(tr('Updated', '?? ??????')); await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisRejectSample = async (id) => {
  const reason = prompt(tr('Rejection reason:', '??? ?????:'));
  if (!reason) return;
  try {
    const out = await API.put(`/api/lab/samples/${id}`, { action: 'reject', rejected_reason: reason });
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    showToast(tr('Sample rejected', '?? ??? ??????')); await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisOpenResultForm = (sampleId) => {
  const f = document.getElementById('lisResultForm');
  if (!f) return;
  f.innerHTML = `<div class="card glass-card-premium" style="border:1px solid var(--accent)">
    <div class="card-title">? ${tr('Enter Result for Sample', '????? ????? ??????')} #${safeId(sampleId)}</div>
  >
    <div class="flex gap-8 mb-8" style="flex-wrap:wrap">
      <input class="form-input" id="resTestName" placeholder="${tr('Test name', '??? ?????')}" style="flex:1;min-width:140px">
        <input class="form-input" id="resLoinc" placeholder="LOINC" style="width:120px">
        <input class="form-input" id="resValue" placeholder="${tr('Value', '?????')}" style="width:110px">
        <input class="form-input" id="resUnit" placeholder="${tr('Unit', '?????')}" style="width:90px">
        <input class="form-input" id="resLow" type="number" step="any" placeholder="${tr('Ref low', '???? ??????')}" style="width:100px">
        <input class="form-input" id="resHigh" type="number" step="any" placeholder="${tr('Ref high', '???? ??????')}" style="width:100px">
        <button class="btn btn-success" onclick="lisSubmitResult(${safeId(sampleId)})">? ${tr('Save & Auto-verify', '??? ?????? ???')}</button>
      </div>
    <div style="font-size:11px;color:var(--text-dim)">${tr('Auto-verify HOLDS on out-of-range, critical, missing range, or significant delta.', '????? ?????? ??? ?????? ?? ?????? ?? ?????? ?????? ?? ?????? ?????? ??')}</div>
  `;
};

```

```

??? ????.')}</div>
</div>`;
};
window.lisSubmitResult = async (sampleId) => {
  const body = {
    lab_sample_id: sampleId,
    test_name: document.getElementById('resTestName').value.trim(),
    loinc: document.getElementById('resLoinc').value.trim(),
    value: document.getElementById('resValue').value.trim(),
    unit: document.getElementById('resUnit').value.trim(),
    ref_low: document.getElementById('resLow').value.trim(),
    ref_high: document.getElementById('resHigh').value.trim(),
  };
  if (!body.test_name || body.value === '') { showToast(tr('Test name and value required', '????? ??????? ????'
  ???'), 'error'); return; }
  try {
    const out = await API.post('/api/lab/results', body);
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    const v = out.verdict || {};
    if (v.status === 'held') showToast('?? ' + tr('HELD for review', '??????? ?????????') + (v.is_critical ? '
    ? ' + tr('CRITICAL', '????') : '') + ' ? ' + (v.reasons || []).join(', '), 'error');
    else showToast('?? ' + tr('Auto-verified', '?? ????????? ??????'));
    document.getElementById('lisResultForm').innerHTML = '';
    await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisVerifyResult = async (id) => {
  try {
    const out = await API.put(`/api/lab/results/${id}/verify`, {});
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    showToast(tr('Verified', '?? ?????????')); await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisCriticalCallback = async (id) => {
  const notified = prompt(tr('Notified whom (physician name/contact)?', '?? ????? ?? (???/??? ??????)?'));
  if (!notified) return;
  try {
    const out = await API.post(`/api/lab/results/${id}/callback`, { notified_to: notified, ack: 1 });
    const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
    showToast(tr('Call-back documented', '?? ?????? ?????????'));
    await renderLisPanel();
  } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};
window.lisReportResult = async (id, isCritical) => {
  try {
    const out = await API.put(`/api/lab/results/${id}/report`, {});
    const err = lisErr(out);
    if (err) {
      if (err.includes('CRITICAL_CALLBACK_REQUIRED') || err.includes('call-back')) {
        showToast('? ' + tr('Critical result requires a documented call-back first', '??????? ?????? ?????? ???
        ?? ?????? ?????? ??????'), 'error');
      } else { showToast(tr('Error', '???') + ': ' + err, 'error'); }
      return;
    }
  }
}

```

```

        showToast(tr('Reported', '?? ??????'));
        await renderLisPanel();
    } catch (e) { showToast(tr('Error', '???') + ': ' + ((e && e.message) || String(e)), 'error'); }
};

window.lisEnterQc = async () => {
    const body = {
        analyzer: document.getElementById('qcAnalyzer').value.trim(),
        analyte: document.getElementById('qcAnalyte').value.trim(),
        level: document.getElementById('qcLevel').value.trim(),
        value: document.getElementById('qcValue').value.trim(),
        target: document.getElementById('qcTarget').value.trim(),
        sd: document.getElementById('qcSd').value.trim(),
    };
    try {
        const out = await API.post('/api/lab/qc', body);
        const err = lisErr(out); if (err) { showToast(tr('Error', '???') + ': ' + err, 'error'); return; }
        if (out.flag && out.flag.breach) showToast('?? ' + tr('QC BREACH', '????? ????') + ' (' + (out.flag
.rule || '') + ')', 'error');
        else showToast(tr('QC recorded', '?? ????? ?????'));
        await renderLisPanel();
    } catch (e) { showToast(tr('Error', '???') + ': ' + (e.message || e), 'error'); }
};

window.printLabBarcode = (orderId, patientName, testType, patientId) => {
    const labelId = 'LAB-' + orderId;
    const timestamp = new Date().toLocaleString();
    const testCode = testType.substring(0, 10).toUpperCase().replace(/^[^A-Z]/g, '') || 'GEN';

    const modalHtml = `
        <div style="display:flex;gap:20px;align-items:start">
            <div id="thermalLabelContainer" style="width:260px;background:#fff;color:#000;border:2px solid #000;border-radius:4px;padding:12px;font-family:'Courier New', monospace;box-shadow:0 4px 10px rgba(0,0,0,0.15)">
                <div style="text-align:center;border-bottom:1px dashed #000;padding-bottom:4px;margin-bottom:8px;font-weight:bold;font-size:14px">
                    NAMA LAB RECEIPT
                </div>
                <div style="font-size:11px;line-height:1.3">
                    <strong>PATIENT ID:</strong> P-${patientId || '1001'}<br>
                    <strong>PATIENT:</strong> ${escapeHTML(patientName.toUpperCase())}<br>
                    <strong>TEST CODE:</strong> ${testCode}<br>
                    <strong>ORDER ID:</strong> ${orderId}<br>
                    <strong>DATE:</strong> ${timestamp}<br>
                </div>
                <div style="text-align:center;margin:10px 0">
                    <svg id="thermalLabelBarcode" style="width:100%;height:45px"></svg>
                </div>
                <div style="border-top:1px dashed #000;padding-top:4px;margin-top:8px;font-size:10px;text-align:center">
                    <strong>SCAN STATUS:</strong> <span id="lblScanStatus" style="background:#f3f4f6;padding:2px 4px;border:1px solid #000;font-weight:bold">READY_TO_SCAN</span>
                </div>
            </div>

            <div style="flex:1;display:flex;flex-direction:column;gap:10px">
                <h4 style="margin:0">${tr('Thermal Label Simulator', '????? ?????? ??????')}</h4>

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<p style="font-size:12px;color:var(--text-muted)">
    ${tr('This simulates the direct printing of a ZPL/thermal barcode label for clinical specimens. You
can scan this barcode to automatically route the sample into the analyzer.', '??? ????? ??????? ??????? ?????
???????? ??????? ??????? ??????. ????? ?????? ?????? ????????? ?????? ?????? ????????? ??? ??? ???????.')}}
</p>
<button class="btn btn-secondary w-full" onclick="window.simulateBarcodeScan('${labelId}', ${orderId})
">
    ?? ${tr('Simulate Barcode Scan', '????? ?? ?????????')}
</button>
<button class="btn btn-primary w-full" onclick="window.printThermalLabelContent('${labelId}')">
    ?? ${tr('Print to Label Printer', '????? ?? ?????? ?????????')}
</button>
</div>
</div>
`;

showModal(tr('LIS Barcode Simulator', '????? ?????? ?????????'), modalHtml);

setTimeout(() => {
    try {
        JsBarcode('#thermalLabelBarcode', labelId, {
            format: 'CODE128',
            width: 1.5,
            height: 35,
            fontSize: 9,
            displayValue: true,
            margin: 2
        });
    } catch (e) { }
}, 100);
};

window.updateLabStatus = async (id, status) => {
    try { await API.put(`/api/lab/orders/${id}`, { status }); showToast(tr('Updated', '?? ?????????')); await navi
gateTo(4); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.saveLabReport = async (id) => {
    const rpt = document.getElementById('labRpt' + id).value.trim();
    if (!rpt) { showToast(tr('Write the report first', '???? ??????? ??????'), 'error'); return; }
    try {
        // Get test name for critical value check
        const orderRow = document.getElementById('labRpt' + id)?.closest('tr') || document.getElementById('labRpt'
+ id)?.closest('.card');
        const testName = orderRow?.querySelector('td')?.textContent || orderRow?.querySelector('.badge')?.textCont
ent || '';
        const critical = checkCriticalLabValue(testName, rpt);
        if (critical) {
            alert('?? ' + tr('CRITICAL VALUE ALERT!', '????? ??? ????!') + '\n\n' + critical.test + ': ' + critical
.value + ' ' + (critical.range.unit || '') + '\n' + tr('Status: ', '?????: ') + critical.status + '\n' + tr('
Normal range: ', '????? ??????: ') + critical.range.low + ' - ' + critical.range.high + ' ' + (critical.rang
e.unit || '') + '\n\n' + tr('Please notify the attending physician immediately!', '???? ?????? ?????? ??????? ?
????!'));
        }
        await API.put(`/api/lab/orders/${id}`, { results: rpt });
    }
};

```

```

    showToast(critical ? tr('?? Report saved - CRITICAL VALUE!', '?? ?? ??? ??????? - ??? ????!') : tr('Report saved!', '?? ??? ???????!'), critical ? 'error' : 'success');
    await navigateTo(4);
  }
  catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.sendDirectLab = async () => {
  try {
    await API.post('/api/lab/orders/direct', { patient_id: document.getElementById('labPatientId')?.value || ' ', order_type: document.getElementById('labDirectType').value, description: document.getElementById('labDirectDesc')?.value || ' ' });
    showToast(tr('Lab order created!', '?? ????? ??????!')); await navigateTo(4);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

async function loadPendingPaymentOrders() {
  try {
    const orders = await API.get('/api/orders/pending-payment');
    const container = document.getElementById('pendingPaymentTable');
    if (!container) return;
    if (!orders.length) {
      container.innerHTML = `<div style="text-align:center;padding:20px;color:var(--text-dim)">? ${tr('No pending payment orders', '?? ??? ???? ?????? ??????')}</div>`;
      return;
    }
    container.innerHTML = `<div class="table-wrapper"><table class="data-table"><thead><tr>
      <th>#</th><th>${tr('Patient', '????')}</th><th>${tr('File #', '?? ????')}</th>
      <th>${tr('Dept', '????')}</th><th>${tr('Test/Scan', '????/????')}</th>
      <th>${tr('Details', '????')}</th><th>${tr('Date', '????')}</th>
      <th>${tr('Action', '????')}</th>
    </tr></thead><tbody>
      ${orders.map(o => `<tr style="background:${o.is_radiology ? '#fef9c3' : '#dbeafe'}">
        <td>${escapeHTML(o.id)}</td>
        <td><strong>${escapeHTML(o.patient_name || o.name_en || '')}</strong></td>
        <td>${escapeHTML(o.file_number || '')}</td>
        <td>${o.is_radiology ? `<span class="badge badge-warning">? ${tr('Radiology', '????')}</span>` : `<span class="badge badge-info">? ${tr('Lab', '????')}</span>`}</td>
        <td>${escapeHTML(o.order_type || '')}</td>
        <td>${escapeHTML(o.description || '')}</td>
        <td>${escapeHTML(o.created_at?.split('T')[0] || '')}</td>
        <td>
          <button class="btn btn-sm btn-success" onclick="approveOrderPayment(${safeId(o.id)}, '${jsStr(o.patient_name || o.name_en || '')}', '${jsStr(o.order_type || '')}', ${o.is_radiology ? 'true' : 'false'})">
            ? ${tr('Pay & Approve', '???? ?????')}
          </button>
        </td>
      </tr>`).join('')}
    </tbody></table></div>`;
  } catch (e) { console.error(e); }
}

window.approveOrderPayment = async (orderId, patientName, testType, isRad) => {
  const deptName = isRad ? tr('Radiology', '????') : tr('Lab', '????');
  const price = prompt(`${tr('Enter price for', '????')} "${testType}" ${tr('for patient', '????')} ${patientName}: \n${tr('Enter 0 for free', '???? 0 ? ? ?????')}`);
  if (price === null) return;

```



```
const priceNum = parseFloat(price) || 0;
const payMethod = priceNum > 0 ? (prompt(`${tr('Payment method', '????? ??????')}`):\n1 = ${tr('Cash', '???')}\n2 = ${tr('Card/POS', '????')}\n3 = ${tr('Transfer', '????')}`) || '1') : '1';
const methods = { '1': 'Cash', '2': 'Card', '3': 'Transfer' };
try {
  await API.put(`/api/orders/${orderId}/approve-payment`, { price: priceNum, payment_method: methods[payMethod] || 'Cash' });
  showToast(`${tr('Paid & sent to', '?? ?????? ?????????? ???')}` ${deptName}!`, 'success');
  loadPendingPaymentOrders();
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.scanLabBarcode = async () => {
  const code = document.getElementById('labBarcodeInput').value.trim(); if (!code) return;
  const m = code.match(/LAB-(\d+)/); const oid = m ? m[1] : code;
  try {
    const orders = await API.get('/api/lab/orders'); const o = orders.find(x => x.id == oid);
    document.getElementById('labScanResult').innerHTML = o ? `
```

```

    <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
</div>
<div class="split-layout">
  <div class="card glass-card-premium">
    <div class="skeleton-bar title mb-12"></div>
    <div class="skeleton-bar mb-8"></div>
  </div>
  <div class="card glass-card-premium" style="flex:2">
    <div class="skeleton-bar title mb-12"></div>
    <div class="skeleton-bar mb-8"></div>
  </div>
</div>
`;

let orders = [];
let patients = [];
try {
  [orders, patients] = await Promise.all([
    API.get('/api/radiology/orders'),
    API.get('/api/patients')
  ]);
} catch (e) {
  el.innerHTML = `
    <div class="page-title">? ${tr('Radiology', '??????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load radiology data', '??? ?? ????? ?????? ??????')}</h3>
      <p>${escapeHTML(e.message || e)}</p>
      <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
button>
    </div>
  `;
  return;
}
el.innerHTML = `<div class="page-title">? ${tr('Radiology', '??????')}</div>
<div class="stats-grid">
  <div class="stat-card" style="--stat-color:#f59e0b"><div class="stat-label">${tr('Pending', '????????')}
</div><div class="stat-value">${orders.filter(o => o.status === 'Requested').length}</div></div>
  <div class="stat-card" style="--stat-color:#3b82f6"><div class="stat-label">${tr('In Progress', '??? ???
??')}</div><div class="stat-value">${orders.filter(o => o.status === 'In Progress').length}</div></div>
  <div class="stat-card" style="--stat-color:#4ade80"><div class="stat-label">${tr('Completed', '????')}<
/div><div class="stat-value">${orders.filter(o => o.status === 'Done').length}</div></div>
</div>
<div class="split-layout">
  <div class="card glass-card-premium" style="flex:1">
    <div class="card-title">? ${tr('Direct Radiology Order', '????? ??? ?????')}</div>
    <div class="form-group mb-12"><label>${tr('Select Patient (Optional)', '???? ???? (???????)')}</label>
    <select class="form-input" id="radPatientId"><option value="">--</option>${patients.map(p => `<optio
n value="${safeId(p.id)}">${escapeHTML(p.name_ar || p.name_en)}</option>`).join('')}</select>
    </div>
    <div class="form-group mb-12"><label>${tr('Scan Type', '??? ?????')}</label>
    <select class="form-input" id="radDirectType">
      <optgroup label="${tr('X-Ray', '?????? ?????????')}">
        <option>X-Ray Chest (PA/LAT)</option>

```

```

<option>X-Ray Abdomen (Erect/Supine)</option>
<option>X-Ray KUB (Kidney, Ureter, Bladder)</option>
<option>X-Ray Cervical Spine (AP/LAT/Open Mouth)</option>
<option>X-Ray Thoracic Spine</option>
<option>X-Ray Lumbar Spine (AP/LAT)</option>
<option>X-Ray Pelvis (AP)</option>
<option>X-Ray Skull / Facial Bones / PNS</option>
<option>X-Ray Shoulder / Clavicle</option>
<option>X-Ray Arm (Humerus/Radius/Ulna)</option>
<option>X-Ray Hand / Wrist</option>
<option>X-Ray Hip/Femur</option>
<option>X-Ray Knee (AP/LAT/Skyline)</option>
<option>X-Ray Ankle / Foot</option>
<option>X-Ray Bone Age</option>
</optgroup>
<optgroup label="{tr('Ultrasound', '?????? ???? / ??????')}">
  <option>Ultrasound Abdomen (Whole)</option>
  <option>Ultrasound Pelvis (Transabdominal/Transvaginal)</option>
  <option>Ultrasound Abdomen & Pelvis</option>
  <option>Ultrasound KUB / Prostate</option>
  <option>Ultrasound Thyroid / Neck</option>
  <option>Ultrasound Breast</option>
  <option>Ultrasound Scrotum / Testicular</option>
  <option>Obstetric Ultrasound (1st Trimester/Viability)</option>
  <option>Obstetric Ultrasound (Anomaly Scan 2nd Trimester)</option>
  <option>Obstetric Ultrasound (Growth 3rd Trimester)</option>
  <option>Folliculometry (Ovulation Tracking)</option>
  <option>Ultrasound Soft Tissue / Swelling</option>
  <option>Doppler Ultrasound - Carotid</option>
  <option>Doppler Ultrasound - Lower Limb Venous (DVT)</option>
  <option>Doppler Ultrasound - Lower Limb Arterial</option>
  <option>Doppler Ultrasound - Renal Artery</option>
  <option>Doppler Ultrasound - Obstetrics / Umbilical Artery</option>
  <option>Echocardiogram (Echo - Heart)</option>
</optgroup>
<optgroup label="{tr('CT Scan', '????? ?????????')}">
  <option>CT Brain / Head (Without Contrast)</option>
  <option>CT Brain / Head (With Contrast)</option>
  <option>CT PNS (Paranasal Sinuses)</option>
  <option>CT Neck (With Contrast)</option>
  <option>CT Chest (HRCT) Without Contrast</option>
  <option>CT Chest / Lungs (With Contrast)</option>
  <option>CT Abdomen & Pelvis (Without Contrast - Triphasic)</option>
  <option>CT Abdomen & Pelvis (With Contrast)</option>
  <option>CT KUB (Stone Protocol - Non Contrast)</option>
  <option>CT Urography (With Contrast)</option>
  <option>CT Cervical Spine</option>
  <option>CT Lumbar Spine</option>
  <option>CT Angiography - Pulmonary (CTPA)</option>
  <option>CT Angiography - Brain</option>
  <option>CT Angiography - Aorta / Lower Limbs</option>
  <option>CT Virtual Colonoscopy</option>
</optgroup>
<optgroup label="{tr('MRI', '????? ?????????')}">

```

☐MRI Brain (Without Contrast)

☐MRI Brain (With Contrast)

☐MRI Pituitary Fossa

☐MRI Cervical Spine

☐MRI Thoracic Spine

☐MRI Lumbar Spine

☐MRI Whole Spine

☐MRI Pelvis (Male/Female)

☐MRI Prostate (Multiparametric)

☐MRI Shoulder Joint

☐MRI Knee Joint

☐MRI Ankle / Wrist Joint

☐MRI Abdomen

☐MRCP (Magnetic Resonance Cholangiopancreatography)

☐MR Venography (MRV)

☐MRA (Magnetic Resonance Angiography) - Brain

☐Specialized Imaging & Scans

☐Mammogram (Bilateral/Unilateral)

☐DEXA Scan (Bone Density)

☐Fluoroscopy - Barium Swallow

☐Fluoroscopy - Barium Meal / Follow Through

☐Fluoroscopy - Barium Enema

☐Fluoroscopy - HSG (Hysterosalpingography)

☐Fluoroscopy - IVP (Intravenous Pyelogram)

☐Panoramic Dental X-Ray (OPG)

☐Cephalometric X-Ray

☐CBCT (Cone Beam CT for Dentistry)

☐PET Scan (Positron Emission Tomography)

☐Cardiology & Neuro

☐ECG (Electrocardiogram)

☐Holter Monitor (24/48 Hours)

☐Ambulatory Blood Pressure Monitoring (ABPM)

☐Treadmill Stress Test (TMT)

☐EEG (Electroencephalogram)

☐EMG (Electromyography) / NCS

☐Spirometry / Lung Function Test

☐Upper GI Endoscopy (OGD)

☐Colonoscopy

☐Other

☐Other Scan (Specify in details)

☐Details

☐Barcode Scanner

```


<input class="form-input" id="radBarcodeInput" placeholder="{tr('Scan barcode or enter order ID...', '???? ??????? ?? ???? ?? ??????..')}" style="flex:3" onkeydown="if(event.key==='Enter')scanRadBarcode()"><button class="btn btn-primary" onclick="scanRadBarcode()" style="flex:1">? {tr('Search', '???')}</button></div>


```

```

T</option><option>MR</option><option>US</option><option>MG</option><option>OTHER</option></select></div>
    <button class="btn btn-success" onclick="risAddToWorklist()">? ${tr('Add to Worklist', '????? ?????? ?
????')}</button>
</div>
<div class="split-layout">
    <div id="risWorklistBox" style="flex:2"></div>
    <div class="card" id="risReportBox" style="flex:1.5"><div style="color:var(--text-dim);font-size:12px"
>${tr('Select an exam to report.', '???? ?????? ?????? ??????')}</div></div>
</div>
</div>`;
window.loadRisWorklist();
setTimeout(() => { orders.forEach(o => { try { JsBarcode('#radBC' + o.id, 'RAD-' + o.id + '-' + (o.patient_n
ame || '').replace(/^[a-zA-Z0-9]/g, '').substring(0, 8), { format: 'CODE128', width: 1.2, height: 35, fontSize
: 9, displayValue: true, margin: 2, textMargin: 1 }); } catch (e) { } }); }, 100);
}
window.updateRadStatus = async (id, status) => {
    try { await API.put(`/api/radiology/orders/${id}`, { status }); showToast(tr('Updated', '?? ??????')); awai
t navigateTo(5); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.saveRadReport = async (id) => {
    const rpt = document.getElementById('radRpt' + id).value.trim();
    if (!rpt) { showToast(tr('Write the report first', '???? ?????? ??????'), 'error'); return; }
    try { await API.put(`/api/radiology/orders/${id}`, { result: rpt }); showToast(tr('Report saved!', '?? ??? ?
?????')); await navigateTo(5); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.uploadRadImage = async (id, input) => {
    if (!input.files[0]) return;
    const fd = new FormData();
    fd.append('image', input.files[0]);
    try {
        const res = await fetch(`/api/radiology/orders/${id}/upload`, { method: 'POST', body: fd, credentials: 'sa
me-origin' });
        if (res.status === 401) { window.location.href = '/login.html'; return; }
        const data = await res.json();
        if (data.success) { showToast(tr('Image uploaded!', '?? ??? ??????')); await navigateTo(5); }
        else showToast(tr('Upload failed', '??? ??????'), 'error');
    } catch (e) { showToast(tr('Error uploading', '??? ?? ??????'), 'error'); }
};
window.scanRadBarcode = async () => {
    const code = document.getElementById('radBarcodeInput').value.trim(); if (!code) return;
    const m = code.match(/RAD-(\d+)/); const oid = m ? m[1] : code;
    try {
        const orders = await API.get('/api/radiology/orders'); const o = orders.find(x => x.id == oid);
        document.getElementById('radScanResult').innerHTML = o ? `<div class="card" style="border:2px solid var(--
accent);margin-top:12px"><div class="card-title">? ${tr('Order Found', '?? ?????? ??? ??????')} #${escapeHTML(o
.id)}</div><div class="flex gap-8" style="flex-wrap:wrap"><span class="badge badge-info">? </span><span class=
"badge badge-purple">? ${escapeHTML(o.order_type)}</span>${statusBadge(o.status)}</div>${o.results ? `<div cla
ss="mt-16">${renderRadResults(o.results)}</div>` : ''}</div>` : `<div class="badge badge-danger mt-16">${tr('O
rder not found', '????? ??? ??????')}</div>`;
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

```

```
// ===== E4: RIS WORKLIST + STRUCTURED REPORTING + CRITICAL + PRIOR COMPARE =====
const RIS_STATES = ['Scheduled', 'Arrived', 'InProgress', 'Completed', 'Reported'];
const RIS_NEXT = { Scheduled: 'Arrived', Arrived: 'InProgress', InProgress: 'Completed', Completed: 'Reported', Reported: null };
window.__risExams = [];
window.__risReports = {};

// Render the RIS worklist board into #risWorklistBox (called from renderRadiology)
window.loadRisWorklist = async () => {
  const box = document.getElementById('risWorklistBox');
  if (!box) return;
  let exams = [];
  try { exams = await API.get('/api/radiology/worklist'); }
  catch (e) { box.innerHTML = `<div class="badge badge-danger">${tr('Failed to load worklist', '??? ????? ????'
? ?????')}</div>`; return; }
  window.__risExams = exams || [];
  if (!exams.length) { box.innerHTML = `<div class="empty-state-card"><div class="empty-state-icon">??</div><div>
  ${tr('No scheduled exams. Use \\"Add to Worklist\\" on an order.', '?? ????' ?????? ??????. ?????? "????? ????'
  ?? ?????? "?? ??????')}</div></div>`; return; }
  box.innerHTML = `<div class="table-wrapper"><table class="data-table"><thead><tr>
    <th>#</th><th>${tr('Patient', '?????')}</th><th>${tr('Modality', '?????')}</th><th>${tr('Exam', '????')}</th><th>${tr('State', '????')}</th><th>${tr('Actions', '????')}</th>
  </tr></thead><tbody>
    ${exams.map(e => `<tr>
      <td>${safeId(e.id)}</td>
      <td>${escapeHTML(e.patient_name || ('#' + (e.patient_id || '')))}</td>
      <td>${escapeHTML(e.modality || '')}</td>
      <td>${escapeHTML(e.exam_name || '')}</td>
      <td>${statusBadge(e.state)}</td>
      <td>
        ${RIS_NEXT[e.state]} ? `<button class="btn btn-sm btn-primary" onclick="risAdvance(${safeId(e.id)},'${escapeHTML(RIS_NEXT[e.state])}'>? ${escapeHTML(RIS_NEXT[e.state])}</button>` : `<span class="badge badge-success">?</span>`
      <button class="btn btn-sm btn-success" onclick="risOpenReport(${safeId(e.id)})>? ${tr('Report', '????')}</button>
      </td>
    </tr>`).join('')}
  </tbody></table></div>`;
};

window.risAddToWorklist = async () => {
  const orderId = safeId(document.getElementById('risOrderId')?.value);
  const modality = document.getElementById('risModality')?.value || '';
  if (!orderId) { showToast(tr('Select an order', '???? ?????'), 'error'); return; }
  try {
    await API.post('/api/radiology/worklist', { rad_order_id: orderId, modality });
    showToast(tr('Added to worklist', '??? ?????? ?????? ??????'));
    await window.loadRisWorklist();
  } catch (e) { showToast(tr('Error', '???') + ': ' + escapeHTML(e.message || ''), 'error'); }
};

window.risAdvance = async (examId, state) => {
  try { await API.put(`/api/radiology/worklist/${examId}/state`, { state }); showToast(tr('State updated', '??
  ????? ??????')); await window.loadRisWorklist(); }
}
```

```

    catch (e) { showToast(tr('Error', '???') + ': ' + escapeHTML(e.message || ''), 'error'); }
};

// Open the structured report editor for an exam (loads priors + existing reports)
window.risOpenReport = async (examId) => {
    const exam = (window.__risExams || []).find(x => x.id == examId);
    const box = document.getElementById('risReportBox');
    if (!box) return;
    let priors = [], reports = [];
    try {
        if (exam && exam.patient_id) priors = await API.get(`/api/radiology/reports/priors?patient_id=${encodeURIComponent(exam.patient_id)}&modality=${encodeURIComponent(exam.modality || '')}`);
        reports = await API.get(`/api/radiology/reports?rad_exam_id=${encodeURIComponent(examId)}`);
    } catch (e) { }
    window.__risReports[examId] = reports || [];
    const priorHtml = (priors && priors.length)
        ? priors.map(p => `
```



```

<path d="M105,30 Q140,35 160,55" stroke="#ffffff" stroke-width="3" fill="none" opacity="0.4" />
<path d="M105,50 Q150,55 170,80" stroke="#ffffff" stroke-width="3" fill="none" opacity="0.4" />
<path d="M105,70 Q155,75 175,105" stroke="#ffffff" stroke-width="3" fill="none" opacity="0.4" />
<path d="M105,90 Q160,95 180,130" stroke="#ffffff" stroke-width="3" fill="none" opacity="0.4" />
<path d="M105,110 Q160,115 180,155" stroke="#ffffff" stroke-width="3" fill="none" opacity="0.4" />

<path d="M95,20 Q60,10 30,22" stroke="#ffffff" stroke-width="4" fill="none" opacity="0.5" />
<path d="M105,20 Q140,10 170,22" stroke="#ffffff" stroke-width="4" fill="none" opacity="0.5" />
<rect x="70" y="40" width="60" height="2" fill="#ffffff" opacity="0.6" />
<path d="M95,80 Q120,100 95,140 Z" fill="#ffffff" opacity="0.25" />
<path d="M10,180 Q50,165 95,180 Q140,165 190,180 Z" fill="#ffffff" opacity="0.35" />
</svg>

```

```

<div style="position:absolute;bottom:6px;left:8px;font-size:9px;color:#aaa;pointer-events:none">
  PATIENT: ${escapeHTML(exam.patient_name || ('#' + (exam.patient_id || '')))}<br>
  MODALITY: ${escapeHTML(exam.modality || 'XR')}<br>
  STUDY: CHEST PA
</div>
</div>

```

```

<div style="display:flex;flex-direction:column;gap:4px;font-size:10px">
  <div style="display:flex;justify-content:space-between;align-items:center">
    <label>Contrast (Window Width)</label>
    <input type="range" id="dicomContrast" min="0.5" max="3" step="0.1" value="1" style="width:120px"
oninput="window.adjustDicom()">
  </div>
  <div style="display:flex;justify-content:space-between;align-items:center">
    <label>Brightness (Window Level)</label>
    <input type="range" id="dicomBrightness" min="0.5" max="2" step="0.1" value="1" style="width:120px"
" oninput="window.adjustDicom()">
  </div>
  <div style="display:flex;justify-content:space-between;align-items:center;margin-top:4px">
    <span style="font-weight:bold">Tools:</span>
    <div style="display:flex;gap:4px">
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.zo
omDicom(0.2)">? Zoom In</button>
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.zo
omDicom(-0.2)">? Zoom Out</button>
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.pa
nDicom(-10, 0)">??</button>
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.pa
nDicom(10, 0)">??</button>
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.pa
nDicom(0, -10)">??</button>
      <button class="btn btn-xs" style="background:#333;color:#fff;padding:2px 6px" onclick="window.pa
nDicom(0, 10)">??</button>
      <button class="btn btn-xs" style="background:#991b1b;color:#fff;padding:2px 6px" onclick="window
.resetDicom()">Reset</button>
    </div>
  </div>
</div>
</div>
</div>

```

```

<div style="display:flex;flex-direction:column;gap:8px">

```

```

        <div class="card-title" style="margin:0">? ${tr('Structured Report', '?????')} ? ${tr('Exam', '??
?')} #${safeId(examId)}</div>
        <div class="form-group"><label>${tr('Template', '????')}</label>
            <select class="form-input" id="rrTemplate" onchange="window.applyRadReportTemplate(this.value)"><opt
ion value="generic">${tr('Generic', '???')}</option><option value="BI-RADS">BI-RADS</option></select></div>
            <div class="form-group"><label>${tr('Findings', '????')}</label><textarea class="form-input form-te
xtarea" id="rrFindings" rows="3"></textarea></div>
            <div class="form-group"><label>${tr('Impression', '????')}</label><textarea class="form-input form
-textarea" id="rrImpression" rows="2"></textarea></div>
            <div class="form-group"><label>BI-RADS</label><input class="form-input" id="rrBirads" placeholder="0-6
"></div>

            <label class="flex gap-8" style="align-items:center"><input type="checkbox" id="rrCritical"> <span sty
le="color:#ef4444;font-weight:600">${tr('Critical finding (requires notification before signing)', '????')}
(????) ?</span></label>
            <button class="btn btn-primary w-full" onclick="window.risSaveReport(${safeId(examId)})">? ${tr('Save
Draft', '???')}</button>
        </div>
    </div>
    <div style="margin-top:12px;border-top:1px solid var(--outline);padding-top:12px">
        <div class="card-title" style="font-size:13px">? ${tr('Prior Signed Reports', '????')}</di
v>
        <div class="mb-12">${priorHtml}</div>
        <div class="card-title" style="font-size:13px">? ${tr('Reports for this exam', '????')}</div>
    >
        <div id="rrList">${reportsHtml}</div>
    </div>
`;
};

window.risSaveReport = async (examId) => {
    const body = {
        rad_exam_id: examId,
        template: document.getElementById('rrTemplate')?.value || 'generic',
        findings: document.getElementById('rrFindings')?.value || '',
        impression: document.getElementById('rrImpression')?.value || '',
        birads: document.getElementById('rrBirads')?.value || '',
        is_critical: !!document.getElementById('rrCritical')?.checked
    };
    if (!body.findings && !body.impression) { showToast(tr('Enter findings or impression', '????'), 'error'); return; }
    try { await API.post('/api/radiology/reports', body); showToast(tr('Draft saved', '??'), 'success'); await window.risOpenReport(examId); }
    catch (e) { showToast(tr('Error', '???') + ': ' + escapeHTML(e.message || ''), 'error'); }
};

window.risCriticalNotify = async (reportId, examId) => {
    const note = prompt(tr('Critical finding note / who was called:', '???? / ??'), '');
    if (note === null) return;
    try { await API.post(`/api/radiology/reports/${reportId}/critical-notify`, { note }); showToast(tr('Critical
notification documented', '??'), 'success'); await window.risOpenReport(examId); }
    catch (e) { showToast(tr('Error', '???') + ': ' + escapeHTML(e.message || ''), 'error'); }
};

```

```

window.risSignReport = async (reportId, examId) => {
  try { await API.put(`/api/radiology/reports/${reportId}/sign`, {}); showToast(tr('Report signed', '?? ?????
???????')); await window.risOpenReport(examId); await window.loadRisWorklist(); }
  catch (e) {
    showToast(tr('Cannot sign: ', '????? ????????': ') + escapeHTML(e.message || ''), 'error');
  }
};

// ===== PHARMACY =====
async function renderPharmacy(el) {
  // Show skeletons instantly
  el.innerHTML = `
    <div class="page-title">? ${tr('Pharmacy', '????????')}</div>
    <div class="stats-grid">
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
      <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
    </div>
    <div class="card glass-card-premium mt-16">
      <div class="skeleton-bar title mb-12"></div>
      <div class="skeleton-bar mb-8"></div>
    </div>
  `;

  let drugs = [];
  let queue = [];
  try {
    [drugs, queue] = await Promise.all([
      API.get('/api/pharmacy/drugs'),
      API.get('/api/pharmacy/queue')
    ]);
  } catch (e) {
    el.innerHTML = `
      <div class="page-title">? ${tr('Pharmacy', '????????')}</div>
      <div class="error-card-premium">
        <div class="error-card-icon">??</div>
        <h3>${tr('Failed to load pharmacy data', '??? ?? ????? ?????? ?????????')}</h3>
        <p>${escapeHTML(e.message || e)}</p>
        <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
button>
      </div>
    `;
    return;
  }

  const stockBadge = (qty) => {
    const q = parseInt(qty) || 0;
    if (q <= 5) return rawHtml(`<span class="badge badge-danger">?? ${q} (${tr('Critical Low', '???? ?????')}</span>`);
    if (q <= 20) return rawHtml(`<span class="badge badge-warning">?? ${q} (${tr('Low Stock', '?????')}</span>
>`);
    return rawHtml(`<span class="badge badge-success">? ${q}</span>`);
  };

  // Helper to find drug price from catalog
  const findDrug = (medName) => {

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```

    if (!medName) return null;
    return drugs.find(x => x.drug_name && medName.toLowerCase().includes(x.drug_name.toLowerCase())) || null;
};
const findDrugPrice = (medName) => {
    const d = findDrug(medName);
    return d ? (d.selling_price || 0) : 0;
};
el.innerHTML = `<div class="page-title">? ${tr('Pharmacy', '???????')}</div>
<div class="stats-grid">
    <div class="stat-card" style="--stat-color:#f59e0b"><div class="stat-label">${tr('Pending Prescriptions'
, '????? ?????????')}</div><div class="stat-value">${queue.filter(q => q.status === 'Pending').length}</div></div>
<div class="stat-card" style="--stat-color:#4ade80"><div class="stat-label">${tr('Dispensed Today', '??
????')}</div><div class="stat-value">${queue.filter(q => q.status === 'Dispensed').length}</div></div>
    <div class="stat-card" style="--stat-color:#3b82f6"><div class="stat-label">${tr('Total Drugs', '?????
????????')}</div><div class="stat-value">${drugs.length}</div></div>
</div>
<div class="card glass-card-premium mb-16"><div class="card-title">? ${tr('Prescription Queue', '????? ??
????')}</div>
<div id="rxQueue"><div class="table-wrapper"><table class="data-table"><thead><tr>
    <th>${tr('Barcode', '????????')}</th><th>${tr('Patient', '???????')}</th><th>${tr('Doctor', '???????')}</t
h><th>${tr('Prescription', '???????')}</th><th>${tr('Price', '?????')}</th><th>${tr('Status', '???????')}</th><t
h>${tr('Date', '????????')}</th><th>${tr('Actions', '????????')}</th>
</tr></thead><tbody>
    ${queue.length === 0 ? `<tr><td colspan="8"><div class="empty-state-card"><div class="empty-state-icon"><
/div><div>${tr('No prescriptions found in queue', '?? ???? ????? ?? ?????????')}</div></div></td></tr>` : queue.
map(q => {
    // Use individual columns if available, fallback to text parsing
    const txt = q.prescription_text || '';
    let parts = txt.includes(' | ') ? txt.split(' | ') : txt.split(' - ');
    const med = (q.medication_name && q.medication_name.trim()) || parts[0] || '';
    const dose = (q.dosage && q.dosage.trim()) || parts[1] || '';
    const qty = (q.quantity_per_day && q.quantity_per_day !== '1' && q.quantity_per_day.trim()) || '1';
    const freq = (q.frequency && q.frequency.trim()) || parts[2] || '';
    const dur = (q.duration && q.duration.trim()) || parts[3] || '';
    const matchedDrug = findDrug(med);
    const autoPrice = q.price > 0 ? q.price : (matchedDrug ? (matchedDrug.selling_price || 0) : 0);
    const drugId = matchedDrug ? matchedDrug.id : 0;
    const isControlled = matchedDrug && Number(matchedDrug.is_controlled) > 0 ? 1 : 0;
    return `<tr>
        <td><svg id="rxBC${safeId(q.id)}" class="barcode-svg"></svg><br>
            <button class="btn btn-sm btn-info" onclick="printRxLabel(${safeId(q.id)}, '${jsStr(q.patient_name |
| ')}', '${jsStr((q.age || '').toString())}', '${jsStr(q.department || ')}', '${jsStr(med)}', '${jsStr(dose)
}', '${jsStr(qty.toString())}', '${jsStr(freq)}', '${jsStr(dur)}')" style="margin-top:4px;font-size:11px">? $
${tr('Print Label', '?????')}</button>
        </td>
        <td><strong>${escapeHTML(q.patient_name || '#' + q.patient_id)}</strong>${q.age ? '<br><small>? ' + es
capeHTML(q.age) + '</small>' : ''}${q.department ? '<br><small>? ' + escapeHTML(q.department) + '</small>' : '
'}</td>
        <td style="color:var(--accent);font-weight:600">${escapeHTML(q.doctor || q.doctor_name || '?')}</td>
        <td><strong>${escapeHTML(med)}</strong>${dose ? '<br>? ' + escapeHTML(dose) : ''}${freq ? '<br>? ' + e
scapeHTML(freq) : ''}${dur ? '<br>? ' + escapeHTML(dur) : ''}</td>
        <td style="font-weight:bold;color:var(--accent)">${autoPrice > 0 ? escapeHTML(autoPrice) + ' ' + tr('S
AR', '?.?') : '-'}</td>

```

```

<td>${statusBadge(q.status)}</td>
<td>${escapeHTML(q.created_at?.split('T')[0] || '')}</td>
<td>${q.status === 'Pending'
  ? `<button class="btn btn-sm btn-warning" onclick="pharmacyVerify(${safeId(q.id)}, '${jsStr(med)}')`
  >? ${tr('Verify (CDS)', '???? (CDS)')}</button>`
  : q.status === 'Verified'
  ? `<span class="badge badge-success" style="display:inline-block;margin-bottom:4px">? ${tr('Verifi
ed', '?? ?????')}</span><br><button class="btn btn-sm btn-success" onclick="showDispensePanel(${safeId(q.id)}
, '${jsStr(q.patient_name || '')}', '${jsStr(med)}', '${jsStr(dose)}', '${jsStr(qty.toString())}', '${jsStr(fr
eq)}', '${jsStr(dur)}', ${safeId(q.patient_id) || 0}, ${safeId(autoPrice) || 0}, '${jsStr((q.age || '').toStri
ng())}', '${jsStr(q.department || '')}', ${safeId(drugId) || 0}, ${isControlled})">? ${tr('Dispense (FEF0)', '
??? (FEF0)')}</button>`
  : `<button class="btn btn-sm btn-info" onclick="printPharmacyInvoice(${safeId(q.id)}, '${jsStr(q.p
atient_name || '')}', '${jsStr(med)}', '${jsStr(dose)}', '${jsStr(freq)}', '${jsStr(dur)}', ${safeId(q.price)
|| 0}, '${jsStr(q.payment_method || '')}')">? ${tr('Print Invoice', '????? ?????')}</button>`</td>
</tr>`;
}).join(''))
</tbody></table></div></div>
<div id="dispensePanel" style="display:none"></div>
</div>
<div class="card glass-card-premium mb-16"><div class="card-title">? ${tr('Batches & Expiry (FEF0)', '????
??? ????????? (FEF0)')}</div>
<div class="flex gap-8 mb-12" style="flex-wrap:wrap;align-items:flex-end">
  <div class="form-group" style="flex:0.6;min-width:90px"><label>${tr('Drug ID', '??? ?????')}</label><
input class="form-input" id="batDrugId" type="number" placeholder="#"></div>
  <div class="form-group" style="flex:1;min-width:120px"><label>${tr('Drug name', '??? ?????')}</label>
<input class="form-input" id="batDrugName" placeholder="${tr('Drug name', '??? ?????')}></div>
  <div class="form-group" style="flex:0.8;min-width:110px"><label>${tr('Lot', '??? ?????')}</label><i
nput class="form-input" id="batLot" placeholder="LOT"></div>
  <div class="form-group" style="flex:0.8;min-width:140px"><label>${tr('Expiry', '????????')}</label><in
put class="form-input" id="batExpiry" type="date"></div>
  <div class="form-group" style="flex:0.6;min-width:90px"><label>${tr('Qty', '?????')}</label><input cl
ass="form-input" id="batQty" type="number" min="1"></div>
  <button class="btn btn-primary" onclick="addBatch()">? ${tr('Receive Batch', '?????? ?????')}</button>
</div>
<div id="batchesTable"><div class="empty-state-card"><div>${tr('Loading batches...', '???? ????? ?????')}
...')</div></div></div>
</div>
<div class="card glass-card-premium mb-16"><div class="card-title">? ${tr('Scan-to-Dispense (FEF0)', '????
???????? (FEF0)')}</div>
<div class="flex gap-8 mb-8" style="flex-wrap:wrap;align-items:flex-end">
  <div class="form-group" style="flex:0.8;min-width:110px"><label>RX #</label><input class="form-input"
id="scanRxId" type="number" placeholder="${tr('Verified queue #', '??? ??? ?????????')}></div>
  <div class="form-group" style="flex:1.4;min-width:160px"><label>${tr('Drug Barcode', '?????? ?????')}<
/label><input class="form-input" id="scanBarcode" placeholder="${tr('Scan / type barcode', '????/???? ??????
?')}></div>
  <div class="form-group" style="flex:0.6;min-width:80px"><label>${tr('Qty', '?????')}</label><input cl
ass="form-input" id="scanQty" type="number" min="1" value="1"></div>
  <div class="form-group" style="flex:0.9;min-width:130px"><label>${tr('Witness (controlled)', '???? (??
???)')}</label><input class="form-input" id="scanWitness" type="number" placeholder="${tr('User ID', '??? ???
????')}></div>
  <div class="form-group" style="flex:0.6;min-width:90px"><label>${tr('Price', '????')}</label><input c
lass="form-input" id="scanPrice" type="number" min="0" value="0"></div>
  <button class="btn btn-success" onclick="dispenseByScan()">? ${tr('Dispense', '???')}</button>

```

```

    </div>
    <small style="color:var(--muted,#888)">${tr('FEFO: earliest non-expired batch is used first. Expired batches are never dispensed. Controlled drugs require a witness.', 'FEFO: ????? ???? ???? ??? ?????? ?????? ?? ?? ??? ?????? ???????? ??????. ??????? ????????? ?????? ??????.')}</small>
  </div>
  <div class="card glass-card-premium mb-16"><div class="card-title">? ${tr('Drug Catalog', '???? ?????')}</div>
  <div class="flex gap-8 mb-12"><input class="form-input" id="phName" placeholder="${tr('Drug name', '??? ? ?????')}" style="flex:2"><input class="form-input" id="phPrice" placeholder="${tr('Price', '????')}" type="number" style="flex:1"><input class="form-input" id="phStock" placeholder="${tr('Stock', '??????')}" type="number" style="flex:1"><button class="btn btn-primary" onclick="addDrug()"></button></div>
  <input class="search-filter" placeholder="${tr('Search drugs...', '??? ? ???????.')}" oninput="filterTable(this, 'phTable')">
  <div id="phTable">${makeTable([tr('Name', '????'), tr('Category', '???????'), tr('Price', '????'), tr('Stock', '??????')], drugs.map(d => ({ cells: [d.drug_name, d.category, d.selling_price, stockBadge(d.stock_qty)] })))}</div></div>`;
  // Generate barcodes for prescriptions
  setTimeout(() => { queue.forEach(q => { try { JsBarcode('#rxBC' + q.id, 'RX-' + q.id, { format: 'CODE128', width: 1.2, height: 35, fontSize: 9, displayValue: true, margin: 2, textMargin: 1 }); } catch (e) { } }); }, 1000);
  loadBatches();
}

// ===== E5: load + render drug batches (FEFO view) =====
async function loadBatches() {
  const elB = document.getElementById('batchesTable');
  if (!elB) return;
  try {
    const batches = await API.get('/api/pharmacy/batches');
    if (!Array.isArray(batches) || batches.length === 0) {
      elB.innerHTML = `<div class="empty-state-card"><div>${tr('No batches recorded', '?? ???? ????? ?????')}</div></div>`;
      return;
    }
    const rows = batches.map(b => {
      const expBadge = b.is_expired
        ? rawHtml(`<span class="badge badge-danger">? ${tr('Expired', '????')}</span>`)
        : b.is_near_expiry
          ? rawHtml(`<span class="badge badge-warning">?? ${tr('Near expiry', '??? ??????')}</span>`)
          : rawHtml(`<span class="badge badge-success">? ${tr('OK', '????')}</span>`);
      return { cells: [b.drug_name || ('#' + b.drug_id), b.lot || '?', (b.expiry_date || '').toString().split('T')[0], b.qty_on_hand, expBadge] };
    });
    elB.innerHTML = makeTable([tr('Drug', '?????'), tr('Lot', '????????'), tr('Expiry', '????????'), tr('On Hand', '?????'), tr('Status', '?????')], rows);
  } catch (e) {
    elB.innerHTML = `<div class="empty-state-card"><div>${tr('Failed to load batches', '??? ????? ?????')}</div>${escapeHTML(e.message || e)}</div></div>`;
  }
}

window.addBatch = async () => {
  const drug_id = parseInt(document.getElementById('batDrugId').value, 10) || null;
  const drug_name = document.getElementById('batDrugName').value.trim();

```

```

const lot = document.getElementById('batLot').value.trim();
const expiry_date = document.getElementById('batExpiry').value;
const qty_received = parseInt(document.getElementById('batQty').value, 10) || 0;
if (!expiry_date) { showToast(tr('Expiry date required', '????? ????????? ??????'), 'error'); return; }
if (qty_received <= 0) { showToast(tr('Quantity must be > 0', '?????? ??? ?? ??? ???? ?? ???'), 'error'); r
return; }
try {
  const r = await API.post('/api/pharmacy/batches', { drug_id, drug_name, lot, expiry_date, qty_received });
  if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
  showToast(tr('Batch received', '?? ?????? ??????'), 'success');
  loadBatches();
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// E5: pharmacist VERIFY ? re-runs server-side CDS; CRITICAL hard-stop captures an override reason (audited).
window.pharmacyVerify = async (rxId, med, overrideReason) => {
  try {
    const r = await API.put(`/api/pharmacy/queue/${rxId}/verify`, { override_reason: overrideReason || undefin
ed });
    // API.request resolves the JSON body even for non-2xx (no throw); a 422 hard-stop arrives as {blocked:tru
e, alerts}.
    if (r && (r.blocked || r.requires_override_reason)) {
      const txt = (r.alerts || []).map(a => '? ' + (a.message_ar || a.message_en || a.message)).join('\n');
      const reason = window.prompt('? ' + tr('CRITICAL CDS alert ? hard stop. To override you MUST enter a cli
nical reason (audited):',
        '????? CDS ??? ? ?????? ????????. ??????? ??? ?????? ??? ?????? (??????):') + '\n\n' + txt, '');
      if (reason && reason.trim()) return window.pharmacyVerify(rxId, med, reason.trim());
      showToast(tr('Verify blocked by CDS (no override reason)', '?? ??? ?????? ?????? CDS (???? ???? ?????)'),
'error');
      return;
    }
    if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
    showToast(tr('Verified ? ready to dispense', '?? ?????? ? ???? ??????') + (overrideReason ? ' (override)' :
'), 'success');
    if (typeof navigateTo === 'function') navigateTo(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// E5: scan-to-dispense ? FEFO server-side; controlled drugs require a witness id (server fail-closed).
window.dispenseByScan = async () => {
  const prescription_id = parseInt(document.getElementById('scanRxId').value, 10) || 0;
  const barcode = document.getElementById('scanBarcode').value.trim();
  const quantity = parseInt(document.getElementById('scanQty').value, 10) || 0;
  const witness_user_id = parseInt(document.getElementById('scanWitness').value, 10) || undefined;
  const price = parseFloat(document.getElementById('scanPrice').value) || 0;
  if (!prescription_id || !barcode || quantity <= 0) { showToast(tr('RX #, barcode and quantity required', '??
? ?????? ?????????? ??????? ??????'), 'error'); return; }
  try {
    const r = await API.post('/api/pharmacy/dispense', { prescription_id, barcode, quantity, witness_user_id,
price, payment_method: 'Cash' });
    if (r && r.requires_witness) { showToast(tr('Controlled drug ? a second witness (User ID) is required', '?
??? ????? ? ???? ???? ???? (??? ?????????)'), 'error'); return; }
    if (r && r.error) { showToast(escapeHTML(r.error) + (r.available !== undefined ? ' (' + tr('available', '?
?????') + ': ' + r.available + ')': ''), 'error'); return; }

```

```

        showToast(tr('Dispensed (FEF0)', '?? ????? (FEF0)') + ' ? ' + (r.dispensed || quantity), 'success');
        if (typeof navigateTo === 'function') navigateTo(currentPage);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.printRxLabel = (rxId, patientName, age, dept, med, dose, qty, freq, dur) => {
    const svgEl = document.getElementById('rxBC' + rxId);
    const svgData = svgEl ? new XMLSerializer().serializeToString(svgEl) : '';
    // Clean dose field from embedded qty if present
    const pureDose = dose.replace(/\\s\\(x\\d+\\)/, '').trim();
    const w = window.open('', '_blank', 'width=520,height=500');
    w.document.write(`<!DOCTYPE html><html lang="ar" dir="rtl"><head><meta charset="UTF-8"><title>Rx Label</title>
e>
<style>
@page{size:80mm auto;margin:3mm}
*{margin:0;padding:0;box-sizing:border-box}
body{font-family:'Segoe UI',Tahoma,Arial,sans-serif;padding:10px;direction:rtl;font-size:13px}
.label{border:2px solid #333;border-radius:10px;padding:14px;max-width:420px;margin:0 auto}
.clinic{font-size:15px;font-weight:bold;color:#1a365d;text-align:center;margin-bottom:8px;border-bottom:2px solid #1a365d;padding-bottom:6px}
.barcode-area{text-align:center;margin:8px 0;direction:ltr}
.info-grid{display:grid;grid-template-columns:auto 1fr;gap:4px 10px;font-size:12px;margin:8px 0;padding:8px;background:#f7f8fa;border-radius:8px}
.info-grid .lk{font-weight:700;color:#1a365d;white-space:nowrap}
.med-table{width:100%;border-collapse:collapse;margin:10px 0;font-size:11px}
.med-table th{background:#1a365d;color:#fff;padding:5px 4px;text-align:center;font-size:10px}
.med-table td{border:1px solid #ccc;padding:5px 4px;text-align:center;font-weight:600}
.footer{text-align:center;font-size:10px;color:#999;margin-top:8px;border-top:1px dashed #ccc;padding-top:6px}
.no-print{text-align:center;margin-bottom:12px}
@media print{.no-print{display:none!important}body{padding:2px}}
</style></head><body>
<div class="no-print">
    <button onclick="window.print()" style="padding:10px 30px;font-size:14px;background:#1a365d;color:#fff;border:none;border-radius:8px;cursor:pointer">?? ????? / Print</button>
    <button onclick="window.close()" style="padding:10px 20px;font-size:14px;background:#dc3545;color:#fff;border:none;border-radius:8px;cursor:pointer;margin-right:8px">?</button>
</div>
<div class="label">
    <div class="clinic">? ????? ????? ? ????????<br><small style="font-size:11px;color:#666">jumanaMedical ? Pharmacy</small></div>
    <div class="barcode-area">${svgData}</div>
    <div class="info-grid">
        <span class="lk">? ?????? / Patient:</span><span>${escapeHTML(patientName)}</span>
        <span class="lk">? ?????? / Age:</span><span>${escapeHTML(age || '-')}</span>
        <span class="lk">? ?????? / Dept:</span><span>${escapeHTML(dept || '-')}</span>
        <span class="lk">? ???????? / Date:</span><span>${new Date().toLocaleDateString('ar-SA')}</span>
    </div>
    <table class="med-table">
        <thead><tr>
            <th>? ??????<br>Drug</th>
            <th>? ??????<br>Dose</th>
            <th>? ??????/?<br>Qty/Day</th>
            <th>? ??????<br>Freq</th>
            <th>? ??????<br>Days</th>
        </tr></thead>

```



```

<tbody><tr>
  <td style="font-size:12px;color:#4338ca">${escapeHTML(med)}</td>
  <td>${escapeHTML(pureDose || '-')}</td>
  <td style="font-size:14px;font-weight:bold;color:#e74c3c">${escapeHTML(qty)}</td>
  <td>${escapeHTML(freq || '-')}</td>
  <td>${escapeHTML(dur || '-')}</td>
</tr></tbody>
</table>
<div class="footer">Rx #${escapeHTML(rxId)} | ${new Date().toLocaleDateString('en-CA')} | ????? ?????</div>
</div>
<script>setTimeout(()=>{window.print();},400);<\/script>
</body></html>`);
w.document.close();
};
window.showDispensePanel = (id, patientName, med, dose, qty, freq, dur, patientId, autoPrice, age, dept, drugId, isControlled) => {
  const panel = document.getElementById('dispensePanel');
  panel.style.display = 'block';
  const defaultQty = parseInt(qty, 10) > 0 ? parseInt(qty, 10) : 1;
  const controlled = Number(isControlled) > 0;
  panel.innerHTML = `<div class="card mt-16" style="border:2px solid var(--accent);background:var(--hover)">
    <div class="card-title">? ${tr('Confirm Dispense & Sale (FEFO)', '????? ?????? ?????? (FEFO)')} ? RX-${escapeHTML(id)}</div>
    <div class="flex gap-16" style="flex-wrap:wrap;align-items:flex-end">
      <div style="flex:1;min-width:150px">
        <div style="font-size:13px;margin-bottom:4px"><strong>? ${tr('Patient', '?????')}</strong> ${escapeHTML(patientName)}</div>
        <div style="font-size:13px"><strong>? ${tr('Drug', '?????')}</strong> ${escapeHTML(med)} ${dose ? '?' + escapeHTML(dose) : ''} ${controlled ? '<span class="badge badge-danger">?' + tr('Controlled', '?????') + '</span>' : ''}</div>
        <div style="font-size:13px"><strong>? ${tr('Qty/Day', '?????/?')}</strong> ${escapeHTML(qty)} | <strong>?</strong> ${escapeHTML(freq)} | <strong>?</strong> ${escapeHTML(dur)}</div>
      </div>
      <div class="form-group" style="flex:0.4;min-width:90px">
        <label>${tr('Qty to dispense', '????? ?????')}</label>
        <input class="form-input" id="dispQty" type="number" value="${defaultQty}" min="1" step="1" style="font-size:16px;font-weight:bold;text-align:center">
      </div>
      ${controlled ? `<div class="form-group" style="flex:0.5;min-width:120px">
        <label>${tr('Witness (User ID)', '???? (??? ??????)')}</label>
        <input class="form-input" id="dispWitness" type="number" placeholder="${tr('Required', '?????')}" min="1" style="font-size:15px;text-align:center">
      </div>` : ''}
      <div class="form-group" style="flex:0.5;min-width:120px">
        <label>${tr('Price', '????')} (${tr('SAR', '?.?.')})</label>
        <input class="form-input" id="dispPrice" type="number" value="${escapeHTML(autoPrice)}" min="0" step="0.5" style="font-size:16px;font-weight:bold;text-align:center">
      </div>
      <div class="form-group" style="flex:1;min-width:250px">
        <label>${tr('Payment Method', '????? ?????')}</label>
        <div class="flex gap-16" style="margin-top:6px">
          <label style="display:flex;align-items:center;gap:6px;cursor:pointer;font-size:14px"><input type="radio" name="dispPay" value="Cash" checked style="width:18px;height:18px;accent-color:var(--accent,#6c5ce7)"> ? ${tr('Cash', '???')}</label>

```

```

        <label style="display:flex;align-items:center;gap:6px;cursor:pointer;font-size:14px"><input type="radio" name="dispPay" value="Card" style="width:18px;height:18px;accent-color:var(--accent,#6c5ce7)"> ? ${tr('POS/Card', '????')}</label>
        <label style="display:flex;align-items:center;gap:6px;cursor:pointer;font-size:14px"><input type="radio" name="dispPay" value="Transfer" style="width:18px;height:18px;accent-color:var(--accent,#6c5ce7)"> ? ${tr('Transfer', '????')}</label>
    </div>
</div>
<div class="flex gap-8">
    <button class="btn btn-success" onclick="confirmDispense(${safeId(id)}, '${jsStr(patientName)}', '${jsStr(med)}', '${jsStr(dose)}', '${jsStr(qty.toString())}', '${jsStr(freq)}', '${jsStr(dur)}', ${safeId(patientId)}, '${jsStr((age || '').toString())}', '${jsStr(dept || '')}', ${safeId(drugId) || 0}, ${controlled ? 1 : 0})">? ${tr('Confirm & Print', '???? ????')}</button>
    <button class="btn btn-danger" onclick="document.getElementById('dispensePanel').style.display='none'">? ${tr('Cancel', '????')}</button>
</div>
</div>`;
panel.scrollToView({ behavior: 'smooth' });
};
// E5: primary confirm now routes through the full FEFO/CDS/controlled safety stack
// (POST /api/pharmacy/dispense) ? NOT the legacy PUT /queue/:id which bypassed FEFO,
// the witness gate, pharmacy_dispense and controlled_drug_log.
window.confirmDispense = async (id, patientName, med, dose, qty, freq, dur, patientId, age, dept, drugId, isControlled) => {
    const priceNum = parseFloat(document.getElementById('dispPrice').value) || 0;
    const payMethod = document.querySelector('input[name="dispPay"]:checked').value || 'Cash';
    const quantity = parseInt(document.getElementById('dispQty')?.value, 10) || 0;
    if (quantity <= 0) { showToast(tr('Quantity must be > 0', '???? ? ? ? ? ? ? ?'), 'error'); return; }
    if (!drugId) { showToast(tr('Drug not matched in catalog ? dispense by barcode below', '?? ????? ???? ? ? ? ? ? ? ?'), 'error'); return; }
    const witness_user_id = (Number(isControlled) > 0)
        ? (parseInt(document.getElementById('dispWitness')?.value, 10) || undefined)
        : undefined;
    if (Number(isControlled) > 0 && !witness_user_id) {
        showToast(tr('Controlled drug ? a second witness (User ID) is required', '???? ???? ? ? ? ? ? ? ? (?? ?????)'), 'error');
        return;
    }
    try {
        const r = await API.post('/api/pharmacy/dispense', { prescription_id: id, drug_id: drugId, quantity, witness_user_id, price: priceNum, payment_method: payMethod });
        // API.request resolves the JSON body even for non-2xx; surface FEFO shortfall (409) /
        // CDS hard-stop or missing-witness (422) errors to the pharmacist.
        if (r && r.requires_witness) { showToast(tr('Controlled drug ? a valid second witness (User ID) is required', '???? ???? ? ? ? ? ? ? ? ? ? ? (?? ?????)'), 'error'); return; }
        if (r && r.error) { showToast(escapeHTML(r.error) + (r.available !== undefined ? ' (' + tr('available', '????') + ': ' + r.available + ')': ''), 'error'); return; }
        showToast(`? ${tr('Dispensed & sold!', '?? ????? ????')} ${priceNum > 0 ? priceNum + ' ' + tr('SAR', '?.?') : tr('Free', '????')}`, 'success');
        document.getElementById('dispensePanel').style.display = 'none';
        // Auto-print barcode label with all doctor data
        printRxLabel(id, patientName, age, dept, med, dose, qty, freq, dur);
    }
};

```

```

// Auto-print invoice
setTimeout(() => { printPharmacyInvoice(id, patientName, med, dose, freq, dur, priceNum, payMethod); }, 800);
setTimeout(() => navigateTo(6), 1200);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.printPharmacyInvoice = (rxId, patientName, med, dose, freq, dur, price, payMethod) => {
  const w = window.open('', '_blank', 'width=500,height=600');
  const payAr = payMethod === 'Card' ? '????' : payMethod === 'Transfer' ? '?????' : '???';
  w.document.write(`<!DOCTYPE html><html lang="ar" dir="rtl"><head><meta charset="UTF-8"><title>Pharmacy Invoice</title>
<style>
@page{size:80mm auto;margin:3mm}
*{margin:0;padding:0;box-sizing:border-box}
body{font-family:'Segoe UI',Tahoma,Arial,sans-serif;padding:10px;direction:rtl;font-size:13px}
.inv{border:2px solid #333;border-radius:10px;padding:16px;max-width:400px;margin:0 auto}
.header{text-align:center;border-bottom:2px solid #1a365d;padding-bottom:8px;margin-bottom:10px}
.header h2{color:#1a365d;margin:0;font-size:16px}
.header small{color:#666;font-size:11px}
.row{display:flex;justify-content:space-between;padding:4px 0;font-size:12px;border-bottom:1px dotted #ddd}
.row .k{font-weight:700;color:#1a365d}
.med-tbl{width:100%;border-collapse:collapse;margin:10px 0;font-size:12px}
.med-tbl th{background:#1a365d;color:#fff;padding:6px;text-align:center}
.med-tbl td{border:1px solid #ccc;padding:6px;text-align:center;font-weight:600}
.total-box{background:#eef2ff;border:2px solid #6366f1;border-radius:8px;padding:10px;text-align:center;margin:10px 0;font-size:16px;font-weight:bold;color:#4338ca}
.footer{text-align:center;font-size:10px;color:#999;margin-top:10px;border-top:1px dashed #ccc;padding-top:6px}
.no-print{text-align:center;margin-bottom:12px}
@media print{.no-print{display:none!important}body{padding:2px}}
</style></head><body>
<div class="no-print">
  <button onclick="window.print()" style="padding:10px 30px;font-size:14px;background:#1a365d;color:#fff;border:none;border-radius:8px;cursor:pointer">?? ????? / Print</button>
  <button onclick="window.close()" style="padding:10px 20px;font-size:14px;background:#dc3545;color:#fff;border:none;border-radius:8px;cursor:pointer;margin-right:8px">?</button>
</div>
<div class="inv">
  <div class="header"><h2> ????? ???? ? ?????? ??????</h2><div style="margin-bottom:12px"><button class="btn btn-sm" onclick="toggleCalendarView()" id="calToggleBtn" style="background:#e3f2fd;color:#1565c0">? ${tr("Calendar View", "??? ??????")}</button></div><div style="display:flex;gap:8px;margin-bottom:12px;flex-wrap:wrap"><button class="btn btn-primary" onclick="callNextPatient()" style="padding:8px 20px;font-size:15px;animation:pulse 2s infinite">? ${tr("Next Patient", "????? ??????")}</button><button class="btn btn-sm" onclick="loadMyQueue()" style="background:#e3f2fd;color:#1565c0">? ${tr("My Queue", "?????")}</button></div><small>jumanaMedical ? Pharmacy Invoice</small></div>
  <div class="row"><span class="k">? ? ? ? ??????</span><span>RX-${escapeHTML(rxId)}</span></div>
  <div class="row"><span class="k">? ??????</span><span>${escapeHTML(patientName)}</span></div>
  <div class="row"><span class="k">? ??????</span><span>${new Date().toLocaleDateString('ar-SA')} ? ${new Date().toLocaleTimeString('ar-SA', { hour: '2-digit', minute: '2-digit' })}</span></div>
  <table class="med-tbl">
    <thead><tr><th> ??????</th><th> ??????</th><th> ??????/?</th><th> ??????</th></tr></thead>
    <tbody><tr><td>${escapeHTML(med)}</td><td>${escapeHTML(dose || '-')}</td><td>${escapeHTML(freq || '-')}</td><td>${escapeHTML(dur || '-')}</td></tr></tbody>
  </table>

```

```

<div class="total-box">? ??????: ${escapeHTML(price || 0)} ?.?</div>
<div class="row"><span class="k">? ?????</span><span>${payAr} (${escapeHTML(payMethod || 'Cash')})</span></div>
<div class="row"><span class="k">? ?????</span><span style="color:green;font-weight:bold">???? ? Paid</span></div>
<div class="footer">???? | ${new Date().toLocaleDateString('en-CA')} | ????</div>
</div>
<script>setTimeout(()=>{window.print();},400);</script>
</body></html>`);
w.document.close();
};
window.addDrug = async () => {
const name = document.getElementById('phName').value.trim();
if (!name) { showToast(tr('Enter drug name', '??? ????'), 'error'); return; }
try {
await API.post('/api/pharmacy/drugs', { drug_name: name, selling_price: document.getElementById('phPrice').value, stock_qty: document.getElementById('phStock').value });
showToast(tr('Drug added!', '??? ?????!'));
await navigateTo(6);
} catch (e) { showToast(tr('Error adding', '??? ????'), 'error'); }
};

// ===== HR =====
async function renderHR(el) {
const content = el;
if (!window.hrTab) window.hrTab = 'employees';

// Fetch data from backend API
const [employees, salaries, leaves, attendance] = await Promise.all([
API.get('/api/employees').catch(() => []),
API.get('/api/hr/salaries').catch(() => []),
API.get('/api/hr/leave-requests').catch(() => []), // E18 leave state machine (requested/approved/denied)
API.get('/api/hr/attendance').catch(() => [])
]);

const totalEmps = employees.length;
const doctors = employees.filter(e => e.role === 'Doctor').length;
const nurses = employees.filter(e => e.role === 'Nurse').length;
const staff = totalEmps - doctors - nurses;

let htmlContent = `
<div class="mb-xl flex flex-col md:flex-row md:items-end justify-between gap-md">
<div>
<nav class="flex text-on-surface-variant font-caption text-caption mb-xs">
<span>${tr('Operations', '???????')}</span>
<span class="mx-xs"></span>
<span class="text-primary font-bold">${tr('HR & Payroll', '?????? ?????? ???????')}</span>
</nav>
<h3 class="font-display-lg text-display-lg text-primary tracking-tight">${tr('Human Resources & Payroll', '?????? ?????? ???????')}</h3>
<p class="font-body-md text-body-md text-on-surface-variant max-w-2xl">
${tr('Manage employee directory, payroll runs, leaves, and attendance with Saudi labor compliance.', '???? ????? ???? ?????????? ?????????? ?????????? ????? ?????? ????? ?????? ???? ?????? ???? ??????')}
`

```

```

    </p>
  </div>
</div>

<!-- Tabs Bar -->
<div class="tab-bar mb-md flex border-b border-outline-variant/30 overflow-x-auto gap-2">
  <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.hrTab === 'employees' ?
'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}" o
nclick="window.hrTab='employees'; navigateTo(currentPage)">
    ? ${tr('Employees Directory', '???? ?????')}
  </button>
  <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.hrTab === 'payroll' ? 'a
ctive border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}" oncl
ick="window.hrTab='payroll'; navigateTo(currentPage)">
    ? ${tr('Payroll & Salaries', '????? ?????')}
  </button>
  <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.hrTab === 'leaves' ? 'ac
tive border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}" oncl
ick="window.hrTab='leaves'; navigateTo(currentPage)">
    ?? ${tr('Leaves & Requests', '??????? ?????')}
  </button>
  <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.hrTab === 'attendance' ?
'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.hrTab='attendance'; navigateTo(currentPage)">
    ? ${tr('Attendance Registry', '????? ?????')}
  </button>
</div>
`;

if (window.hrTab === 'employees') {
  htmlContent += `
    <!-- KPI Bento Grid -->
    <div class="grid grid-cols-1 md:grid-cols-4 gap-gutter mb-xl">
      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined">groups</span>
          <span class="text-secondary text-xs font-bold">${totalEmps} ${tr('Employees', '????')}</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Total Employees', '????? ??
????')}</p>
          <h4 class="text-display-lg text-primary font-bold mt-xs">${totalEmps}</h4>
        </div>
      </div>

      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">medical_ser
vices</span>
          <span class="text-secondary text-xs font-bold">${doctors} ${tr('Doctors', '????')}</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Medical Doctors', '????? ??
????')}</p>

```

```

        <h4 class="text-display-lg text-primary font-bold mt-xs">${doctors}</h4>
    </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
    <div class="flex justify-between items-start relative z-10">
        <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">person</spa
n>

        <span class="text-secondary text-xs font-bold">${nurses} ${tr('Nurses', '????')}</span>
    </div>
    <div class="mt-lg relative z-10">
        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Nursing Staff', '???? ??????
')}}</p>

        <h4 class="text-display-lg text-primary font-bold mt-xs">${nurses}</h4>
    </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
    <div class="flex justify-between items-start relative z-10">
        <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">badge</span
>

        <span class="text-secondary text-xs font-bold">${staff} ${tr('Administrative', '??????')}</span>
    </div>
    <div class="mt-lg relative z-10">
        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Support & Admin', '???? ????
????')}}</p>

        <h4 class="text-display-lg text-primary font-bold mt-xs">${staff}</h4>
    </div>
</div>

<!-- Add Employee & Table section -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 flex flex-col gap-md">
        <h4 class="font-title-lg text-title-lg text-primary">${tr('Register New Employee', '???? ???? ????
')}}</h4>

        <div class="form-group">
            <label>${tr('Name (Arabic)', '???? ???????')}</label>
            <input class="form-input" id="hrNameAr" placeholder="${tr('Arabic name', '???? ???????')}>
        </div>
        <div class="form-group">
            <label>${tr('Name (English)', '???? ?????????')}</label>
            <input class="form-input" id="hrNameEn" placeholder="${tr('English name', '???? ?????????')}>
        </div>
        <div class="grid grid-cols-2 gap-sm">
            <div class="form-group">
                <label>${tr('Role', '??????')}</label>
                <select class="form-input" id="hrRole">
                    <option value="Staff">${tr('Staff', '???? ????)}</option>
                    <option value="Doctor">${tr('Doctor', '????')}</option>
                    <option value="Nurse">${tr('Nurse', '????')}</option>
                    <option value="Admin">${tr('Admin', '???? ????')}</option>
                    <option value="Receptionist">${tr('Receptionist', '???? ???????')}</option>
                </select>
            </div>

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```

</div>
<div class="form-group">
  <label>${tr('Basic Salary', '????? ?????')}</label>
  <input class="form-input" id="hrSalary" type="number" placeholder="5000">
</div>
</div>
<div class="form-group">
  <label>${tr('Department', '????')}</label>
  <select class="form-input" id="hrDept">
    <option value="" data-ar="???? ??">-- ${tr('Department', '????')} --</option>
    <optgroup label="${tr('Medical Departments', '?????? ?????')}">
      <option value="General Practice" data-ar="???? ?????">${tr('General Practice', '???? ?????')}<
/option>

      <option value="Dentistry" data-ar="?? ??????">${tr('Dentistry', '?? ?????')}</option>
      <option value="Endocrinology & Diabetes" data-ar="???? ????? ??????">${tr('Endocrinology &
Diabetes', '???? ????? ?????')}</option>
      <option value="Pediatrics" data-ar="?? ??????">${tr('Pediatrics', '?? ?????')}</option>
      <option value="Orthopedics" data-ar="???? ??????">${tr('Orthopedics', '???? ?????')}</optio
n>

      <option value="Dermatology" data-ar="?????">${tr('Dermatology', '????')}</option>
      <option value="ENT" data-ar="???? ????? ??????">${tr('ENT', '???? ????? ?????')}</opt
ion>

      <option value="Ophthalmology" data-ar="?????">${tr('Ophthalmology', '????')}</option>
      <option value="Cardiology" data-ar="?????">${tr('Cardiology', '????')}</option>
      <option value="Internal Medicine" data-ar="?????">${tr('Internal Medicine', '????')}</o
ption>

      <option value="Obstetrics & Gynecology" data-ar="????? ??????">${tr('Obstetrics & Gynecolog
y', '????? ?????')}</option>
      <option value="Neurology" data-ar="???? ??????">${tr('Neurology', '???? ?????')}</option>
      <option value="Psychiatry" data-ar="???? ??????">${tr('Psychiatry', '???? ?????')}</option>
    </optgroup>
    <optgroup label="${tr('Other Departments', '????')}">
      <option value="Radiology" data-ar="?????">${tr('Radiology', '????')}</option>
      <option value="Laboratory" data-ar="?????">${tr('Laboratory', '????')}</option>
      <option value="Administration" data-ar="?????">${tr('Administration', '????')}</option>
      <option value="Reception" data-ar="?????">${tr('Reception', '????')}</option>
      <option value="Pharmacy" data-ar="?????">${tr('Pharmacy', '????')}</option>
    </optgroup>
  </select>
</div>
<div class="form-group" id="hrCommRow" style="display:none">
  <label>${tr('Commission Type & Value', '??? ????? ?????')}</label>
  <div class="flex gap-2">
    <select class="form-input" id="hrCommType" style="flex:1">
      <option value="percentage">${tr('Percentage %', '???? ????? %')}</option>
      <option value="fixed">${tr('Fixed Value', '???? ?????')}</option>
    </select>
    <input class="form-input" id="hrCommValue" type="number" step="0.5" value="0" style="flex:1">
  </div>
</div>
<button class="btn btn-primary w-full py-3 rounded-xl font-bold flex items-center justify-center gap
-2" onclick="addEmp()">
  <span class="material-symbols-outlined">person_add</span>
  <span>${tr('Add Employee', '????? ?????')}</span>

```

```

        </button>
    </div>

    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 lg:col-span-2">
        <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Employee Roster', '??? ???????')}<
    /h4>

    <div id="hrTableDiv" class="overflow-x-auto custom-scrollbar"></div>
    </div>
</div>
`;

setTimeout(() => {
    const hTableEl = document.getElementById('hrTableDiv');
    if (hTableEl) {
        createTable(hTableEl, 'hrTbl',
            [tr('Name', '?????'), tr('Role', '???????'), tr('Department', '?????'), tr('Salary', '???????'), tr('
Commission', '???????'), tr('Status', '???????'), tr('Delete', '???')],
            employees.map(e => ({
                cells: [
                    isArabic ? e.name_ar : e.name_en,
                    e.role,
                    isArabic ? e.department_ar : e.department_en,
                    (e.salary || 0).toLocaleString() + ' ' + tr('SAR', '??.?'),
                    e.role === 'Doctor' ? `${e.commission_value || 0}${e.commission_type === 'percentage' ? '%' : '
SAR'}` : '- ',
                    statusBadge(e.status)
                ],
                id: e.id
            })),
            r => `<button class="btn btn-danger btn-sm" onclick="delEmp(${safeId(r.id)})">?</button>`
        );
    }
}, 50);

} else if (window.hrTab === 'payroll') {
    // I2: removed employees.map fallback ? fabricated net_salary*1.25 with hardcoded status:'Paid'
    // is a falsely-reassuring figure. Real net pay comes only from POST /api/hr/payroll-slips.
    // Render empty state when no payroll slips exist.
    const displaySalaries = salaries.length ? salaries : [];

    htmlContent += `
    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
        <div class="flex items-center justify-between mb-md flex-wrap gap-2">
            <h4 class="font-title-lg text-title-lg text-primary">${tr('Payroll Run & Slips', '?????? ??????? ??
?? ???????')}</h4>
            <button class="btn btn-secondary btn-sm" onclick="showToast('${tr('WPS File generated successfully!'
, '?? ????? ?? ????? ?????? WPS ?????!')}')">
                ? ${tr('Export WPS File (Wages Protection)', '????? ?? ????? ??????')}
            </button>
        </div>
        <div id="payrollTable" class="overflow-x-auto custom-scrollbar"></div>
    </div>
    `;
};

```



```

setTimeout(() => {
  const pTableEl = document.getElementById('payrollTable');
  if (pTableEl) {
    createTable(pTableEl, 'payrollTbl',
      [tr('Employee Name', '??? ?????'), tr('Basic Salary', '????? ??????'), tr('Allowances', '??????'),
        tr('Deductions', '????????????'), tr('Net Salary', '???? ?????'), tr('Status', '?????'), tr('Actions', '??
        ??????')],
      displaySalaries.map(s => ({
        cells: [
          s.employee_name,
          parseFloat(s.basic || 0).toLocaleString() + ' ' + tr('SAR', '??.?'),
          parseFloat(s.allowances || 0).toLocaleString() + ' ' + tr('SAR', '??.?'),
          parseFloat(s.deductions || 0).toLocaleString() + ' ' + tr('SAR', '??.?'),
          parseFloat(s.net_salary || 0).toLocaleString() + ' ' + tr('SAR', '??.?'),
          statusBadge(s.status || 'Paid'),
        ],
        id: s.employee_id || s.id // C1: pass employee id, not salary-record PK
      })),
      r => `<button class="btn btn-sm" style="background:rgba(0,105,112,0.1);color:#006970" onclick="showP
      ayslip(${safeId(r.id)})">? ${tr('Payslip', '????? ??????')}</button>`
    );
  }
}, 50);

} else if (window.hrTab === 'leaves') {
  const displayLeaves = leaves.length ? leaves : [
    { id: 1, employee_name: '??. ???? ??????', leave_type: tr('Annual Leave', '????? ??????'), start_date: '20
    26-06-20', end_date: '2026-07-05', days: 15, status: 'Approved' },
    { id: 2, employee_name: '???? ??????', leave_type: tr('Sick Leave', '????? ??????'), start_date: '2026-0
    6-12', end_date: '2026-06-14', days: 2, status: 'Approved' },
    { id: 3, employee_name: '???? ??????', leave_type: tr('Maternity Leave', '????? ??????'), start_date: '2
    026-07-01', end_date: '2026-08-15', days: 45, status: 'Pending' }
  ];

  htmlContent += `
  <div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 flex flex-col gap-md">
      <h4 class="font-title-lg text-title-lg text-primary">${tr('Submit Leave Request', '????? ??? ?????')}
    </h4>

    <div class="form-group">
      <label>${tr('Select Employee', '???? ??????')}</label>
      <select class="form-input" id="leaveEmpId">
        ${employees.map(e => `<option value="${safeId(e.id)}">${escapeHTML(isArabic ? e.name_ar : e.name
        _en)}</option>`).join('')}
      </select>
    </div>

    <div class="form-group">
      <label>${tr('Leave Type', '??? ??????')}</label>
      <select class="form-input" id="leaveType">
        <option value="Annual">${tr('Annual Leave', '????? ??????')}</option>
        <option value="Sick">${tr('Sick Leave', '????? ??????')}</option>
        <option value="Maternity">${tr('Maternity Leave', '????? ??????')}</option>
        <option value="Emergency">${tr('Emergency Leave', '????? ?????????')}</option>
      </select>
    </div>
  </div>
  `;
}

```

```

</div>
<div class="grid grid-cols-2 gap-sm">
  <div class="form-group">
    <label>${tr('From Date', '?? ?????')}</label>
    <input type="date" class="form-input" id="leaveFrom" value="${new Date().toISOString().split('T')
)[0]}">
  </div>
  <div class="form-group">
    <label>${tr('To Date', '??? ?????')}</label>
    <input type="date" class="form-input" id="leaveTo" value="${new Date(Date.now() + 5*24*60*60*100
0).toISOString().split('T')[0]}">
  </div>
</div>
<button class="btn btn-primary w-full py-3 rounded-xl font-bold flex items-center justify-center gap
-2" onclick="requestLeave()">
  <span class="material-symbols-outlined">event_note</span>
  <span>${tr('Request Leave', '????? ??? ???????')}</span>
</button>
</div>

<div class="glass-card p-lg rounded-2xl border border-outline-variant/30 lg:col-span-2">
  <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Leaves Log', '??? ????? ???????')}<
/h4>
  <div id="leavesTable" class="overflow-x-auto custom-scrollbar"></div>
</div>
</div>
`;

setTimeout(() => {
  const lTableEl = document.getElementById('leavesTable');
  if (lTableEl) {
    createTable(lTableEl, 'leavesTbl',
      [tr('Employee Name', '??? ??????'), tr('Leave Type', '??? ???????'), tr('Duration', '?????'), tr('D
ays', '?????'), tr('Status', '?????'), tr('Actions', '????????')],
      displayLeaves.map(l => ({
        cells: [
          l.employee_name,
          l.leave_type,
          `${l.start_date} ? ${l.end_date}`,
          l.days + ' ' + tr('Days', '????'),
          statusBadge(l.status)
        ],
        id: l.id,
        status: l.status
      })),
      r => (r.status === 'requested' || r.cells[4].includes('Pending') || r.cells[4].includes('??? ??????
?')) ? `<button class="btn btn-success btn-sm" onclick="approveLeave(${safeId(r.id)})"> ${tr('Approve', '????
?')}</button>` : '-'
    );
  }
}, 50);

} else if (window.hrTab === 'attendance') {
  const displayAttendance = attendance.length ? attendance : employees.map(e => ({

```

```

        id: e.id,
        employee_name: isArabic ? e.name_ar : e.name_en,
        date: new Date().toISOString().split('T')[0],
        check_in: '08:00',
        check_out: '16:00',
        total_hours: 8.0,
        status: '????'
    }));

    htmlContent += `
        <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
            <div class="flex items-center justify-between mb-md flex-wrap gap-2">
                <h4 class="font-title-lg text-title-lg text-primary">${tr('Staff Daily Attendance Log', '??? ?????
                ???????? ??????')}</h4>
                <button class="btn btn-primary btn-sm" onclick="showToast('${tr('Clock-in registered!', '?? ????? ??
                ?? ?????? ??????!')}')">
                    ? ${tr('Self Clock-in', '????? ??? ????')}
                </button>
            </div>
            <div id="attendanceTable" class="overflow-x-auto custom-scrollbar"></div>
        </div>
    `;

    setTimeout(() => {
        const aTableEl = document.getElementById('attendanceTable');
        if (aTableEl) {
            createTable(aTableEl, 'attendanceTbl',
                [tr('Employee Name', '??? ??????'), tr('Date', '???????'), tr('Check In', '????? ??????'), tr('Check
                Out', '????? ??????'), tr('Total Hours', '????? ??????'), tr('Status', '?????')],
                displayAttendance.map(a => ({
                    cells: [
                        a.employee_name,
                        a.date,
                        a.check_in || '-',
                        a.check_out || '-',
                        a.total_hours ? parseFloat(a.total_hours).toFixed(1) + ' ' + tr('Hours', '????') : '-',
                        statusBadge(a.status || 'Present')
                    ],
                    id: a.id
                })))
        );
    }, 50);
}

content.innerHTML = htmlContent;

// Add the reactive listeners and setup GOSI payslip helper
const hrRoleEl = document.getElementById('hrRole');
if (hrRoleEl) {
    const showCommRow = () => {
        const commRow = document.getElementById('hrCommRow');
        if (commRow) commRow.style.display = hrRoleEl.value === 'Doctor' ? 'block' : 'none';
    };
}

```

```

    hrRoleEl.addEventListener('change', showCommRow);
    showCommRow();
}

// Payslip: NET PAY is computed SERVER-SIDE (anti-spoof). We POST to the safe route to
// generate/recompute the draft slip and render ONLY the server-returned figures.
window.showPayslip = async (empId) => {
    const emp = employees.find(e => e.id === empId) || { name_ar: tr('Employee', '????'), name_en: 'Employee',
    role: 'Staff' };
    const payMonth = new Date().toISOString().slice(0, 7); // YYYY-MM
    let slip;
    try {
        const resp = await API.post('/api/hr/payroll-slips', { employee_id: empId, pay_month: payMonth });
        slip = resp; // server-authoritative computed figures
    } catch (e) {
        showToast(tr('Could not generate payslip', '????? ????? ????? ??????'), 'error');
        return;
    }
    const num = (v) => (parseFloat(v) || 0).toFixed(2);

    const modal = document.createElement('div');
    modal.className = 'modal-overlay';
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
    modal.innerHTML = `
        <div class="glass-card p-xl rounded-2xl shadow-2xl max-w-lg w-full bg-white relative text-right" style="
direction:rtl">
            <h3 class="font-display-lg text-primary border-b pb-md mb-md">? ${tr('Detailed Salary Slip', '????? ??
?? ?????')} (${escapeHTML(slip.status || 'draft')})</h3>
            <div class="grid grid-cols-2 gap-sm mb-md text-sm text-on-surface-variant">
                <div><strong>${tr('Employee Name:', '??? ??????:')}</strong> ${escapeHTML(isArabic ? emp.name_ar : e
mp.name_en)}</div>
                <div><strong>${tr('Salary Month:', '??? ??????:')}</strong> ${escapeHTML(slip.pay_month || payMonth)}
</div>
            </div>
            <div class="border rounded-xl p-md bg-surface-lowest flex flex-col gap-xs mb-md text-sm text-on-surfac
e">
                <div class="flex justify-between"><span>${tr('Basic Salary', '????? ??????')}</span><strong>${num(
slip.basic)} ${tr('SAR', '?.?.')}</strong></div>
                <div class="flex justify-between"><span>${tr('Housing Allowance', '??? ?????')}</span><strong>${num(
slip.housing_allowance)} ${tr('SAR', '?.?.')}</strong></div>
                <div class="flex justify-between"><span>${tr('Transport Allowance', '??? ??????')}</span><strong>${
num(slip.transport_allowance)} ${tr('SAR', '?.?.')}</strong></div>
                <div class="flex justify-between text-error"><span>${tr('GOSI Deduction', '??? ????????? ?????????')}
)</span><strong>-${num(slip.gosi_deduction)} ${tr('SAR', '?.?.')}</strong></div>
                <div class="flex justify-between text-error"><span>${tr('Advances / Other Deductions', '???? / ?????
???? ?????')}</span><strong>-${num((parseFloat(slip.advances_deducted)||0)+(parseFloat(slip.other_deductions)||
0))} ${tr('SAR', '?.?.')}</strong></div>
                <div class="border-t pt-xs flex justify-between font-bold text-primary text-base"><span>${tr('Net Sa
lary', '???? ?????? ??????')}</span><strong>${num(slip.net_salary)} ${tr('SAR', '?.?.')}</strong></div>
            </div>
            <div class="flex gap-2">
                <button class="btn btn-primary flex-1 py-2 rounded-xl" onclick="window.print()">${tr('Print Slip', '
????? ??????')}</button>
    `

```

```

        <button class="btn btn-secondary flex-1 py-2 rounded-xl" onclick="this.closest('.modal-overlay').remove()">${tr('Close', '????')}</button>
    </div>
</div>
`;
document.body.appendChild(modal);
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };
};

// Leave request: POSTs to the safe route ? server forces status='requested' and computes days.
window.requestLeave = async () => {
    const empSel = document.getElementById('leaveEmpId');
    if (!empSel) return;
    const employee_id = parseInt(empSel.value, 10);
    const leave_type = document.getElementById('leaveType')?.value || 'Annual';
    const start_date = document.getElementById('leaveFrom')?.value;
    const end_date = document.getElementById('leaveTo')?.value;
    if (!Number.isInteger(employee_id) || !start_date || !end_date) { showToast(tr('Complete the leave form', '???? ????? ?????'), 'error'); return; }
    try {
        await API.post('/api/hr/leave-requests', { employee_id, leave_type, start_date, end_date });
        showToast(tr('Leave request submitted', '?? ????? ?? ?????????'), 'success');
        navigateTo(currentPage);
    } catch (e) { showToast(tr('Error submitting request', '??? ?? ????? ?????'), 'error'); }
};

// Approve leave: PUTs the state transition through the server-side state machine (409 on invalid).
window.approveLeave = async (id) => {
    try {
        await API.put(`/api/hr/leave-requests/${id}/status`, { status: 'approved' });
        showToast(tr('Leave request approved', '?? ????? ?? ?????????'), 'success');
        navigateTo(currentPage);
    } catch (e) { showToast(tr('Could not approve (invalid state?)', '????? ?????? (???? ?? ??????)'), 'error'); }
};

window.addEmp = async () => {
    const nameEn = document.getElementById('hrNameEn').value.trim();
    const nameAr = document.getElementById('hrNameAr').value.trim();
    const deptSel = document.getElementById('hrDept');
    const opt = deptSel.options[deptSel.selectedIndex];

    if (!nameEn && !nameAr) { showToast(tr('Enter employee name', '???? ?? ??????'), 'error'); return; }
    try {
        const role = document.getElementById('hrRole').value;
        const commType = role === 'Doctor' ? (document.getElementById('hrCommType')?.value || 'percentage') : 'percentage';
        const commValue = role === 'Doctor' ? (parseFloat(document.getElementById('hrCommValue')?.value) || 0) : 0;
        await API.post('/api/employees', {
            name_ar: nameAr,
            name_en: nameEn,
            role,

```

```

    department_en: deptSel.value,
    department_ar: opt ? (opt.getAttribute('data-ar') || '') : '',
    salary: document.getElementById('hrSalary').value,
    commission_type: commType,
    commission_value: commValue
  });
  showToast(tr('Employee added!', '??? ??????'));
  await navigateTo(7);
} catch (e) { showToast(tr('Error adding', '??? ?? ??????'), 'error'); }
};

window.delEmp = async (id) => {
  if (!confirm(tr('Delete this employee?', '??? ??? ??????'))) return;
  try { await API.del(`/api/employees/${id}`); showToast(tr('Deleted', '?? ?????')); await navigateTo(7); }
  catch (e) { showToast(tr('Error deleting', '??? ?? ?????'), 'error'); }
};

// ===== FINANCE =====
async function renderFinance(el) {
  const content = el;
  if (!window.financeTab) window.financeTab = 'dashboard';

  const today = new Date().toISOString().slice(0, 10);
  const monthStart = new Date(new Date().getFullYear(), new Date().getMonth(), 1).toISOString().slice(0, 10);

  let htmlContent = `
    <div class="mb-xl flex flex-col md:flex-row md:items-end justify-between gap-md">
      <div>
        <nav class="flex text-on-surface-variant font-caption text-caption mb-xs">
          <span>${tr('Operations', '????????')}</span>
          <span class="mx-xs">/</span>
          <span class="text-primary font-bold">${tr('Finance & General Ledger', '??????? ????????? ??????')}</
span>
        </nav>
        <h3 class="font-display-lg text-display-lg text-primary tracking-tight">${tr('Financial Performance',
'??????? ???????? ??????')}</h3>
        <p class="font-body-md text-body-md text-on-surface-variant max-w-2xl">
          ${tr('Track revenue, general ledger accounts, vouchers, daily closing, and ZATCA audit reports.', '?
??? ???????? ????????? ?????? ?????? ?????? ????????? ???????? ?????? ?????? ???????? ????????? ??????????')}
        </p>
      </div>
    </div>

    <!-- Tabs Bar -->
    <div class="tab-bar mb-md flex border-b border-outline-variant/30 overflow-x-auto gap-2">
      <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'dashboar
d' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary
'}" onclick="window.financeTab='dashboard'; navigateTo(currentPage)">
        ? ${tr('Dashboard & Invoices', '???? ?????? ??????')}
      </button>
      <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'accounts
' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary
'}" onclick="window.financeTab='accounts'; navigateTo(currentPage)">
        ? ${tr('Chart of Accounts', '???? ?????????')}

```

```

</button>
<button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'vouchers'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.financeTab='vouchers'; navigateTo(currentPage)">
  ? ${tr('Vouchers & Daily Close', '?????? ???????')}
</button>
<button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'denials'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.financeTab='denials'; navigateTo(currentPage)">
  ?? ${tr('Revenue Cycle & Denials', '???? ????????? ?????????')}
</button>
<button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'ledger'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.financeTab='ledger'; navigateTo(currentPage)">
  ? ${tr('General Ledger', '????? ?????')}
</button>
<button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.financeTab === 'aging' ?
'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.financeTab='aging'; navigateTo(currentPage)">
  ? ${tr('AR Aging', '????? ?????? ?????????')}
</button>
</div>
`;

if (window.financeTab === 'dashboard') {
  htmlContent += `
    <!-- Filters Card -->
    <div class="glass-card p-md rounded-xl mb-xl border border-outline-variant/20">
      <div class="flex flex-wrap gap-md items-center justify-between">
        <div class="flex items-center gap-sm flex-wrap">
          <span class="font-bold text-primary">${tr('Filter Period:', '???? ??????')}</span>
          <input type="date" class="form-input w-auto" id="finFrom" value="${monthStart}">
          <span class="text-on-surface-variant">?</span>
          <input type="date" class="form-input w-auto" id="finTo" value="${today}">
          <button class="btn btn-primary btn-sm flex items-center gap-1" onclick="loadFinance()">
            <span class="material-symbols-outlined text-sm">search</span>
            <span>${tr('Apply', '????')}</span>
          </button>
        </div>
        <div class="flex gap-2">
          <button class="btn btn-secondary btn-sm flex items-center gap-1" onclick="exportToCSV(window._finI
nvoices||[], 'finance')">
            <span class="material-symbols-outlined text-sm">download</span>
            <span>${tr('Export', '????')}</span>
          </button>
          <button class="btn btn-secondary btn-sm flex items-center gap-1" onclick="window.print()">
            <span class="material-symbols-outlined text-sm">print</span>
            <span>${tr('Print', '????')}</span>
          </button>
        </div>
      </div>
    </div>
  `;
  <!-- Stats Grid -->

```

```

<div id="finStats" class="mb-xl"></div>

<!-- Charts Section -->
<div class="grid grid-cols-1 lg:grid-cols-2 gap-gutter mb-xl">
  <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
    <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Revenue Trend', '????? ?????????')}
  </h4>

    <div style="height: 250px; position: relative;"><canvas id="finRevenueChart"></canvas></div>
  </div>

  <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
    <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Revenue by Service', '????? ????????
    ?? ??? ??????')}</h4>

    <div style="height: 250px; position: relative;"><canvas id="finServiceChart"></canvas></div>
  </div>
</div>

<!-- Recent Invoices Table -->
<div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
  <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Recent Invoices', '??? ????????? ?????
  ??')}</h4>

  <div id="finTable" class="overflow-x-auto custom-scrollbar"></div>
</div>
`;

setTimeout(() => {
  loadFinance();
}, 50);

} else if (window.financeTab === 'accounts') {
  htmlContent += `
    <div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
      <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 flex flex-col gap-md">
        <h4 class="font-title-lg text-title-lg text-primary">${tr('Create Ledger Account', '??? ???? ??????
        ?????')}</h4>

        <div class="form-group">
          <label>${tr('Account Code', '??? ?????')}</label>
          <input class="form-input" id="coaCode" placeholder="e.g. 110101">
        </div>

        <div class="form-group">
          <label>${tr('Account Name (Arabic)', '??? ?????? ?????????')}</label>
          <input class="form-input" id="coaNameAr" placeholder="?????: ?????? ??????">
        </div>

        <div class="form-group">
          <label>${tr('Account Name (English)', '??? ?????? ?????????')}</label>
          <input class="form-input" id="coaNameEn" placeholder="e.g. Main Cash Box">
        </div>

        <div class="form-group">
          <label>${tr('Account Type', '??? ?????')}</label>
          <select class="form-input" id="coaType">
            <option value="Asset">${tr('Asset', '????')}</option>
            <option value="Liability">${tr('Liability', '????')}</option>
            <option value="Equity">${tr('Equity', '???? ?????')}</option>
            <option value="Revenue">${tr('Revenue', '????????')}</option>
            <option value="Expense">${tr('Expense', '????????')}</option>
          </select>
        </div>
      </div>
    </div>
  `;
}

```



```

        </select>
    </div>
    <button class="btn btn-primary w-full py-3 rounded-xl font-bold flex items-center justify-center gap
-2" onclick="createLedgerAccount()">
        <span class="material-symbols-outlined">account_balance_wallet</span>
        <span>${tr('Create Account', '????? ?????')}</span>
    </button>
</div>

<div class="glass-card p-lg rounded-2xl border border-outline-variant/30 lg:col-span-2">
    <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Chart of Accounts Directory', '????
???? ?????????')}</h4>
    <div id="coaTable" class="overflow-x-auto custom-scrollbar"></div>
</div>
</div>
`;

setTimeout(async () => {
    const coaTableEl = document.getElementById('coaTable');
    if (coaTableEl) {
        const accounts = await API.get('/api/finance/accounts').catch(() => []);
        const displayAccounts = accounts.length ? accounts : [
            { account_code: '100000', account_name_ar: '??????', account_name_en: 'Assets', account_type: 'Asset
' },
            { account_code: '110000', account_name_ar: '?????? ??????????', account_name_en: 'Current Assets', ac
count_type: 'Asset' },
            { account_code: '110101', account_name_ar: '???? ??? ????????? ?????????', account_name_en: 'Al Rajhi Ma
in Bank', account_type: 'Asset' },
            { account_code: '200000', account_name_ar: '?????? ??????????????', account_name_en: 'Liabilities', acc
ount_type: 'Liability' },
            { account_code: '300000', account_name_ar: '???? ?????????', account_name_en: 'Equity', account_type:
'Equity' },
            { account_code: '400000', account_name_ar: '?????????? ??????????', account_name_en: 'Operating Revenu
e', account_type: 'Revenue' },
            { account_code: '500000', account_name_ar: '?????????? ??????? ??????????', account_name_en: 'General E
xpenses', account_type: 'Expense' }
        ];

        createTable(coaTableEl, 'coaTbl',
            [tr('Account Code', '??? ??????'), tr('Account Name (AR)', '??? ?????? (????)'), tr('Account Name (E
N)', '??? ?????? (????????)'), tr('Type', '?????')],
            displayAccounts.map(a => ({
                cells: [
                    rawHtml(`<span class="font-mono text-secondary">${escapeHTML(a.account_code)}</span>`),
                    a.account_name_ar,
                    a.account_name_en,
                    rawHtml(`<span class="text-caption bg-surface-container px-sm py-1 rounded-full">${escapeHTML(a.
account_type)}</span>`)
                ],
                id: a.account_code
            })))
        );
    }
}, 500);

```

```

} else if (window.financeTab === 'vouchers') {
  htmlContent += `
    <div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
      <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 flex flex-col gap-md">
        <h4 class="font-title-lg text-title-lg text-primary">${tr('Register Daily Close', '????? ????? ?????
        ?? ??????')}</h4>
        <div class="form-group">
          <label>${tr('Opening Balance (SAR)', '????? ????????? (?.?)')}</label>
          <input class="form-input" id="dcOpen" type="number" value="1000">
        </div>
        <div class="form-group">
          <label>${tr('Closing Balance (SAR)', '????? ?????? ?? ?????? (?.?)')}</label>
          <input class="form-input" id="dcClose" type="number" placeholder="5000">
        </div>
        <div class="form-group">
          <label>${tr('Closing Notes', '????? ??????')}</label>
          <textarea class="form-input" id="dcNotes" rows="3" placeholder="????? ???????? ??????..."></text
area>
        </div>
        <button class="btn btn-primary w-full py-3 rounded-xl font-bold flex items-center justify-center gap
-2" onclick="performDailyClose()">
          <span class="material-symbols-outlined">lock_open</span>
          <span>${tr('Perform Daily Close', '????? ????? ??????')}</span>
        </button>
      </div>

      <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 lg:col-span-2">
        <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Journal Vouchers', '????? ?????? ??
        ?????? ??????')}</h4>
        <div id="vouchersTable" class="overflow-x-auto custom-scrollbar"></div>
      </div>
    </div>
  `;

  setTimeout(async () => {
    const vTableEl = document.getElementById('vouchersTable');
    if (vTableEl) {
      const vouchers = await API.get('/api/finance/vouchers').catch(() => []);
      const displayVouchers = vouchers.length ? vouchers : [
        { voucher_number: 'JV-2026-001', voucher_type: 'Receipt', amount: 850.00, voucher_date: today, desc
ription: '??? ????' },
        { voucher_number: 'JV-2026-002', voucher_type: 'Payment', amount: 1500.00, voucher_date: today, desc
ription: '???? ???????? ????? ??????' },
        { voucher_number: 'JV-2026-003', voucher_type: 'Journal', amount: 12450.00, voucher_date: today, des
cription: '??? ?????? ?????? ?????? ????' }
      ];

      createTable(vTableEl, 'vouchersTbl',
        [tr('Voucher #', '??? ?????'), tr('Type', '?????'), tr('Amount', '?????'), tr('Date', '???????'), t
r('Description', '?????')],
        displayVouchers.map(v => ({
          cells: [
            rawHtml(`<span class="font-mono text-secondary font-bold">${escapeHTML(v.voucher_number)}</span>

```

```

    ),
        v.voucher_type,
        parseFloat(v.amount).toFixed(2) + ' ' + tr('SAR', '??.?'),
        v.voucher_date,
        v.description
    ],
    id: v.voucher_number
  )))
  );
}
}, 50);

} else if (window.financeTab === 'denials') {
  htmlContent += `
    <div class="grid grid-cols-1 md:grid-cols-3 gap-gutter mb-xl">
      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-error/10 rounded-lg text-error material-symbols-outlined">cancel</span>
          <span class="text-error text-xs font-bold">2.4%</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Claims Denial Rate', '???? ??
          ?????? ?????????')}</p>
          <h4 class="text-display-lg text-error font-bold mt-xs">2.4%</h4>
        </div>
      </div>

      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">payments</s
pan>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Expected Recovery', '???????
          ???????? ?????????')}</p>
          <h4 class="text-display-lg text-primary font-bold mt-xs">98,540 <span class="text-title-lg">${tr('
          SAR', '??.?')}</span></h4>
        </div>
      </div>

      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">published_w
ith_changes</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Collection Efficiency', '????
          ? ?????? ?????????')}</p>
          <h4 class="text-display-lg text-primary font-bold mt-xs">97.6%</h4>
        </div>
      </div>
    </div>

    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">

```

```

<h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Insurance Denial Logs', '??? ??????
??????? ?????? ??????')}</h4>

<div id="denialClaimsTable" class="overflow-x-auto custom-scrollbar"></div>
</div>
`;

setTimeout(async () => {
  const dcTableEl = document.getElementById('denialClaimsTable');
  if (dcTableEl) {
    const claims = await API.get('/api/insurance/claims').catch(() => []);
    const displayClaims = claims.length ? claims : [
      { id: 1, patient_name: '????? ??????', insurance_company: '????????? ???????', claim_amount: 1450.00, status: 'Rejected', waseel_status: 'Error: Code 23' },
      { id: 2, patient_name: '??? ?????????', insurance_company: '???? ?????????', claim_amount: 3200.00, status: 'Approved', waseel_status: 'Sent Successfully' },
      { id: 3, patient_name: '??? ?????????', insurance_company: '?????? ?????????', claim_amount: 880.00, status: 'Pending', waseel_status: 'Under Review' }
    ];

    createTable(dcTableEl, 'dcClaimsTbl',
      [tr('Patient Name', '??? ??????'), tr('Insurance Company', '???? ?????????'), tr('Claim Amount', '????
????????'), tr('Status', '??????'), tr('Waseel Status', '????? ?????')],
      displayClaims.map(c => ({
        cells: [
          c.patient_name,
          c.insurance_company,
          parseFloat(c.claim_amount).toFixed(2) + ' ' + tr('SAR', '??.?'),
          statusBadge(c.status),
          rawHtml(`<span class="font-mono text-secondary">${escapeHTML(c.waseel_status)}</span>`)
        ],
        id: c.id
      })))
    );
  }
}, 50);

} else if (window.financeTab === 'ledger') {
  // E10 General Ledger: balanced double-entry journal. Posting to the ledger is GATED OFF by
  // default (server flag ACCOUNTING_POSTING_ENABLED). Entries are created as DRAFT.
  htmlContent += `
<div id="glGateBanner" class="mb-md"></div>
<div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
  <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 flex flex-col gap-md">
    <h4 class="font-title-lg text-title-lg text-primary">${tr('New Journal Entry', '??? ?????? ?????')}</h4>

    <div class="form-group">
      <label>${tr('Entry Date', '????? ?????')}</label>
      <input class="form-input" id="jeDate" type="date" value="${today}">
    </div>

    <div class="form-group">
      <label>${tr('Description', '???????')}</label>
      <input class="form-input" id="jeDesc" placeholder="${tr('e.g. Cash sale', '?????: ?????? ?????')}">
    </div>

    <div id="jeLines" class="flex flex-col gap-sm"></div>
  </div>
</div>
4>

```

```

        <button class="btn btn-secondary btn-sm" onclick="e10AddJournalLine()">+ ${tr('Add Line', '????? ???
')}}</button>

        <div class="text-caption text-on-surface-variant" id="jeBalanceHint"></div>
        <button class="btn btn-primary w-full py-3 rounded-xl font-bold" onclick="postJournalEntry()">
            <span class="material-symbols-outlined">post_add</span>
            <span>${tr('Create Draft Entry', '????? ??? (?????)')}</span>
        </button>
    </div>
    <div class="glass-card p-lg rounded-2xl border border-outline-variant/30 lg:col-span-2">
        <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Journal Entries', '???? ???????')}<
/h4>

        <div id="glTable" class="overflow-x-auto custom-scrollbar"></div>
    </div>
</div>
`;
setTimeout(() => { if (typeof e10ResetJournalForm === 'function') e10ResetJournalForm(); loadGeneralLedger
(); }, 50);

} else if (window.financeTab === 'aging') {
    htmlContent += `
        <div id="agingStats" class="mb-xl"></div>
        <div class="glass-card p-lg rounded-2xl border border-outline-variant/30">
            <h4 class="font-title-lg text-title-lg text-primary mb-md">${tr('Accounts Receivable Aging', '????? ??
??? ?????? ???????')}</h4>
            <div id="agingTable" class="overflow-x-auto custom-scrollbar"></div>
        </div>
    `;
    setTimeout(() => { loadARaging(); }, 50);
}

content.innerHTML = htmlContent;

window.loadFinance = async () => {
    const from = document.getElementById('finFrom')?.value || '';
    const to = document.getElementById('finTo')?.value || '';
    try {
        const data = await API.get('/api/finance/summary?from=' + from + '&to=' + to);
        const statsEl = document.getElementById('finStats');
        if (statsEl) {
            statsEl.innerHTML = `
                <div class="grid grid-cols-1 md:grid-cols-4 gap-gutter">
                    <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relati
ve">
                        <div class="flex justify-between items-start relative z-10">
                            <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined">payments</s
pan>
                        </div>
                        <div class="mt-lg relative z-10">
                            <p class="text-on-surface-variant font-label-md text-label-md">${tr('Total Revenue', '?????? ?
????????')}</p>
                            <h4 class="text-display-lg text-primary font-bold mt-xs">${parseFloat(data.revenue || 0).toLoc
aleString()} <span class="text-title-lg">${tr('SAR', '?.?.')}</span></h4>
                        </div>
                    </div>
                `;
            }
        }
    }
}

```

```

        <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">

            <div class="flex justify-between items-start relative z-10">
                <span class="p-xs bg-success/10 rounded-lg text-success material-symbols-outlined">check_circle</span>
            </div>

            <div class="mt-lg relative z-10">
                <p class="text-on-surface-variant font-label-md text-label-md">${tr('Collected Amount', '?????
                ?? ???????')}</p>

                <h4 class="text-display-lg text-success font-bold mt-xs">${parseFloat(data.paid || 0).toLocale
                String()} <span class="text-title-lg">${tr('SAR', '?.?')}</span></h4>
            </div>
        </div>

        <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">

            <div class="flex justify-between items-start relative z-10">
                <span class="p-xs bg-error/10 rounded-lg text-error material-symbols-outlined">pending</span>
            </div>

            <div class="mt-lg relative z-10">
                <p class="text-on-surface-variant font-label-md text-label-md">${tr('Outstanding Balance', '??
                ??? ?????? ???????')}</p>

                <h4 class="text-display-lg text-error font-bold mt-xs">${parseFloat(data.unpaid || 0).toLocale
                String()} <span class="text-title-lg">${tr('SAR', '?.?')}</span></h4>
            </div>
        </div>

        <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">

            <div class="flex justify-between items-start relative z-10">
                <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">receipt
                _long</span>
            </div>

            <div class="mt-lg relative z-10">
                <p class="text-on-surface-variant font-label-md text-label-md">${tr('Invoices Issued', '?????
                ??????? ???????')}</p>

                <h4 class="text-display-lg text-primary font-bold mt-xs">${escapeHTML(data.count || 0)}</h4>
            </div>
        </div>
    `;
}

// Revenue chart
if (typeof Chart !== 'undefined' && data.daily?.length > 0) {
    const revCtx = document.getElementById('finRevenueChart');
    if (revCtx) { Chart.getChart(revCtx)?.destroy(); new Chart(revCtx, { type: 'line', data: { labels: dat
    a.daily.map(d => new Date(d.day).toLocaleDateString('ar-SA', { day: 'numeric', month: 'short' })), datasets: [
    { label: tr('Revenue', '??????'), data: data.daily.map(d => parseFloat(d.amount)), borderColor: '#006970', ba
    ckgroundColor: 'rgba(0,105,112,0.1)', fill: true, tension: 0.4 } ] }, options: { responsive: true, maintainAspe
    ctRatio: false, plugins: { legend: { display: false } }, scales: { y: { beginAtZero: true } } } }); }
    const svcCtx = document.getElementById('finServiceChart');
    if (svcCtx && data.byService?.length > 0) { Chart.getChart(svcCtx)?.destroy(); new Chart(svcCtx, { typ
    e: 'doughnut', data: { labels: data.byService.map(s => s.service), datasets: [{ data: data.byService.map(s =>

```

```

parseFloat(s.amount)), backgroundColor: ['#001629', '#006970', '#ac8c44', '#ea4335', '#ff6d01', '#46bdc6', '#7
baaf7', '#f07b72', '#fcd04f', '#71c287'] ] } }, options: { responsive: true, maintainAspectRatio: false, plugin
s: { legend: { position: 'right', labels: { font: { size: 11 } } } } } }); }
}
// Load invoices table
const invoices = await API.get('/api/invoices');
window._finInvoices = invoices;
const ft = document.getElementById('finTable');
if (ft && invoices.length) {
  createTable(ft, 'finTbl',
    [tr('#', '#'), tr('Patient', '?????'), tr('Service', '?????'), tr('Amount', '?????'), tr('Status'
, '?????'), tr('Date', '???????')],
    invoices.slice(0, 50).map(i => ({ cells: [i.invoice_number || i.id, i.patient_name || '', i.descript
ion || i.service_type || '', parseFloat(i.total || 0).toFixed(2) + ' ' + tr('SAR', '????'), statusBadge(i.paid
? 'Paid' : 'Unpaid'), i.created_at ? new Date(i.created_at).toLocaleDateString('ar-SA') : ''], id: i.id })))
  );
}
} catch (e) { console.error(e); }
};

window.createLedgerAccount = async () => {
  const code = document.getElementById('coaCode').value.trim();
  const nameAr = document.getElementById('coaNameAr').value.trim();
  const nameEn = document.getElementById('coaNameEn').value.trim();
  const type = document.getElementById('coaType').value;

  if (!code || !nameAr || !nameEn) { showToast(tr('Fill all fields', '????? ??? ???? ??????'), 'error'); re
turn; }
  try {
    await API.post('/api/finance/accounts', { account_code: code, account_name_ar: nameAr, account_name_en:
nameEn, account_type: type });
    showToast(tr('Ledger account created!', '?? ????? ?????? ?????!'));
    navigateTo(currentPage);
  } catch (e) { showToast(tr('Error creating account', '??? ?? ?????? ??????'), 'error'); }
};

}

window.performDailyClose = async function () {
  try {
    const result = await API.post('/api/finance/daily-close', {
      opening_balance: document.getElementById('dcOpen').value || 0,
      closing_balance: document.getElementById('dcClose').value || 0,
      notes: document.getElementById('dcNotes').value
    });
    showToast(tr('Day closed! Variance: ' + result.variance + ' SAR', '?? ????????! ??????: ' + result.variance
+ ' ?.'));
    navigateTo(8);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== E10 GENERAL LEDGER (double-entry) + AR AGING UI handlers =====
// All server values rendered via escapeHTML/safeId; the balanced-entry rule + posting gate are
// enforced SERVER-SIDE ? these handlers are convenience only and never assert authority.
window._e10JournalLines = [];

```

```

window.e10ResetJournalForm = function () {
  window._e10JournalLines = [{ account_id: '', debit: '', credit: '' }, { account_id: '', debit: '', credit: '' }];
  e10RenderJournalLines();
};

window.e10AddJournalLine = function () {
  window._e10JournalLines.push({ account_id: '', debit: '', credit: '' });
  e10RenderJournalLines();
};

window.e10RenderJournalLines = function () {
  const host = document.getElementById('jeLines');
  if (!host) return;
  host.innerHTML = window._e10JournalLines.map((ln, i) => `
    <div class="flex gap-1 items-center">
      <input class="form-input flex-1" placeholder="${tr('Acct ID', '??? ?????')}" value="${escapeHTML(ln.account_id)}" oninput="window._e10JournalLines[${safeId(i)}].account_id=this.value">
      <input class="form-input w-24" type="number" placeholder="${tr('Debit', '????')}" value="${escapeHTML(ln.debit)}" oninput="window._e10JournalLines[${safeId(i)}].debit=this.value; e10UpdateBalanceHint()">
      <input class="form-input w-24" type="number" placeholder="${tr('Credit', '????')}" value="${escapeHTML(ln.credit)}" oninput="window._e10JournalLines[${safeId(i)}].credit=this.value; e10UpdateBalanceHint()">
    </div>` ).join('');
  e10UpdateBalanceHint();
};

window.e10UpdateBalanceHint = function () {
  const hint = document.getElementById('jeBalanceHint');
  if (!hint) return;
  const d = window._e10JournalLines.reduce((s, l) => s + (parseFloat(l.debit) || 0), 0);
  const c = window._e10JournalLines.reduce((s, l) => s + (parseFloat(l.credit) || 0), 0);
  const balanced = Math.round(d * 100) === Math.round(c * 100) && d > 0;
  hint.innerHTML = `${tr('Debit', '????')}: ${escapeHTML(d.toFixed(2))} | ${tr('Credit', '????')}: ${escapeHTML(c.toFixed(2))} ` +
    (balanced ? `<span class="text-success font-bold"> ${tr('Balanced', '??????')}</span>` : `<span class="text-error font-bold"> ${tr('Unbalanced', '??? ?????')}</span>`);
};

window.postJournalEntry = async function () {
  const lines = (window._e10JournalLines || []).
    .map(l => ({ account_id: parseInt(l.account_id, 10), debit: parseFloat(l.debit) || 0, credit: parseFloat(l.credit) || 0 }));
  .filter(l => Number.isInteger(l.account_id) && (l.debit > 0 || l.credit > 0));
  try {
    const r = await API.post('/api/finance/journal', {
      entry_date: document.getElementById('jeDate')?.value || '',
      description: document.getElementById('jeDesc')?.value || '',
      lines
    });
    if (r && r.error) { showToast(tr('Rejected: ' + r.error, '?????: ' + r.error), 'error'); return; }
    showToast(tr('Draft journal entry created', '?? ????? ?? ?????'));
    e10ResetJournalForm();
    loadGeneralLedger();
  } catch (e) { showToast(tr('Error creating entry', '??? ?? ????? ?????'), 'error'); }
};

window.loadGeneralLedger = async function () {
  const banner = document.getElementById('glGateBanner');
  if (banner) {

```



```

const st = await API.get('/api/finance/posting-status').catch(() => ({ posting_enabled: false }));
banner.innerHTML = st.posting_enabled
? `
```

```

        parseFloat(inv.balance || 0).toFixed(2) + ' ' + tr('SAR', '?.?'),
        String(parseInt(inv.age_days, 10) || 0)
    ],
    id: inv.id
  })))
);
};

// ===== INSURANCE =====
async function renderInsurance(el) {
  if (!window.insuranceTab) window.insuranceTab = 'claims';

  const [claims, companies, policies] = await Promise.all([
    API.get('/api/insurance/claims'),
    API.get('/api/insurance/companies').catch(() => []),
    API.get('/api/insurance/policies').catch(() => [])
  ]);

  const approved = claims.filter(c => c.status === 'Approved' || c.lifecycle_status === 'adjudicated').reduce(
    (s, c) => s + (c.claim_amount || 0), 0);
  const pending = claims.filter(c => c.status === 'Pending' || c.lifecycle_status === 'submitted' || c.lifecyc
le_status === 'draft').reduce((s, c) => s + (c.claim_amount || 0), 0);

  el.innerHTML = `
    <div class="mb-xl flex flex-col md:flex-row md:items-end justify-between gap-md" style="margin-bottom:24px
">

      <div>
        <nav class="flex text-on-surface-variant font-caption text-caption mb-xs" style="font-size:11px;color:
var(--text-dim);margin-bottom:4px">
          <span>${tr('Revenue Cycle', '???? ?????????')}</span>
          <span style="margin:0 4px"></span>
          <span class="text-primary font-bold" style="color:var(--primary);font-weight:700">${tr('Insurance &
NPHIES Gateway', '?????? ???? NPHIES ???????')}</span>
        </nav>
        <h3 class="font-display-lg text-display-lg text-primary tracking-tight" style="font-size:24px;color:va
r(--primary);font-weight:800;margin:0 0 8px">${tr('Insurance & NPHIES Management', '???? ???? NPHIE
S')}</h3>
        <p class="font-body-md text-body-md text-on-surface-variant max-w-2xl" style="font-size:13px;color:var
(--text-dim);margin:0">
          ${tr('Verify patient eligibility, manage prior authorizations, and track electronic claims submissio
n via FHIR gateway.', '????? ? ???? ?????? ?????? ?????????? ?????? ?????????? ?????? ?????? ??????????
?? ??? ????? FHIR ??????.')}
        </p>
      </div>
    </div>

    <!-- NPHIES Tabs Bar -->
    <div class="flex gap-4 mb-16" style="display:flex;gap:12px;margin-bottom:20px;border-bottom:1px solid var(
--border);padding-bottom:12px;flex-wrap:wrap">
      <button class="btn ${window.insuranceTab === 'claims' ? 'btn-primary' : 'btn-secondary'}" onclick="windo
w.insuranceTab='claims';navigateTo(9)" style="font-size:12px">? ${tr('Claims Management', '????? ?????????')}<
/button>
      <button class="btn ${window.insuranceTab === 'eligibility' ? 'btn-primary' : 'btn-secondary'}" onclick="
window.insuranceTab='eligibility';navigateTo(9)" style="font-size:12px">? ${tr('Eligibility (NPHIES)', '?????
????? (NPHIES)')}</button>

```

```

        <button class="btn ${window.insuranceTab === 'priorauth' ? 'btn-primary' : 'btn-secondary'}" onclick="window.insuranceTab='priorauth';navigateTo(9)" style="font-size:12px">?? ${tr('Prior Authorization', '????????????????')}</button>

        <button class="btn ${window.insuranceTab === 'companies' ? 'btn-primary' : 'btn-secondary'}" onclick="window.insuranceTab='companies';navigateTo(9)" style="font-size:12px">? ${tr('Payer Companies', '????? ?????')}</button>
    </div>

    <div id="insuranceTabContent"></div>
    `;

    const cont = document.getElementById('insuranceTabContent');
    if (!cont) return;

    if (window.insuranceTab === 'claims') {
        cont.innerHTML = `
            <div class="stats-grid" style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px;margin-bottom:24px">
                <div class="stat-card" style="--stat-color:#3b82f6;border-left:4px solid #3b82f6;background:var(--surface-container,#f8fafc);padding:14px;border-radius:12px">
                    <div class="stat-label" style="font-size:12px;color:var(--text-dim)">${tr('Total Claims', '????? ?? ?????')}</div>
                    <div class="stat-value" style="font-size:22px;font-weight:800">${claims.length}</div>
                </div>
                <div class="stat-card" style="--stat-color:#4ade80;border-left:4px solid #4ade80;background:var(--surface-container,#f8fafc);padding:14px;border-radius:12px">
                    <div class="stat-label" style="font-size:12px;color:var(--text-dim)">${tr('Approved / Paid', '????? ? ?????')}</div>
                    <div class="stat-value" style="font-size:22px;font-weight:800;color:#16a34a">${approved.toLocaleString()} SAR</div>
                </div>
                <div class="stat-card" style="--stat-color:#f59e0b;border-left:4px solid #f59e0b;background:var(--surface-container,#f8fafc);padding:14px;border-radius:12px">
                    <div class="stat-label" style="font-size:12px;color:var(--text-dim)">${tr('Pending Claims', '????? ?????')}</div>
                    <div class="stat-value" style="font-size:22px;font-weight:800;color:#d97706">${pending.toLocaleString()} SAR</div>
                </div>
                <div class="stat-card" style="--stat-color:#8b5cf6;border-left:4px solid #8b5cf6;background:var(--surface-container,#f8fafc);padding:14px;border-radius:12px">
                    <div class="stat-label" style="font-size:12px;color:var(--text-dim)">${tr('Active Payers', '????? ?????')}</div>
                    <div class="stat-value" style="font-size:22px;font-weight:800;color:#7c3aed">${companies.length}</div>
            </div>

            <div class="grid-equal" style="display:grid;grid-template-columns:1fr 1fr;gap:20px;margin-bottom:20px">
                <div class="card">
                    <div class="card-title">? ${tr('New Insurance Claim', '????? ?????')}</div>
                    <div class="form-group mb-12" style="margin-bottom:12px">
                        <label style="font-weight:600;font-size:12px">${tr('Patient Name', '??? ?????')}</label>
                        <input class="form-input" id="insPatient" placeholder="${tr('Patient name', '??? ?????')}>
                    </div>
                </div>
            </div>
        `;
    }

```

```

<div class="form-group mb-12" style="margin-bottom:12px">
  <label style="font-weight:600;font-size:12px">${tr('Insurance Payer', '???? ?????')}</label>
  <select class="form-input" id="insCompany">
    <option value="Bupa Arabia">Bupa Arabia</option>
    <option value="Tawuniya">Tawuniya</option>
    <option value="MedGulf">MedGulf</option>
    <option value="Alrajhi Takaful">Alrajhi Takaful</option>
    <option value="Walaa">Walaa</option>
    ${companies.map(c => `<option value="${escapeHTML(c.name_en || c.name_ar)}">${escapeHTML(c.name_
en || c.name_ar)}</option>`).join('')}
  </select>
</div>

<div class="form-group mb-12" style="margin-bottom:16px">
  <label style="font-weight:600;font-size:12px">${tr('Claim Amount', '???? ?????')}</label>
  <input class="form-input" id="insAmount" type="number" placeholder="0.00">
</div>

<button class="btn btn-primary w-full" onclick="addClaim()" style="height:40px;width:100%">? ${tr('S
ubmit Claim', '???? ?????')}</button>
</div>

<div class="card">
  <div class="card-title">? ${tr('Recent Claims Log', '??? ?????')}</div>
  <input class="search-filter" placeholder="${tr('Search...', '???...')}" oninput="filterTable(this,'i
nsClaimsT')" style="margin-bottom:12px;padding:8px;font-size:12px;width:100%">
  <div id="insClaimsT" style="overflow-x:auto">
    ${makeTable(
      [tr('Patient', '?????'), tr('Company', '?????'), tr('Amount', '?????'), tr('Status', '?????')
), tr('Actions', '?????')],
      claims.map(c => ({
        cells: [
          c.patient_name,
          c.insurance_company,
          (c.claim_amount || 0) + ' SAR',
          statusBadge(c.lifecycle_status || c.status)
        ],
        id: c.id,
        lifecycle: c.lifecycle_status || 'draft'
      })),
      (row) => insClaimActions(row)
    )}
  </div>
</div>
</div>
`;
} else if (window.insuranceTab === 'eligibility') {
  cont.innerHTML = `
    <div class="grid-equal" style="display:grid;grid-template-columns:1fr 1fr;gap:20px">
      <div class="card">
        <div class="card-title">? ${tr('Check Patient Eligibility (NPHIES Gateway)', '?????? ?????? ??????
??? NPHIES')}</div>
        <p style="font-size:11px;color:var(--text-dim);margin-bottom:12px">
          ${tr('Query national insurance database using patient national ID or policy details to get instant
coverage breakdown.', '????? ?????? ?? ????? ??????? ??????? ??????? ?????? ??????? ?????? ?????? ?? ?????
?? ??????.')}

```

```

</p>
<div class="form-group mb-12" style="margin-bottom:12px">
  <label style="font-weight:600;font-size:12px">${tr('Patient National ID / IQAMA', '??? ???? ??????
/ ???????')}</label>
  <input class="form-input" id="eligPatientId" placeholder="e.g. 1024567890">
</div>
<div class="form-group mb-12" style="margin-bottom:12px">
  <label style="font-weight:600;font-size:12px">${tr('Insurance Company', '???? ???????')}</label>
  <select class="form-input" id="eligCompany">
    <option value="Bupa Arabia">Bupa Arabia</option>
    <option value="Tawuniya">Tawuniya</option>
    <option value="MedGulf">MedGulf</option>
    <option value="Alrajhi Takaful">Alrajhi Takaful</option>
    ${companies.map(c => `<option value="${escapeHTML(c.name_en || c.name_ar)}">${escapeHTML(c.name_
en || c.name_ar)}</option>`).join('')}
  </select>
</div>
<div class="form-group mb-12" style="margin-bottom:16px">
  <label style="font-weight:600;font-size:12px">${tr('Policy / Card Number', '??? ??????? / ???????
')}</label>
  <input class="form-input" id="eligPolicyNum" placeholder="e.g. POL-99234">
</div>
  <button class="btn btn-primary w-full" onclick="triggerNphiesEligibility()" style="height:42px;width
:100%">? ${tr('Check Eligibility', '????? ? ? ?????? ????')}</button>
</div>

<div class="card">
  <div class="card-title">?? ${tr('Eligibility Verification History', '??? ?????? ?????? ???????')}</d
iv>
  <div style="font-size:12px;color:var(--text-dim);margin-bottom:8px">${tr('Showing logs of requests r
outed through NPHIES gateway.', '???? ???? ???? ??????? ??????? ??????? NPHIES.')}</div>
  <div style="overflow-x:auto">
    <table class="data-table">
      <thead>
        <tr>
          <th>${tr('Patient ID', '??? ??????')}</th>
          <th>${tr('Company', '???? ???????')}</th>
          <th>${tr('Policy', '??? ???????')}</th>
          <th>${tr('Response', '????????')}</th>
          <th>${tr('Status', '?????')}</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>1087456723</td>
          <td>Bupa Arabia</td>
          <td>BP-90032</td>
          <td>${tr('Active (100% covered)', '??? (???? 100%')}</td>
          <td><span class="badge badge-success">? ${tr('Eligible', '????')}</span></td>
        </tr>
        <tr>
          <td>1098345122</td>
          <td>Tawuniya</td>
          <td>TW-10229</td>

```

```

        <td>${tr('Co-pay: 20% max 100 SAR', '????? 20% ?? ???? 100')}</td>
        <td><span class="badge badge-success">? ${tr('Eligible', '????')}</span></td>
    </tr>
    <tr>
        <td>2034981765</td>
        <td>MedGulf</td>
        <td>MG-0021</td>
        <td>${tr('Card Expired on 2026-06-30', '?????? ???? ? 2026-06-30')}</td>
        <td><span class="badge badge-danger">? ${tr('Ineligible', '?? ????')}</span></td>
    </tr>
</tbody>
</table>
</div>
</div>
</div>
`;
} else if (window.insuranceTab === 'priorauth') {
    cont.innerHTML = `
        <div class="grid-equal" style="display:grid;grid-template-columns:1fr 1fr;gap:20px">
            <div class="card">
                <div class="card-title">?? ${tr('Request Prior Authorization (NPHIES Claim)', '?? ????? ???? ????
                ?????')}</div>
                <p style="font-size:11px;color:var(--text-dim);margin-bottom:12px">
                    ${tr('Submit prior auth request before executing surgical procedures, expensive imaging, or high-c
                    ost therapies.', '???? ? ???? ???? ? ???? ???? ???? ? ? ???? ? ? ???? ? ? ???? ? ? ???? ? ????
                    ')}
                </p>
                <div class="form-group mb-12" style="margin-bottom:12px">
                    <label style="font-weight:600;font-size:12px">${tr('Patient National ID / IQAMA', '?? ???? ?????
                    / ?????')}</label>
                    <input class="form-input" id="paPatientId" placeholder="e.g. 1024567890">
                </div>
                <div class="form-group mb-12" style="margin-bottom:12px">
                    <label style="font-weight:600;font-size:12px">${tr('Insurance Company', '???? ?????')}</label>
                    <select class="form-input" id="paCompany">
                        <option value="Bupa Arabia">Bupa Arabia</option>
                        <option value="Tawuniya">Tawuniya</option>
                        <option value="MedGulf">MedGulf</option>
                        ${companies.map(c => `<option value="${escapeHTML(c.name_en || c.name_ar)}">${escapeHTML(c.name_
                        en || c.name_ar)}</option>`).join('')}
                    </select>
                </div>
                <div class="form-group mb-12" style="margin-bottom:12px">
                    <label style="font-weight:600;font-size:12px">${tr('Requested Amount (SAR)', '????? ?????? (?.)
                    ')}</label>
                    <input class="form-input" id="paAmount" type="number" placeholder="0.00">
                </div>
                <div class="form-group mb-12" style="margin-bottom:16px">
                    <label style="font-weight:600;font-size:12px">${tr('Clinical Justification', '?????? ???? ????
                    ')}</label>
                    <textarea class="form-input" id="paJustification" rows="2" placeholder="${tr('Describe diagnosis a
                    nd clinical necessity...', '???? ?????? ?????? ???? ???? ??????...')}" style="height:60px"></textarea>
                </div>
                <button class="btn btn-primary w-full" onclick="triggerNphiesPriorAuth()" style="height:42px;width:1

```

```
00%">? ${tr('Request Prior Auth', '????? ??? ???????')}</button>
    </div>

    <div class="card">
        <div class="card-title">? ${tr('Prior Authorization Status Board', '???? ?????? ?????? ???????')}</di
v>

        <div style="overflow-x:auto">
            <table class="data-table">
                <thead>
                    <tr>
                        <th>${tr('Patient ID', '??? ??????')}</th>
                        <th>${tr('Company', '??????')}</th>
                        <th>${tr('Amount', '??????')}</th>
                        <th>${tr('Status', '??????')}</th>
                        <th>${tr('Approval Code', '??? ???????')}</th>
                    </tr>
                </thead>
                <tbody>
                    <tr>
                        <td>1087456723</td>
                        <td>Bupa Arabia</td>
                        <td>14,500 SAR</td>
                        <td><span class="badge badge-success">? Approved</span></td>
                        <td><span class="font-mono text-secondary">ATH-99882</span></td>
                    </tr>
                    <tr>
                        <td>1098345122</td>
                        <td>Tawuniya</td>
                        <td>2,800 SAR</td>
                        <td><span class="badge badge-warning">? Pending</span></td>
                        <td>-</td>
                    </tr>
                    <tr>
                        <td>2034981765</td>
                        <td>MedGulf</td>
                        <td>6,200 SAR</td>
                        <td><span class="badge badge-danger">? Denied</span></td>
                        <td>-</td>
                    </tr>
                </tbody>
            </table>
        </div>
    </div>
</div>
`;
} else if (window.insuranceTab === 'companies') {
    cont.innerHTML = `
        <div class="grid-equal" style="display:grid;grid-template-columns:1fr 1fr;gap:20px">
            <div class="card">
                <div class="card-title">? ${tr('Add Insurance Payer Company', '????? ???? ??????')}</div>
                <div class="form-group mb-12" style="margin-bottom:12px">
                    <label style="font-weight:600;font-size:12px">${tr('Arabic name', '????? ???????')}</label>
                    <input class="form-input" id="insCoNameAr" placeholder="${tr('Arabic name', '????? ???????')}">
                </div>
            </div>
        </div>
    `;
}
```

```

    <div class="form-group mb-12" style="margin-bottom:16px">
      <label style="font-weight:600;font-size:12px">${tr('English name', '????? ?????????')}</label>
      <input class="form-input" id="insCoNameEn" placeholder="${tr('English name', '????? ?????????')}>
    ">

    </div>

    <button class="btn btn-primary w-full" onclick="addInsCompany()" style="height:40px;width:100%"> ${
tr('Add Company', '????? ?????')}</button>
  </div>

  <div class="card">
    <div class="card-title">? ${tr('Registered Insurance Companies', '?????? ?????')}</div>
    <div style="overflow-x:auto">
      ${makeTable(
        [tr('Name (AR)', '????? ?????'), tr('Name (EN)', '????? ?????????')],
        companies.map(c => ({ cells: [c.name_ar, c.name_en] })))
      }
    </div>
  </div>
</div>
`;
}
}

// Lifecycle-aware action buttons ? every button calls a server-authoritative route (state machine enforced se
rver-side).
function insClaimActions(row) {
  const id = safeId(row.id);
  const ls = row.lifecycle;
  if (ls === 'draft') return `<button class="btn btn-sm btn-primary" onclick="submitClaimNphies(${id})">? ${tr
('Submit', '????')}</button>`;
  if (ls === 'submitted') return `<div class="flex gap-4"><button class="btn btn-sm btn-success" onclick="tran
sitionClaim(${id}, 'adjudicated')">? ${tr('Adjudicate', '????')}</button><button class="btn btn-sm btn-danger"
onclick="transitionClaim(${id}, 'denied')">? ${tr('Deny', '???')}</button></div>`;
  if (ls === 'adjudicated') return `<button class="btn btn-sm btn-primary" onclick="transitionClaim(${id}, 'rem
ittance_posted')">? ${tr('Post Remittance', '????? ?????')}</button>`;
  if (ls === 'denied') return `<button class="btn btn-sm btn-warning" onclick="appealDeniedClaim(${id})">?? ${
tr('Appeal', '???????')}</button>`;
  return `<span class="badge badge-secondary">${escapeHTML(ls)}</span>`;
}

window.addClaim = async () => {
  const name = document.getElementById('insPatient').value.trim();
  if (!name) { showToast(tr('Enter patient name', '???? ??? ?????'), 'error'); return; }
  try {
    await API.post('/api/insurance/claims', { patient_name: name, insurance_company: document.getElementById('
insCompany').value, claim_amount: parseFloat(document.getElementById('insAmount').value) || 0 });
    showToast(tr('Claim submitted!', '?? ????? ?????????!')); await navigateTo(9);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// Server-authoritative claim transitions. The state machine lives on the server; invalid transitions => 409.
window.transitionClaim = async (id, target) => {
  try {
    const r = await API.put(`/api/insurance/claims/${id}/transition`, { target });
    if (r && r.error) { showToast(r.error, 'error'); return; } // e.g. 409 invalid transition
    showToast(tr('Updated', '?? ?????')); await navigateTo(9);
  }
}

```



```

    } catch (e) { showToast(tr('Error', '???'), 'error'); }
  };
  // Finding 2 fix: Appeal button resolves denial_id from server then routes through
  // POST /api/insurance/denials/:id/appeal ? not transitionClaim ? so the denial record's
  // appeal_status is updated AND the INSURANCE_DENIAL_APPEAL audit is logged correctly.
  window.appealDeniedClaim = async (claimId) => {
    try {
      const denials = await API.get('/api/insurance/denials');
      // find the most recent open denial for this claim
      const denial = (denials || []).filter(d => d.claim_id == claimId && d.appeal_status === 'open')
        .sort((a, b) => b.id - a.id)[0];
      if (!denial) { showToast(tr('No open denial found for this claim', '?? ???? ??? ????? ???? ?????????'), 'error'); return; }
      await window.appealDenial(denial.id, 'appealed', '');
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
  };
  // NPHIES claim submission ? gated server-side (503 stub when NPHIES disabled); records intent.
  window.submitClaimNphies = async (id) => {
    try {
      const r = await API.post(`/api/nphies/submit-claim/${id}`, {});
      if (r && r.gated) showToast(tr('NPHIES disabled ? submission recorded', 'NPHIES ????? ? ????? ?????'), 'warning');
      else if (r && r.error) showToast(r.error, 'error');
      else showToast(tr('Submitted', '?? ?????????'));
      await navigateTo(9);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
  };
  // Eligibility check (NPHIES gated). Records the request; 503 stub when disabled.
  window.checkEligibility = async (patientId, companyId, policyNumber) => {
    try {
      const r = await API.post('/api/insurance/eligibility', { patient_id: patientId, insurance_company_id: companyId, policy_number: policyNumber });
      if (r && r.gated) showToast(tr('NPHIES disabled ? request recorded', 'NPHIES ????? ? ????? ?????'), 'warning');
      else if (r && r.error) showToast(r.error, 'error');
      else showToast(tr('Eligibility requested', '?? ??? ?????????'));
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
  };
  window.triggerNphiesEligibility = async () => {
    const pid = document.getElementById('eligPatientId')?.value.trim();
    const comp = document.getElementById('eligCompany')?.value;
    const pol = document.getElementById('eligPolicyNum')?.value.trim();
    if (!pid) return showToast(tr('Enter Patient ID', '???? ??? ??? ?????'), 'error');
    await window.checkEligibility(pid, comp, pol);
    await navigateTo(9);
  };
  // Pre-authorization request (NPHIES gated). Server creates it in 'requested' state.
  window.requestPreAuth = async (patientId, companyId, requestedAmount, justification) => {
    try {
      const r = await API.post('/api/insurance/pre-auth', { patient_id: patientId, insurance_company_id: companyId, requested_amount: requestedAmount, clinical_justification: justification });
      if (r && r.gated) showToast(tr('NPHIES disabled ? pre-auth recorded', 'NPHIES ????? ? ????? ?????'), 'warning');
      else if (r && r.error) showToast(r.error, 'error');
    }
  }

```

```

    else showToast(tr('Pre-auth requested', '?? ??? ??????'));
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.triggerNphiesPriorAuth = async () => {
  const pid = document.getElementById('paPatientId')?.value.trim();
  const comp = document.getElementById('paCompany')?.value;
  const amt = parseFloat(document.getElementById('paAmount')?.value) || 0;
  const just = document.getElementById('paJustification')?.value.trim();
  if (!pid) return showToast(tr('Enter Patient ID', '???? ??? ??? ??????'), 'error');
  if (!amt) return showToast(tr('Enter requested amount', '???? ?????? ??????'), 'error');
  await window.requestPreAuth(pid, comp, amt, just);
  await navigateTo(9);
};

// Appeal a denied claim's denial record (server moves claim denied->appealed on 'overturned').
window.appealDenial = async (denialId, appealStatus, notes) => {
  try {
    const r = await API.put(`/api/insurance/denials/${denialId}/appeal`, { appeal_status: appealStatus, appeal_status_notes: notes });
    if (r && r.error) { showToast(r.error, 'error'); return; }
    showToast(tr('Appeal updated', '?? ?????? ??????????')); await navigateTo(9);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// LEGACY shim: old callers of updateClaim(id, 'Approved'|'Rejected') now route through the state machine.
window.updateClaim = async (id, status) => {
  const target = status === 'Approved' ? 'adjudicated' : status === 'Rejected' ? 'denied' : null;
  if (!target) { showToast(tr('Unsupported action', '????? ??? ??????'), 'error'); return; }
  return window.transitionClaim(id, target);
};

window.addInsCompany = async () => {
  const ar = document.getElementById('insCoNameAr').value.trim();
  const en = document.getElementById('insCoNameEn').value.trim();
  if (!ar && !en) { showToast(tr('Enter company name', '???? ??? ??????'), 'error'); return; }
  try {
    await API.post('/api/insurance/companies', { name_ar: ar, name_en: en });
    showToast(tr('Company added!', '??? ??????!')); await navigateTo(9);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== INVENTORY =====
async function renderInventory(el) {
  const content = el;

  // Initialize tabs and local data if not present
  if (!window.inventoryTab) window.inventoryTab = 'items';
  if (!window.localSuppliers) {
    window.localSuppliers = [
      { id: 1, name: '????????? ?????????', location: '????????? ????????? ????????? ?????????', email: 'info@medtronics.sa', phone: '011-456-7890', category: '????? ????', outstanding: 124500.00, status: '???' },
      { id: 2, name: '???? ?????? ????????? ??????', location: '????????? ????????? ????????? ?????????', email: 'purchasing@aldawaa.com.sa', phone: '013-890-1234', category: '????? ??????????', outstanding: 85200.00, status: '???' },
      { id: 3, name: '????????? ??????', location: '????????? ????????? ????????? ?????????', email: 'sales@aramcomed.com', phone: '013-876-5432', category: '????????? ?????????', outstanding: 310000.00, status: '???' },
      { id: 4, name: '????????? ?????? ???', location: '????? ?????????? ?????????? ?????????', email: 'service@philips.com' }
    ];
  }
}

```

```
.sa', phone: '012-654-3210', category: '??? ???? ??????', outstanding: 0.00, status: '???' }
    ];
}
if (!window.localPOs) {
    window.localPOs = [
        { id: 'PO-2024-001', supplier: '????????? ???????', date: '2026-06-10', amount: 45200.00, status: '????
?', delivery: '2026-06-12', items: [{ name: '???? ?????? MRI', qty: 2, cost: 22600.00 }] },
        { id: 'PO-2024-002', supplier: '???? ?????? ??????? ??????', date: '2026-06-12', amount: 12850.50, statu
s: '??? ?????????', delivery: '2026-06-18', items: [{ name: '????? ?????? ??? 11', qty: 150, cost: 85.67 }] },
        { id: 'PO-2024-003', supplier: '?????? ??????', date: '2026-06-05', amount: 310000.00, status: '?????',
delivery: '2026-06-14', items: [{ name: '?????????? ???? ????????? (300 ????)', qty: 12, cost: 25833.33 }] },
        { id: 'PO-2024-004', supplier: '?????? ???? ???', date: '2026-06-14', amount: 88400.00, status: '?????',
delivery: '2026-06-15', items: [{ name: '????? ?????? ???? ??????', qty: 5, cost: 17680.00 }] }
    ];
}

// Fetch real items from backend
const items = await API.get('/api/inventory').catch(() => []);
window._invData = items;

const lowStock = items.filter(i => parseInt(i.quantity || 0) <= parseInt(i.reorder_level || 10));
const totalItems = items.length;
const totalValue = items.reduce((acc, i) => acc + (parseFloat(i.cost || 0) * parseInt(i.quantity || 0)), 0);
const avgCost = totalItems > 0 ? (items.reduce((acc, i) => acc + parseFloat(i.cost || 0), 0) / totalItems) :
0;

// Premium Header and Tabs
let htmlContent = `
    <div class="mb-xl flex flex-col md:flex-row md:items-end justify-between gap-md">
        <div>
            <nav class="flex text-on-surface-variant font-caption text-caption mb-xs">
                <span>${tr('Operations', '????????')}</span>
                <span class="mx-xs"></span>
                <span class="text-primary font-bold">${tr('Supply Chain & Inventory', '????? ????????? ????????? ??????')}</span>
            </nav>
            <h3 class="font-display-lg text-display-lg text-primary tracking-tight">${tr('Supply Chain & Medical I
nventory', '????? ????????? ????????? ??????')}</h3>
            <p class="font-body-md text-body-md text-on-surface-variant max-w-2xl">
                ${tr('Premium system to manage medical supplies, pharmaceuticals, spare parts with real-time trackin
g.', '???? ?????? ?????? ?????????? ?????? ?????????? ???? ?????? ?? ???? ???? ?????? ??? ?????? ?????? ??????????
????????? ?????? ??????')}
            </p>
        </div>
        <div class="flex gap-sm">
            <button class="flex items-center gap-2 px-md py-sm border-2 border-secondary text-secondary font-bold
rounded-xl hover:bg-secondary/5 transition-all" onclick="exportToCSV(window._invData|[], 'inventory')">
                <span class="material-symbols-outlined">download</span>
                <span>${tr('Export Inventory', '????? ?????????')}</span>
            </button>
            <button class="flex items-center gap-2 px-md py-sm bg-primary text-on-primary font-bold rounded-xl sha
dow-lg hover:shadow-xl transition-all scale-100 hover:scale-[1.02] active:scale-95" onclick="showAddInvModal()"
">
                <span class="material-symbols-outlined">add</span>
            </button>
        </div>
    </div>
`;
```

```

        <span>${tr('Add New Item', '????? ??? ?????')}</span>
    </button>
</div>
</div>

<!-- Tabs Bar -->
<div class="tab-bar mb-md flex border-b border-outline-variant/30 overflow-x-auto gap-2">
    <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.inventoryTab === 'items'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}
" onclick="window.inventoryTab='items'; navigateTo(currentPage)">
        ? ${tr('Medical Inventory', '??????? ?????')}
    </button>
    <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.inventoryTab === 'procurement'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.inventoryTab='procurement'; navigateTo(currentPage)">
        ? ${tr('Procurement & POs', '????? ?????? ?????????')}
    </button>
    <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.inventoryTab === 'suppliers'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.inventoryTab='suppliers'; navigateTo(currentPage)">
        ? ${tr('Suppliers Registry', '??? ?????????')}
    </button>
    <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.inventoryTab === 'logistics'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.inventoryTab='logistics'; navigateTo(currentPage)">
        ? ${tr('Logistics Tracking', '??????? ?????????')}
    </button>
    <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.inventoryTab === 'intelligence'
? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary'}"
onclick="window.inventoryTab='intelligence'; navigateTo(currentPage)">
        ? ${tr('AI Command Center', '????????? ?????????')}
    </button>
</div>
`;

if (window.inventoryTab === 'items') {
    // -----
    // Tab 1: Medical Inventory Items
    // -----
    htmlContent += `
        <!-- KPI Cards Bento Grid -->
        <div class="grid grid-cols-1 md:grid-cols-4 gap-gutter mb-xl">
            <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
                <div class="flex justify-between items-start relative z-10">
                    <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined">inventory</span>

                    <span class="text-secondary text-xs font-bold">${totalItems} ${tr('Items', '????')}</span>
                </div>
                <div class="mt-lg relative z-10">
                    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Total Active Items', '?????
??????? ?????????')}</p>
                    <h4 class="text-display-lg text-primary font-bold mt-xs">${totalItems}</h4>
                </div>
                <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti

```

```

on-transform">
    <span class="material-symbols-outlined text-[120px]">inventory_2</span>
  </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative ${lowStock.length > 0 ? 'border-r-4 border-r-error' : ''}">
  <div class="flex justify-between items-start relative z-10">
    <span class="p-xs bg-error/10 rounded-lg text-error material-symbols-outlined">warning</span>
    ${lowStock.length > 0 ? `<span class="bg-error text-on-error px-xs py-1 rounded text-caption font-bold animate-pulse">${tr('Low Stock', '???? ????')}</span>` : `<span class="text-secondary text-xs font-bold">${tr('Secure', '??')}</span>`}
  </div>
  <div class="mt-lg relative z-10">
    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Low Stock Alert', '???? ???? ? ?????')}</p>
    <h4 class="text-display-lg ${lowStock.length > 0 ? 'text-error font-bold' : 'text-primary'} mt-xs">${lowStock.length}</h4>
  </div>
  <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transition-transform">
    <span class="material-symbols-outlined text-[120px]">report_problem</span>
  </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
  <div class="flex justify-between items-start relative z-10">
    <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">payments</span>
  </div>
  <div class="mt-lg relative z-10">
    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Total Inventory Value', '???? ? ? ?????? ? ??????')}</p>
    <h4 class="text-display-lg text-primary font-bold mt-xs">${totalValue.toLocaleString(undefined, {minimumFractionDigits: 2, maximumFractionDigits: 2})} <span class="text-title-lg">${tr('SAR', '?.?')}</span></h4>
  </div>
  <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transition-transform">
    <span class="material-symbols-outlined text-[120px]">monetization_on</span>
  </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
  <div class="flex justify-between items-start relative z-10">
    <span class="p-xs bg-tertiary-fixed-dim/20 rounded-lg text-tertiary-fixed-dim material-symbols-outlined">calculate</span>
  </div>
  <div class="mt-lg relative z-10">
    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Average Cost per Item', '???? ? ????? ? ?????')}</p>
    <h4 class="text-display-lg text-primary font-bold mt-xs">${avgCost.toLocaleString(undefined, {minimumFractionDigits: 2, maximumFractionDigits: 2})} <span class="text-title-lg">${tr('SAR', '?.?')}</span></h4>
  </div>

```

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        <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transition-transform">
            <span class="material-symbols-outlined text-[120px]">balance</span>
        </div>
    </div>
</div>

<!-- Critical Stock Alert Section (Only if low stock items exist) -->
${lowStock.length > 0 ? `
<div class="glass-card p-lg rounded-xl mb-xl border-r-4 border-r-error relative overflow-hidden">
    <div class="absolute -top-10 -left-10 w-40 h-40 bg-error/5 rounded-full blur-3xl"></div>
    <div class="z-10">
        <div class="flex items-center gap-sm mb-md text-error">
            <span class="material-symbols-outlined" data-weight="fill">warning</span>
            <span class="font-bold text-headline-md">${tr('Critical Stock Alerts', '?????? ????? ???')}</span>
        </div>
    </div>
    <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-sm">
        ${lowStock.map(i => `
            <div class="flex items-center justify-between p-sm bg-error-container/20 rounded-lg border border-error/10">
                <span class="font-body-md text-primary font-bold">${escapeHTML(i.name)}</span>
                <span class="bg-error text-on-error px-sm py-1 rounded text-caption font-bold">${tr('Remaining', '????????:')} ${escapeHTML(i.quantity || 0)} ${escapeHTML(i.unit || '')}</span>
            </div>
        `).join('')}
    </div>
</div>
` : ''}

<!-- Main Content and Filter Bar -->
<div class="glass-card rounded-xl overflow-hidden flex flex-col p-md">
    <div class="flex flex-wrap items-center justify-between gap-md mb-md pb-md border-b border-outline-variant/30">
        <div class="relative w-full max-w-md">
            <span class="material-symbols-outlined absolute right-3 top-1/2 -translate-y-1/2 text-on-surface-variant">search</span>
            <input class="w-full pr-10 bg-surface-container-low border-none rounded-lg focus:ring-2 focus:ring-secondary font-body-md py-sm" id="invSearch" placeholder="${tr('Search by name or supplier...', '????? ???? ???? ???? ???? ????...')}" type="text" oninput="filterInvTable()">
        </div>
        <div class="flex items-center gap-sm">
            <span class="font-label-md text-on-surface-variant">${tr('Filter by Category:', '????? ??? ?????:')}</span>
            <select class="bg-surface border-outline-variant/30 rounded-lg font-body-md py-1 pr-8 text-on-surface" id="invCatFilter" onchange="filterInvTable()">
                <option value="">${tr('All Categories', '???? ?????')}</option>
                ${Array.from(new Set(items.map(i => i.category || tr('Uncategorized', '??? ?????')))).map(c => `<option value="${escapeHTML(c)}">${escapeHTML(c)}</option>`).join('')}
            </select>
        </div>
    </div>
    <div class="overflow-x-auto custom-scrollbar" id="invTableDiv"></div>

```

```

</div>
`;

// Populate Table when rendered
setTimeout(() => {
  const it = document.getElementById('invTableDiv');
  if (it) {
    if (!items.length) {
      it.innerHTML = `<div class="empty-state py-lg text-center"><div class="empty-icon text-4xl mb-2">?</div><p class="text-on-surface-variant">${tr('No items found in database', '?? ???? ???? ? ? ???? ???????')}</p></div>`;
    }
    return;
  }

  // Table Rendering
  let tableHtml = `
<table class="w-full text-right border-collapse" id="invTbl">
  <thead>
    <tr class="bg-surface-container border-b border-outline-variant/30">
      <th class="p-md font-bold text-sm text-primary">${tr('Item ID', '??? ?????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Item Name', '??? ?????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Category', '????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Current Stock', '?????? ?????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Reorder Level', '?? ????? ?????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Cost (SAR)', '?????? (?.?)')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Supplier', '????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Status', '????')}</th>
      <th class="p-md font-bold text-sm text-primary">${tr('Actions', '????????')}</th>
    </tr>
  </thead>
  <tbody class="divide-y divide-outline-variant/10">
    ${items.map(i => {
      const isLow = parseInt(i.quantity || 0) <= parseInt(i.reorder_level || 10);
      return `
        <tr class="hover:bg-primary/5 transition-all cursor-pointer ${isLow ? 'bg-error-container/5'
: ''}">
          <td class="p-md font-mono text-secondary">#MED-${escapeHTML(String(i.id).padStart(4, '0'))}
        </td>
          <td class="p-md font-bold text-primary">${escapeHTML(i.name)}</td>
          <td class="p-md"><span class="text-caption bg-surface-container px-sm py-1 rounded-full">${
escapeHTML(i.category || tr('Uncategorized', '??? ????'))}</span></td>
          <td class="p-md font-bold ${isLow ? 'text-error' : 'text-secondary'}">${escapeHTML(i.quant
ity)} ${escapeHTML(i.unit || '')}</td>
          <td class="p-md text-on-surface-variant">${escapeHTML(i.reorder_level || 10)}</td>
          <td class="p-md font-bold text-primary">${parseFloat(i.cost || 0).toFixed(2)}</td>
          <td class="p-md text-on-surface-variant">${escapeHTML(i.supplier || tr('Not specified', '?
?? ????'))}</td>
          <td class="p-md">
            ${isLow ? `
              <span class="inline-flex items-center gap-1 px-sm py-1 bg-error/10 text-error rounded-
full text-xs font-bold">
                <span class="w-1.5 h-1.5 bg-error rounded-full animate-pulse"></span>
                ${tr('Critical / Low', '??? / ?????')}
              </span>
            ` : ''}
          </td>
        `;
    })}
  </tbody>
</table>

```

```

        ` : `
        <span class="inline-flex items-center gap-1 px-sm py-1 bg-secondary/10 text-secondary
rounded-full text-xs font-bold">
            <span class="w-1.5 h-1.5 bg-secondary rounded-full"></span>
            ${tr('Stable', '????')}
        </span>
    `}
</td>
<td class="p-md">
    <div class="flex gap-2">
        <button class="p-1 hover:bg-secondary/10 text-secondary rounded transition-colors" tit
le="${tr('Edit Item', '????')}" onclick="showEditInvModal(${escapeHTML(JSON.stringify(i))})">
            <span class="material-symbols-outlined text-sm">edit</span>
        </button>
        <button class="p-1 hover:bg-error/10 text-error rounded transition-colors" title="${tr
('Delete Item', '???')}" onclick="deleteInvItem(${safeId(i.id)})">
            <span class="material-symbols-outlined text-sm">delete</span>
        </button>
    </div>
</td>
</tr>
`;
    }).join('')}
</tbody>
</table>
`;
    it.innerHTML = tableHtml;
}
}, 50);

} else if (window.inventoryTab === 'procurement') {
    // -----
    // Tab 2: Procurement & Purchase Orders
    // -----
    const totalPOAmount = window.localPOs.reduce((acc, po) => acc + po.amount, 0);
    const pendingPOs = window.localPOs.filter(po => po.status !== '????').length;

    htmlContent += `
        <!-- KPI Cards Bento Grid -->
        <div class="grid grid-cols-1 md:grid-cols-3 gap-gutter mb-xl">
            <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
                <div class="flex justify-between items-start relative z-10">
                    <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined">shopping_cart</
span>
                </div>
                <div class="mt-lg relative z-10">
                    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Total Procurement Value', '??
???? ???? ?????????')}</p>
                    <h4 class="text-display-lg text-primary font-bold mt-xs">${totalPOAmount.toLocaleString(undefined,
{minimumFractionDigits: 2, maximumFractionDigits: 2})} <span class="text-title-lg">${tr('SAR', '?.?')}</span>
</h4>
                </div>
                <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">

```



```

        <span class="material-symbols-outlined text-[120px]">payments</span>
    </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
    <div class="flex justify-between items-start relative z-10">
        <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">pending_act
ions</span>
    </div>
    <div class="mt-lg relative z-10">
        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Pending PO approvals', '????
???? ??? ???????/????????')}</p>
        <h4 class="text-display-lg text-primary font-bold mt-xs">${pendingPOs}</h4>
    </div>
    <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">
        <span class="material-symbols-outlined text-[120px]">hourglass_empty</span>
    </div>
</div>

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
    <div class="flex justify-between items-start relative z-10">
        <span class="p-xs bg-on-primary-fixed-variant/10 rounded-lg text-on-primary-fixed-variant material
-symbols-outlined">schedule</span>
    </div>
    <div class="mt-lg relative z-10">
        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Average Delivery Time', '????
? ??? ??????? ?????????')}</p>
        <h4 class="text-display-lg text-primary font-bold mt-xs">4.2 <span class="text-title-lg">${tr('Day
s', '????')}</span></h4>
    </div>
    <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">
        <span class="material-symbols-outlined text-[120px]">local_shipping</span>
    </div>
</div>
</div>

<!-- Action Section -->
<div class="glass-card p-md rounded-2xl flex flex-wrap justify-between items-center gap-md mb-md">
    <div class="flex items-center gap-2">
        <span class="material-symbols-outlined text-secondary">receipt_long</span>
        <strong class="text-primary font-bold text-headline-md">${tr('Purchase Orders Registry', '??? ??????
?????? ???????')}</strong>
    </div>
    <button class="bg-primary text-on-primary font-bold px-md py-sm rounded-xl hover:opacity-90 flex items
-center gap-1" onclick="showAddPOModal()">
        <span class="material-symbols-outlined text-sm">add_shopping_cart</span>
        <span>${tr('Create New Purchase Order', '????? ??? ???? ?????')}</span>
    </button>
</div>

<!-- POs Table -->
<div class="glass-card rounded-xl overflow-hidden p-md">

```

```

<div class="overflow-x-auto custom-scrollbar">
  <table class="w-full text-right border-collapse">
    <thead>
      <tr class="bg-surface-container border-b border-outline-variant/30">
        <th class="p-md font-bold text-sm text-primary">${tr('PO Number', '??? ?????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Supplier', '??? ?????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Order Date', '???? ?????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Delivery Date', '???? ?????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Amount (SAR)', '????? ?????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Status', '????')}</th>
        <th class="p-md font-bold text-sm text-primary">${tr('Actions', '????')}</th>
      </tr>
    </thead>
    <tbody class="divide-y divide-outline-variant/10">
      ${window.localPOs.map(po => `
        <tr class="hover:bg-primary/5 transition-all">
          <td class="p-md font-mono text-secondary">${escapeHTML(po.id)}</td>
          <td class="p-md font-bold text-primary">${escapeHTML(po.supplier)}</td>
          <td class="p-md text-on-surface-variant">${escapeHTML(po.date)}</td>
          <td class="p-md text-on-surface-variant">${escapeHTML(po.delivery)}</td>
          <td class="p-md font-bold text-primary">${parseFloat(po.amount).toLocaleString(undefined, {m
inimumFractionDigits: 2, maximumFractionDigits: 2})}</td>
          <td class="p-md">
            ${po.status === '????' ? `
              <span class="inline-flex items-center gap-1 px-sm py-1 bg-secondary/10 text-secondary ro
unded-full text-xs font-bold">
                <span class="w-1.5 h-1.5 bg-secondary rounded-full"></span>
                ${tr('Completed / Received', '?? ?????? / ?????')}
              </span>
            ` : po.status === '????' || po.status === '????' ? `
              <span class="inline-flex items-center gap-1 px-sm py-1 bg-error/10 text-error rounded-fu
ll text-xs font-bold animate-pulse">
                <span class="w-1.5 h-1.5 bg-error rounded-full"></span>
                ${tr('Delayed', '????')}
              </span>
            ` : `
              <span class="inline-flex items-center gap-1 px-sm py-1 bg-tertiary-fixed/20 text-on-tert
iary-fixed-variant rounded-full text-xs font-bold">
                <span class="w-1.5 h-1.5 bg-on-tertiary-fixed-variant rounded-full"></span>
                ${tr('Pending Approval', '?? ?????')}
              </span>
            `}
          </td>
          <td class="p-md">
            <div class="flex gap-2">
              ${po.status === '??? ?????' || po.status === '????' ? `
                <button class="bg-secondary text-on-secondary px-sm py-1 rounded text-caption hover:op
acity-90" onclick="receivePO('${jsStr(po.id)}')">
                  ? ${tr('Receive items', '????')}
                </button>
              ` : `
                <span class="text-caption text-on-surface-variant font-bold">? ${tr('Archived', '????
')}}</span>
              `}
            </div>
          </td>
        `
      )}]
    </tbody>
  </table>

```

```

        </div>
      </td>
    </tr>
  `).join('')
</tbody>
</table>
</div>
</div>
`;

} else if (window.inventoryTab === 'suppliers') {
  // -----
  // Tab 3: Suppliers Registry
  // -----
  const outstandingInvoices = window.localSuppliers.reduce((acc, s) => acc + s.outstanding, 0);

  htmlContent += `
    <!-- KPI Cards Bento Grid -->
    <div class="grid grid-cols-1 md:grid-cols-3 gap-gutter mb-xl">
      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined">group</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Active Suppliers', '????????
????????')}</p>
          <h4 class="text-display-lg text-primary font-bold mt-xs">${window.localSuppliers.length}</h4>
        </div>
        <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">
          <span class="material-symbols-outlined text-[120px]">corporate_fare</span>
        </div>
      </div>

      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">
          <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined">account_bal
ance_wallet</span>
        </div>
        <div class="mt-lg relative z-10">
          <p class="text-on-surface-variant font-label-md text-label-md">${tr('Outstanding Invoices', '????
? ???????? ?????????? ?????????')}</p>
          <h4 class="text-display-lg text-primary font-bold mt-xs">${outstandingInvoices.toLocaleString(unde
fined, {minimumFractionDigits: 2, maximumFractionDigits: 2})} <span class="text-title-lg">${tr('SAR', '?.?')}<
/span></h4>
        </div>
        <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">
          <span class="material-symbols-outlined text-[120px]">account_balance</span>
        </div>
      </div>

      <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
        <div class="flex justify-between items-start relative z-10">

```

```

        <span class="p-xs bg-tertiary-fixed-dim/20 rounded-lg text-tertiary-fixed-dim material-symbols-out
lined">verified</span>
    </div>
    <div class="mt-lg relative z-10">
        <p class="text-on-surface-variant font-label-md text-label-md">${tr('MOH Compliance Rate', '???? ?
????? ???? ???????')}</p>
        <h4 class="text-display-lg text-primary font-bold mt-xs">94%</h4>
    </div>
    <div class="absolute -right-4 -bottom-4 opacity-5 transform scale-150 group-hover:scale-110 transiti
on-transform">
        <span class="material-symbols-outlined text-[120px]">fact_check</span>
    </div>
</div>
</div>

<!-- Action Section -->
<div class="glass-card p-md rounded-2xl flex flex-wrap justify-between items-center gap-md mb-md">
    <div class="flex items-center gap-2">
        <span class="material-symbols-outlined text-secondary">contact_page</span>
        <strong class="text-primary font-bold text-headline-md">${tr('Suppliers Management Directory', '????
????????? ?????????')}</strong>
    </div>
    <button class="bg-primary text-on-primary font-bold px-md py-sm rounded-xl hover:opacity-90 flex items
-center gap-1" onclick="showAddSupplierModal()">
        <span class="material-symbols-outlined text-sm">person_add</span>
        <span>${tr('Add New Supplier', '????? ???? ?????')}</span>
    </button>
</div>

<!-- Suppliers Table -->
<div class="glass-card rounded-xl overflow-hidden p-md">
    <div class="overflow-x-auto custom-scrollbar">
        <table class="w-full text-right border-collapse">
            <thead>
                <tr class="bg-surface-container border-b border-outline-variant/30">
                    <th class="p-md font-bold text-sm text-primary">${tr('Supplier Name', '??? ??????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Location / HQ', '?????? ?????????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Email Address', '?????? ??????????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Phone', '??? ??????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Supply Category', '??? ?????????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Outstanding Bal (SAR)', '?????? ?????????
????????')}</th>
                    <th class="p-md font-bold text-sm text-primary">${tr('Status', '??????')}</th>
                </tr>
            </thead>
            <tbody class="divide-y divide-outline-variant/10">
                <script>
                    ${window.localSuppliers.map(s => `
                        <tr class="hover:bg-primary/5 transition-all">
                            <td class="p-md font-bold text-primary flex items-center gap-2">
                                <div class="w-8 h-8 rounded-full bg-secondary-container/30 flex items-center justify-cente
r font-bold text-secondary text-xs">${escapeHTML(s.name.charAt(0))}</div>
                                <span>${escapeHTML(s.name)}</span>
                            </td>

```

```

        <td class="p-md text-on-surface-variant font-mono text-sm">${escapeHTML(s.location)}</td>
        <td class="p-md text-on-surface-variant font-mono text-sm">${escapeHTML(s.email)}</td>
        <td class="p-md text-on-surface-variant font-mono text-sm">${escapeHTML(s.phone)}</td>
        <td class="p-md"><span class="text-caption bg-surface-container px-sm py-1 rounded-full">${e
scapeHTML(s.category)}</span></td>
        <td class="p-md font-bold text-primary">${parseFloat(s.outstanding).toLocaleString()}</td>
        <td class="p-md">
            <span class="inline-flex items-center gap-1 px-sm py-1 bg-secondary/10 text-secondary roun
ded-full text-xs font-bold">
                <span class="w-1.5 h-1.5 bg-secondary rounded-full"></span>
                ${escapeHTML(s.status)}
            </span>
        </td>
    </tr>
    `).join('')}
</tbody>
</table>
</div>
</div>
`;

} else if (window.inventoryTab === 'logistics') {
    // -----
    // Tab 4: Logistics & Auto-Ordering
    // -----
    htmlContent += `
        <div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">

            <!-- Auto Ordering and Suggestions (Interactive Widget) -->
            <div class="glass-card p-lg rounded-xl col-span-2 border-r-4 border-r-secondary flex flex-col justify-
between">
                <div>
                    <div class="flex items-center justify-between mb-lg">
                        <div>
                            <h4 class="font-headline-md text-headline-md text-primary font-bold">${tr('Automated Replenish
ment Dashboard', '????????? ?????? ?????? ?????????')}</h4>
                            <p class="font-body-md text-on-surface-variant mt-xs">${tr('System recommends purchase orders
based on real-time reorder thresholds.', '???? ?????? ?????????? ?????????? ?????? ?????? ?????? ?????? ??????
????? ??? ??????.')}</p>
                        </div>
                        <span class="material-symbols-outlined text-secondary text-display-lg" style="font-size: 48px;">
insights</span>
                    </div>

                    <h5 class="font-bold text-title-lg text-primary mb-md">${tr('Recommended Restock Orders', '????? ?
????? ?????????? ??????')}</h5>

                    <div class="space-y-sm">
                        ${lowStock.length === 0 ? `
                            <div class="p-sm bg-secondary-container/20 rounded-lg text-secondary text-center">
                                ? ${tr('All stock levels are completely healthy. No restocks suggested.', '???? ?????????
???? ?????? ??????. ?? ??? ?????????? ??????.')}
                            </div>
                        ` : lowStock.map(i => {

```

```
const suggestQty = (i.reorder_level * 5) - i.quantity;
const estCost = suggestQty * (i.cost || 50);
return `
    <div class="p-md bg-surface-container rounded-lg border border-outline-variant/30 flex flex-
col md:flex-row md:items-center justify-between gap-sm">
        <div>
            <div class="font-bold text-primary">${escapeHTML(i.name)}</div>
            <div class="text-caption text-on-surface-variant">${tr('Category:', '?????:')} ${escapeH
TML(i.category)} | ${tr('Current Stock:', '?????? ??????:')} ${escapeHTML(i.quantity)} ${escapeHTML(i.unit ||
''')}</div>
        </div>
        <div class="flex items-center gap-md">
            <div class="text-right">
                <div class="text-sm font-bold text-primary">${tr('Suggested Order:', '????? ??????:')}
} ${suggestQty} ${escapeHTML(i.unit || '')}</div>
                <div class="text-caption text-secondary font-bold">${tr('Estimated Cost:', '??????? ??
?????:')} ${estCost.toLocaleString()} ${tr('SAR', '?.?')}</div>
            </div>
            <button class="bg-secondary text-on-secondary px-md py-sm rounded-lg font-bold hover:opa
city-90 text-sm transition-all" onclick="approveAutoOrder('${jsStr(i.name)}', ${suggestQty}, ${estCost}, '${js
Str(i.supplier || tr('Aldawaa', '???? ?????? ??????? ???????')})')">
                ? ${tr('Approve PO', '?????? ??????')}
            </button>
        </div>
    </div>
    `;
    }).join('')}
</div>
</div>

    <div class="mt-lg pt-md border-t border-outline-variant/30 text-caption text-on-surface-variant flex
items-center gap-2">
        <span class="material-symbols-outlined text-sm">schedule</span>
        <span>${tr('Last consumption algorithms check: 15 mins ago', '?? ??? ?????? ?????????? ?????????? ??
? 15 ??????')}</span>
    </div>
</div>

<!-- Logistics Transit and Map Simulation -->
<div class="glass-card p-lg rounded-xl flex flex-col justify-between">
    <div>
        <h4 class="font-headline-md text-headline-md text-primary font-bold mb-md">${tr('Active Deliveries
in Transit', '???? ??????? ?????????')}</h4>

        <div class="space-y-md">
            <div class="flex items-start gap-sm">
                <span class="p-xs bg-secondary/10 text-secondary rounded-lg material-symbols-outlined mt-1">lo
cal_shipping</span>
                <div class="flex-1">
                    <div class="font-bold text-primary text-sm">${tr('Metronic Delivery (PO-2024-002)', '???? ??
???????? (PO-2024-002)')}</div>
                    <div class="text-caption text-on-surface-variant">${tr('Status: In Transit (On Route)', '???
? ??????: ?? ?????? ??? ?????????')}</div>
                    <div class="w-full bg-surface-container rounded-full h-1.5 mt-2">
```

```
        <div class="bg-secondary h-1.5 rounded-full animate-pulse" style="width: 75%"></div>
    </div>
    <div class="text-caption text-secondary font-bold mt-1">${tr('ETA: 4 hours', '????? ???????
?????: 4 ?????')}</div>
    </div>
</div>

<div class="flex items-start gap-sm pt-sm border-t border-outline-variant/10">
    <span class="p-xs bg-tertiary-fixed-dim/20 text-tertiary-fixed-dim rounded-lg material-symbols
-outlined mt-1">schedule</span>
    <div class="flex-1">
        <div class="font-bold text-primary text-sm">${tr('Aldawaa Medical Cargo', '???? ???? ??????')}</div>

        <div class="text-caption text-on-surface-variant">${tr('Status: Supplier Delays Expected', '
???? ??????: ?????? ??? ??????')}</div>
        <div class="w-full bg-surface-container rounded-full h-1.5 mt-2">
            <div class="bg-error h-1.5 rounded-full" style="width: 20%"></div>
            </div>
            <div class="text-caption text-error font-bold mt-1">${tr('ETA: 48 hours', '????? ??????? ????
???: 48 ?????')}</div>
        </div>
    </div>
</div>
</div>

<div class="bg-surface-container-low p-sm rounded-lg text-center mt-lg border border-outline-variant
/20">
    <span class="text-caption text-on-surface-variant block mb-1">${tr('MOH Integrated Fleet', '?????
?????? ??????')}</span>
    <span class="text-headline-md font-bold text-primary">4.2 ${tr('hrs', '????')}</span>
    <span class="text-caption text-secondary block mt-1">${tr('Average urgent delivery lead time', '??
??? ??? ???? ?????? ?????????? ?????????')}</span>
    </div>
</div>

</div>
`;

} else if (window.inventoryTab === 'intelligence') {
    // -----
    // Tab 5: AI Command Center (Intelligence)
    // -----
    htmlContent += `
        <div class="grid grid-cols-1 md:grid-cols-2 gap-gutter">

            <div class="glass-card p-lg rounded-xl border-r-4 border-r-secondary flex flex-col justify-between">
                <div>
                    <h4 class="font-headline-md text-headline-md text-primary font-bold mb-md">${tr('Consumption Intel
ligence Analytics', '??????? ???? ?????????')}</h4>
                    <p class="font-body-md text-on-surface-variant mb-lg">${tr('AI predicts stock depletion rates base
d on weekly patient admission volume.', '???? ?????? ?????????? ??????? ?????? ???? ?????????? ?????? ??? ?????? ?
????????? ?????????? ?????????? ?????????')}</p>

                    <div class="space-y-md">
```

```
<div class="flex items-center justify-between p-sm bg-surface-container rounded-lg">
  <span class="font-body-md text-primary">${tr('Sterilization Kits usage speed:', '???? ??????
?????????:')}</span>
  <span class="text-error font-bold">+18% ${tr('increase', '?????')}</span>
</div>
<div class="flex items-center justify-between p-sm bg-surface-container rounded-lg">
  <span class="font-body-md text-primary">${tr('Disposable syringe usage:', '??????? ????? ?????
?:')}</span>
  <span class="text-secondary font-bold">-2% ${tr('decrease', '?????')}</span>
</div>
</div>
</div>

<div class="mt-lg p-sm bg-primary text-on-primary rounded-lg text-center">
  <div class="font-label-md">${tr('AI Predicted Wastage Reduction', '???? ??? ????? ????? ??????')}</div>
  <div class="text-headline-md font-bold">-0.8% ${tr('this quarter', '??? ?????')}</div>
  <p class="text-caption opacity-80 mt-1">${tr('Calculated using smart inventory optimization rules'
, '???? ????? ????? ????? ????? ????? ?????? ?????? ?????????')}</p>
</div>
</div>

<div class="glass-card p-lg rounded-xl flex flex-col justify-between relative overflow-hidden">
  <div>
    <h4 class="font-headline-md text-headline-md text-primary font-bold mb-md">${tr('Unified Inventory
Intelligence Command', '???? ????? ?????? ??????')}</h4>
    <p class="font-body-md text-on-surface-variant mb-lg">${tr('Live tracking dashboard for all pharma
cy warehouses and clinical supplies.', '??? ????? ?????? ?????? ???????? ?????????? ?????????? ?? ??? ??????????')}</p>

  <div class="grid grid-cols-2 gap-sm">
    <div class="bg-surface-container-low p-sm rounded-lg text-center">
      <span class="text-caption text-on-surface-variant block">${tr('Annual Waste Rate', '???? ??????
??????')}</span>
      <span class="text-headline-md font-bold text-primary">0.8%</span>
      <span class="text-caption text-secondary block mt-1">? 0.2% ${tr('YoY', '?? ?????? ??????')}</s
pan>
    </div>
    <div class="bg-surface-container-low p-sm rounded-lg text-center">
      <span class="text-caption text-on-surface-variant block">${tr('Optimal stock availability', '?
??? ??????? ???????')}</span>
      <span class="text-headline-md font-bold text-primary">92%</span>
      <span class="text-caption text-secondary block mt-1">${tr('Optimal', '?????? ??????')}</span>
    </div>
  </div>
</div>

<!-- TODO: Backend integration placeholder widget -->
<div class="p-sm bg-surface-container-low border border-outline-variant/30 rounded-lg text-center mt
-lg">
  <p class="text-caption text-on-surface-variant mb-1">? TODO: Integrate live Chart.js or D3.js pred
iction curves on next phase release</p>
  <span class="text-xs text-secondary font-bold">Predictive AI Model: V2.1 Active</span>
</div>
```



```

    </div>

    </div>
    `;
}

content.innerHTML = htmlContent;

// Search function local to page
window.filterInvTable = () => {
    const txt = (document.getElementById('invSearch')?.value || '').toLowerCase();
    const cat = document.getElementById('invCatFilter')?.value || '';

    document.querySelectorAll('#invTbl tbody tr').forEach(r => {
        const rowText = r.textContent.toLowerCase();
        const matchText = rowText.includes(txt);
        const matchCat = cat === '' || rowText.includes(cat.toLowerCase());

        r.style.display = (matchText && matchCat) ? '' : 'none';
    });
};

// Automated order execution handler
window.approveAutoOrder = (name, qty, cost, supplier) => {
    const newId = `PO-2024-00${window.localPOs.length + 1}`;
    window.localPOs.unshift({
        id: newId,
        supplier: supplier,
        date: new Date().toISOString().split('T')[0],
        delivery: new Date(Date.now() + 5*24*60*60*1000).toISOString().split('T')[0],
        amount: cost,
        status: '??? ???????',
        items: [{ name: name, qty: qty, cost: cost / qty }]
    });
    showToast(tr(`Auto-Order approved and PO created!`, `?? ?????? ??? ?????? ??????? ?? ${name}!`), 'success');
    window.inventoryTab = 'procurement';
    navigateTo(currentPage);
};

// Receive Purchase Order handler
window.receivePO = async (poId) => {
    const poIndex = window.localPOs.findIndex(po => po.id === poId);
    if (poIndex === -1) return;

    const po = window.localPOs[poIndex];
    po.status = '?????';

    // Update the item quantity in the database if the item exists in real database
    if (po.items && po.items.length) {
        for (const poItem of po.items) {
            // Find existing item by name
            const matchItem = items.find(i => i.name.toLowerCase() === poItem.name.toLowerCase());
            if (matchItem) {

```

```

const newQty = parseInt(matchItem.quantity || 0) + parseInt(poItem.qty);
try {
  await API.put(`/api/inventory/${matchItem.id}`, {
    ...matchItem,
    quantity: newQty
  });
  showToast(tr(`Received and added ${poItem.qty} units to ${matchItem.name}`, `?? ?????? ?????? ???
?? ${poItem.qty} ??? ? ${matchItem.name}`), 'success');
} catch (e) {
  console.error('Error auto-updating inventory quantity', e);
}
} else {
  // If not found in real database, add it as a new inventory item
  try {
    await API.post('/api/inventory', {
      name: poItem.name,
      category: tr('Supplies', '???????? ????'),
      quantity: poItem.qty,
      unit: tr('Units', '???'),
      reorder_level: 10,
      location: tr('A1 Warehouse', '????? ?1'),
      supplier: po.supplier,
      cost: poItem.cost,
      expiry_date: new Date(Date.now() + 365*24*60*60*1000).toISOString().split('T')[0]
    });
  } catch(e) {
    console.error('Error auto-inserting new item', e);
  }
}
}
}

showToast(tr(`PO ${poId} successfully received and inventory updated!`, `?? ?????? ?????? ${poId} ?????? ??
????? ?????!`), 'success');
navigateTo(currentPage);
};

// -----
// MODALS & ACTIONS IMPLEMENTATIONS
// -----
window.showAddInvModal = () => {
  const modal = document.createElement('div');
  modal.className = 'modal-overlay';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-i
tems:center;justify-content:center';
  modal.innerHTML = `
    <div class="glass-card p-lg rounded-2xl w-full max-w-xl max-h-[90vh] overflow-y-auto" style="direction:r
tl">
      <div class="flex justify-between items-center mb-md">
        <h4 class="font-bold text-headline-md text-primary">? ${tr('Add New Inventory Item', '????? ??? ???
')}</h4>
        <button class="text-on-surface-variant hover:text-error" onclick="this.closest('.modal-overlay').rem
ove()">?</button>
      </div>

```

```

<div class="grid grid-cols-1 md:grid-cols-2 gap-sm">
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Item Name (AR/EN)', '??? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="invName" required placeholder="e.g. ??????????"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Category', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="invCat" placeholder="e.g. ?????"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Quantity', '????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" id="invQty" value="100"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Unit', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="invUnit" value="????"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Reorder Level', '?? ????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" id="invReorder" value="10"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Warehouse Location', '???? ?????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="invLoc" placeholder="e.g. ????? ? - ??? 3"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Supplier Name', '?? ?????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="invSupplier" placeholder="e.g. ??? ??????"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Cost per Unit (SAR)', '??????? ?? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" step="0.01" id="invCost" value="15.50"></div>
  <div class="form-group md:col-span-2"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Expiry Date', '????? ?????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="date" id="invExpiry" value="2027-12-31"></div>
</div>
<button class="bg-primary text-on-primary font-bold w-full py-3 rounded-xl mt-md hover:opacity-90 transition-all" onclick="saveInvItem()">${tr('Add to Database', '??? ?????')}</button>
</div>
`;
document.body.appendChild(modal);
modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

window.saveInvItem = async () => {
  const name = document.getElementById('invName').value.trim();
  if (!name) { showToast(tr('Enter item name', '???? ?? ?????'), 'error'); return; }
  try {
    await API.post('/api/inventory', {
      name: name,
      category: document.getElementById('invCat').value.trim(),
      quantity: parseInt(document.getElementById('invQty').value) || 0,
      unit: document.getElementById('invUnit').value.trim() || '????',
      reorder_level: parseInt(document.getElementById('invReorder').value) || 10,
      location: document.getElementById('invLoc').value.trim() || tr('Not specified', '??? ?????'),
      supplier: document.getElementById('invSupplier').value.trim() || tr('Not specified', '??? ?????'),
      cost: parseFloat(document.getElementById('invCost').value) || 0,
      expiry_date: document.getElementById('invExpiry').value || null
    });
  }
  showToast(tr('Item added successfully!', '??? ????? ????? ?????!'), 'success');
  document.querySelector('.modal-overlay')?.remove();
}

```

```

        navigateTo(currentPage);
    } catch (e) {
        showToast(tr('Error adding item', '??? ?? ????? ?????'), 'error');
    }
};

window.showEditInvModal = (i) => {
    const modal = document.createElement('div');
    modal.className = 'modal-overlay';
    modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
    modal.innerHTML = `
        <div class="glass-card p-lg rounded-2xl w-full max-w-xl max-h-[90vh] overflow-y-auto" style="direction:rtl">
            <div class="flex justify-between items-center mb-md">
                <h4 class="font-bold text-headline-md text-primary">? ${tr('Edit Inventory Item', '????? ???')}</h4>
                <button class="text-on-surface-variant hover:text-error" onclick="this.closest('.modal-overlay').remove()">?</button>
            </div>
            <div class="grid grid-cols-1 md:grid-cols-2 gap-sm">
                <input type="hidden" id="editInvId" value="${safeId(i.id)}">
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Item Name (AR/EN)', '??? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="editInvName" value="${escapeHTML(i.name || '')}" required></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Category', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="editInvCat" value="${escapeHTML(i.category || '')}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Quantity', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" id="editInvQty" value="${escapeHTML(i.quantity || 0)}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Unit', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="editInvUnit" value="${escapeHTML(i.unit || '')}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Reorder Level', '?? ????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" id="editInvReorder" value="${escapeHTML(i.reorder_level || 10)}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Warehouse Location', '???? ??????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="editInvLoc" value="${escapeHTML(i.location || '')}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Supplier Name', '??? ?????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="editInvSupplier" value="${escapeHTML(i.supplier || '')}"></div>
                <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Cost per Unit (SAR)', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" step="0.01" id="editInvCost" value="${escapeHTML(i.cost || 0)}"></div>
                <div class="form-group md:col-span-2"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Expiry Date', '????? ?????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="date" id="editInvExpiry" value="${escapeHTML(i.expiry_date ? i.expiry_date.split('T')[0] : '')}"></div>
            </div>
            <button class="bg-primary text-on-primary font-bold w-full py-3 rounded-xl mt-md hover:opacity-90 transition-all" onclick="updateInvItem()">${tr('Update Item', '??? ?????')}</button>
        </div>
    `;
};

```

```

document.body.appendChild(modal);
modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

window.updateInvItem = async () => {
  const id = document.getElementById('editInvId').value;
  const name = document.getElementById('editInvName').value.trim();
  if (!name) { showToast(tr('Enter item name', '???? ??? ?????'), 'error'); return; }
  try {
    await API.put(`/api/inventory/${id}`, {
      name: name,
      category: document.getElementById('editInvCat').value.trim(),
      quantity: parseInt(document.getElementById('editInvQty').value) || 0,
      unit: document.getElementById('editInvUnit').value.trim() || '????',
      reorder_level: parseInt(document.getElementById('editInvReorder').value) || 10,
      location: document.getElementById('editInvLoc').value.trim(),
      supplier: document.getElementById('editInvSupplier').value.trim(),
      cost: parseFloat(document.getElementById('editInvCost').value) || 0,
      expiry_date: document.getElementById('editInvExpiry').value || null
    });
    showToast(tr('Item updated successfully!', '?? ????? ????? ?????!'), 'success');
    document.querySelector('.modal-overlay')?.remove();
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error updating item', '??? ?? ????? ?????'), 'error');
  }
};

window.deleteInvItem = async (id) => {
  if (!confirm(tr('Are you sure you want to delete this item?', '?? ??? ????? ?? ????? ?? ??? ??? ????? ???
????')))) return;
  try {
    await API.delete(`/api/inventory/${id}`);
    showToast(tr('Item deleted!', '?? ??? ?????!'), 'success');
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error deleting item', '??? ?? ??? ?????'), 'error');
  }
};

// Add Supplier Modal
window.showAddSupplierModal = () => {
  const modal = document.createElement('div');
  modal.className = 'modal-overlay';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-i
tems:center;justify-content:center';
  modal.innerHTML = `
    <div class="glass-card p-lg rounded-2xl w-full max-w-lg" style="direction:rtl">
      <div class="flex justify-between items-center mb-md">
        <h4 class="font-bold text-headline-md text-primary">? ${tr('Add New Supplier', '????? ??? ????')}</
h4>
        <button class="text-on-surface-variant hover:text-error" onclick="this.closest('.modal-overlay').rem
ove()">?</button>
      </div>

```

```

<div class="space-y-sm">
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Supplier Name', '??? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="supName" required></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Location', '????? / ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="supLoc"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Email', '????? ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="email" id="supEmail"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Phone', '??? ????? / ?????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="supPhone"></div>
  <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Supply Category', '????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="supCat" placeholder="e.g. ?????? ???? "></div>
</div>
<button class="bg-primary text-on-primary font-bold w-full py-3 rounded-xl mt-md hover:opacity-90 transition-all" onclick="saveSupplier()">${tr('Add Supplier', '??? ?????')}</button>
</div>
`;
document.body.appendChild(modal);
modal.onclick = e => { if (e.target === modal) modal.remove(); };
};

window.saveSupplier = () => {
  const name = document.getElementById('supName').value.trim();
  if (!name) { showToast(tr('Enter supplier name', '???? ????'), 'error'); return; }
  window.localSuppliers.unshift({
    id: window.localSuppliers.length + 1,
    name: name,
    location: document.getElementById('supLoc').value.trim() || tr('Riyadh, Saudi Arabia', '????? ?????? ?????? ??????'),
    email: document.getElementById('supEmail').value.trim() || 'info@supplier.sa',
    phone: document.getElementById('supPhone').value.trim() || '011-000-0000',
    category: document.getElementById('supCat').value.trim() || tr('Supplies', '????? ????'),
    outstanding: 0.00,
    status: '???'
  });
  showToast(tr('Supplier added successfully!', '??? ????? ?????? !'), 'success');
  document.querySelector('.modal-overlay')?.remove();
  navigateTo(currentPage);
};

// Add Purchase Order Modal
window.showAddPOModal = () => {
  const modal = document.createElement('div');
  modal.className = 'modal-overlay';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
  modal.innerHTML = `
    <div class="glass-card p-lg rounded-2xl w-full max-w-lg" style="direction:rtl">
      <div class="flex justify-between items-center mb-md">
        <h4 class="font-bold text-headline-md text-primary">? ${tr('Create Purchase Order', '????? ???? ????

```

```

????')}</h4>
    <button class="text-on-surface-variant hover:text-error" onclick="this.closest('.modal-overlay').remove()">?</button>
  </div>
  <div class="space-y-sm">
    <div class="form-group">
      <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Select Supplier', '????????')}</label>
      <select class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="poSupplierSelect">
        ${window.localSuppliers.map(s => `<option value="${escapeHTML(s.name)}">${escapeHTML(s.name)}</option>`).join('')}
      </select>
    </div>
    <div class="form-group">
      <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Select Medical Item', '????????? ?? ?????????')}</label>
      <select class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" id="poItemSelect">
        ${items.map(i => `<option value="${escapeHTML(i.name)}" data-cost="${escapeHTML(i.cost)}">${escapeHTML(i.name)} (????????: ${escapeHTML(i.cost)} ??)</option>`).join('')}
      </select>
    </div>
    <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Quantity Required', '?????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="number" id="poQty" value="10" oninput="calcPOTotal()"></div>
    <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Estimated PO Amount (SAR)', '?????? ?????????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2 font-bold text-secondary text-lg" type="number" id="poTotalAmt" readonly value="0"></div>
    <div class="form-group"><label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Expected Delivery Date', '????? ?????????? ?????????')}</label><input class="w-full bg-surface-container-low border border-outline-variant/30 rounded-lg p-2" type="date" id="poDeliveryDate" value="2026-06-25"></div>
  </div>
  <button class="bg-primary text-on-primary font-bold w-full py-3 rounded-xl mt-md hover:opacity-90 transition-all" onclick="savePurchaseOrder()">${tr('Issue Purchase Order', '????? ??? ?????????')}</button>
</div>
`;
document.body.appendChild(modal);
modal.onclick = e => { if (e.target === modal) modal.remove(); };
calcPOTotal();

// Event listener to update cost on select change
document.getElementById('poItemSelect').addEventListener('change', calcPOTotal);
};

window.calcPOTotal = () => {
  const poSelect = document.getElementById('poItemSelect');
  if (!poSelect) return;
  const selectedOpt = poSelect.options[poSelect.selectedIndex];
  if (!selectedOpt) return;
  const cost = parseFloat(selectedOpt.dataset.cost || 0);
  const qty = parseInt(document.getElementById('poQty').value) || 0;
  document.getElementById('poTotalAmt').value = (cost * qty).toFixed(2);
}

```

```

};

window.savePurchaseOrder = () => {
  const supplier = document.getElementById('poSupplierSelect').value;
  const itemName = document.getElementById('poItemSelect').value;
  const qty = parseInt(document.getElementById('poQty').value) || 0;
  const total = parseFloat(document.getElementById('poTotalAmt').value) || 0;
  if (qty <= 0) { showToast(tr('Enter a valid quantity', '???? ?????'), 'error'); return; }

  const newId = `PO-2024-00${window.localPOs.length + 1}`;
  window.localPOs.unshift({
    id: newId,
    supplier: supplier,
    date: new Date().toISOString().split('T')[0],
    delivery: document.getElementById('poDeliveryDate').value || new Date(Date.now() + 5*24*60*60*1000).toISOString().split('T')[0],
    amount: total,
    status: '??? ??????',
    items: [{ name: itemName, qty: qty, cost: total / qty }]
  });

  showToast(tr('Purchase Order issued successfully!', '?? ????? ???? ???? ???? ???? ???? ???? ????'),
'success');
  document.querySelector('.modal-overlay')?.remove();
  navigateTo(currentPage);
};

}

// ===== SIMPLE MODULE PAGES =====
let nurseTab = 'vitals';
async function renderNursing(el) {
  return renderNursingStation(el);
  // Show premium loading skeletons instantly
  el.innerHTML = `
    <div class="page-title">???? ${tr('Nursing Station', '???? ?????')}</div>
    <div class="tab-bar">
      <button class="tab-btn active">?? ${tr('Vitals', '???????? ?????')}</button>
    </div>
    <div class="split-layout">
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
    </div>
  `;

  let patients = [];
  let vitals = [];
  let emarOrders = [];

```



```

let carePlans = [];
let assessments = [];
let riskAssessments = [];
try {
  [patients, vitals, emarOrders, carePlans, assessments, riskAssessments] = await Promise.all([
    API.get('/api/patients'),
    API.get('/api/nursing/vitals').catch(() => []),
    API.get('/api/emar/orders').catch(() => []),
    API.get('/api/nursing/care-plans').catch(() => []),
    API.get('/api/nursing/assessments').catch(() => []),
    API.get('/api/nursing/risk-assessments').catch(() => [])
  ]);
  window.nursingPatients = patients; // Store globally for access by modal dialogs
} catch (e) {
  el.innerHTML = `
    <div class="page-title">???? ${tr('Nursing Station', '???? ???????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load nursing data', '??? ?? ????? ?????? ????? ???????')}</h3>
      <p>${escapeHTML(e.message || e)}</p>
      <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ???????')}</
button>
    </div>
  `;
  return;
}
el.innerHTML = `
  <div class="page-title">???? ${tr('Nursing Station', '???? ???????')}</div>
  <div class="tab-bar">
    <button class="tab-btn ${nurseTab === 'vitals' ? 'active' : ''}" onclick="nurseTab='vitals';navigateTo(11)">?? ${tr('Vitals', '???????? ???????')}</button>
    <button class="tab-btn ${nurseTab === 'emar' ? 'active' : ''}" onclick="nurseTab='emar';navigateTo(11)">
? ${tr('eMAR', '????? ???????')}</button>
    <button class="tab-btn ${nurseTab === 'careplans' ? 'active' : ''}" onclick="nurseTab='careplans';naviga
teTo(11)">? ${tr('Care Plans', '??? ???????')}</button>
    <button class="tab-btn ${nurseTab === 'assess' ? 'active' : ''}" onclick="nurseTab='assess';navigateTo(11)">? ${tr('Assessments', '?????????')}</button>
  </div>`;
  if (nurseTab === 'emar') {
    el.innerHTML += `<div class="card glass-card-premium"><h3>? ${tr('Electronic Medication Administration Rec
ord', '??? ?????? ??????? ??????????')}</h3>
    ${emarOrders.length ? makeTable(
      [tr('Patient', '??????'), tr('Medication', '??????'), tr('Dose', '??????'), tr('Route', '????????'), tr('
Frequency', '????????'), tr('Status', '??????'), tr('Actions', '????????')],
      emarOrders.map(o => ({
        cells: [o.patient_name, o.medication, o.dose, o.route, o.frequency, statusBadge(o.status),
        rawHtml(`<button class="btn btn-sm btn-success" onclick="administerMed(${parseInt(o.id, 10)},${parseInt(
o.patient_id, 10)},'${jsStr(o.medication)}','${jsStr(o.dose)}','${jsStr(o.route || '')}')">? ${tr('Give', '?
????')}</button>`)
      })
    ) : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No active orders', '??
???? ?????? ??????')}</div></div>`
  }
  </div>`;

```

```

    } else if (nurseTab === 'careplans') {
      el.innerHTML += `<div class="card glass-card-premium"><h3>? ${tr('Nursing Care Plans', '??? ?????? ??????
      ???')}</h3>
      <button class="btn btn-primary" onclick="nurseTab='newplan';navigateTo(11)" style="margin-bottom:12px">? $
      {tr('New Plan', '??? ?????')}</button>
      ${carePlans.length ? makeTable(
        [tr('Patient', '??????'), tr('Diagnosis', '????????'), tr('Priority', '????????'), tr('Goals', '????????')
        , tr('Status', '??????')],
        carePlans.map(c => ({ cells: [c.patient_name, c.diagnosis, c.priority === 'High' ? rawHtml(`<span class=
        "badge badge-danger">? ${tr('High', '?????')}</span>`) : c.priority === 'Low' ? rawHtml(`<span class="badge ba
        dge-success">? ${tr('Low', '??????')}</span>`) : rawHtml(`<span class="badge badge-warning">? ${tr('Medium', '
        ??????')}</span>`), c.goals?.substring(0, 60) || '-', statusBadge(c.status)] }))
        ) : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No care plans', '?? ??
        ? ?? ????')}</div></div>`}
      </div>`;
    } else if (nurseTab === 'assess') {
      el.innerHTML += `<div class="card glass-card-premium"><h3>? ${tr('Nursing Assessments (Legacy)', '????????
      ? ?????????')}</h3>
      ${assessments.length ? makeTable(
        [tr('Patient', '??????'), tr('Type', '?????'), tr('Fall Risk', '??? ??????'), tr('Braden', 'Braden'), tr
        ('Pain', '???'), tr('GCS', 'GCS'), tr('Nurse', '??????'), tr('Shift', '????????')],
        assessments.map(a => ({
          cells: [a.patient_name, a.assessment_type,
            rawHtml(`<span style="color:${a.fall_risk_score} >= 45 ? '#ef4444' : a.fall_risk_score >= 25 ? '#f59e0b
            ' : '#22c55e'`)>${escapeHTML(a.fall_risk_score)}</span>`),
            rawHtml(`<span style="color:${a.braden_score} <= 12 ? '#ef4444' : a.braden_score <= 18 ? '#f59e0b' : '#
            22c55e'`)>${escapeHTML(a.braden_score)}/23</span>`),
            rawHtml(`<span style="color:${a.pain_score} >= 7 ? '#ef4444' : a.pain_score >= 4 ? '#f59e0b' : '#22c55e
            '`)>${escapeHTML(a.pain_score)}/10</span>`),
            a.gcs_score + '/15', a.nurse, a.shift
          ]
        }
      }
      ) : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No assessments', '?? ??
      ?? ?????')}</div></div>`}
      </div>

      <div class="card glass-card-premium" style="margin-top:20px">
      <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
      <h3>? ${tr('Braden Scale & Morse Fall Risk Assessments', '?????? ????? ????? ????? ?????')}</
      h3>
      <div style="display:flex;gap:8px">
      <button class="btn btn-sm btn-primary" onclick="openBradenModal()">? ${tr('Record Braden Scale', '??
      ??? ?????')}</button>
      <button class="btn btn-sm btn-warning" onclick="openMorseModal()">? ${tr('Record Morse Fall Risk', '
      ????? ?????')}</button>
      </div>
      </div>
      ${riskAssessments.length ? makeTable(
        [tr('Patient', '??????'), tr('Type', '??? ??????'), tr('Total Score', '?????? ?????????'), tr('Risk L
        evel', '????? ??????'), tr('Assessed By', '??????'), tr('Date', '????????')],
        riskAssessments.map(r => ({
          cells: [
            `${r.file_number || ''} - ${isArabic ? (r.name_ar || r.name_en) : (r.name_en || r.name_ar)}`,
            r.assessment_type,

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        r.total_score,
        rawHtml(`<span class="badge ${r.risk_level.toLowerCase().includes('high')} || r.risk_level.toLowerCase().includes('severe')} ? 'badge-danger' : r.risk_level.toLowerCase().includes('moderate')} || r.risk_level.toLowerCase().includes('mild')} ? 'badge-warning' : 'badge-success'`>${escapeHTML(r.risk_level)}</span>`),
        r.assessed_by || '-',
        new Date(r.created_at).toLocaleString()
    ]
    }))
    ) : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No Braden or Morse as
sessments recorded', '?? ???? ?????? ?????? ?????? ?? ?????')}</div></div>`
    </div>`;
    } else if (nurseTab === 'newplan') {
        el.innerHTML += `<div class="card glass-card-premium"><h3>? ${tr('New Care Plan', '??? ????? ?????')}</h3>
<div class="form-grid">
    <div><label>${tr('Patient', '??????')}</label><select id="cpPatientN" class="form-input">${patients.map(
p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.file_nu
mber)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}</select></div>
    <div><label>${tr('Priority', '????????')}</label><select id="cpPriorityN" class="form-input"><option val
ue="Low">${tr('Low', '??????')}</option><option value="Medium" selected>${tr('Medium', '??????')}</option><opt
ion value="High">${tr('High', '????')}</option></select></div>
    <div style="grid-column:1/-1"><label>${tr('Diagnosis', '????????')}</label><input id="cpDiagn" class="for
m-input"></div>
    <div style="grid-column:1/-1"><label>${tr('Goals', '????????')}</label><textarea id="cpGoalsN" class="for
m-input" rows="2"></textarea></div>
    <div style="grid-column:1/-1"><label>${tr('Interventions', '????????')}</label><textarea id="cpIntN" cla
ss="form-input" rows="2"></textarea></div>
</div>
    <button class="btn btn-primary" onclick="saveCarePlan()" style="margin-top:8px">? ${tr('Save', '???')}</bu
tton></div>`;
    } else {
        el.innerHTML += `<div class="split-layout">
<div>
    <div class="card glass-card-premium mb-16">
        <div class="card-title">?? ${tr('Record Patient Vitals', '????? ????????? ?????????')}</div>
        <div class="form-group mb-12"><label>${tr('Patient', '??????')}</label><select class="form-input" id
="nsPatient"><option value="">${tr('-- Select --', '-- ???? ???? --')}</option>${patients.map(p => `<option va
lue="${safeId(p.id)}" data-name="">${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? (p.name_ar || p.name
_en) : (p.name_en || p.name_ar))}</option>`).join('')}</select></div>
        <div class="flex gap-8 mb-12">
            <div class="form-group" style="flex:1"><label>? ${tr('Blood Pressure', '??? ????')}</label><input
class="form-input" id="nsBp" placeholder="120/80"></div>
            <div class="form-group" style="flex:1"><label>?? ${tr('Temp (°C)', '????????')}</label><input class
="form-input" id="nsTemp" type="number" step="0.1" placeholder="37.0"></div>
        </div>
        <div class="flex gap-8 mb-12">
            <div class="form-group" style="flex:1"><label>?? ${tr('Pulse (bpm)', '????')}</label><input class
="form-input" id="nsPulse" type="number" placeholder="75"></div>
            <div class="form-group" style="flex:1"><label>? ${tr('O2 Sat (%)', '????????')}</label><input clas
s="form-input" id="nsO2" type="number" placeholder="98"></div>
        </div>
        <div class="flex gap-8 mb-12">
            <div class="form-group" style="flex:1"><label>? ${tr('Weight (kg)', '????')}</label><input class=
"form-input" id="nsWeight" type="number" step="0.1" placeholder="70.5"></div>
            <div class="form-group" style="flex:1"><label>? ${tr('Height (cm)', '????')}</label><input class=

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"form-input" id="nsHeight" type="number" placeholder="170"></div>
</div>
<div class="flex gap-8 mb-12">
  <div class="form-group" style="flex:1"><label>?? ${tr('Respiratory Rate', '???? ?????')}</label><
input class="form-input" id="nsResp" type="number" placeholder="18"></div>
  <div class="form-group" style="flex:1"><label>? ${tr('Blood Sugar', '????')}</label><input class=
"form-input" id="nsSugar" type="number" placeholder="100"></div>
</div>
</div>
<div class="card glass-card-premium mb-16">
  <div class="card-title">? ${tr('Medical History', '?????? ?????')}</div>
  <div class="form-group mb-12"><label>? ${tr('Chronic Diseases', '?????? ?????')}</label><textarea
class="form-input form-textarea" id="nsChronic" placeholder="${tr('e.g. Diabetes, Hypertension, Asthma...', '
?????: ????? ???? ???...')}"></textarea></div>
  <div class="form-group mb-12"><label>? ${tr('Current Medications', '?????? ?????')}</label><textare
a class="form-input form-textarea" id="nsMeds" placeholder="${tr('e.g. Metformin 500mg, Aspirin 100mg...', '
?????: ????????? 500??? ????? 100??...')}"></textarea></div>
  <div class="form-group mb-12"><label>?? ${tr('Allergies', '????????')}</label><textarea class="form-
input form-textarea" id="nsAllergies" placeholder="${tr('e.g. Penicillin, Peanuts, Latex...', '?????: ??????
??? ?????? ?????...')}"></textarea></div>
  <div class="form-group mb-16"><label>? ${tr('Notes / Triage', '?????? / ???')}</label><textarea cla
ss="form-input form-textarea" id="nsNotes"></textarea></div>
  <button class="btn btn-primary w-full" style="height:44px" onclick="saveVitals()">? ${tr('Save Vital
s & Send to Doctor', '??? ?????? ?????')}</button>
</div>
</div>
<div class="card glass-card-premium">
  <div class="card-title">? ${tr('Recent Vitals Registry', '??? ????????? ?????')}</div>
  <input class="search-filter" id="nsSearch" placeholder="${tr('Search...', '????...')}">
  <div id="nsTable">${vitals.length === 0 ? `<div class="empty-state-card"><div class="empty-state-icon"
>?</div><div>${tr('No vitals data found', '?? ???? ?????? ????? ?????')}</div></div>` : vitals.map(v => `
  <div class="card mb-12" style="padding:12px;border:1px solid var(--border-color,#e5e7eb);border-radi
us:10px;background:var(--card-bg,#fff)">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:8px">
      <strong style="font-size:14px">? ${escapeHTML(v.patient_name || v.patient_id)}</strong>
      <span style="font-size:12px;color:var(--text-muted,#999)">? ${escapeHTML(v.created_at?.split('T'
)[0] || '')}</span>
    </div>
    <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:6px;font-size:13px">
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">? ${tr('BP', '????')}<br><strong>${escapeHTML(v.bp || '-')}</strong></div>
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">?? ${tr('Temp', '????')}<br><strong>${v.temp ? escapeHTML(v.temp) + '°' : '-'}</strong></div>
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">?? ${tr('Pulse', '???')}<br><strong>${escapeHTML(v.pulse || '-')}</strong></div>
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">? ${tr('O2', '????')}<br><strong>${v.o2_sat ? escapeHTML(v.o2_sat) + '%' : '-'}</strong></div>
    </div>
    <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:6px;font-size:13px;margin-top:6px
">
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">? ${tr('Weight', '???')}<br><strong>${v.weight ? escapeHTML(v.weight) + ' kg' : '-'}</strong></div>
      <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">? ${tr('Height', '???')}<br><strong>${v.height ? escapeHTML(v.height) + ' cm' : '-'}</strong></div>

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        <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">?? ${tr('Resp', '????')}<br><strong>${escapeHTML(v.respiratory_rate || '-')}</strong></div>
        <div style="background:var(--bg-secondary,#f8f9fa);padding:6px 8px;border-radius:6px;text-align:
center">? ${tr('Sugar', '???')}<br><strong>${escapeHTML(v.blood_sugar || '-')}</strong></div>
        </div>
        ${v.allergies ? `<div style="margin-top:6px"><span class="badge badge-danger">?? ${escapeHTML(v.allergies)}</span></div>` : ''}
        ${v.chronic_diseases ? `<div style="margin-top:4px;font-size:12px;color:var(--text-muted,#888)">?
${escapeHTML(v.chronic_diseases)}</div>` : ''}
        </div>
        `).join('')</div>
    </div>`;
// Search filter for vitals cards
document.getElementById('nsSearch')?.addEventListener('input', (e) => {
    const txt = e.target.value.toLowerCase();
    document.querySelectorAll('#nsTable .card').forEach(c => {
        c.style.display = c.textContent.toLowerCase().includes(txt) ? '' : 'none';
    });
});
}
}

window.saveVitals = async () => {
    const sel = document.getElementById('nsPatient');
    const pid = sel.value;
    if (!pid) { showToast(tr('Select patient first', '???? ?????? ??????'), 'error'); return; }
    const pname = sel.options[sel.selectedIndex].getAttribute('data-name');
    try {
        await API.post('/api/nursing/vitals', {
            patient_id: pid, patient_name: pname,
            bp: document.getElementById('nsBp').value,
            temp: parseFloat(document.getElementById('nsTemp').value) || 0,
            weight: parseFloat(document.getElementById('nsWeight').value) || 0,
            height: parseFloat(document.getElementById('nsHeight').value) || 0,
            pulse: parseInt(document.getElementById('nsPulse').value) || 0,
            o2_sat: parseInt(document.getElementById('nsO2').value) || 0,
            respiratory_rate: parseInt(document.getElementById('nsResp').value) || 0,
            blood_sugar: parseInt(document.getElementById('nsSugar').value) || 0,
            chronic_diseases: document.getElementById('nsChronic').value,
            current_medications: document.getElementById('nsMeds').value,
            allergies: document.getElementById('nsAllergies').value,
            notes: document.getElementById('nsNotes').value
        });
        showToast(tr('Vitals recorded and patient routed to doctor!', '?? ?????? ?????????? ?????????? ?????????? !'))
    }
    ;
    await navigateTo(11);
} catch (e) { showToast(tr('Error saving', '??? ?? ??????'), 'error'); }
};

// High-alert medications (mirror of the server MAR_HIGH_ALERT list) ? used only to PRE-show the
// witness field; the SERVER is the authority (it re-checks and fail-closes if witness is missing).
const MAR_HIGH_ALERT_CLIENT = ['insulin', 'heparin', 'warfarin', 'morphine', 'hydromorphone', 'fentanyl',
    'methadone', 'oxycodone', 'potassium chloride', 'kcl', 'magnesium sulfate', 'digoxin', 'epinephrine',
    'norepinephrine', 'chemotherapy', 'methotrexate', 'oxytocin'];

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function isHighAlertClient(name) {
  const n = String(name || '').trim().toLowerCase();
  return !!n && MAR_HIGH_ALERT_CLIENT.some(h => n.includes(h));
}

// PRIMARY nurse "Give" path ? posts to the SAFE /api/mar/administer route (server-enforced 5 rights
// + CDS + witness). The server resolves drug/dose/route from the order; we send the scanned values so
// the server can verify them. Errors (422/409 ? {error}) are surfaced via showToast (no silent pass).
window.administerMed = async function (orderId, patientId, med, dose, route) {
  const highAlert = isHighAlertClient(med);
  let witness_user_id = null;
  if (highAlert) {
    // High-alert drug: a distinct second-nurse witness user id is required (server re-verifies).
    const w = prompt(tr('High-alert medication ? enter WITNESS user ID (a different nurse):',
      '???? ???? ?????? ? ???? ????? ??????? ????? (???? ????):'));
    if (!w) { showToast(tr('Witness required for high-alert medication', '???? ???? ??????? ????? ???????'), '
error'); return; }
    witness_user_id = parseInt(w, 10);
  }
  try {
    const resp = await API.post('/api/mar/administer', {
      emar_order_id: orderId,
      patient_id: patientId,
      scanned_patient_id: patientId,
      scanned_drug: med,
      scanned_dose: dose,
      scanned_route: route || '',
      scheduled_at: new Date().toISOString(),
      witness_user_id: witness_user_id
    });
    if (resp && resp.error) {
      // 422 (right/CDS/witness block) or 409 ? server returns {error, right?, requires_*?}.
      showToast(resp.error + (resp.right ? ` (${resp.right})` : ''), 'error');
      return;
    }
    showToast(tr('Medication administered', '?? ????? ??????'));
    navigateTo(11);
  } catch (e) {
    showToast(tr('Error administering medication', '??? ?? ????? ??????'), 'error');
  }
};

window.saveCarePlan = async function () {
  const sel = document.getElementById('cpPatientN');
  const patient_name = sel.options[sel.selectedIndex].dataset.name;
  await API.post('/api/nursing/care-plans', { patient_id: sel.value, patient_name, diagnosis: document.getElementBy
Id('cpDiagn').value, priority: document.getElementById('cpPriorityN').value, goals: document.getElementById('cpGoalsN').value, interventions: document.getElementById('cpIntN').value });
  showToast(tr('Care plan saved', '?? ??????')); nurseTab = 'careplans'; navigateTo(11);
};

async function renderWaitingQueue(el) {
  const content = el;

  const [patients, appointments] = await Promise.all([
    API.get('/api/queue/patients').catch(() => []),

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    API.get('/api/appointments').catch(() => [])
  });
  const today = new Date().toISOString().slice(0, 10);
  const todayApts = appointments.filter(a => (a.appt_date || a.date || '').includes(today));

  // Group by doctor
  const byDoctor = {};
  patients.forEach(p => {
    const d = p.doctor || p.doctor_name || tr('Unassigned', '??? ???');
    if (!byDoctor[d]) byDoctor[d] = [];
    byDoctor[d].push(p);
  });

  const getAcuityStyle = (level) => {
    switch(parseInt(level, 10)) {
      case 1: return { color: '#ef4444', bg: '#fef2f2', border: '#fca5a5', name_ar: '????? ??? (ESI-1)', name_en: 'Resuscitation (ESI-1)' };
      case 2: return { color: '#f97316', bg: '#fff7ed', border: '#ffedd5', name_ar: '???? ????? (ESI-2)', name_en: 'Emergent (ESI-2)' };
      case 3: return { color: '#eab308', bg: '#fef9c3', border: '#fef08a', name_ar: '???? (ESI-3)', name_en: 'Urgent (ESI-3)' };
      case 4: return { color: '#3b82f6', bg: '#eff6ff', border: '#bdfbfe', name_ar: '??? ??? (ESI-4)', name_en: 'Less Urgent (ESI-4)' };
      default: return { color: '#10b981', bg: '#ecfdf5', border: '#a7f3d0', name_ar: '??? ??? (ESI-5)', name_en: 'Non-Urgent (ESI-5)' };
    }
  };

  content.innerHTML = `
    <div style="display:flex;justify-content:between;align-items:center;margin-bottom:20px;flex-wrap:wrap;gap:12px">
      <h2>${tr('Waiting Queue Dashboard', '???? ??? ?????????')}</h2>
      <div style="display:flex;gap:8px;align-items:center">
        <span style="font-size:12px;color:var(--text-dim,#666)" id="queueTimer">?? ${tr('Auto-refresh: 15s', '????? ?????: 15 ?????')}</span>
        <button class="btn btn-sm" onclick="navigateTo(currentPage)" style="background:#e3f2fd;color:#1565c0">? ${tr('Refresh', '????')}</button>
      </div>
    </div>

    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px;margin-bottom:24px">
      <div class="card glass-card-premium" style="padding:20px;text-align:center;border-left:5px solid #1565c0;background:rgba(21,101,192,0.05)">
        <h3 style="margin:0;color:#1565c0;font-size:28px">${patients.length}</h3>
        <p style="margin:6px 0 0;font-size:13px;font-weight:bold">${tr('Patients in Queue', '????? ?? ??????')}</p>
      </div>
      <div class="card glass-card-premium" style="padding:20px;text-align:center;border-left:5px solid #2e7d32;background:rgba(46,125,50,0.05)">
        <h3 style="margin:0;color:#2e7d32;font-size:28px">${todayApts.length}</h3>
        <p style="margin:6px 0 0;font-size:13px;font-weight:bold">${tr('Today Appointments', '????? ?????')}</p>
    </div>
  `

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```

<div class="card glass-card-premium" style="padding:20px;text-align:center;border-left:5px solid #e65100
;background:rgba(230,81,0,0.05)">
  <h3 style="margin:0;color:#e65100;font-size:28px">${Object.keys(byDoctor).length}</h3>
  <p style="margin:6px 0 0;font-size:13px;font-weight:bold">${tr('Active Doctors', '?????? ?????')}</
p>
  </div>
</div>

<div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(400px,1fr));gap:20px">
  ${Object.entries(byDoctor).map(([doc, pts]) => {
    return `
      <div class="card glass-card-premium" style="padding:20px">
        <h4 style="margin:0 0 12px;color:var(--primary);display:flex;justify-content:space-between;align-i
tems:center;border-bottom:1px solid var(--border,#e0e0e0);padding-bottom:8px">
          <span>???? ${escapeHTML(doc)}</span>
          <span class="badge" style="background:var(--primary-light,#e3f2fd);color:var(--primary);font-siz
e:12px">${pts.length} ${tr('Patients', '????')}</span>
        </h4>
        <div style="display:flex;flex-direction:column;gap:10px">
          ${pts.map((p, i) => {
            const acuity = getAcuityStyle(p.triage_level);
            return `
              <div style="padding:14px;border-radius:12px;background:${acuity.bg};border:1px solid ${acuit
y.border};border-left:5px solid ${acuity.color};display:flex;flex-direction:column;gap:8px">
                <div style="display:flex;justify-content:space-between;align-items:center">
                  <span style="font-weight:bold;font-size:15px;color:var(--text,#333)">
                    ${i + 1}. ${escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}
                  </span>
                  <span style="font-size:12px;font-weight:bold;color:${acuity.color};padding:2px 8px;borde
r-radius:20px;background:white;border:1px solid ${acuity.color}">
                    ${isArabic ? acuity.name_ar : acuity.name_en}
                  </span>
                </div>
                <div style="display:flex;justify-content:space-between;align-items:center;font-size:12px;c
olor:#666">
                  <span>? MRN: <strong>${escapeHTML(String(p.file_number || p.patient_id))}</strong></span>
                  <span>? ${tr('Room:', '????:')} <strong>${escapeHTML(p.exam_room_id || tr('Not assigned'
, '?? ?????'))}</strong></span>
                  <span>?? ${tr('Status:', '?????:')} <strong style="color:var(--primary)">${escapeHTML(p
.status)}</strong></span>
                </div>

                ${p.acuity_notes ? `<div style="font-size:12px;background:rgba(255,255,255,0.6);padding:6p
x 10px;border-radius:6px;color:#555;border:1px dashed ${acuity.border}"> ${escapeHTML(p.acuity_notes)}</div>`
: ''}

                <div style="display:flex;gap:6px;margin-top:6px;flex-wrap:wrap">
                  <button class="btn btn-sm" onclick="callPatientNotification(${p.id})" style="background:
#e3f2fd;color:#1565c0;flex:1;min-width:70px">? ${tr('Call', '????')}</button>
                  <button class="btn btn-sm" onclick="triagePatientModal(${p.id}, ${p.triage_level}, '${js
Str(p.acuity_notes || '')}', '${jsStr(p.exam_room_id || '')}')" style="background:#fff3e0;color:#e65100;flex:1

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;min-width:70px">? ${tr('Triage', '???')}</button>

    ${p.status === 'CheckedIn' || p.status === 'WaitingForProvider' ? `
        <button class="btn btn-sm btn-primary" onclick="updateQueueStatus(${p.id}, 'InConsulta
tion')" style="flex:1.5;min-width:100px">? ${tr('Consult', '???? ???????')}</button>
        ` : ''}

    ${p.status === 'InConsultation' ? `
        <button class="btn btn-sm" onclick="updateQueueStatus(${p.id}, 'WaitingForResults')" s
tyle="background:#f3e5f5;color:#8e24aa;flex:1.2;min-width:80px">?? ${tr('Hold', '????? ???????')}</button>
        <button class="btn btn-sm btn-success" onclick="updateQueueStatus(${p.id}, 'ReadyForDi
scharge')" style="background:#e8f5e9;color:#2e7d32;flex:1.2;min-width:80px">? ${tr('Discharge', '????? ?????')
}</button>
        ` : ''}

    ${p.status === 'WaitingForResults' ? `
        <button class="btn btn-sm btn-primary" onclick="updateQueueStatus(${p.id}, 'InConsulta
tion')" style="flex:1.5;min-width:100px">? ${tr('Resume', '??????')}</button>
        ` : ''}
    </div>
</div>
`;
}).join('')}
</div>
</div>
`;
}).join('')}
${Object.keys(byDoctor).length === 0 ? `
    <div class="card glass-card-premium" style="padding:60px;text-align:center;color:#999;grid-column:1/-1
">

        <div style="font-size:48px;margin-bottom:12px">?</div>
        <h3>${tr('No patients in waiting queue', '?? ???? ???? ?? ????? ????????? ??????')}</h3>
    </div>
    ` : ''}
</div>
`;

// Auto-refresh every 15 seconds
if (window._queueInterval) clearInterval(window._queueInterval);
window._queueInterval = setInterval(() => {
    if (currentPage === NAV_ITEMS.findIndex(n => n.en === 'Waiting Queue')) navigateTo(currentPage);
}, 15000);
}

window.triagePatientModal = async (queueId, currentLevel, currentNotes, currentRoom) => {
    let activeRooms = [];
    try {
        activeRooms = await API.get('/api/settings/rooms/active').catch(() => []);
    } catch (e) { console.error(e); }

    const modal = document.createElement('div');
    modal.id = 'triageModal';
    modal.className = 'modal-backdrop';
    modal.innerHTML = `

```

```

<div class="card glass-card-premium" style="max-width:450px;margin:10% auto;padding:24px;position:relative
">
    <h3 style="margin:0 0 16px;color:var(--primary)">? ${tr('Clinical Triage & Prioritization', '????? ?????
?????? ?????????')}</h3>

    <div class="form-group" style="margin-bottom:16px">
        <label>${tr('Acuity / Triage Level (ESI)', '????? ?????? ?????? ????????? (ESI)')}</label>
        <select id="triageLevelSelect" class="form-input">
            <option value="1" ${currentLevel == 1 ? 'selected' : ''}>? ESI-1: Resuscitation (????? ???)</option>
            <option value="2" ${currentLevel == 2 ? 'selected' : ''}>? ESI-2: Emergent (???? ?????)</option>
            <option value="3" ${currentLevel == 3 ? 'selected' : ''}>? ESI-3: Urgent (????)</option>
            <option value="4" ${currentLevel == 4 ? 'selected' : ''}>? ESI-4: Less Urgent (??? ?????)</option>
            <option value="5" ${currentLevel == 5 ? 'selected' : ''}>? ESI-5: Non-Urgent (??? ?????)</option>
        </select>
    </div>

    <div class="form-group" style="margin-bottom:16px">
        <label>${tr('Assign Exam Room / Clinic', '????? ?????? ?????? / ??????')}</label>
        <select id="triageRoomSelect" class="form-input">
            <option value="">-- ${tr('Select Clinic/Room', '???? ???????/??????')} --</option>
            ${activeRooms.map(r => `
                <option value="${escapeHTML(r.room_number)}" ${currentRoom === r.room_number ? 'selected' : ''}>
                    ? ${escapeHTML(r.room_number)} - ${escapeHTML(isArabic ? (r.name_ar || r.name_en) : (r.name_en |
| r.name_ar))}
                </option>
            `).join('')}
        </select>
    </div>

    <div class="form-group" style="margin-bottom:20px">
        <label>${tr('Triage / Acuity Notes', '???????? ?????? ?????? ??????')}</label>
        <textarea id="triageNotesArea" class="form-input" style="height:80px" placeholder="${tr('Describe pati
ent condition, pain score, or vitals...', '???? ??? ??????? ?????? ?????? ?? ????????? ?????????...')}">${escapeHT
ML(currentNotes || '')}</textarea>
    </div>

    <div style="display:flex;gap:12px">
        <button class="btn btn-secondary" style="flex:1" onclick="document.getElementById('triageModal').remov
e()">${tr('Cancel', '????')}</button>
        <button class="btn btn-primary" style="flex:2" onclick="saveTriageData(${queueId})">${tr('Save & Route
to Doctor', '??? ?????? ??????')}</button>
    </div>
</div>
`;
document.body.appendChild(modal);
};

window.saveTriageData = async (queueId) => {
    const triage_level = parseInt(document.getElementById('triageLevelSelect').value, 10);
    const acuity_notes = document.getElementById('triageNotesArea').value;
    const exam_room_id = document.getElementById('triageRoomSelect').value;
    try {
        await API.put(`/api/queue/patients/${queueId}/triage`, { triage_level, acuity_notes, exam_room_id });
        showToast(tr('Triage assessment updated!', '?? ?????? ?????? ?????? ?????? ?????? ??????'));
    }

```

```

    document.getElementById('triageModal').remove();
    navigateTo(currentPage);
  } catch (e) {
    showToast(e.message || tr('Error saving triage', '??? ?? ??? ?????'), 'error');
  }
};

window.callPatientNotification = async (queueId) => {
  try {
    const res = await API.put(`/api/queue/patients/${queueId}/call`);
    showToast(tr('Patient calling signal sent!', '?? ????? ????? ?????? ??????!') + ' ?');

    if ('speechSynthesis' in window) {
      window.speechSynthesis.cancel();
      const pName = res.patient_name || '';
      const pRoom = res.room || '';
      const text = isArabic
        ? `????? ? ???? ${pName} ????? ? ???? ???? ${pRoom}`
        : `Please, patient ${pName} proceed to exam room ${pRoom}`;
      const utterance = new SpeechSynthesisUtterance(text);
      utterance.lang = isArabic ? 'ar-SA' : 'en-US';
      utterance.rate = 0.9;
      window.speechSynthesis.speak(utterance);
    }
  } catch (e) {
    showToast(e.message || tr('Error calling patient', '??? ?? ?????? ??????'), 'error');
  }
};

window.updateQueueStatus = async (queueId, status) => {
  try {
    await API.put(`/api/queue/patients/${queueId}/status`, { status });
    showToast(tr('Queue status updated!', '?? ????? ???? ?????? ?? ??????!!'));
    navigateTo(currentPage);
  } catch (e) {
    showToast(e.message || tr('Error updating queue status', '??? ?? ????? ??????'), 'error');
  }
};

window.callPatient = async function (id) {
  await API.put('/api/patients/' + id, { status: 'With Doctor' });
  showToast(tr('Patient called', '?? ?????? ??????'));
  navigateTo(12);
};

async function renderPatientAccounts(el) {
  const content = el;

  content.innerHTML = `
    <h2>${tr('Patient Accounts', '????? ??????')}</h2>
    <div style="display:flex;gap:12px;margin-bottom:16px;flex-wrap:wrap">
      <input class="form-input" id="paSearch" placeholder="${tr('Search patient (name/MRN)...', '??? ?????? ??
      ??? ??????...')}" style="max-width:300px" oninput="searchPatientAccounts()">
  `;

```

```

        <button class="btn btn-sm" onclick="exportToCSV(window._paData||[], 'patient_accounts')" style="background-color: #e0f7fa; color: #00838f">? ${tr('Export', '????')}</button>
    </div>
    <div id="paResults"></div>`;

window.searchPatientAccounts = async () => {
    const search = document.getElementById('paSearch')?.value || '';
    const patients = await API.get('/api/patients?search=' + encodeURIComponent(search));
    const el = document.getElementById('paResults');
    if (!el) return;
    if (!patients.length) { el.innerHTML = '<p style="color: #999; text-align: center; padding: 40px">' + tr('No patients found', '?? ?? ???? ????') + '</p>'; return; }

    let invoiceData = [];
    try { invoiceData = await API.get('/api/invoices'); } catch (e) { }

    window._paData = patients.map(p => {
        const pInvoices = invoiceData.filter(i => i.patient_id === p.id);
        const totalBilled = pInvoices.reduce((s, i) => s + parseFloat(i.total || 0), 0);
        const totalPaid = pInvoices.filter(i => i.paid).reduce((s, i) => s + parseFloat(i.total || 0), 0);
        return { ...p, total_billed: totalBilled.toFixed(2), total_paid: totalPaid.toFixed(2), balance: (totalBilled - totalPaid).toFixed(2), invoice_count: pInvoices.length };
    });

    createTable(el, 'paTable',
        [tr('MRN', '????'), tr('Name', '????'), tr('Phone', '????'), tr('Invoices', '????'), tr('Billed', '????'), tr('Paid', '????'), tr('Balance', '????')],
        window._paData.map(p => ({
            cells: [p.mrn || p.file_number, isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar), p.phone, p.invoice_count,
                p.total_billed + ' ' + tr('SAR', '????'), p.total_paid + ' ' + tr('SAR', '????'),
                rawHtml('<span style="color: ' + (parseFloat(p.balance) > 0 ? '#cc0000; font-weight: bold' : '#2e7d32') + '>' + escapeHTML(p.balance) + ' ' + tr('SAR', '????') + '</span>']],
            id: p.id
        }))),
        (row) => `<button class="btn btn-sm" onclick="viewPatientInvoices(${safeId(row.id)})" style="background-color: #e3f2fd; color: #1565c0">? ${tr('Invoices', '????')}</button>`
    );
};

await window.searchPatientAccounts();

window.viewPatientInvoices = async (pid) => {
    const invoices = await API.get('/api/invoices?patient_id=' + pid);
    const modal = document.createElement('div');
    modal.style.cssText = 'position: fixed; inset: 0; background: rgba(0, 0, 0, 0.5); z-index: 9999; display: flex; align-items: center; justify-content: center';
    let rows = invoices.map(i => '<tr><td style="padding: 6px; border-bottom: 1px solid #eee">' + escapeHTML(i.invoice_number || i.id) + '</td><td style="padding: 6px; border-bottom: 1px solid #eee">' + escapeHTML(i.description || i.service_type || '') + '</td><td style="padding: 6px; border-bottom: 1px solid #eee">' + parseFloat(i.total || 0).toFixed(2) + '</td><td style="padding: 6px; border-bottom: 1px solid #eee"><span style="color: ' + (i.paid ? '#2e7d32' : '#cc0000') + '>' + (i.paid ? tr('Paid', '????') : tr('Unpaid', '???')) + '</span></td></tr>').join('');
    modal.innerHTML = '<div style="background: #fff; border-radius: 16px; padding: 24px; width: 600px; direction: rtl; margin: 0 auto">';
};

```

```

ax-height:80vh;overflow:auto"><h3 style="margin:0 0 16px">' + tr('Patient Invoices', '????? ?????') + '</h3>
<table style="width:100%"><thead><tr style="background:#f5f5f5"><th style="padding:8px;text-align:right">#</th>
<th style="padding:8px;text-align:right">' + tr('Service', '?????') + '</th><th style="padding:8px;text-align:right">' + tr('Amount', '?????') + '</th><th style="padding:8px;text-align:right">' + tr('Status', '?????')
) + '</th></tr></thead><tbody>' + rows + '</tbody></table><button class="btn btn-secondary" onclick="this.parentElement.parentElement.remove()" style="margin-top:16px;width:100%">' + tr('Close', '?????') + '</button></div>';

document.body.appendChild(modal);
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };
};

}

window.loadPatientAccount = async () => {
const pid = document.getElementById('paPatient').value;
if (!pid) return;
try {
const data = await API.get(`/api/billing/summary/${pid}`);
const pInfo = await API.get(`/api/patients/${pid}/account`);
const p = pInfo.patient;
// Build billing breakdown by service type
let breakdownHtml = '';
const typeIcons = { 'File Opening': '?', 'Lab Test': '?', 'Radiology': '?', 'Consultation': '?', 'Pharmacy': '?', 'Appointment': '?', 'Medical Services': '?', 'Other': '?' };
const typeNameNames = { 'File Opening': tr('File Opening', '??? ???'), 'Lab Test': tr('Lab Tests', '????? ???'), 'Radiology': tr('Radiology', '?????'), 'Consultation': tr('Consultation', '???????'), 'Pharmacy': tr('Pharmacy/Drugs', '???????/???????'), 'Appointment': tr('Appointments', '???????'), 'Medical Services': tr('Medical Services', '????? ???') };
for (const [type, info] of Object.entries(data.byType)) {
breakdownHtml += `<div style="display:flex;justify-content:space-between;align-items:center;padding:10px 12px;background:var(--hover);border-radius:8px;margin:6px 0">
<span>${typeIcons[type] || '?'} <strong>${escapeHTML(typeNames[type] || type)}</strong> <span class="badge badge-info" style="font-size:11px">${escapeHTML(info.count)}</span></span>
<span style="font-weight:600">${info.total.toLocaleString()} $<tr('SAR', '?.?')</span>
</div>`;
}
document.getElementById('paResult').innerHTML = `
<div class="card mb-16">
<div class="card-title">? ${escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))} - #${escapeHTML(p.file_number)}</div>
<div class="stats-grid">
<div class="stat-card" style="--stat-color:#3b82f6"><div class="stat-label">${tr('Total Billed', '????? ?????')}</div><div class="stat-value">${data.totalBilled.toLocaleString()} SAR</div></div>
<div class="stat-card" style="--stat-color:#4ade80"><div class="stat-label">${tr('Total Paid', '?????')}</div><div class="stat-value">${data.totalPaid.toLocaleString()} SAR</div></div>
<div class="stat-card" style="--stat-color:${data.balance > 0 ? '#f87171' : '#4ade80'}"><div class="stat-label">${tr('Balance Due', '???????')}</div><div class="stat-value">${data.balance.toLocaleString()} SAR</div></div>
</div>
</div>
<div class="card mb-16">
<div class="card-title">? ${tr('Billing Breakdown', '????? ?????')}</div>
${breakdownHtml || `<div class="empty-state"><p>${tr('No billing data', '?? ???? ?????')}</p></div>`}
}
</div>

```

```

<div class="card mb-16"><div class="card-title">? ${tr('All Invoices', '???? ?????')} (${data.invoices.length})</div>
    ${makeTable([tr('Type', '????'), tr('Description', '????'), tr('Amount', '????'), tr('Status', '??
????'), tr('Date', '?????'), tr('Actions', '?????')],
    data.invoices.map(i => ({ cells: [i.service_type || '', i.description || '', `${i.total} SAR`, i.paid ?
badge(tr('Paid', '????'), 'success') : badge(tr('Unpaid', '?? ?????'), 'danger'), i.created_at?.split('T')[0
] || ''], id: i.id, paid: i.paid })),
    (row) => !row.paid ? `
        <button class="btn btn-sm btn-success" onclick="payInvoicePA(${safeId(row.id)})">? ${tr('Cash', '???')
}</button>
        <button class="btn btn-sm btn-info" onclick="payInvoiceMoyasar(${safeId(row.id)})">? ${tr('Moyasar', '
????')}</button>
        ` : `<span class="badge badge-success">?</span>`
    )}</div>
    <div style="display:flex;gap:8px;margin-top:12px">
        <button class="btn btn-primary" onclick="printPatientStatement(${safeId(pid)})">?? ${tr('Print State
ment', '????? ?? ?????')}</button>
        <button class="btn" onclick="exportTableCSV('patient_account')">? ${tr('Export CSV', '????? CSV')}</
button>
    </div>`;
    } catch (e) { showToast(tr('Error loading account', '??? ?? ????? ?????'), 'error'); }
};
window.payInvoicePA = async (id) => {
    try { await API.put(`/api/invoices/${id}/pay`, { payment_method: 'Cash' }); showToast(tr('Paid!', '?? ?????!
')); loadPatientAccount(); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.payInvoiceMoyasar = async (id) => {
    try {
        showToast(tr('Initiating payment...', '???? ????? ????? ?????...'));
        const res = await API.post('/api/payments/moyasar/initiate', { invoiceId: id });
        if (res && res.id) {
            const width = 500, height = 600;
            const left = (screen.width - width) / 2;
            const top = (screen.height - height) / 2;
            const win = window.open('/api/payments/moyasar/callback?payment_id=' + res.id, 'MoyasarPayment', `width=
${width},height=${height},top=${top},left=${left}`);
            const timer = setInterval(() => {
                if (!win || win.closed) {
                    clearInterval(timer);
                    showToast(tr('Payment completed or window closed', '?? ????? ?? ?? ????? ????? ?????'));
                    loadPatientAccount();
                }
            }, 1000);
        } else {
            showToast(tr('Failed to initiate payment', '??? ?? ?? ????? ?????'), 'error');
        }
    } catch (e) {
        showToast(tr('Error', '???'), 'error');
    }
};
async function renderReports(el) {
    const content = el;

```

```

content.innerHTML = `
  <div class="flex justify-between items-end mb-6">
    <div>
      <h2 class="font-headline-lg text-headline-lg text-primary">? ${tr('Reports', '????????')}</h2>
      <p class="text-on-surface-variant text-sm mt-1">${tr('Generate patient, financial, clinical, and pharm
acy audit reports', '????? ?????? ???????? ????????? ????????? ????????? ????????? ?????????')}</p>
    </div>
  </div>
  <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-md mb-6">
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('patients')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Patient Report', '????? ?????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('All registered patients', '???? ?????? ?????????')}</p>
    >
  </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('invoices')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Financial Report', '??????? ?????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Revenue and invoices', '????????? ?????????')}</p>
    </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('appointments')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Appointments Report', '????? ?????????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Bookings and attendance', '????????? ?????????')}</p>
    </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('lab')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Lab Report', '????? ?????????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Test orders and results', '????????? ?????????')}</p>
    </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('pharmacy')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Pharmacy Report', '????? ?????????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Dispensing and stock', '????? ?????????')}</p>
    </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('inventory')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Inventory Report', '????? ?????????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Stock levels and low items', '????????? ?????????')}</p>
    </div>
    <div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('radiology')">
      <div class="text-4xl mb-3">?</div>
      <h4 class="font-title-lg text-primary mb-1">${tr('Radiology Report', '????? ?????')}</h4>
      <p class="text-xs text-on-surface-variant">${tr('Imaging orders and results', '????? ?????? ?????????')}</p>
  </div>
</div>

```

```

<div class="card glass-card-premium p-6 cursor-pointer flex flex-col items-center text-center transition
-all duration-300" onclick="genReport('medical_history')">
  <div class="text-4xl mb-3">?</div>
  <h4 class="font-title-lg text-primary mb-1">${tr('Medical History', '????? ?????')}</h4>
  <p class="text-xs text-on-surface-variant">${tr('Previous reports, tests & prescriptions', '????????? ?
????????? ?????????? ?????????')}</p>
</div>
</div>
<div id="reportOutput" class="card glass-card-premium p-6 mt-6" style="display:none">
  <div class="flex justify-between items-center mb-4 flex-wrap gap-2">
    <h4 id="reportTitle" class="font-title-lg text-primary margin-0"></h4>
    <div class="flex gap-2">
      <button class="btn btn-secondary btn-sm flex items-center gap-1" onclick="exportToCSV(window._report
Data||[],'report')">? ${tr('Export CSV', '????? CSV')}</button>
      <button class="btn btn-primary btn-sm flex items-center gap-1" onclick="window.print()">?? ${tr('Pri
nt', '?????')}</button>
    </div>
  </div>
  <div id="reportTable"></div>
</div>`;

```

```

window.genReport = async (type) => {
  const output = document.getElementById('reportOutput');
  const title = document.getElementById('reportTitle');
  const table = document.getElementById('reportTable');
  if (!output || !table) return;
  output.style.display = '';
  // Show premium skeletons instantly
  table.innerHTML = `
    <div class="skeleton-bar title mb-16"></div>
    <div class="skeleton-bar mb-12"></div>
    <div class="skeleton-bar mb-12"></div>
    <div class="skeleton-bar mb-12"></div>
  `;

  try {
    let data, headers, rows;
    switch (type) {
      case 'patients':
        data = await API.get('/api/patients');
        title.textContent = tr('Patient Report', '????? ?????') + ' (' + data.length + ')';
        headers = [tr('MRN', '?????'), tr('Name', '?????'), tr('Phone', '??????'), tr('ID', '??????'), tr('N
ationality', '??????'), tr('Registered', '??????')];
        rows = data.map(p => ({ cells: [p.mrn || p.file_number, isArabic ? (p.name_ar || p.name_en) : (p.nam
e_en || p.name_ar), p.phone, p.national_id, p.nationality, p.created_at ? new Date(p.created_at).toLocaleDates
tring('ar-SA') : ''], id: p.id }));
        break;
      case 'invoices':
        data = await API.get('/api/invoices');
        title.textContent = tr('Financial Report', '??????? ?????') + ' (' + data.length + ')';
        headers = [tr('#', '#'), tr('Patient', '??????'), tr('Service', '??????'), tr('Amount', '??????'), t
r('Paid', '?????'), tr('Date', '??????')];
        rows = data.map(i => ({ cells: [i.invoice_number || i.id, i.patient_name, i.description || i.service
_type, parseFloat(i.total || 0).toFixed(2), i.paid ? '?' : '?', i.created_at ? new Date(i.created_at).toLocale

```



```

DateString('ar-SA') : ''], id: i.id }));
    break;
    case 'appointments':
        data = await API.get('/api/appointments');
        title.textContent = tr('Appointments Report', '????? ?????????') + ' (' + data.length + ')';
        headers = [tr('Patient', '??????'), tr('Doctor', '??????'), tr('Department', '?????'), tr('Date', '?
??????'), tr('Time', '?????'), tr('Status', '??????')];
        rows = data.map(a => ({ cells: [a.patient_name, a.doctor_name || a.doctor, a.department, a.appt_date
|| a.date, a.appt_time || a.time, statusBadge(a.status)], id: a.id }));
        break;
    case 'lab':
        data = await API.get('/api/lab/orders');
        title.textContent = tr('Lab Report', '????? ?????????') + ' (' + data.length + ')';
        headers = [tr('Patient', '??????'), tr('Test', '?????'), tr('Doctor', '??????'), tr('Status', '?????
?'), tr('Date', '????????')];
        rows = data.map(l => ({ cells: [l.patient_name, l.test_name || l.test_type, l.doctor, statusBadge(l
.status), l.created_at ? new Date(l.created_at).toLocaleDateString('ar-SA') : '', id: l.id }));
        break;
    case 'pharmacy':
        data = await API.get('/api/pharmacy/prescriptions');
        title.textContent = tr('Pharmacy Report', '????? ?????????') + ' (' + data.length + ')';
        headers = [tr('Patient', '??????'), tr('Medication', '??????'), tr('Doctor', '??????'), tr('Status',
'??????'), tr('Date', '????????')];
        rows = data.map(p => ({ cells: [p.patient_name, p.medication || p.drug_name, p.doctor, statusBadge(p
.status), p.created_at ? new Date(p.created_at).toLocaleDateString('ar-SA') : '', id: p.id }));
        break;
    case 'inventory':
        data = await API.get('/api/inventory');
        title.textContent = tr('Inventory Report', '????? ?????????') + ' (' + data.length + ')';
        headers = [tr('Item', '?????'), tr('Category', '?????'), tr('Qty', '??????'), tr('Reorder', '????? ?
????'), tr('Unit', '??????'), tr('Status', '??????')];
        rows = data.map(i => ({ cells: [i.name, i.category, i.quantity, i.reorder_level || 10, i.unit, parse
Int(i.quantity) <= parseInt(i.reorder_level || 10) ? rawHtml('<span style="color:#cc0000;font-weight:bold">??
' + tr('Low', '?????') + '</span>') : rawHtml('<span style="color:#2e7d32">? ' + tr('OK', '???') + '</span>')]
, id: i.id }));
        break;
    case 'radiology':
        data = await API.get('/api/lab/orders');
        data = data.filter(o => o.order_type === 'Radiology' || o.test_type === 'Radiology');
        title.textContent = tr('Radiology Report', '????? ?????????') + ' (' + data.length + ')';
        headers = [tr('Patient', '??????'), tr('Exam', '?????'), tr('Doctor', '?????? ?????????'), tr('Status',
'??????'), tr('Date', '????????')];
        rows = data.map(r => ({ cells: [r.patient_name, r.test_name || r.test_type, r.doctor || '?', statusB
adge(r.status), r.created_at ? new Date(r.created_at).toLocaleDateString('ar-SA') : '', id: r.id }));
        break;
    case 'medical_history':
        const [labH, rxH, apptH] = await Promise.all([
            API.get('/api/lab/orders').catch(() => []),
            API.get('/api/pharmacy/prescriptions').catch(() => []),
            API.get('/api/appointments').catch(() => [])
        ]);
        // Merge all into single timeline
        const allRecords = [];
        labH.forEach(l => allRecords.push({ type: '? ' + tr('Lab', '?????'), patient: l.patient_name, detail

```

```

: l.test_name || l.test_type, doctor: l.doctor || '?', status: l.status, date: l.created_at }));
    rxH.forEach(p => allRecords.push({ type: '?' + tr('Pharmacy', '?????'), patient: p.patient_name, d
etail: p.medication || p.drug_name, doctor: p.doctor || '?', status: p.status, date: p.created_at }));
    apptH.forEach(a => allRecords.push({ type: '?' + tr('Visit', '?????'), patient: a.patient_name, det
ail: a.department || a.speciality || '', doctor: a.doctor_name || a.doctor || '?', status: a.status, date: a.a
ppt_date || a.created_at }));
    allRecords.sort((a, b) => new Date(b.date || 0) - new Date(a.date || 0));
    data = allRecords;
    title.textContent = tr('Medical History', '????? ?????') + ' (' + data.length + ')';
    headers = [tr('Type', '?????'), tr('Patient', '???????'), tr('Detail', '?????????'), tr('Doctor', '???
????'), tr('Status', '???????'), tr('Date', '????????')];
    rows = allRecords.map((r, i) => ({ cells: [r.type, r.patient, r.detail, r.doctor, statusBadge(r.stat
us), r.date ? new Date(r.date).toLocaleDateString('ar-SA') : ''], id: i }));
    break;
}
window._reportData = data;
if (!data || data.length === 0) {
    table.innerHTML = `
        <div class="empty-state-card">
            <div class="empty-state-icon">?</div>
            <div>${tr('No report data found', '?? ???? ?????? ????? ???? ?????????')}</div>
        </div>
    `;
} else {
    createTable(table, 'rptTbl', headers, rows);
    // Wrap table for responsiveness
    const tblEl = table.querySelector('table');
    if (tblEl) {
        const wrapper = document.createElement('div');
        wrapper.className = 'table-wrapper';
        tblEl.parentNode.insertBefore(wrapper, tblEl);
        wrapper.appendChild(tblEl);
    }
}
} catch (e) {
    table.innerHTML = `
        <div class="error-card-premium">
            <div class="error-card-icon">?</div>
            <h3>${tr('Failed to load report data', '??? ?? ?????? ?????? ?????????')}</h3>
            <p>${escapeHTML(e.message || e)}</p>
            <button class="btn btn-primary" onclick="genReport('${jsStr(type)}')">? ${tr('Retry', '????? ???????
?')}</button>
        </div>
    `;
}
};

let msgTab = 'inbox';
async function renderMessaging(el) {
    const content = el;

    const messages = await API.get('/api/messages').catch(() => []);

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const users = await API.get('/api/users').catch(() => []);
const myId = window.currentUser?.id;
const inbox = messages.filter(m => m.to_user_id == myId);
const sent = messages.filter(m => m.from_user_id == myId);
const unread = inbox.filter(m => !m.read_at).length;

content.innerHTML = `
  <h2>${tr('Messaging', '???????')}</h2>
  <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:16px">
    <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1565c0">${inbox.length}</h3><p style="margin:4px 0;font-size:12px">${tr('Inbox', '???????')}</p></div>
    <div class="card" style="padding:16px;text-align:center;background:${unread > 0 ? '#fce4ec' : '#e8f5e9'}"><h3 style="margin:0;color:${unread > 0 ? '#c62828' : '#2e7d32'}">${unread}</h3><p style="margin:4px 0;font-size:12px">${tr('Unread', '??? ?????')}</p></div>
    <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${sent.length}</h3><p style="margin:4px 0;font-size:12px">${tr('Sent', '???????')}</p></div>
  </div>
  <div style="display:flex;gap:8px;margin-bottom:16px">
    <button class="btn btn-primary" onclick="showComposeModal()">?? ${tr('Compose', '????? ?????')}</button>
    <button class="btn btn-sm msgTab" onclick="showMsgTab('inbox')" style="background:#1a73e8;color:#fff">${tr('Inbox', '???????')}</button>
    <button class="btn btn-sm msgTab" onclick="showMsgTab('sent')" style="background:#e0e0e0;color:#333">${tr('Sent', '???????')}</button>
  </div>
  <div id="msgInbox" class="card" style="padding:16px"></div>
  <div id="msgSent" class="card" style="padding:16px;display:none"></div>`;

// Render inbox
const ib = document.getElementById('msgInbox');
if (ib) { ib.innerHTML = inbox.length ? inbox.map(m => '<div style="padding:12px;margin:4px 0;border-radius:8px;background:' + (m.read_at ? '#fff' : '#e3f2fd') + ';cursor:pointer;border-left:4px solid ' + (m.read_at ? '#ccc' : '#1a73e8') + '" onclick="readMsg(' + safeId(m.id) + ')"><div style="display:flex;justify-content:space-between"><strong>' + escapeHTML(m.from_name || tr('System', '???????')) + '</strong><span style="font-size:11px;color:#666">' + (m.created_at ? new Date(m.created_at).toLocaleString('ar-SA') : '') + '</span></div><p style="margin:4px 0;font-size:13px;color:#666">' + escapeHTML((m.subject || m.content || '').substring(0, 80)) + '</p></div>').join('') : '<p style="text-align:center;color:#999">' + tr('No messages', '?? ???? ?????') + '</p>'; }

const st = document.getElementById('msgSent');
if (st) { st.innerHTML = sent.length ? sent.map(m => '<div style="padding:12px;margin:4px 0;border-radius:8px;background:#f5f5f5;border-left:4px solid #4caf50"><div style="display:flex;justify-content:space-between"><strong>' + escapeHTML(m.to_name || '') + '</strong><span style="font-size:11px;color:#666">' + (m.created_at ? new Date(m.created_at).toLocaleString('ar-SA') : '') + '</span></div><p style="margin:4px 0;font-size:13px;color:#666">' + escapeHTML((m.subject || m.content || '').substring(0, 80)) + '</p></div>').join('') : '<p style="text-align:center;color:#999">' + tr('No sent messages', '?? ???? ????? ?????') + '</p>'; }

window.showMsgTab = (tab) => { document.getElementById('msgInbox').style.display = tab === 'inbox' ? '' : 'none'; document.getElementById('msgSent').style.display = tab === 'sent' ? '' : 'none'; };

window.readMsg = async (id) => { try { await API.put('/api/messages/' + id + '/read', {}); const m = messages.find(x => x.id === id); if (m) { const modal = document.createElement('div'); modal.className = 'modal-overlay'; modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center'; modal.innerHTML = '<div style="background:#fff;border-radius:16px;padding:24px;width:500px;max-height:80vh;overflow:auto"><h3 style="margin:0 0 8px">' + escapeHTML(m.subject || tr('M

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message', '?????')) + '</h3><p style="color:#666;font-size:12px;margin:0 0 16px">' + tr('From', '??') + ': ' +
escapeHTML(m.from_name || '') + ' ? ' + (m.created_at ? new Date(m.created_at).toLocaleString('ar-SA') : '') +
'</p><div style="padding:12px;background:#f5f5f5;border-radius:8px;white-space:pre-wrap">' + escapeHTML(m.cont
ent || '') + '</div><button class="btn btn-secondary" onclick="this.parentElement.parentElement.remove()" styl
e="width:100%;margin-top:16px">' + tr('Close', '?????') + '</button></div>'; document.body.appendChild(modal);
modal.onclick = e => { if (e.target === modal) modal.remove(); } } } catch (e) { } };
window.showComposeModal = () => {
    const modal = document.createElement('div'); modal.className = 'modal-overlay'; modal.style.cssText = 'pos
ition:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:ce
nter';
    modal.innerHTML = '<div style="background:#fff;border-radius:16px;padding:24px;width:500px"><h3 style="mar
gin:0 0 16px">?? ' + tr('New Message', '????? ?????') + '</h3><div class="form-group"><label>' + tr('To', '???
') + '</label><select class="form-input" id="msgTo">' + users.map(u => '<option value="' + escapeHTML(u.id) +
'">' + escapeHTML(u.display_name) + '</option>').join('') + '</select></div><div class="form-group"><label>' +
tr('Subject', '???????') + '</label><input class="form-input" id="msgSubject"></div><div class="form-group"><
label>' + tr('Message', '???????') + '</label><textarea class="form-input" id="msgContent" rows="4"></textarea
></div><div style="display:flex;gap:8px"><button class="btn btn-primary" onclick="sendMessage()" style="flex:1
">? ' + tr('Send', '?????') + '</button><button class="btn btn-secondary" onclick="this.parentElement.parentEl
ement.parentElement.remove()" style="flex:1">' + tr('Cancel', '?????') + '</button></div></div>';
    document.body.appendChild(modal); modal.onclick = e => { if (e.target === modal) modal.remove(); };
};
window.sendMessage = async () => { try { await API.post('/api/messages', { to_user_id: document.getElementById
Id('msgTo').value, subject: document.getElementById('msgSubject').value, content: document.getElementById('msg
Content').value }); document.querySelector('.modal-overlay')?.remove(); showToast(tr('Message sent!', '?? ????
? ???????!')); navigateTo(currentPage); } catch (e) { showToast(tr('Error', '???'), 'error'); } };
}
window.sendMsg = async function () {
    const receiver_id = document.getElementById('msgTo').value;
    const subject = document.getElementById('msgSubject').value;
    const body = document.getElementById('msgBody').value;
    const priority = document.getElementById('msgPriority').value;
    if (!subject) return showToast(tr('Subject required', '??????? ?????'), 'error');
    await API.post('/api/messages', { receiver_id, subject, body, priority });
    showToast(tr('Message sent', '?? ???????!')); msgTab = 'sent'; navigateTo(15);
};
window.markRead = async function (id) { await API.put('/api/messages/' + id + '/read', {}); navigateTo(15); };
window.deleteMsg = async function (id) { if (confirm(tr('Delete?', '????'))) { await API.delete('/api/messages
/' + id); navigateTo(15); } };

// ===== SETTINGS =====
let settingsUsersList = [];
let editingUserId = null;

async function loadDashboardCharts() {
    try {
        if (typeof Chart === 'undefined') return;
        const data = await API.get('/api/dashboard/charts');

        // Revenue trend line chart
        const revCtx = document.getElementById('revenueChart');
        if (revCtx && data.revenueTrend) {

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const existingChart = Chart.getChart(revCtx);
if (existingChart) existingChart.destroy();

new Chart(revCtx, {
  type: 'line',
  data: {
    labels: data.revenueTrend.map(d => new Date(d.day).toLocaleDateString('ar-SA', { day: 'numeric', month: 'short' })),
    datasets: [{
      label: isArabic ? '????????' : 'Revenue',
      data: data.revenueTrend.map(d => parseFloat(d.revenue)),
      borderColor: '#006970',
      backgroundColor: 'rgba(0,105,112,0.06)',
      fill: true, tension: 0.4, pointRadius: 3
    }]
  },
  options: { responsive: true, maintainAspectRatio: false, plugins: { legend: { display: false } }, scales: { y: { beginAtZero: true } } }
});
}

// Department pie chart
const deptCtx = document.getElementById('deptChart');
if (deptCtx && data.byDepartment && data.byDepartment.length > 0) {
  const existingChart = Chart.getChart(deptCtx);
  if (existingChart) existingChart.destroy();

  const colors = ['#001629', '#006970', '#ac8c44', '#ba1a1a', '#22c55e', '#8b5cf6', '#ec4899', '#06b6d4'];
  new Chart(deptCtx, {
    type: 'doughnut',
    data: {
      labels: data.byDepartment.map(d => d.dept),
      datasets: [{ data: data.byDepartment.map(d => parseInt(d.count)), backgroundColor: colors.slice(0, data.byDepartment.length) }]
    },
    options: { responsive: true, maintainAspectRatio: false, plugins: { legend: { display: false } } }
  });
}

// Payment methods pie
const payCtx = document.getElementById('paymentChart');
if (payCtx && data.paymentMethods && data.paymentMethods.length > 0) {
  const existingChart = Chart.getChart(payCtx);
  if (existingChart) existingChart.destroy();

  const colors = ['#006970', '#001629', '#ac8c44', '#ba1a1a', '#22c55e'];
  new Chart(payCtx, {
    type: 'pie',
    data: {
      labels: data.paymentMethods.map(d => d.method),
      datasets: [{ data: data.paymentMethods.map(d => parseFloat(d.total)), backgroundColor: colors.slice(0, data.paymentMethods.length) }]
    },
    options: { responsive: true, maintainAspectRatio: false, plugins: { legend: { display: false } } }
  });
}

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    });
}

} catch (e) { console.error('Charts load error:', e); }
}

let settingsTab = 'hospital';
async function renderSettings(el) {
    const content = el;

    const role = getCurrentUserRole();
    const isAdmin = role === 'Admin';
    const permissions = getCurrentUserPermissions();
    const canReadAudit = isAdmin || permissions.includes('audit.read') || permissions.includes('46');

    // Show premium loading skeletons instantly
    content.innerHTML = `
        <div class="flex justify-between items-end mb-6">
            <div>
                <h2 class="font-headline-lg text-headline-lg text-primary">${tr('System Settings & Governance', '????
                ?? ?????? ?????????')}</h2>
                <p class="text-on-surface-variant text-sm mt-1">${tr('Configure hospital records, manage users, and mo
                nitor cybersecurity compliance', '????? ?????? ?????????? ?????? ?????????? ?????????? ?????? ??????????')}</p>
            </div>
        </div>
        <div class="flex gap-4 mb-6 border-b border-outline-variant/30 pb-2">
            <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'hospital' ? 'bg-primary
            text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}">
                ? ${tr('Hospital Settings', '??????? ?????????')}
            </button>
            <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'users' ? 'bg-primary tex
            t-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}">
                ? ${tr('User Manager', '????? ?????????????')}
            </button>
            <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'cybersecurity' ? 'bg-pri
            mary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}">
                ?? ${tr('Cybersecurity & Governance', '????? ?????????? ?????????')}
            </button>
        </div>
        <div class="glass-card-premium p-6">
            <div class="skeleton-bar title mb-16"></div>
            <div class="skeleton-bar mb-12"></div>
            <div class="skeleton-bar mb-12"></div>
        </div>
    `;

    const getActionBadge = (act) => {
        const action = act.toUpperCase();
        if (action.includes('CREATE') || action.includes('INSERT') || action.includes('ADD')) return `<span class=
        "badge badge-success">${escapeHTML(act)}</span>`;
        if (action.includes('UPDATE') || action.includes('EDIT') || action.includes('SAVE')) return `<span class="
        badge badge-warning">${escapeHTML(act)}</span>`;
        if (action.includes('DELETE') || action.includes('REMOVE') || action.includes('DESTROY')) return `<span cl

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ass="badge badge-danger">${escapeHTML(act)}</span>`;
    if (action.includes('LOGIN') || action.includes('AUTH')) return `<span class="badge badge-info">${escapeHT
ML(act)}</span>`;
    return `<span class="badge">${escapeHTML(act)}</span>`;
};

let backupInfo = { database: 'nama_medical_web', totalSizeMB: '0.00', backupCommand: '', tables: [] };
let backupList = [];
let auditLogs = [];

try {
    if (settingsTab === 'users') {
        settingsUsersList = await API.get('/api/settings/users');
    } else if (settingsTab === 'cybersecurity') {
        [backupInfo, backupList, auditLogs] = await Promise.all([
            API.get('/api/admin/backup-info'),
            API.get('/api/admin/backups'),
            API.get('/api/admin/audit-trail?limit=15')
        ]);
    } else if (settingsTab === 'compliance') {
        window.integrationsList = await API.get('/api/settings/integrations').catch(() => []);
    } else if (settingsTab === 'audit') {
        auditLogs = await API.get('/api/admin/audit-trail?limit=1000').catch(() => []);
        window._fullAuditLogs = auditLogs;
    }
} catch (e) {
    content.innerHTML = `
        <div class="flex justify-between items-end mb-6">
            <div>
                <h2 class="font-headline-lg text-headline-lg text-primary">${tr('System Settings & Governance', '???
???? ?????? ?????????')}</h2>
            </div>
        </div>
        <div class="error-card-premium">
            <div class="error-card-icon">??</div>
            <h3>${tr('Failed to load settings data', '??? ?? ?????? ?????? ?????????')}</h3>
            <p>${escapeHTML(e.message || e)}</p>
            <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
button>
        </div>
    `;
    return;
}

content.innerHTML = `
    <div class="flex justify-between items-end mb-6">
        <div>
            <h2 class="font-headline-lg text-headline-lg text-primary">${tr('System Settings & Governance', '?????
?? ?????? ?????????')}</h2>
            <p class="text-on-surface-variant text-sm mt-1">${tr('Configure hospital records, manage users, and mo
nitor cybersecurity compliance', '????? ?????? ?????????? ?????? ?????????? ?????????? ?????????? ?????? ?????????')}</p>
        </div>
    </div>

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<!-- Tabs Navigation -->
<div class="flex gap-4 mb-6 border-b border-outline-variant/30 pb-2">
  <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'hospital' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="settingsTab='hospital';navigateTo(42)">
    ? ${tr('Hospital Settings', '?????? ??????')}
  </button>
  <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'users' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="settingsTab='users';navigateTo(42)">
    ? ${tr('User Manager', '???? ?????????')}
  </button>
  <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'cybersecurity' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="settingsTab='cybersecurity';navigateTo(42)">
    ?? ${tr('Cybersecurity & Governance', '???? ????????? ?????????')}
  </button>
  <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'compliance' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="settingsTab='compliance';navigateTo(42)">
    ?? ${tr('Saudi Compliance & Integrations', '???????? ????????? ?????????')}
  </button>
  ${canReadAudit ? `
    <button class="px-4 py-2 font-bold rounded-lg transition-all ${settingsTab === 'audit' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="settingsTab='audit';navigateTo(42)">
      ? ${tr('Audit Trail Viewer', '???? ??? ?????????')}
    </button>
  ` : ''}
</div>

<div class="tab-pane-content">
  ${settingsTab === 'hospital' ? `
    <div class="grid grid-cols-1 lg:grid-cols-2 gap-md">
      <div class="glass-card glass-card-premium p-6 rounded-2xl">
        <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Hospital Information', '???????? ?????????')}</h4>
        <div class="form-group mb-4">
          <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Hospital Name (AR)', '??? ????????? (????)')}</label>
          <input class="form-input w-full" id="sNameAr" value="???? ??????">
        </div>
        <div class="form-group mb-4">
          <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Hospital Name (EN)', '??? ????????? (?????)')}</label>
          <input class="form-input w-full" id="sNameEn" value="jumanaMedical">
        </div>
        <div class="form-group mb-4">
          <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Phone', '?????')}</label>
          <input class="form-input w-full" id="sPhone" value="0112345678">
        </div>
        <div class="form-group mb-4">
          <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Email', '????? ??????')}

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????')}</label>
    <input class="form-input w-full" id="sEmail" value="info@namamedical.com">
</div>
<div class="form-group mb-4">
    <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Address', '??????')}<
/label>
    <textarea class="form-input w-full" id="sAddr" rows="2">?????? ???? ???? ????</textare
ea>
</div>
<div class="grid grid-cols-2 gap-4 mb-4">
    <div class="form-group">
        <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('CR Number', '???? ?
?????')}</label>
        <input class="form-input w-full" id="sCr" value="1010000234">
        </div>
        <div class="form-group">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('VAT Number', '????
?????')}</label>
            <input class="form-input w-full" id="sTax" value="300012345600003">
            </div>
        </div>
        <button class="btn btn-primary w-full py-2.5 rounded-xl font-bold flex items-center justify-center
gap-2" onclick="saveSettings()">
            ? ${tr('Save Changes', '?? ?')}</button>
    </div>
    <div class="space-y-md">
        <div class="glass-card glass-card-premium p-6 rounded-2xl">
            <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Appearance', '????')}</h4>
            <div class="form-group mb-4">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Language', '????')}<
/label>
                <select class="form-input w-full" onchange="setLang(this.value)">
                    <option value="en" ${!isArabic ? 'selected' : ''}>English</option>
                    <option value="ar" ${isArabic ? 'selected' : ''}>????</option>
                </select>
            </div>
            <div class="form-group">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Theme Select', '????
??')}</label>
                <select class="form-input w-full" id="setTheme" onchange="if(typeof changeTheme==='function')c
hangeTheme(this.value)">
                    <option value="light-blue">${tr('Light Blue', '????')}</option>
                    <option value="dark">${tr('Dark', '????')}</option>
                    <option value="green">${tr('Green', '????')}</option>
                </select>
            </div>
        </div>
        <div class="glass-card glass-card-premium p-6 rounded-2xl">
            <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('System Notifications', '????
?')}</h4>
            <div class="space-y-3">
                <label class="flex items-center gap-3 text-sm text-on-surface-variant cursor-pointer">
                    <input type="checkbox" checked class="rounded border-outline-variant text-primary focus:ring
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-primary">
    ${tr('Alert when lab results are ready', '????? ??? ?????? ????? ?????????')}
</label>
<label class="flex items-center gap-3 text-sm text-on-surface-variant cursor-pointer">
    <input type="checkbox" checked class="rounded border-outline-variant text-primary focus:ring
-primary">

    ${tr('Alert for new appointments booked', '????? ??? ????????? ?????????')}
</label>
<label class="flex items-center gap-3 text-sm text-on-surface-variant cursor-pointer">
    <input type="checkbox" checked class="rounded border-outline-variant text-primary focus:ring
-primary">

    ${tr('Alert for low inventory stock level', '????? ??? ????????? ????? ?????????????')}
</label>
<label class="flex items-center gap-3 text-sm text-on-surface-variant cursor-pointer">
    <input type="checkbox" class="rounded border-outline-variant text-primary focus:ring-primary
">

    ${tr('Enable system email logs dispatch', '????? ?????? ?????? ?????? ?????????')}
</label>
</div>
</div>
</div>
</div>
` : settingsTab === 'users' ? `
<div class="grid grid-cols-1 lg:grid-cols-3 gap-md">
    <!-- User Manager Form -->
    <div class="glass-card glass-card-premium p-6 rounded-2xl h-fit">
        <h4 class="font-title-lg text-title-lg text-primary mb-4" id="suFormTitle">? ${tr('User Details',
'??????? ?????????')}</h4>
        <div class="form-group mb-3">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Full Display Name', '?
???? ?????????')}</label>
            <input class="form-input w-full" id="suName" placeholder="e.g. ?. ???? ??">
        </div>
        <div class="form-group mb-3">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Username', '??? ??????
???')}</label>
            <input class="form-input w-full" id="suUser" placeholder="e.g. dr_mohammed">
        </div>
        <div class="form-group mb-3">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Password', '???? ??????
?')}</label>
            <input type="password" class="form-input w-full" id="suPass" placeholder="?????????">
        </div>
        <div class="form-group mb-3">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('System Role', '??? ???
???')}</label>
            <select class="form-input w-full" id="suRole" onchange="toggleSuRoleFields(this.value)">
                <option value="Reception">${tr('Receptionist', '???? ?????????')}</option>
                <option value="Doctor">${tr('Doctor', '????')}</option>
                <option value="Nurse">${tr('Nurse', '????/?')}</option>
                <option value="Pharmacist">${tr('Pharmacist', '????')}</option>
                <option value="Lab Technician">${tr('Lab Tech', '??? ?????')}</option>
                <option value="Radiologist">${tr('Radiologist', '??? ?????')}</option>
                <option value="Finance">${tr('Finance', '????? ?????????')}</option>

```

```

        <option value="HR">${tr('HR', '????? ?????')}</option>
        <option value="Admin">${tr('Administrator', '???? ?????')}</option>
    </select>
</div>
<div id="suSpecDiv" class="form-group mb-3" style="display:none">
    <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Medical Speciality', '
?????? ?????')}</label>
    <input class="form-input w-full" id="suSpec" placeholder="e.g. Cardiology">
</div>
<div id="suCommDiv" class="mb-3" style="display:none">
    <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Commission Rules', '??
??? ?????')}</label>
    <div class="grid grid-cols-2 gap-2">
        <select class="form-input" id="suCommType">
            <option value="percentage">${tr('Percentage %', '???? ????? %')}</option>
            <option value="fixed">${tr('Fixed Amount', '???? ?????')}</option>
        </select>
        <input type="number" class="form-input" id="suCommValue" value="0" step="0.1">
    </div>
</div>
<div class="form-group mb-4">
    <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Permissions', '???????
???????')}</label>
    <div id="suPerms" style="display:grid;grid-template-columns:1fr;gap:6px;max-height:160px;overflo
w-y:auto;border:1px solid rgba(0,0,0,0.1);padding:8px;border-radius:8px">
        ${NAV_ITEMS.map((item, i) => `
            <label style="display:flex;align-items:center;gap:6px;font-size:12px" class="cursor-pointer
hover:bg-surface-container-low p-1 rounded">
                <input type="checkbox" id="perm_${i}" value="${i}" checked class="rounded border-outline-v
ariant text-primary focus:ring-primary">
                <span>${item.icon}</span> <span>${tr(item.en, item.ar)}</span>
            </label>
        `).join('')}
    </div>
</div>
<div class="flex gap-2">
    <button class="btn btn-primary flex-1 py-2.5 rounded-xl font-bold" id="suAddBtn" onclick="addOrU
pdateUser()">
        ? ${tr('Save User', '??? ?????????')}
    </button>
    <button class="btn btn-secondary py-2.5 rounded-xl font-bold" id="suCancelBtn" onclick="cancelEd
itUser()" style="display:none">
        ${tr('Cancel', '????')}
    </button>
</div>
</div>

<!-- User Registry Table -->
<div class="glass-card glass-card-premium p-6 rounded-2xl lg:col-span-2 overflow-x-auto">
    <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('User Accounts Directory', '??? ?
????????? ?????')}</h4>
    <table class="w-full text-right divide-y divide-outline-variant/30">
        <thead class="bg-surface-container-low font-label-md text-label-md text-on-surface-variant">
            <tr>

```

```

        <th class="p-3">${tr('Name', '????')}</th>
        <th class="p-3">${tr('Username', '????????')}</th>
        <th class="p-3">${tr('Role', '????')}</th>
        <th class="p-3">${tr('Specialty', '????')}</th>
        <th class="p-3 text-center">${tr('Actions', '????????')}</th>
    </tr>
</thead>
<tbody class="font-body-md text-body-md divide-y divide-outline-variant/10">
    ${settingsUsersList.map(u => `
        <tr class="hover:bg-primary/5 transition-colors">
            <td class="p-3 font-bold text-primary">${escapeHTML(u.display_name || '?')}</td>
            <td class="p-3 font-mono text-sm">${escapeHTML(u.username)}</td>
            <td class="p-3"><span class="px-2 py-0.5 rounded text-xs bg-primary-container text-on-primary-fixed-variant font-bold">${escapeHTML(u.role)}</span></td>
            <td class="p-3 text-on-surface-variant">${escapeHTML(u.speciality || '?')}</td>
            <td class="p-3 text-center">
                <div class="flex justify-center gap-2">
                    <button class="btn btn-sm bg-secondary/10 text-secondary hover:bg-secondary/20 p-2 rounded-lg" onclick="editUser(${safeId(u.id)})">??</button>
                    <button class="btn btn-sm bg-error/10 text-error hover:bg-error/20 p-2 rounded-lg" onclick="deleteUser(${safeId(u.id)})">??</button>
                </div>
            </td>
        </tr>
    `).join('')}
</tbody>
</table>
</div>
</div>
` : settingsTab === 'cybersecurity' ? `
<!-- Cybersecurity Governance -->
<div class="space-y-md">
    <!-- Staging Warning banner inside tab -->
    <div class="p-4 rounded-xl border border-amber-300 bg-amber-50 text-amber-900 text-xs md:text-sm mb-4 flex items-center gap-2">
        <span class="material-symbols-outlined text-sm">warning</span>
        <span><strong>????? ?????? ??????????:</strong> ???? ??? ?????? ?????? ????? Staging ?????? ??? ?????
        ????? HTTP ??? ??????. ??? ?????? ?????? ?????? ?? ?????? ?? ?????? ???? ?????? ?? ?????? ??????</span>
    </div>
    <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-md">
        <!-- Threat Level Circular Gauge -->
        <div class="glass-card glass-card-premium p-6 rounded-2xl flex flex-col items-center justify-center min-h-[260px]">
            <h3 class="font-title-lg text-title-lg text-primary mb-4 w-full text-right">?? ${tr('Current Cybersecurity Threat Level', '????? ???????? ?????????? ????????')}</h3>
            <div class="relative w-40 h-40 flex items-center justify-center">
                <svg class="w-full h-full transform -rotate-90">
                    <circle class="text-surface-container-highest" cx="80" cy="80" fill="transparent" r="68" stroke="currentColor" stroke-width="10"></circle>
                    <circle class="text-secondary" cx="80" cy="80" fill="transparent" r="68" stroke="currentColor" stroke-dasharray="427.2" stroke-dashoffset="299.0" stroke-width="10"></circle>
                </svg>
                <div class="absolute inset-0 flex flex-col items-center justify-center">
                    <span class="font-display-lg text-3xl font-bold text-primary leading-none">30%</span>
                </div>
            </div>
        </div>
    </div>
`

```

```

        <span class="font-label-md text-label-md text-secondary mt-1">${tr('Low / Secure', '???? /
????')}</span>
    </div>
</div>
    <p class="text-[11px] text-on-surface-variant mt-4 text-center">${tr('Firewall rules active. Sin
gle session enforcement enabled.', '???? ?????? ???. ??? ?????? ??????? ??????? ?????'.')}</p>
</div>

<!-- Encryption & Session Controls -->
<div class="glass-card glass-card-premium p-6 rounded-2xl flex flex-col justify-between">
    <div>
        <h3 class="font-title-lg text-title-lg text-primary mb-4 flex items-center gap-2">? ${tr('Data
Encryption Status', '???? ?????? ?????????')}</h3>
        <div class="space-y-3">
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-on-surface-variant font-bold">${tr('Patients Database', '???? ?
???? ?????')}</span>
                <span class="text-xs font-bold text-green-600">AES-256 (Active)</span>
            </div>
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-on-surface-variant font-bold">${tr('Radiology Logs', '???? ????
??')}</span>
                <span class="text-xs font-bold text-green-600">RSA-4096 (Active)</span>
            </div>
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-on-surface-variant font-bold">${tr('Cloud Backup Transmission',
'?? ????? ?????????? ??????')}</span>
                <span class="text-xs font-bold text-secondary">SSL/TLS 1.3</span>
            </div>
        </div>
    </div>
    <div class="pt-4 border-t border-outline-variant/30 flex justify-between items-center">
        <span class="text-xs text-on-surface-variant font-medium">${tr('Session Enforcement', '?????
????? ?????????')}</span>
        <span class="px-2 py-0.5 rounded bg-green-100 text-green-700 text-xs font-bold">${tr('Complianc
e', '????')}</span>
    </div>
</div>

<!-- Compliance Authority Status -->
<div class="glass-card glass-card-premium p-6 rounded-2xl flex flex-col justify-between">
    <div>
        <h3 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('National Authority Complianc
e', '????????? ?????? ?????????? ?????????')}</h3>
        <div class="space-y-3">
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-on-surface-variant font-bold">???? ?????? ?????????? ????????? (PDPL)
            </div>
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-green-700 font-bold">????? 100%</span>
            </div>
            <div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
                <span class="text-xs text-on-surface-variant font-bold">?????? ?????? ?????????? ????????? (SDA
IA)</span>
                <span class="text-xs text-secondary font-bold">??? ?????????</span>
            </div>
        </div>
    </div>
</div>

```

```
</div>
<div class="flex justify-between items-center p-2 bg-surface-container-low rounded-xl">
  <span class="text-xs text-on-surface-variant font-bold">???? ???? ???? ???? ????
?</span>

  <span class="text-xs text-green-700 font-bold">???? 100%</span>
</div>
</div>
<div class="pt-2 text-center">
  <p class="text-[10px] text-on-surface-variant">???? ???? ???? ???? ? ? ???? ????
? ???? ???? ????</p>
</div>
</div>
</div>

<!-- Backup Manager & Database Info -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-md">
  <div class="glass-card glass-card-premium p-6 rounded-2xl flex flex-col justify-between">
    <div>
      <h4 class="font-title-lg text-title-lg text-primary mb-3">? ${tr('Database Infrastructure', '?
??? ???? ????')}</h4>
      <div class="space-y-2 text-xs">
        <div class="flex justify-between p-1 border-b border-outline-variant/20"><span class="text-o
n-surface-variant">??? ???? ????</span><span class="font-bold text-primary">${escapeHTML(backupInfo.data
base)}</span></div>
        <div class="flex justify-between p-1 border-b border-outline-variant/20"><span class="text-o
n-surface-variant">???? ????</span><span class="font-bold text-primary">${escapeHTML(backupInfo.totalSiz
eMB)} MB</span></div>
        <div class="flex justify-between p-1 border-b border-outline-variant/20"><span class="text-o
n-surface-variant">???? ???? ????</span><span class="font-mono text-[10px] text-secondary">pg_dump</spa
n></div>
      </div>
    </div>
    <div class="mt-4">
      <button onclick="startBackup()" class="w-full btn btn-primary py-2.5 rounded-xl font-bold flex
items-center justify-center gap-2 shadow-md">
        ? ${tr('Run Database Backup Now', '???? ???? ????')}
      </button>
    </div>
  </div>
</div>

<!-- Existing Backups Registry -->
<div class="glass-card glass-card-premium p-6 rounded-2xl lg:col-span-2 overflow-x-auto">
  <h4 class="font-title-lg text-title-lg text-primary mb-3">? ${tr('Database Backup Registry', '?
? ???? ????')}</h4>
  <div class="max-h-48 overflow-y-auto">
    <table class="w-full text-right divide-y divide-outline-variant/30 text-xs">
      <thead class="bg-surface-container-low font-label-md text-on-surface-variant">
        <tr>
          <th class="p-2">${tr('Filename', '???')}</th>
          <th class="p-2">${tr('Size', '???')}</th>
          <th class="p-2">${tr('Date Created', '????')}</th>
        </tr>
      </thead>
    </table>
  </div>
</div>
```

```

        <tbody class="divide-y divide-outline-variant/10">
          ${backupList.length ? backupList.map(b => `
            <tr class="hover:bg-primary/5 transition-colors">
              <td class="p-2 font-mono text-[11px] text-primary">${escapeHTML(b.name)}</td>
              <td class="p-2 font-bold">${escapeHTML(b.size)}</td>
              <td class="p-2 text-on-surface-variant">${new Date(b.date).toLocaleString('ar-SA')}</td>
            </tr>
          `).join('')} : `<tr><td colspan="3"><div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No backups found', '?? ??? ???? ????')}</div></div></td></tr>`)
        </tbody>
      </table>
    </div>
  </div>
</div>

<!-- Administrative Security Audit Trail -->
<div class="glass-card glass-card-premium p-6 rounded-2xl overflow-x-auto">
  <h4 class="font-title-lg text-title-lg text-primary mb-4">${tr('Security & Governance Audit Trail', '??? ?????? ?????? ??????')}</h4>
  <div class="max-h-60 overflow-y-auto">
    <table class="w-full text-right divide-y divide-outline-variant/30 text-xs">
      <thead class="bg-surface-container-low font-label-md text-on-surface-variant">
        <tr>
          <th class="p-3">${tr('User', '????????')}</th>
          <th class="p-3">${tr('Module', '????')}</th>
          <th class="p-3">${tr('Action', '????')}</th>
          <th class="p-3">${tr('Context / Details', '???????? ?????')}</th>
          <th class="p-3">${tr('IP Address', '???? IP')}</th>
          <th class="p-3">${tr('Timestamp', '???? ?????')}</th>
        </tr>
      </thead>
      <tbody class="divide-y divide-outline-variant/10">
        ${auditLogs.length ? auditLogs.map(log => `
          <tr class="hover:bg-primary/5 transition-colors">
            <td class="p-3 font-bold text-primary">${escapeHTML(log.user_name || '?')}</td>
            <td class="p-3"><span class="px-2 py-0.5 rounded text-[10px] bg-secondary/10 text-secondary font-bold">${escapeHTML(log.module)}</span></td>
            <td class="p-3">${getActionBadge(log.action)}</td>
            <td class="p-3 text-on-surface-variant max-w-xs truncate" title="${escapeHTML(log.details || '')}">${escapeHTML(log.details || '?')}</td>
            <td class="p-3 font-mono text-[11px]">${escapeHTML(log.ip_address || '?')}</td>
            <td class="p-3 text-on-surface-variant">${new Date(log.created_at).toLocaleString('ar-SA')}</td>
          </tr>
        `).join('')} : `<tr><td colspan="6"><div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('No audit trail events logged', '?? ??? ???? ????')}</div></div></td></tr>`)
      </tbody>
    </table>
  </div>
</div>
</div>
` : settingsTab === 'compliance' ? `
  <div class="space-y-md">

```

```

<div class="glass-card glass-card-premium p-6 rounded-2xl">
  <h4 class="font-title-lg text-title-lg text-primary mb-4">?? ${tr('Saudi Compliance & Integration
Modules', '????? ???? ???? ???? ???? ????')}</h4>
  <p class="text-sm text-on-surface-variant mb-6">${tr('Configure local regulatory integration adapt
ers (ZATCA Phase 2, NPHIES unified billing, CBAHI quality indicators, and PDPL protection protocols).', '????
???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ????
??? ???? ???? ????').')}</p>

  <div class="grid grid-cols-1 md:grid-cols-2 gap-6">
    ${window.integrationsList || []}.map(item => `
      <div class="border border-outline-variant/40 rounded-xl p-5 bg-surface-container-low/40 relati
ve">
        <div class="flex justify-between items-start mb-4">
          <div>
            <span class="font-bold text-lg text-primary">${escapeHTML(item.integration_name)}</span>
            <span class="text-xs px-2 py-0.5 bg-surface-container-high rounded-full block text-on-su
rface-variant mt-1">${escapeHTML(item.provider)}</span>
          </div>
          <label class="relative inline-flex items-center cursor-pointer" style="margin-right: auto;
margin-left: 0;">
            <input type="checkbox" value="" class="sr-only peer" ${item.is_enabled === 1 ? 'checked'
: ''} onchange="toggleIntegration('${escapeHTML(item.integration_name)}', this.checked)">
            <div class="w-11 h-6 bg-outline-variant/50 peer-focus:outline-none rounded-full peer pee
r-checked:after:translate-x-full peer-checked:after:border-white after:content-[''] after:absolute after:top-[
2px] after:left-[2px] after:bg-white after:border-gray-300 after:border after:rounded-full after:height-5 afte
r:width-5 after:transition-all peer-checked:bg-primary font-bold"></div>
          </label>
        </div>

        <div class="space-y-3 font-body-sm text-sm">
          <div class="flex justify-between">
            <span class="text-on-surface-variant">${tr('Endpoint URL', '??? ???? ????')}</span>
            <span class="font-mono text-xs truncate max-w-[200px]" title="${escapeHTML(item.endpoint
_url)}">${escapeHTML(item.endpoint_url || '?')}</span>
          </div>
          <div class="flex justify-between">
            <span class="text-on-surface-variant">${tr('Last Sync', '?? ????')}</span>
            <span>${item.last_sync ? new Date(item.last_sync).toLocaleDateString('ar-SA') : tr('Neve
r', '?? ????')}</span>
          </div>
          <div class="flex justify-between items-center mt-1">
            <span class="text-on-surface-variant text-xs">${tr('Connection State', '??? ?????')}:
</span>
            <span id="ping_state_${escapeHTML(item.integration_name)}" class="px-2 py-0.5 rounded te
xt-[10px] bg-outline-variant/30 text-on-surface-variant font-bold">${tr('Not Checked', '?? ?? ????')}</span>
          </div>
          <div class="flex gap-2 mt-3">
            <button class="btn btn-sm btn-secondary flex-1" onclick="openIntegrationConfig('${escape
HTML(item.integration_name)}')">
              ?? ${tr('Configure', '??? ?????')}</button>
            <button class="btn btn-sm flex-1" style="background:var(--primary-container);color:var(-
on-primary-container)" onclick="pingIntegration('${escapeHTML(item.integration_name)}')">
              ? ${tr('Ping', '???')}</button>
          </div>
        </div>
      `
    )
  </div>

```



```

        </button>
      </div>
    </div>
  </div>
</div>
` : settingsTab === 'audit' ? `
  <div class="space-y-md">
    <div class="glass-card glass-card-premium p-6 rounded-2xl">
      <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Audit Trail Search & Export', '?
      </h4>
      <div style="display:flex;gap:12px;margin-bottom:16px;flex-wrap:wrap">
        <input class="form-input" id="auditSearchInput" placeholder="${tr('Search by user, action or det
        ails...', '?? ?????????? ?????? ?? ?????????...')}" style="max-width:300px" oninput="window.filterAuditLogs()"
        >
        <select class="form-input" id="auditModuleFilter" style="max-width:200px" onchange="window.filte
        rAuditLogs()">
          <option value="">${tr('All Modules', '?? ???????')}</option>
          ${[...new Set((window._fullAuditLogs || []).map(l => l.module))].map(mod => `<option value="${$
          escapeHTML(mod)}">${escapeHTML(mod)}</option>`).join('')}}
        </select>
        <button class="btn btn-sm" onclick="exportToCSV(window._filteredAuditLogs||window._fullAuditLogs
        ||[], 'system_audit_trail')" style="background:#e0f7fa;color:#00838f">? ${tr('Export to CSV', '????? ?? CSV')}
        </button>
      </div>
      <div class="overflow-x-auto max-h-[500px]">
        <table class="w-full text-right divide-y divide-outline-variant/30 text-xs" id="auditLogsTable">
          <thead class="bg-surface-container-low font-label-md text-on-surface-variant">
            <tr>
              <th class="p-3">${tr('User', '????????')}</th>
              <th class="p-3">${tr('Module', '????')}</th>
              <th class="p-3">${tr('Action', '????')}</th>
              <th class="p-3">${tr('Context / Details', '???????? ????????')}</th>
              <th class="p-3">${tr('IP Address', '???? IP')}</th>
              <th class="p-3">${tr('Timestamp', '????? ?????????')}</th>
            </tr>
          </thead>
          <tbody class="divide-y divide-outline-variant/10" id="auditLogsTableBody">
            <!-- Dynamic content -->
          </tbody>
        </table>
      </div>
    </div>
  </div>
  </div>
  ` : ''}
</div>
`;

// Global action to filter logs
window.filterAuditLogs = () => {
  const q = (document.getElementById('auditSearchInput')?.value || '').toLowerCase();

```

```

const mod = document.getElementById('auditModuleFilter')?.value || '';
const logs = window._fullAuditLogs || [];
const filtered = logs.filter(log => {
  const uMatch = (log.user_name || log.user_id || '').toLowerCase().includes(q);
  const aMatch = (log.action || '').toLowerCase().includes(q);
  const dMatch = (typeof log.details === 'object' ? JSON.stringify(log.details) : String(log.details || ''
)).toLowerCase().includes(q);
  const mMatch = !mod || log.module === mod;
  return (uMatch || aMatch || dMatch) && mMatch;
});
window._filteredAuditLogs = filtered;

const tbody = document.getElementById('auditLogsTableBody');
if (!tbody) return;
if (!filtered.length) {
  tbody.innerHTML = `<tr><td colspan="6" class="p-4 text-center text-on-surface-variant">${tr('No matching
logs found', '?? ??? ?????? ??? ?????? ??????')}</td></tr>`;
  return;
}

tbody.innerHTML = filtered.map(log => `
<tr class="hover:bg-primary/5 transition-colors">
  <td class="p-3 font-bold text-primary">${escapeHTML(log.user_name || log.user_id || '?')}</td>
  <td class="p-3"><span class="px-2 py-0.5 rounded text-[10px] bg-secondary/10 text-secondary font-bold"
>${escapeHTML(log.module)}</span></td>
  <td class="p-3">${getActionBadge(log.action)}</td>
  <td class="p-3 text-on-surface-variant max-w-xs truncate" title="${escapeHTML(typeof log.details === '
object' ? JSON.stringify(log.details) : String(log.details || ''))}">${escapeHTML(typeof log.details === 'obje
ct' ? JSON.stringify(log.details) : String(log.details || ''))}</td>
  <td class="p-3 font-mono text-[11px]">${escapeHTML(log.ip_address || '?')}</td>
  <td class="p-3 text-on-surface-variant">${new Date(log.created_at).toLocaleString('ar-SA')}</td>
</tr>
`).join('');
};

if (settingsTab === 'audit') {
  window.filterAuditLogs();
}
}

window.toggleIntegration = async (name, enabled) => {
  const item = (window.integrationsList || []).find(i => i.integration_name === name);
  if (!item) return;
  try {
    const body = {
      integration_name: name,
      provider: item.provider,
      endpoint_url: item.endpoint_url,
      is_enabled: enabled ? 1 : 0,
      config_json: item.config_json
    };
    body['api_key'] = item.api_key;
    body['api_secret'] = item.api_secret;

```

```

    await API.post('/api/settings/integrations', body);
    showToast(tr('Settings updated', '?? ????? ?????????'));
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

window.pingIntegration = async (name) => {
  const badge = document.getElementById(`ping_state_${name}`);
  if (badge) {
    badge.className = 'px-2 py-0.5 rounded text-[10px] bg-warning-container text-on-warning-container font-bol
d';
    badge.innerText = tr('Checking...', '???? ?????...');
  }
  try {
    const res = await API.post('/api/settings/integrations/ping', { integration_name: name });
    if (badge) {
      if (res.success) {
        badge.className = 'px-2 py-0.5 rounded text-[10px] bg-success text-white font-bold';
        badge.innerText = tr('Connected', '????');
      } else {
        badge.className = 'px-2 py-0.5 rounded text-[10px] bg-error text-white font-bold';
        badge.innerText = tr('Failed', '??? ??????');
        showToast(res.message || tr('Connection failed', '??? ??????'), 'error');
      }
    }
  } catch (e) {
    if (badge) {
      badge.className = 'px-2 py-0.5 rounded text-[10px] bg-error text-white font-bold';
      badge.innerText = tr('Error', '???');
    }
    showToast(e.message || tr('Connection error', '??? ?? ??????'), 'error');
  }
};

window.openIntegrationConfig = (name) => {
  const item = (window.integrationsList || []).find(i => i.integration_name === name) || { integration_name: n
ame, provider: '', api_key: '', api_secret: '', endpoint_url: '', is_enabled: 0, config_json: '{}' };
  const modal = document.createElement('div');
  modal.className = 'modal-overlay';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center';
  modal.innerHTML = '<div style="background:var(--bg-card,#fff);border-radius:16px;padding:24px;width:500px;ma
x-height:90vh;overflow-y:auto;direction:rtl">' +
    '<h3 style="margin:0 0 16px;color:var(--primary)">' + tr(`Configure ${name} Integration`, `??????? ???? ${n
ame}`) + '</h3>' +
    '<div class="form-group mb-3"><label class="block text-xs font-bold text-on-surface-variant mb-1">' + tr('
Provider Name', '??? ?????') + '</label>' +
    '<input class="form-input w-full" id="intProvider" value="' + (item.provider || '') + '"></div>' +
    '<div class="form-group mb-3"><label class="block text-xs font-bold text-on-surface-variant mb-1">' + tr('
Endpoint URL', '???? ?????') + '</label>' +
    '<input class="form-input w-full" id="intUrl" value="' + (item.endpoint_url || '') + '"></div>' +
    '<div class="form-group mb-3"><label class="block text-xs font-bold text-on-surface-variant mb-1">' + tr('

```

```

API Key / Client ID', '????? API') + '</label>' +
  '<input class="form-input w-full" id="intKey" value="' + (item.api_key || '') + '" type="password"></div>'
+
  '<div class="form-group mb-3"><label class="block text-xs font-bold text-on-surface-variant mb-1">' + tr('
API Secret / Client Secret', '???? ???????') + '</label>' +
  '<input class="form-input w-full" id="intSecret" value="' + (item.api_secret || '') + '" type="password"><
/div>' +
  '<div class="form-group mb-3"><label class="block text-xs font-bold text-on-surface-variant mb-1">' + tr('
Custom Configurations (JSON)', '??????? ?????? (JSON)') + '</label>' +
  '<textarea class="form-input w-full" id="intConfig" rows="3">' + (item.config_json || '{}') + '</textarea>
</div>' +
  '<div style="display:flex;gap:12px;margin-top:16px">' +
  '<button class="btn btn-primary" id="btnSaveInt" style="flex:1">' + tr('Save Settings', '??? ?????????')
+ '</button>' +
  '<button class="btn btn-secondary" onclick="this.closest(\'\.modal-overlay\').remove()" style="flex:1">' +
tr('Cancel', '????') + '</button>' +
  '</div></div>';
document.body.appendChild(modal);
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };

document.getElementById('btnSaveInt').onclick = async () => {
  try {
    const body = {
      integration_name: name,
      provider: document.getElementById('intProvider').value,
      endpoint_url: document.getElementById('intUrl').value,
      is_enabled: item.is_enabled,
      config_json: document.getElementById('intConfig').value
    };
    body['api_key'] = document.getElementById('intKey').value;
    body['api_secret'] = document.getElementById('intSecret').value;
    JSON.parse(body.config_json);

    await API.post('/api/settings/integrations', body);
    showToast(tr('Integration settings saved successfully', '?? ?? ??????? ????? ?????????'));
    modal.remove();
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error saving (check JSON format)', '??? ?? ????? (???? ?? ????? JSON)'), 'error');
  }
};
};

window.toggleSuRoleFields = (role) => {
  const isDoc = role === 'Doctor';
  const spec = document.getElementById('suSpecDiv');
  const comm = document.getElementById('suCommDiv');
  if (spec) spec.style.display = isDoc ? 'block' : 'none';
  if (comm) comm.style.display = isDoc ? 'block' : 'none';
};

window.startBackup = async () => {
  try {
    showToast(tr('Running database backup...', '???? ????? ????? ??????????'));
  }
};

```

```

const response = await fetch('/api/admin/backup', {
  method: 'POST',
  headers: {
    'Content-Type': 'application/json'
  }
});
if (!response.ok) throw new Error('Backup failed');
const blob = await response.blob();
const url = window.URL.createObjectURL(blob);
const a = document.createElement('a');
a.href = url;
const disposition = response.headers.get('Content-Disposition');
let filename = 'nama_backup.sql';
if (disposition && disposition.indexOf('attachment') !== -1) {
  const filenameRegex = /filename[^;=\n]*=((['"]).*?\2|[^;\n]*)/;
  const matches = filenameRegex.exec(disposition);
  if (matches != null && matches[1]) filename = matches[1].replace(/['"]/g, '');
}
a.download = filename;
document.body.appendChild(a);
a.click();
a.remove();
showToast(tr('Backup complete and downloaded!', '???? ???? ????????? ???? ??????!!'));
setTimeout(() => navigateTo(currentPage), 1000);
} catch (e) {
  showToast(tr('Backup failed', '??? ????? ?????????'), 'error');
}
};

window.saveSettings = async () => {
  try {
    await API.put('/api/settings', {
      company_name_ar: document.getElementById('sNameAr').value,
      company_name_en: document.getElementById('sNameEn').value,
      tax_number: document.getElementById('sTax').value,
      cr_number: document.getElementById('sCr').value,
      phone: document.getElementById('sPhone').value,
      address: document.getElementById('sAddr').value
    });
    showToast(tr('Settings saved!', '?? ?? ?????????!!'));
  } catch (e) { showToast(tr('Error saving', '??? ?? ??????'), 'error'); }
};

window.addOrUpdateUser = async () => {
  const username = document.getElementById('suUser').value.trim();
  const password = document.getElementById('suPass').value.trim();

  if (!username) { showToast(tr('Enter username', '???? ?????????'), 'error'); return; }
  if (!editingUserId && !password) { showToast(tr('Enter password for new user', '???? ???? ?????? ????????? ??
?????'), 'error'); return; }

  try {
    const role = document.getElementById('suRole').value;
    const spec = role === 'Doctor' ? document.getElementById('suSpec').value : '';

```

```

const perms = Array.from(document.querySelectorAll('#suPerms input:checked')).map(cb => cb.value).join(', '
);
const commType = role === 'Doctor' ? (document.getElementById('suCommType')?.value || 'percentage') : 'per
centage';
const commValue = role === 'Doctor' ? (parseFloat(document.getElementById('suCommValue')?.value) || 0) : 0
;

if (editingUserId) {
  await API.put(`/api/settings/users/${editingUserId}`, {
    username,
    password: password || undefined,
    display_name: document.getElementById('suName').value,
    role,
    speciality: spec,
    permissions: perms,
    commission_type: commType,
    commission_value: commValue,
    is_active: 1
  });
  showToast(tr('User updated!', '?? ????? ?????????!'));
} else {
  await API.post('/api/settings/users', {
    username,
    password,
    display_name: document.getElementById('suName').value,
    role,
    speciality: spec,
    permissions: perms,
    commission_type: commType,
    commission_value: commValue
  });
  showToast(tr('User added!', '?? ????? ?????????!'));
}

editingUserId = null;
await navigateTo(42);
} catch (e) { showToast(e.message || tr('Error saving user', '??? ?? ????? ?????'), 'error'); }
};

window.editUser = (id) => {
  const user = settingsUsersList.find(u => u.id === id);
  if (!user) return;
  editingUserId = id;
  document.getElementById('suUser').value = user.username || '';
  document.getElementById('suName').value = user.display_name || '';
  document.getElementById('suRole').value = user.role || 'Reception';
  document.getElementById('suPass').value = '';

  toggleSuRoleFields(user.role);

  if (user.role === 'Doctor') {
    document.getElementById('suSpec').value = user.speciality || 'General Clinic';
    document.getElementById('suCommType').value = user.commission_type || 'percentage';
    document.getElementById('suCommValue').value = user.commission_value || 0;
  }
}

```

```

}

document.querySelectorAll('#suPerms input').forEach(cb => cb.checked = false);
const perms = (user.permissions || '').split(',');
perms.forEach(p => {
  const cb = document.getElementById(`perm_${p}`);
  if (cb) cb.checked = true;
});

document.getElementById('suCancelBtn').style.display = 'inline-block';
document.getElementById('suAddBtn').innerHTML = `? ${tr('Update User', '????? ?????????')}`;
};

window.cancelEditUser = () => {
  editingUserId = null;
  document.getElementById('suUser').value = '';
  document.getElementById('suName').value = '';
  document.getElementById('suPass').value = '';
  document.getElementById('suRole').value = 'Reception';
  toggleSuRoleFields('Reception');
  document.querySelectorAll('#suPerms input').forEach(cb => cb.checked = true);
  document.getElementById('suCancelBtn').style.display = 'none';
  document.getElementById('suAddBtn').innerHTML = `? ${tr('Save User', '??? ?????????')}`;
};

window.deleteUser = async (id) => {
  if (!confirm(tr('Are you sure you want to delete this user? This cannot be undone.', '?? ??? ?????? ?? ??? ??
? ?????????? ??? ????????? ?? ????? ????????? ???.'))) return;
  try {
    await API.delete(`/api/settings/users/${id}`);
    showToast(tr('User deleted!', '?? ?????? ??????!'));
    await navigateTo(42);
  } catch (e) { showToast(e.message || tr('Error deleting', '??? ?? ??????'), 'error'); }
};

// ===== CATALOG MODULE =====
async function renderCatalog(el) {
  const [labTests, radExams, services] = await Promise.all([
    API.get('/api/catalog/lab'),
    API.get('/api/catalog/radiology'),
    API.get('/api/medical/services')
  ]);

  // Group lab tests by category
  const labGroups = {};
  labTests.forEach(t => { if (!labGroups[t.category]) labGroups[t.category] = []; labGroups[t.category].push(t)
}); });

  // Group radiology by modality
  const radGroups = {};
  radExams.forEach(r => { if (!radGroups[r.modality]) radGroups[r.modality] = []; radGroups[r.modality].push(r)
}); });

  // Group services by specialty then category

```

```

const svcGroups = {};
services.forEach(s => {
  if (!svcGroups[s.specialty]) svcGroups[s.specialty] = {};
  if (!svcGroups[s.specialty][s.category]) svcGroups[s.specialty][s.category] = [];
  svcGroups[s.specialty][s.category].push(s);
});

const specNames = {
  'General Practice': '???? ?????', 'Dentistry': '?? ??????', 'Internal Medicine': '?????????',
  'Cardiology': '?????', 'Dermatology': '????????', 'Ophthalmology': '?????',
  'ENT': '????? ??????', 'Orthopedics': '?????', 'Obstetrics': '?????? ?????????',
  'Pediatrics': '????????', 'Neurology': '????????', 'Psychiatry': '???? ??????',
  'Urology': '???????? ?????????', 'Endocrinology': '????? ??????', 'Gastroenterology': '?????? ??????',
  'Pulmonology': '????????', 'Nephrology': '?????', 'Surgery': '???????? ??????',
  'Oncology': '????????', 'Physiotherapy': '?????? ?????????', 'Nutrition': '????????',
  'Emergency': '????????'
};

const catIcons = { 'Consultation': '?', 'Procedure': '?', 'Diagnostic': '?', 'Therapy': '?', 'Service': '?'
};

el.innerHTML = `
<div class="page-title">? ${tr('Service Catalog', '???????? ?????????')}</div>
<div class="flex gap-8 mb-16">
  <button class="btn btn-primary" id="catTabLab" onclick="switchCatTab('lab')" style="flex:1">? ${tr('Lab
Tests', '?????? ?????????')} (${labTests.length})</button>
  <button class="btn btn-secondary" id="catTabRad" onclick="switchCatTab('rad')" style="flex:1">? ${tr('Ra
diology', '??????')} (${radExams.length})</button>
  <button class="btn btn-secondary" id="catTabSvc" onclick="switchCatTab('svc')" style="flex:1">? ${tr('Pr
ocedures', '????????? ??????')} (${services.length})</button>
</div>
<input class="form-input mb-12" id="catSearch" placeholder="${tr('Search...', '???...')}" oninput="filterC
atalog()">

<div id="catLabContent">
  ${Object.entries(labGroups).map(([cat, tests]) => `
    <div class="card mb-12 cat-item">
      <div class="card-title" style="cursor:pointer" onclick="this.nextElementSibling.style.display=this.n
extElementSibling.style.display==='none'?'block':'none'">
        ? ${escapeHTML(cat)} <span class="badge badge-info">${tests.length}</span> <span style="float:left
;font-size:12px;color:var(--text-dim)">?</span>
      </div>
      <div style="display:none">
        <table class="data-table"><thead><tr>
          <th style="width:40%">${tr('Test Name', '??? ?????')}</th>
          <th>${tr('Normal Range', '?????? ?????????')}</th>
          <th style="width:100px">${tr('Price', '?????')}</th>
          <th style="width:60px"></th>
        </tr></thead><tbody>
          ${tests.map(t => `<tr class="cat-row" data-name="${escapeHTML(t.test_name.toLowerCase())}">
            <td>${escapeHTML(t.test_name)}</td>
            <td style="font-size:11px;color:var(--text-dim)">${escapeHTML(t.normal_range || '-')}</td>
            <td><input type="number" class="form-input" value="${escapeHTML(t.price)}" id="labP${safeId(t.id
)}" style="width:80px;text-align:center;padding:4px 6px;font-size:12px"></td>

```



```

        <td><button class="btn btn-sm btn-success" onclick="saveCatPrice('lab',{safeId(t.id)})">?</button></td>

    </tr>`).join('')
  </tbody></table>
</div>
</div>
`).join('')
</div>

<div id="catRadContent" style="display:none">
  ${Object.entries(radGroups).map(([mod, exams]) => `
    <div class="card mb-12 cat-item">
      <div class="card-title" style="cursor:pointer" onclick="this.nextElementSibling.style.display=this.nextElementSibling.style.display==='none'?'block':'none'"
        ? ${escapeHTML(mod)} <span class="badge badge-info">${exams.length}</span> <span style="float:left;font-size:12px;color:var(--text-dim)">?</span>
      </div>
      <div style="display:none">
        <table class="data-table"><thead><tr>
          <th style="width:60%">${tr('Exam Name', '????')}</th>
          <th style="width:100px">${tr('Price', '????')}</th>
          <th style="width:60px"></th>
        </tr></thead><tbody>
          ${exams.map(r => `<tr class="cat-row" data-name="${escapeHTML(r.exact_name.toLowerCase())}">
            <td>${escapeHTML(r.exact_name)}</td>
            <td><input type="number" class="form-input" value="${escapeHTML(r.price)}" id="radP${safeId(r.id)}" style="width:80px;text-align:center;padding:4px 6px;font-size:12px"></td>
            <td><button class="btn btn-sm btn-success" onclick="saveCatPrice('rad',{safeId(r.id)})">?</button></td>
          </tr>`).join('')
          </tbody></table>
        </div>
      </div>
    `).join('')
  </div>

<div id="catSvcContent" style="display:none">
  <div class="flex gap-8 mb-12" style="flex-wrap:wrap" id="catSpecFilter">
    <button class="btn btn-sm btn-primary" onclick="filterSpec('all')">? ${tr('All', '????')}</button>
    ${Object.keys(svcGroups).map(s => `<button class="btn btn-sm btn-secondary" onclick="filterSpec('${jsStr(s)}')">${escapeHTML(specNames[s] || s)}</button>`).join('')}
  </div>
  ${Object.entries(svcGroups).map(([spec, cats]) => `
    <div class="spec-group" data-spec="${escapeHTML(spec)}">
      <div class="card mb-12">
        <div class="card-title" style="cursor:pointer;background:var(--hover);border-radius:8px;padding:12px" onclick="this.nextElementSibling.style.display=this.nextElementSibling.style.display==='none'?'block':'none'">
          ? ${escapeHTML(specNames[spec] || spec)} ? ${escapeHTML(spec)} <span class="badge badge-info">${Object.values(cats).flat().length}</span> <span style="float:left;font-size:12px;color:var(--text-dim)">?</span>
        </div>
        <div style="display:none">
          ${Object.entries(cats).map([cat, items]) => `

```

```

    <div style="margin:12px 0">
      <div style="font-weight:600;margin-bottom:8px;padding:6px 12px;background:var(--hover);border-radius:6px">${catIcons[cat] || '?'} ${escapeHTML(cat)} <span class="badge badge-info" style="font-size:10px">${items.length}</span></div>
      <table class="data-table"><thead><tr>
        <th>${tr('Procedure (EN)', '?????? (??????)')}</th>
        <th>${tr('Procedure (AR)', '?????? (????)')}</th>
        <th style="width:100px">${tr('Price', '????')}</th>
        <th style="width:60px"></th>
      </tr></thead><tbody>
        ${items.map(s => `<tr class="cat-row" data-name="${escapeHTML(s.name_en.toLowerCase())} ${escapeHTML(s.name_ar)}">
          <td style="font-size:12px">${escapeHTML(s.name_en)}</td>
          <td style="font-size:12px">${escapeHTML(s.name_ar)}</td>
          <td><input type="number" class="form-input" value="${escapeHTML(s.price)}" id="svcP${safeId(s.id)}" style="width:80px;text-align:center;padding:4px 6px;font-size:12px"></td>
          <td><button class="btn btn-sm btn-success" onclick="saveCatPrice('svc',${safeId(s.id)})">?</button></td>
        </tr>`).join('')}
      </tbody></table>
    </div>
  `).join('')}
</div>
</div>
</div>
</div>`;
```

```

window.switchCatTab = (tab) => {
  document.getElementById('catLabContent').style.display = tab === 'lab' ? 'block' : 'none';
  document.getElementById('catRadContent').style.display = tab === 'rad' ? 'block' : 'none';
  document.getElementById('catSvcContent').style.display = tab === 'svc' ? 'block' : 'none';
  document.getElementById('catTabLab').className = `btn ${tab === 'lab' ? 'btn-primary' : 'btn-secondary'}`;
  document.getElementById('catTabRad').className = `btn ${tab === 'rad' ? 'btn-primary' : 'btn-secondary'}`;
  document.getElementById('catTabSvc').className = `btn ${tab === 'svc' ? 'btn-primary' : 'btn-secondary'}`;
};
```

```

window.filterCatalog = () => {
  const q = document.getElementById('catSearch').value.toLowerCase();
  document.querySelectorAll('.cat-row').forEach(row => {
    row.style.display = row.dataset.name.includes(q) ? '' : 'none';
  });
};
```

```

window.filterSpec = (spec) => {
  document.querySelectorAll('.spec-group').forEach(g => {
    g.style.display = (spec === 'all' || g.dataset.spec === spec) ? 'block' : 'none';
  });
};
```

```

window.saveCatPrice = async (type, id) => {
  try {
    let url, price;
```

```

    if (type === 'lab') { url = `/api/catalog/lab/${id}`; price = parseFloat(document.getElementById(`labP${id}`
    `).value); }
    else if (type === 'rad') { url = `/api/catalog/radiology/${id}`; price = parseFloat(document.getElementById(`radP${id}`
    `).value); }
    else { url = `/api/medical/services/${id}`; price = parseFloat(document.getElementById(`svcP${id}`).value)
    ; }
    await API.put(url, { price });
    showToast(tr('Price saved!', '?? ??? ?????!'));
  } catch (e) { showToast(tr('Error saving', '??? ?? ?????'), 'error'); }
};

// ===== DEPARTMENT RESOURCE REQUESTS =====
async function renderDeptRequests(el) {
  const content = el;

  const requests = await API.get('/api/dept-requests').catch(() => []);
  const pending = requests.filter(r => r.status === 'pending').length;

  content.innerHTML = `
    <h2>${tr('Department Requests', '???? ???????')}</h2>
    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:1
    6px">
      <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1
    565c0">${requests.length}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Total', '????????')}</p></div>
      <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e
    65100">${pending}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Pending Approval', '?????? ???????')}</
    p></div>
    </div>
    <div style="display:grid;grid-template-columns:1fr 2fr;gap:16px">
      <div class="card" style="padding:20px">
        <h4 style="margin:0 0 12px">${tr('New Request', '??? ?????')}</h4>
        <div class="form-group"><label>${tr('Type', '????')}</label>
          <select class="form-input" id="drType"><option value="supplies">${tr('Supplies', '????????')}</optio
          n><option value="equipment">${tr('Equipment', '????')}</option><option value="maintenance">${tr('Maintenance'
          , '????')}</option><option value="staffing">${tr('Staffing', '????')}</option><option value="other">${tr('Ot
          her', '????')}</option></select></div>
          <div class="form-group"><label>${tr('Department', '????')}</label><input class="form-input" id="drDep
          t"></div>
          <div class="form-group"><label>${tr('Item/Description', '????/????')}</label><input class="form-inpu
          t" id="drItem"></div>
          <div class="form-group"><label>${tr('Quantity', '????')}</label><input type="number" class="form-inp
          ut" id="drQty" value="1"></div>
          <div class="form-group"><label>${tr('Priority', '????????')}</label>
            <select class="form-input" id="drPriority"><option value="low">${tr('Low', '????')}</option><optio
            n value="medium" selected>${tr('Medium', '????')}</option><option value="high">${tr('High', '????')}</optio
            n><option value="urgent">${tr('Urgent', '????')}</option></select></div>
            <div class="form-group"><label>${tr('Notes', '????????')}</label><textarea class="form-input" id="drNot
            es" rows="2"></textarea></div>
            <button class="btn btn-primary w-full" onclick="saveDeptReq()">? ${tr('Submit', '????')}</button>
          </div>
      <div class="card" style="padding:20px">
        <h4 style="margin:0 0 12px">${tr('Requests', '????????')}</h4>
        <div id="drTable"></div>
      </div>
    </div>
  `

```

```

</div>`;

const dt = document.getElementById('drTable');
if (dt) {
  createTable(dt, 'drTbl',
    [tr('Type', '?????'), tr('Dept', '?????'), tr('Item', '?????'), tr('Qty', '?????'), tr('Priority', '???
?????'), tr('Status', '?????'), tr('Date', '???????'), tr('Actions', '')],
    requests.map(r => ({ cells: [r.request_type || r.type || '', r.department || '', r.item || r.description
|| '', r.quantity || '', r.priority || '', statusBadge(r.status), r.created_at ? new Date(r.created_at).toLoc
aleDateString('ar-SA') : '', r.status === 'pending' ? rawHtml('<button class="btn btn-sm" style="background:#e
8f5e9;color:#2e7d32" onclick="approveDeptReq(' + parseInt(r.id, 10) + ')">? ' + tr('Approve', '?????') + '</b
utton> <button class="btn btn-sm" style="background:#fce4ec;color:#c62828" onclick="rejectDeptReq(' + parseInt
(r.id, 10) + ')">?</button>') : ''], id: r.id })))
  );
}

window.saveDeptReq = async () => { try { await API.post('/api/dept-requests', { request_type: document.getEl
ementById('drType').value, department: document.getElementById('drDept').value, item: document.getElementById(
'drItem').value, quantity: document.getElementById('drQty').value, priority: document.getElementById('drPriori
ty').value, notes: document.getElementById('drNotes').value, requested_by: window.currentUser?.display_name ||
'' }); showToast(tr('Request submitted!', '?? ????? ?????!')); navigateTo(currentPage); } catch (e) { showToa
st(tr('Error', '???'), 'error'); } };

window.approveDeptReq = async (id) => { try { await API.put('/api/dept-requests/' + id, { status: 'approved'
}); showToast('?'); navigateTo(currentPage); } catch (e) { } };

window.rejectDeptReq = async (id) => { try { await API.put('/api/dept-requests/' + id, { status: 'rejected'
}); showToast('?'); navigateTo(currentPage); } catch (e) { } };

}

let drqItems = [];
window.addDrqItem = () => {
  const sel = document.getElementById('drqItem');
  const itemId = parseInt(sel.value);
  const itemName = sel.options[sel.selectedIndex]?.text || '';
  const qty = parseInt(document.getElementById('drqQty').value) || 1;
  if (!itemId) return;
  if (drqItems.find(x => x.item_id === itemId)) { showToast(tr('Item already added', '????? ???? ??????'), 'er
ror'); return; }
  drqItems.push({ item_id: itemId, name: itemName, qty });
  renderDrqItems();
};

function renderDrqItems() {
  const c = document.getElementById('drqItemsList');
  if (!drqItems.length) { c.innerHTML = `<span style="color:var(--text-dim);font-size:13px">${tr('No items add
ed', '?? ??? ?????? ?????')}</span>`; return; }
  c.innerHTML = drqItems.map((item, i) => `<div style="display:flex;justify-content:space-between;align-items:
center;padding:8px 12px;background:var(--hover);border-radius:8px;margin:4px 0">
  <span>${escapeHTML(item.name)} × <strong>${escapeHTML(item.qty)}</strong></span>
  <button class="btn btn-danger btn-sm" onclick="drqItems.splice(${i},1);renderDrqItems()">?</button>
</div>`).join('');
}

window.submitDrq = async () => {
  if (!drqItems.length) { showToast(tr('Add items first', '??? ?????? ??????'), 'error'); return; }
  try {
    await API.post('/api/dept-requests', {
      department: document.getElementById('drqDept').value,

```

```

        requested_by: currentUser?.name || '',
        items: drqItems,
        notes: document.getElementById('drqNotes').value
    });
    showToast(tr('Request submitted!', '?? ????? ?????!'));
    drqItems = [];
    await navigateTo(17);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.approveDrq = async (id) => {
    try { await API.put(`/api/dept-requests/${id}`, { status: 'Approved' }); showToast(tr('Approved!', '?? ?????
????!')); await navigateTo(17); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.rejectDrq = async (id) => {
    try { await API.put(`/api/dept-requests/${id}`, { status: 'Rejected' }); showToast(tr('Rejected', '?? ?????'
)); await navigateTo(17); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== SURGERY & PRE-OP =====
let surgeryTab = 'schedule';
let orWorkflowSurgery = null; // E12: currently-selected surgery id for the OR Workflow tab
async function renderSurgery(el) {
    const [surgeries, patients, ors, emps] = await Promise.all([
        API.get('/api/surgeries'), API.get('/api/patients'),
        API.get('/api/operating-rooms'), API.get('/api/employees')
    ]);
    const doctors = emps.filter(e => e.role === 'Doctor' || e.name);
    const priorityBadge = p => p === 'Urgent' ? badge(p, 'danger') : p === 'Emergency' ? badge(p, 'danger') : ba
dge(p, 'info');
    const surgStatusBadge = s => ({ Scheduled: 'info', 'In Progress': 'warning', Completed: 'success', Cancelled
: 'danger' }[s] || 'info');

    el.innerHTML = `
        <div class="page-title">? ${tr('Surgery & Pre-Op Management', '???????? ? ? ?')}</div>
        <div class="flex gap-8 mb-16" style="flex-wrap:wrap">
            <button class="btn ${surgeryTab === 'schedule' ? 'btn-primary' : 'btn-secondary'}" onclick="surgeryTab='
schedule';navigateTo(18)">? ${tr('Surgery Schedule', '???? ?????????')}</button>
            <button class="btn ${surgeryTab === 'preop' ? 'btn-primary' : 'btn-secondary'}" onclick="surgeryTab='pre
op';navigateTo(18)">? ${tr('Pre-Op Assessment', '???? ? ? ? ?')}</button>
            <button class="btn ${surgeryTab === 'anesthesia' ? 'btn-primary' : 'btn-secondary'}" onclick="surgeryTab
='anesthesia';navigateTo(18)">? ${tr('Anesthesia', '????????')}</button>
            <button class="btn ${surgeryTab === 'orworkflow' ? 'btn-primary' : 'btn-secondary'}" onclick="surgeryTab
='orworkflow';navigateTo(18)">? ${tr('OR Workflow (WHO/PACU)', '??? ???????? (WHO/????????)')}</button>
            <button class="btn ${surgeryTab === 'rooms' ? 'btn-primary' : 'btn-secondary'}" onclick="surgeryTab='roo
ms';navigateTo(18)">? ${tr('Operating Rooms', '??? ?????????')}</button>
        </div>
        <div id="surgeryContent"></div>`;

    const cont = document.getElementById('surgeryContent');
    if (surgeryTab === 'schedule') {
        cont.innerHTML = `
            <div class="split-layout"><div class="card">

```

```

<div class="card-title">? ${tr('Schedule New Surgery', '????? ????? ?????')}</div>
<div class="form-group mb-12"><label>${tr('Patient', '??????')}</label><select class="form-input" id="sr
gPatient">${patients.map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.file_number)} - ${e
scapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</option>`).join('')}</select></div>
<div class="form-group mb-12"><label>${tr('Procedure', '????????')}</label><input class="form-input" id="
srgProc" placeholder="${tr('e.g. Appendectomy', '????: ??????? ???????')}"></div>
<div class="form-group mb-12"><label>${tr('Procedure (Arabic)', '???????? ?????????')}</label><input class
="form-input" id="srgProcAr"></div>
<div class="form-group mb-12"><label>${tr('Surgeon', '??????')}</label><select class="form-input" id="sr
gSurgeon"><option value="">${tr('Select', '????')}</option>${doctors.map(d => `<option value="${safeId(d.id)}"
data-name="${escapeHTML(d.name)}">${escapeHTML(d.name)}</option>`).join('')}</select></div>
<div class="form-group mb-12"><label>${tr('Anesthetist', '???? ?????????')}</label><select class="form-inp
ut" id="srgAnesth"><option value="">${tr('Select', '????')}</option>${doctors.map(d => `<option value="${safeI
d(d.id)}" data-name="${escapeHTML(d.name)}">${escapeHTML(d.name)}</option>`).join('')}</select></div>
<div class="form-group mb-12"><label>${tr('Type', '????')}</label><select class="form-input" id="srgTyp
e"><option value="Elective">${tr('Elective', '????????')}</option><option value="Urgent">${tr('Urgent', '?????
')}</option><option value="Emergency">${tr('Emergency', '????')}</option></select></div>
<div class="form-group mb-12"><label>${tr('Operating Room', '???? ?????????')}</label><select class="form
-input" id="srgOR">${ors.map(o => `<option value="${escapeHTML(isArabic ? o.room_name_ar : o.room_name)}">${es
capeHTML(isArabic ? o.room_name_ar : o.room_name)} (${escapeHTML(o.location)})</option>`).join('')}</select></
div>
<div class="form-group mb-12"><label>${tr('Date', '????????')}</label><input class="form-input" type="dat
e" id="srgDate" value="${new Date().toISOString().split('T')[0]}"></div>
<div class="form-group mb-12"><label>${tr('Time', '????')}</label><input class="form-input" type="time"
id="srgTime" value="08:00"></div>
<div class="form-group mb-12"><label>${tr('Duration (min)', '????? (????)')}</label><input class="form-
input" type="number" id="srgDur" value="60"></div>
<div class="form-group mb-12"><label>${tr('Priority', '????????')}</label><select class="form-input" id=
"srgPriority"><option value="Normal">${tr('Normal', '????')}</option><option value="Urgent">${tr('Urgent', '?
????')}</option><option value="Emergency">${tr('Emergency', '????')}</option></select></div>
<div class="form-group mb-12"><label>${tr('Notes', '????????')}</label><textarea class="form-input form-t
extarea" id="srgNotes"></textarea></div>
<button class="btn btn-primary w-full" onclick="scheduleSurgery()" style="height:44px">? ${tr('Schedule
Surgery', '????? ?????????')}</button>
</div><div class="card">
<div class="card-title">? ${tr('Surgery Schedule', '???? ?????????')}</div>
<input class="search-filter" placeholder="${tr('Search...', '????...')}" oninput="filterTable(this, 'srgTa
ble')">
<div id="srgTable">${makeTable(
[tr('ID', '#'), tr('Patient', '??????'), tr('Procedure', '????????'), tr('Surgeon', '??????'), tr('Date',
'????????'), tr('Time', '?????'), tr('OR', '????????'), tr('Priority', '????????'), tr('Pre-Op', '?? ???'), tr('
Status', '??????'), tr('Actions', '????????')],
surgeries.map(s => ({ cells: [s.id, s.patient_name, isArabic ? (s.procedure_name_ar || s.procedure_name)
: s.procedure_name, s.surgeon_name, s.scheduled_date, s.scheduled_time, s.operating_room, priorityBadge(s.pri
ority), badge(s.preop_status, s.preop_status === 'Complete' ? 'success' : s.preop_status === 'In Progress' ? '
warning' : 'danger'), badge(s.status, surgStatusBadge(s.status))], id: s.id })),
row => `<div class="flex gap-4" style="flex-wrap:wrap">
${row.cells[9]?.includes('Scheduled') || row.cells[9]?.includes('info')} ? `<button class="btn bt
n-warning btn-sm" onclick="updateSurgStatus(${safeId(row.id)}, 'InProgress')" style="font-size:11px">? ${tr('St
art', '????')}</button>` : ''}
<button class="btn btn-secondary btn-sm" onclick="surgeryTab='orworkflow';orWorkflowSurgery=${sa
feId(row.id)};navigateTo(18)" style="font-size:11px">? ${tr('OR Workflow', '??? ?????????')}</button>
<button class="btn btn-danger btn-sm" onclick="deleteSurgery(${safeId(row.id)})" style="font-siz
e:11px">?</button>

```

```

        </div>`
    })</div>
</div></div>`;
} else if (surgeryTab === 'preop') {
    cont.innerHTML = `
```

```

    } else if (surgeryTab === 'anesthesia') {
        cont.innerHTML = `<div class="card">
            <div class="card-title">? ${tr('Anesthesia Record', '??? ?????')}</div>
            <div class="form-group mb-12"><label>${tr('Select Surgery', '???? ?????')}</label>
                <select class="form-input" id="anesthSurgery" onchange="loadAnesthRecord()">
                    <option value="">${tr('-- Select --', '-- ????' --)}</option>
                    ${surgeries.map(s => `<option value="${safeId(s.id)}">${escapeHTML(s.id)} - ${escapeHTML(s.patient_name)} - ${escapeHTML(s.procedure_name)} (${escapeHTML(s.scheduled_date)})</option>`).join('')}
                </select></div>
            <div id="anesthForm" style="display:none">
                <div class="grid-equal"><div>
                    <div class="form-group mb-12"><label>${tr('Anesthetist', '???? ?????')}</label><input class="form-input" id="anName"></div>
                    <div class="form-group mb-12"><label>ASA ${tr('Classification', '???????')}</label><select class="form-input" id="anASA"><option>ASA I</option><option>ASA II</option><option>ASA III</option><option>ASA IV</option><option>ASA V</option><option>ASA VI</option></select></div>
                    <div class="form-group mb-12"><label>${tr('Anesthesia Type', '??? ?????')}</label><select class="form-input" id="anType"><option value="General">${tr('General', '???')}</option><option value="Spinal">${tr('Spinal', '????')}</option><option value="Epidural">${tr('Epidural', '??? ?????')}</option><option value="Regional">${tr('Regional', '????')}</option><option value="Local">${tr('Local', '????')}</option><option value="Sedation">${tr('Sedation', '???? ????')}</option></select></div>
                    <div class="form-group mb-12"><label>${tr('Airway Assessment', '???? ???? ????')}</label><input class="form-input" id="anAirway"></div>
                    <div class="form-group mb-12"><label>Mallampati Score</label><select class="form-input" id="anMallampati"><option value=""></option><option>Class I</option><option>Class II</option><option>Class III</option><option>Class IV</option></select></div>
                </div><div>
                    <div class="form-group mb-12"><label>${tr('Premedication', '???? ?????')}</label><input class="form-input" id="anPremed"></div>
                    <div class="form-group mb-12"><label>${tr('Induction Agents', '???? ?????')}</label><input class="form-input" id="anInduction"></div>
                    <div class="form-group mb-12"><label>${tr('Maintenance', '???? ?????')}</label><input class="form-input" id="anMaint"></div>
                    <div class="form-group mb-12"><label>${tr('Muscle Relaxants', '???? ????')}</label><input class="form-input" id="anRelax"></div>
                    <div class="form-group mb-12"><label>${tr('IV Access', '???? ????')}</label><input class="form-input" id="anIV"></div>
                </div><div>
                    <div class="grid-equal"><div>
                        <div class="form-group mb-12"><label>${tr('Fluid Given', '???? ????')}</label><input class="form-input" id="anFluid"></div>
                        <div class="form-group mb-12"><label>${tr('Blood Loss (ml)', '???? ???? (??)')}</label><input class="form-input" type="number" id="anBloodLoss" value="0"></div>
                    </div><div>
                        <div class="form-group mb-12"><label>${tr('Complications', '????')}</label><input class="form-input" id="anComp"></div>
                        <div class="form-group mb-12"><label>${tr('Recovery Notes', '???? ????')}</label><textarea class="form-input form-textarea" id="anRecovery"></textarea></div>
                    </div></div>
                    <button class="btn btn-primary w-full" onclick="saveAnesthRecord()" style="height:44px">? ${tr('Save Anesthesia Record', '??? ??? ?????')}</button>
                </div></div>`;
    } else if (surgeryTab === 'orworkflow') {
        cont.innerHTML = `<div class="card">

```



```

<div class="card-title">? ${tr('OR Workflow ? WHO Checklist, PACU & Operative Note', '?? ???? ? ????
? WHO ?????? ?????? ??????')}</div>
<div class="form-group mb-12"><label>${tr('Select Surgery', '??? ?????')}</label>
  <select class="form-input" id="orwfSurgery" onchange="orWorkflowSurgery=this.value;loadOrWorkflow()">
    <option value="">${tr('-- Select --', '-- ??? --')}</option>
    ${surgeries.map(s => `<option value="${safeId(s.id)}" ${String(s.id) === String(orWorkflowSurgery) ?
'selected' : ''}>${escapeHTML(s.id)} - ${escapeHTML(s.patient_name)} - ${escapeHTML(s.procedure_name)} [${esc
apeHTML(s.status)}]</option>`).join('')}
  </select></div>
<div id="orwfBody"></div>
</div>`;
if (orWorkflowSurgery) loadOrWorkflow();
} else if (surgeryTab === 'rooms') {
  cont.innerHTML = `<div class="card">
    <div class="card-title">? ${tr('Operating Rooms', '?? ?????')}</div>
    <div class="stats-grid" style="margin-bottom:16px">${ors.map(o => `<div class="stat-card" style="--stat-
color:${o.status === 'Available' ? '#4ade80' : '#f87171'}">
      <span class="stat-icon">?</span><div class="stat-label">${escapeHTML(isArabic ? o.room_name_ar : o.roo
m_name)}</div>
      <div class="stat-value" style="font-size:14px">${escapeHTML(o.location)}</div>
      <div>${badge(o.status, o.status === 'Available' ? 'success' : 'danger')}</div>
    </div>`).join('')}</div>
  </div>`;
}
}

window.scheduleSurgery = async () => {
  const pSel = document.getElementById('srgPatient');
  try {
    await API.post('/api/surgeries', {
      patient_id: pSel.value, patient_name: pSel.options[pSel.selectedIndex]?.dataset?.name || '',
      surgeon_id: document.getElementById('srgSurgeon').value, surgeon_name: document.getElementById('srgSurge
on').options[document.getElementById('srgSurgeon').selectedIndex]?.dataset?.name || '',
      anesthetist_id: document.getElementById('srgAnesth').value, anesthetist_name: document.getElementById('s
rgAnesth').options[document.getElementById('srgAnesth').selectedIndex]?.dataset?.name || '',
      procedure_name: document.getElementById('srgProc').value, procedure_name_ar: document.getElementById('sr
gProcAr').value,
      surgery_type: document.getElementById('srgType').value, operating_room: document.getElementById('srgOR')
.value,
      scheduled_date: document.getElementById('srgDate').value, scheduled_time: document.getElementById('srgTi
me').value,
      estimated_duration: document.getElementById('srgDur').value, priority: document.getElementById('srgPrior
ity').value,
      notes: document.getElementById('srgNotes').value
    });
    showToast(tr('Surgery scheduled!', '?? ????? ??????')); surgeryTab = 'schedule'; await navigateTo(18);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// E12: status changes flow through the server-side state machine (rejects invalid transitions / WHO gating wi
th 409).
window.updateSurgStatus = async (id, status) => {
  try {
    const r = await API.put(`/api/or/surgeries/${id}/status`, { status });
    if (r && r.error) { showToast(r.error, 'error'); return; }
  }
}

```

```

    showToast(tr('Updated', '?? ??????')); await navigateTo(18);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.deleteSurgery = async (id) => {
  if (!confirm(tr('Delete this surgery?', '??? ??? ?????????'))) return;
  try { await API.del(`/api/surgeries/${id}`); showToast(tr('Deleted', '?? ??????')); await navigateTo(18); }
  catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.loadPreopAssessment = async () => {
  const sid = document.getElementById('preopSurgery').value;
  if (!sid) { document.getElementById('preopForm').style.display = 'none'; return; }
  document.getElementById('preopForm').style.display = 'block';
  try {
    const data = await API.get(`/api/surgeries/${sid}/preop`);
    if (data) {
      const map = { npo: 'npo_confirmed', allergies: 'allergies_reviewed', medications: 'medications_reviewed',
        labs: 'labs_reviewed', imaging: 'imaging_reviewed', blood_type: 'blood_type_confirmed', consent: 'consent_signed',
        anesthesia_clr: 'anesthesia_clearance', nursing: 'nursing_assessment', cardiac: 'cardiac_clearance', pulmonary: 'pulmonary_clearance',
        infection: 'infection_screening', dvt: 'dvt_prophylaxis' };
      Object.entries(map).forEach(([k, v]) => { const el = document.getElementById('preop_' + k); if (el) el.checked = !!data[v]; });
      ['allergies', 'medications', 'labs', 'imaging', 'nursing', 'cardiac'].forEach(k => { const el = document.getElementById('preop_' + k + '_notes'); if (el) el.value = data[k + '_notes'] || ''; });
      document.getElementById('preop_blood_reserved').value = data.blood_reserved ? '1' : '0';
    }
    const tests = await API.get(`/api/surgeries/${sid}/preop-tests`);
    const tl = document.getElementById('preopTestsList');
    tl.innerHTML = tests.length ? makeTable([tr('Type', '?????'), tr('Test', '?????'), tr('Status', '???????'), tr('Result', '????????'), tr('Action', '?????')],
      tests.map(t => ({ cells: [t.test_type, t.test_name, t.is_completed ? badge(tr('Done', '??'), 'success') : badge(tr('Pending', '????'), 'warning'), t.result_summary || '-'], id: t.id })),
      row => `<button class="btn btn-success btn-sm" onclick="markTestDone(${safeId(row.id)})">?</button>`) :
    `<p style="color:var(--text-dim)">${tr('No tests added', '?? ??? ?????? ??????')}</p>`;
  } catch (e) { console.error(e); }
};

window.savePreopAssessment = async () => {
  const sid = document.getElementById('preopSurgery').value;
  if (!sid) return;
  try {
    await API.post(`/api/surgeries/${sid}/preop`, {
      npo_confirmed: document.getElementById('preop_npo').checked, allergies_reviewed: document.getElementById('preop_allergies').checked,
      allergies_notes: document.getElementById('preop_allergies_notes')?.value || '', medications_reviewed: document.getElementById('preop_medications').checked,
      medications_notes: document.getElementById('preop_medications_notes')?.value || '', labs_reviewed: document.getElementById('preop_labs').checked,
      labs_notes: document.getElementById('preop_labs_notes')?.value || '', imaging_reviewed: document.getElementById('preop_imaging').checked,
      imaging_notes: document.getElementById('preop_imaging_notes')?.value || '', blood_type_confirmed: document.getElementById('preop_blood_type').checked,
      blood_reserved: document.getElementById('preop_blood_reserved').value === '1', consent_signed: document.getElementById('preop_consent').checked,
      anesthesia_clearance: document.getElementById('preop_anesthesia_clr').checked, nursing_assessment: document.getElementById('preop_nursing').checked,

```

```

        nursing_notes: document.getElementById('preop_nursing_notes')?.value || '', cardiac_clearance: document.
getElementById('preop_cardiac').checked,
        cardiac_notes: document.getElementById('preop_cardiac_notes')?.value || '', pulmonary_clearance: documen
t.getElementById('preop_pulmonary').checked,
        infection_screening: document.getElementById('preop_infection').checked, dvt_prophylaxis: document.getEl
ementById('preop_dvt').checked
    });
    showToast(tr('Assessment saved!', '?? ??? ??????!!'));
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.addPreopTest = async () => {
    const sid = document.getElementById('preopSurgery').value;
    if (!sid) return;
    try {
        await API.post(`/api/surgeries/${sid}/preop-tests`, { test_type: document.getElementById('preopTestType').
value, test_name: document.getElementById('preopTestName').value });
        showToast(tr('Test added', '?? ????? ??????')); loadPreopAssessment();
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.markTestDone = async (id) => {
    try { await API.put(`/api/surgery-preop-tests/${id}`, { is_completed: 1 }); showToast(tr('Done', '??')); loa
dPreopAssessment(); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.loadAnesthRecord = async () => {
    const sid = document.getElementById('anesthSurgery').value;
    if (!sid) { document.getElementById('anesthForm').style.display = 'none'; return; }
    document.getElementById('anesthForm').style.display = 'block';
    try {
        const d = await API.get(`/api/surgeries/${sid}/anesthesia`);
        if (d) {
            document.getElementById('anName').value = d.anesthetist_name || ''; document.getElementById('anASA').val
ue = d.asa_class || 'ASA I';
            document.getElementById('anType').value = d.anesthesia_type || 'General'; document.getElementById('anAir
way').value = d.airway_assessment || '';
            document.getElementById('anMallampati').value = d.mallampati_score || ''; document.getElementById('anPre
med').value = d.premedication || '';
            document.getElementById('anInduction').value = d.induction_agents || ''; document.getElementById('anMain
t').value = d.maintenance_agents || '';
            document.getElementById('anRelax').value = d.muscle_relaxants || ''; document.getElementById('anIV').val
ue = d.iv_access || '';
            document.getElementById('anFluid').value = d.fluid_given || ''; document.getElementById('anBloodLoss').v
alue = d.blood_loss_ml || 0;
            document.getElementById('anComp').value = d.complications || ''; document.getElementById('anRecovery').v
alue = d.recovery_notes || '';
        }
    } catch (e) { }
};

window.saveAnesthRecord = async () => {
    const sid = document.getElementById('anesthSurgery').value;
    if (!sid) return;
    try {
        await API.post(`/api/surgeries/${sid}/anesthesia`, {
            anesthetist_name: document.getElementById('anName').value, asa_class: document.getElementById('anASA').v

```

```

    alue,
    anesthesia_type: document.getElementById('anType').value, airway_assessment: document.getElementById('an
Airway').value,
    mallampati_score: document.getElementById('anMallampati').value, premedication: document.getElementById(
'anPremed').value,
    induction_agents: document.getElementById('anInduction').value, maintenance_agents: document.getElementB
yId('anMaint').value,
    muscle_relaxants: document.getElementById('anRelax').value, iv_access: document.getElementById('anIV').v
alue,
    fluid_given: document.getElementById('anFluid').value, blood_loss_ml: parseInt(document.getElementById('
anBloodLoss').value) || 0,
    complications: document.getElementById('anComp').value, recovery_notes: document.getElementById('anRecov
ery').value
  });
  showToast(tr('Anesthesia record saved!', '?? ??? ??? ???????!'));
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== E12: OR WORKFLOW (WHO checklist + PACU + operative note) =====
window.loadOrWorkflow = async () => {
  const sid = window.safeId(orWorkflowSurgery);
  const body = document.getElementById('orwfBody');
  if (!body) return;
  if (!sid) { body.innerHTML = ''; return; }
  try {
    const [surgery, who, pacu, opData] = await Promise.all([
      API.get(`/api/surgeries/${sid}`),
      API.get(`/api/or/surgeries/${sid}/who-checklist`),
      API.get(`/api/or/surgeries/${sid}/pacu`),
      API.get(`/api/or/surgeries/${sid}/operative-note`)
    ]);
    if (surgery && surgery.error) { body.innerHTML = `
```

```

        <div>3) ${tr('Sign-Out (before leaving OR)', '????? ?????? (??? ?????? ??????)')} ${phaseDone('sign_
out')}}

        ${who && who['sign_out_completed'] ? '' : `<button class="btn btn-primary btn-sm" onclick="complet
eWhoPhase('sign-out')">${tr('Complete', '????')}</button>`}</div>

    </div>

    <div class="flex gap-8 mt-12">
        <button class="btn btn-warning btn-sm" onclick="updateSurgStatus(${safeId(sid)}, 'InProgress')">? ${tr
('Start Incision (InProgress)', '??? ?????')}</button>

        <button class="btn btn-secondary btn-sm" onclick="updateSurgStatus(${safeId(sid)}, 'PACU')">?? ${tr('
Move to PACU', '??? ??????')}</button>

        <button class="btn btn-success btn-sm" onclick="updateSurgStatus(${safeId(sid)}, 'Completed')">? ${tr
('Complete Surgery', '????? ??????')}</button>

        <button class="btn btn-danger btn-sm" onclick="updateSurgStatus(${safeId(sid)}, 'Cancelled')">? ${tr(
'Cancel', '????')}</button>

    </div>

</div>

<div class="card mb-12"><div class="card-title">?? ${tr('PACU / Recovery (Aldrete)', '??????? (?????)')}
</div>

    <div class="grid-equal"><div>
        ${{[['activity', tr('Activity', '?????')], ['respiration', tr('Respiration', '?????')], ['circulati
on', tr('Circulation', '?????')]].map(([k, l]) => `<div class="form-group mb-8"><label>${l} (0-2)</label><sel
ect class="form-input" id="pacu_${k}"><option value=""></option><option>0</option><option>1</option><option>2
</option></select></div>`).join('')}}

    </div><div>
        ${{[['consciousness', tr('Consciousness', '?????')], ['oxygen', tr('O2 Saturation', '????????')]].map
(([k, l]) => `<div class="form-group mb-8"><label>${l} (0-2)</label><select class="form-input" id="pacu_${k}">
<option value=""></option><option>0</option><option>1</option><option>2</option></select></div>`).join('')}}

        <div class="form-group mb-8"><label>${tr('Pain Score (0-10)', '????? ?????')}</label><input class="f
orm-input" type="number" id="pacu_pain" value="${escapeHTML(pacu && pacu.pain_score != null ? pacu.pain_score
: 0)}"></div>

        <div class="form-group mb-8"><label>${tr('Discharge?', '????????')}</label><select class="form-input"
id="pacu_discharge"><option value="In Recovery">${tr('In Recovery', '?? ??????')}</option><option value="Dis
charged" ${pacu && pacu.discharge_status === 'Discharged' ? 'selected' : ''}>${tr('Discharge from PACU', '????
?? ?? ??????')}</option></select></div>

    </div></div>

    <div class="flex gap-8" style="align-items:center"><button class="btn btn-primary btn-sm" onclick="sav
ePacu()">? ${tr('Save PACU', '??? ??????')}</button>

    ${pacu && pacu.aldrete_score != null ? `<span class="badge badge-info">${tr('Aldrete', '?????')}: $
${escapeHTML(pacu.aldrete_score)} / 10 ? ${escapeHTML(pacu.discharge_status || '')}</span>` : ''}</div>

</div>

<div class="card"><div class="card-title">? ${tr('Operative Note + Consumption', '???????? ??????? + ???
?????')}</div>

    <div class="form-group mb-8"><label>${tr('Procedure Description', '??? ??????')}</label><textarea cla
ss="form-input form-textarea" id="op_desc">${escapeHTML(op && op.procedure_description || '')}</textarea></div>
>

    <div class="form-group mb-8"><label>${tr('Findings', '????????')}</label><input class="form-input" id="
op_findings" value="${escapeHTML(op && op.findings || '')}"></div>

    <div class="form-group mb-8"><label>${tr('Complications', '????????')}</label><input class="form-inpu
t" id="op_comp" value="${escapeHTML(op && op.complications || '')}"></div>

    <div class="flex gap-8"><div class="form-group mb-8" style="flex:1"><label>${tr('Blood Loss (ml)', '??
??? ?????')}</label><input class="form-input" type="number" id="op_blood" value="${escapeHTML(op && op.blood_lo
ss_final || 0)}"></div>

    <div class="form-group mb-8" style="flex:1"><label>${tr('Counts Verified?', '????? ?????')}</label><
select class="form-input" id="op_counts"><option value="0">${tr('No / Incomplete', '?? / ??? ?????')}</option>

```

```

<option value="1" ${op && op.counts_verified === 'Verified' ? 'selected' : ''}>${tr('Yes ? Verified', '??? ? ?
???')}</option></select></div></div>

    <div style="margin-bottom:12px">
        <button class="btn btn-outline btn-sm w-full" type="button" onclick="openSurgicalCountModal(${safeId
(orWorkflowSurgery)})">? ${tr('Record/Verify Surgical Count Sheet', '????/???? ??? ?????? ??????? ??????')}
</button>
    </div>

    <div class="card mt-8"><div class="card-title" style="font-size:13px">? ${tr('Consumption (decrements
inventory)', '???????? (???? ?? ??????)')}</div>
        <div id="op_cons_rows">${cons.map(c => orConsRowHtml(c.item_id, c.qty_used)).join('')}</div>
        <button class="btn btn-secondary btn-sm mt-8" onclick="addConsumptionRow()">? ${tr('Add Item', '????
? ???')}</button>
    </div>
    <button class="btn btn-primary mt-12" onclick="saveOperativeNote()">? ${tr('Save Operative Note', '???
????')}</button>
</div>`;
    } catch (e) { console.error(e); body.innerHTML = `<p style="color:var(--danger)">${tr('Error loading workflo
w', '??? ?? ?????')}</p>`; }
};
window.orConsRowHtml = (itemId, qty) => `<div class="flex gap-8 mb-4 op-cons-row"><input class="form-input op-
cons-item" placeholder="${tr('Item ID', '??? ?????')}" type="number" value="${escapeHTML(itemId != null ? item
Id : '')}" style="flex:1"><input class="form-input op-cons-qty" placeholder="${tr('Qty', '????')}" type="num
ber" value="${escapeHTML(qty != null ? qty : '')}" style="flex:1"></div>`;
window.addConsumptionRow = () => { const w = document.getElementById('op_cons_rows'); if (w) w.insertAdjacentH
TML('beforeend', orConsRowHtml(null, null)); };
window.completeWhoPhase = async (phase) => {
    const sid = window.safeId(orWorkflowSurgery); if (!sid) return;
    try {
        const r = await API.post(`/api/or/surgeries/${sid}/who-checklist/${phase}`, {});
        if (r && r.error) { showToast(r.error, 'error'); return; }
        showToast(tr('Checklist updated', '?? ????? ?????')); loadOrWorkflow();
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.savePacu = async () => {
    const sid = window.safeId(orWorkflowSurgery); if (!sid) return;
    const v = id => { const el = document.getElementById(id); return el ? el.value : ''; };
    try {
        const r = await API.post(`/api/or/surgeries/${sid}/pacu`, {
            activity: v('pacu_activity'), respiration: v('pacu_respiration'), circulation: v('pacu_circulation'),
            consciousness: v('pacu_consciousness'), oxygen: v('pacu_oxygen'),
            pain_score: v('pacu_pain'), discharge_status: v('pacu_discharge')
        });
        if (r && r.error) { showToast(r.error, 'error'); return; }
        showToast(tr('PACU saved', '?? ??? ?????')); loadOrWorkflow();
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.saveOperativeNote = async () => {
    const sid = window.safeId(orWorkflowSurgery); if (!sid) return;
    const v = id => { const el = document.getElementById(id); return el ? el.value : ''; };
    const consumption = Array.from(document.querySelectorAll('#op_cons_rows .op-cons-row')).map(r => ({
        item_id: r.querySelector('.op-cons-item')?.value, qty_used: r.querySelector('.op-cons-qty')?.value
    })).filter(c => c.item_id && c.qty_used);

```

```

try {
  const r = await API.post(`/api/or/surgeries/${sid}/operative-note`, {
    procedure_description: v('op_desc'), findings: v('op_findings'), complications: v('op_comp'),
    blood_loss_final: v('op_blood'), counts_verified: v('op_counts') === '1', consumption
  });
  if (r && r.error) { showToast(r.error, 'error'); return; }
  showToast(tr('Operative note saved', '?? ??? ??????')); loadOrWorkflow();
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== BLOOD BANK =====
let bbTab = 'inventory';
async function renderBloodBank(el) {
  const results = await Promise.allSettled([
    API.get('/api/bloodbank/stats'), API.get('/api/bloodbank/units'),
    API.get('/api/bloodbank/donors'), API.get('/api/bloodbank/crossmatch'),
    API.get('/api/bloodbank/transfusions'), API.get('/api/patients')
  ]);

  const stats = results[0].status === 'fulfilled' ? results[0].value : { total: 0, expiring: 0, totalDonors: 0
, todayTransfusions: 0, pendingCrossmatch: 0, byType: [] };
  const units = results[1].status === 'fulfilled' ? results[1].value : [];
  const donors = results[2].status === 'fulfilled' ? results[2].value : [];
  const crossmatches = results[3].status === 'fulfilled' ? results[3].value : [];
  const transfusions = results[4].status === 'fulfilled' ? results[4].value : [];
  const patients = results[5].status === 'fulfilled' ? results[5].value : [];

  // Track and display partial load errors if any of the promises failed
  const failedCalls = [];
  if (results[0].status === 'rejected') failedCalls.push('Stats');
  if (results[1].status === 'rejected') failedCalls.push('Inventory');
  if (results[2].status === 'rejected') failedCalls.push('Donors');
  if (results[3].status === 'rejected') failedCalls.push('Crossmatches');
  if (results[4].status === 'rejected') failedCalls.push('Transfusions');
  if (results[5].status === 'rejected') failedCalls.push('Patients');

  let bbLoadErrors = '';
  if (failedCalls.length > 0) {
    bbLoadErrors = `
```

```

<div class="stat-card" style="--stat-color:#f59e0b"><span class="stat-icon">?</span><div class="stat-label">${tr('Expiring Soon', '????? ?????')}</div><div class="stat-value">${stats.expiring}</div></div>
<div class="stat-card" style="--stat-color:#3b82f6"><span class="stat-icon">?</span><div class="stat-label">${tr('Total Donors', '????? ?????')}</div><div class="stat-value">${stats.totalDonors}</div></div>
<div class="stat-card" style="--stat-color:#8b5cf6"><span class="stat-icon">?</span><div class="stat-label">${tr('Today Transfusions', '??? ?? ?????')}</div><div class="stat-value">${stats.todayTransfusions}</div></div>
<div class="stat-card" style="--stat-color:#06b6d4"><span class="stat-icon">?</span><div class="stat-label">${tr('Pending Cross-Match', '????? ?????')}</div><div class="stat-value">${stats.pendingCrossmatch}</div></div>
</div>
<div class="stats-grid mt-16" style="grid-template-columns:repeat(8,1fr)">
  ${['A+', 'A-', 'B+', 'B-', 'AB+', 'AB-', 'O+', 'O-'].map(t => {
    const bt = t.replace(/[+-]/, ''), rh = t.includes('+') ? '+' : '-';
    const cnt = stats.byType?.find(b => b.blood_type === bt && b.rh_factor === rh)?.cnt || 0;
    return `<div class="stat-card" style="--stat-color:${btColors[bt]} || '#888';text-align:center;padding:12px"><div style="font-size:24px;font-weight:800">${t}</div><div style="font-size:18px;font-weight:600">${cnt}</div><div style="font-size:10px">${tr('units', '????')}</div></div>`;
  })}.join('')}
</div>
<div class="flex gap-8 mt-16 mb-16" style="flex-wrap:wrap">
  <button class="btn ${bbTab === 'inventory' ? 'btn-primary' : 'btn-secondary'}" onclick="bbTab='inventory';navigateTo(19)">? ${tr('Inventory', '???????')}</button>
  <button class="btn ${bbTab === 'donors' ? 'btn-primary' : 'btn-secondary'}" onclick="bbTab='donors';navigateTo(19)">? ${tr('Donors', '???????')}</button>
  <button class="btn ${bbTab === 'crossmatch' ? 'btn-primary' : 'btn-secondary'}" onclick="bbTab='crossmatch';navigateTo(19)">? ${tr('Cross-Match', '???????')}</button>
  <button class="btn ${bbTab === 'transfusions' ? 'btn-primary' : 'btn-secondary'}" onclick="bbTab='transfusions';navigateTo(19)">? ${tr('Transfusions', '??? ?????')}</button>
</div>
<div id="bbContent"></div>`;
const cont = document.getElementById('bbContent');
if (bbTab === 'inventory') {
  cont.innerHTML = `<div class="split-layout"><div class="card">
    <div class="card-title">? ${tr('Add Blood Unit', '????? ?????')}</div>
    <div class="form-group mb-12"><label>${tr('Bag Number', '??? ?????')}</label><input class="form-input" id="bbBag"></div>
    <div class="form-group mb-12"><label>${tr('Blood Type', '????? ?????')}</label><select class="form-input" id="bbType"><option>A</option><option>B</option><option>AB</option><option>O</option></select></div>
    <div class="form-group mb-12"><label>Rh</label><select class="form-input" id="bbRh"><option value="+">+</option><option value="-">-</option></select></div>
    <div class="form-group mb-12"><label>${tr('Component', '?????')}</label><select class="form-input" id="bbComp"><option>Whole Blood</option><option>Packed RBC</option><option>FFP</option><option>Platelets</option><option>Cryoprecipitate</option></select></div>
    <div class="form-group mb-12"><label>${tr('Collection Date', '????? ?????')}</label><input class="form-input" type="date" id="bbCollDate" value="${new Date().toISOString().split('T')[0]}"></div>
    <div class="form-group mb-12"><label>${tr('Expiry Date', '????? ?????')}</label><input class="form-input" type="date" id="bbExpDate"></div>
    <div class="form-group mb-12"><label>${tr('Volume (ml)', '????? (??)')}</label><input class="form-input" type="number" id="bbVol" value="450"></div>
    <button class="btn btn-primary w-full" onclick="addBloodUnit()" style="height:44px">? ${tr('Add Unit', '????? ?????')}</button>
  </div><div class="card">
    <div class="card-title">? ${tr('Blood Units', '????? ?????')}</div>

```



```

        <div id="bbUnitsTable">${makeTable([tr('Bag#', '??? ?????'), tr('Type', '???????'), tr('Component', '???
????'), tr('Collection', '???????'), tr('Expiry', '????????'), tr('Status', '?????')]),
        units.map(u => ({ cells: [u.bag_number, u.blood_type + u.rh_factor, u.component, u.collection_date, u.ex
piry_date, statusBadge(u.status)] })))</div>
    </div></div>`;
    } else if (bbTab === 'donors') {
        cont.innerHTML = `<div class="split-layout"><div class="card">
            <div class="card-title">? ${tr('Register Donor', '????? ?????')}</div>
            <div class="form-group mb-12"><label>${tr('Name (EN)', '????? (??????')}</label><input class="form-inp
ut" id="bbDonorName"></div>
            <div class="form-group mb-12"><label>${tr('Name (AR)', '????? (????')}</label><input class="form-input"
id="bbDonorNameAr"></div>
            <div class="form-group mb-12"><label>${tr('National ID', '?????')}</label><input class="form-input" id=
"bbDonorNID"></div>
            <div class="form-group mb-12"><label>${tr('Phone', '?????')}</label><input class="form-input" id="bbDon
orPhone"></div>
            <div class="form-group mb-12"><label>${tr('Blood Type', '????? ?????')}</label><select class="form-input"
id="bbDonorBT"><option>A</option><option>B</option><option>AB</option><option>O</option></select></div>
            <div class="form-group mb-12"><label>Rh</label><select class="form-input" id="bbDonorRh"><option value="
+">+</option><option value="-">-</option></select></div>
            <div class="form-group mb-12"><label>${tr('Age', '????')}</label><input class="form-input" type="number
" id="bbDonorAge"></div>
            <button class="btn btn-primary w-full" onclick="addDonor()" style="height:44px">? ${tr('Register', '????
?')}</button>
        </div><div class="card">
            <div class="card-title">? ${tr('Donors List', '????? ?????????')}</div>
            <div id="bbDonorsTable">${makeTable([tr('Name', '?????'), tr('ID', '?????'), tr('Blood Type', '????????'
), tr('Phone', '?????'), tr('Last Donation', '??? ?????'), tr('Eligible', '????')]),
            donors.map(d => ({ cells: [isArabic ? (d.donor_name_ar || d.donor_name) : d.donor_name, d.national_id, d
.blood_type + d.rh_factor, d.phone, d.last_donation_date, d.is_eligible ? badge(tr('Yes', '???'), 'success') :
badge(tr('No', '??'), 'danger')] })))</div>
        </div></div>`;
    } else if (bbTab === 'crossmatch') {
        cont.innerHTML = `<div class="split-layout"><div class="card">
            <div class="card-title">? ${tr('Request Cross-Match', '??? ??? ?????')}</div>
            <div class="form-group mb-12"><label>${tr('Patient', '?????')}</label><select class="form-input" id="bb
CMPatient">${patients.map(p => `<option value="${safeId(p.id)}">${escapeHTML(p.file_number)} - ${escapeHTML(is
Arabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</option>`).join('')}</select></div>
            <div class="form-group mb-12"><label>${tr('Unit (optional ? validates ABO/Rh)', '????? (??????? ? ?????
?? ??????')}</label><select class="form-input" id="bbCMUnit"><option value="">${escapeHTML(tr('? Pending (n
o unit) ?', '? ???? (???? ???? ?))}</option>${units.filter(u => u.status === 'Available' && !u.is_expired).m
ap(u => `<option value="${safeId(u.id)}">${escapeHTML(u.bag_number)} (${escapeHTML(u.blood_type)}${escapeHTML(
u.rh_factor)} - ${escapeHTML(u.component)})</option>`).join('')}</select></div>
            <div class="form-group mb-12"><label>${tr('Units Needed', '??????? ?????????')}</label><input class="form
-input" type="number" id="bbCMUnits" value="1"></div>
            <div class="text-muted" style="font-size:12px;margin-bottom:8px">${escapeHTML(tr('Patient blood type and
compatibility are determined by the server.', '????? ?? ?????? ????????? ????????? ?? ??????'))}</div>
            <button class="btn btn-primary w-full" onclick="requestCrossmatch()" style="height:44px">? ${tr('Request
', '????')}</button>
        </div><div class="card">
            <div class="card-title">? ${tr('Cross-Match Results', '????? ?????????')}</div>
            <div id="bbCMTable">${makeTable([tr('Patient', '?????'), tr('Type', '????????'), tr('Units', '????????'),
tr('Technician', '?????'), tr('Result', '????????'), tr('Action', '?????')]),
            crossmatches.map(c => ({ cells: [c.patient_name, c.patient_blood_type, c.units_needed, c.lab_technician,

```

```

c.result === 'Pending' ? badge(c.result, 'warning') : c.result === 'Compatible' ? badge(c.result, 'success')
: badge(c.result, 'danger')]], id: c.id, result: c.result })),
    row => row.result === 'Pending'
      ? `<select class="form-input" id="cmVU_${safeId(row.id)}" style="display:inline-block;width:auto;margi
n-${isArabic ? 'left' : 'right'}:6px">${units.filter(u => u.status === 'Available' && !u.is_expired).map(u =>
`<option value="${safeId(u.id)}">${escapeHTML(u.bag_number)} (${escapeHTML(u.blood_type)}${escapeHTML(u.rh_fac
tor)})</option>`.join('')}</select><button class="btn btn-primary btn-sm" onclick="validateCrossmatch(${safeI
d(row.id)})">? ${escapeHTML(tr('Validate', '????'))}</button>`
      : `<span class="text-muted">${escapeHTML(tr('Finalized', '????'))}</span>`}</div>
</div></div>`;
} else if (bbTab === 'transfusions') {
  cont.innerHTML = `<div class="split-layout"><div class="card">
    <div class="card-title">? ${tr('Record Transfusion', '????? ??? ?')}</div>
    <div class="form-group mb-12"><label>${tr('Patient', '?????')}</label><select class="form-input" id="bb
TrPatient">${patients.map(p => `<option value="${safeId(p.id)}" data-name="">${escapeHTML(p.file_number)} - ${
escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</option>`).join('')}</select></div>
    <div class="form-group mb-12"><label>${tr('Blood Unit', '???? ????')}</label><select class="form-input"
id="bbTrUnit">${units.filter(u => u.status === 'Available' && !u.is_expired).map(u => `<option value="${safeId
(u.id)}" data-bag="${escapeHTML(u.bag_number)}" data-bt="${escapeHTML(u.blood_type + u.rh_factor)}" data-comp=
"${escapeHTML(u.component)}">${escapeHTML(u.bag_number)} (${escapeHTML(u.blood_type)}${escapeHTML(u.rh_factor)
} - ${escapeHTML(u.component)})</option>`).join('')}</select></div>
    <div class="form-group mb-12"><label>${tr('Volume (ml)', '????? (??)')}</label><input class="form-input"
type="number" id="bbTrVol" value="450"></div>
    <button class="btn btn-primary w-full" onclick="recordTransfusion()" style="height:44px">? ${tr('Record'
, '????')}</button>
  </div><div class="card">
    <div class="card-title">? ${tr('Transfusion Records', '??? ??? ????')}</div>
    <div id="bbTrTable">${makeTable([tr('Patient', '?????'), tr('Bag#', '?????'), tr('Type', '???????'), tr
('Component', '?????'), tr('By', '?????'), tr('Time', '?????'), tr('Reaction', '?????')],
    transfusions.map(t => ({ cells: [t.patient_name, t.bag_number, t.blood_type, t.component, t.administered
_by, t.start_time?.split('T')[0] || '', t.adverse_reaction ? badge(tr('Yes', '???'), 'danger') : badge(tr('No'
, '???'), 'success')] })))}</div>
  </div></div>`;
}
}
// E13: PRIMARY blood-bank handlers now call the safe /api/bloodbank/* routes.
// Server is authoritative for ABO/Rh compatibility, unit status, expiry and tenant scope;
// the client never marks a crossmatch compatible. The API client resolves (does not throw)
// on non-2xx, so we inspect the response: an { error } body means the server blocked/refused
// (422/409/410) and we surface the server reason instead of a falsely-reassuring success.
function bbBlocked(resp) { return resp && typeof resp === 'object' && (resp.error || resp.compatible === false
); }
function bbShowReason(resp, fallbackEn, fallbackAr) {
  const reason = resp && (resp.reason || resp.error);
  showToast(reason ? String(reason) : tr(fallbackEn, fallbackAr), 'error');
}
window.addBloodUnit = async () => {
  try {
    const r = await API.post('/api/bloodbank/units', { bag_number: document.getElementById('bbBag').value, blo
od_type: document.getElementById('bbType').value, rh_factor: document.getElementById('bbRh').value, component:
document.getElementById('bbComp').value, collection_date: document.getElementById('bbCollDate').value, expiry
_date: document.getElementById('bbExpDate').value, volume_ml: document.getElementById('bbVol').value });
    if (bbBlocked(r)) return bbShowReason(r, 'Error', '???');
  }
}

```

```

        showToast(tr('Unit added!', '?? ????? ??????!')); bbTab = 'inventory'; await navigateTo(19);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.addDonor = async () => {
    try {
        const r = await API.post('/api/bloodbank/donors', { donor_name: document.getElementById('bbDonorName').value, donor_name_ar: document.getElementById('bbDonorNameAr').value, national_id: document.getElementById('bbDonorNID').value, phone: document.getElementById('bbDonorPhone').value, blood_type: document.getElementById('bbDonorBT').value, rh_factor: document.getElementById('bbDonorRh').value, age: document.getElementById('bbDonorAge').value });
        if (bbBlocked(r)) return bbShowReason(r, 'Error', '???');
        showToast(tr('Donor registered!', '?? ????? ??????!')); bbTab = 'donors'; await navigateTo(19);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.requestCrossmatch = async () => {
    const sel = document.getElementById('bbCMPatient');
    const unitSel = document.getElementById('bbCMUnit');
    try {
        // Server reads patient blood type from the chart and validates ABO/Rh if a unit is chosen.
        const r = await API.post('/api/bloodbank/crossmatch', { patient_id: sel.value, unit_id: unitSel.value || undefined, units_needed: document.getElementById('bbCMUnits').value });
        if (bbBlocked(r)) return bbShowReason(r, 'Cross-match blocked', '?? ??? ??? ??????');
        showToast(tr('Cross-match requested!', '?? ??? ??? ??????!')); bbTab = 'crossmatch'; await navigateTo(19);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// validate a pending crossmatch against a selected unit (server-side ABO/Rh; client cannot force Compatible)
window.validateCrossmatch = async (id) => {
    const us = document.getElementById('cmVU_' + id);
    const unitId = us ? us.value : '';
    try {
        const r = await API.put(`/api/bloodbank/crossmatch/${id}/validate`, { unit_id: unitId });
        if (bbBlocked(r)) { bbShowReason(r, 'Incompatible / blocked', '??? ?????? / ??????'); await navigateTo(19); return; }
        showToast(tr('Compatible ? validated', '????? ? ?? ??????')); await navigateTo(19);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.recordTransfusion = async () => {
    const pSel = document.getElementById('bbTrPatient'), uSel = document.getElementById('bbTrUnit');
    try {
        // Transactional, ABO/Rh + expiry enforced server-side; double-issue -> 409.
        const r = await API.post('/api/bloodbank/transfuse', { patient_id: pSel.value, unit_id: uSel.value, volume_ml: document.getElementById('bbTrVol').value });
        if (bbBlocked(r)) return bbShowReason(r, 'Transfusion blocked', '?? ??? ??? ?????');
        showToast(tr('Transfusion recorded!', '?? ????? ??? ?????!')); bbTab = 'transfusions'; await navigateTo(19);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== CONSENT FORMS =====

window.printConsentForm = async (formId) => {
    try {
        const [form, settings] = await Promise.all([

```

```

    API.get('/api/consent-forms/' + formId),
    API.get('/api/settings')
  ]);
  // Try to use rich HTML template if available
  if (form.form_type) {
    const renderUrl = `/api/consent-forms/render/${form.form_type}?patient_id=${form.patient_id || ''}&doctor_name=${encodeURIComponent(form.doctor_name || '')}`;
    try {
      const resp = await fetch(renderUrl);
      if (resp.ok) {
        const w = window.open('', '_blank');
        const html = await resp.text();
        // Add print/close buttons at top
        const printBar = `
```

```

.header h2{font-size:16px;color:#555;font-weight:500;margin-bottom:4px}
.header .hospital-info{font-size:11px;color:#888;margin-top:8px}
.patient-box{display:grid;grid-template-columns:1fr 1fr;gap:10px;border:1px solid #ccc;border-radius:8px;padding:16px;margin-bottom:20px;background:#fafbfc}
.patient-box .field{font-size:13px}
.patient-box .field label{font-weight:700;color:#1a365d}
.consent-section{margin:20px 0;padding:18px;border:1px solid #ddd;border-radius:10px}
.consent-section h3{color:#1a365d;font-size:16px;text-align:center;margin-bottom:14px;border-bottom:1px solid #eee;padding-bottom:8px}
.consent-text-ar{font-size:14px;line-height:2;text-align:justify;margin-bottom:16px;padding:12px;background:#f7f8fa;border-radius:8px}
.consent-text-en{font-size:12px;line-height:1.8;text-align:left;direction:ltr;color:#555;font-style:italic;padding:12px;background:#f0f4f8;border-radius:8px;margin-top:10px}
.sig-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:30px;margin-top:35px;padding-top:15px;border-top:2px solid #eee}
.sig-box{text-align:center}
.sig-box .sig-label{font-weight:700;font-size:13px;color:#1a365d;margin-bottom:4px}
.sig-box .sig-label-en{font-size:11px;color:#888}
.sig-box .sig-area{margin-top:8px;min-height:70px;display:flex;align-items:flex-end;justify-content:center}
.sig-box .sig-line{border-top:2px solid #333;width:100%;margin-top:60px;padding-top:4px}
.status-badge{display:inline-block;padding:3px 14px;border-radius:12px;font-size:12px;font-weight:700}
.status-signed{background:#d4edda;color:#155724}
.status-pending{background:#fff3cd;color:#856404}
.footer{text-align:center;margin-top:30px;padding-top:12px;border-top:1px solid #ccc;font-size:10px;color:#999}
}
@media print{body{padding:15px}.no-print{display:none!important}}
</style></head><body>
<div class="no-print" style="text-align:center;margin-bottom:20px">
  <button onclick="window.print()" style="padding:12px 40px;font-size:16px;background:#1a365d;color:#fff;border:none;border-radius:8px;cursor:pointer">?? ????? / Print</button>
  <button onclick="window.close()" style="padding:12px 30px;font-size:16px;background:#dc3545;color:#fff;border:none;border-radius:8px;cursor:pointer;margin-right:10px">?????</button>
</div>
<div class="header">
  <h1>${escapeHTML(hospitalAr)}</h1>
  <h2>${escapeHTML(hospitalEn)}</h2><div style="display:flex;gap:12px;margin-bottom:16px;flex-wrap:wrap"><button class="btn" onclick="showChangePassword()" style="background:#fff3e0;border:1px solid #ff9800;color:#e65100">? ${tr("Change Password", "????? ???? ??????")}</button><button class="btn" onclick="startBackup()" style="background:#e8f5e9;border:1px solid #4caf50;color:#2e7d32">? ${tr("Database Backup", "???? ?????????")}</button>
</div>
  <div class="hospital-info">
    ${phone ? '?' + escapeHTML(phone) + ' | ' : ''}${address ? '?' + escapeHTML(address) + ' | ' : ''}${taxNum ? '????? ??????: ' + escapeHTML(taxNum) : ''}
  </div>
</div>
<h3 style="text-align:center;color:#1a365d;font-size:18px;margin-bottom:5px">? ${escapeHTML(title)}</h3>
<p style="text-align:center;color:#777;font-size:13px;margin-bottom:20px">${escapeHTML(titleEn)}</p>
<div class="patient-box">
  <div class="field"><label>??? ?????? / Patient Name:</label> ${escapeHTML(form.patient_name || '')}</div>
  <div class="field"><label>????? ???????? / Date:</label> ${signedDate}</div>
  <div class="field"><label>??????? ???????? / Doctor:</label> ${escapeHTML(form.doctor_name || '')}</div>
  <div class="field"><label>??????? / Status:</label> <span class="status-badge ${form.status === 'Signed' ? 'status-signed' : 'status-pending'}">${form.status === 'Signed' ? '?' : '???? Signed' : '?' : '???? Pending'}</span></div>
</div>

```

```

</div>
<div class="consent-section">
  <h3>? ? ???? ? Consent Declaration</h3>
  <div class="consent-text-ar">${arabicContent.map(escapeHTML).join('<br>')}</div>
  ${englishContent.length ? `<div class="consent-text-en">${englishContent.map(escapeHTML).join('<br>')}</div>` : ''}
</div>
<div class="sig-grid">
  <div class="sig-box">
    <div class="sig-label">???? ????</div>
    <div class="sig-label-en">Patient Signature</div>
    <div class="sig-area">${sigImg}</div>
  </div>
  <div class="sig-box">
    <div class="sig-label">???? ????</div>
    <div class="sig-label-en">Physician Signature</div>
    <div class="sig-area"><div class="sig-line">${escapeHTML(form.doctor_name || '')}</div></div>
  </div>
  <div class="sig-box">
    <div class="sig-label">???? ????</div>
    <div class="sig-label-en">Witness</div>
    <div class="sig-area"><div class="sig-line">${escapeHTML(form.witness_name || '')}</div></div>
  </div>
</div>
<div class="footer">
  ${escapeHTML(hospitalAr)} ? ${escapeHTML(hospitalEn)} | ${tr('Form #', '???? ?')} ${escapeHTML(form.id)}
  | ${tr('Printed on', '???? ?')} ${new Date().toLocaleDateString('ar-SA')}
</div>
</body></html>`);
  w.document.close();
} catch (e) { console.error(e); showToast(tr('Print error', '?? ? ????'), 'error'); }
};
window.loadConsentTemplate = () => {
  const sel = document.getElementById('cfTemplate');
  const opt = sel.options[sel.selectedIndex];
  if (opt && opt.value) {
    document.getElementById('cfTitle').value = isArabic ? (opt.dataset.titleAr || opt.dataset.title) : opt.dataset.title;
    document.getElementById('cfContent').value = opt.dataset.content || '';
  }
};
window.createConsentForm = async () => {
  const pSel = document.getElementById('cfPatient');
  const tSel = document.getElementById('cfTemplate');
  const opt = tSel.options[tSel.selectedIndex];
  try {
    await API.post('/api/consent-forms', {
      patient_id: pSel.value, patient_name: pSel.options[pSel.selectedIndex]?.dataset?.name || '',
      form_type: tSel.value || 'general', form_title: document.getElementById('cfTitle').value,
      form_title_ar: opt?.dataset?.titleAr || document.getElementById('cfTitle').value,
      content: document.getElementById('cfContent').value, doctor_name: document.getElementById('cfDoctor').value
    });
    showToast(tr('Form created!', '?? ???? ????'), 'success'); await navigateTo(20);
  } catch (e) { console.error(e); showToast(tr('Print error', '?? ? ????'), 'error'); }
};

```

```

    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.loadConsentForSign = async () => {
    const fid = document.getElementById('cfSignSelect').value;
    if (!fid) { document.getElementById('cfSignArea').style.display = 'none'; return; }
    document.getElementById('cfSignArea').style.display = 'block';
    try {
        const f = await API.get(`/api/consent-forms/${fid}`);
        document.getElementById('cfSignContent').innerHTML = `

### > ${escapeHTML(isArabic ? (f.form_title_ar || f .form_title) : f.form_title)}</h3><p>${escapeHTML(f.content)}</p><p><strong>${tr('Patient', '?????')}</stro ng> ${escapeHTML(f.patient_name)}<br><strong>${tr('Doctor', '?????')}</strong> ${escapeHTML(f.doctor_name)}< /p>`; // Setup canvas setTimeout(() => { const canvas = document.getElementById('cfSigCanvas'); if (!canvas) return; const ctx = canvas.getContext('2d'); let drawing = false; canvas.onpointerdown = (e) => { drawing = true; ctx.beginPath(); ctx.moveTo(e.offsetX, e.offsetY); }; canvas.onpointermove = (e) => { if (!drawing) return; ctx.lineTo(e.offsetX, e.offsetY); ctx.strokeStyle = '#000'; ctx.lineWidth = 2; ctx.stroke(); }; canvas.onpointerup = () => drawing = false; canvas.onpointerout = () => drawing = false; }, 100); } catch (e) { console.error(e); } }; window.clearSigCanvas = () => { const c = document.getElementById('cfSigCanvas'); if (c) c.getContext('2d').clearRect(0, 0, c.width, c.height); }; window.signConsentForm = async () => { const fid = document.getElementById('cfSignSelect').value; if (!fid) return; const canvas = document.getElementById('cfSigCanvas'); const sig = canvas ? canvas.toDataURL('image/png') : ''; try { await API.put(`/api/consent-forms/${fid}/sign`, { patient_signature: sig, witness_name: document.getElementById('cfWitness').value }); showToast(tr('Consent signed!', '?? ????? ???????!')); await navigateTo(20); } catch (e) { showToast(tr('Error', '???'), 'error'); } }; // ===== EMERGENCY DEPARTMENT ===== let erTab = 'board'; async function renderEmergency(el) { // Show premium loading skeletons instantly el.innerHTML = ` <div class="page-title">? ${tr('Emergency Department', '???????')}</div> <div class="stats-grid" style="grid-template-columns:repeat(5,1fr)"> <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div> <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div> <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div> <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div> </div> `; }


```

```

    <div class="stat-card glass-card-premium"><div class="skeleton-bar"></div></div>
  </div>
  <div class="card glass-card-premium mt-16">
    <div class="skeleton-bar title mb-16"></div>
    <div class="skeleton-bar mb-12"></div>
  </div>
`;

let stats = {};
let visits = [];
let beds = [];
let patients = [];
let doctors = [];
let board = [];
try {
  [stats, visits, beds, patients, doctors, board] = await Promise.all([
    API.get('/api/emergency/stats'),
    API.get('/api/emergency/visits'),
    API.get('/api/emergency/beds'),
    API.get('/api/patients'),
    API.get('/api/employees'),
    API.get('/api/er/board')
  ]);
} catch (e) {
  el.innerHTML = `
    <div class="page-title">? ${tr('Emergency Department', '???????')}</div>
    <div class="error-card-premium">
      <div class="error-card-icon">??</div>
      <h3>${tr('Failed to load emergency data', '??? ?? ????? ?????? ??????')}</h3>
      <p>${escapeHTML(e.message || e)}</p>
      <button class="btn btn-primary" onclick="navigateTo(currentPage)">? ${tr('Retry', '????? ?????????')}</
button>
    </div>
  `;
  return;
}
window.erVisits = visits;
const drs = (doctors || []).filter(d => d.role === 'Doctor' || d.department_en === 'Emergency');
const triageColors = { Red: '#e74c3c', Orange: '#e67e22', Yellow: '#f1c40f', Green: '#2ecc71', Blue: '#3498db' };
const active = (visits || []).filter(v => v.status === 'Active');
const discharged = (visits || []).filter(v => v.status === 'Discharged');
const admitted = (visits || []).filter(v => v.status === 'Admitted');
el.innerHTML = `<div class="page-title">? ${tr('Emergency Department', '???????')}</div>
  <div class="stats-grid" style="grid-template-columns:repeat(5,1fr)">
    <div class="stat-card"><div class="stat-icon" style="background:#e74c3c22;color:#e74c3c">?</div><div class="stat-value" style="color:#e74c3c">${stats.active}</div><div class="stat-label">${tr('Active Cases', '?????')}</div></div>
    <div class="stat-card"><div class="stat-icon" style="background:#e67e2222;color:#e67e22">?</div><div class="stat-value" style="color:#e67e22">${stats.critical}</div><div class="stat-label">${tr('Critical', '????')}</div></div>
    <div class="stat-card"><div class="stat-icon" style="background:#3498db22;color:#3498db">?</div><div class="stat-value" style="color:#3498db">${stats.today}</div><div class="stat-label">${tr('Today', '????')}</div>
  </div>

```



```

        <div class="stat-card"><div class="stat-icon" style="background:#2ecc7122;color:#2ecc71">?</div><div cla
ss="stat-value" style="color:#2ecc71">${discharged.length}</div><div class="stat-label">${tr('Discharged', '??
????')}</div></div>

        <div class="stat-card"><div class="stat-icon" style="background:#9b59b622;color:#9b59b6">?</div><div cla
ss="stat-value" style="color:#9b59b6">${admitted.length}</div><div class="stat-label">${tr('Transferred', '???
????')}</div></div>
    </div>

    <div class="tab-bar"><button class="tab-btn ${erTab === 'board' ? 'active' : ''}" onclick="erTab='board';n
avigateTo(21)">? ${tr('ER Board', '???? ?????')}</button>

    <button class="tab-btn ${erTab === 'register' ? 'active' : ''}" onclick="erTab='register';navigateTo(21)
">? ${tr('Register', '???? ?????')}</button>

    <button class="tab-btn ${erTab === 'triage' ? 'active' : ''}" onclick="erTab='triage';navigateTo(21)">?
${tr('Triage (ESI)', '???? ESI')}</button>

    <button class="tab-btn ${erTab === 'discharged' ? 'active' : ''}" onclick="erTab='discharged';navigateTo
(21)">? ${tr('Discharged', '????????')}</button>

    <button class="tab-btn ${erTab === 'transferred' ? 'active' : ''}" onclick="erTab='transferred';navigate
To(21)">? ${tr('Transferred', '???????? ?????')}</button>

    <button class="tab-btn ${erTab === 'beds' ? 'active' : ''}" onclick="erTab='beds';navigateTo(21)">?? ${t
r('Bed Map', '???? ?????')}</button></div>

    <div class="card glass-card-premium" id="erContent"></div>

    <div id="erDischargeModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;backgrou
nd:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">

        <div style="background:#fff;padding:30px;border-radius:16px;width:500px;max-width:90%;max-height:80vh;ov
erflow-y:auto;direction:rtl">

            <h3 style="margin-bottom:16px">? ${tr('Discharge from ER', '???? ?? ?????')}</h3>

            <input type="hidden" id="erDischargeId">

            <div class="form-grid" style="gap:12px">

                <div><label>${tr('Diagnosis', '????????')}</label><textarea id="erDischargeDiag" class="form-control"
rows="2"></textarea></div>

                <div><label>${tr('Instructions', '???????? ?????')}</label><textarea id="erDischargeInst" class="for
m-control" rows="2"></textarea></div>

                <div><label>${tr('Medications', '????????')}</label><input id="erDischargeMeds" class="form-control">
</div>

                <div><label>${tr('Follow-up Date', '???? ?????')}</label><input id="erDischargeFollowup" type="da
te" class="form-control"></div>

            </div>

            <div style="display:flex;gap:10px;margin-top:16px">

                <button class="btn btn-primary" onclick="confirmERDischarge()" style="flex:1">? ${tr('Confirm Discha
rge', '???? ?????')}</button>

                <button class="btn" onclick="document.getElementById('erDischargeModal').style.display='none'" style
="flex:1">? ${tr('Cancel', '????')}</button>

            </div>
        </div>
    </div>

    <div id="erDispositionModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;backgr
ound:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">

        <div style="background:#fff;padding:30px;border-radius:16px;width:520px;max-width:90%;max-height:85vh;ov
erflow-y:auto;direction:rtl">

            <h3 style="margin-bottom:16px">? ${tr('ED Disposition', '????? ?????')} ? <span id="erDispPatient"><
/span></h3>

            <input type="hidden" id="erDispId">

            <div class="form-grid" style="gap:12px">

                <div><label>${tr('Disposition', '????????')}</label><select id="erDispType" class="form-control">

                    <option value="Discharged">${tr('Discharge home', '???? ?????')}</option>

```

```

        <option value="Admitted">${tr('Admit (inpatient ADT)', '????')}</option>
        <option value="Transferred">${tr('Transfer out', '???? ????')}</option>
        <option value="LWBS">${tr('Left without being seen (LWBS)', '??? ? ?')}</option></select></div>
v>

        <div><label>${tr('Diagnosis', '????')}</label><input id="erDispDiag" class="form-control"></div>
        <div><label>${tr('Instructions', '????')}</label><textarea id="erDispInst" class="form-control"
rows="2"></textarea></div>
        <div><label>${tr('Medications', '????')}</label><input id="erDispMeds" class="form-control"></div>
>

        <div><label>${tr('Follow-up Date', '??? ????')}</label><input id="erDispFollowup" type="date" class="form-control"></div>
        </div>
        <div style="display:flex;gap:10px;margin-top:16px">
            <button class="btn btn-primary" onclick="confirmERDisposition()" style="flex:1">? ${tr('Confirm', '???')}</button>
            <button class="btn" onclick="document.getElementById('erDispositionModal').style.display='none'" style="flex:1">? ${tr('Cancel', '???')}</button>
        </div>
    </div>`;

const c = document.getElementById('erContent');
if (erTab === 'board') {
    // ESI-sorted tracking board (server-ordered: ESI priority then arrival). Columns:
    // patient, ESI, complaint, location/bed, time-since-arrival, time-to-provider, phase, actions.
    const rows = (board || []);
    const phaseLabel = (p) => ({ Arrival: tr('Arrival', '???'), Triage: tr('Triage', '???'), Waiting: tr('Waiting', '????'), InTreatment: tr('In Treatment', '???'), Disposition: tr('Disposition', '???') }[p] || escapeHTML(p));
    c.innerHTML = `<h3>? ${tr('ED Tracking Board', '???')} ($${rows.length}) ? ${tr('sorted by ESI', '??? ESI')}</h3>
        <input class="form-control" placeholder="${tr('Search...', '???')}" oninput="filterTable(this, 'erTable')" style="margin-bottom:12px">
        $${rows.length} ? <table class="data-table" id="erTable"><thead><tr><th>#</th><th>${tr('Patient', '????')}</th><th>ESI</th><th>${tr('Complaint', '????')}</th><th>${tr('Location', '????')}</th><th>${tr('Wait', '????')}</th><th>${tr('To Provider', '???')}</th><th>${tr('Phase', '????')}</th><th>${tr('Doctor', '????')}</th><th>${tr('Actions', '????')}</th></tr></thead><tbody>${rows.map(v => {
            const tc = triageColors[v.triage_color] || '#999';
            const esiTxt = v.esi_level ? ('ESI ' + v.esi_level) : tr('Untriaged', '???');
            const ttp = v.time_to_provider_min != null && v.provider_assigned_at ? (v.time_to_provider_min + 'm') : (v.time_to_provider_breach ? `<span style="color:#e74c3c;font-weight:700">? ${tr('BREACH', '???')}</span>` : '-');
            return `<tr><td>${escapeHTML(v.id)}</td><td>${escapeHTML(v.patient_name)}</td>
                <td><span class="badge" style="background:${tc};color:#fff;font-weight:700">? ${escapeHTML(esiTxt)}</span></td>
                <td>${escapeHTML(v.chief_complaint_ar || v.chief_complaint || '-')}</td>
                <td>${escapeHTML(v.location || '-')}</td>
                <td>${escapeHTML(String(v.minutes_since_arrival != null ? v.minutes_since_arrival : 0))}m</td>
                <td>${rawHtml(tpp)}</td>
                <td>${rawHtml(phaseLabel(v.phase))}</td>
                <td>${escapeHTML(v.assigned_doctor || '-')}</td>
                <td>
                    <button class="btn btn-sm" onclick="showERTriageModal(${safeId(v.id)},'${jsStr(v.chief_complaint || '-')}', '${jsStr(v.chief_complaint_ar || '-')}')">? ${tr('Triage', '???')}</button>
                    <button class="btn btn-sm" onclick="assignERProvider(${safeId(v.id)})">???? ${tr('Assign', '???')}
                </td>
            `;
        })}`;
}

```

```

    }</button>
        <button class="btn btn-sm btn-warning" onclick="showERDispositionModal(${safeId(v.id)}, '${jsStr(v.
patient_name || ' ')}')">? ${tr('Disposition', '????')}</button>
        </td></tr>`;
    }).join('')</tbody></table>` : `<div class="empty-state-card"><div class="empty-state-icon"></div><div>${
tr('No active emergency cases', '?? ???? ????? ?????')}</div></div>`;
    } else if (erTab === 'triage') {
        c.innerHTML = `<h3>? ${tr('ESI Triage (server computes the level)', '??? ESI (?????? ??????)')}</h
3>
        <p style="color:#888;font-size:.9em">${tr('Enter vitals and presentation. The ESI level is computed on t
he server from danger-zone vitals, high-risk presentation, and anticipated resources ? it cannot be set by han
d.', '???? ????????? ??????? ?????? ????????. ????? ?????? ESI ??? ?????? ?? ????????? ?????? ?????? ???? ???????
????????? ????????? ? ??? ????? ?????? ??????')}</p>

        <div id="pediatricTriageTip" class="hidden mb-12 p-12 rounded border-l-4" style="background:rgba(230,126
,34,0.1);border-left-color:#e67e22;font-size:12px"></div>

        <div class="form-grid">
            <div><label>${tr('Active ER Visit', '???? ?????')}</label><select id="trVisit" class="form-contr
ol" onchange="window.onERTriageVisitChange(this.value)"><option value="">${tr('Select', '????')}</option>${act
ive.map(v => `<option value="${safeId(v.id)}">#${escapeHTML(v.id)} - ${escapeHTML(v.patient_name)} (${escapeHT
ML(v.chief_complaint_ar || v.chief_complaint || ' ')}</option>`).join('')}</select></div>
            <div><label>${tr('Heart Rate (HR)', '????')}</label><input id="trHr" type="number" class="form-contr
ol"></div>
            <div><label>${tr('Resp Rate (RR)', '????')}</label><input id="trRr" type="number" class="form-contr
ol"></div>
            <div><label>${tr('SpO2 %', '?????? %')}</label><input id="trSpo2" type="number" class="form-control"
></div>
            <div><label>${tr('Systolic BP', '???? ?????')}</label><input id="trSbp" type="number" class="form-
control"></div>
            <div><label>${tr('Temp °C', '????')}</label><input id="trTemp" type="number" step="0.1" class="form-
control"></div>
            <div><label>${tr('Age (years)', '???? (????)')}</label><input id="trAge" type="number" class="form-c
ontrol"></div>
            <div><label>${tr('LOC (AVPU)', '???? ?????')}</label><select id="trLoc" class="form-control"><option
value="A">${tr('Alert', '????')}</option><option value="V">${tr('Verbal', '?????')}</option><option va
lue="P">${tr('Pain', '?????')}</option><option value="U">${tr('Unresponsive', '?? ?????')}</option></s
elect></div>
            <div><label>${tr('Pain (0-10)', '???? (0-10)')}</label><input id="trPain" type="number" min="0" max="
10" class="form-control"></div>
            <div><label>${tr('Anticipated Resources', '????? ?????')}</label><input id="trResources" type="nu
mber" min="0" class="form-control" placeholder="${tr('e.g. labs+imaging = 2', '????: ?????+???? = 2')}</div>
            >
            <div><label>${tr('High-risk presentation', '??? ???? ?????')}</label><select id="trHighRisk" class="
form-control"><option value="">${tr('No', '??')}</option><option value="1">${tr('Yes', '??')}</option></selec
t></div>
            <div><label>${tr('Requires life-saving intervention', '???? ?????? ??????')}</label><select id
="trLifesaving" class="form-control"><option value="">${tr('No', '??')}</option><option value="1">${tr('Yes',
'??')}</option></select></div>
        </div>
        <button class="btn btn-primary" onclick="submitERTriage()" style="margin-top:16px">? ${tr('Compute ESI &
Triage', '???? ESI ?????')}</button>
        <div id="trResult" style="margin-top:16px"></div>`;
    } else if (erTab === 'register') {

```

```

c.innerHTML = `<h3>? ${tr('Register ER Visit', '????? ???? ?????')}</h3>
<div class="form-grid">
  <div><label>${tr('Patient', '?????')}</label><select id="erPatient" class="form-control"><option valu
e="">${tr('Select', '????')}</option>${(patients || []).map(p => `<option value="${safeId(p.id)}" data-name="$
${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.name_ar || p.name_en)} (${escapeHTML(p.file_number)})</op
tion>`).join('')}</select></div>
  <div><label>${tr('Arrival Mode', '????? ?????')}</label><select id="erArrival" class="form-control"><
option value="Walk-in">${tr('Walk-in', '????')}</option><option value="Ambulance">${tr('Ambulance', '????')}</
option><option value="Referred">${tr('Referred', '????')}</option><option value="Police">${tr('Police', '????
')}</option></select></div>
  <div><label>${tr('Chief Complaint', '?????? ???????')}</label><input id="erComplaint" class="form-con
trol"></div>
  <div><label>${tr('Complaint (AR)', '?????? ???????')}</label><input id="erComplaintAr" class="form-con
trol"></div>
  <div><label>${tr('Triage Level', '????? ?????')}</label><select id="erTriage" class="form-control" onc
hange="document.getElementById('erTriageColor').value=['','Red','Orange','Yellow','Green','Blue'][this.value]"
>
    <option value="1">1 - ${tr('Resuscitation', '????')}</option><option value="2">2 - ${tr('Emergent',
'????')}</option><option value="3" selected>3 - ${tr('Urgent', '????')}</option><option value="4">4 - ${tr('L
ess Urgent', '??? ?????')}</option><option value="5">5 - ${tr('Non-Urgent', '??? ?????')}</option></select></
div>
  <div><label>${tr('Triage Color', '??? ?????')}</label><select id="erTriageColor" class="form-control">
<option value="Red">${tr('Red', '????')}</option><option value="Orange">${tr('Orange', '??????')}</option><op
tion value="Yellow" selected>${tr('Yellow', '????')}</option><option value="Green">${tr('Green', '????')}</opt
ion><option value="Blue">${tr('Blue', '????')}</option></select></div>
  <div><label>${tr('Doctor', '?????')}</label><select id="erDoctor" class="form-control"><option value=
"">${tr('Select', '????')}</option>${(drs.map(d => `<option value="${escapeHTML(d.name_ar || d.name)}">${escape
HTML(d.name_ar || d.name)}</option>`).join('')}</select></div>
  <div><label>${tr('ER Bed', '???? ???????')}</label><select id="erBed" class="form-control"><option val
ue="">${tr('None', '????')}</option>${(beds || []).filter(b => b.status === 'Available').map(b => `<option val
ue="${escapeHTML(b.bed_name)}">${escapeHTML(b.bed_name_ar)} (${escapeHTML(b.zone_ar)})</option>`).join('')}</s
elect></div>
</div>
<button class="btn btn-primary" onclick="registerERVisit()" style="margin-top:16px">? ${tr('Register', '
????')}</button>`;
} else if (erTab === 'discharged') {
c.innerHTML = `<h3>? ${tr('Discharged from ER', '???????? ?? ???????')} (${discharged.length})</h3>
<input class="form-control" placeholder="${tr('Search...', '????...')}" oninput="filterTable(this, 'erDisc
hTable')" style="margin-bottom:12px">
  ${discharged.length ? `<table class="data-table" id="erDischTable"><thead><tr><th>#</th><th>${tr('Patien
t', '?????')}</th><th>${tr('Complaint', '?????')}</th><th>${tr('Triage', '????')}</th><th>${tr('Arrival', '
?????')}</th><th>${tr('Discharge', '?????')}</th><th>${tr('Doctor', '?????')}</th><th>${tr('Diagnosis', '??
????')}</th></tr></thead><tbody>${discharged.map(v => {
  const tc = triageColors[v.triage_color] || '#999';
  return `<tr><td>${escapeHTML(v.id)}</td><td>${escapeHTML(v.patient_name)}</td><td>${escapeHTML(v.chief_c
omplaint_ar || v.chief_complaint || '-')}</td>
    <td><span style="background:${tc};color:#fff;padding:2px 8px;border-radius:12px;font-size:.85em">${esc
apeHTML(v.triage_color)}</span></td>
    <td>${v.arrival_time ? new Date(v.arrival_time).toLocaleString('ar-SA', { month: 'short', day: 'numeri
c', hour: '2-digit', minute: '2-digit' }) : '-'}</td>
    <td>${v.discharge_time ? new Date(v.discharge_time).toLocaleString('ar-SA', { month: 'short', day: 'nu
meric', hour: '2-digit', minute: '2-digit' }) : '-'}</td>
    <td>${escapeHTML(v.assigned_doctor || '-')}</td><td>${escapeHTML(v.discharge_diagnosis || '-')}</td></
tr>`;
}

```

```

    }).join('')}</tbody></table>` : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>$
{tr('No discharged emergency patients', '?? ???? ???? ?????? ?? ???????')}</div></div>`;
    } else if (erTab === 'transferred') {
        c.innerHTML = `<h3>? ${tr('Transferred to Inpatient', '???????? ???????')} (${admitted.length})</h3>
        <input class="form-control" placeholder="${tr('Search...', '????...')}" oninput="filterTable(this, 'erTran
sTable')" style="margin-bottom:12px">
        ${admitted.length ? `<table class="data-table" id="erTransTable"><thead><tr><th>#</th><th>${tr('Patient'
, '??????')}</th><th>${tr('Complaint', '??????')}</th><th>${tr('Triage', '?????')}</th><th>${tr('ER Doctor', '
???? ???????')}</th><th>${tr('Arrival', '??????')}</th><th>${tr('Status', '??????')}</th></tr></thead><tbody>$
{admitted.map(v => {
            const tc = triageColors[v.triage_color] || '#999';
            return `<tr><td>${escapeHTML(v.id)}</td><td>${escapeHTML(v.patient_name)}</td><td>${escapeHTML(v.chief_c
omplaint_ar || v.chief_complaint || '-')}</td>
            <td><span style="background:${tc};color:#fff;padding:2px 8px;border-radius:12px;font-size:.85em">${esc
apeHTML(v.triage_color)}</span></td>
            <td>${escapeHTML(v.assigned_doctor || '-')}</td>
            <td>${v.arrival_time ? new Date(v.arrival_time).toLocaleString('ar-SA', { month: 'short', day: 'numeri
c', hour: '2-digit', minute: '2-digit' }) : '-'}</td>
            <td>${badge(tr('Admitted', '?? ???????'), 'success')}</td></tr>`;
        }).join('')}</tbody></table>` : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>$
{tr('No transferred patients', '?? ???? ???? ???????')}</div></div>`;
    } else {
        const zones = ['Resuscitation', 'Critical', 'Acute', 'Observation'];
        c.innerHTML = `<h3>?? ${tr('ER Bed Map', '????? ?????? ???????')}</h3>
        ${zones.map(z => `<h4 style="margin:16px 0 8px">${tr(z, z === 'Resuscitation' ? '???????' : z === 'Criti
cal' ? '???????' : z === 'Acute' ? '???????' : '?????????')}</h4>
            <div style="display:grid;grid-template-columns:repeat(4,1fr);gap:10px">${(beds || []).filter(b => b.zo
ne === z).map(b => `<div style="padding:16px;border-radius:12px;text-align:center;background:${b.status === 'A
vailable' ? '#d4edda' : '#f8d7da'};border:2px solid ${b.status === 'Available' ? '#28a745' : '#dc3545'}">
                <div style="font-size:1.4em;font-weight:700">${escapeHTML(b.bed_name_ar)}</div><div style="font-size
:.85em;margin-top:4px">${statusBadge(b.status)}</div></div>`).join('')}</div>`).join('')}`;
        }
    }
}
window.registerERVisit = async function () {
    const ps = document.getElementById('erPatient'); if (!ps.value) return showToast(tr('Select patient', '????
??????'), 'error');
    try {
        await API.post('/api/emergency/visits', { patient_id: ps.value, patient_name: ps.options[ps.selectedIndex]
.dataset.name, arrival_mode: document.getElementById('erArrival').value, chief_complaint: document.getElementB
yId('erComplaint').value, chief_complaint_ar: document.getElementById('erComplaintAr').value, triage_level: do
cument.getElementById('erTriage').value, triage_color: document.getElementById('erTriageColor').value, assigne
d_doctor: document.getElementById('erDoctor').value, assigned_bed: document.getElementById('erBed').value });
        showToast(tr('ER visit registered!', '?? ?????? ???? ???????!')); erTab = 'board'; await navigateTo(21);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.showERDischargeModal = function (id) {
    document.getElementById('erDischargeId').value = id;
    document.getElementById('erDischargeDiag').value = '';
    document.getElementById('erDischargeInst').value = '';
    document.getElementById('erDischargeMeds').value = '';
    document.getElementById('erDischargeFollowup').value = '';
    document.getElementById('erDischargeModal').style.display = 'flex';
};
window.confirmERDischarge = async function () {

```

```

const id = document.getElementById('erDischargeId').value;
try {
  // Terminal disposition MUST go through the guarded /api/er/disposition route (server-authoritative
  // state machine + ESI/triage gating). The legacy PUT no longer accepts terminal statuses.
  const r = await API.post('/api/er/disposition', {
    visit_id: Number(id),
    disposition_type: 'Discharged',
    diagnosis: document.getElementById('erDischargeDiag').value,
    instructions: document.getElementById('erDischargeInst').value,
    medications: document.getElementById('erDischargeMeds').value,
    followup_date: document.getElementById('erDischargeFollowup').value
  });
  if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
  document.getElementById('erDischargeModal').style.display = 'none';
  showToast(tr('Patient discharged from ER!', '?? ??? ???? ? ? ??????'));
  erTab = 'discharged'; await navigateTo(21);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.transferERTOInpatient = async function (visitId, patientName, patientId, doctor, complaint) {
  if (!confirm(tr('Transfer this patient to inpatient?', '?? ??? ???? ? ? ????? ????')) return;
  try {
    // The disposition route performs the ER->ADT admission handoff internally; do NOT also POST
    // /api/admissions here (would double-admit) nor use the legacy PUT.
    const r = await API.post('/api/er/disposition', {
      visit_id: Number(visitId),
      disposition_type: 'Admitted',
      admission_department: 'Emergency',
      admitting_doctor: doctor,
      diagnosis: complaint
    });
    if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
    showToast(tr('Patient transferred to inpatient!', '?? ??? ???? ????'));
    erTab = 'transferred'; await navigateTo(21);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== E7: ESI triage / assign-provider / disposition (call the GUARDED /api/er/* routes) =====
window._erTriagePreselect = null;
window.showERTriageModal = function (visitId) {
  // Jump to the triage tab with this visit pre-selected (server computes ESI from inputs).
  window._erTriagePreselect = visitId;
  erTab = 'triage';
  navigateTo(21).then(() => {
    const sel = document.getElementById('trVisit');
    if (sel && visitId) {
      sel.value = String(visitId);
      window.onERTriageVisitChange(visitId);
    }
  });
};

window.onERTriageVisitChange = async function(visitId) {
  if (!visitId) {
    const tip = document.getElementById('pediatricTriageTip');

```

```

    if (tip) { tip.innerHTML = ''; tip.classList.add('hidden'); }
    return;
}
try {
    const visit = (window.erVisits || []).find(v => String(v.id) === String(visitId));
    if (visit && visit.patient_id) {
        const patient = await API.get('/api/patients/' + visit.patient_id);
        if (patient && patient.birth_date) {
            const birth = new Date(patient.birth_date);
            const ageDifMs = Date.now() - birth.getTime();
            const ageDate = new Date(ageDifMs);
            const ageYears = Math.abs(ageDate.getUTCFullYear() - 1970);

            const ageInput = document.getElementById('trAge');
            if (ageInput) ageInput.value = ageYears;

            const tip = document.getElementById('pediatricTriageTip');
            if (tip) {
                if (ageYears < 1) {
                    tip.innerHTML = `? <strong>${tr('Pediatric Neonate/Infant Danger Thresholds:', '???? ??????? ??????
? ??????? ??????? (??? ?? ???):')}</strong><br>
                    ? ${tr('Heart Rate (HR) > 180 bpm', '???? ?????? ?????? > 180 ???/????')}<br>
                    ? ${tr('Respiratory Rate (RR) > 50 bpm', '???? ??????? > 50 ???/????')}<br>
                    ? ${tr('Temperature > 38.0°C (100.4°F) is a critical sepsis indicator', '???????
> 38.0°? (???? ????)')}`;
                    tip.className = 'mb-12 p-12 rounded border-l-4 block';
                    tip.style.background = 'rgba(239, 68, 68, 0.1)';
                    tip.style.borderLeftColor = 'red';
                } else if (ageYears <= 3) {
                    tip.innerHTML = `? <strong>${tr('Pediatric Toddler Danger Thresholds (1-3 years):', '???? ???????
??????? ???????? (1-3 ?????):')}</strong><br>
                    ? ${tr('Heart Rate (HR) > 160 bpm', '???? ?????? ?????? > 160 ???/????')}<br>
                    ? ${tr('Respiratory Rate (RR) > 40 bpm', '???? ??????? > 40 ???/????')}`;
                    tip.className = 'mb-12 p-12 rounded border-l-4 block';
                    tip.style.background = 'rgba(245, 158, 11, 0.1)';
                    tip.style.borderLeftColor = '#f59e0b';
                } else if (ageYears <= 8) {
                    tip.innerHTML = `? <strong>${tr('Pediatric Child Danger Thresholds (4-8 years):', '???? ??????? ??
????? (4-8 ?????):')}</strong><br>
                    ? ${tr('Heart Rate (HR) > 140 bpm', '???? ?????? ?????? > 140 ???/????')}<br>
                    ? ${tr('Respiratory Rate (RR) > 30 bpm', '???? ??????? > 30 ???/????')}`;
                    tip.className = 'mb-12 p-12 rounded border-l-4 block';
                    tip.style.background = 'rgba(245, 158, 11, 0.1)';
                    tip.style.borderLeftColor = '#f59e0b';
                } else {
                    tip.innerHTML = `? <strong>${tr('Adult ESI Danger Thresholds (>8 years):', '???? ??????? ????????
(>8 ?????):')}</strong><br>
                    ? ${tr('Heart Rate (HR) > 100 bpm or Respiratory Rate (RR) > 20 bpm', '????? > 10
0 ?? ?????? > 20 ???/????')}<br>
                    ? ${tr('Oxygen Saturation SpO2 < 92%', '???? ???????? SpO2 < 92%')}`;
                    tip.className = 'mb-12 p-12 rounded border-l-4 block';
                    tip.style.background = 'rgba(59, 130, 246, 0.1)';
                    tip.style.borderLeftColor = '#3b82f6';
                }
            }
}

```

```

    }
  }
}
} catch (e) {
  console.error('Failed to pre-populate patient triage details:', e);
}
};

window.submitERTriage = async function () {
  const visitId = document.getElementById('trVisit').value;
  if (!visitId) return showToast(tr('Select an ER visit', '???? ????? ?????'), 'error');
  const num = (id) => { const v = document.getElementById(id).value; return v === '' ? undefined : Number(v); };

  // NOTE: no esi_level is sent ? the server is authoritative and computes it from these inputs.
  const payload = {
    visit_id: Number(visitId),
    vitals: { hr: num('trHr'), rr: num('trRr'), spo2: num('trSpo2'), sbp: num('trSbp'), temp: num('trTemp'), l
oc: document.getElementById('trLoc').value },
    age: num('trAge'),
    pain_score: num('trPain'),
    resource_count: num('trResources'),
    high_risk: document.getElementById('trHighRisk').value === '1',
    requires_lifesaving: document.getElementById('trLifesaving').value === '1'
  };

  try {
    const r = await API.post('/api/er/triage', payload);
    if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
    const colors = { 1: '#e74c3c', 2: '#e67e22', 3: '#f1c40f', 4: '#2ecc71', 5: '#3498db' };
    const col = colors[r.esi_level] || '#999';
    const rationale = Array.isArray(r.rationale) ? r.rationale : [];
    document.getElementById('trResult').innerHTML = `
      <div class="card" style="border-left:6px solid ${col};padding:16px">
        <div style="font-size:1.3em;font-weight:700;color:${col}">? ESI ${escapeHTML(r.esi_level)} ? ${escapeH
TML(r.triage_color)}${r.danger_zone ? ' ?? ' + tr('DANGER ZONE', '????? ????') : ''}${r.fail_safe ? ' (' + tr('
fail-safe', '??????? ????') + ')' : ''}</div>
        <div style="margin-top:8px"><b>${tr('Decision point', '???? ?????')}</b> ${escapeHTML(r.decision_poi
nt)}</div>
        <ul style="margin-top:8px">${rationale.map(x => `<li>${escapeHTML(x)}</li>`).join('')}</ul>
      </div>`;
    showToast(tr('Triage recorded (server-computed ESI)', '?? ????? (ESI ????? ??????)'));
  } catch (e) {
    showToast((e && e.message) ? e.message : tr('Triage failed', '??? ?????'), 'error');
  }
};

window.assignERProvider = async function (visitId) {
  const provider = prompt(tr('Provider name (leave blank = you):', '??? ?????? (???? = ???):')) || '';
  try {
    const r = await API.post('/api/er/assign-provider', { visit_id: Number(visitId), provider });
    if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
    showToast(tr('Provider assigned', '?? ?????? ??????') + ' (TTP ' + (r.time_to_provider_min != null ? r.time
_to_provider_min : '-') + 'm)');
    await navigateTo(21);
  } catch (e) {
    showToast((e && e.message) ? e.message : tr('Assign failed', '??? ??????'), 'error');
  }
}

```



```

};
window.showERDispositionModal = function (visitId, patientName) {
    document.getElementById('erDispId').value = visitId;
    document.getElementById('erDispPatient').textContent = patientName || '';
    document.getElementById('erDispType').value = 'Discharged';
    document.getElementById('erDispDiag').value = '';
    document.getElementById('erDispInst').value = '';
    document.getElementById('erDispMeds').value = '';
    document.getElementById('erDispFollowup').value = '';
    document.getElementById('erDispositionModal').style.display = 'flex';
};
window.confirmERDisposition = async function () {
    const id = document.getElementById('erDispId').value;
    const payload = {
        visit_id: Number(id),
        disposition_type: document.getElementById('erDispType').value,
        diagnosis: document.getElementById('erDispDiag').value,
        instructions: document.getElementById('erDispInst').value,
        medications: document.getElementById('erDispMeds').value,
        followup_date: document.getElementById('erDispFollowup').value
    };
    try {
        const r = await API.post('/api/er/disposition', payload);
        if (r && r.error) { showToast(escapeHTML(r.error), 'error'); return; }
        document.getElementById('erDispositionModal').style.display = 'none';
        showToast(tr('Disposition recorded', '?? ????? ???????') + ': ' + payload.disposition_type + (r.admission_
id ? ' ? ADT #' + r.admission_id : ''));
        await navigateTo(21);
    } catch (e) {
        showToast((e && e.message) ? e.message : tr('Disposition failed', '??? ???????'), 'error');
    }
};

// ===== INPATIENT ADT =====
let adtTab = 'census';
async function renderInpatient(el) {
    const [census, activeAdm, dischargedAdm, patients, doctors, wards] = await Promise.all([
        API.get('/api/beds/census'), API.get('/api/admissions?status=Active'),
        API.get('/api/admissions?status=Discharged'),
        API.get('/api/patients'), API.get('/api/employees'), API.get('/api/wards')
    ]);
    const drs = (doctors || []).filter(d => d.role === 'Doctor');
    el.innerHTML = `<div class="page-title">?? ${tr('Inpatient ADT', '???????')}</div>
<div class="stats-grid" style="grid-template-columns:repeat(5,1fr)">
    <div class="stat-card"><div class="stat-value" style="color:#2ecc71">${census.available || 0}</div><div
class="stat-label">${tr('Available Beds', '????? ?????')}</div></div>
    <div class="stat-card"><div class="stat-value" style="color:#e74c3c">${census.occupied || 0}</div><div c
lass="stat-label">${tr('Occupied', '???????')}</div></div>
    <div class="stat-card"><div class="stat-value" style="color:#3498db">${(activeAdm || []).length}</div><d
iv class="stat-label">${tr('Current Patients', '????????')}</div></div>
    <div class="stat-card"><div class="stat-value" style="color:#27ae60">${(dischargedAdm || []).length}</di
v><div class="stat-label">${tr('Discharged', '????????')}</div></div>
    <div class="stat-card"><div class="stat-value" style="color:#9b59b6">${census.occupancyRate || 0}%</div>
<div class="stat-label">${tr('Occupancy', '???? ???????')}</div></div>

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</div>
<div class="tab-bar"><button class="tab-btn ${adtTab === 'census' ? 'active' : ''}" onclick="adtTab='census';navigateTo(22)">?? ${tr('Census', '???????')}</button>
  <button class="tab-btn ${adtTab === 'admit' ? 'active' : ''}" onclick="adtTab='admit';navigateTo(22)">?
  ${tr('Admit', '????')}</button>
  <button class="tab-btn ${adtTab === 'patients' ? 'active' : ''}" onclick="adtTab='patients';navigateTo(22)">? ${tr('Patients', '????????')}</button>
  <button class="tab-btn ${adtTab === 'discharged' ? 'active' : ''}" onclick="adtTab='discharged';navigateTo(22)">? ${tr('Discharged', '????????')}</button></div>
<div class="card" id="adtContent"></div>
<div id="adtDischargeModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;background:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">
  <div style="background:#fff;padding:30px;border-radius:16px;width:550px;max-width:90%;max-height:85vh;overflow-y:auto;direction:rtl">
    <h3 style="margin-bottom:16px">? ${tr('Discharge Patient', '???? ?????')}</h3>
    <input type="hidden" id="adtDischargeId">
    <div class="form-grid" style="gap:12px">
      <div><label>${tr('Discharge Type', '??? ?????')}</label><select id="adtDischargeType" class="form-control">
        <option value="Regular">${tr('Regular', '????')}</option><option value="AMA">${tr('Against Medical Advice', '?? ?????? ?????')}</option>
        <option value="Transfer">${tr('Transfer', '????')}</option><option value="Death">${tr('Death', '????')}</option></select></div>
      <div style="grid-column:span 2"><label>${tr('Discharge Summary', '???? ?????')}</label><textarea id="adtDischargeSummary" class="form-control" rows="3"></textarea></div>
      <div style="grid-column:span 2"><label>${tr('Instructions', '??????? ?????')}</label><textarea id="adtDischargeInst" class="form-control" rows="2"></textarea></div>
      <div><label>${tr('Medications', '????? ?????')}</label><textarea id="adtDischargeMeds" class="form-control" rows="2"></textarea></div>
      <div><label>${tr('Follow-up Date', '???? ?????????')}</label><input id="adtFollowupDate" type="date" class="form-control"></div>
      <div><label>${tr('Follow-up Doctor', '???? ?????????')}</label><select id="adtFollowupDoctor" class="form-control"><option value="">${tr('Select', '????')}</option>${drs.map(d => `<option value="${escapeHTML(d.name_ar || d.name)}">${escapeHTML(d.name_ar || d.name)}</option>`).join('')}</select></div>
    </div>
    <div style="display:flex;gap:10px;margin-top:16px">
      <button class="btn btn-primary" onclick="confirmInpatientDischarge()" style="flex:1">? ${tr('Confirm Discharge', '????? ?????')}</button>
      <button class="btn" onclick="document.getElementById('adtDischargeModal').style.display='none'" style="flex:1">? ${tr('Cancel', '????')}</button>
    </div>
  </div>
</div>
<div id="adtTransferModal" style="display:none;position:fixed;top:0;left:0;width:100%;height:100%;background:rgba(0,0,0,.5);z-index:9999;align-items:center;justify-content:center">
  <div style="background:#fff;padding:30px;border-radius:16px;width:480px;max-width:90%;direction:rtl">
    <h3 style="margin-bottom:16px">? ${tr('Transfer Patient', '??? ?????')}</h3>
    <input type="hidden" id="adtTransferAdmId">
    <div id="adtTransferName" style="margin-bottom:12px;font-weight:600"></div>
    <div class="form-grid" style="gap:12px">
      <div><label>${tr('Ward', '???????')}</label><select id="adtTransferWard" class="form-control" onchange="loadTransferBeds(this.value)"><option value="">${tr('Select', '????')}</option>${(wards || []).map(w => `<option value="${safeId(w.id)}">${escapeHTML(w.ward_name_ar)}</option>`).join('')}</select></div>
      <div><label>${tr('Bed', '???????')}</label><select id="adtTransferBed" class="form-control"><option v

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value="">${tr('Select ward first', '???? ?????? ??????')}</option></select></div>
    <div style="grid-column:span 2"><label>${tr('Reason', '??? ?????')}</label><textarea id="adtTransfer
Reason" class="form-control" rows="2"></textarea></div>
    </div>
    <div style="display:flex;gap:10px;margin-top:16px">
        <button class="btn btn-primary" onclick="confirmInpatientTransfer()" style="flex:1">? ${tr('Confirm
Transfer', '????? ?????')}</button>
        <button class="btn" onclick="document.getElementById('adtTransferModal').style.display='none'" style
="flex:1">? ${tr('Cancel', '?????')}</button>
    </div>
</div>`;
const c = document.getElementById('adtContent');
if (adtTab === 'census') {
    const legendHtml = `
        <div class="bed-legend">
            <div class="legend-item"><div class="legend-color available"></div><span>${tr('Available Bed', '???? ?
???')}</span></div>
            <div class="legend-item"><div class="legend-color occupied"></div><span>${tr('Occupied Bed', '???? ???
??')}</span></div>
        </div>
    `;

    const wardsMapHtml = (census.wards || []).map(w => {
        const wBeds = (census.beds || []).filter(b => b.ward_id === w.id);
        const occupiedCount = wBeds.filter(b => b.status === 'Occupied').length;
        const totalCount = wBeds.length;
        const isHighOccupancy = totalCount > 0 && (occupiedCount / totalCount) > 0.8;

        const badgeClass = isHighOccupancy ? 'ward-occupancy-high' : 'ward-occupancy-normal';
        const badgeText = `${occupiedCount} / ${totalCount}`;

        return `
            <div class="ward-card-premium">
                <div class="ward-header">
                    <div class="ward-title-ar">? ${escapeHTML(isArabic ? w.ward_name_ar : w.ward_name)}</div>
                    <div class="ward-occupancy-badge ${badgeClass}">
                        ? ${tr('Occupancy:', '???? ???????:')} ${badgeText}
                    </div>
                </div>
                <div class="beds-grid">
                    ${wBeds.map(b => {
                        const isOccupied = b.status === 'Occupied';
                        const cardClass = isOccupied ? 'occupied' : 'available';
                        const bedIcon = isOccupied ? '?' : '??';
                        const tooltipTitle = isOccupied
                            ? `${tr('Patient:', '?????:')} ${escapeHTML(b.patient_name || '')}\\n${tr('Diagnosis:', '?????
?:')} ${escapeHTML(b.diagnosis || '')}`
                            : tr('Available Bed', '???? ?????');

                        const clickAction = isOccupied
                            ? `showBedActionsModal(${safeId(b.id)}, '${jsStr(b.patient_name || '')}')`
                            : `quickAdmitToBed(${safeId(w.id)}, ${safeId(b.id)})`;

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        return `
            <div class="bed-card-interactive ${cardClass}"
                onclick="${clickAction}"
                title="${tooltipTitle}">
                <div class="bed-icon-label">${bedIcon}</div>
                <div class="bed-number">${tr('Bed', '????')} ${escapeHTML(b.bed_number)}</div>
                <div class="bed-room">${tr('Room', '????')} ${escapeHTML(b.room_number)}</div>
                ${isOccupied ? `<div class="bed-patient-name" title="${escapeHTML(b.patient_name)}">? ${escapeHTML(b.patient_name)}</div>` : `<div class="bed-patient-name" style="color:#2ecc71; font-size:10px;">${tr('Free', '????')}</div>`}
            </div>
        `;
    }).join('')
</div>
</div>
`;
}).join('');

c.innerHTML = legendHtml + `<div class="ward-map-container">${wardsMapHtml}</div>`;
} else if (adtTab === 'admit') {
    c.innerHTML = `<h3>? ${tr('New Admission', '?????')}</h3><div class="form-grid">
        <div><label>${tr('Patient', '????')}</label><select id="admPatient" class="form-control"><option value="">${tr('Select', '????')}</option>${(patients || []).map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.name_ar || p.name_en)} (${escapeHTML(p.file_number)})</option>`).join('')}</select></div>
        <div><label>${tr('Type', '????')}</label><select id="admType" class="form-control"><option value="Regular">${tr('Regular', '????')}</option><option value="Emergency">${tr('Emergency', '????')}</option><option value="Transfer">${tr('Transfer', '????')}</option></select></div>
        <div><label>${tr('Attending Doctor', '?????')}</label><select id="admDoctor" class="form-control"><option value="">${tr('Select', '????')}</option>${(drs.map(d => `<option value="${escapeHTML(d.name_ar || d.name)}">${escapeHTML(d.name_ar || d.name)}</option>`).join('')}</select></div>
        <div><label>${tr('Department', '????')}</label><input id="admDept" class="form-control"></div>
        <div><label>${tr('Ward', '????')}</label><select id="admWard" class="form-control" onchange="loadWardBeds(this.value)"><option value="">${tr('Select', '????')}</option>${(wards || []).map(w => `<option value="${safeId(w.id)}">${escapeHTML(w.ward_name_ar)}</option>`).join('')}</select></div>
        <div><label>${tr('Bed', '????')}</label><select id="admBed" class="form-control"><option value="">${tr('Select ward first', '????')}</option></select></div>
        <div style="grid-column:span 2"><label>${tr('Diagnosis', '????')}</label><textarea id="admDiagnosis" class="form-control" rows="2"></textarea></div>
        <div><label>${tr('Diet', '????')}</label><select id="admDiet" class="form-control"><option value="Regular">${tr('Regular', '????')}</option><option value="Diabetic">${tr('Diabetic', '????')}</option><option value="Renal">${tr('Renal', '????')}</option><option value="Cardiac">${tr('Cardiac', '????')}</option><option value="NPO">${tr('NPO', '????')}</option><option value="Liquid">${tr('Liquid', '????')}</option></select></div>
        <div><label>${tr('Expected LOS', '???')}</label><input id="admLOS" type="number" value="3" class="form-control"></div>
    </div><button class="btn btn-primary" onclick="admitPatient()" style="margin-top:16px">? ${tr('Admit', '????')}</button>`;
} else if (adtTab === 'patients') {
    c.innerHTML = `<h3>? ${tr('Current Inpatients', '????')} ${((activeAdm || []).length)}</h3>
        <input class="form-control" placeholder="${tr('Search...', '???')}" oninput="filterTable(this, 'adtpatientsTable')" style="margin-bottom:12px">
        ${((activeAdm || []).length ? `<table class="data-table" id="adtpatientsTable"><thead><tr><th>#</th><th>${tr('Patient', '????')}</th><th>${tr('Type', '????')}</th><th>${tr('Doctor', '????')}</th><th>${tr('Diagnosis', '????')}</th><th>${tr('Admission Date', '????')}</th><th>${tr('Days', '????')}</th><th>${tr`

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('Actions', '???????')}</th></tr></thead><tbody>${(activeAdm || []).map(a => {
  const days = Math.floor((new Date() - new Date(a.admission_date)) / 86400000);
  const typeBadge = a.admission_type === 'Emergency' ? badge(tr('ER', '????'), 'danger') : a.admission_type === 'Transfer' ? badge(tr('Transfer', '????'), 'warning') : badge(tr('Regular', '????'), 'info');
  return `<tr><td>${escapeHTML(a.id)}</td><td><strong>${escapeHTML(a.patient_name)}</strong></td><td>${typeBadge}</td><td>${escapeHTML(a.attending_doctor || '-')}</td><td>${escapeHTML(a.diagnosis || '-')}</td><td>${new Date(a.admission_date).toLocaleDateString('ar-SA')}</td><td><span style="font-weight:700;color:${days > 7 ? '#e74c3c' : '#2ecc71'}">${days}</span></td>
    <td><button class="btn btn-sm" onclick="showAdtTransferModal(${safeId(a.id)}, '${jsStr(a.patient_name)}')">? ${tr('Transfer', '????')}</button> <button class="btn btn-sm" onclick="showInpatientDischargeModal(${safeId(a.id)})">? ${tr('Discharge', '????')}</button></td></tr>`;
  }).join('')}</tbody></table>` : `<div class="empty-state"><div class="empty-icon">??</div><p>${tr('No inpatients', '?? ???? ?????')}</p></div>`;
} else if (adtTab === 'discharged') {
  c.innerHTML = `<h3>? ${tr('Discharged Patients', '???????? ?? ?????')} (${(dischargedAdm || []).length})</h3>
    <input class="form-control" placeholder="${tr('Search...', '????...')}" oninput="filterTable(this, 'adtDischTable')" style="margin-bottom:12px">
    ${((dischargedAdm || []).length ? `<table class="data-table" id="adtDischTable"><thead><tr><th>#</th><th>${tr('Patient', '????')}</th><th>${tr('Type', '??? ?????')}</th><th>${tr('Doctor', '????')}</th><th>${tr('Diagnosis', '????')}</th><th>${tr('Admitted', '????')}</th><th>${tr('Discharged', '????')}</th><th>${tr('LOS', '????')}</th></tr></thead><tbody>${(dischargedAdm || []).map(a => {
      const los = a.discharge_date && a.admission_date ? Math.floor((new Date(a.discharge_date) - new Date(a.admission_date)) / 86400000) : '-';
      const dtBadge = a.discharge_type === 'AMA' ? badge(tr('AMA', '?? ?????'), 'danger') : a.discharge_type === 'Death' ? badge(tr('Death', '????'), 'danger') : a.discharge_type === 'Transfer' ? badge(tr('Transfer', '????'), 'warning') : badge(tr('Regular', '????'), 'success');
      return `<tr><td>${escapeHTML(a.id)}</td><td>${escapeHTML(a.patient_name)}</td><td>${dtBadge}</td><td>${escapeHTML(a.attending_doctor || '-')}</td><td>${escapeHTML(a.diagnosis || '-')}</td>
        <td>${a.admission_date ? new Date(a.admission_date).toLocaleDateString('ar-SA') : '-'}</td>
        <td>${a.discharge_date ? new Date(a.discharge_date).toLocaleDateString('ar-SA') : '-'}</td>
        <td><strong>${los}</strong> ${tr('days', '???')}</td></tr>`;
      }).join('')}</tbody></table>` : `<div class="empty-state-card"><div class="empty-state-icon">??</div><div>${tr('No discharged emergency patients', '?? ???? ???? ????? ?? ?????')}</div></div>`;
    }
  }
}

window.loadWardBeds = async function (wardId) {
  if (!wardId) return;
  // E8 fix I1: use the fail-closed, tenant-scoped ADT bed board (returns { beds: [...] }),
  // not the legacy /api/beds route which has an unscoped fallback when tenantId is null.
  const r = await API.get('/api/adt/beds?ward_id=' + wardId);
  const beds = (r && r.beds) ? r.beds : [];
  const s = document.getElementById('admBed');
  s.innerHTML = `<option value="">${tr('Select', '????')}</option>${beds.filter(b => b.status === 'Available').map(b => `<option value="${safeId(b.id)}">${tr('Bed', '????')} ${escapeHTML(b.bed_number)} - ${tr('Room', '?? ??')} ${escapeHTML(b.room_number)}</option>`).join('')}`;
};

window.showBedActionsModal = function (bedId, patientName) {
  API.get('/api/admissions?status=Active').then(admissions => {
    const adm = admissions.find(a => a.bed_id === bedId);
    if (!adm) return;
    const confirmAction = confirm(
      isArabic
        ? `???? ?????: ${patientName}\n\n???? "?????" ???? ?????? ??? ???? ???.\n???? "?????" ?????? ???? ????`

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        : `Patient Bed: ${patientName}\n\nClick "OK" to TRANSFER the patient.\nClick "Cancel" to DISCHARGE the
patient.`
    );
    if (confirmAction) {
        document.getElementById('adtTransferAdmId').value = adm.id;
        document.getElementById('adtTransferName').innerText = `${tr('Patient:', '?????:')} ${patientName}`;
        document.getElementById('adtTransferModal').style.display = 'flex';
    } else {
        document.getElementById('adtDischargeId').value = adm.id;
        document.getElementById('adtDischargeModal').style.display = 'flex';
    }
});
};
window.quickAdmitToBed = function(wardId, bedId) {
    adtTab = 'admit';
    navigateTo(22);
    setTimeout(() => {
        const wardSelect = document.getElementById('admWard');
        if (wardSelect) {
            wardSelect.value = wardId;
            window.loadWardBeds(wardId).then(() => {
                const bedSelect = document.getElementById('admBed');
                if (bedSelect) {
                    bedSelect.value = bedId;
                }
            });
        }
    }, 100);
};
window.admitPatient = async function () {
    const ps = document.getElementById('admPatient'); if (!ps.value) return showToast(tr('Select patient', '????
?????'), 'error');
    const bedSel = document.getElementById('admBed');
    if (!bedSel.value) return showToast(tr('Select a bed', '???? ??????'), 'error');
    try {
        // E8: admit via the state-machine route which atomically occupies the (locked) bed.
        await API.post('/api/adt/admit', { patient_id: ps.value, patient_name: ps.options[ps.selectedIndex].datase
t.name, admission_type: document.getElementById('admType').value, attending_doctor: document.getElementById('a
dmDoctor').value, admitting_doctor: document.getElementById('admDoctor').value, department: document.getElemen
tById('admDept').value, ward_id: document.getElementById('admWard').value, bed_id: bedSel.value, diagnosis: do
cument.getElementById('admDiagnosis').value, diet_order: document.getElementById('admDiet').value, expected_lo
s: document.getElementById('admLOS').value });
        showToast(tr('Patient admitted!', '?? ???????!')); adtTab = 'patients'; await navigateTo(22);
    } catch (e) { showToast((e && e.message) ? e.message : tr('Error', '???'), 'error'); }
};
window.showInpatientDischargeModal = function (id) {
    document.getElementById('adtDischargeId').value = id;
    document.getElementById('adtDischargeSummary').value = '';
    document.getElementById('adtDischargeInst').value = '';
    document.getElementById('adtDischargeMeds').value = '';
    document.getElementById('adtFollowupDate').value = '';
    document.getElementById('adtDischargeModal').style.display = 'flex';
};
```

```

window.confirmInpatientDischarge = async function () {
  const id = document.getElementById('adtDischargeId').value;
  try {
    // E8: discharge via the state-machine route (frees the bed -> Cleaning atomically,
    // rejects discharge of an already-discharged admission server-side).
    await API.post('/api/adt/discharge', {
      admission_id: id,
      discharge_type: document.getElementById('adtDischargeType').value,
      discharge_summary: document.getElementById('adtDischargeSummary').value,
      discharge_instructions: document.getElementById('adtDischargeInst').value,
      discharge_medications: document.getElementById('adtDischargeMeds').value,
      followup_date: document.getElementById('adtFollowupDate').value,
      followup_doctor: document.getElementById('adtFollowupDoctor').value
    });
    document.getElementById('adtDischargeModal').style.display = 'none';
    showToast(tr('Patient discharged!', '?? ???? ??????!'));
    adtTab = 'discharged'; await navigateTo(22);
  } catch (e) { showToast((e && e.message) ? e.message : tr('Error', '???'), 'error'); }
};

// NOTE: Do NOT redefine window.dischargePatient here. The doctor-station "Patient Done"
// button (defined ~app.js:3490) passes a PATIENT id and marks the patient Done. The ADT
// patients table calls showInpatientDischargeModal(admission_id) DIRECTLY (see above), so no
// ADT alias is needed. A prior collision here overwrote the doctor-station handler, mis-routing
// patient ids to ADT discharge as admission ids (-> 404). (E8 adversarial fix C1.)
window.showAdtTransferModal = function (admissionId, patientName) {
  document.getElementById('adtTransferAdmId').value = admissionId;
  document.getElementById('adtTransferName').textContent = patientName || '';
  document.getElementById('adtTransferReason').value = '';
  document.getElementById('adtTransferBed').innerHTML = `<option value="">${tr('Select ward first', '???? ????
?? ?????')}</option>`;
  const ws = document.getElementById('adtTransferWard'); if (ws) ws.value = '';
  document.getElementById('adtTransferModal').style.display = 'flex';
};

window.loadTransferBeds = async function (wardId) {
  if (!wardId) return;
  // Use the E8 bed board so only genuinely-free beds (Available) are offered as destinations.
  const r = await API.get('/api/adt/beds?ward_id=' + wardId);
  const beds = (r && r.beds) ? r.beds : [];
  const s = document.getElementById('adtTransferBed');
  s.innerHTML = `<option value="">${tr('Select', '????')}</option>${beds.filter(b => b.status === 'Available')
.map(b => `<option value="${safeId(b.id)}">${tr('Bed', '????')} ${escapeHTML(b.bed_number)} - ${tr('Room', '??
??')} ${escapeHTML(b.room_number)}</option>`).join('')}`;
};

window.confirmInpatientTransfer = async function () {
  const admissionId = document.getElementById('adtTransferAdmId').value;
  const toBed = document.getElementById('adtTransferBed').value;
  if (!toBed) return showToast(tr('Select destination bed', '???? ???? ??????'), 'error');
  try {
    // E8: transfer via the state-machine route (atomic free-source + occupy-dest, rejects occupied dest).
    await API.post('/api/adt/transfer', {
      admission_id: admissionId,
      to_bed: toBed,
      transfer_reason: document.getElementById('adtTransferReason').value
    });
  }
};

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    document.getElementById('adtTransferModal').style.display = 'none';
    showToast(tr('Patient transferred!', '?? ??? ??????!'));
    adtTab = 'patients'; await navigateTo(22);
  } catch (e) { showToast((e && e.message) ? e.message : tr('Error', '???'), 'error'); }
};
// ===== ICU =====
// ===== ICU =====
let icuTab = 'board';
async function renderICU(el) {
  const [icuPatients, allAdmissions] = await Promise.all([
    API.get('/api/icu/patients'),
    API.get('/api/admissions')
  ]);
  const discharged = (allAdmissions || []).filter(a => a.status === 'Discharged' && a.department === 'ICU');
  const totalICU = (icuPatients || []).length;
  const onVent = (icuPatients || []).filter(p => p.activity_level === 'Ventilated' || p.dvt_prophylaxis).length;
  el.innerHTML = `<div class="page-title">? ${tr('ICU / Critical Care', '??????? ???????')}</div>
    <div class="stats-grid" style="grid-template-columns:repeat(4,1fr)">
      <div class="stat-card"><div class="stat-icon" style="background:#e74c3c22;color:#e74c3c">?</div><div class="stat-value" style="color:#e74c3c">${totalICU}</div><div class="stat-label">${tr('Current Patients', '????? ? ???????')}</div></div>
      <div class="stat-card"><div class="stat-icon" style="background:#3498db22;color:#3498db">?</div><div class="stat-value" style="color:#3498db">${onVent}</div><div class="stat-label">${tr('On Ventilator', '??? ??????')}</div></div>
      <div class="stat-card"><div class="stat-icon" style="background:#2ecc7122;color:#2ecc71">?</div><div class="stat-value" style="color:#2ecc71">${discharged.length}</div><div class="stat-label">${tr('Discharged', '??? ???')}</div></div>
      <div class="stat-card"><div class="stat-icon" style="background:#9b59b622;color:#9b59b6">?</div><div class="stat-value" style="color:#9b59b6">${totalICU > 0 ? Math.round((totalICU / (totalICU + discharged.length || 1)) * 100) : 0}%</div><div class="stat-label">${tr('Occupancy', '???????')}</div></div>
    </div>
    <div class="tab-bar"><button class="tab-btn ${icuTab === 'board' ? 'active' : ''}" onclick="icuTab='board';navigateTo(23)">? ${tr('Board', '???? ???????')}</button>
      <button class="tab-btn ${icuTab === 'patients' ? 'active' : ''}" onclick="icuTab='patients';navigateTo(23)">? ${tr('Patients', '??????')}</button>
      <button class="tab-btn ${icuTab === 'monitor' ? 'active' : ''}" onclick="icuTab='monitor';navigateTo(23)">? ${tr('Flowsheet', '?????')}</button>
      <button class="tab-btn ${icuTab === 'ventilator' ? 'active' : ''}" onclick="icuTab='ventilator';navigateTo(23)">? ${tr('Ventilator', '??????')}</button>
      <button class="tab-btn ${icuTab === 'infusions' ? 'active' : ''}" onclick="icuTab='infusions';navigateTo(23)">? ${tr('Infusions', '????????')}</button>
      <button class="tab-btn ${icuTab === 'scores' ? 'active' : ''}" onclick="icuTab='scores';navigateTo(23)">? ${tr('Scores', '????????')}</button>
      <button class="tab-btn ${icuTab === 'fluid' ? 'active' : ''}" onclick="icuTab='fluid';navigateTo(23)">? ${tr('Fluid Balance', '????? ???????')}</button>
      <button class="tab-btn ${icuTab === 'bundles' ? 'active' : ''}" onclick="icuTab='bundles';navigateTo(23)">?? ${tr('Prevention Bundles', '??? ???????/??? ??????')}</button>
      <button class="tab-btn ${icuTab === 'discharged' ? 'active' : ''}" onclick="icuTab='discharged';navigateTo(23)">? ${tr('Discharged', '????????')}</button></div>
    <div class="card" id="icuContent"></div>`;
  const c = document.getElementById('icuContent');
  if (icuTab === 'board') {
    let board = [];

```



```

try { const r = await API.get('/api/icu/board'); board = (r && r.board) || []; } catch (e) { board = []; }
const bandColor = b => b === 'Critical' || b === 'Very High' ? '#e74c3c' : b === 'High' ? '#e67e22' : b ===
= 'Moderate' ? '#f1c40f' : b === 'Low' ? '#2ecc71' : '#95a5a6';
c.innerHTML = `<h3> ${tr('ICU Acuity Board', '???? ???? ?????? ??????')} (${board.length})</h3>
<p style="color:#7f8c8d;font-size:13px">${tr('Sorted by acuity (sickest first). SOFA/GCS/APACHE are serv
er-computed.', '???? ???? ???? ???? ???? ????). ????? SOFA/GCS/APACHE ?? ??????')}</p>
${board.length ? `<table class="data-table"><thead><tr><th>${tr('Patient', '?????')}</th><th>${tr('Ward
/Bed', '?????/????')}</th><th>SOFA</th><th>GCS</th><th>APACHE</th><th>${tr('Vent', '????')}</th><th>${tr('D
octor', '?????')}</th></tr></thead><tbody>${board.map(p => `<tr>
<td><strong>${escapeHTML(p.patient_name || '-')}</strong></td>
<td>${escapeHTML(p.ward_name_ar || p.ward_name || '-')} / ${tr('Bed', '????')} ${escapeHTML(p.bed_numbe
r || '-')}</td>
<td><span style="font-weight:700;color:${bandColor(p.sofa_band)}">${p.latest_sofa === null || p.latest
_sofa === undefined ? '?' : escapeHTML(p.latest_sofa)}</span>${p.sofa_band ? ` <small>(${escapeHTML(p.sofa_ban
d)})</small>` : ''}</td>
<td>${p.latest_gcs === null || p.latest_gcs === undefined ? '?' : escapeHTML(p.latest_gcs)}</td>
<td>${p.latest_apache === null || p.latest_apache === undefined ? '?' : escapeHTML(p.latest_apache)}</
td>
<td>${p.on_ventilator ? '?' + escapeHTML(p.vent_mode || tr('Yes', '???')) : ''}</td>
<td>${escapeHTML(p.attending_doctor || '-')}</td></tr>`).join('')}</tbody></table>`
: `<div class="empty-state"><div class="empty-icon"></div><p>${tr('No ICU patients', '?? ???? ???? ?
?????')}</p></div>`;
} else if (icuTab === 'patients') {
c.innerHTML = `<h3> ${tr('ICU Patients', '???? ?????? ??????')} (${totalICU})</h3>
<input class="form-control" placeholder="${tr('Search...', '???...')}" oninput="filterTable(this, 'icuPTa
ble')}" style="margin-bottom:12px">
${totalICU ? `<table class="data-table" id="icuPTable"><thead><tr><th>#</th><th>${tr('Patient', '?????')}
</th><th>${tr('Ward', '?????')}</th><th>${tr('Bed', '?????')}</th><th>${tr('Doctor', '?????')}</th><th>${
tr('Diagnosis', '?????')}</th><th>${tr('Days', '????')}</th><th>${tr('Actions', '?????')}</th></tr></thead>
<tbody>${(icuPatients || []).map(p => {
const days = Math.floor((new Date() - new Date(p.admission_date)) / 86400000);
return `<tr><td>${escapeHTML(p.id)}</td><td><strong>${escapeHTML(p.patient_name)}</strong></td><td>${esc
apeHTML(p.ward_name_ar || '-')}</td><td>${tr('Bed', '????')} ${escapeHTML(p.bed_number || '-')}</td><td>${esca
peHTML(p.attending_doctor || '-')}</td><td>${escapeHTML(p.diagnosis || '-')}</td>
<td><span style="font-weight:700;color:${days > 7 ? '#e74c3c' : '#2ecc71'}">${days}</span></td>
<td><button class="btn btn-sm" onclick="showInpatientDischargeModal(${safeId(p.id)})">? ${tr('Discharg
e', '????')}</button></td></tr>`;
}).join('')}</tbody></table>` : `<div class="empty-state"><div class="empty-icon"></div><p>${tr('No ICU p
atients', '?? ???? ???? ??????')}</p></div>`;
} else if (icuTab === 'monitor') {
c.innerHTML = `<h3> ${tr('Record Vitals', '????? ?????? ??????')}</h3>
<div class="form-grid">
<div><label>${tr('Patient', '?????')}</label><select id="icuPatientMon" class="form-control"><option
value="">${tr('Select', '????')}</option>${(icuPatients || []).map(p => `<option value="${safeId(p.id)}" data-
pid="${safeId(p.patient_id)}">${escapeHTML(p.patient_name)} - ${escapeHTML(p.ward_name_ar || '')} ${tr('Bed',
'????')} ${escapeHTML(p.bed_number || '')}</option>`).join('')}</select></div>
<div><label>HR</label><input id="icuHR" type="number" class="form-control" placeholder="bpm"></div>
<div><label>SBP/DBP</label><div style="display:flex;gap:4px"><input id="icuSBP" type="number" class="f
orm-control" placeholder="SBP"><input id="icuDBP" type="number" class="form-control" placeholder="DBP"></div><
div>
<div><label>SpO2</label><input id="icuSpO2" type="number" class="form-control" placeholder="%"></div>
<div><label>RR</label><input id="icuRR" type="number" class="form-control" placeholder="/min"></div>
<div><label>Temp</label><input id="icuTemp" type="number" step="0.1" class="form-control" placeholder=
"°C"></div>

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        <div><label>FiO2</label><input id="icuFiO2" type="number" class="form-control" placeholder="%"></div>
        <div><label>Urine (ml)</label><input id="icuUrine" type="number" class="form-control" placeholder="ml"
    ></div>
    </div><button class="btn btn-primary" onclick="saveICUMonitor()" style="margin-top:12px">? ${tr('Save',
'???')}</button>;
    } else if (icuTab === 'ventilator') {
        c.innerHTML = `<h3>? ${tr('Ventilator Settings', '??????? ?????? ??????')}</h3>
        <div class="form-grid">
            <div><label>${tr('Patient', '?????')}</label><select id="icuPatientVent" class="form-control"><option
value="">${tr('Select', '????')}</option>${(icuPatients || []).map(p => `<option value="${safeId(p.id)}" data
-pid="${safeId(p.patient_id)}">${escapeHTML(p.patient_name)}</option>`).join('')}</select></div>
            <div><label>${tr('Mode', '????')}</label><select id="ventMode" class="form-control"><option>CMV</opti
on><option>SIMV</option><option>PSV</option><option>CPAP</option><option>BiPAP</option><option>APRV</option></
select></div>
            <div><label>FiO2 %</label><input id="ventFiO2" type="number" value="21" class="form-control"></div>
            <div><label>TV (ml)</label><input id="ventTV" type="number" class="form-control"></div>
            <div><label>RR</label><input id="ventRR" type="number" class="form-control"></div>
            <div><label>PEEP</label><input id="ventPEEP" type="number" class="form-control"></div>
            <div><label>PIP</label><input id="ventPIP" type="number" class="form-control"></div>
            <div><label>PS</label><input id="ventPS" type="number" class="form-control"></div>
            <div><label>ETT Size</label><input id="ventETT" class="form-control"></div>
        </div><button class="btn btn-primary" onclick="saveVentilator()" style="margin-top:12px">? ${tr('Save',
'???')}</button>;
    } else if (icuTab === 'infusions') {
        c.innerHTML = `<h3>? ${tr('Continuous Infusions / Drips', '????????? ?????????? ??????????')}</h3>
        <div class="form-grid">
            <div><label>${tr('Patient', '?????')}</label><select id="icuPatientInfusion" class="form-control"><op
tion value="">${tr('Select', '????')}</option>${(icuPatients || []).map(p => `<option value="${safeId(p.id)}"
data-pid="${safeId(p.patient_id)}">${escapeHTML(p.patient_name)}</option>`).join('')}</select></div>
            <div><label>${tr('Drug', '?????')}</label><input id="infDrug" class="form-control" placeholder="${tr(
'e.g. Noradrenaline', '?????: ?????????????')}"></div>
            <div><label>${tr('Concentration', '???????')}</label><input id="infConc" class="form-control" placehol
der="mg/mL"></div>
            <div><label>${tr('Rate', '???????')}</label><input id="infRate" type="number" step="0.1" class="form-c
ontrol"></div>
            <div><label>${tr('Rate Unit', '????? ?????????')}</label><select id="infRateUnit" class="form-control"><o
ption>mL/hr</option><option>mcg/kg/min</option><option>mcg/min</option><option>units/hr</option></select></div>
            <div><label>${tr('Status', '?????')}</label><select id="infStatus" class="form-control"><option value
="Running">${tr('Running', '????')}</option><option value="Paused">${tr('Paused', '????? ??????')}</option><op
tion value="Stopped">${tr('Stopped', '????')}</option></select></div>
        </div><button class="btn btn-primary" onclick="saveICUInfusion()" style="margin-top:12px">? ${tr('Save',
'???')}</button>
        <div id="infusionResult" style="margin-top:12px"></div>;
    } else if (icuTab === 'scores') {
        c.innerHTML = `<h3>? ${tr('Clinical Scores (server-computed)', '????????? ?????????? (????? ?? ??????)')}</h3>
        <p style="color:#7f8c8d;font-size:13px">${tr('Enter RAW observations ? SOFA / GCS / APACHE-II are comput
ed server-side (anti-spoof).', '???? ?????????? ?????? ? ?????? SOFA / GCS / APACHE-II ?? ?????? (???? ??????).')}<
/p>
        <div class="form-grid">
            <div><label>${tr('Patient', '?????')}</label><select id="icuPatientScore" class="form-control"><optio
n value="">${tr('Select', '????')}</option>${(icuPatients || []).map(p => `<option value="${safeId(p.id)}" dat
a-pid="${safeId(p.patient_id)}">${escapeHTML(p.patient_name)}</option>`).join('')}</select></div>

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<div><label>${tr('Age', '????')}</label><input id="scoreAge" type="number" class="form-control"></div>
>
<div style="grid-column:span 2"><h4 style="color:#3498db">${tr('GCS components', '????? GCS')}</h4></div>
<div>
<div><label>${tr('Eye (1-4)', '???? 1-4')}</label><input id="gcsEye" type="number" min="1" max="4" class="form-control"></div>
<div><label>${tr('Verbal (1-5)', '????? 1-5')}</label><input id="gcsVerbal" type="number" min="1" max="5" class="form-control"></div>
<div><label>${tr('Motor (1-6)', '????? 1-6')}</label><input id="gcsMotor" type="number" min="1" max="6" class="form-control"></div>
<div style="grid-column:span 2"><h4 style="color:#3498db">${tr('Physiology / labs', '?????????/?')}</h4></div>
<div><label>MAP</label><input id="scoreMAP" type="number" class="form-control"></div>
<div><label>HR</label><input id="scoreHR" type="number" class="form-control"></div>
<div><label>RR</label><input id="scoreRR" type="number" class="form-control"></div>
<div><label>SpO2</label><input id="scoreSpO2" type="number" class="form-control"></div>
<div><label>Temp</label><input id="scoreTemp" type="number" step="0.1" class="form-control"></div>
<div><label>PaO2/FiO2</label><input id="scorePF" type="number" class="form-control"></div>
<div><label>${tr('Ventilated', '??? ?????')}</label><select id="scoreVent" class="form-control"><option value="">${tr('No', '??')}</option><option value="1">${tr('Yes', '??')}</option></select></div>
<div><label>${tr('Platelets', '???????')} (x103)</label><input id="scorePlt" type="number" class="form-control"></div>
<div><label>${tr('Bilirubin', '????????')}</label><input id="scoreBili" type="number" step="0.1" class="form-control"></div>
<div><label>${tr('Creatinine', '????????')}</label><input id="scoreCreat" type="number" step="0.1" class="form-control"></div>
<div><label>${tr('Urine 24h (mL)', '???? 24')}</label><input id="scoreUrine24" type="number" class="form-control"></div>
<div style="grid-column:span 2"><h4 style="color:#3498db">${tr('Nursing assessments', '????? ?????')}</h4></div>
<div><label>RASS (-5..4)</label><input id="scoreRASS" type="number" value="0" class="form-control"></div>
<div><label>Braden</label><input id="scoreBraden" type="number" value="23" class="form-control"></div>
<div><label>Morse Fall</label><input id="scoreMorse" type="number" class="form-control"></div>
<div><label>Pain (0-10)</label><input id="scorePain" type="number" class="form-control"></div>
</div><button class="btn btn-primary" onclick="saveICUScores()" style="margin-top:12px">? ${tr('Compute & Save', '??? ?????')}</button>
<div id="scoreResult" style="margin-top:12px"></div>;
} else if (icuTab === 'fluid') {
c.innerHTML = `<h3>? ${tr('Fluid Balance', '???? ?????')}</h3>
<div class="form-grid">
<div><label>${tr('Patient', '????')}</label><select id="icuPatientFluid" class="form-control"><option value="">${tr('Select', '????')}</option>${(icuPatients || []).map(p => `<option value="${safeId(p.id)}" data-pid="${safeId(p.patient_id)}">${escapeHTML(p.patient_name)}</option>`).join('')}</select></div>
<div><label>${tr('Shift', '????')}</label><select id="fluidShift" class="form-control"><option value="Day">${tr('Day', '????')}</option><option value="Night">${tr('Night', '????')}</option></select></div>
<div style="grid-column:span 2"><h4 style="color:#2ecc71">?? ${tr('Intake', '????')}</h4></div>
<div><label>IV Fluids (mL)</label><input id="fluidIV" type="number" class="form-control"></div>
<div><label>Oral (mL)</label><input id="fluidOral" type="number" class="form-control"></div>
<div><label>Blood Products (mL)</label><input id="fluidBlood" type="number" class="form-control"></div>
>
<div><label>IV Meds (mL)</label><input id="fluidMeds" type="number" class="form-control"></div>
<div style="grid-column:span 2"><h4 style="color:#e74c3c">?? ${tr('Output', '????')}</h4></div>
<div><label>Urine (mL)</label><input id="fluidUrine" type="number" class="form-control"></div>

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<div><label>Drains (ml)</label><input id="fluidDrains" type="number" class="form-control"></div>
<div><label>NGT (ml)</label><input id="fluidNGT" type="number" class="form-control"></div>
<div><label>Vomit (ml)</label><input id="fluidVomit" type="number" class="form-control"></div>
</div><button class="btn btn-primary" onclick="saveFluidBalance()" style="margin-top:12px">? ${tr('Save'
, '???')}</button>`;
} else if (icuTab === 'discharged') {
  c.innerHTML = `<h3>? ${tr('Discharged from ICU', '???????? ?? ?????? ??????')} (${discharged.length})</h
3>
  <input class="form-control" placeholder="${tr('Search...', '???...')}" oninput="filterTable(this, 'icuDis
chTable')" style="margin-bottom:12px">
  ${discharged.length ? `<table class="data-table" id="icuDischTable"><thead><tr><th>#</th><th>${tr('Patie
nt', '?????')}</th><th>${tr('Discharge Type', '??? ?????')}</th><th>${tr('Doctor', '?????')}</th><th>${tr('
Diagnosis', '???????')}</th><th>${tr('Admitted', '???????')}</th><th>${tr('Discharged', '???????')}</th><th>${t
r('LOS', '?????')}</th></tr></thead><tbody>${discharged.map(a => {
  const los = a.discharge_date && a.admission_date ? Math.floor((new Date(a.discharge_date) - new Date(a.a
dmission_date)) / 86400000) : '-';
  const dtBadge = a.discharge_type === 'AMA' ? badge(tr('AMA', '?? ?????'), 'danger') : a.discharge_type
=== 'Death' ? badge(tr('Death', '????'), 'danger') : a.discharge_type === 'Transfer' ? badge(tr('Transfer', '
????'), 'warning') : badge(tr('Regular', '????'), 'success');
  return `<tr><td>${escapeHTML(a.id)}</td><td>${escapeHTML(a.patient_name)}</td><td>${dtBadge}</td><td>${e
scapeHTML(a.attending_doctor || '-')}</td><td>${escapeHTML(a.diagnosis || '-')}</td>
<td>${a.admission_date ? new Date(a.admission_date).toLocaleDateString('ar-SA') : '-'}</td>
<td>${a.discharge_date ? new Date(a.discharge_date).toLocaleDateString('ar-SA') : '-'}</td>
<td><strong>${los}</strong> ${tr('days', '???')}</td></tr>`;
}}).join('')}</tbody></table>` : `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${
tr('No discharged emergency patients', '?? ???? ???? ?????? ?? ??????')}</div></div>`};
} else if (icuTab === 'bundles') {
  c.innerHTML = `<h3>?? ${tr('ICU Prevention Bundles & Infection Control', '??? ?????? ?????? ?????? ?????
??? ??????')}</h3>
  <p style="color:#7f8c8d;font-size:13px">${tr('Daily prevention checklists for VAP, CLABSI, and CAUTI. A
non-compliance reason is required if compliance is under 100%', '????? ?????? ?????? ???? ?????? (VAP, CLAB
SI, CAUTI). ???? ??? ??? ??? ?????? ???? ???? ?????? ??? ?? 100%.')}</p>

  <div style="display:flex;gap:16px;flex-wrap:wrap">
    <!-- Form Column -->
    <div style="flex:1;min-width:320px">
      <div class="form-group">
        <label>${tr('Patient', '?????')}</label>
        <select id="icuBundlePatient" class="form-control" onchange="window.e1LoadBundleLogs()">
          <option value="">${tr('Select', '????')}</option>
          ${icuPatients || []}.map(p => `<option value="${safeId(p.id)}">${escapeHTML(p.patient_name)}</o
ption>`).join('')}
        </select>
      </div>

      <div class="form-group">
        <label>${tr('Bundle Type', '??? ???? ??????')}</label>
        <select id="icuBundleType" class="form-control" onchange="window.e1RenderBundleChecklist()">
          <option value="VAP">VAP (Ventilator-Associated Pneumonia)</option>
          <option value="CLABSI">CLABSI (Central Line-Associated Bloodstream Infection)</option>
          <option value="CAUTI">CAUTI (Catheter-Associated Urinary Tract Infection)</option>
        </select>
      </div>
    </div>
  </div>

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        <div class="form-group">
            <label>${tr('Audit Date', '????? ???????')}</label>
            <input type="date" id="icuBundleDate" class="form-control" value="${new Date().toISOString().split
('T')[0]}">
        </div>

        <!-- Checklist Items Wrapper -->
        <div id="icuBundleChecklistWrapper" class="card" style="padding:16px;margin:12px 0;background:#ffffff
f">

        </div>

        <!-- Compliance Indicator -->
        <div style="margin:12px 0;padding:12px;border-radius:8px;background:var(--hover,#f8f9fa);display:fl
ex;justify-content:space-between;align-items:center">
            <strong>${tr('Compliance Rate', '??? ???????')}</strong>
            <span id="icuBundleComplianceRate" style="font-size:16px;font-weight:bold;color:#22c55e">100.00%</
span>
        </div>

        <!-- Reason for Non-compliance -->
        <div id="icuNonComplianceReasonGroup" class="form-group" style="display:none">
            <label style="color:#ef4444;font-weight:bold">${tr('Reason for Non-compliance', '??? ??? ???????
(?????)')} *</label>
            <textarea id="icuBundleReason" class="form-control" rows="2" placeholder="${tr('e.g. Clinical cont
raindication, patient emergency, etc.', '????: ??? ???? ???? ???? ???? ?????? ??????..')}></textarea>
        </div>

        <button class="btn btn-primary w-full" onclick="window.saveIcuBundle()">${tr('Save Daily Audit', '??
? ?????? ??????')}</button>
    </div>

    <!-- History Column -->
    <div style="flex:1;min-width:320px">
        <h4>? ${tr('Recent Audits History', '??? ?????? ?????? ??????')}</h4>
        <div id="icuBundleHistoryList">${tr('Please select a patient to view history', '???? ?????? ??? ?
? ?????')}</div>
    </div>
</div>
`;

    window.e1RenderBundleChecklist();
}
}

window.e1RenderBundleChecklist = () => {
    const type = document.getElementById('icuBundleType')?.value || 'VAP';
    const wrapper = document.getElementById('icuBundleChecklistWrapper');
    if (!wrapper) return;

    const checklists = {
        VAP: [
            { id: 'head_of_bed_elevation', label: tr('Head of bed elevation (30-45°)', '??? ??? ?????? ?????? 30-45
????') },
            { id: 'sedation_interruption', label: tr('Daily sedation interruption & assessment of readiness to extub

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ate', '????? ?????? ?????? ?????? ?????? ???? ???? ??????') },
  { id: 'pud_prophylaxis', label: tr('Peptic ulcer disease (PUD) prophylaxis', '??????? ?? ???? ??????') }
,
  { id: 'dvt_prophylaxis', label: tr('Deep vein thrombosis (DVT) prophylaxis', '??????? ?? ???? ?????? ?
?????') },
  { id: 'oral_care', label: tr('Daily oral care with Chlorhexidine', '??????? ?????? ?????? ?????? ?????
????????') }
],
CLABSI: [
  { id: 'hand_hygiene', label: tr('Hand hygiene performed', '???? ?????? ??????') },
  { id: 'sterile_barrier', label: tr('Maximal sterile barriers used during insertion', '??????? ?????? ??
????? ?????? ?????? ??????') },
  { id: 'skin_antisepsis', label: tr('Skin antisepsis with Chlorhexidine', '????? ?????? ?????????????????')
},
  { id: 'site_selection', label: tr('Optimal site selection (avoid femoral vein)', '????? ?????? ?????? ?
????? (???? ?????? ??????)') },
  { id: 'daily_review', label: tr('Daily review of line necessity', '????????? ?????? ???? ?????? ?????? ?
?????') }
],
CAUTI: [
  { id: 'hand_hygiene', label: tr('Hand hygiene performed', '???? ?????? ??????') },
  { id: 'proper_indication', label: tr('Proper indication for insertion', '????? ?????? ?????????? ??????')
} ],
  { id: 'closed_drainage', label: tr('Closed drainage system maintained', '??????? ??? ???? ?????? ?????') },
  { id: 'unobstructed_flow', label: tr('Unobstructed urine flow maintained', '??????? ??? ???? ?????? ??? ?
???') },
  { id: 'daily_review', label: tr('Daily review of catheter necessity', '????????? ?????? ???? ?????? ?????
?? ??????') }
]
};

const items = checklists[type];
wrapper.innerHTML = `
<h5 style="margin-top:0;color:var(--primary)">?? ${type} Checklist</h5>
${items.map(item => `
  <div style="display:flex;align-items:flex-start;gap:8px;margin-bottom:10px">
    <input type="checkbox" id="chk_${item.id}" checked style="margin-top:3px" onchange="window.e1Calculate
BundleCompliance()">
    <label for="chk_${item.id}" style="font-size:13px;line-height:1.4;cursor:pointer">${item.label}</label
>
  </div>
`).join('')}
`;
window.e1CalculateBundleCompliance();
};

window.e1CalculateBundleCompliance = () => {
  const wrapper = document.getElementById('icuBundleChecklistWrapper');
  if (!wrapper) return;
  const checkboxes = wrapper.querySelectorAll('input[type="checkbox"]');
  let checked = 0;
  checkboxes.forEach(c => { if (c.checked) checked++; });

  const total = checkboxes.length || 1;

```

```

const rate = parseFloat(((checked / total) * 100).toFixed(2));

const rateEl = document.getElementById('icuBundleComplianceRate');
if (rateEl) {
  rateEl.textContent = `${rate.toFixed(2)}%`;
  if (rate === 100.0) {
    rateEl.style.color = '#22c55e';
  } else {
    rateEl.style.color = '#ef4444';
  }
}

const reasonGroup = document.getElementById('icuNonComplianceReasonGroup');
if (reasonGroup) {
  if (rate < 100.0) {
    reasonGroup.style.display = 'block';
  } else {
    reasonGroup.style.display = 'none';
    const reasonInput = document.getElementById('icuBundleReason');
    if (reasonInput) reasonInput.value = '';
  }
}

};

window.saveIcuBundle = async () => {
  const patientSelect = document.getElementById('icuBundlePatient');
  if (!patientSelect || !patientSelect.value) {
    return showToast(tr('Please select a patient', '???? ?????? ??? ??????'), 'error');
  }
  const type = document.getElementById('icuBundleType').value;
  const date = document.getElementById('icuBundleDate').value;
  const reason = document.getElementById('icuBundleReason')?.value || '';

  const wrapper = document.getElementById('icuBundleChecklistWrapper');
  const checkboxes = wrapper.querySelectorAll('input[type="checkbox"]');
  const checkedItems = {};
  let checkedCount = 0;
  checkboxes.forEach(c => {
    const key = c.id.replace('chk_', '');
    checkedItems[key] = c.checked;
    if (c.checked) checkedCount++;
  });

  const total = checkboxes.length || 1;
  const compliance = parseFloat(((checkedCount / total) * 100).toFixed(2));

  if (compliance < 100.0 && (!reason || !reason.trim())) {
    return showToast(tr('Reason for non-compliance is required!', '??? ?????? ??? ????? ?????????!'), 'error');
  }

  try {
    await API.post('/api/icu/prevention-bundles', {
      admission_id: patientSelect.value,
      bundle_type: type,

```

```

        audit_date: date,
        checked_items: checkedItems,
        non_compliance_reason: reason
    });
    showToast(tr('Daily prevention audit saved successfully!', '?? ??? ????? ?????? ?????? ?????!'));
    window.e1LoadBundleLogs();
} catch (err) {
    showToast(tr('Error saving prevention audit', '??? ?? ??? ????? ??????'), 'error');
}
};

window.e1LoadBundleLogs = async () => {
    const patientSelect = document.getElementById('icuBundlePatient');
    const historyList = document.getElementById('icuBundleHistoryList');
    if (!historyList) return;

    if (!patientSelect || !patientSelect.value) {
        historyList.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('Please select a patient to view history', '???? ?????? ??? ???? ?????')}</div>`;
        return;
    }

    try {
        const logs = await API.get('/api/icu/prevention-bundles?admission_id=' + patientSelect.value);
        if (!logs || !logs.length) {
            historyList.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No infection control logs found for this patient', '?? ??? ???? ????? ???? ?????? ????? ???? ??????')}</div>`;
        } else {
            historyList.innerHTML = logs.map(l => {
                const rate = parseFloat(l.compliance_rate);
                const rateColor = rate === 100.0 ? '#22c55e' : '#ef4444';
                return `
                    <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid ${rateColor};font-size:12px">
                        <div style="display:flex;justify-content:space-between;font-weight:700">
                            <span style="color:var(--primary)">?? ${l.bundle_type} Audit</span>
                            <span>? ${new Date(l.audit_date).toLocaleDateString('ar-SA')}</span>
                        </div>
                        <div style="margin-top:6px">
                            <strong>${tr('Compliance Rate', '???? ??????')}</strong>
                            <span style="color:${rateColor};font-weight:bold">${rate.toFixed(2)}%</span>
                        </div>
                        ${l.non_compliance_reason ? `
                            <div style="margin-top:4px;color:#ef4444;font-size:11px">
                                <strong>?? ${tr('Reason', '????')}</strong> ${escapeHTML(l.non_compliance_reason)}
                            </div>
                            ` : ''}
                        <div style="font-size:10px;color:var(--text-dim);margin-top:4px">
                            ??? ${tr('By', '????')}: ${escapeHTML(l.recorded_by_name || l.recorded_by || '')}
                        </div>
                    </div>
                `;
            }).join('');
        }
    }
};

```



```

    } catch (err) {
        historyList.innerHTML = `
```

```
// and returns {sofa, sofa_band, gcs, apache, ...}.
const s = document.getElementById('icuPatientScore'); if (!s.value) return showToast(tr('Select patient', '?
??? ??????'), 'error');

const val = id => { const el = document.getElementById(id); return el ? el.value : ''; };

try {
  const r = await API.post('/api/icu/score', {
    admission_id: s.value,
    age: val('scoreAge'),
    gcs_eye: val('gcsEye'), gcs_verbal: val('gcsVerbal'), gcs_motor: val('gcsMotor'),
    map: val('scoreMAP'), hr: val('scoreHR'), rr: val('scoreRR'), spo2: val('scoreSpO2'), temp: val('scoreTem
mp'),
    pao2_fio2: val('scorePF'), ventilated: val('scoreVent') === '1',
    platelets: val('scorePlt'), bilirubin: val('scoreBili'), creatinine: val('scoreCreat'), urine_output_24h
: val('scoreUrine24'),
    rass: val('scoreRASS'), braden: val('scoreBraden'), morse_fall: val('scoreMorse'), pain_score: val('scor
ePain'),
    calculated_by: currentUser?.display_name
  });
  const box = document.getElementById('scoreResult');
  if (box && r) {
    box.innerHTML = `
```

```

content.innerHTML = `
    <h2>${tr('CSSD - Sterilization', '?????? ??????')}</h2>
    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:16px">
        <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1565c0">${batches.length}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Total Batches', '????? ??????')}</p></div>
        <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e65100">${processing}</h3><p style="margin:4px 0 0;font-size:12px">${tr('In Process', '??? ??????')}</p></div>
        <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${done}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Completed', '?????')}</p></div>
        <div class="card" style="padding:16px;text-align:center;background:#f3e5f5"><h3 style="margin:0;color:#6a1b9a">${sterileTrays}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Sterile Trays', '????? ??????')}</p></div>
    </div>
    <div style="display:grid;grid-template-columns:1fr 2fr;gap:16px">
        <div class="card" style="padding:20px">
            <h4 style="margin:0 0 12px">${tr('New Batch', '???? ??????')}</h4>
            <div class="form-group"><label>${tr('Batch #', '??? ??????')}</label><input class="form-input" id="cssdBatch" value="CSSD-${Date.now().toString().slice(-6)}"></div>
            <div class="form-group"><label>${tr('Items', '????????')}</label><textarea class="form-input" id="cssdItems" rows="2" placeholder="${tr('Surgical instruments, trays...', '????? ???????? ??????...')}></div>
            <div class="form-group"><label>${tr('Department', '?????')}</label><input class="form-input" id="cssdDept" placeholder="${tr('Surgery, ER...', '????????? ????????...')}></div>
            <div class="form-group"><label>${tr('Method', '????? ????????')}</label>
            <select class="form-input" id="cssdMethod">
                <option value="autoclave">${tr('Steam Autoclave', '????????? ??????')}</option>
                <option value="eto">${tr('ETO Gas', '??? ETO')}</option>
                <option value="plasma">${tr('H2O2 Plasma', '??????? H2O2')}</option>
                <option value="chemical">${tr('Chemical', '????????')}</option>
            </select></div>
            <div class="form-group"><label>${tr('Temperature', '????????')}</label><input class="form-input" id="cssdTemp" placeholder="134°C"></div>
            <div class="form-group"><label>${tr('Operator', '????????')}</label><input class="form-input" id="cssdOp"></div>
            <button class="btn btn-primary w-full" onclick="e16SaveCssdBatch()">? ${tr('Start Cycle', '??? ??????')}</button>
        </div>
        <div class="card" style="padding:20px">
            <h4 style="margin:0 0 12px">${tr('Batch History', '??? ????????')}</h4>
            <div id="cssdTable"></div>
        </div>
    </div>
    <div class="card" style="padding:20px;margin-top:16px">
        <h4 style="margin:0 0 12px">${tr('Sterilization Cycles ? Biological Indicator (BI) Gate', '????? ???????? ? ?????? ???????? ??????')}</h4>
        <p style="font-size:12px;color:#b71c1c;margin:0 0 12px">${tr('A cycle can only be released for sterile issue after a recorded BI = Pass. The system blocks release on missing/failed/pending BI (fail-closed).', '??? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? = ????. ????? ???? ???? ???? ???? ????/?? ?/????? ???? ???? (???-????).')}</p>
        <div id="cssdCyclesTable"></div>
    </div>`;

```

```

const ct = document.getElementById('cssdTable');
if (ct) {
  createTable(ct, 'cssdTbl',
    [tr('Batch#', '?????'), tr('Items', '???????'), tr('Dept', '?????'), tr('Method', '???????'), tr('Temp', '???????'), tr('Status', '?????'), tr('Date', '???????'), tr('Actions', '???????')],
    batches.map(b => ({
      cells: [b.batch_number, (b.items || '').substring(0, 30), b.department || '', b.method || '', b.temperature || '', statusBadge(b.status), b.created_at ? new Date(b.created_at).toLocaleDateString('ar-SA') : '',
        b.status === 'processing' ? rawHtml('<button class="btn btn-sm" onclick="e16CompleteCssdBatch(' + parseInt(b.id, 10) + ')>? ' + tr('Complete', '????') + '</button>') : '?'], id: b.id
    })))
  );
}

const cct = document.getElementById('cssdCyclesTable');
if (cct) {
  createTable(cct, 'cssdCyclesTbl',
    [tr('Cycle#', '?????'), tr('Type', '?????'), tr('Status', '?????'), tr('BI Result', '????? BI'), tr('Released', '?????'), tr('Actions', '???????')],
    cycles.map(c => {
      const id = parseInt(c.id, 10);
      const bi = (c.bi_test_result || 'Pending');
      const released = (c.released_for_issue === 1 || c.released_for_issue === true);
      let actions = '';
      if (String(c.status).toLowerCase() === 'running') {
        actions += '<button class="btn btn-sm" onclick="e16CompleteCycle(' + id + ')>? ' + tr('Complete', '????') + '</button> ';
      }
      if (String(c.status).toLowerCase() === 'completed' && !released) {
        actions += '<button class="btn btn-sm" onclick="e16RecordBI(' + id + ')>? ' + tr('Record BI', '????? BI') + '</button> ';
        actions += '<button class="btn btn-sm btn-primary" onclick="e16ReleaseSterile(' + id + ')>? ' + tr('Release', '????') + '</button>';
      }
      if (released) actions = '? ' + tr('Released', '????');
      return { cells: [escapeHTML(c.cycle_number || ''), escapeHTML(c.cycle_type || ''), statusBadge(c.status), escapeHTML(bi), released ? '?' : '?', rawHtml(actions || '?')], id: c.id };
    })
  );
}

window.addInstrumentSet = async function () {
  try { await API.post('/api/cssd/instruments', { set_name: document.getElementById('cssdName').value, set_name_ar: document.getElementById('cssdNameAr').value, set_code: document.getElementById('cssdCode').value, category: document.getElementById('cssdCat').value, instrument_count: document.getElementById('cssdCount').value, department: document.getElementById('cssdDept').value }); showToast(tr('Added!', '??? ??????!')); await navigateTo(24); } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== E16 CSSD client handlers (safe routes; BI gate fail-closed) =====
// NOTE (E16): the previous window.completeCssd / window.completeCycle hardcoded
// bi_test_result: 'Pass' and flipped status with no gate (client-asserted authority).
// They are replaced by e16-prefixed handlers that NEVER self-assert a BI pass and
// surface the server's 409 when the biological-indicator gate blocks release.

```

```

// API.request resolves (does NOT throw) on non-2xx, so we inspect the returned object's .error.
window.e16SaveCssdBatch = async () => {
  try {
    const r = await API.post('/api/cssd/batches', { batch_number: document.getElementById('cssdBatch').value,
    items: document.getElementById('cssdItems').value, department: document.getElementById('cssdDept').value, meth
    od: document.getElementById('cssdMethod').value, temperature: document.getElementById('cssdTemp').value, opera
    tor: document.getElementById('cssdOp').value });
    if (r && r.error) return showToast(r.error, 'error');
    showToast(tr('Cycle started!', '???? ??????!')); navigateTo(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.e16CompleteCssdBatch = async (id) => {
  try {
    const r = await API.put('/api/cssd/batches/' + parseInt(id, 10), { status: 'completed' });
    if (r && r.error) return showToast(r.error, 'error');
    showToast('?'); navigateTo(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.e16CompleteCycle = async (id) => {
  try {
    const r = await API.put('/api/cssd/cycles/' + parseInt(id, 10), { status: 'completed' });
    if (r && r.error) return showToast(r.error, 'error');
    showToast(tr('Cycle completed ? record BI result next', '?????? ?????? ? ???? ?????? BI ??????')); navigate
    To(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// Record the biological-indicator result (the value comes from the technician's reading, NOT a hardcoded pass
).
window.e16RecordBI = async (id) => {
  const v = (prompt(tr('Enter BI result: pass / fail / pending', '???? ?????? ???????? ??????: pass / fail / pen
ding')) || '').trim();
  if (!v) return;
  try {
    const r = await API.put('/api/cssd/cycles/' + parseInt(id, 10) + '/bi-result', { bi_test_result: v });
    if (r && r.error) return showToast(r.error, 'error');
    showToast(tr('BI result recorded', '?? ?????? ?????? BI')); navigateTo(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// Attempt to release for sterile issue ? fail-closed: server blocks (409) unless BI = pass.
window.e16ReleaseSterile = async (id) => {
  try {
    const r = await API.put('/api/cssd/cycles/' + parseInt(id, 10) + '/release', {});
    if (r && r.error) return showToast(tr('Release BLOCKED ? BI not passed', '???????? ?????? ? ?? ???? BI') + (
r.reason ? ': ' + r.reason : ''), 'error');
    showToast(tr('Released for sterile issue (BI passed)', '?? ???????? ???????? ???????? (??? BI)')); navigateTo
(currentPage);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// Legacy shims (kept for any cached references) ? now route to the SAFE handlers, never self-assert a pass.
window.completeCssd = window.e16CompleteCssdBatch;
window.completeCycle = window.e16CompleteCycle;

// ===== DIETARY =====
async function renderDietary(el) {

```

```

const patients = await API.get('/api/dietary/orders').catch(() => []);
content.innerHTML = `
  <h2>${tr('Dietary / Nutrition', '???????')}</h2>
  <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:12px;margin-bottom:16px">
    <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${patients.length || 0}</h3><p style="margin:4px 0 0;font-size:13px;color:#666">${tr('Active Orders', '????? ?????')}</p></div>
    <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e65100">${patients.filter?.(p => p.diet_type === 'diabetic')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px;color:#666">${tr('Diabetic Diet', '???? ?????')}</p></div>
    <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1565c0">${patients.filter?.(p => p.diet_type === 'soft')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px;color:#666">${tr('Soft Diet', '???? ?????')}</p></div>
  </div>
  <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px">
    <div class="card" style="padding:20px">
      <h4 style="margin:0 0 12px">${tr('New Diet Order', '??? ????? ?????')}</h4>
      <div class="form-group"><label>${tr('Patient', '???????')}</label><input class="form-input" id="dietPatient" placeholder="${tr('Patient name', '??? ???????')}"></div>
      <div class="form-group"><label>${tr('Ward/Room', '???????/???????')}</label><input class="form-input" id="dietRoom" placeholder="${tr('e.g. 3A-201', '?????: 3?-201')}"></div>
      <div class="form-group"><label>${tr('Diet Type', '??? ???????')}</label>
      <select class="form-input" id="dietType">
        <option value="regular">${tr('Regular', '????')}</option>
        <option value="diabetic">${tr('Diabetic', '????')}</option>
        <option value="soft">${tr('Soft', '????')}</option>
        <option value="liquid">${tr('Liquid', '????')}</option>
        <option value="NPO">${tr('NPO (Nothing by mouth)', '????')}</option>
        <option value="low_sodium">${tr('Low Sodium', '???? ?????')}</option>
        <option value="renal">${tr('Renal', '????')}</option>
        <option value="gluten_free">${tr('Gluten Free', '???? ???????')}</option>
      </select></div>
      <div class="form-group"><label>${tr('Allergies/Notes', '???????/???????')}</label><textarea class="form-input" id="dietNotes" rows="2"></textarea></div>
      <div class="form-group"><label>${tr('Meal Time', '??? ???????')}</label>
      <select class="form-input" id="dietMeal"><option value="breakfast">${tr('Breakfast', '????')}</option>
      <option value="lunch">${tr('Lunch', '????')}</option><option value="dinner">${tr('Dinner', '????')}</option><option value="all">${tr('All Meals', '?? ???????')}</option></select></div>
      <button class="btn btn-primary w-full" onclick="saveDietOrder()">? ${tr('Save Order', '??? ?????')}</button>
    </div>
    <div class="card" style="padding:20px">
      <h4 style="margin:0 0 12px">${tr('Active Diet Orders', '????? ??????? ???????')}</h4>
      <div id="dietOrdersList"></div>
    </div>
  </div>`;

const list = document.getElementById('dietOrdersList');
if (list && patients.length > 0) {
  list.innerHTML = patients.map(p => '<div style="padding:10px;margin:6px 0;background:#f8f9fa;border-radius:8px;border-right:4px solid ' + (p.diet_type === 'NPO' ? '#cc0000' : p.diet_type === 'diabetic' ? '#ff9800' : '#4caf50') + '"><strong>' + escapeHTML(p.patient_name || '') + '</strong> - ' + escapeHTML(p.room || '') + '<b

```

```

r><span style="font-size:13px;color:#666">' + escapeHTML(p.diet_type || '') + ' | ' + escapeHTML(p.meal_time || '') + '</span></div>').join('');
    } else if (list) { list.innerHTML = '<p style="color:#999;text-align:center">' + tr('No active orders', '???
???? ?????') + '</p>'; }

window.saveDietOrder = async () => {
    try {
        await API.post('/api/dietary/orders', { patient_name: document.getElementById('dietPatient')?.value, room: document.getElementById('dietRoom')?.value, diet_type: document.getElementById('dietType')?.value, meal_time: document.getElementById('dietMeal')?.value, notes: document.getElementById('dietNotes')?.value });
        showToast(tr('Diet order saved!', '?? ?? ???? !'));
        renderDietary(content);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}
window.addDietOrder = async function () {
    const s = document.getElementById('dietPatient'); if (!s.value) return showToast(tr('Select patient', '????
?????'), 'error');
    const dt = document.getElementById('dietType');
    try { await API.post('/api/dietary/orders', { admission_id: s.value, patient_id: s.options[s.selectedIndex].dataset.pid, patient_name: s.options[s.selectedIndex].dataset.name, diet_type: dt.value, diet_type_ar: dt.options[dt.selectedIndex].dataset.ar, texture: document.getElementById('dietTexture').value, allergies: document.getElementById('dietAllergies').value, ordered_by: currentUser?.display_name }); showToast(tr('Diet ordered!', '?? ?? ?????!')); await navigateTo(25); } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ===== INFECTION CONTROL =====
async function renderInfectionControl(el) {
    const content = el;
    const reports = await API.get('/api/infection-control/reports').catch(() => []);
    const active = reports.filter(r => r.status === 'active').length;
    const byType = {};
    reports.forEach(r => { const t = r.infection_type || 'Other'; byType[t] = (byType[t] || 0) + 1; });
    const hai = { CLABSI: 0, CAUTI: 0, VAP: 0, SSI: 0, CDI: 0 };
    reports.forEach(r => { if (r.hai_category && hai[r.hai_category] !== undefined) hai[r.hai_category]++; });
    if (!window.icTab) window.icTab = 'dashboard';

    content.innerHTML = `
        <div class="page-title">? ${tr('Infection Control & HAI Surveillance', '????? ?????? ?????? ?????? ??????')}</div>
        <div style="display:flex;gap:8px;margin-bottom:20px;border-bottom:1px solid var(--border);padding-bottom:12px;flex-wrap:wrap">
            <button class="btn ${window.icTab==='dashboard'?'btn-primary':'btn-secondary'}" onclick="window.icTab='dashboard';navigateTo(26)" style="font-size:12px">? ${tr('HAI Dashboard', '???? ?????')}</button>
            <button class="btn ${window.icTab==='report'?'btn-primary':'btn-secondary'}" onclick="window.icTab='report';navigateTo(26)" style="font-size:12px">? ${tr('Report Infection', '???? ?????')}</button>
            <button class="btn ${window.icTab==='hygiene'?'btn-primary':'btn-secondary'}" onclick="window.icTab='hygiene';navigateTo(26)" style="font-size:12px">? ${tr('Hand Hygiene', '??? ?????')}</button>
            <button class="btn ${window.icTab==='isolation'?'btn-primary':'btn-secondary'}" onclick="window.icTab='isolation';navigateTo(26)" style="font-size:12px">? ${tr('Isolation', '????')}</button>
        </div>
        <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(130px,1fr));gap:12px;margin-bottom:20px">

```

```

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #ef4444"><div style="font-size:24px;font-weight:800;color:#ef4444">${reports.length}</div><div style="font-size:11px;color:var(--text-muted)">${tr('Total','????')}</div></div>

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #f97316"><div style="font-size:24px;font-weight:800;color:#f97316">${active}</div><div style="font-size:11px;color:var(--text-muted)">${tr('Active','????')}</div></div>

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #8b5cf6"><div style="font-size:24px;font-weight:800;color:#8b5cf6">${hai.CLABSI}</div><div style="font-size:11px;color:var(--text-muted)">CLABSI</div></div>

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #3b82f6"><div style="font-size:24px;font-weight:800;color:#3b82f6">${hai.CAUTI}</div><div style="font-size:11px;color:var(--text-muted)">CAUTI</div></div>

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #06b6d4"><div style="font-size:24px;font-weight:800;color:#06b6d4">${hai.VAP}</div><div style="font-size:11px;color:var(--text-muted)">VAP</div></div>

        <div class="card" style="padding:14px;text-align:center;border-top:3px solid #22c55e"><div style="font-size:24px;font-weight:800;color:#22c55e">${hai.SSI}</div><div style="font-size:11px;color:var(--text-muted)">SSI</div></div>

    </div>
    <div id="icTabContent"></div>`;

const icCont = document.getElementById('icTabContent');
if (!icCont) return;

if (window.icTab === 'dashboard') {
    icCont.innerHTML = `
        <div class="card" style="padding:20px">
            <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px">
                <h4 style="margin:0">? ${tr('Infection Reports','?? ?')}</h4>
                <button class="btn btn-sm btn-secondary" onclick="exportToCSV([], 'hai_reports')">? ${tr('Export','??')}</button>
            </div>
            <div id="icTable"></div>
        </div>

        ${Object.keys(byType).length > 0 ? `<div class="card" style="padding:16px;margin-top:16px">
            <h4 style="margin:0 0 12px">${tr('By Infection Type','?? ?')}</h4>
            <div style="display:flex;gap:8px;flex-wrap:wrap">
                ${Object.entries(byType).map(([t,c])=>`<span style="padding:6px 14px;border-radius:20px;background:#fce4ec;font-size:12px;font-weight:600">${escapeHTML(t)}: ${c}</span>`).join('')}
            </div>
        </div>` : ''};

    const ict = document.getElementById('icTable');
    if (ict) createTable(ict,'icTbl',
        [tr('Patient','????'),tr('Type','????'),tr('HAI'),tr('Ward','????'),tr('Isolation','????'),tr('Status','????'),tr('Date','????'),tr('')],
        reports.map(r=>({cells:[
            r.patient_name||'',r.infection_type||'',
            r.hai_category?rawHtml(`<span class="badge" style="background:#8b5cf6;color:#fff">${escapeHTML(r.hai_category)}</span>`):'',
            r.ward||'',r.isolation_type||'',statusBadge(r.status),
            r.created_at?new Date(r.created_at).toLocaleDateString('ar-SA'):'',
            r.status==='active'?rawHtml(`<button class="btn btn-sm" onclick="resolveIc(${parseInt(r.id,10)}")>? ${tr('Resolve','??')}</button>`):''
        ],id:r.id}))
    );

```



```

);

} else if (window.icTab === 'report') {
  icCont.innerHTML = `
    <div class="card" style="padding:20px;max-width:600px">
      <h4 style="margin:0 0 16px;color:var(--primary)">? ${tr('Report Infection Case','????? ???? ????')}</h
4>

      <div class="form-group"><label class="form-label">${tr('Patient Name','??? ?????')}</label><input cla
ss="form-input" id="icPatient" required></div>
      <div class="form-group"><label class="form-label">${tr('Infection Type','??? ?????')}</label>
      <select class="form-input" id="icType">
        <option value="MRSA">MRSA</option><option value="VRE">VRE</option><option value="C.diff">C. diffic
ile</option>
        <option value="ESBL">ESBL</option><option value="TB">TB</option><option value="COVID-19">COVID-19<
/option>
        <option value="UTI">UTI</option><option value="SSI">SSI</option><option value="Other">${tr('Other'
,'????')}</option>
      </select></div>
      <div class="form-group"><label class="form-label">HAI ${tr('Category','????')}</label>
      <select class="form-input" id="icHAI">
        <option value="">${tr('None','?? ?????')}</option>
        <option value="CLABSI">CLABSI</option><option value="CAUTI">CAUTI</option>
        <option value="VAP">VAP</option><option value="SSI">SSI</option><option value="CDIFF">C.DIFF</opti
on>
      </select></div>
      <div class="form-group"><label class="form-label">${tr('Ward','????')}</label><input class="form-inp
ut" id="icWard"></div>
      <div class="form-group"><label class="form-label">${tr('Isolation Type','??? ?????')}</label>
      <select class="form-input" id="icIsolation">
        <option value="none">${tr('None','????')}</option>
        <option value="contact">${tr('Contact','????')}</option><option value="droplet">${tr('Droplet','
????')}</option>
        <option value="airborne">${tr('Airborne','????')}</option><option value="protective">${tr('Protec
tive','????')}</option>
      </select></div>
      <div class="form-group"><label class="form-label">${tr('Organism','????? ????')}</label><input clas
s="form-input" id="icOrganism" placeholder="e.g. MRSA, Klebsiella"></div>
      <div class="form-group"><label class="form-label">${tr('Culture Results','????? ?????')}</label><tex
tarea class="form-input" id="icCulture" rows="2"></textarea></div>
      <div class="form-group"><label class="form-label">${tr('Action Taken','????? ????')}</label><texta
rea class="form-input" id="icAction" rows="2"></textarea></div>
      <button class="btn btn-primary" style="width:100%" onclick="window.reportInfection()">? ${tr('Submit R
eport','????? ?????')}</button>
    </div>`;

} else if (window.icTab === 'hygiene') {
  const hhData = await API.get('/api/infection/hand-hygiene').catch(() => []);
  const hhAvg = hhData.length ? Math.round(hhData.reduce((s,h)=>s+(h.compliance_rate||(h.moments_observed?Ma
th.round((h.moments_compliant/h.moments_observed)*100):0)),0)/hhData.length) : 0;
  icCont.innerHTML = `
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:20px">
      <div class="card" style="padding:20px">
        <h4 style="margin:0 0 16px;color:var(--primary)">? ${tr('Hand Hygiene Audit','????? ????? ????')}<
/h4>

```

```

        <div class="form-group"><label class="form-label">${tr('Department','????')}</label><input class="form-input" id="hhDept"></div>
        <div class="form-group"><label class="form-label">${tr('Moments Observed','?????? ?????????')}</label><input class="form-input" id="hhObs" type="number" min="1"></div>
        <div class="form-group"><label class="form-label">${tr('Moments Compliant','?????? ?????????')}</label><input class="form-input" id="hhComp" type="number" min="0"></div>
        <div class="form-group"><label class="form-label">${tr('Shift','????????')}</label><select class="form-input" id="hhShift"><option>Morning</option><option>Afternoon</option><option>Night</option></select></div>
        <button class="btn btn-primary" style="width:100%" onclick="window.addHHAudit()">? ${tr('Record Audit','????')}</button>
    </div>
    <div class="card" style="padding:20px">
        <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
            <h4 style="margin:0">? ${tr('Compliance Trend','????? ?????????')}</h4>
            <div style="font-size:24px;font-weight:800;color:${hhAvg}>80?'#22c55e':hhAvg>60?'#f97316': '#ef4444'}">${hhAvg}%</div>
        </div>
        <div style="background:${hhAvg}>80?'#f0fdf4':hhAvg>60?'#fff7ed': '#fef2f2'};padding:10px;border-radius:8px;margin-bottom:12px;font-size:12px">
            WHO Target ? 80% | CBAHI Min ? 75%
            ${hhAvg}>80?' ? Compliant':hhAvg>75?' ?? Near Target': ' ? Below Target'}
        </div>
        ${hhData.length?`<table class="data-table"><thead><tr><th>${tr('Dept','????')}</th><th>Obs</th><th>Comp</th><th>Rate</th><th>${tr('Date','????????')}</th></tr></thead>
        <tbody>${hhData.slice(0,15).map(h=>{const r2=h.compliance_rate||(h.moments_observed?Math.round((h.moments_compliant/h.moments_observed)*100):0);return`<tr><td>${escapeHTML(h.department||'')}</td><td>${h.moments_observed||0}</td><td>${h.moments_compliant||0}</td><td style="font-weight:700;color:${r2}>80?'#22c55e':r2>=60?'#f97316': '#ef4444'}">${r2}%</td><td style="font-size:11px">${(h.created_at||'').slice(0,10)}</td></tr>`}}.join('')}</tbody></table>`
        :`<p style="text-align:center;color:var(--text-muted)">${tr('No audits yet','? ? ? ? ? ? ? ? ? ?')}</p>`}
    </div>
</div>`;

    } else if (window.icTab === 'isolation') {
        const isolated = reports.filter(r=>r.isolation_type&&r.isolation_type!=='none'&&r.status==='active');
        const isoColors={contact:'#f97316',droplet:'#3b82f6',airborne:'#ef4444',protective:'#22c55e'};
        icCont.innerHTML=`<div class="card" style="padding:20px">
            <h4 style="margin:0 0 16px">? ${tr('Active Isolation Rooms','??? ?????? ??????')} (${isolated.length})</h4>
            ${!isolated.length?`<div style="text-align:center;padding:40px;color:var(--text-muted)"><div style="font-size:48px">?</div><p>${tr('No active isolations','? ? ? ? ? ? ? ? ? ?')}</p></div>`
            :`<div style="display:grid;grid-template-columns:repeat(auto-fill,minmax(240px,1fr));gap:12px">${isolate
d.map(r=>`<div style="padding:14px;border-radius:10px;border:2px solid ${isoColors[r.isolation_type]}|#999'};
background:${isoColors[r.isolation_type]}|#999'}15"><div style="display:flex;justify-content:space-between;margin-bottom:6px"><strong>${escapeHTML(r.patient_name||'')}</strong><span class="badge" style="background:${iso
Colors[r.isolation_type]}|#999'};color:#fff;font-size:10px">${(r.isolation_type||'').toUpperCase()}</span></div><div style="font-size:12px;color:var(--text-muted)">? ${escapeHTML(r.infection_type||'')} ${r.hai_category?'
. <strong>'+escapeHTML(r.hai_category)+'</strong>':''}</div><div style="font-size:12px;color:var(--text-muted)">? ${escapeHTML(r.ward||'')}</div><div style="font-size:11px;color:var(--text-muted)">? ${(r.created_at||'')
.slice(0,10)}</div><button class="btn btn-sm" style="width:100%;margin-top:8px" onclick="resolveIc(${parseInt
(r.id,10)})">? ${tr('Resolve','????')}</button></div>`}}.join('')}</div>`}
        </div>`;
    }
}

```

```

window.resolveIc = async (id) => {
  try { await API.put('/api/infection-control/reports/'+id,{status:'resolved'}); showToast('? '+tr('Case resolved','?? ?????')); navigateTo(currentPage); }
  catch(e) { showToast(tr('Error','???'),'error'); }
};
}

window.reportInfection = async function() {
  const p=document.getElementById('icPatient')?.value?.trim();
  const t=document.getElementById('icType')?.value;
  if(!p||!t) return showToast(tr('Patient name and type required','????? ??? ?????? ??????'),'error');
  try {
    await API.post('/api/infection/surveillance',{patient_name:p,infection_type:t,organism:document.getElementById('icOrganism')?.value,ward:document.getElementById('icWard')?.value,hai_category:document.getElementById('icHAI')?.value,isolation_type:document.getElementById('icIsolation')?.value,culture_results:document.getElementById('icCulture')?.value,action_taken:document.getElementById('icAction')?.value,reported_by:currentUser?.display_name});
    showToast(tr('Reported!','?? ???????!'));
    await navigateTo(26);
  } catch(e){showToast(tr('Error','???'),'error');}
};

window.addHHAudit = async function() {
  const dept=document.getElementById('hhDept')?.value?.trim();
  const obs=parseInt(document.getElementById('hhObs')?.value)||0;
  const comp=parseInt(document.getElementById('hhComp')?.value)||0;
  if(!dept||!obs) return showToast(tr('Enter department and observations','???? ?????? ??? ??????'),'error');
  if(comp>obs) return showToast(tr('Compliant cannot exceed observed','?????? ?? ?????? ??????'),'error');
  try {
    await API.post('/api/infection/hand-hygiene',{department:dept,moments_observed:obs,moments_compliant:comp,shift:document.getElementById('hhShift')?.value,auditor:currentUser?.display_name});
    showToast(tr('Audit recorded!','?? ?????? ???????!'));
    await navigateTo(26);
  } catch(e){showToast(tr('Error','???'),'error');}
};

// ===== OVR (Occurrence Variance Reports) =====
async function renderOVR(el) {
  const content = el;
  if (!window.ovrTab) window.ovrTab = 'list';
  const ovr = await API.get('/api/incidents/ovr').catch(() => []);
  const openCount = ovr.filter(r=>r.status==='open').length;
  content.innerHTML = `
    <div class="page-title">? ${tr('OVR - Occurrence Variance Reports','????? ?????????')}</div>
    <div style="display:flex;gap:8px;margin-bottom:20px;flex-wrap:wrap">
      <button class="btn ${window.ovrTab==='list'?'btn-primary':'btn-secondary'}" onclick="window.ovrTab='list';navigateTo(currentPage)">? ${tr('Reports List','????? ?????????')}</button>
      <button class="btn ${window.ovrTab==='new'?'btn-primary':'btn-secondary'}" onclick="window.ovrTab='new';navigateTo(currentPage)">? ${tr('New Report','????? ?????')}</button>
    </div>
    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:20px">
      <div class="card" style="padding:14px;text-align:center;border-top:3px solid #3b82f6"><div style="font-s

```

```

size:24px;font-weight:800;color:#3b82f6">${ovrs.length}</div><div style="font-size:11px;color:var(--text-muted)
">${tr('Total OVRs','????')}</div></div>

    <div class="card" style="padding:14px;text-align:center;border-top:3px solid #f97316"><div style="font-s
ize:24px;font-weight:800;color:#f97316">${openCount}</div><div style="font-size:11px;color:var(--text-muted)">
${tr('Open','????')}</div></div>

    <div class="card" style="padding:14px;text-align:center;border-top:3px solid #22c55e"><div style="font-s
ize:24px;font-weight:800;color:#22c55e">${ovrs.length-openCount}</div><div style="font-size:11px;color:var(--t
ext-muted)">${tr('Closed','????')}</div></div>

</div>

<div id="ovrContent"></div>`;
const ovrCont = document.getElementById('ovrContent');
if (!ovrCont) return;
if (window.ovrTab === 'list') {
    ovrCont.innerHTML = `<div class="card" style="padding:20px"><div id="ovrTable"></div></div>`;
    const ot = document.getElementById('ovrTable');
    if (ot) createTable(ot,'ovrTbl',
        [tr('ID','????'),tr('Type','????'),tr('Severity','?????'),tr('Status','?????'),tr('Reporter','????
?'),tr('Date','?????')],
        ovrs.map(r=>({cells:[r.id,r.incident_type||'',r.severity||'',statusBadge(r.status||''),r.reporter_name||
'',r.created_at?(new Date(r.created_at)).toLocaleDateString('ar-SA':'')]}))
    );
} else if (window.ovrTab === 'new') {
    ovrCont.innerHTML = `<div class="card" style="padding:20px;max-width:600px">
    <h4 style="margin:0 0 16px">? ${tr('New OVR Report','???? ???? ????')}</h4>
    <div class="form-group"><label class="form-label">${tr('Incident Type','??? ?????')}</label>
        <select class="form-input" id="ovrType"><option value="medication">${tr('Medication Error','??? ?????'
)}</option><option value="fall">${tr('Patient Fall','???? ????')}</option><option value="procedure">${tr('Proc
edure Error','??? ?????')}</option><option value="equipment">${tr('Equipment Failure','??? ????')}</option><o
ption value="other">${tr('Other','????')}</option></select></div>
        <div class="form-group"><label class="form-label">${tr('Severity','?????')}</label>
            <select class="form-input" id="ovrSeverity"><option value="low">Low</option><option value="moderate">M
oderate</option><option value="high">High</option><option value="critical">Critical</option></select></div>
            <div class="form-group"><label class="form-label">${tr('Description','????')}</label><textarea class="f
orm-input" id="ovrDesc" rows="3"></textarea></div>
            <div class="form-group"><label class="form-label">${tr('Immediate Action','?????? ????')}</label><tex
tarea class="form-input" id="ovrAction" rows="2"></textarea></div>
            <div class="form-group"><label class="form-label">${tr('Patient Involved','???? ????')}</label><input cl
ass="form-input" id="ovrPatient"></div>
            <button class="btn btn-primary" style="width:100%" onclick="window.submitOVR()">? ${tr('Submit OVR','???
?? ?????')}</button>
        </div>`;
}
window.submitOVR = async () => {
    try { await API.post('/api/incidents/ovr',{incident_type:document.getElementById('ovrType')?.value,severit
y:document.getElementById('ovrSeverity')?.value,description:document.getElementById('ovrDesc')?.value,immediat
e_action:document.getElementById('ovrAction')?.value,patient_name:document.getElementById('ovrPatient')?.value
,reporter_name:currentUser?.display_name}); showToast(tr('OVR submitted','?? ?????')); window.ovrTab='list';
    navigateTo(currentPage); } catch(e){showToast(tr('Error','???'),'error');}
};
}

// ===== AUDIT LOG =====
async function renderAuditLog(el) {
    const content = el;

```

```

const [logs, modules] = await Promise.all([
  API.get('/api/admin/audit-trail?limit=100').catch(()=>[]),
  API.get('/api/admin/audit-trail/modules').catch(()=>[])
]);
const modFilter = window.auditModFilter||'';
const filtered = modFilter ? logs.filter(l=>l.module===modFilter) : logs;
content.innerHTML = `
  <div class="page-title">? ${tr('Audit Trail','??? ?????')}</div>
  <div style="display:flex;gap:8px;margin-bottom:20px;align-items:center">
    <select class="form-input" style="max-width:200px" onchange="window.auditModFilter=this.value;navigateTo
(currentPage)">
      <option value="">${tr('All Modules','???? ?????')}</option>
      ${Array.isArray(modules)?modules:[]}.map(m=>`<option value="${escapeHTML(m)}" ${m===modFilter?'select
ed':''}>${escapeHTML(m)}</option>`).join('')}
    </select>
    <span style="color:var(--text-muted);font-size:13px">${filtered.length} ${tr('records','???')}</span>
    <button class="btn btn-sm btn-secondary" style="margin-right:auto" onclick="exportToCSV(filtered,'audit_
log')">? ${tr('Export','????')}</button>
  </div>
  <div class="card" style="padding:20px">
    <div id="auditTable"></div>
  </div>`;
const at = document.getElementById('auditTable');
if (at) createTable(at, 'auditTbl',
  [tr('Time','????'),tr('User','???????'),tr('Module','?????'),tr('Action','?????'),tr('Details','???????
?'),tr('IP','???????')],
  filtered.map(l=>({cells:[
    l.created_at?new Date(l.created_at).toLocaleString('ar-SA':''),
    l.display_name||l.user_id||'',
    l.module||'',
    l.action||'',
    typeof l.details==='object'?JSON.stringify(l.details).slice(0,80):String(l.details||'').slice(0,80),
    l.ip_address||''
  ]}))
);
}

let specialtiesTab = 'cardiology';
let specialtiesSelectedPatientId = null;

async function renderSpecialties(el) {
  const [patients, cathReports, patientRegimens] = await Promise.all([
    API.get('/api/patients').catch(() => []),
    specialtiesSelectedPatientId ? API.get('/api/cardiology/cath-reports/' + specialtiesSelectedPatientId).cat
ch(() => []) : Promise.resolve([]),
    specialtiesSelectedPatientId ? API.get('/api/oncology/patient-regimens/' + specialtiesSelectedPatientId).c
atch(() => []) : Promise.resolve([])
  ]);

  if (!specialtiesSelectedPatientId && patients.length > 0) {
    specialtiesSelectedPatientId = patients[0].id;
  }

  const selectedPatient = patients.find(p => parseInt(p.id, 10) === parseInt(specialtiesSelectedPatientId, 10))

```

```

);

el.innerHTML = `
  <div class="page-title">? ${tr('Specialized Medical Services (HIS/EMR Expansion)', '?????? ???? ????')}</div>

  <div class="card mb-16">
    <div style="display:flex;align-items:center;gap:12px;flex-wrap:wrap">
      <label style="font-weight:bold">${tr('Select Patient File:', '??? ? ????')}</label>
      <select id="specPatientSelect" class="form-input" style="max-width:300px" onchange="window.selectSpecPatient(this.value)">
        ${patients.map(p => `<option value="${safeId(p.id)}" ${parseInt(p.id, 10) === parseInt(specialtiesSelectedPatientId, 10) ? 'selected' : ''}>${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}
      </select>
      ${selectedPatient ? `<div class="badge badge-info">${tr('MRN: ', '?? ????')}${selectedPatient.file_number}</div>` : ''}
    </div>
  </div>

  <div class="tab-bar">
    <button class="tab-btn ${specialtiesTab === 'cardiology' ? 'active' : ''}" onclick="specialtiesTab='cardiology';navigateTo(44)">?? ${tr('Cardiology Cath Lab', '???? ????')}</button>
    <button class="tab-btn ${specialtiesTab === 'oncology' ? 'active' : ''}" onclick="specialtiesTab='oncology';navigateTo(44)">?? ${tr('Oncology Chemotherapy Planner', '???? ???? ????')}</button>
  </div>

  <div id="specTabContent"></div>
`;

const content = document.getElementById('specTabContent');

if (specialtiesTab === 'cardiology') {
  const latestReport = cathReports[0] || { blockage_lad: 0, blockage_lcx: 0, blockage_rca: 0, findings: '' };
;
  content.innerHTML = `
    <div style="display:grid;grid-template-columns:1.2fr 1fr;gap:20px;align-items:start">
      <div class="card">
        <h3>?? ${tr('Coronary Artery Blockage Simulator', '???? ????')}</h3>

        <div style="margin:20px 0;display:flex;flex-direction:column;align-items:center;gap:16px">
          <!-- Coronary Tree SVG Map -->
          <svg width="220" height="220" viewBox="0 0 100 100" style="background:#0f172a;border-radius:12px;padding:10px">
            <!-- Aorta (Main Root) -->
            <path d="M50,10 C50,25 50,30 50,35" stroke="#ef4444" stroke-width="8" fill="none" />
            <text x="50" y="8" font-size="4" fill="#fff" text-anchor="middle">Aorta</text>

            <!-- LAD (Left Anterior Descending) -->
            <path d="M50,35 C35,45 25,60 20,85" stroke="${getCathColor(latestReport.blockage_lad)}" stroke-width="4" fill="none" id="svg_lad" />
            <text x="18" y="92" font-size="4" fill="#94a3b8" text-anchor="middle">LAD (${latestReport.blockage_lad}%)</text>
          </svg>
        </div>
      </div>
    </div>
  `;

```

```

        <!-- LCx (Left Circumflex) -->
        <path d="M50,35 C65,45 75,55 80,75" stroke="${getCathColor(latestReport.blockage_lcx)}" stroke-w
idth="4" fill="none" id="svg_lcx" />
        <text x="82" y="82" font-size="4" fill="#94a3b8" text-anchor="middle">LCx (${latestReport.blocka
ge_lcx}%)</text>

        <!-- RCA (Right Coronary Artery) -->
        <path d="M50,30 C30,30 40,55 50,85" stroke="${getCathColor(latestReport.blockage_rca)}" stroke-w
idth="4" fill="none" id="svg_rca" />
        <text x="50" y="92" font-size="4" fill="#94a3b8" text-anchor="middle">RCA (${latestReport.blocka
ge_rca}%)</text>
    </svg>

    <!-- Sliders to Simulate Blockages -->
    <div style="width:100%;display:flex;flex-direction:column;gap:12px">
        <div class="form-group">
            <label>LAD Blockage (Left Anterior Descending): <span id="lad_label" font-weight="bold">${late
stReport.blockage_lad}%</span></label>
            <input type="range" id="slider_lad" class="form-input" min="0" max="100" value="${latestReport
.blockage_lad}" oninput="document.getElementById('lad_label').textContent=this.value+'%' />
        </div>
        <div class="form-group">
            <label>LCx Blockage (Left Circumflex): <span id="lcx_label" font-weight="bold">${latestReport.
blockage_lcx}%</span></label>
            <input type="range" id="slider_lcx" class="form-input" min="0" max="100" value="${latestReport
.blockage_lcx}" oninput="document.getElementById('lcx_label').textContent=this.value+'%' />
        </div>
        <div class="form-group">
            <label>RCA Blockage (Right Coronary Artery): <span id="rca_label" font-weight="bold">${latestR
eport.blockage_rca}%</span></label>
            <input type="range" id="slider_rca" class="form-input" min="0" max="100" value="${latestReport
.blockage_rca}" oninput="document.getElementById('rca_label').textContent=this.value+'%' />
        </div>
        <div class="form-group">
            <label>${tr('Clinical Findings & Recommendation:', '?????? ????????? ????????? ?????????:')}</l
abel>
            <textarea id="cathFindings" class="form-input" rows="2" placeholder="${tr('Enter stenosis deta
ils and recommendations...', '???? ?????? ?????? ????????? ?????????...')}">${latestReport.findings}</textarea>
        </div>
    </div>
    <button class="btn btn-primary" onclick="window.saveCathReport()" style="width:100%">? ${tr('Save Ca
th Lab Report', '??? ?????? ??????? ?????? ??????')}</button>
</div>

<div class="card">
    <h3>? ${tr('Cath Lab Report History', '????? ????????? ????????? ?????????')}</h3>
    ${cathReports.length ? makeTable(
        [tr('Date', '????????'), 'LAD', 'LCx', 'RCA', tr('Findings', '????????')],
        cathReports.map(r => ({
            cells: [
                new Date(r.created_at).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US'),
                r.blockage_lad + '%',
                r.blockage_lcx + '%',

```

```

        r.blockage_rca + '%',
        r.findings || '-'
    ]
    )))
    ) : `<div class="empty-state"><p>${tr('No previous cath lab reports recorded', '?? ???? ?????? ?????
?????')}</p></div>`
</div>
</div>
`;
} else {
    content.innerHTML = `
<div style="display:grid;grid-template-columns:1fr 1fr;gap:20px;align-items:start">
<div class="card">
<h3>?? ${tr('Schedule Chemotherapy Protocol', '????? ????????? ?????? ?????????')}</h3>
<div class="form-grid" style="grid-template-columns:1fr;gap:12px;margin-top:12px">
<div class="form-group">
<label>${tr('Select Regimen Template:', '???? ???? ?????????')}</label>
<select id="oncoRegimenSelect" class="form-input">
<option value="FOLFOX (Colorectal Cancer)">FOLFOX (Oxaliplatin, Leucovorin, 5-Fluorouracil)</o
ption>
<option value="AC-T (Breast Cancer)">AC-T (Doxorubicin, Cyclophosphamide, Paclitaxel)</option>
<option value="FEC (Breast Cancer)">FEC (Fluorouracil, Epirubicin, Cyclophosphamide)</option>
</select>
</div>
<div class="form-group">
<label>${tr('Target Cycles count:', '??? ?????? ?????????')}</label>
<input type="number" id="oncoCyclesCount" class="form-input" min="1" max="12" value="6" />
</div>
<div class="form-group">
<label>${tr('Protocol Start Date:', '????? ??? ?????????')}</label>
<input type="date" id="oncoStartDate" class="form-input" value="${new Date().toISOString().split
('T')[0]}" />
</div>
</div>
<button class="btn btn-primary" onclick="window.scheduleChemoProtocol()" style="margin-top:16px;widt
h:100%">? ${tr('Generate & Schedule Regimen', '????? ?????? ????????? ?????? ?????????')}</button>
</div>

<div class="card">
<h3>? ${tr('Scheduled Chemotherapy Cycles', '????? ?????? ????????? ????????? ??????')}</h3>
${patientRegimens.length ? makeTable(
    [tr('Date', '???????'), tr('Protocol Name', '??? ?????????'), tr('Cycle#', '??? ??????'), tr('Sta
tus', '?????')],
    patientRegimens.map(r => ({
        cells: [
            new Date(r.start_date).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US'),
            r.regimen_name,
            r.cycle_number + ' / 6',
            r.status === 'Completed' ? '? Completed / ?????' : '? Scheduled / ?????'
        ]
    }))
    ) : `<div class="empty-state"><p>${tr('No chemotherapy protocols scheduled', '?? ???? ?????????? ?
? ? ?????? ?????? ???? ??????')}</p></div>`}
</div>

```



```

        </div>
    `;
}
}

window.selectSpecPatient = function(val) {
    specialtiesSelectedPatientId = val;
    navigateTo(44);
};

window.saveCathReport = async function() {
    const lad = document.getElementById('slider_lad').value;
    const lcx = document.getElementById('slider_lcx').value;
    const rca = document.getElementById('slider_rca').value;
    const findings = document.getElementById('cathFindings').value;
    try {
        await API.post('/api/cardiology/cath-reports', {
            patient_id: specialtiesSelectedPatientId,
            blockage_lad: lad,
            blockage_lcx: lcx,
            blockage_rca: rca,
            findings
        });
        showToast(tr('Cath report saved successfully', '?? ??? ????? ??????? ??????'));
        navigateTo(44);
    } catch (e) {
        showToast(tr('Error saving report', '??? ?? ?????'), 'error');
    }
};

window.scheduleChemoProtocol = async function() {
    const name = document.getElementById('oncoRegimenSelect').value;
    const cycles = parseInt(document.getElementById('oncoCyclesCount').value, 10);
    const start = document.getElementById('oncoStartDate').value;
    try {
        for (let i = 1; i <= cycles; i++) {
            const cycleDate = new Date(start);
            cycleDate.setDate(cycleDate.getDate() + (i - 1) * 14); // 2-week cycle interval
            await API.post('/api/oncology/patient-regimens', {
                patient_id: specialtiesSelectedPatientId,
                regimen_name: name,
                cycle_number: i,
                status: i === 1 ? 'Completed' : 'Scheduled',
                start_date: cycleDate.toISOString().split('T')[0]
            });
        }
        showToast(tr('Chemotherapy protocol generated and scheduled', '?? ????? ??????? ????????? ?????? ??????'));
        navigateTo(44);
    } catch (e) {
        showToast(tr('Error scheduling chemotherapy', '??? ?? ????? ??????? ?????????'), 'error');
    }
};

function getCathColor(blockage) {

```

```

const val = parseInt(blockage, 10);
if (val >= 70) return '#ef4444'; // Red (Severe blockage)
if (val >= 40) return '#eab308'; // Yellow (Moderate blockage)
return '#22c55e'; // Green (Healthy / Mild)
}

// ===== SPECIALTIES PANEL (INDEX 47) =====
async function renderSpecialtiesPanel(el) {
  // 100+ Master Specialties list
  const specialtiesList = [
    // 1. ?????? ?????? ????????? (Medical & Clinical Family)
    { id: 'cardiology', en: 'Cardiology', ar: '????? ??????', category: 'medical', icon: '??', status: 'Active' },
    { id: 'pulmonology', en: 'Pulmonology', ar: '????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'gastroenterology', en: 'Gastroenterology', ar: '????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'nephrology', en: 'Nephrology', ar: '????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'endocrinology', en: 'Endocrinology & Diabetes', ar: '????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'neurology', en: 'Neurology', ar: '????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'dermatology', en: 'Dermatology', ar: '????????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'psychiatry', en: 'Psychiatry & Mental Health', ar: '????? ?????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'rheumatology', en: 'Rheumatology', ar: '????? ?????????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'oncology', en: 'Oncology', ar: '????????', category: 'medical', icon: '??', status: 'Active' },
    { id: 'hematology', en: 'Hematology', ar: '????? ?????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'allergy_immunology', en: 'Allergy & Immunology', ar: '????????? ?????????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'infectious_diseases', en: 'Infectious Diseases', ar: '????????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'family_medicine', en: 'Family Medicine', ar: '?? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'internal_medicine', en: 'Internal Medicine', ar: '????? ?????? ??????', category: 'medical', icon: '?', status: 'Active' },
    { id: 'geriatrics', en: 'Geriatric Medicine', ar: '?? ?????????? ?????? ?????', category: 'medical', icon: '?', status: 'Active' },

    // 2. ?????? ????????? (Surgical Family)
    { id: 'general_surgery', en: 'General Surgery', ar: '????????? ??????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'cardiothoracic_surgery', en: 'Cardiothoracic Surgery', ar: '????? ?????? ??????', category: 'surgical', icon: '????', status: 'Active' },
    { id: 'vascular_surgery', en: 'Vascular Surgery', ar: '????? ?????????? ??????', category: 'surgical', icon: '??', status: 'Active' },
    { id: 'neurosurgery', en: 'Neurosurgery', ar: '????? ?????? ?????????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'urology', en: 'Urology & Andrology', ar: '????? ?????????? ?????????? ??????', category: 'surgical', icon: '?', status: 'Active' },
  ];
}

```

```

    { id: 'orthopedics', en: 'Orthopedic Surgery', ar: '????? ?????? ?????????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'plastic_surgery', en: 'Plastic & Reconstructive Surgery', ar: '????? ?????? ?????????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'ophthalmology', en: 'Ophthalmology & Eye Surgery', ar: '?? ?????? ??????', category: 'surgical', icon: '??', status: 'Active' },
    { id: 'ent', en: 'Otolaryngology (ENT)', ar: '?? ?????? ?????? ?????? ?????????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'pediatric_surgery', en: 'Pediatric Surgery', ar: '????? ?????????', category: 'surgical', icon: '?', status: 'Active' },
    { id: 'bariatric_surgery', en: 'Bariatric & Obesity Surgery', ar: '????? ?????? ??????????', category: 'surgical', icon: '??', status: 'Active' },

    // 3. ????? ??? ?????? ?????? (Women & Children Family)
    { id: 'pediatrics', en: 'Pediatrics', ar: '?? ?????? ??????', category: 'women_children', icon: '?', status: 'Active' },
    { id: 'neonatology', en: 'Neonatology & NICU', ar: '?? ?????? ?????? ??????', category: 'women_children', icon: '?', status: 'Active' },
    { id: 'developmental_pediatrics', en: 'Developmental Pediatrics', ar: '?? ?????? ?????? ?????????', category: 'women_children', icon: '?', status: 'Active' },
    { id: 'obgyn', en: 'Obstetrics & Gynecology', ar: '????? ?????????', category: 'women_children', icon: '?', status: 'Active' },
    { id: 'maternal_fetal', en: 'Maternal-Fetal Medicine (MFM)', ar: '?? ?????? ?????? ??? ?????????', category: 'women_children', icon: '?', status: 'Active' },
    { id: 'reproductive_endocrinology', en: 'Reproductive Endocrinology (IVF)', ar: '????? ?????????? ?????? (????? ?????????)', category: 'women_children', icon: '?', status: 'Active' },

    // 4. ?????? ?????? ?????? (Emergency & Critical Care)
    { id: 'emergency', en: 'Emergency Medicine', ar: '?? ?????? ?????????', category: 'critical', icon: '?', status: 'Active' },
    { id: 'pediatric_emergency', en: 'Pediatric Emergency', ar: '????? ?????????', category: 'critical', icon: '?', status: 'Active' },
    { id: 'icu', en: 'Intensive Care Unit (ICU)', ar: '????? ?????? ??????', category: 'critical', icon: '?', status: 'Active' },
    { id: 'picu', en: 'Pediatric Intensive Care (PICU)', ar: '????? ?????? ??????', category: 'critical', icon: '?', status: 'Active' },
    { id: 'burns', en: 'Burn Unit', ar: '???? ?????? ????????? ??????', category: 'critical', icon: '?', status: 'Active' },

    // 5. ?????? ?????? ?????? ?????? (Support & Diagnostic Services)
    { id: 'anesthesiology', en: 'Anesthesiology', ar: '????? ?????? ??????', category: 'support', icon: '?', status: 'Active' },
    { id: 'radiology', en: 'Diagnostic Radiology', ar: '????? ?????????? ?????????', category: 'support', icon: '?', status: 'Active' },
    { id: 'laboratory', en: 'Clinical Laboratory', ar: '????? ?????? ?????????', category: 'support', icon: '?', status: 'Active' },
    { id: 'pathology', en: 'Anatomical Pathology', ar: '??? ?????? ?????????', category: 'support', icon: '?', status: 'Active' },
    { id: 'blood_bank', en: 'Blood Bank', ar: '??? ??? ??????', category: 'support', icon: '?', status: 'Active' },
    { id: 'rehabilitation', en: 'Physical Medicine & Rehabilitation', ar: '???? ?????? ?????? ??????', category: 'support', icon: '?', status: 'Active' },
    { id: 'clinical_dietary', en: 'Clinical Nutrition & Dietary', ar: '????? ?????? ?????????', category: 'support', icon: '?', status: 'Active' },

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    { id: 'clinical_pharmacy', en: 'Clinical Pharmacy', ar: '???????? ????????? ?????????', category:
'support', icon: '?', status: 'Active' },
    { id: 'infection_control', en: 'Infection Prevention & Control', ar: '?????? ?????? ????????? ?????????', ca
teory: 'support', icon: '?', status: 'Active' },

    // 6. ?? ??????? ??????? (Dental & Oral Health Family)
    { id: 'general_dentistry', en: 'General Dentistry', ar: '?? ??????? ??????', category: 'dental', icon: '?',
status: 'Active' },
    { id: 'orthodontics', en: 'Orthodontics', ar: '????? ????????? ?????????', category: 'dental', icon: '?', stat
us: 'Active' },
    { id: 'endodontics', en: 'Endodontics (Root Canal)', ar: '???? ???? ???? ?????????', category: 'dental', ico
n: '?', status: 'Active' },
    { id: 'maxillofacial', en: 'Oral & Maxillofacial Surgery', ar: '????? ???? ??????? ?????????', category: 'den
tal', icon: '?', status: 'Active' }
];

    // 100+ departments simulation (generating dynamically if needed, or listing the key 100 to show complete sy
stem scale)
    const categories = {
        medical: { en: 'Medical Specialties', ar: '????????? ?????????? ?????????' },
        surgical: { en: 'Surgical Specialties', ar: '????????? ??????????' },
        women_children: { en: 'Women & Children Health', ar: '??? ?????? ??????' },
        critical: { en: 'Emergency & Critical Care', ar: '????????? ?????????? ?????????' },
        support: { en: 'Diagnostic & Clinical Support', ar: '????????? ?????????? ??????????' },
        dental: { en: 'Dental & Oral Health', ar: '?? ??????? ???? ??????????' }
    };

    // Add more dynamic sub-specialties to reach 100+
    const totalNeeded = 105;
    let currentCount = specialtiesList.length;
    const categoriesKeys = Object.keys(categories);
    for (let i = 1; i <= (totalNeeded - currentCount); i++) {
        const cat = categoriesKeys[i % categoriesKeys.length];
        specialtiesList.push({
            id: `sub_spec_${i}`,
            en: `Sub-specialty Unit ${i}`,
            ar: `???? ?????? ?????? ${i}`,
            category: cat,
            icon: '?',
            status: 'Active',
            isDynamic: true
        });
    }

    // Handle local state or search query
    if (window.specialtiesSearchQuery === undefined) {
        window.specialtiesSearchQuery = '';
    }
    if (window.selectedSpecialtyCategory === undefined) {
        window.selectedSpecialtyCategory = 'all';
    }

    // Filter list
    const filtered = specialtiesList.filter(s => {

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const matchesSearch = s.en.toLowerCase().includes(window.specialtiesSearchQuery.toLowerCase()) ||
    s.ar.includes(window.specialtiesSearchQuery);
const matchesCategory = window.selectedSpecialtyCategory === 'all' || s.category === window.selectedSpecialtyCategory;
return matchesSearch && matchesCategory;
});

el.innerHTML = `
<div class="page-title">? ${tr('Master Specialties Panel (100+ HIS/EMR Units)', '???? ??????? ?????? ??? (100+ ????)')}</div>

<div class="card mb-16" style="padding: 16px;">
    <div style="display:flex;align-items:center;gap:12px;flex-wrap:wrap;justify-content:space-between">
        <div style="display:flex;align-items:center;gap:12px;flex:1;min-width:300px">
            <input type="text" id="specialtiesSearchInput" class="form-input"
                placeholder="${tr('Search specialties/departments...', '???? ?? ??????? ?? ?????? ????????. ..')}"
                value="${escapeHTML(window.specialtiesSearchQuery)}"
                style="flex:1">
            <button class="btn btn-primary" onclick="window.triggerSpecialtiesSearch()">${tr('Search', '???')}</button>
        </div>
        <div style="display:flex;align-items:center;gap:8px">
            <label style="font-weight:bold">${tr('Category:', '?????:')}</label>
            <select class="form-input" style="width:200px" onchange="window.filterSpecialtyCategory(this.value)">
                <option value="all" ${window.selectedSpecialtyCategory === 'all' ? 'selected' : ''}>${tr('All Categories', '???? ???????')}</option>
                ${Object.entries(categories).map(([k, v]) => `<option value="${k}" ${window.selectedSpecialtyCategory === k ? 'selected' : ''}>${escapeHTML(tr(v.en, v.ar))}</option>`).join('')}
            </select>
        </div>
    </div>

<!-- Analytics Stats -->
<div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px;margin-bottom:20px">
    <div class="card" style="padding:16px;display:flex;align-items:center;gap:16px;border-left:4px solid var(--primary-color)">
        <div style="font-size:32px">?</div>
        <div>
            <div style="font-size:24px;font-weight:bold;color:var(--text-main)">${specialtiesList.length}</div>
            <div style="font-size:12px;color:var(--text-muted)">${tr('Total Master Specialties', '????? ?????? ? ??????')}</div>
        </div>
    </div>
    <div class="card" style="padding:16px;display:flex;align-items:center;gap:16px;border-left:4px solid var(--success-color)">
        <div style="font-size:32px">?</div>
        <div>
            <div style="font-size:24px;font-weight:bold;color:var(--text-main)">${specialtiesList.filter(s=s.status==='Active').length}</div>
            <div style="font-size:12px;color:var(--text-muted)">${tr('Active & Operational', '????? ?????? ??

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?????')}</div>
    </div>
  </div>
  <div class="card" style="padding:16px;display:flex;align-items:center;gap:16px;border-left:4px solid var(--accent-color)">
    <div style="font-size:32px">?</div>
    <div>
      <div style="font-size:24px;font-weight:bold;color:var(--text-main)">${filtered.length}</div>
      <div style="font-size:12px;color:var(--text-muted)">${tr('Filtered Results', '???? ???? ????')}
    </div>
  </div>
  </div>

  <div class="card" style="padding:20px">
    <div style="display:grid;grid-template-columns:repeat(auto-fill,minmax(260px,1fr));gap:16px">
      ${filtered.map(s => `
        <div class="specialty-card" style="padding:16px;border:1px solid var(--border-color);border-radius:8px;background:var(--bg-light);transition:all 0.3s;display:flex;flex-direction:column;justify-content:space-between;cursor:pointer" onclick="window.viewSpecialtyDetails('${s.id}')">
          <div style="display:flex;align-items:center;gap:12px;margin-bottom:12px">
            <span style="font-size:24px">${s.icon}</span>
            <div>
              <div style="font-weight:bold;color:var(--text-main);font-size:15px">${escapeHTML(tr(s.en, s.ar))}</div>
              <div style="font-size:11px;color:var(--text-muted)">${escapeHTML(tr(categories[s.category]?.en || 'Sub Unit', categories[s.category]?.ar || '??? ????'))}</div>
            </div>
          </div>
          <div style="display:flex;align-items:center;justify-content:space-between">
            <span class="badge ${s.status === 'Active' ? 'badge-success' : 'badge-secondary'}" style="font-size:10px">${tr(s.status, s.status === 'Active' ? '???' : '??? ???')}</span>
            ${s.isDynamic ? `<span style="font-size:10px;color:var(--primary-color)">${tr('System Generated', '???? ????')}</span>` : `<span style="font-size:10px;color:var(--accent-color)">${tr('Core EHR', '??? ???')}</span>`}
          </div>
        </div>
      `).join('')}
    </div>
  </div>
`;

// Listeners
const searchInput = document.getElementById('specialtiesSearchInput');
if (searchInput) {
  searchInput.addEventListener('keyup', (e) => {
    if (e.key === 'Enter') {
      window.triggerSpecialtiesSearch();
    }
  });
}

// Global actions for page UI
window.triggerSpecialtiesSearch = () => {

```

```

    window.specialtiesSearchQuery = document.getElementById('specialtiesSearchInput')?.value || '';
    navigateTo(47);
};

window.filterSpecialtyCategory = (val) => {
    window.selectedSpecialtyCategory = val;
    navigateTo(47);
};

window.viewSpecialtyDetails = (id) => {
    const spec = specialtiesList.find(s => s.id === id);
    if (!spec) return;

    // Log simulation audit trail log
    API.post('/api/admin/audit-trail', {
        module: 'Specialties Panel',
        action: 'View Specialty',
        details: { specialty_id: id, en: spec.en, ar: spec.ar }
    }).catch(()=>{});

    showToast(`${tr('Opening clinical console for', '??? ???? ?????? ????????? ??')} ${tr(spec.en, spec.ar)}`);

    // If it's oncology or cardiology, go to specialized clinical page (index 44)
    if (id === 'oncology' || id === 'cardiology' || id === 'pediatrics') {
        window.specialtiesTab = id === 'oncology' ? 'oncology' : (id === 'cardiology' ? 'cardiology' : 'cardiology');
        navigateTo(44);
    }
};

}

// ===== QUALITY =====
let qTab = 'incidents';
async function renderQuality(el) {
    const content = el;
    if (!window.qualityTab) window.qualityTab = 'dashboard';

    const selectedModule = window.auditModuleFilter || '';

    // Fetch data
    const [incidents, satisfaction, kpis, stats, auditLogs] = await Promise.all([
        API.get('/api/quality/incidents').catch(() => []),
        API.get('/api/quality/satisfaction').catch(() => []),
        API.get('/api/quality/kpis').catch(() => []),
        API.get('/api/quality/stats').catch(() => ({ openIncidents: 0, totalIncidents: 0, avgSatisfaction: 0, kpiOnTrack: 0, kpiTotal: 0 })),
        API.get(selectedModule ? `/api/admin/audit-trail?limit=50&module=${selectedModule}` : '/api/admin/audit-trail?limit=50').catch(() => [])
    ]);

    const avgSat = stats.avgSatisfaction || 'N/A';

    // Calculate compliance checklist score from localStorage
    const checklistItems = [

```

```

    { id: 'zatca', cat: 'ZATCA', title_ar: '????????? ????? ?? ??? ?????? ????????? (Phase 2)', title_en: 'ZATC
A Phase 2 Integration Compliance', desc_ar: '????? ?????? ?????? ?????? ?????? XML ?????? ?????????? ????????.', d
esc_en: 'Digital stamp generation, XML layout, real-time ZATCA clearance.', weight: 25 },
    { id: 'cbahi_consent', cat: 'CBAHI', title_ar: '????? ??? ???? ???? ?????????? ??????????', title_en: 'Patient
Rights & Informed Consent Policy', desc_ar: '????? ?? ????? ?????? ?????? ?????? ?????? ??? ????????.',
desc_en: 'Verification of medical consent signatures for all patients pre-operation.', weight: 15 },
    { id: 'cbahi_hygiene', cat: 'CBAHI', title_ar: '???? ?????????? ?????? ?????? ??????', title_en: 'Hand Hygie
ne Compliance Observational Audits', desc_ar: '????? ?????? ?? ?????? ?????? ?????? ?????? ?????? ?? ??????? ?
?????', desc_en: 'Monitoring staff compliance rate with hand hygiene protocols.', weight: 15 },
    { id: 'cbahi_meds', cat: 'CBAHI', title_ar: '????????????? ?????? ?????? ??????', title_en: 'High-Alert Medi
cations Verification Protocol', desc_ar: '???? ?????? ?????? ?????? ?????? ?????? ?????? ?????? ?????? ??????', desc_en
: 'Double-verification system for narcotics and controlled high-alert drugs.', weight: 15 },
    { id: 'moh_records', cat: 'MOH', title_ar: '???? ?????????? ?????? ?????? ?????????? ??????', title_en: 'Elec
tronic Health Record Confidentiality', desc_ar: '????? ?????? ?????? ??? ?????? ?? ?????? ?? ? ?????? ??? ?
?????', desc_en: 'Medical record encryption and unauthorized access prevention logs.', weight: 15 },
    { id: 'jci_comm', cat: 'JCI', title_ar: '????? ?????? ??? ?????????? ?????? (ISBAR)', title_en: 'Effecti
ve Communication & Clinical Handover (ISBAR)', desc_ar: '????? ?????? ISBAR ??? ?????? ?????????? ?????? ???
????? ??????', desc_en: 'Standardized ISBAR structure for handovers and critical notifications.', weight: 15
}
];

let completedWeight = 0;
let totalWeight = 0;
checklistItems.forEach(item => {
  const isDone = localStorage.getItem('compliance_' + item.id) === 'true';
  if (isDone) completedWeight += item.weight;
  totalWeight += item.weight;
});

const complianceScore = totalWeight ? Math.round((completedWeight / totalWeight) * 100) : 100;

let htmlContent = `
<div class="mb-xl flex flex-col md:flex-row md:items-end justify-between gap-md">
  <div>
    <nav class="flex text-on-surface-variant font-caption text-caption mb-xs">
      <span>${tr('Operations', '????????')}</span>
      <span class="mx-xs"></span>
      <span class="text-primary font-bold">${tr('Quality & Compliance', '????? ?????????')}</span>
    </nav>
    <h3 class="font-display-lg text-display-lg text-primary tracking-tight">${tr('Quality Assurance & Comp
liance', '????? ?????? ?????????')}</h3>
    <p class="font-body-md text-body-md text-on-surface-variant max-w-2xl">
      ${tr('Monitor clinical indicators, patient safety incident reports, Saudi ZATCA & CBAHI compliance c
hecklist, and system audit trails.', '????? ?????????? ?????????? ?????? ?????? ?????? ?????? ?????? ?? ??????
????? ?????? ?????? ?????????? ?????????? (ZATCA)? ?????? ?????? ??????')}
    </p>
  </div>
</div>

<!-- Tabs Bar -->
<div class="tab-bar mb-md flex border-b border-outline-variant/30 overflow-x-auto gap-2">
  <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.qualityTab === 'dashboar
d' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary
'}" onclick="window.qualityTab='dashboard'; navigateTo(currentPage)">
    ? ${tr('Dashboard & KPIs', '???? ?????????')}
  </button>
</div>

```



```

        </button>
        <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.qualityTab === 'incident
s' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary
'}" onclick="window.qualityTab='incidents'; navigateTo(currentPage)">
            ?? ${tr('Incident Log', '??? ?????? ?????????')}
        </button>
        <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.qualityTab === 'checklis
t' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primary
'}" onclick="window.qualityTab='checklist'; navigateTo(currentPage)">
            ? ${tr('Compliance Checklist', '????? ?????? ?? ?????????')}
        </button>
        <button class="tab-btn px-lg py-sm border-b-2 font-bold transition-all ${window.qualityTab === 'audit_lo
gs' ? 'active border-secondary text-secondary' : 'border-transparent text-on-surface-variant hover:text-primar
y'}" onclick="window.qualityTab='audit_logs'; navigateTo(currentPage)">
            ? ${tr('System Audit Logs', '????? ?????? ?????????')}
        </button>
    </div>
    `;

    if (window.qualityTab === 'dashboard') {
        htmlContent += `
            <!-- KPI Bento Grid -->
            <div class="grid grid-cols-1 md:grid-cols-4 gap-gutter mb-xl">
                <!-- Open Incidents -->
                <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
                    <div class="flex justify-between items-start relative z-10">
                        <span class="p-xs bg-error/10 rounded-lg text-error material-symbols-outlined font-icon">warning</
span>
                        <span class="text-error text-xs font-bold">${stats.openIncidents} ${tr('Open', '????')}</span>
                    </div>
                    <div class="mt-lg relative z-10">
                        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Open Incidents', '????? ????
??? ?????????')}</p>
                        <h4 class="text-display-lg text-primary font-bold mt-xs">${stats.openIncidents} <span class="text-
title-lg text-on-surface-variant"> ${stats.totalIncidents}</span></h4>
                    </div>
                </div>

                <!-- Patient Satisfaction -->
                <div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
                    <div class="flex justify-between items-start relative z-10">
                        <span class="p-xs bg-tertiary-container/30 rounded-lg text-on-tertiary-container material-symbols-
outlined font-icon">sentiment_very_satisfied</span>
                        <span class="text-on-tertiary-container text-xs font-bold">? ${avgSat} / 5</span>
                    </div>
                    <div class="mt-lg relative z-10">
                        <p class="text-on-surface-variant font-label-md text-label-md">${tr('Avg Patient Satisfaction', '?
???? ??? ?????')}</p>
                        <h4 class="text-display-lg text-primary font-bold mt-xs">${avgSat} <span class="text-title-lg text
-on-surface-variant">?</span></h4>
                    </div>
                </div>

            <!-- KPIs On Track -->

```

```

<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
  <div class="flex justify-between items-start relative z-10">
    <span class="p-xs bg-secondary/10 rounded-lg text-secondary material-symbols-outlined font-icon">t
rendering_up</span>
    <span class="text-secondary text-xs font-bold">${stats.kpiOnTrack} / ${stats.kpiTotal}</span>
  </div>
  <div class="mt-lg relative z-10">
    <p class="text-on-surface-variant font-label-md text-label-md">${tr('KPIs On Track', '???????? ???
???? ?????')}</p>
    <h4 class="text-display-lg text-primary font-bold mt-xs">${stats.kpiOnTrack} <span class="text-tit
le-lg text-on-surface-variant">/ ${stats.kpiTotal}</span></h4>
  </div>
</div>

<!-- Compliance Score -->
<div class="glass-card p-md rounded-2xl flex flex-col justify-between group overflow-hidden relative">
  <div class="flex justify-between items-start relative z-10">
    <span class="p-xs bg-primary/10 rounded-lg text-primary material-symbols-outlined font-icon">verif
ied_user</span>
    <span class="text-primary text-xs font-bold">${complianceScore}%</span>
  </div>
  <div class="mt-lg relative z-10">
    <p class="text-on-surface-variant font-label-md text-label-md">${tr('Compliance Score', '???? ???
???? ?????????')}</p>
    <h4 class="text-display-lg text-secondary font-bold mt-xs">${complianceScore}%</h4>
  </div>
</div>

<!-- Main Layout -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
  <!-- Add KPI Form -->
  <div class="glass-card p-md rounded-2xl lg:col-span-1">
    <h4 class="text-title-lg font-bold text-primary mb-md">${tr('Add Quality KPI', '????? ???
???? ???
')}</h4>

    <div class="form-group mb-sm">
      <label class="text-label-md text-on-surface-variant block mb-xs">${tr('KPI Name (English)', '??? ?
????? (???????)')}</label>
      <input type="text" class="form-input w-full" id="kpiName" placeholder="e.g. Patient Wait Time">
    </div>

    <div class="form-group mb-sm">
      <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Target Value', '????? ???
????? (%)')}</label>
      <input type="number" class="form-input w-full" id="kpiTarget" value="90">
    </div>

    <div class="form-group mb-sm">
      <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Actual Value', '????? ???
??? (%)')}</label>
      <input type="number" class="form-input w-full" id="kpiActual" value="85">
    </div>
  </div>
</div>

```

```

<div class="form-group mb-sm">
  <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Period', '?????')}</label>
  <select class="form-input w-full" id="kpiPeriod">
    <option value="Q1 2026">Q1 2026</option>
    <option value="Q2 2026">Q2 2026</option>
    <option value="Monthly">Monthly</option>
  </select>
</div>

<button class="btn btn-primary w-full mt-md flex items-center justify-center gap-xs" onclick="window
.addKPI()">
  <span class="material-symbols-outlined text-body-lg">add</span>
  <span>${tr('Add KPI Indicator', '????? ???? ?????')}</span>
</button>
</div>

<!-- KPIs List Table -->
<div class="glass-card p-md rounded-2xl lg:col-span-2">
  <h4 class="text-title-lg font-bold text-primary mb-md">${tr('Active Indicators List', '????? ?????
????? ?????')}</h4>
  <div class="overflow-x-auto custom-scrollbar">
    <table class="w-full text-right border-collapse">
      <thead>
        <tr class="border-b border-outline-variant/30 text-on-surface-variant font-label-md">
          <th class="py-sm px-md">${tr('Indicator Name', '??? ?????')}</th>
          <th class="py-sm px-md">${tr('Target', '????????')}</th>
          <th class="py-sm px-md">${tr('Actual', '?????')}</th>
          <th class="py-sm px-md">${tr('Period', '?????')}</th>
          <th class="py-sm px-md">${tr('Status', '?????')}</th>
        </tr>
      </thead>
      <tbody class="divide-y divide-outline-variant/10 text-body-md text-on-surface">
        ${kpis.length ? kpis.map(k => {
          let statusClass = 'bg-error/10 text-error';
          if (k.status === 'On Track') statusClass = 'bg-success/10 text-success';
          else if (k.status === 'At Risk') statusClass = 'bg-tertiary-container text-on-tertiary-conta
iner';

          return `
            <tr class="hover:bg-surface-variant/20 transition-all">
              <td class="py-sm px-md font-bold">${escapeHTML(k.kpi_name || k.kpi_name_ar)}</td>
              <td class="py-sm px-md">${escapeHTML(k.target_value)}${escapeHTML(k.unit || '%')}</td>
              <td class="py-sm px-md">${escapeHTML(k.actual_value)}${escapeHTML(k.unit || '%')}</td>
              <td class="py-sm px-md">${escapeHTML(k.period || '')}</td>
              <td class="py-sm px-md">
                <span class="px-sm py-xs rounded-full font-label-md ${statusClass}">
                  ${escapeHTML(tr(k.status, k.status === 'On Track' ? '?????' : k.status === 'At Risk'
? '???? ?????' : '??? ?????'))}
                </span>
              </td>
            </tr>
          `;
        }).join('') : `
            <tr>

```

```

        <td colspan="5" class="py-lg text-center text-on-surface-variant font-body-md">${tr('No KP
Is registered yet.', '?? ???? ????? ???? ????? ???.')}</td>
    </tr>
    `}
</tbody>
</table>
</div>
</div>
</div>
`;
} else if (window.qualityTab === 'incidents') {
    htmlContent += `
        <div class="grid grid-cols-1 lg:grid-cols-3 gap-gutter">
            <!-- Report Form -->
            <div class="glass-card p-md rounded-2xl lg:col-span-1">
                <h4 class="text-title-lg font-bold text-primary mb-md">${tr('Report Safety Incident', '?????? ?? ??
??? ?????')}</h4>

                <div class="form-group mb-sm">
                    <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Incident Type', '??? ?????
?')}</label>
                    <select class="form-input w-full" id="qiType">
                        <option value="medication_error">${tr('Medication Error', '??? ?????')}</option>
                        <option value="fall">${tr('Patient Fall', '???? ?????')}</option>
                        <option value="infection">${tr('Healthcare Infection', '???? ?????')}</option>
                        <option value="equipment">${tr('Equipment Malfunction', '??? ?? ?????? ?????')}</option>
                        <option value="complaint">${tr('Patient Complaint', '???? ?????')}</option>
                        <option value="other">${tr('Other', '????')}</option>
                    </select>
                </div>

                <div class="form-group mb-sm">
                    <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Severity Level', '????? ??
????')}</label>
                    <select class="form-input w-full" id="qiSeverity">
                        <option value="low">? ${tr('Low / Minor', '????? / ?????')}</option>
                        <option value="medium">? ${tr('Medium / Moderate', '?????')}</option>
                        <option value="high">? ${tr('High / Severe', '????? / ?????')}</option>
                        <option value="critical">? ${tr('Critical / Sentinel', '???? / ????? ?????')}</option>
                    </select>
                </div>

                <div class="form-group mb-sm">
                    <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Harm Level', '????? ?????')}
                </label>
                    <select class="form-input w-full" id="qiHarm">
                        <option value="None">${tr('No Harm', '???? ???')}</option>
                        <option value="Mild">${tr('Mild', '????')}</option>
                        <option value="Moderate">${tr('Moderate', '????')}</option>
                        <option value="Severe">${tr('Severe', '????')}</option>
                        <option value="Death">${tr('Death', '????')}</option>
                    </select>
                </div>
            </div>
        </div>
    `;
}

```

```

        <div class="form-group mb-sm flex items-center gap-md flex-wrap">
            <label class="text-label-md text-on-surface flex items-center gap-xs"><input type="checkbox" id="qiNearMiss"> ${tr('Near Miss', '???? ???')}</label>
            <label class="text-label-md text-on-surface flex items-center gap-xs"><input type="checkbox" id="qiConfidential"> ? ${tr('Confidential', '???')}</label>
        </div>

        <div class="form-group mb-sm">
            <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Department', '????')}</label>

            <input type="text" class="form-input w-full" id="qiDept" placeholder="e.g. ICU / ??????">
        </div>

        <div class="form-group mb-sm">
            <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Incident Description', '????
            ?????? ??????')}</label>
            <textarea class="form-input w-full" id="qiDesc" rows="3" placeholder="${tr('Describe what happened
            ...', '???? ?????? ?? ??? ?????...')}" required></textarea>
        </div>

        <div class="form-group mb-md">
            <label class="text-label-md text-on-surface-variant block mb-xs">${tr('Immediate Corrective Action
            ', '?????? ?????? ??????')}</label>
            <input type="text" class="form-input w-full" id="qiAction" placeholder="e.g. Notify doctor, check
            vitals">
        </div>

        <button class="btn btn-primary w-full flex items-center justify-center gap-xs" onclick="window.report
        tIncident()">
            <span class="material-symbols-outlined text-body-lg">report_problem</span>
            <span>${tr('Submit Incident Report', '????? ??????')}</span>
        </button>
    </div>

    <!-- Incidents List -->
    <div class="glass-card p-md rounded-2xl lg:col-span-2">
        <h4 class="text-title-lg font-bold text-primary mb-md">${tr('Reported Incidents Log', '??? ??????
        ??????')}</h4>
        <div class="overflow-x-auto custom-scrollbar">
            <table class="w-full text-right border-collapse">
                <thead>
                    <tr class="border-b border-outline-variant/30 text-on-surface-variant font-label-md">
                        <th class="py-sm px-md">${tr('Type', '????')}</th>
                        <th class="py-sm px-md">${tr('Severity', '???????')}</th>
                        <th class="py-sm px-md">${tr('Dept', '????')}</th>
                        <th class="py-sm px-md">${tr('Date', '???????')}</th>
                        <th class="py-sm px-md">${tr('Status', '???????')}</th>
                        <th class="py-sm px-md">${tr('Actions', '????????')}</th>
                    </tr>
                </thead>
                <tbody class="divide-y divide-outline-variant/10 text-body-md text-on-surface">
                    ${incidents.length ? incidents.map(i => {
                        let sevColor = 'bg-success/10 text-success';
                        if (i.severity === 'critical') sevColor = 'bg-primary text-white';
                    }} : ''}
                </tbody>
            </table>
        </div>
    </div>

```

```

else if (i.severity === 'high') sevColor = 'bg-error/10 text-error';
else if (i.severity === 'medium') sevColor = 'bg-tertiary-container text-on-tertiary-contain
er';

const typeName = tr(i.type || i.incident_type || '', i.type || i.incident_type || '');

return `
<tr class="hover:bg-surface-variant/20 transition-all">
  <td class="py-sm px-md font-bold">${escapeHTML(typeName)}</td>
  <td class="py-sm px-md">
    <span class="px-sm py-xs rounded-full font-label-md ${sevColor}">
      ${escapeHTML(tr(i.severity, i.severity === 'low' ? '?????' : i.severity === 'medium
' ? '?????' : i.severity === 'high' ? '?????' : '????'))}
    </span>
  </td>
  <td class="py-sm px-md">${escapeHTML(i.department || '')}</td>
  <td class="py-sm px-md">${i.created_at ? new Date(i.created_at).toLocaleDateString('ar-S
A') : ''}</td>
  <td class="py-sm px-md">
    <span class="px-sm py-xs rounded-full font-label-md ${i.status === 'Closed' ? 'bg-succ
ess/10 text-success' : 'bg-error/10 text-error'}">
      ${escapeHTML(tr(i.status, i.status === 'Closed' ? '????' : '????'))}
    </span>
  </td>
  <td class="py-sm px-md">
    <div class="flex gap-xs flex-wrap">
      <button class="px-sm py-xs bg-primary/10 text-primary rounded hover:opacity-90 font-
label-md inline-flex items-center gap-xs" onclick="window.manageCAPA(${safeId(i.id)})">
        <span class="material-symbols-outlined text-body-sm">task_alt</span>
        <span>${tr('CAPA', '??????')}</span>
      </button>
      ${i.workflow_state || i.status) !== 'Closed' ? `
      <button class="px-sm py-xs bg-secondary text-white rounded hover:opacity-90 font-lab
el-md inline-flex items-center gap-xs" onclick="window.closeIncident(${safeId(i.id)})">
        <span class="material-symbols-outlined text-body-sm">check</span>
        <span>${tr('Close', '????')}</span>
      </button>
      ` : '<span class="text-on-surface-variant text-xs">? Done</span>'}
    </div>
  </td>
</tr>
`;
}).join('') : `
<tr>
  <td colspan="6" class="py-lg text-center text-on-surface-variant font-body-md">${tr('No in
cidents reported.', '?? ???? ????? ??????.')}</td>
</tr>
`
}
</tbody>
</table>
</div>
</div>
</div>
`;

```

```

} else if (window.qualityTab === 'checklist') {
  htmlContent += `
    <div class="glass-card p-md rounded-2xl mb-xl">
      <div class="flex justify-between items-center mb-md border-b border-outline-variant/30 pb-sm">
        <div>
          <h4 class="text-title-lg font-bold text-primary">${tr('Hospital Compliance Checklist', '????? ????
?? ?????? ?? ?????????')}</h4>
          <p class="text-body-md text-on-surface-variant">${tr('Verify systems readiness for Saudi ZATCA Pha
se 2, CBAHI Clinical Standards, and MOH Regulations.', '????? ?? ?????? ??????? ?????? ?????????? ?? ?????? ?
????? ?????? ?????????? ?????????? ?????? ??????.')}</p>
        </div>
        <div class="text-left">
          <span class="text-caption text-on-surface-variant block font-label-md">${tr('Total Compliance Scor
e', '???? ?????????? ??????????')}</span>
          <span class="text-display-lg text-secondary font-bold font-display-lg">${complianceScore}%</span>
        </div>
      </div>

      <div class="w-full bg-surface-variant/30 h-3 rounded-full overflow-hidden mb-lg relative">
        <div class="bg-secondary h-full transition-all duration-500" style="width: ${complianceScore}%></di
v>

      </div>

      <div class="grid grid-cols-1 md:grid-cols-2 gap-md">
        ${checklistItems.map(item => {
          const isDone = localStorage.getItem('compliance_' + item.id) === 'true';
          const catColor = item.cat === 'ZATCA' ? 'bg-secondary/10 text-secondary' : item.cat === 'CBAHI' ?
'bg-primary/10 text-primary' : 'bg-tertiary-container text-on-tertiary-container';
          return `
            <div class="p-md border border-outline-variant/30 rounded-2xl hover:border-secondary/50 transiti
on-all flex items-start justify-between gap-md ${isDone ? 'bg-success/5 border-success/30' : 'bg-transparent'}
">

              <div class="flex items-start gap-sm">
                <div class="pt-1">
                  <input type="checkbox" class="w-5 h-5 rounded border-outline cursor-pointer focus:ring-sec
ondary text-secondary"
                    ${isDone ? 'checked' : ''}
                    onchange="localStorage.setItem('compliance_${item.id}', this.checked); navigateTo(curren
tPage)">
                </div>
                <div>
                  <span class="px-sm py-xs rounded font-label-md text-caption block w-fit mb-xs ${catColor}"
>${item.cat}</span>
                  <h5 class="font-bold text-body-lg text-primary ${isDone ? 'line-through text-on-surface-va
riant/60' : ''}>${tr(item.title_en, item.title_ar)}</h5>
                  <p class="text-body-md text-on-surface-variant mt-xs">${tr(item.desc_en, item.desc_ar)}</p>
                </div>
              </div>

              <div class="text-left font-label-md text-caption text-on-surface-variant font-bold">
                +${item.weight}%
              </div>
            </div>
          `;
        })}
      </div>
    `;
  };

```

```

    }).join('')}
  </div>
</div>
`;
} else if (window.qualityTab === 'audit_logs') {
  htmlContent += `
    <div class="glass-card p-md rounded-2xl">
      <div class="flex flex-col md:flex-row md:items-center justify-between gap-md mb-md border-b border-out
line-variant/30 pb-sm">
        <div>
          <h4 class="text-title-lg font-bold text-primary">${tr('System Audit Trail', '??? ????? ??????')}</h4>
          <p class="text-body-md text-on-surface-variant">${tr('Real-time security and administrative operat
ion tracking.', '????? ???????? ???????? ???????? ?????? ?? ?????? ??? ????')}</p>
          </div>
          <div class="flex items-center gap-xs">
            <label class="text-label-md text-on-surface-variant font-bold whitespace-nowrap">${tr('Filter Modu
le:', '???? ?????:')}</label>
            <select class="form-input" onchange="window.auditModuleFilter=this.value; navigateTo(currentPage)"
>
              <option value="" ${!selectedModule ? 'selected' : ''}>${tr('All Modules', '?? ??????')}</option>
              <option value="Auth" ${selectedModule === 'Auth' ? 'selected' : ''}>${tr('Authentication', '????
?? ??????')}</option>
              <option value="Finance" ${selectedModule === 'Finance' ? 'selected' : ''}>${tr('Finance', '????
??')}</option>
              <option value="Pharmacy" ${selectedModule === 'Pharmacy' ? 'selected' : ''}>${tr('Pharmacy', '??
????')}</option>
              <option value="Patients" ${selectedModule === 'Patients' ? 'selected' : ''}>${tr('Patients Regis
try', '??? ?????')}</option>
              <option value="OB/GYN" ${selectedModule === 'OB/GYN' ? 'selected' : ''}>${tr('OB/GYN & Clinical'
, '????? ?????')}</option>
            </select>
          </div>
        </div>
      <div class="overflow-x-auto custom-scrollbar">
        <table class="w-full text-right border-collapse">
          <thead>
            <tr class="border-b border-outline-variant/30 text-on-surface-variant font-label-md">
              <th class="py-sm px-md">${tr('User', '???????')}</th>
              <th class="py-sm px-md">${tr('Action', '????')}</th>
              <th class="py-sm px-md">${tr('Module', '????')}</th>
              <th class="py-sm px-md">${tr('Details', '???????')}</th>
              <th class="py-sm px-md">${tr('IP Address', '???? IP')}</th>
              <th class="py-sm px-md">${tr('Timestamp', '???? ??????')}</th>
            </tr>
          </thead>
          <tbody class="divide-y divide-outline-variant/10 text-body-md text-on-surface">
            ${auditLogs.length ? auditLogs.map(l => `
              <tr class="hover:bg-surface-variant/20 transition-all">
                <td class="py-sm px-md font-bold">${escapeHTML(l.user_name || l.user_id || 'System')}</td>
                <td class="py-sm px-md">
                  <span class="px-sm py-xs rounded font-label-md text-caption $f

```



```

        l.action === 'DELETE' || l.action === 'SOFT_DELETE' ? 'bg-error/10 text-error' :
        l.action === 'CHANGE_PASSWORD' || l.action === 'LOGIN' ? 'bg-primary/10 text-primary' :
        'bg-secondary/10 text-secondary'
    }">
        ${escapeHTML(l.action || '')}
    </span>
</td>
<td class="py-sm px-md font-label-md text-caption text-on-surface-variant">${escapeHTML(l.module || '')}</td>
    <td class="py-sm px-md text-body-sm text-on-surface-variant">${escapeHTML(l.details || l.description || '')}</td>
    <td class="py-sm px-md font-mono text-caption text-on-surface-variant">${escapeHTML(l.ip_address || '')}</td>
    <td class="py-sm px-md text-caption">${l.created_at ? new Date(l.created_at).toLocaleString('ar-SA') : ''}</td>
</tr>
`}).join('') : `
<tr>
    <td colspan="6" class="py-lg text-center text-on-surface-variant font-body-md">${tr('No audit logs recorded for this module.', '?? ??? ???? ???? ???? ???? ????.')}</td>
</tr>
`
</tbody>
</table>
</div>
</div>
`;
}

content.innerHTML = htmlContent;
}

window.reportIncident = async function () {
    try {
        const r = await API.post('/api/quality/incidents', {
            incident_type: document.getElementById('qiType').value,
            severity: document.getElementById('qiSeverity').value,
            harm_level: document.getElementById('qiHarm') ? document.getElementById('qiHarm').value : 'None',
            near_miss: document.getElementById('qiNearMiss') ? document.getElementById('qiNearMiss').checked : false,
            confidential: document.getElementById('qiConfidential') ? document.getElementById('qiConfidential').checked : false,
            department: document.getElementById('qiDept').value,
            description: document.getElementById('qiDesc').value,
            immediate_action: document.getElementById('qiAction').value
            // reported_by is stamped server-side from the session (anti-spoof)
        });
        if (r && r.error) { showToast(r.error, 'error'); return; }
        showToast(tr('Reported!', '?? ??????!!'));
        await navigateTo(27);
    } catch (e) {
        showToast(tr('Error', '???'), 'error');
    }
};

```

```

window.closeIncident = async function (id) {
  try {
    // Server enforces the incident state machine; surface 409/validation errors to the user.
    const r = await API.put('/api/quality/incidents/' + parseInt(id, 10), { status: 'Closed' });
    if (r && r.error) { showToast(r.error, 'error'); return; }
    showToast(tr('Closed!', '?? ???????!'));
    await navigateTo(27);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

// E17: per-incident CAPA management (corrective/preventive actions + state machine).
if (!window.manageCAPA) window.manageCAPA = async function (id) {
  const incidentId = parseInt(id, 10);
  if (!Number.isInteger(incidentId)) return;
  let list = [];
  try {
    list = await API.get('/api/quality/incidents/' + incidentId + '/capa');
    if (list && list.error) { showToast(list.error, 'error'); return; }
  } catch (e) { showToast(tr('Error', '???'), 'error'); return; }
  if (!Array.isArray(list)) list = [];

  const CAPA_NEXT = { 'Pending': ['InProgress', 'Cancelled'], 'InProgress': ['Completed', 'Cancelled'], 'Completed': ['Verified'], 'Verified': [], 'Cancelled': [] };
  const rows = list.map(c => {
    const nexts = (CAPA_NEXT[c.status] || []).map(s =>
      `<button class="px-xs py-xxs bg-secondary/10 text-secondary rounded text-xs" onclick="window.capaTransition(${safeId(c.id)}, '${jsStr(s)}', ${safeId(incidentId)})">${escapeHTML(s)}</button>`
    ).join(' ');
    return `<tr class="border-b border-outline-variant/10">
      <td class="py-xs px-sm">${escapeHTML(c.capa_type || '')}</td>
      <td class="py-xs px-sm">${escapeHTML(c.title || '')}</td>
      <td class="py-xs px-sm">${escapeHTML(c.owner_name || '')}</td>
      <td class="py-xs px-sm">${escapeHTML(c.due_date || '')}</td>
      <td class="py-xs px-sm"><span class="px-xs py-xxs rounded-full text-xs bg-surface-variant/40">${escapeHTML(c.status || '')}</span></td>
      <td class="py-xs px-sm">${nexts}</td>
    </tr>`;
  }).join('');

  const html = `
<div class="glass-card p-md rounded-2xl" style="max-width:760px">
  <h4 class="text-title-lg font-bold text-primary mb-md">${tr('CAPA ? Incident', '????????? ?????????? ?????????')} #${escapeHTML(incidentId)}</h4>
  <div class="grid grid-cols-2 gap-sm mb-sm">
    <select class="form-input" id="capaType"><option value="Corrective">${tr('Corrective', '??????')}</option><option value="Preventive">${tr('Preventive', '?????')}</option></select>
    <input class="form-input" id="capaTitle" placeholder="${tr('Title', '????????')}">
    <input class="form-input" id="capaOwner" placeholder="${tr('Owner', '????????')}">
    <input class="form-input" type="date" id="capaDue">
  </div>
  <textarea class="form-input w-full mb-sm" id="capaDesc" rows="2" placeholder="${tr('Action description',

```

```

'??? ??????')}]></textarea>
    <button class="btn btn-primary mb-md" onclick="window.saveCAPA(${safeId(incidentId)})">${tr('Add CAPA',
'???? ????')}</button>
    <div class="overflow-x-auto">
        <table class="w-full text-right text-body-md">
            <thead><tr class="text-on-surface-variant font-label-md border-b border-outline-variant/30">
                <th class="py-xs px-sm">${tr('Type', '????')}</th><th class="py-xs px-sm">${tr('Title', '??????'
)}</th><th class="py-xs px-sm">${tr('Owner', '??????')}</th><th class="py-xs px-sm">${tr('Due', '????????')}
</th><th class="py-xs px-sm">${tr('Status', '????')}</th><th class="py-xs px-sm">${tr('Actions', '????????')}
</th>
            </tr></thead>
            <tbody>${rows || `<tr><td colspan="6" class="py-md text-center text-on-surface-variant">${tr('No CAP
A yet.', '?? ???? ???????.')}</td></tr>`}</tbody>
        </table>
    </div>`;
    showModal(tr('CAPA Management', '????? ?????????? ?????????? ??????????'), html);
};

if (!window.saveCAPA) window.saveCAPA = async function (incidentId) {
    try {
        const r = await API.post('/api/quality/incidents/' + parseInt(incidentId, 10) + '/capa', {
            capa_type: document.getElementById('capaType').value,
            title: document.getElementById('capaTitle').value,
            description: document.getElementById('capaDesc').value,
            owner_name: document.getElementById('capaOwner').value,
            due_date: document.getElementById('capaDue').value || null
        });
        if (r && r.error) { showToast(r.error, 'error'); return; }
        showToast(tr('Added!', '?? ???????!'));
        window.manageCAPA(incidentId);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

if (!window.capaTransition) window.capaTransition = async function (capaId, status, incidentId) {
    try {
        // Server rejects invalid CAPA transitions with 409.
        const r = await API.put('/api/quality/capa/' + parseInt(capaId, 10), { status: status });
        if (r && r.error) { showToast(r.error, 'error'); return; }
        showToast(tr('Updated!', '?? ???????!'));
        window.manageCAPA(incidentId);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

window.addKPI = async function () {
    try {
        await API.post('/api/quality/kpis', {
            kpi_name: document.getElementById('kpiName').value,
            target_value: document.getElementById('kpiTarget').value,
            actual_value: document.getElementById('kpiActual').value,
            period: document.getElementById('kpiPeriod').value
        });
        showToast(tr('Added!', '?? ???????!'));
        await navigateTo(27);
    }

```

```

    } catch (e) {
      showToast(tr('Error', '???'), 'error');
    }
  };
// ===== MAINTENANCE =====
let mntTab = 'dashboard';
async function renderMaintenance(el) {
  const content = el;

  let stats = { openW0: 0, inProgressW0: 0, overduePM: 0, totalEquipment: 0 };
  let mntOrders = [];
  let workOrders = [];
  let equipment = [];

  try {
    stats = await API.get('/api/maintenance/stats').catch(() => ({ openW0: 0, inProgressW0: 0, overduePM: 0, totalEquipment: 0 }));
    mntOrders = await API.get('/api/maintenance/orders').catch(() => []);
    workOrders = await API.get('/api/maintenance/work-orders').catch(() => []);
    equipment = await API.get('/api/maintenance/equipment').catch(() => []);
  } catch (e) {
    console.error('Error fetching maintenance data:', e);
  }

  const activeW0 = (parseInt(stats.openW0) || 0) + (parseInt(stats.inProgressW0) || 0);
  const overduePM = parseInt(stats.overduePM) || 0;
  const totalEquip = parseInt(stats.totalEquipment) || equipment.length;

  content.innerHTML = `
    <div class="flex justify-between items-end mb-6">
      <div>
        <h2 class="font-headline-lg text-headline-lg text-primary">${tr('Biomedical & Facility Maintenance', '
        ????? ??????? ??????? ??????')}</h2>
        <p class="text-on-surface-variant text-sm mt-1">${tr('Monitor medical devices safety, schedule preventive calibrations, and log repair tickets', '?????? ????? ??????? ??????? ????? ?????????? ??????? ?????
        ???????')}</p>
      </div>
    </div>

    <!-- Tabs Navigation -->
    <div class="flex gap-4 mb-6 border-b border-outline-variant/30 pb-2">
      <button class="px-4 py-2 font-bold rounded-lg transition-all ${mntTab === 'dashboard' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="mntTab='dashboard';navigateTo(28)">
        ? ${tr('Maintenance Dashboard', '???? ???????')}
      </button>
      <button class="px-4 py-2 font-bold rounded-lg transition-all ${mntTab === 'orders' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="mntTab='orders';navigateTo(28)">
        ? ${tr('Work Orders', '????? ???????')}
      </button>
      <button class="px-4 py-2 font-bold rounded-lg transition-all ${mntTab === 'assets' ? 'bg-primary text-white shadow-md' : 'text-on-surface-variant hover:bg-surface-container-high/50'}" onclick="mntTab='assets';navigateTo(28)">

```

```
? ${tr('Biomedical Assets', '??? ?????? ??????')}
</button>
</div>

<div class="tab-pane-content">
  ${mntTab === 'dashboard' ? `
    <!-- Metrics Bento Grid -->
    <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-md mb-lg">
      <div class="glass-card p-6 rounded-2xl flex flex-col justify-between">
        <div class="flex justify-between items-start">
          <div class="p-3 bg-primary/5 rounded-xl text-primary flex items-center justify-center">
            <span class="material-symbols-outlined">precision_manufacturing</span>
          </div>
          <span class="text-green-600 text-xs font-bold">+2.4%</span>
        </div>
        <div class="mt-4">
          <p class="text-on-surface-variant text-xs font-bold">${tr('Total Biomedical Assets', '?????? ???
          ??? ??????')}</p>
          <h3 class="font-display-lg text-3xl font-bold text-primary mt-1">${totalEquip}</h3>
        </div>
      </div>

      <div class="glass-card p-6 rounded-2xl flex flex-col justify-between border-r-4 border-r-error">
        <div class="flex justify-between items-start">
          <div class="p-3 bg-error/5 rounded-xl text-error flex items-center justify-center">
            <span class="material-symbols-outlined">error</span>
          </div>
          <span class="text-error text-xs font-bold">${tr('Critical', '????')}</span>
        </div>
        <div class="mt-4">
          <p class="text-on-surface-variant text-xs font-bold">${tr('Active Work Orders', '????? ??????? ?
          ?????')}</p>
          <h3 class="font-display-lg text-3xl font-bold text-error mt-1">${activeW0}</h3>
        </div>
      </div>

      <div class="glass-card p-6 rounded-2xl flex flex-col justify-between border-r-4 border-r-warning">
        <div class="flex justify-between items-start">
          <div class="p-3 bg-warning/5 rounded-xl text-warning flex items-center justify-center">
            <span class="material-symbols-outlined">calendar_today</span>
          </div>
          <span class="text-warning text-xs font-bold">${overduePM} ${tr('Tasks', '????')}</span>
        </div>
        <div class="mt-4">
          <p class="text-on-surface-variant text-xs font-bold">${tr('Overdue PM Schedules', '????? ???????
          ??????')}</p>
          <h3 class="font-display-lg text-3xl font-bold text-primary mt-1">${overduePM}</h3>
        </div>
      </div>

      <div class="glass-card p-6 rounded-2xl flex flex-col justify-between">
        <div class="flex justify-between items-start">
          <div class="p-3 bg-secondary/5 rounded-xl text-secondary flex items-center justify-center">
            <span class="material-symbols-outlined">timer_off</span>
```

```
</div>
<span class="text-green-600 text-xs font-bold">-0.8%</span>
</div>
<div class="mt-4">
  <p class="text-on-surface-variant text-xs font-bold">${tr('Equipment Downtime Rate', '???? ????
??????')}</p>
  <h3 class="font-display-lg text-3xl font-bold text-primary mt-1">1.2%</h3>
</div>
</div>
</div>

<div class="grid grid-cols-1 lg:grid-cols-12 gap-gutter items-start mb-lg">
  <!-- Calendar view -->
  <div class="lg:col-span-8 glass-card rounded-2xl overflow-hidden">
    <div class="p-4 border-b border-outline-variant/30 flex justify-between items-center bg-surface-co
ntainer-low/50">
      <h4 class="font-title-lg font-bold text-primary">${tr('Preventive Maintenance Calendar', '????
?????? ???? ???? ????')}</h4>
      <div class="flex items-center gap-2">
        <button class="w-8 h-8 flex items-center justify-center rounded-full border border-outline-var
iant hover:bg-surface-container transition-colors">
          <span class="material-symbols-outlined text-sm">chevron_right</span>
        </button>
        <span class="font-label-md px-2 text-primary font-bold">???? 2026</span>
        <button class="w-8 h-8 flex items-center justify-center rounded-full border border-outline-var
iant hover:bg-surface-container transition-colors">
          <span class="material-symbols-outlined text-sm">chevron_left</span>
        </button>
      </div>
    </div>
    <div class="grid grid-cols-7 border-b border-outline-variant/30 text-center py-2 bg-surface-contai
ner-low/20">
      <div class="text-xs font-bold text-on-surface-variant">${tr('Sun', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Mon', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Tue', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Wed', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Thu', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Fri', '????')}</div>
      <div class="text-xs font-bold text-on-surface-variant">${tr('Sat', '????')}</div>
    </div>
    <div class="grid grid-cols-7 grid-rows-2">
      <div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 relative hover:bg-prim
ary/5 transition-colors">
        <span class="text-xs text-on-surface-variant font-bold">14</span>
        <div class="mt-2">
          <div class="bg-secondary/10 border-r-2 border-secondary p-1 rounded text-[10px] text-on-seco
ndary-container truncate" title="????? ???? ????>????? ???? (3)</div>
        </div>
      </div>
      <div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 relative hover:bg-prim
ary/5 transition-colors">
        <span class="text-xs text-on-surface-variant font-bold">15</span>
        <div class="mt-2 space-y-1">
          <div class="bg-error/10 border-r-2 border-error p-1 rounded text-[10px] text-error truncate"
```

```
title="????? ECG">????? ECG - ??? 204</div>
    <div class="bg-primary/10 border-r-2 border-primary p-1 rounded text-[10px] text-primary truncate" title="????? ?????">????? ???? PB 980</div>
    </div>
</div>
<div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 bg-secondary-container/10">
    <span class="text-xs text-secondary font-bold">16</span>
    <p class="text-[10px] text-secondary text-center mt-6 font-bold">${tr('Today', '????')}</p>
</div>
<div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 hover:bg-primary/5 transition-colors"><span class="text-xs text-on-surface-variant">17</span></div>
<div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 hover:bg-primary/5 transition-colors"><span class="text-xs text-on-surface-variant">18</span></div>
<div class="min-h-[100px] p-2 border-l border-b border-outline-variant/10 hover:bg-primary/5 transition-colors"><span class="text-xs text-on-surface-variant">19</span></div>
<div class="min-h-[100px] p-2 border-b border-outline-variant/10 hover:bg-primary/5 transition-colors"><span class="text-xs text-on-surface-variant">20</span></div>
</div>
</div>

<!-- Calibration and Compliance Tracker -->
<div class="lg:col-span-4 space-y-md">
    <div class="glass-card p-6 rounded-2xl bg-gradient-to-br from-white to-surface-container-low">
        <div class="flex items-center justify-between mb-4">
            <h4 class="font-title-lg font-bold text-primary">${tr('Calibration Tracking', '???? ?????? ????')}</h4>
            <span class="material-symbols-outlined text-secondary">verified</span>
        </div>
        <div class="space-y-4">
            <div class="p-3 bg-white rounded-xl border border-outline-variant/30 flex justify-between items-center">
                <div>
                    <p class="text-xs font-bold text-primary">????? ???? ????</p>
                    <p class="text-[10px] text-on-surface-variant mt-0.5">????? ??????????: 12 ???</p>
                </div>
                <div class="w-10 h-10 relative flex items-center justify-center">
                    <svg class="w-full h-full transform -rotate-90">
                        <circle class="text-outline/10" cx="20" cy="20" fill="transparent" r="16" stroke="currentColor" stroke-width="3"></circle>
                        <circle class="text-secondary" cx="20" cy="20" fill="transparent" r="16" stroke="currentColor" stroke-dasharray="100.5" stroke-dashoffset="15" stroke-width="3"></circle>
                    </svg>
                    <span class="absolute text-[9px] font-bold">85%</span>
                </div>
            </div>
            <div class="p-3 bg-white rounded-xl border border-outline-variant/30 flex justify-between items-center">
                <div>
                    <p class="text-xs font-bold text-primary">????? ????</p>
                    <p class="text-[10px] text-on-surface-variant mt-0.5">????? - ?????</p>
                </div>
                <span class="material-symbols-outlined text-green-500">check_circle</span>
            </div>
        </div>
    </div>
</div>
```

```

    </div>
</div>

<!-- Quick Action Shortcuts -->
<div class="glass-card p-6 rounded-2xl">
    <h4 class="font-title-lg font-bold text-primary mb-4">${tr('Quick Actions', '?????? ?????')}</h4>

    <div class="space-y-2.5">
        <button onclick="mntTab='orders';navigateTo(28)" class="w-full flex items-center justify-between p-3.5 bg-primary text-white rounded-xl hover:bg-primary-container transition-all group">
            <span class="font-label-md">? ${tr('Open Work Orders Tab', '???? ???? ????? ?????')}</span>
            <span class="material-symbols-outlined text-sm">arrow_back</span>
        </button>
        <button onclick="mntTab='assets';navigateTo(28)" class="w-full flex items-center justify-between p-3.5 border border-secondary text-secondary rounded-xl hover:bg-secondary/5 transition-all">
            <span class="font-label-md">? ${tr('Open Asset Register', '???? ???? ?????')}</span>
            <span class="material-symbols-outlined text-sm">arrow_back</span>
        </button>
    </div>
</div>
</div>
</div>
` : mntTab === 'orders' ? `
<!-- Work Orders Tab -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-md">
    <!-- Forms Column -->
    <div class="space-y-md">
        <!-- Legacy Maintenance Orders Form -->
        <div class="glass-card p-6 rounded-2xl">
            <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('New Maintenance Ticket', '????
????? ??????')}</h4>
            <div class="form-group mb-3">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Equipment Name/ID',
'??????/??????')}</label>
                <input class="form-input w-full" id="mntEquip" placeholder="e.g. Ventilator PB 980">
            </div>
            <div class="form-group mb-3">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Location/Room', '???
??? / ??????')}</label>
                <input class="form-input w-full" id="mntLoc" placeholder="e.g. ICU Room 3">
            </div>
            <div class="form-group mb-3">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Maintenance Type', '
??? ??????')}</label>
                <select class="form-input w-full" id="mntType">
                    <option value="corrective">${tr('Corrective', '??????')}</option>
                    <option value="preventive">${tr('Preventive', '??????')}</option>
                    <option value="emergency">${tr('Emergency', '?????')}</option>
                    <option value="calibration">${tr('Calibration', '?????')}</option>
                </select>
            </div>
            <div class="form-group mb-3">
                <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Priority', '????????
?')}</label>

```



```

        <select class="form-input w-full" id="mntPriority">
            <option value="low">${tr('Low', '?????')}</option>
            <option value="medium">${tr('Medium', '?????')}</option>
            <option value="high">${tr('High', '?????')}</option>
            <option value="urgent">${tr('Urgent', '?????')}</option>
        </select>
    </div>
    <div class="form-group mb-4">
        <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Issue Description',
'??? ??????')}</label>
        <textarea class="form-input w-full" id="mntDesc" rows="2" placeholder="????? ??????..."></text
area>
    </div>
    <button class="btn btn-primary w-full py-2.5 rounded-xl font-bold flex items-center justify-cent
er gap-2" onclick="saveMntOrder()">
        ? ${tr('Submit Order', '????? ?????')}
    </button>
</div>

<!-- Legacy Work Orders Form -->
<div class="glass-card p-6 rounded-2xl">
    <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Detailed Work Order', '??? ???
?? ??????')}</h4>
    <div class="form-group mb-3">
        <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Request Type', '???
?????')}</label>
        <select class="form-input w-full" id="woType">
            <option value="Corrective">${tr('Corrective', '???????')}</option>
            <option value="Preventive">${tr('Preventive', '???????')}</option>
            <option value="Calibration">${tr('Calibration', '???????')}</option>
        </select>
    </div>
    <div class="form-group mb-3">
        <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Priority', '?????????
')}</label>
        <select class="form-input w-full" id="woPriority">
            <option value="Low">${tr('Low', '???????')}</option>
            <option value="Normal">${tr('Normal', '?????')}</option>
            <option value="High">${tr('High', '?????')}</option>
            <option value="Emergency">${tr('Emergency', '?????')}</option>
        </select>
    </div>
    <div class="grid grid-cols-2 gap-2 mb-3">
        <div class="form-group">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Department', '????
?')}</label>
            <input class="form-input w-full" id="woDept" placeholder="e.g. ICU">
        </div>
        <div class="form-group">
            <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Location', '???????
')}</label>
            <input class="form-input w-full" id="woLocation" placeholder="e.g. F2-Room12">
        </div>
    </div>

```



```

    </table>
</div>

<!-- Active Work Orders Queue Table -->
<div class="glass-card p-6 rounded-2xl overflow-x-auto">
    <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('System Work Orders Queue', '??
??? ????? ????? ?????')}</h4>
    <table class="w-full text-right divide-y divide-outline-variant/30 text-xs">
        <thead class="bg-surface-container-low font-label-md text-on-surface-variant">
            <tr>
                <th class="p-3">${tr('WO#', '??? ?????')}</th>
                <th class="p-3">${tr('Dept / Location', '????? / ?????')}</th>
                <th class="p-3">${tr('Type', '????')}</th>
                <th class="p-3">${tr('Priority', '????????')}</th>
                <th class="p-3">${tr('Status', '????')}</th>
                <th class="p-3 text-center">${tr('Actions', '????')}</th>
            </tr>
        </thead>
        <tbody class="divide-y divide-outline-variant/10 font-body-md">
            ${workOrders.length ? workOrders.map(w => `
                <tr class="hover:bg-primary/5 transition-colors">
                    <td class="p-3 font-mono font-bold text-primary">${escapeHTML(w.wo_number)}</td>
                    <td class="p-3 text-on-surface-variant">${escapeHTML(w.department || '')} / ${escapeHTML
(w.location || '')}</td>
                    <td class="p-3"><span class="px-2 py-0.5 rounded text-[10px] bg-secondary/10 text-second
ary font-bold">${escapeHTML(w.request_type)}</span></td>
                    <td class="p-3">
                        <span class="px-2 py-0.5 rounded text-[10px] font-bold ${w.priority === 'High' || w.pr
iority === 'Emergency' ? 'bg-error-container text-on-error-container' : 'bg-primary-container text-on-primary-
fixed-variant'}">
                            ${escapeHTML(w.priority)}
                        </span>
                    </td>
                    <td class="p-3">${statusBadge(w.status)}</td>
                    <td class="p-3 text-center">
                        ${w.status !== 'Completed' ? `<button class="btn btn-sm bg-green-100 text-green-700 ho
ver:bg-green-200 px-2 py-1 rounded" onclick="completeWO(${safeId(w.id)})">? ${tr('Resolve', '?? ?????')}</bu
tton>` : '?'}
                    </td>
                </tr>
            `).join('') : `<tr><td colspan="6" class="text-center py-4 text-on-surface-variant">${tr('No
records found', '?? ???? ????? ???? ?????')}</td></tr>`}
        </tbody>
    </table>
</div>
</div>
</div>
` : `
<!-- Asset Register Tab -->
<div class="grid grid-cols-1 lg:grid-cols-3 gap-md">
    <!-- Asset Form -->
    <div class="glass-card p-6 rounded-2xl h-fit">
        <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Register Medical Device', '????
??? ???? ?????')}</h4>

```

```

<div class="form-group mb-3">
  <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Device Name (EN)', '??
? ?????? (???????)')}</label>
  <input class="form-input w-full" id="eqName" placeholder="e.g. GE ECG Monitor B850">
</div>
<div class="form-group mb-3">
  <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Device Name (AR)', '??
? ?????? (?????)')}</label>
  <input class="form-input w-full" id="eqNameAr" placeholder="e.g. ??? GE">
</div>
<div class="form-group mb-3">
  <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Category', '???? ??
??')}</label>
  <input class="form-input w-full" id="eqCat" placeholder="e.g. Diagnostic / Life Support">
</div>
<div class="form-group mb-3">
  <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Manufacturer', '?????
??????')}</label>
  <input class="form-input w-full" id="eqMfg" placeholder="e.g. GE Healthcare">
</div>
<div class="grid grid-cols-2 gap-2 mb-4">
  <div class="form-group">
    <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Serial Number', '??
?? ???????')}</label>
    <input class="form-input w-full" id="eqSerial" placeholder="e.g. SN-8829402">
    </div>
    <div class="form-group">
      <label class="block text-xs font-bold text-on-surface-variant mb-1">${tr('Department', '????
??????')}</label>
      <input class="form-input w-full" id="eqDept" placeholder="e.g. Cardiology">
      </div>
    </div>
    <button class="btn btn-primary w-full py-2.5 rounded-xl font-bold flex items-center justify-center
gap-2" onclick="addEquipment()">
      ? ${tr('Register Asset', '???? ??????')}
    </button>
  </div>

<!-- Assets Table -->
<div class="glass-card p-6 rounded-2xl lg:col-span-2 overflow-x-auto">
  <h4 class="font-title-lg text-title-lg text-primary mb-4">? ${tr('Biomedical Assets Registry', '??
? ?? ?????? ?????? ??????')}</h4>
  <table class="w-full text-right divide-y divide-outline-variant/30 text-xs">
    <thead class="bg-surface-container-low font-label-md text-on-surface-variant">
      <tr>
        <th class="p-3">${tr('Code', '????')}</th>
        <th class="p-3">${tr('Device Name', '?? ?????')}</th>
        <th class="p-3">${tr('Category', '????')}</th>
        <th class="p-3">${tr('Manufacturer', '????')}</th>
        <th class="p-3">${tr('Serial#', '???? ?????')}</th>
        <th class="p-3">${tr('Status', '????')}</th>
      </tr>
    </thead>
    <tbody class="divide-y divide-outline-variant/10 font-body-md">

```

```

    ${equipment.length ? equipment.map(e => `
      <tr class="hover:bg-primary/5 transition-colors">
        <td class="p-3 font-mono font-bold text-primary">${escapeHTML(e.equipment_code || `EQ-${e.
id}`)}}</td>

        <td class="p-3">
          <div class="font-bold text-primary">${escapeHTML(isArabic ? (e.equipment_name_ar || e.eq
uipment_name) : (e.equipment_name || e.equipment_name_ar))}</div>
          <div class="text-[10px] text-on-surface-variant">${escapeHTML(e.department || '')} - ${e
scapeHTML(e.location || '')}</div>
        </td>
        <td class="p-3 text-on-surface-variant">${escapeHTML(e.category || '')}</td>
        <td class="p-3 text-on-surface-variant">${escapeHTML(e.manufacturer || '')}</td>
        <td class="p-3 font-mono text-[11px]">${escapeHTML(e.serial_number || '?')}</td>
        <td class="p-3">${statusBadge(e.status || 'Active')}</td>
      </tr>
    `).join('') : `<tr><td colspan="6" class="text-center py-4 text-on-surface-variant">${tr('No e
quipment registered', '?? ???? ???? ???? ?????')}</td></tr>`}
  </tbody>
</table>
</div>
</div>
`}
</div>
`;

```

```

window.saveMntOrder = async () => {
  try {
    await API.post('/api/maintenance/orders', {
      equipment: document.getElementById('mntEquip').value,
      location: document.getElementById('mntLoc').value,
      maintenance_type: document.getElementById('mntType').value,
      priority: document.getElementById('mntPriority').value,
      description: document.getElementById('mntDesc').value,
      requested_by: window.currentUser?.display_name || ''
    });
    showToast(tr('Order submitted!', '?? ????? ?????!'));
    mntTab = 'orders';
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

window.completeMnt = async (id) => {
  try {
    await API.put('/api/maintenance/orders/' + id, { status: 'completed' });
    showToast('?');
    mntTab = 'orders';
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};
}

```

```

window.addWorkOrder = async function () {
  try {
    await API.post('/api/maintenance/work-orders', {
      request_type: document.getElementById('woType').value,
      priority: document.getElementById('woPriority').value,
      department: document.getElementById('woDept').value,
      location: document.getElementById('woLocation').value,
      description: document.getElementById('woDesc').value,
      requested_by: currentUser?.display_name
    });
    showToast(tr('Created!', '?? ??????!!'));
    mntTab = 'orders';
    await navigateTo(28);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

window.completeWO = async function (id) {
  try {
    await API.put('/api/maintenance/work-orders/' + id, { status: 'Completed' });
    showToast(tr('Completed!', '?????!'));
    mntTab = 'orders';
    await navigateTo(28);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

window.addEquipment = async function () {
  try {
    await API.post('/api/maintenance/equipment', {
      equipment_name: document.getElementById('eqName').value,
      equipment_name_ar: document.getElementById('eqNameAr').value,
      category: document.getElementById('eqCat').value,
      manufacturer: document.getElementById('eqMfg').value,
      serial_number: document.getElementById('eqSerial').value,
      department: document.getElementById('eqDept').value
    });
    showToast(tr('Added!', '??? ??????!!'));
    mntTab = 'assets';
    await navigateTo(28);
  } catch (e) {
    showToast(tr('Error', '???'), 'error');
  }
};

// ===== TRANSPORT =====
async function renderTransport(el) {

  const requests = await API.get('/api/transport/requests').catch(() => []);
  content.innerHTML = `
    <h2>${tr('Transport', '????')}</h2>

```



```

window.updateTransport = async (id, status) => {
  try { await API.put('/api/transport/requests/' + id, { status }); showToast(tr('Updated', '?? ?????'));
renderTransport(content); } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}

window.addTransport = async function () {
  const s = document.getElementById('trPatient'); if (!s.value) return showToast(tr('Select patient', '???? ??
????'), 'error');
  try { await API.post('/api/transport/requests', { patient_id: s.value, patient_name: s.options[s.selectedInd
ex].dataset.name, from_location: document.getElementById('trFrom').value, to_location: document.getElementById
('trTo').value, transport_type: document.getElementById('trType').value, priority: document.getElementById('tr
Priority').value, special_needs: document.getElementById('trNeeds').value, requested_by: currentUser?.display_
name }); showToast(tr('Requested!', '?? ?????!')); await navigateTo(29); } catch (e) { showToast(tr('Error', '
???'), 'error'); }
};

window.completeTransport = async function (id) {
  try { await API.put('/api/transport/requests/' + id, { status: 'Completed', dropoff_time: new Date().toISOSt
ring() }); showToast(tr('Done!', '??!')); await navigateTo(29); } catch (e) { showToast(tr('Error', '???'), 'e
rror'); }
};

// ===== MEDICAL RECORDS / HIM =====
let mrTab = 'records';
async function renderMedicalRecords(el) {
  const content = el;

  const patients = await API.get('/api/patients').catch(() => []);
  const isHimAdmin = ['Admin', 'HIM'].includes(currentUser?.role);

  const tabs = [
    { k: 'records', en: 'Patient Records', ar: '????? ??????', icon: '?' },
    { k: 'coding', en: 'Coding', ar: '???????', icon: '?' },
    { k: 'deficiencies', en: 'Deficiencies', ar: '???????', icon: '?' },
    { k: 'roi', en: 'Release of Info', ar: '??????? ?? ??????????', icon: '?' }
  ];
  if (isHimAdmin) tabs.push({ k: 'access', en: 'Access Log', ar: '??? ??????', icon: '?' });

  const tabBar = tabs.map(t => `<button class="btn btn-sm" onclick="mrTab='${t.k}';navigateTo(30)" style="${mr
Tab === t.k ? 'background:#1565c0;color:#fff' : 'background:#e3f2fd;color:#1565c0'}">${t.icon} ${tr(t.en, t.ar
)}</button>`).join(' ');

  // tabBar is composed only from a fixed allowlist (no user/PHI data) ? no injection surface.
  content.innerHTML = `
    <h2>${tr('Medical Records', '??????? ??????')}</h2>
    <div style="display:flex;gap:8px;flex-wrap:wrap;margin-bottom:16px">${tabBar}</div>
    <div id="mrTabBody"></div>`;

  const body = document.getElementById('mrTabBody');
  window._mrData = patients;

  window.filterMedRecords = () => { const n = (document.getElementById('mrSearchName')?.value || '').toLowerCa
se(); const mrn = (document.getElementById('mrSearchMRN')?.value || ''); const sid = (document.getElementById(
'mrSearchID')?.value || ''); document.querySelectorAll('#mrTbl tbody tr').forEach(r => { const t = r.textConte

```



```

nt.toLowerCase(); r.style.display = (t.includes(n) && (!mrn || t.includes(mrn)) && (!sid || t.includes(sid)))
? '' : 'none'; }); });

// ---- TAB: PATIENT RECORDS (longitudinal record entry) ----
if (mrTab === 'records') {
  body.innerHTML = `
    <div class="card" style="padding:20px;margin-bottom:16px">
      <h4 style="margin:0 0 12px">? ${tr('Search Medical Records', '??? ?? ?????? ??????')}</h4>
      <div style="display:grid;grid-template-columns:1fr 1fr 1fr auto;gap:12px;align-items:end">
        <div class="form-group"><label>${tr('Name', '????')}</label><input class="form-input" id="mrSearchName" oninput="filterMedRecords()"></div>
        <div class="form-group"><label>${tr('MRN / File #', '??? ?????')}</label><input class="form-input" id="mrSearchMRN" oninput="filterMedRecords()"></div>
        <div class="form-group"><label>${tr('ID / Iqama', '?????/???????')}</label><input class="form-input" id="mrSearchID" oninput="filterMedRecords()"></div>
        <button class="btn btn-primary" onclick="filterMedRecords()">?</button>
      </div>
    </div>
    <div class="card" style="padding:20px">
      <div style="display:flex;justify-content:space-between;margin-bottom:12px">
        <h4 style="margin:0">${tr('Patient Records', '????? ?????')}</h4>
        <button class="btn btn-sm" onclick="exportToCSV(window._mrData||[], 'medical_records')" style="background-color:#e0f7fa;color:#00838f">? ${tr('Export', '????')}</button>
      </div>
      <div id="mrTable"></div>
    </div>`;
  const mrt = document.getElementById('mrTable');
  if (mrt) {
    createTable(mrt, 'mrTbl',
      [tr('File #', '??? ?????'), tr('Name (AR)', '????? ????????'), tr('Name (EN)', '????? ??????????'), tr('ID', '?????'), tr('Phone', '?????'), tr('DOB', '????? ????????'), tr('Nationality', '???????'), tr('Actions', '')],
      patients.slice(0, 100).map(p => ({ cells: [p.file_number || p.id, p.name_ar || '', p.name_en || '', p.id_number || '', p.phone || '', p.dob || '', p.nationality || '', rawHtml('<button class="btn btn-sm" onclick="viewLongitudinalRecord(' + (safeId(p.id) || 0) + ')">? ' + tr('Open Chart', '??? ?????') + '</button> <button class="btn btn-sm" onclick="breakGlassRecord(' + (safeId(p.id) || 0) + ')" style="background:#ffebee;color:#c62828">? ' + tr('Break-Glass', '???? ?????') + '</button>'), id: p.id })));
  });
}
}

// ---- TAB: CODING ----
if (mrTab === 'coding') {
  const codes = await API.get('/api/him/coding').catch(() => []);
  body.innerHTML = `
    <div class="card" style="padding:20px;margin-bottom:16px">
      <h4 style="margin:0 0 12px">? ${tr('Add Code', '????? ???')}</h4>
      <div style="display:grid;grid-template-columns:1fr 1fr 1fr 2fr auto;gap:12px;align-items:end">
        <div class="form-group"><label>${tr('Patient', '?????')}</label><select class="form-input" id="codePatient">${patients.slice(0, 300).map(p => '<option value="' + (safeId(p.id) || '') + '"'>? ' + escapeHTML(p.name_ar || p.name_en || ('#' + p.id)) + '</option>').join('')}</select></div>
        <div class="form-group"><label>${tr('System', '?????')}</label><select class="form-input" id="codeSystem"><option>ICD10</option><option>SNOMED</option><option>CPT</option></select></div>
        <div class="form-group"><label>${tr('Code', '?????')}</label><input class="form-input" id="codeValue`

```

```

"></div>
    <div class="form-group"><label>${tr('Description', '????')}</label><input class="form-input" id="codeDesc"></div>
    <button class="btn btn-primary" onclick="addCoding()">${tr('Add', '????')}</button>
  </div>
</div>
<div class="card" style="padding:20px"><h4 style="margin:0 0 12px">${tr('Codes', '??????')}</h4><div id="codeTable"></div></div>`;
const ct = document.getElementById('codeTable');
if (ct) createTable(ct, 'codeTbl',
  [tr('System', '????'), tr('Code', '????'), tr('Description', '????'), tr('Patient', '????'), tr('Date', '????')],
  codes.map(c => ({ cells: [c.code_system || '', c.code || '', c.description || '', c.patient_id || '', c.created_at ? new Date(c.created_at).toLocaleDateString('en-GB') : ''], id: c.id })));
}

// ---- TAB: DEFICIENCIES ----
if (mrTab === 'deficiencies') {
  const defs = await API.get('/api/him/deficiencies').catch(() => []);
  body.innerHTML = `<div class="card" style="padding:20px"><h4 style="margin:0 0 12px">?? ${tr('Record Deficiencies', '???? ????')} (${defs.length})</h4><div id="defTable"></div></div>`;
  const dt = document.getElementById('defTable');
  if (dt) createTable(dt, 'defTbl',
    [tr('Type', '????'), tr('Record', '????'), tr('Patient', '????'), tr('Detail', '????'), tr('Date', '????')],
    defs.map(d => ({ cells: [d.type || '', (d.record_type || '') + ' #' + (d.record_id || ''), d.patient_id || '', d.detail || '', d.date ? new Date(d.date).toLocaleDateString('en-GB') : ''] })));
}

// ---- TAB: ROI ----
if (mrTab === 'roi') {
  const roi = await API.get('/api/him/roi').catch(() => []);
  body.innerHTML = `
    <div class="card" style="padding:20px;margin-bottom:16px">
      <h4 style="margin:0 0 12px">? ${tr('New ROI Request', '??? ????')}</h4>
      <div style="display:grid;grid-template-columns:1fr 1fr 2fr auto;gap:12px;align-items:end">
        <div class="form-group"><label>${tr('Patient', '????')}</label><select class="form-input" id="roiPatient">${patients.slice(0, 300).map(p => '<option value="' + (safeId(p.id) || '') + '"'> + escapeHTML(p.name_ar || p.name_en || ('#' + p.id)) + '</option>').join('')}</select></div>
        <div class="form-group"><label>${tr('Requester', '???? ????')}</label><input class="form-input" id="roiRequester"></div>
        <div class="form-group"><label>${tr('Purpose', '????')}</label><input class="form-input" id="roiPurpose"></div>
        <button class="btn btn-primary" onclick="createROI()">${tr('Submit', '????')}</button>
      </div>
    </div>
    <div class="card" style="padding:20px"><h4 style="margin:0 0 12px">${tr('ROI Requests', '???? ????')}</h4><div id="roiTable"></div></div>`;
  const rt = document.getElementById('roiTable');
  if (rt) createTable(rt, 'roiTbl',
    [tr('Patient', '????'), tr('Requester', '????'), tr('Purpose', '????'), tr('Status', '????'), tr('Actions', '')],
    roi.map(r => {
      let actions = '';

```

```

        if (r.status === 'pending') actions = '<button class="btn btn-sm" onclick="actROI(' + (safeId(r.id) || 0) + ", 'approve'" + ')" style="background:#c8e6c9;color:#1b5e20">? ' + tr('Approve', '?????') + '</button> <button class="btn btn-sm" onclick="actROI(' + (safeId(r.id) || 0) + ", 'deny'" + ')" style="background:#ffcdd2;color:#b71c1c">? ' + tr('Deny', '???') + '</button>';
        else if (r.status === 'approved') actions = '<button class="btn btn-sm" onclick="actROI(' + (safeId(r.id) || 0) + ", 'release'" + ')" style="background:#bbdefb;color:#0d47a1">? ' + tr('Release', '?????') + '</button>';

        return { cells: [r.patient_id || '', r.requester || '', r.purpose || '', r.status || '', rawHtml(actions)], id: r.id };
    });
}

// ---- TAB: ACCESS LOG (Admin/HIM only) ----
if (mrTab === 'access') {
    if (!isHimAdmin) { body.innerHTML = '<div class="card" style="padding:20px;color:#c62828">' + tr('Access denied', '????? ?????') + '</div>'; return; }
    const logs = await API.get('/api/him/access-log').catch(() => []);
    body.innerHTML = `<div class="card" style="padding:20px"><h4 style="margin:0 0 12px">? ${tr('Record Access Log', '??? ?????? ???????')} (${logs.length})</h4><div id="accTable"></div></div>`;
    const at = document.getElementById('accTable');
    if (at) createTable(at, 'accTbl',
        [tr('Patient', '?????'), tr('Accessor', '????????'), tr('Type', '?????'), tr('Reason', '?????'), tr('When', '?????')],
        logs.map(l => ({ cells: [l.patient_id || '', l.accessor_id || '', l.access_type === 'break_glass' ? rawHtml('<span class="badge" style="background:#ffcdd2;color:#b71c1c">? break-glass</span>') : 'normal', l.reason || '', l.at ? new Date(l.at).toLocaleString('en-GB') : '', id: l.id })));
    }
}

// ---- Longitudinal record viewer (access-logged server-side) ----
window.viewLongitudinalRecord = async (id) => {
    try {
        const pid = safeId(id); if (pid === '') return;
        const data = await API.get('/api/him/record/' + pid).catch(() => null);
        if (!data) { showToast(tr('Cannot open record', '????? ??? ?????')); return; }
        const p = data.patient || {};
        const events = data.events || [];
        const modal = document.createElement('div');
        modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center';
        const evHtml = events.length ? events.map(e =>
            '<div style="padding:10px;margin:6px 0;background:#f5f5f5;border-radius:8px;border-left:3px solid #1565c0">' +
            '<div style="font-size:11px;color:#777">' + escapeHTML(e.type || '') + ' ? ' + escapeHTML(e.date ? new Date(e.date).toLocaleString('en-GB') : '') + '</div>' +
            '<strong>' + escapeHTML((e.icon || '') + ' ' + (e.title || '')) + '</strong>' +
            (e.subtitle ? '<div style="font-size:12px;color:#555">' + escapeHTML(e.subtitle) + '</div>' : '') +
            '</div>').join('') : '<p style="color:#999">' + tr('No records', '?? ???? ?????') + '</p>';
        modal.innerHTML = '<div style="background:#fff;border-radius:16px;padding:24px;width:760px;max-height:85vh;overflow:auto">' +
            '<h3 style="margin:0 0 8px">? ' + escapeHTML(p.name_ar || p.name_en || '') + '</h3>' +
            '<div style="font-size:12px;color:#666;margin-bottom:12px">' + tr('File', '???') + ': ' + escapeHTML(String(p.file_number || p.id || '')) + ' ? ' + tr('Longitudinal Record', '????? ??????') + ' (' + events.length +
            ')</div>' +

```

```

        evHtml +
        '<button class="btn btn-secondary w-full" onclick="this.parentElement.parentElement.remove()" style="margin-top:16px">' + tr('Close', '????') + '</button></div>';
        document.body.appendChild(modal);
        modal.onclick = e => { if (e.target === modal) modal.remove(); };
    } catch (e) { }
};

// ---- Break-glass (requires reason) ----
window.breakGlassRecord = async (id) => {
    const pid = safeId(id); if (pid === '') return;
    const reason = prompt(tr('Break-glass requires a reason. Enter reason:', '????? ?????? ?????? ?????? ?????? ???? ???? :'));
    if (!reason || !reason.trim()) { showToast(tr('Reason required', '???? ?????')); return; }
    const r = await API.post('/api/him/break-glass', { patient_id: pid, reason: reason.trim() }).catch(() => null);
    if (r && r.success) { showToast(tr('Break-glass recorded', '?? ?????? ?????? ??????')); window.viewLongitudinalRecord(pid); }
    else showToast(tr('Break-glass failed', '?? ?????? ??????'));
};

window.addCoding = async () => {
    const patient_id = document.getElementById('codePatient')?.value;
    const code = (document.getElementById('codeValue')?.value || '').trim();
    if (!code) { showToast(tr('Code required', '???? ?????')); return; }
    await API.post('/api/him/coding', { patient_id, code_system: document.getElementById('codeSystem')?.value, code, description: document.getElementById('codeDesc')?.value || '' }).catch(() => null);
    showToast(tr('Code added', '?? ??????')); navigateTo(30);
};

window.createROI = async () => {
    const patient_id = document.getElementById('roiPatient')?.value;
    const requester = (document.getElementById('roiRequester')?.value || '').trim();
    if (!requester) { showToast(tr('Requester required', '???? ?????? ??????')); return; }
    await API.post('/api/him/roi', { patient_id, requester, purpose: document.getElementById('roiPurpose')?.value || '' }).catch(() => null);
    showToast(tr('ROI submitted', '?? ?????? ??????')); navigateTo(30);
};

window.actROI = async (id, action) => {
    const rid = safeId(id); if (rid === '') return;
    await API.put('/api/him/roi/' + rid, { action }).catch(() => null);
    showToast(tr('Updated', '?? ??????')); navigateTo(30);
};

window.submitMRRequest = async function () {
    const sel = document.getElementById('mrPatient');
    const patient_id = sel.value;
    const file_number = sel.options[sel.selectedIndex].dataset.fn;
    await API.post('/api/medical-records/requests', { patient_id, file_number, department: document.getElementById('mrDept').value, purpose: document.getElementById('mrPurpose').value, notes: document.getElementById('mrNotes').value });
    showToast(tr('Request submitted', '?? ??????')); mrTab = 'requests'; navigateTo(30);
};

window.updateMRRequest = async function (id, status) {

```

```

    await API.put('/api/medical-records/requests/' + id, { status });
    showToast(tr('Updated', '?? ??????')); navigateTo(30);
  };

// ===== CLINICAL PHARMACY =====
let cpTab = 'prescriptions'; // default to prescriptions list
async function renderClinicalPharmacy(el) {
  const content = el;

  // Show premium loading skeletons instantly
  content.innerHTML = `
    <div class="page-title">? ${tr('Clinical Pharmacy', '?????? ?????????')}</div>
    <div class="tab-bar">
      <button class="tab-btn active">? ${tr('Prescriptions Review', '????? ?????')}</button>
    </div>
    <div class="split-layout">
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
      <div class="card glass-card-premium">
        <div class="skeleton-bar title mb-16"></div>
        <div class="skeleton-bar mb-12"></div>
      </div>
    </div>
  `;

  let prescriptions = [];
  let reviews = [];
  let interactions = [];
  let educationLogs = [];
  let patients = [];

  try {
    [prescriptions, reviews, interactions, educationLogs, patients] = await Promise.all([
      API.get('/api/pharmacy/prescriptions').catch(() => []),
      API.get('/api/clinical-pharmacy/reviews').catch(() => []),
      API.get('/api/clinical-pharmacy/interactions').catch(() => []),
      API.get('/api/clinical-pharmacy/education').catch(() => []),
      API.get('/api/patients').catch(() => [])
    ]);
  } catch (e) {
    content.innerHTML = `
      <div class="page-title">? ${tr('Clinical Pharmacy', '?????? ?????????')}</div>
      <div class="error-card-premium">
        <h3>${tr('Failed to load clinical pharmacy data', '??? ?? ????? ?????? ????????? ?????????')}</h3>
        <p>${escapeHTML(e.message || e)}</p>
        <button class="btn btn-primary" onclick="navigateTo(31)">? ${tr('Retry', '????? ?????')}</button>
      </div>
    `;
    return;
  }

  const pendingCount = prescriptions.filter(p => p.status === 'pending' || p.status === 'Pending').length;

```

```

const activeReviewsCount = reviews.filter(r => r.status === 'Open').length;

content.innerHTML = `
  <div class="page-title">? ${tr('Clinical Pharmacy', '?????? ?????????')}</div>
  <div class="stats-grid" style="grid-template-columns:repeat(4,1fr); margin-bottom:16px;">
    <div class="stat-card" style="--stat-color:#3b82f6"><span class="stat-icon">?</span><div class="stat-label">${tr('Total Prescriptions', '?????? ?????????')}</div><div class="stat-value">${prescriptions.length}</div></div>
    <div class="stat-card" style="--stat-color:#f59e0b"><span class="stat-icon">?</span><div class="stat-label">${tr('Pending Review', '?????? ?????????')}</div><div class="stat-value">${pendingCount}</div></div>
    <div class="stat-card" style="--stat-color:#ef4444"><span class="stat-icon">?</span><div class="stat-label">${tr('Active Reviews', '???????? ?????????')}</div><div class="stat-value">${activeReviewsCount}</div></div>
  >
  <div class="stat-card" style="--stat-color:#10b981"><span class="stat-icon">?</span><div class="stat-label">${tr('Patient Educations', '????? ?????????')}</div><div class="stat-value">${educationLogs.length}</div></div>
</div>
<div class="tab-bar">
  <button class="tab-btn ${cpTab === 'prescriptions' ? 'active' : ''}" onclick="cpTab='prescriptions';navigateTo(31)">? ${tr('Prescriptions Review', '?????? ?????????')}</button>
  <button class="tab-btn ${cpTab === 'checker' ? 'active' : ''}" onclick="cpTab='checker';navigateTo(31)">? ${tr('Interaction Checker', '??? ?????????')}</button>
  <button class="tab-btn ${cpTab === 'reviews' ? 'active' : ''}" onclick="cpTab='reviews';navigateTo(31)">? ${tr('Review Ledger', '??? ?????????')}</button>
  <button class="tab-btn ${cpTab === 'education' ? 'active' : ''}" onclick="cpTab='education';navigateTo(31)">? ${tr('Patient Education', '????? ?????????')}</button>
</div>
<div class="card glass-card-premium" id="cpTabContent"></div>
`;

const tabContainer = document.getElementById('cpTabContent');

if (cpTab === 'prescriptions') {
  tabContainer.innerHTML = `
    <h3>? ${tr('Prescriptions Queue for Clinical Review', '????? ???????? ????????? ?????????????')}</h3>
    <div id="rxReviewTableContainer"></div>
  `;
  const tblContainer = document.getElementById('rxReviewTableContainer');
  if (prescriptions.length) {
    createTable(tblContainer, 'rxReviewTbl',
      [tr('Patient ID', '??? ??????'), tr('Patient', '??????'), tr('Medication', '??????'), tr('Dosage', '??
?????'), tr('Frequency', '????????'), tr('Status', '??????'), tr('Actions', '????????')],
      prescriptions.map(p => ({
        cells: [
          p.patient_id || '',
          p.patient_name || '',
          p.medication || p.medication_name || '',
          p.dosage || '',
          p.frequency || '',
          statusBadge(p.status),
          rawHtml(`
            <button class="btn btn-sm btn-primary" onclick="openClinicalReviewModal(${p.id}, ${p.patient_id}
, '${jsStr(p.patient_name || '')}', '${jsStr(p.medication || p.medication_name || '')}')">? ${tr('Review', '??
????')}</button>

```

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        ${p.status === 'pending' || p.status === 'Pending') ? `<button class="btn btn-sm btn-success" o
nclick="verifyPrescription(${p.id})">? ${tr('Verify', '???? ????')}</button>` : ''}
    `)
  ]
  })))
);
} else {
  tblContainer.innerHTML = `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('
No prescriptions found', '?? ???? ???? ????')}</div></div>`;
}
}
else if (cpTab === 'checker') {
  tabContainer.innerHTML = `
    <div class="split-layout" style="display:grid;grid-template-columns:1fr 1fr;gap:20px;">
      <div class="card" style="padding:20px;background:rgba(255,255,255,0.01);border:1px solid rgba(255,255,
255,0.05)">
        <h4 style="margin:0 0 16px;color:var(--primary)">? ${tr('Drug Interaction Checker', '??? ???? ????
??')}</h4>
        <div class="form-group mb-12">
          <label class="label">${tr('Drug 1', '????? 1')}</label>
          <input class="input w-full" id="cpDrug1" placeholder="${tr('e.g. Warfarin', '??? : ?????')}>
        </div>
        <div class="form-group mb-16">
          <label class="label">${tr('Drug 2', '????? 2')}</label>
          <input class="input w-full" id="cpDrug2" placeholder="${tr('e.g. Aspirin', '??? : ?????')}>
        </div>
        <button class="btn btn-primary w-full" onclick="checkInteraction()">? ${tr('Check Interaction', '???
????')}</button>
        <div id="cpResult" style="margin-top:16px"></div>
      </div>
      <div class="card" style="padding:20px;background:rgba(255,255,255,0.01);border:1px solid rgba(255,255,
255,0.05)">
        <h4 style="margin:0 0 16px;color:var(--primary)">?? ${tr('Seeded Clinical Interactions', '??????? ?
??????? ???? ????')}</h4>
        <div style="max-height:380px;overflow-y:auto;padding-inline-end:4px">
          ${interactions.length ? interactions.map(i => `
            <div style="padding:10px;margin-bottom:8px;border-radius:8px;background:${i.severity === 'High'
? 'rgba(239,68,68,0.1)' : 'rgba(245,158,11,0.1)'};border-inline-start:4px solid ${i.severity === 'High' ? '#ef
4444' : '#f59e0b'}">
              <strong>${escapeHTML(i.drug_a)} + ${escapeHTML(i.drug_b)}</strong>
              <div style="font-size:12px;color:var(--text-dim);margin:4px 0">${escapeHTML(i.description)}</d
iv>
              <div style="font-size:11px;color:var(--primary)">Action: ${escapeHTML(i.clinical_action || '')}
            </div>
            <span class="badge ${i.severity === 'High' ? 'badge-danger' : 'badge-warning'}" style="margin-
top:6px">${escapeHTML(i.severity)}</span>
          </div>
          `).join('') : `<div style="color:var(--text-dim)">${tr('No interactions in catalog', '?? ???? ????
??? ????')}</div>`}
        </div>
      </div>
    </div>
  `;
};

```

```

window.checkInteraction = () => {
  const d1 = (document.getElementById('cpDrug1')?.value || '').toLowerCase().trim();
  const d2 = (document.getElementById('cpDrug2')?.value || '').toLowerCase().trim();
  const res = document.getElementById('cpResult');
  if (!d1 || !d2) { res.innerHTML = '<p style="color:#ef4444">' + tr('Please enter both drug names', '????
  ?? ????? ?? ?????????') + '</p>'; return; }

  const found = interactions.find(i =>
    (i.drug_a.toLowerCase().includes(d1) && i.drug_b.toLowerCase().includes(d2)) ||
    (i.drug_a.toLowerCase().includes(d2) && i.drug_b.toLowerCase().includes(d1))
  );
  if (found) {
    res.innerHTML = `
      <div style="padding:12px;background:rgba(239,68,68,0.1);border-radius:8px;border-inline-start:4px so
lid #ef4444">
        <strong style="color:#ef4444">?? ${tr('INTERACTION FOUND', '?? ?????? ?????? ?????!')}</strong><br>
        <strong style="display:block;margin:4px 0">${escapeHTML(found.drug_a)} + ${escapeHTML(found.drug_b
)}}</strong>
        <span style="font-size:12px;color:var(--text-dim)">${escapeHTML(found.description)}</span><br>
        <span style="font-size:12px;color:var(--primary)">Action: ${escapeHTML(found.clinical_action || ''
)}}</span><br>
        <span class="badge badge-danger" style="margin-top:6px">${escapeHTML(found.severity.toUpperCase())
}</span>
      </div>
    `;
  } else {
    res.innerHTML = `
      <div style="padding:12px;background:rgba(16,185,129,0.1);border-radius:8px;border-inline-start:4px s
olid #10b981;color:#10b981">
        <strong>? ${tr('No known interaction found in database', '?? ??? ?????? ??? ?????? ?????? ?? ?????? ?
????????')}</strong>
      </div>
    `;
  }
};

}

else if (cpTab === 'reviews') {
  tabContainer.innerHTML = `
    <h3>? ${tr('Pharmacist Clinical Review Ledger', '??? ?????? ?????? ??????')}</h3>
    <div id="reviewsLedgerTableContainer"></div>
  `;
  const revContainer = document.getElementById('reviewsLedgerTableContainer');
  if (reviews.length) {
    createTable(revContainer, 'reviewsLedgerTbl',
      [tr('Patient', '?????'), tr('Pharmacist', '???????'), tr('Severity', '???????'), tr('Findings', '????
?????'), tr('Recommendations', '????????'), tr('Outcome', '????????'), tr('Status', '???????'), tr('Actions', '?
???????')],
      reviews.map(r => ({
        cells: [
          r.patient_name || '',
          r.pharmacist || '',
          r.severity === 'High' ? rawHtml('<span class="badge badge-danger">? High</span>') : r.severity ===
'Medium' ? rawHtml('<span class="badge badge-warning">? Medium</span>') : rawHtml('<span class="badge badge-s
uccess">? Low</span>'),

```



```

        r.findings || '',
        r.recommendations || '',
        r.outcome || '-',
        statusBadge(r.status),
        rawHtml(r.status === 'Open' ? `<button class="btn btn-sm btn-success" onclick="resolveCPReview(${r
.id})">? ${tr('Resolve', '????? ???????')}</button>` : '-')
    ]
    })))
    );
    } else {
        revContainer.innerHTML = `<div class="empty-state-card"><div class="empty-state-icon">?</div><div>${tr('
No clinical reviews found', '?? ???? ??????? ?????? ??????')}</div></div>`;
    }
    }
    else if (cpTab === 'education') {
        tabContainer.innerHTML = `
        <div class="split-layout" style="display:grid;grid-template-columns:1fr 1fr;gap:20px;">
            <div class="card" style="padding:20px;background:rgba(255,255,255,0.01);border:1px solid rgba(255,255,
255,0.05)">
                <h4 style="margin:0 0 16px;color:var(--primary)">? ${tr('New Patient Drug Education', '????? ???? ??
?? ?????')}</h4>
                <div class="form-group mb-12">
                    <label class="label">${tr('Select Patient', '???? ??????')}</label>
                    <select class="input w-full" id="cpePatient">
                        ${patients.map(p => `<option value="${p.id}" data-name="${escapeHTML(p.name_en || p.name_ar || '
')}">${escapeHTML(p.name_en || p.name_ar || ' ')} (ID: ${p.id})</option>`).join('')}
                    </select>
                </div>
                <div class="form-group mb-12">
                    <label class="label">${tr('Medication Name', '??? ??????')}</label>
                    <input class="input w-full" id="cpeMedication" placeholder="e.g. Warfarin 5mg">
                </div>
                <div class="form-group mb-12">
                    <label class="label">${tr('Usage Instructions', '??????? ?????????')}</label>
                    <textarea class="input w-full" id="cpeInstructions" rows="2" placeholder="Take once daily at eveni
ng..."></textarea>
                </div>
                <div class="form-group mb-12">
                    <label class="label">${tr('Potential Side Effects', '??????? ????????? ?????????')}</label>
                    <textarea class="input w-full" id="cpeSideEffects" rows="2" placeholder="Bleeding, bruising..."></
textarea>
                </div>
                <div class="form-group mb-16">
                    <label class="label">${tr('Precautions', '??????????? ?????????')}</label>
                    <textarea class="input w-full" id="cpePrecautions" rows="2" placeholder="Avoid heavy green vegetab
les..."></textarea>
                </div>
                <button class="btn btn-primary w-full" onclick="submitCPEducation()">? ${tr('Save Counseling Log', '
??? ??? ???????? ?????????')}</button>
            </div>
            <div class="card" style="padding:20px;background:rgba(255,255,255,0.01);border:1px solid rgba(255,255,
255,0.05)">
                <h4 style="margin:0 0 16px;color:var(--primary)">? ${tr('Previous Patient Counseling Logs', '????? ?
???? ?????? ?????????')}</h4>

```

```

<div style="max-height:480px;overflow-y:auto;padding-inline-end:4px">
  ${educationLogs.length ? educationLogs.map(e => `
    <div class="card" style="margin-bottom:8px;padding:12px;background:rgba(255,255,255,0.01);border
:1px solid rgba(255,255,255,0.04)">
      <div style="font-weight:600;color:var(--primary)">${escapeHTML(e.patient_name)}</div>
      <div style="font-size:13px;font-weight:bold;margin:4px 0">${escapeHTML(e.medication)}</div>
      <div style="font-size:12px;color:var(--text-dim)"><strong style="color:var(--text-normal)">Ins
truct:</strong> ${escapeHTML(e.instructions)}</div>
      <div style="font-size:12px;color:var(--text-dim)"><strong style="color:var(--text-normal)">Sid
e Effects:</strong> ${escapeHTML(e.side_effects)}</div>
      <div style="font-size:12px;color:var(--text-dim)"><strong style="color:var(--text-normal)">Edu
cator:</strong> ${escapeHTML(e.educated_by)} | ${e.created_at ? new Date(e.created_at).toLocaleDateString('ar-
SA') : ''}</div>
    </div>
  `).join('') : `<div style="color:var(--text-dim)">${tr('No counseling logs yet', '?? ???? ????? ??
??? ?????')}</div>`}
</div>
</div>
</div>
`;

```

```

window.submitCPEducation = async () => {
  const sel = document.getElementById('cpePatient');
  const patient_name = sel.options[sel.selectedIndex].dataset.name;
  const medication = document.getElementById('cpeMedication').value;
  const instructions = document.getElementById('cpeInstructions').value;
  const side_effects = document.getElementById('cpeSideEffects').value;
  const precautions = document.getElementById('cpePrecautions').value;

  if (!medication) { showToast(tr('Medication is required', '??? ?????? ??????'), 'error'); return; }

  try {
    await API.post('/api/clinical-pharmacy/education', {
      patient_id: sel.value,
      patient_name,
      medication,
      instructions,
      side_effects,
      precautions
    });
    showToast(tr('Education log saved successfully', '?? ??? ??? ??????? ??????? ??????'));
    navigateTo(31);
  } catch (err) {
    showToast(tr('Error saving', '??? ??? ?????? ??????'), 'error');
  }
};

```

```

// Helper functions exposed globally
window.openClinicalReviewModal = (prescriptionId, patientId, patientName, medicationName) => {
  const modalHtml = `
    <div class="modal-backdrop" id="cpReviewModal">
      <div class="modal-content glass-card-premium" style="max-width:600px;padding:24px">
        <h3 style="margin-top:0">? ${tr('Create Clinical Pharmacy Review', '????? ?????? ?????? ??????')}</h

```

3>

```
<p style="margin-bottom:16px;color:var(--text-dim)">
  <strong>${tr('Patient', '?????')}</strong> ${escapeHTML(patientName)} (ID: ${patientId})<br>
  <strong>${tr('Medication', '?????')}</strong> ${escapeHTML(medicationName)} (Queue #${prescripti
onId})
</p>
<div class="form-group mb-12">
  <label class="label">${tr('Review Type', '??? ???????')}</label>
  <select class="input w-full" id="cpModalType">
    <option value="Medication Review">${tr('Medication Review', '????? ?????? ??????')}</option>
    <option value="Dose Adjustment">${tr('Dose Adjustment', '????? ??????')}</option>
    <option value="Allergy Check">${tr('Allergy Check', '??? ???????')}</option>
    <option value="Drug-Drug Interaction">${tr('Drug-Drug Interaction', '????? ??????')}</option>
  </select>
</div>
<div class="form-group mb-12">
  <label class="label">${tr('Severity', '????? ???????')}</label>
  <select class="input w-full" id="cpModalSeverity">
    <option value="Low">${tr('Low', '?????')}</option>
    <option value="Medium">${tr('Medium', '?????')}</option>
    <option value="High">${tr('High', '????? ?????')}</option>
  </select>
</div>
<div class="form-group mb-12">
  <label class="label">${tr('Findings', '????????? ?????????')}</label>
  <textarea class="input w-full" id="cpModalFindings" rows="2" placeholder="Describe clinical findin
gs..."></textarea>
</div>
<div class="form-group mb-12">
  <label class="label">${tr('Recommendations', '?????????')}</label>
  <textarea class="input w-full" id="cpModalRecs" rows="2" placeholder="Pharmacist recommendations..
."></textarea>
</div>
<div class="form-group mb-16">
  <label class="label">${tr('Interventions', '????????? ????????? ??????')}</label>
  <textarea class="input w-full" id="cpModalInterventions" rows="2" placeholder="Actions taken (e.g.
called prescribing doctor)..."></textarea>
</div>
<div class="flex-row" style="justify-content:flex-end;gap:12px">
  <button class="btn" onclick="document.getElementById('cpReviewModal').remove()">${tr('Cancel', '??
??')}</button>
  <button class="btn btn-primary" onclick="submitModalClinicalReview(${patientId}, '${jsStr(patientN
ame)}', ${prescriptionId})">? ${tr('Submit Review', '??? ?????????')}</button>
</div>
</div>
</div>
`;
const div = document.createElement('div');
div.innerHTML = modalHtml;
document.body.appendChild(div.firstElementChild);
};

window.submitModalClinicalReview = async (patientId, patientName, prescriptionId) => {
  const review_type = document.getElementById('cpModalType').value;
```

```

const severity = document.getElementById('cpModalSeverity').value;
const findings = document.getElementById('cpModalFindings').value;
const recommendations = document.getElementById('cpModalRecs').value;
const interventions = document.getElementById('cpModalInterventions').value;

try {
  await API.post('/api/clinical-pharmacy/reviews', {
    patient_id: patientId,
    patient_name: patientName,
    prescription_id: prescriptionId,
    review_type,
    findings,
    recommendations,
    interventions,
    severity
  });
  showToast(tr('Clinical Review submitted successfully', '?? ????? ????????? ????????? ?????'));
  document.getElementById('cpReviewModal')?.remove();
  navigateTo(31);
} catch (e) {
  showToast(tr('Error submitting review', '??? ????? ?????????'), 'error');
}
};

window.verifyPrescription = async (id) => {
  const r = confirm(tr('Are you sure you want to verify and approve this prescription?', '?? ??? ????? ?? ??
  ??? ??? ?????? ?????????'));
  if (!r) return;
  try {
    await API.put('/api/pharmacy/queue/' + id + '/verify', {});
    showToast(tr('Prescription verified successfully', '?? ????? ?????? ?????'));
    navigateTo(31);
  } catch (err) {
    showToast(tr('Error verifying prescription', '??? ?? ?????? ?????????'), 'error');
  }
};

window.resolveCPReview = async (id) => {
  try {
    await API.put('/api/clinical-pharmacy/reviews/' + id, { outcome: 'Resolved', status: 'Closed' });
    showToast(tr('Review marked as Resolved and Closed', '?? ?????? ?????????? ?????? ?????'));
    navigateTo(31);
  } catch (err) {
    showToast(tr('Error resolving review', '??? ??? ?????? ?????'), 'error');
  }
};

// ===== REHABILITATION / PT =====
let rehabTab = 'patients';
async function renderRehabilitation(el) {
  const [rehabPatients, sessions, allPatients, assessments] = await Promise.all([
    API.get('/api/rehab/patients').catch(() => []),
    API.get('/api/rehab/sessions').catch(() => []),
    API.get('/api/patients').catch(() => []),

```

```

    API.get('/api/rehab/assessments').catch(() => [])
  ]);
  const active = rehabPatients.filter(r => r.status === 'Active').length;
  el.innerHTML = `
    <div class="page-title">?? ${tr('Rehabilitation / Physical Therapy', '????? ??????? / ?????? ???????')}</div>

    <div class="stats-grid" style="grid-template-columns:repeat(4,1fr)">
      <div class="stat-card" style="--stat-color:#3b82f6"><span class="stat-icon">?</span><div class="stat-label">${tr('Active Patients', '????? ??????')}</div><div class="stat-value">${active}</div></div>
      <div class="stat-card" style="--stat-color:#4ade80"><span class="stat-icon">?</span><div class="stat-label">${tr('Total Sessions', '?????? ???????')}</div><div class="stat-value">${sessions.length}</div></div>
      <div class="stat-card" style="--stat-color:#f59e0b"><span class="stat-icon">?</span><div class="stat-label">${tr('Total Patients', '?????? ??????')}</div><div class="stat-value">${rehabPatients.length}</div></div>
      <div class="stat-card" style="--stat-color:#a78bfa"><span class="stat-icon">?</span><div class="stat-label">${tr('Discharged', '?????')}</div><div class="stat-value">${rehabPatients.filter(r => r.status === 'Discharged').length}</div></div>
    </div>

    <div class="tab-bar">
      <button class="tab-btn ${rehabTab === 'patients' ? 'active' : ''}" onclick="rehabTab='patients';navigateTo(32)">? ${tr('Patients', '??????')}</button>
      <button class="tab-btn ${rehabTab === 'new' ? 'active' : ''}" onclick="rehabTab='new';navigateTo(32)">? ${tr('New Referral', '????? ?????')}</button>
      <button class="tab-btn ${rehabTab === 'sessions' ? 'active' : ''}" onclick="rehabTab='sessions';navigateTo(32)">? ${tr('Sessions', '????????')}</button>
      <button class="tab-btn ${rehabTab === 'assessments' ? 'active' : ''}" onclick="rehabTab='assessments';navigateTo(32)">? ${tr('Assessments', '????????')}</button>
    </div>

    <div class="card" id="rehabContent"></div>`;
  const mc = document.getElementById('rehabContent');
  if (rehabTab === 'patients') {
    mc.innerHTML = rehabPatients.length ? makeTable(
      [tr('Patient', '??????'), tr('Diagnosis', '????????'), tr('Therapy', '??????'), tr('Therapist', '????????'), tr('Status', '??????'), tr('Sessions', '????????')],
      rehabPatients.map(r => ({
        cells: [r.patient_name, r.diagnosis, r.therapy_type, r.therapist, statusBadge(r.status),
          rawHtml(`<button class="btn btn-sm" onclick="viewRehabSessions(${parseInt(r.id, 10)})">? ${tr('View', '????')}</button>`)
        ])
      }))
    ) : `<div class="empty-state"><span style="font-size:48px">??</span><p>${tr('No rehab patients', '?? ????? ????? ?????')}</p></div>`;
  } else if (rehabTab === 'new') {
    const therapyTypes = ['Physical Therapy', 'Occupational Therapy', 'Speech Therapy', 'Cardiac Rehab', 'Pulmonary Rehab', 'Neurological Rehab'];
    mc.innerHTML = `<h3>? ${tr('New Rehabilitation Referral', '????? ????? ?????')}</h3>
    <div class="form-grid">
      <div><label>${tr('Patient', '??????')}</label><select id="rehabPatient" class="form-input">${allPatients.map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}</select></div>
      <div><label>${tr('Therapy Type', '??? ??????')}</label><select id="rehabType" class="form-input">${therapyTypes.map(t => `<option>${t}</option>`).join('')}</select></div>
      <div><label>${tr('Therapist', '????????')}</label><input id="rehabTherapist" class="form-input"></div>
      <div><label>${tr('Referral Source', '???? ???????')}</label><input id="rehabSource" class="form-input" placeholder="${tr('Dr. Name / Dept', '??? ?????? / ??????')}></div>
    </div>
  `;
  }

```

```

<div style="grid-column:1/-1"><label>${tr('Diagnosis', '???????')}</label><input id="rehabDiag" class="form-input"></div>

<div style="grid-column:1/-1"><label>${tr('Notes', '???????')}</label><textarea id="rehabNotes" class="form-input" rows="3"></textarea></div>
</div>

<button class="btn btn-primary" onclick="submitRehab()" style="margin-top:8px">?? ${tr('Add Patient', '???
?? ???')}</button>;

} else if (rehabTab === 'assessments') {
  mc.innerHTML = `
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
      <h3>? ${tr('Clinical Assessments', '????????? ????????? ?????????')}</h3>
      <button class="btn btn-primary btn-sm" onclick="showNewRehabAssessmentForm()">${tr('? New Assessment',
'? ????? ?????')}</button>
    </div>
    <div id="rehabAssessmentContainer">
      ${assessments.length ? makeTable(
        [tr('Date', '????????'), tr('Patient', '???????'), tr('Type', '??? ?????????'), tr('ROM Scores', '?????
?????'), tr('Strength', '????? ?????????'), tr('Pain Level', '????? ?????'), tr('Assessor', '????????')],
        assessments.map(a => {
          const pat = rehabPatients.find(rp => parseInt(rp.id, 10) === parseInt(a.rehab_patient_id, 10));
          const name = pat ? pat.patient_name : (a.patient_name || tr('Unknown', '??? ?????'));
          return {
            cells: [
              new Date(a.created_at).toLocaleDateString(isArabic ? 'ar-SA' : 'en-US'),
              name,
              a.assessment_type,
              a.rom_scores || '- ',
              a.strength_scores || '- ',
              rawHtml(`<span class="badge ${a.pain_level > 5 ? 'badge-danger' : 'badge-success'}">${a.pain_l
evel}/10</span>`),
              a.assessor
            ]
          };
        })
      ) : `<div class="empty-state"><p>${tr('No clinical assessments registered', '?? ????? ????????? ?????? ?
????')}</p></div>`}
    </div>
  `;
} else {
  mc.innerHTML = sessions.length ? makeTable(
    [tr('Session#', '????#'), tr('Date', '????????'), tr('Therapist', '????????'), tr('Duration', '?????'), tr
('Pain Before', '??? ???'), tr('Pain After', '??? ???'), tr('Notes', '????????')],
    sessions.map(s => ({
      cells: [s.session_number, s.session_date, s.therapist, s.duration_minutes + ' ' + tr('min', '?'),
        rawHtml(`<span style="color:${s.pain_before > 5 ? '#ef4444' : '#22c55e'}">${escapeHTML(s.pain_before)}
/10</span>`),
        rawHtml(`<span style="color:${s.pain_after > 5 ? '#ef4444' : '#22c55e'}">${escapeHTML(s.pain_after)}/1
0</span>`),
        s.progress_notes?.substring(0, 50) || '- '
      ]
    }))
  ) : `<div class="empty-state"><p>${tr('No sessions', '?? ????? ?????')}</p></div>`;
}
}

```

```

window.submitRehab = async function () {
  const sel = document.getElementById('rehabPatient');
  const patient_name = sel.options[sel.selectedIndex].dataset.name;
  await API.post('/api/rehab/patients', { patient_id: sel.value, patient_name, diagnosis: document.getElementById('rehabDiag').value, referral_source: document.getElementById('rehabSource').value, therapist: document.getElementById('rehabTherapist').value, therapy_type: document.getElementById('rehabType').value, notes: document.getElementById('rehabNotes').value });
  showToast(tr('Patient added', '??? ??????')); rehabTab = 'patients'; navigateTo(32);
};

window.viewRehabSessions = async function (id) {
  rehabTab = 'sessions'; navigateTo(32);
};

window.showNewRehabAssessmentForm = async function() {
  const [rehabPatients] = await Promise.all([
    API.get('/api/rehab/patients').catch(() => [])
  ]);
  const mc = document.getElementById('rehabContent');
  mc.innerHTML = `
    <h3> ${tr('New Clinical PT Assessment', '????? ?????? ?????? ??????')}</h3>
    <div class="form-grid">
      <div>
        <label>${tr('Referral Patient', '?????? ?????? ???????')}</label>
        <select id="assessRehabPatient" class="form-input">
          ${rehabPatients.map(rp => `<option value="${safeId(rp.id)}" data-pid="${safeId(rp.patient_id)}">${escapeHTML(rp.patient_name)} (${escapeHTML(rp.therapy_type)})</option>`).join('')}
        </select>
      </div>
      <div>
        <label>${tr('Assessment Type', '??? ??????')}</label>
        <select id="assessType" class="form-input">
          <option>Initial Evaluation</option>
          <option>Progress Re-Evaluation</option>
          <option>Discharge Assessment</option>
        </select>
      </div>
      <div>
        <label>${tr('Pain Level (0-10)', '????? ?????? (0-10)')}</label>
        <input type="number" id="assessPain" class="form-input" min="0" max="10" value="0">
      </div>
      <div>
        <label>${tr('Range of Motion (ROM) Scores', '???? ?????? ?????? (ROM)')}</label>
        <input type="text" id="assessROM" class="form-input" placeholder="e.g., Shoulder Flexion: 120°">
      </div>
      <div>
        <label>${tr('Muscle Strength Scores (0-5)', '????? ?????? ??????? (0-5)')}</label>
        <input type="text" id="assessStrength" class="form-input" placeholder="e.g., Quadriceps: 4/5">
      </div>
      <div>
        <label>${tr('Functional Score', '?????? ??????')}</label>
        <input type="text" id="assessFunctional" class="form-input" placeholder="e.g., Independent Ambulation">
      </div>
      <div>
        <label>${tr('Balance & Gait Scores', '????? ?????? ??????')}</label>

```

```

        <input type="text" id="assessBalance" class="form-input" placeholder="e.g., Berg Balance: 48/56">
    </div>
    <div>
        <label>${tr('Assessor', '???????')}</label>
        <input type="text" id="assessAssessor" class="form-input" value="${currentUser.name}">
    </div>
</div>
<div style="margin-top:12px;display:flex;gap:8px">
    <button class="btn btn-primary" onclick="submitRehabAssessment()">?? ${tr('Save Assessment', '?? ?????
?')}</button>
    <button class="btn btn-secondary" onclick="rehabTab='assessments';navigateTo(32)">${tr('Cancel', '?????
')}</button>
</div>
`;
};
window.submitRehabAssessment = async function() {
    const sel = document.getElementById('assessRehabPatient');
    if (!sel || sel.selectedIndex === -1) {
        showToast(tr('No patient selected', '?? ??? ?????? ?????'), 'error');
        return;
    }
    const rehab_patient_id = sel.value;
    const patient_id = sel.options[sel.selectedIndex].dataset.pid;
    const assessment_type = document.getElementById('assessType').value;
    const pain_level = document.getElementById('assessPain').value;
    const rom_scores = document.getElementById('assessROM').value;
    const strength_scores = document.getElementById('assessStrength').value;
    const functional_scores = document.getElementById('assessFunctional').value;
    const balance_scores = document.getElementById('assessBalance').value;
    const assessor = document.getElementById('assessAssessor').value;

    await API.post('/api/rehab/assessments', {
        rehab_patient_id,
        patient_id,
        assessment_type,
        pain_level,
        rom_scores,
        strength_scores,
        functional_scores,
        balance_scores,
        assessor
    });

    showToast(tr('Assessment saved successfully', '?? ??? ??????? ?????'));
    rehabTab = 'assessments';
    navigateTo(32);
};

// ===== PATIENT PORTAL =====
async function renderPatientPortal(el) {
    const content = el;

    const patients = await API.get('/api/patients').catch(() => []);
    const appointments = await API.get('/api/appointments').catch(() => []);

```



```
const recentCount = patients.filter(p => { const d = new Date(p.created_at); const week = new Date(); week.setDate(week.getDate() - 7); return d > week; }).length;
```

```
content.innerHTML = `
  <div class="page-title">? ${tr('Patient Portal', '????? ?????? ??????')}</div>
  <div class="flex flex-col md:flex-row gap-24 items-start mb-24 bg-white/40 backdrop-blur-md p-24 rounded-2xl border border-outline/10 shadow-sm w-full">
    <div class="flex-1 text-right">
      <h2 class="text-2xl font-bold text-primary mb-8">${tr('Welcome back, ' + currentUser.name, '????? ??' + currentUser.name)}</h2>
      <p class="text-sm text-on-surface-variant mb-16">${tr('Access your electronic health records, schedule virtual visits, and review recent lab updates.', '????? ?? ?? ?????? ?????? ???????. ??? ???? ????????? ??? ?????? ??? ??? ??????')}</p>
      <div class="flex gap-12">
        <button class="bg-primary hover:bg-primary/90 text-white px-24 py-10 rounded-xl text-sm font-semibold transition-all shadow-md flex items-center gap-8" onclick="navigateTo(2)">
          <span class="material-symbols-outlined text-lg">calendar_month</span>
          ${tr('Book Appointment', '??? ??? ????')}
        </button>
        <button class="border border-secondary text-secondary hover:bg-secondary/10 px-24 py-10 rounded-xl text-sm font-semibold transition-all" onclick="navigateTo(35)">
          ${tr('Virtual Visit', '?????? ?????????')}
        </button>
      </div>
    </div>
  </div>

  <div class="split-layout">
    <!-- Right Side: Appointments & Labs -->
    <div class="space-y-16">
      <!-- Upcoming Appointments -->
      <div class="card p-24">
        <div class="card-title justify-between">
          <span>? ${tr('Upcoming Appointment', '????? ??????')}</span>
          <span class="bg-secondary-container text-on-secondary-container text-xs px-12 py-4 rounded-full font-medium">${tr('Confirmed', '????')}</span>
        </div>
        ${appointments.length ? `
          <div class="flex items-center gap-16 mb-16">
            <div class="w-12 h-12 rounded-full bg-secondary/10 flex items-center justify-center text-secondary">
              <span class="material-symbols-outlined text-2xl">medical_services</span>
            </div>
            <div class="text-right flex-1">
              <h4 class="font-bold text-primary">${escapeHTML(appointments[appointments.length - 1].doctor_name)}</h4>
              <p class="text-xs text-on-surface-variant">${escapeHTML(appointments[appointments.length - 1].department || tr('General Specialist', '????? ???'))}</p>
            </div>
            <div class="text-left">
              <p class="font-bold text-secondary">${escapeHTML(appointments[appointments.length - 1].appt_date)}</p>
              <p class="text-xs text-on-surface-variant">${escapeHTML(appointments[appointments.length - 1].appt_time)}</p>
            </div>
          </div>
        ` : ''}
      </div>
    </div>
  </div>
`
```

```

        </div>
    </div>
    ` : `
    <div class="empty-state">
        <div class="empty-icon">?</div>
        <p>${tr('No upcoming appointments', '?? ???? ?????? ?????')}</p>
    </div>
    `}
    <div class="flex gap-12 pt-12 border-t border-outline/10">
        <button class="btn btn-sm btn-primary" onclick="navigateTo(2)">${tr('Manage Appointments', '????
????????')}</button>
    </div>
</div>

<!-- Recent Lab Results -->
<div class="card p-24">
    <div class="card-title justify-between">
        <span>? ${tr('Recent Lab Results', '??? ????? ???????')}</span>
        <button class="btn btn-sm" onclick="navigateTo(4)">${tr('View All', '??? ?????')}</button>
    </div>
    <div class="space-y-12">
        <div class="flex justify-between items-center p-12 bg-surface-container-low hover:bg-surface-conta
iner rounded-xl transition-all cursor-pointer" onclick="navigateTo(4)">
            <div class="flex items-center gap-12">
                <div class="w-10 h-10 rounded-lg bg-secondary/10 flex items-center justify-center text-seconda
ry">
                    <span class="material-symbols-outlined text-lg">science</span>
                </div>
                <div class="text-right">
                    <h4 class="text-xs font-bold text-primary">${tr('Complete Blood Count (CBC)', '??? ???? ????
????????')}</h4>
                    <p class="text-[10px] text-on-surface-variant">${tr('Normal', '????')} - 12/10/2026</p>
                </div>
            </div>
            <span class="badge badge-success">${tr('Normal', '????')}</span>
        </div>
        <div class="flex justify-between items-center p-12 bg-surface-container-low hover:bg-surface-conta
iner rounded-xl transition-all cursor-pointer" onclick="navigateTo(4)">
            <div class="flex items-center gap-12">
                <div class="w-10 h-10 rounded-lg bg-secondary/10 flex items-center justify-center text-seconda
ry">
                    <span class="material-symbols-outlined text-lg">science</span>
                </div>
                <div class="text-right">
                    <h4 class="text-xs font-bold text-primary">${tr('Fast Blood Glucose (FBS)', '??? ???? ?????
????????')}</h4>
                    <p class="text-[10px] text-on-surface-variant">98 mg/dL - 08/10/2026</p>
                </div>
            </div>
            <span class="badge badge-success">${tr('Normal', '????')}</span>
        </div>
    </div>
</div>

```

```

    <!-- Telemedicine Widget -->
    <div class="card p-24 bg-gradient-to-br from-primary-container to-primary text-white border-none relative overflow-hidden">
        <h3 class="text-lg font-bold mb-8 text-white">${tr('Virtual Smart Clinic', '?????? ?????????? ??????')}</h3>
        <p class="text-xs text-white/80 mb-16">${tr('Consult with our international doctors virtually from anywhere.', '????? ?????? ?????? ?????? ?? ????? ?? ?? ??? ?????????? ????.')}</p>
        <button class="bg-secondary text-white hover:bg-secondary-container px-16 py-8 rounded-lg text-xs font-semibold transition-all border border-secondary" onclick="navigateTo(35)">
            ${tr('Join Waiting Room', '???? ??? ?????????')}
        </button>
    </div>
</div>

<!-- Left Side: Health metrics, Insurance, Emergency -->
<div class="space-y-16">
    <!-- Stats Bento Grid -->
    <div class="grid grid-cols-2 gap-12">
        <div class="card p-16 text-right">
            <span class="material-symbols-outlined text-secondary mb-4">favorite</span>
            <p class="text-[10px] text-on-surface-variant">${tr('Heart Rate', '??? ?????')}</p>
            <p class="text-lg font-extrabold text-primary">72 bpm</p>
        </div>
        <div class="card p-16 text-right">
            <span class="material-symbols-outlined text-secondary mb-4">blood_pressure</span>
            <p class="text-[10px] text-on-surface-variant">${tr('Blood Pressure', '??? ?????')}</p>
            <p class="text-lg font-extrabold text-primary">120/80</p>
        </div>
        <div class="card p-16 text-right">
            <span class="material-symbols-outlined text-secondary mb-4">steps</span>
            <p class="text-[10px] text-on-surface-variant">${tr('Daily Steps', '??????? ?????????')}</p>
            <p class="text-lg font-extrabold text-primary">8,420</p>
        </div>
        <div class="card p-16 text-right">
            <span class="material-symbols-outlined text-secondary mb-4">sleep</span>
            <p class="text-[10px] text-on-surface-variant">${tr('Sleep Quality', '???? ?????')}</p>
            <p class="text-lg font-extrabold text-primary">7.5 hr</p>
        </div>
    </div>

    <!-- Insurance Progress Card -->
    <div class="card p-24">
        <div class="card-title justify-between">
            <span>?? ${tr('Insurance Claims', '????? ???????? (?????????)')}</span>
        </div>
        <div class="mb-16">
            <div class="flex justify-between text-xs mb-4">
                <span>${tr('Limit Used', '?????? ?????????')}</span>
                <span class="font-bold">12,400 / 30,000 SAR</span>
            </div>
            <div class="w-full bg-surface-container-high h-2 rounded-full">
                <div class="bg-secondary h-full rounded-full" style="width: 41.3%"></div>
            </div>
        </div>
    </div>

```

```

        <button class="btn btn-sm btn-primary w-full" onclick="navigateTo(9)">${tr('Manage Claims', '????? ?
????????')}</button>
    </div>

    <!-- Emergency Card -->
    <div class="card p-16 bg-error/5 border border-error/20 flex items-center justify-between">
        <div class="flex items-center gap-12">
            <div class="w-10 h-10 rounded-full bg-error text-white flex items-center justify-center animate-pu
lse">
                <span class="material-symbols-outlined text-lg">emergency</span>
            </div>
            <div class="text-right">
                <h4 class="text-xs font-bold text-error">${tr('Emergency Contact', '???? ??????? ?????????')}</h
4>
                <p class="text-[10px] text-on-surface-variant">${tr('Immediate assistance', '????????? ??????? ??
?? ??????')}</p>
            </div>
        </div>
        <a href="tel:997" class="w-10 h-10 bg-error hover:bg-error/90 text-white rounded-full flex items-cen
ter justify-center shadow-md">
            <span class="material-symbols-outlined text-lg">call</span>
        </a>
    </div>
</div>
`;

}

window.approvePortalAppt = async function (id) { await API.put('/api/portal/appointments/' + id, { status: 'Ap
proved' }); showToast(tr('Approved', '?? ?????????')); navigateTo(33); };
window.rejectPortalAppt = async function (id) { await API.put('/api/portal/appointments/' + id, { status: 'Rej
ected' }); showToast(tr('Rejected', '?? ??????')); navigateTo(33); };

// ===== A1: PORTAL PATIENT SELF-SERVICE VIEW =====
window.renderPortalPatientView = async function(containerId, patientId) {
    const el = document.getElementById(containerId);
    if (!el || !patientId) return;
    el.innerHTML = `<div class="flex gap-8 mb-16 flex-wrap">
        <button id="portalTabLab" class="btn btn-sm bg-primary text-white" onclick="portalSwitchTab('lab', ${patien
tId})">? ${tr('Lab Results', '????? ?????????')}</button>
        <button id="portalTabMeds" class="btn btn-sm" onclick="portalSwitchTab('meds', ${patientId})">? ${tr('Medic
ations', '????????')}</button>
        <button id="portalTabVisits" class="btn btn-sm" onclick="portalSwitchTab('visits', ${patientId})">? ${tr('V
isit Summary', '????? ?????????')}</button>
        <button id="portalTabMsg" class="btn btn-sm" onclick="portalSwitchTab('msg', ${patientId})">?? ${tr('Messag
es', '????????')}</button>
    </div><div id="portalTabContent"></div>`;
    await portalSwitchTab('lab', patientId);
};

window.portalSwitchTab = async function(tab, patientId) {
    const el = document.getElementById('portalTabContent');
    if (!el) return;
    ['Lab', 'Meds', 'Visits', 'Msg'].forEach(t => {
        const btn = document.getElementById('portalTab' + t);

```

```

    if (btn) btn.className = t.toLowerCase() === tab ? 'btn btn-sm bg-primary text-white' : 'btn btn-sm';
  });
  el.innerHTML = `<div class="text-center py-24"><span class="material-symbols-outlined">progress_activity</span></div>`;
  if (tab === 'lab') {
    const results = await API.get('/api/portal/lab-results?patient_id=' + patientId).catch(() => []);
    el.innerHTML = results.length ? `<div class="overflow-x-auto"><table class="w-full text-sm"><thead><tr class="bg-surface-variant">
      <th class="p-8 text-right">${tr('Test','????')}</th><th class="p-8 text-right">${tr('Result','??????')}</th><th class="p-8 text-right">${tr('Reference','????')}</th><th class="p-8 text-right">${tr('Date','?????')}</th><th class="p-8">${tr('Status','????')}</th>
    </tr></thead><tbody>${results.map(r => `<tr class="border-b hover:bg-surface-variant/30">
      <td class="p-8 font-medium">${escapeHTML(r.test_name||'')}</td>
      <td class="p-8 font-bold ${r.critical_flag?'text-error':''}>${escapeHTML(r.result_value||'')} ${escapeHTML(r.unit||'')}</td>
      <td class="p-8 text-on-surface-variant text-xs">${escapeHTML(r.reference_range||'')}</td>
      <td class="p-8 text-xs">${r.created_at ? new Date(r.created_at).toLocaleDateString('ar-SA') : ''}</td>
      <td class="p-8">${r.critical_flag ? '<span class="badge bg-error text-white">?? </span>' : '<span class="badge bg-secondary-container">?</span>'}</td>
    </tr>`).join('')}</tbody></table></div>` :
    `<p class="text-center text-on-surface-variant py-24">${tr('No verified lab results yet','? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ? ?')}</p>`;
  } else if (tab === 'meds') {
    const meds = await API.get('/api/portal/medications?patient_id=' + patientId).catch(() => []);
    el.innerHTML = meds.length ? meds.map(m => `<div class="card p-16 mb-8 flex gap-12 items-start">
      <span class="material-symbols-outlined text-primary mt-4">medication</span>
      <div class="flex-1">
        <div class="font-bold text-primary">${escapeHTML(m.medication_name||'')}</div>
        <div class="text-sm text-on-surface-variant">${escapeHTML(m.dosage||'')} ? ${escapeHTML(m.frequency||'')}
      </div>
        <div class="text-xs text-on-surface-variant mt-4">${tr('Prescribed by','????')}: ${escapeHTML(m.prescribed_by||'')} | ${m.prescribed_date ? new Date(m.prescribed_date).toLocaleDateString('ar-SA') : ''}</div>
      </div>
      <span class="badge ${m.status==='Dispensed'? 'bg-secondary-container': 'bg-surface-variant'}">${escapeHTML(m.status||'')}</span>
    </div>`).join('') : `<p class="text-center text-on-surface-variant py-24">${tr('No active medications','? ? ? ? ? ? ? ? ? ? ? ?')}</p>`;
  } else if (tab === 'visits') {
    const data = await API.get('/api/portal/visit-summary?patient_id=' + patientId).catch(() => ({ visits:[], admissions:[] }));
    const visits = data.visits || []; const adm = data.admissions || [];
    el.innerHTML = `<div class="space-y-8">
      <h4 class="font-bold text-primary">? ${tr('Recent Visits','???????? ????')}</h4>
      ${visits.length ? visits.map(v => `<div class="card p-12 border-r-4 border-primary">
        <div class="flex justify-between"><span class="font-medium">${escapeHTML(v.visit_type||'????')}</span>
        <span class="text-xs">${v.visit_date?new Date(v.visit_date).toLocaleDateString('ar-SA'):''}</span></div>
        <div class="text-sm mt-4">${escapeHTML(v.chief_complaint||'')}</div>
        <div class="text-xs text-on-surface-variant">${tr('Doctor','????')}: ${escapeHTML(v.attending_doctor||'')} | ${escapeHTML(v.department||'')}</div>
      </div>`).join('') : `<p class="text-sm text-on-surface-variant">${tr('No visits','? ? ? ? ? ? ? ?')}</p>`
    }
    ${adm.length ? `<h4 class="font-bold text-primary mt-16">? +tr('Admissions','????????')</h4>` + adm.map(a => `<div class="card p-12 border-r-4 border-secondary">
      <div class="flex justify-between"><span class="font-medium">${escapeHTML(a.ward_name||'')}</span><span`

```

```

class="text-xs">${a.admission_date?new Date(a.admission_date).toLocaleDateString('ar-SA'):''}}</span></div>
    <div class="text-sm mt-4">${escapeHTML(a.diagnosis||'')}</div></div>`).join('') : ''}
</div>`;
} else if (tab === 'msg') {
    const msgs = await API.get('/api/portal/messages?patient_id=' + patientId).catch(() => []);
    el.innerHTML = `<div class="mb-12">
        <h4 class="font-bold mb-8">?? ${tr('Send Message','???? ?????? ?????? ??????')}</h4>
        <input id="portalMsgSubject" class="input w-full mb-8" placeholder="${tr('Subject','????????')}">
        <textarea id="portalMsgBody" class="input w-full mb-8" rows="3" placeholder="${tr('Your message...','???
        ???...')}"></textarea>
        <button class="btn btn-primary" onclick="sendPortalMessage(${patientId})">? ${tr('Send','????')}</butto
n>
    </div>
    <div class="space-y-8">${msgs.map(m => `<div class="card p-12 ${m.sender_type==='patient'?'border-r-4 bord
er-primary':'border-r-4 border-secondary'}">
        <div class="flex justify-between"><span class="font-medium">${escapeHTML(m.subject||'')}</span><span cla
ss="text-xs">${m.created_at?new Date(m.created_at).toLocaleDateString('ar-SA'):''}}</span></div>
        <div class="text-sm mt-4">${escapeHTML(m.body||'')}</div>
        ${m.reply_body?`<div class="mt-8 bg-secondary-container rounded p-8 text-sm"><strong>${tr('Team Reply','
?? ??????')}</strong> ${escapeHTML(m.reply_body)}</div>`:''}
        </div>`).join('') || `<p class="text-sm text-on-surface-variant">${tr('No messages yet','?? ???? ????? ??
')}</p>`}</div>`;
    }
};
window.sendPortalMessage = async function(patientId) {
    const subject = document.getElementById('portalMsgSubject')?.value?.trim();
    const body = document.getElementById('portalMsgBody')?.value?.trim();
    if (!body) { showToast(tr('Message required','???? ??????? ??????'), 'error'); return; }
    try {
        await API.post('/api/portal/messages', { patient_id: patientId, subject, body });
        showToast(tr('Message sent','?? ????? ???????'));
        await portalSwitchTab('msg', patientId);
    } catch(e) { showToast(tr('Error','???'), 'error'); }
};

// ===== A2: PEDIATRIC IMMUNIZATION UI =====
window.renderPediatricImmunization = async function(containerId, patientId) {
    const el = document.getElementById(containerId);
    if (!el) return;
    const schedule = await API.get('/api/pediatrics/immunization-schedule').catch(() => []);
    const recData = patientId ? await API.get('/api/pediatrics/immunization-records/' + patientId).catch(() => (
{ immunizations:[] }))) : { immunizations:[] };
    const imm = recData.immunizations || [];
    el.innerHTML = `<div class="card p-20 mb-16">
        <h3 class="card-title">? ${tr('Saudi MOH Immunization Schedule 2024','???? ?????????? ?????? ??????? 2024')}
    </h3>
        <div class="overflow-x-auto"><table class="w-full text-sm"><thead><tr class="bg-surface-variant">
            <th class="p-8 text-right">${tr('Age','????')}</th><th class="p-8 text-right">${tr('Vaccines','????????
?')}</th><th class="p-8">${tr('Status','????')}</th>
        </tr></thead><tbody>
            ${schedule.map(s => {
                const given = imm.filter(i => s.vaccines.some(v => i.vaccine_name.toLowerCase().includes(v.split(' ')[0]
.toLowerCase()))).length;
                return `<tr class="border-b hover:bg-surface-variant/30">

```

```

        <td class="p-8 font-medium">${escapeHTML(s.age)}</td>
        <td class="p-8">${s.vaccines.map(v=><span class="badge bg-primary-container text-xs mr-4">${escapeHTML(v)}</span>`).join('')}</td>
        <td class="p-8">${given>0?'<span class="badge bg-secondary-container">? <span>':<span class="badge bg-surface-variant">?</span>'}</td>
    </tr>`;
    }).join('')}
</tbody></table></div>
</div>
${patientId ? `<div class="card p-20 mb-16">
    <h3 class="card-title">? ${tr('Record New Vaccine','???? ???? ????')}</h3>
    <div class="grid grid-cols-2 gap-12">
        <div><label class="label">${tr('Vaccine Name','?? ?????')}</label><input id="immVaccineName" class="input w-full" placeholder="BCG / DTaP / MMR..."></div>
        <div><label class="label">${tr('Dose #','?? ?????')}</label><input id="immDoseNum" class="input w-full" type="number" value="1" min="1" max="5"></div>
        <div><label class="label">${tr('Date Given','???? ?????')}</label><input id="immDate" class="input w-full" type="date" value="${new Date().toISOString().slice(0,10)}"></div>
        <div><label class="label">${tr('Batch #','?? ?????')}</label><input id="immBatch" class="input w-full" placeholder="LOT12345"></div>
        <div><label class="label">${tr('Site','????')}</label><select id="immSite" class="input w-full"><option value="Left arm">???? </option><option value="Right arm">???? </option><option value="Left thigh">???? </option><option value="Right thigh">???? </option><option value="Oral">???? </option></select></div>
        <div><label class="label">${tr('Route','????')}</label><select id="immRoute" class="input w-full"><option value="IM">???? IM</option><option value="SC">???? SC</option><option value="ID">???? ID</option><option value="PO">???? PO</option></select></div>
        <div><label class="label">${tr('Next Due','????')}</label><input id="immNextDue" class="input w-full" type="date"></div>
        <div><label class="label">${tr('Notes','????')}</label><input id="immNotes" class="input w-full" placeholder="${tr('Any reaction...','? ? ? ? ?')}"></div>
    </div>
    <button class="btn btn-primary mt-12" onclick="saveImmunization(${patientId})">? ${tr('Save Vaccination','? ? ? ? ?')}</button>
</div>
<div class="card p-20">
    <h3 class="card-title">? ${tr('Vaccination History','?? ?????')}</h3>
    ${imm.length ? `<div class="overflow-x-auto"><table class="w-full text-sm"><thead><tr class="bg-surface-variant">
        <th class="p-8 text-right">${tr('Vaccine','????')}</th><th class="p-8">${tr('Dose','????')}</th><th class="p-8 text-right">${tr('Date','????')}</th><th class="p-8 text-right">${tr('Given By','????')}</th>
    </tr></thead><tbody>
        ${imm.map(i=><tr class="border-b"><td class="p-8 font-medium">${escapeHTML(i.vaccine_name)}</td><td class="p-8 text-center">${i.dose_number}</td><td class="p-8">${i.given_date?new Date(i.given_date).toLocaleDateString('ar-SA'):''}</td><td class="p-8 text-xs">${escapeHTML(i.given_by||'')}</td></tr>`).join('')}
    </tbody></table></div>` : `<p class="text-sm text-on-surface-variant">${tr('No vaccination records','? ? ? ? ?')}</p>`}
</div>` : ''`;
};

window.saveImmunization = async function(patientId) {
    const vaccine_name = document.getElementById('immVaccineName')?.value?.trim();
    if (!vaccine_name) { showToast(tr('Vaccine name required','? ? ? ? ?'), 'error'); return; }
    try {
        await API.post('/api/pediatrics/immunization', {
            patient_id: patientId, vaccine_name,

```

```

    dose_number: document.getElementById('immDoseNum')?.value||1,
    given_date: document.getElementById('immDate')?.value,
    batch_number: document.getElementById('immBatch')?.value,
    site: document.getElementById('immSite')?.value,
    route: document.getElementById('immRoute')?.value,
    next_due: document.getElementById('immNextDue')?.value||null,
    notes: document.getElementById('immNotes')?.value
  });
  showToast(tr('Vaccination recorded','?? ????? ???????'));
  renderPediatricImmunization(document.getElementById('immContainer')?.id||'immContainer', patientId);
} catch(e) { showToast(tr('Error','???'), 'error'); }
};

// ===== A3: NURSING PAIN ASSESSMENT UI (NRS/VAS/FLACC) =====
window.renderNursingPainAssessment = async function(containerId, patientId, admissionId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  const history = patientId ? await API.get('/api/nursing/pain-history/' + patientId).catch(() => []) : [];
  const lastScore = history.length ? history[0].pain_score : null;
  window._selectedPainScore = null;
  el.innerHTML = `<div class="card p-20 mb-16">
    <h3 class="card-title">? ${tr('Pain Assessment','????? ?????')} (NRS/VAS/FLACC)</h3>
    ${lastScore !== null ? `<div class="p-8 rounded-lg mb-12 ${lastScore}>=?'bg-error/10 border border-error t
ext-error':lastScore>=4?'bg-warning-container':'bg-secondary-container'}`>
      ${tr('Last pain score','??? ????? ?????')}: <strong>${lastScore}/10</strong> (${history[0].pain_scale||'NRS
'})
    </div>` : ''}
    <div class="mb-16">
      <label class="label font-bold">${tr('Select Pain Score (0=No Pain, 10=Worst)','???? ????? ????? (0=? ? ?
? 10=??? ???)')}</label>
      <div class="flex gap-2 mt-8">
        ${Array.from({length:11}, (_,i)=>`<button id="painBtn${i}" onclick="selectPainScore(${i})" class="flex-
1 py-8 rounded-lg text-sm font-bold border-2 transition-all ${i<=3?'border-green-200 bg-green-50 hover:bg-gree
n-100':i<=6?'border-yellow-200 bg-yellow-50 hover:bg-yellow-100':'border-red-200 bg-red-50 hover:bg-red-100'}`
        >${i}</button>`).join('')}
      </div>
      <div class="flex justify-between text-xs text-on-surface-variant mt-4">
        <span>? ${tr('No Pain','?? ???')}</span><span>? ${tr('Moderate','????')}</span><span>? ${tr('Worst','
??? ???')}</span>
      </div>
    </div>
    <div class="grid grid-cols-2 gap-12 mb-12">
      <div><label class="label">${tr('Scale','???????')}</label><select id="painScale" class="input w-full"><o
ption value="NRS">NRS</option><option value="VAS">VAS</option><option value="FLACC">FLACC (?????)</option><opt
ion value="FACES">FACES</option><option value="BPS">BPS (???????)</option></select></div>
      <div><label class="label">${tr('Location','???????')}</label><input id="painLocation" class="input w-full
" placeholder="${tr('Head, Chest...','???? ???...')}"></div>
      <div><label class="label">${tr('Character','???????')}</label><input id="painChar" class="input w-full"
placeholder="${tr('Sharp, Burning...','???? ?????...')}"></div>
      <div><label class="label">${tr('Relieving Factors','????? ???????')}</label><input id="painRel" class="i
nput w-full" placeholder="${tr('Rest, Medication...','???????...')}"></div>
      <div><label class="label">${tr('Current Analgesia','????????? ???????')}</label><input id="painMeds" clas
s="input w-full" placeholder="Paracetamol, Morphine..."></div>
      <div><label class="label">${tr('Pain Goal (0-10)','??? ?????')}</label><input id="painGoal" class="input

```



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w-full" type="number" value="3" min="0" max="10"></div>
</div>
<button class="btn btn-primary" onclick="savePainAssessment(${patientId}||'null',{admissionId}||'null'})">
? ${tr('Save Pain Assessment','??? ????? ?????')}</button>
</div>
<div class="card p-20">
<h3 class="card-title">? ${tr('Pain History','????? ?????')}</h3>
${history.length ? `<div class="overflow-x-auto"><table class="w-full text-sm"><thead><tr class="bg-surface
e-variant">
<th class="p-8">${tr('Score','?????')}</th><th class="p-8">${tr('Scale','???????')}</th><th class="p-8
text-right">${tr('Location','?????')}</th><th class="p-8 text-right">${tr('Nurse','?????')}</th><th class="p
-8 text-right">${tr('Time','????')}</th>
</tr></thead><tbody>
${history.map(h=>`<tr class="border-b"><td class="p-8 text-center"><span class="badge font-bold ${h.pain_s
core}>=?'bg-error text-white':h.pain_score>=4?'bg-warning-container':'bg-secondary-container'}`>${h.pain_score
}/10</span></td><td class="p-8 text-center">${escapeHTML(h.pain_scale||'')}</td><td class="p-8">${escapeHTML(h
.pain_location||'')}</td><td class="p-8 text-xs">${escapeHTML(h.assessed_by||'')}</td><td class="p-8 text-xs">
${h.assessed_at?new Date(h.assessed_at).toLocaleString('ar-SA'):''}</td></tr>`)}.join('')}
</tbody></table></div>` : `<p class="text-sm text-on-surface-variant">${tr('No pain assessments recorded',
'?? ?????? ?? ????? ???')}</p>`}
</div>`;
window.selectPainScore = function(score) {
window._selectedPainScore = score;
for (let i=0;i<=10;i++) {
const b = document.getElementById('painBtn'+i);
if (b) { b.style.transform = i===score?'scale(1.15)':''; b.style.fontWeight = i===score?'900':'700'; b.s
tyle.boxShadow = i===score?'0 0 0 3px var(--md-primary)':''; }
}
};
};
window.savePainAssessment = async function(patientId, admissionId) {
const score = window._selectedPainScore;
if (score === null || score === undefined) { showToast(tr('Select pain score first','???? ???? ????? ?????')
, 'error'); return; }
try {
const r = await API.post('/api/nursing/pain-assessment', {
patient_id: patientId, admission_id: admissionId||null,
pain_scale: document.getElementById('painScale')?.value||'NRS', pain_score: score,
pain_location: document.getElementById('painLocation')?.value,
pain_character: document.getElementById('painChar')?.value,
relieving_factors: document.getElementById('painRel')?.value,
current_analgesia: document.getElementById('painMeds')?.value,
pain_goal: document.getElementById('painGoal')?.value||3
});
showToast(r.critical ? tr('?? CRITICAL PAIN ? Physician notified','?? ??? ??? ???? ? ?? ????? ???????') : t
r('Pain assessment saved','?? ??? ????? ?????'), r.critical?'error':'success');
} catch(e) { showToast(tr('Error','???'), 'error'); }
};

// ===== A4: ICU DAILY GOALS CHECKLIST UI (CBAHI / ABCDEF Bundle) =====
window.renderIcuDailyGoals = async function(containerId, admissionId, patientId) {
const el = document.getElementById(containerId);
if (!el || !admissionId) return;
const today = new Date().toISOString().slice(0,10);

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```

const goals = await API.get('/api/icu/daily-goals/' + admissionId + '?date=' + today).catch(() => []);
const g = goals[0] || {};
el.innerHTML = `<div class="card p-20">
  <div class="card-title justify-between mb-16">
    <span>? ${tr('ICU Daily Goals','????? ????? ? ??????? ???????')} ? ${today}</span>
    <span class="badge bg-primary-container">ABCDEF Bundle | CBAHI</span>
  </div>
  <!-- A & B: Awakening + Breathing -->
  <div class="bg-blue-50 rounded-xl p-16 border border-blue-200 mb-12">
    <h4 class="font-bold text-blue-800 mb-12">? A+B ? ${tr('Awakening & Breathing Trials','???????? ?????? ?
??????')}</h4>
    <div class="grid grid-cols-2 gap-8 mb-8">
      <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuDailySat" ${g.daily_sat?'checked':''}><span>${tr('Daily SAT (Sedation Awakening Trial)','????? ?????? ?????? SAT')}</span></label>
      <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuDailySbt" ${g.daily_sbt?'checked':''}><span>${tr('Daily SBT (Spontaneous Breathing Trial)','????? ?????? ???????? SBT')}</span></label>
    </div>
    <div class="grid grid-cols-3 gap-8">
      <div><label class="label text-xs">Target RASS</label><input id="icuRass" class="input w-full" type="number" value="${g.sedation_target_rass??'-2'}" min="-5" max="4"></div>
      <div><label class="label text-xs">FiO2 % Goal</label><input id="icuFio2" class="input w-full" type="number" value="${g.vent_goal_fio2||''}" min="21" max="100" placeholder="40"></div>
      <div><label class="label text-xs">PEEP cmH2O</label><input id="icuPeep" class="input w-full" type="number" value="${g.vent_goal_peep||''}" placeholder="5"></div>
    </div>
    <div class="mt-8"><label class="label text-xs">${tr('Weaning Plan','?? ?')}</label><input id="icuWean" class="input w-full" value="${escapeHTML(g.vent_wean_plan||'')}" placeholder="${tr('Wean FiO2 / Extubation planned...','????? FiO2 / ?????? ??????...')}"></div>
  </div>
  <!-- C & D: Sedation + Delirium -->
  <div class="bg-purple-50 rounded-xl p-16 border border-purple-200 mb-12">
    <h4 class="font-bold text-purple-800 mb-12">? C+D ? ${tr('Sedation & Delirium','??????? ???????')}</h4>
    <div class="grid grid-cols-2 gap-8">
      <div><label class="label text-xs">${tr('Pain Goal (NRS)','?? ?')}</label><input id="icuPainGoal" class="input w-full" type="number" value="${g.pain_goal_nrs||3}" min="0" max="10"></div>
      <div><label class="label text-xs">CAM-ICU ${tr('(Delirium)','?????')}</label>
      <select id="icuCam" class="input w-full">
        <option value="">---</option>
        <option value="Negative" ${g.delirium_cam_icu==='Negative'? 'selected':''}>Negative (?? ?)</option>
        <option value="Positive" ${g.delirium_cam_icu==='Positive'? 'selected':''}>Positive (????)</option>
        <option value="Unable to assess" ${g.delirium_cam_icu==='Unable to assess'? 'selected':''}>Unable to assess</option>
      </select></div>
    </div>
  </div>
  <!-- E: Mobility + F: Family -->
  <div class="bg-green-50 rounded-xl p-16 border border-green-200 mb-12">
    <h4 class="font-bold text-green-800 mb-12">? E+F ? ${tr('Early Mobility & Family','??????? ?????? ??????')}</h4>
    <div class="grid grid-cols-2 gap-8">

```

```

<div><label class="label text-xs">${tr('Mobility Goal','??? ?????')}</label>
  <select id="icuMobility" class="input w-full">
    <option value="">---</option>
    ${['Passive ROM','Sitting in bed','Sitting edge','Standing','Ambulation'].map(v=>`<option value="$
{v}" ${g.mobility_goal===v?'selected':''}>${v}</option>`).join('')}
  </select></div>
  <label class="flex items-center gap-8 cursor-pointer mt-8"><input type="checkbox" id="icuFamilyUpdate"
  ${g.family_update_done?'checked':''}><span>${tr('Family meeting done','????? ?????? ?')}</span></label>
</div>
</div>
<!-- Prevention Bundle -->
<div class="bg-yellow-50 rounded-xl p-16 border border-yellow-200 mb-12">
  <h4 class="font-bold text-yellow-800 mb-12">?? ${tr('Prevention Bundle (VAP / CLABSI / CAUTI / DVT)','??
?? ??????')}</h4>
  <div class="grid grid-cols-2 gap-8">
    <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuOralCare" ${g.oral
_care_done?'checked':''}><span>${tr('Oral care done ? VAP prevention','??????? ??????? ? ????? VAP')}</span></
label>
    <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuHob" ${g.hob_eleva
tion!==false?'checked':''}><span>${tr('HOB 30-45° (head elevation)','??? ?????? 30-45°')}</span></label>
    <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuLineReview" ${g.li
ne_necessity_reviewed?'checked':''}><span>${tr('Central line necessity reviewed','?????? ?????? ?????????
??????')}</span></label>
    <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuFoley" ${g.foley_n
ecessity_reviewed?'checked':''}><span>${tr('Foley necessity reviewed','?????? ?????? ?????????')}</span><
/label>
  <div><label class="label text-xs">DVT ${tr('Prophylaxis','????????')}</label>
    <select id="icuDvt" class="input w-full">
      ${['','Heparin SQ','Mechanical','Both','Contraindicated'].map(v=>`<option value="$${v}" ${g.dvt_pro
phylaxis===v?'selected':''}>${v}||'---'</option>`).join('')}
    </select></div>
    <div><label class="label text-xs">${tr('Stress Ulcer Prophylaxis','????? ??????')}</label>
      <select id="icuSup" class="input w-full">
        ${['','PPI','H2 blocker','Not indicated'].map(v=>`<option value="$${v}" ${g.stress_ulcer_prophy===v
?'selected':''}>${v}||'---'</option>`).join('')}
      </select></div>
    </div>
  </div>
<!-- Nutrition -->
<div class="bg-orange-50 rounded-xl p-16 border border-orange-200 mb-12">
  <h4 class="font-bold text-orange-800 mb-12">?? ${tr('Nutrition Goals','????? ?????????')}</h4>
  <div class="grid grid-cols-3 gap-8">
    <div><label class="label text-xs">${tr('Route','???????')}</label>
      <select id="icuNutRoute" class="input w-full">
        ${['','PO','NG','NJ','TPN','NPO'].map(v=>`<option value="$${v}" ${g.nutrition_route===v?'selected':
''}>${v}||'---'</option>`).join('')}
      </select></div>
      <div><label class="label text-xs">${tr('Caloric Goal (kcal)','????????')}</label><input id="icuKcal" cl
ass="input w-full" type="number" value="${g.caloric_goal_kcal}||'" placeholder="1800"></div>
      <div><label class="label text-xs">${tr('Protein Goal (g)','????????')}</label><input id="icuProtein" c
lass="input w-full" type="number" value="${g.protein_goal_g}||'" placeholder="100"></div>
    </div>
  </div>
<!-- Goals text -->

```

```

<div class="bg-gray-50 rounded-xl p-16 border mb-12">
  <h4 class="font-bold mb-12"> ${tr('Daily Goals','?????? ?????')}</h4>
  <label class="label">${tr('Medical Goals for Today','?????? ?????')}</label>
  <textarea id="icuMedGoals" class="input w-full mb-8" rows="2" placeholder="${tr('Wean sedation, target extubation...','????? ?????????? ??? ??????..')}">${escapeHTML(g.medical_goals||'')}</textarea>
  <label class="label">${tr('Nursing Goals for Today','?????? ?????????? ?????')}</label>
  <textarea id="icuNursGoals" class="input w-full mb-8" rows="2" placeholder="${tr('Hourly neuro checks, Q 2H turning...','??? ???? ????? ????? ?? ??????..')}">${escapeHTML(g.nursing_goals||'')}</textarea>
  <label class="flex items-center gap-8 cursor-pointer"><input type="checkbox" id="icuGoalsDiscussed" ${g.goals_discussed_with_team?'checked':''}><span class="font-medium">${tr('Goals discussed in morning rounds','??? ?????? ?? ?????? ?????????')}</span></label>
</div>

<button class="btn btn-primary w-full" onclick="saveIcuDailyGoals(${admissionId},${patientId}||'null')">
  ? ${tr('Save Daily Goals','??? ????? ?????')} ? ${today}
</button>
</div>`;
};

window.saveIcuDailyGoals = async function(admissionId, patientId) {
  try {
    await API.post('/api/icu/daily-goals', {
      admission_id: admissionId, patient_id: patientId,
      daily_sat: document.getElementById('icuDailySat')?.checked,
      daily_sbt: document.getElementById('icuDailySbt')?.checked,
      sedation_target_rass: document.getElementById('icuRass')?.value||null,
      vent_goal_fio2: document.getElementById('icuFio2')?.value||null,
      vent_goal_peep: document.getElementById('icuPeep')?.value||null,
      vent_wean_plan: document.getElementById('icuWean')?.value,
      pain_goal_nrs: document.getElementById('icuPainGoal')?.value||3,
      delirium_cam_icu: document.getElementById('icuCam')?.value,
      mobility_goal: document.getElementById('icuMobility')?.value,
      family_update_done: document.getElementById('icuFamilyUpdate')?.checked,
      oral_care_done: document.getElementById('icuOralCare')?.checked,
      hob_elevation: document.getElementById('icuHob')?.checked,
      line_necessity_reviewed: document.getElementById('icuLineReview')?.checked,
      foley_necessity_reviewed: document.getElementById('icuFoley')?.checked,
      dvt_prophylaxis: document.getElementById('icuDvt')?.value,
      stress_ulcer_prophy: document.getElementById('icuSup')?.value,
      nutrition_route: document.getElementById('icuNutRoute')?.value,
      caloric_goal_kcal: document.getElementById('icuKcal')?.value||null,
      protein_goal_g: document.getElementById('icuProtein')?.value||null,
      medical_goals: document.getElementById('icuMedGoals')?.value,
      nursing_goals: document.getElementById('icuNursGoals')?.value,
      goals_discussed_with_team: document.getElementById('icuGoalsDiscussed')?.checked
    });
    showToast(tr('ICU daily goals saved','?? ??? ????? ?????'));
  } catch(e) { showToast(tr('Error saving ICU goals','??? ?? ??? ?????????'), 'error'); }
};

// ===== ZATCA E-INVOICING =====
async function renderZATCA(el) {
  const content = el; // C-1 fix: bind content so downstream uses are consistent (el is the container)

  const invoices = await API.get('/api/zatca/invoices').catch(() => []);
  content.innerHTML = `

```

```

<h2>${tr('ZATCA E-Invoice', '???????? ???? - ???')}</h2>
<div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:12px;margin-bottom:16px">
  <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${invoices.filter?.(i => i.zatca_status === 'submitted')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Submitted', '????')}</p></div>
  <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e65100">${invoices.filter?.(i => !i.zatca_status || i.zatca_status === 'pending')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Pending', '???? ????')}</p></div>
  <div class="card" style="padding:16px;text-align:center;background:#fce4ec"><h3 style="margin:0;color:#c62828">${invoices.filter?.(i => i.zatca_status === 'rejected')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Rejected', '????')}</p></div>
</div>
<div class="card" style="padding:20px;margin-bottom:16px">
  <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
    <h4 style="margin:0">${tr('Invoices', '????')}</h4>
    <div style="display:flex;gap:8px">
      <button class="btn btn-sm" onclick="zatcaBulkInfo()" style="background:#e8f5e9;color:#2e7d32">? ${tr('Submit', '????')}</button>
      <button class="btn btn-sm" onclick="exportToCSV(window._zatcaData||[],'zatca_invoices')" style="background:#e0f7fa;color:#00838f">? ${tr('Export', '????')}</button>
    </div>
  </div>
  <div id="zatcaTable"></div>
</div>`;

window._zatcaData = invoices;
const zt = document.getElementById('zatcaTable');
if (zt && invoices.length > 0) {
  createTable(zt, 'ztbl',
    [tr('Invoice #', '??? ?????'), tr('Patient', '????'), tr('Amount', '????'), tr('VAT', '????'), tr('Date', '????'), tr('Status', '????'), tr('Action', '????')],
    invoices.map(i => ({
      // i.invoice_id is the source invoice (FK). Generate/submit are tenant-scoped & gated server-side.
      cells: [i.invoice_number || i.id, i.buyer_name || i.patient_name || '', parseFloat(i.total_with_vat || i.total || 0).toFixed(2), parseFloat(i.vat_amount || 0).toFixed(2), i.created_at ? new Date(i.created_at).toLocaleDateString('ar-SA') : '', statusBadge(i.clearance_status || i.submission_status || 'pending'),
      rawHtml(`<button class="btn btn-sm" onclick="generateZatcaInvoice(${safeId(i.invoice_id)})">? ${tr('Regenerate', '???? ????')}</button> <button class="btn btn-sm" onclick="submitZatcaInvoice(${safeId(i.invoice_id)})">? ${tr('Submit', '????')}</button>`)],
      id: i.id
    })))
  );
} else if (zt) { zt.innerHTML = '<p style="color:#999;text-align:center;padding:20px">' + tr('No invoices', '?? ??? ????') + '</p>'; }

// E10: generate a ZATCA e-invoice (UBL+QR) for a SPECIFIC invoice. Server computes VAT + artifacts.
window.generateZatcaInvoice = async (invoiceId) => {
  const id = parseInt(invoiceId, 10);
  if (!Number.isInteger(id) || id <= 0) { showToast(tr('Invalid invoice', '???? ????'), 'error'); return; }
  try {
    const r = await API.post('/api/zatca/generate', { invoice_id: id });
    if (r && r.error) { showToast(tr('Error: ' + r.error, '??? : ' + r.error), 'error'); return; }
  }
}

```

```

        showToast(tr('E-invoice generated (UBL + QR)', '?? ????? ????????? ??????????'));
        renderZATCA(content);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// Legacy bulk button: per-invoice generate/submit is the safe flow (each row has its own buttons).
window.zatcaBulkInfo = () => { showToast(tr('Use the Generate/Submit buttons on each invoice row', '????? ?
???? ???????/??????? ??? ??????'), 'info'); };

// E10: record submission intent. Clearance is GATED off (no CSID) ? server returns 503, intent logged.
window.submitZatcaInvoice = async (invoiceId) => {
    const id = parseInt(invoiceId, 10);
    if (!Number.isInteger(id) || id <= 0) { showToast(tr('Invalid invoice', '????? ??? ?????'), 'error'); ret
urn; }
    try {
        const r = await API.post('/api/zatca/submit', { invoice_id: id });
        // 503 (gated) returns {error, clearance_status:'RECORDED'} ? treat as informational, not a crash.
        if (r && r.clearance_status === 'RECORDED') { showToast(tr('Submission intent recorded (clearance gated
off)', '?? ????? ??? ?????? (?????? ??????)')); }
        else if (r && r.error) { showToast(tr('Error: ' + r.error, '??? : ' + r.error), 'error'); }
        renderZATCA(content);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}

// ===== TELEMEDICINE =====
let teleTab = 'sessions';
async function renderTelemedicine(el) {
    const content = el;

    const sessions = await API.get('/api/telemedicine/sessions').catch(() => []);
    const active = sessions.filter(s => s.status === 'active').length;

    content.innerHTML = `
        <h2>${tr('Telemedicine', '???? ?? ?????')}</h2>
        <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:1
6px">
            <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1
565c0">${sessions.length}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Total Sessions', '????? ??????')}
)</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2
e7d32">${active}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Active Now', '???? ?????')}</p></div>
        </div>
        <div style="display:grid;grid-template-columns:1fr 2fr;gap:16px">
            <div class="card" style="padding:20px">
                <h4 style="margin:0 0 12px">${tr('Schedule Session', '????? ?????')}</h4>
                <div class="form-group"><label>${tr('Patient', '??????')}</label><input class="form-input" id="telePat
ient"></div>
                <div class="form-group"><label>${tr('Doctor', '??????')}</label><input class="form-input" id="teleDoct
or"></div>
                <div class="form-group"><label>${tr('Date & Time', '??????? ??????')}</label><input type="datetime-loc
al" class="form-input" id="teleDate"></div>
                <div class="form-group"><label>${tr('Platform', '??????')}</label>
                <select class="form-input" id="telePlatform"><option value="zoom">Zoom</option><option value="teams"
>Microsoft Teams</option><option value="meet">Google Meet</option><option value="internal">${tr('Internal', '?

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```

????')}</option></select></div>
    <div class="form-group"><label>${tr('Meeting Link', '???? ?????????')}</label><input class="form-input"
id="teleLink" placeholder="https://..."></div>
    <div class="form-group"><label>${tr('Notes', '????????')}</label><textarea class="form-input" id="teleN
otes" rows="2"></textarea></div>
    <button class="btn btn-primary w-full" onclick="saveTeleSession()">? ${tr('Schedule', '????')}</butto
n>

</div>
<div class="card" style="padding:20px">
    <h4 style="margin:0 0 12px">${tr('Sessions', '????????')}</h4>
    <div id="teleTable"></div>
</div>
</div>`;

const tt = document.getElementById('teleTable');
if (tt) {
    createTable(tt, 'teleTbl',
        [tr('Patient', '?????'), tr('Doctor', '?????'), tr('Date', '?????'), tr('Platform', '?????'), tr('L
ink', '?????'), tr('Status', '?????')],
        sessions.map(s => {
            const sessionDateTime = s.session_date || [s.scheduled_date, s.scheduled_time].filter(Boolean).join('
');
            return { cells: [s.patient_name || '', s.doctor || '', sessionDateTime ? new Date(sessionDateTime).toL
ocaleString('ar-SA') : '', s.platform || s.session_type || '', s.meeting_link ? rawHtml('<a href="' + escapeHT
ML(s.meeting_link) + '" target="_blank" style="color:#1a73e8">? ' + tr('Join', '?????') + '</a>') : '', statu
sBadge(s.status)], id: s.id };
        })
    );
}
window.saveTeleSession = async () => {
    try {
        const rawDateTime = document.getElementById('teleDate').value || '';
        const [scheduledDate, scheduledTime = ''] = rawDateTime.split('T');
        await API.post('/api/telemedicine/sessions', {
            patient_name: document.getElementById('telePatient').value,
            speciality: '',
            session_type: document.getElementById('telePlatform').value || 'Video',
            scheduled_date: scheduledDate || '',
            scheduled_time: scheduledTime || '',
            duration_minutes: 15,
            notes: document.getElementById('teleNotes').value
        });
        showToast(tr('Session scheduled!', '?? ????? ??????!'));
        navigateTo(currentPage);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}
window.scheduleTele = async function () {
    const sel = document.getElementById('telePatient');
    await API.post('/api/telemedicine/sessions', { patient_id: sel.value, patient_name: sel.options[sel.selected
Index].dataset.name, session_type: document.getElementById('teleType').value, scheduled_date: document.getElem
entById('teleDate').value, scheduled_time: document.getElementById('teleTime').value, duration_minutes: docume
nt.getElementById('teleDur').value });

```

```

    showToast(tr('Session scheduled', '??? ??????')); teleTab = 'sessions'; navigateTo(35);
};

// ===== PATHOLOGY =====
async function renderPathology(el) {
    const content = el;

    // E15: specimen->block->slide->report workflow with server-authoritative state machine.
    const [specimens, patients] = await Promise.all([
        API.get('/api/pathology/specimens').catch(() => []),
        API.get('/api/patients').catch(() => [])
    ]);

    const cnt = (st) => specimens.filter(s => s.state === st).length;
    const STATES = ['Received', 'Grossing', 'Processing', 'Reported', 'SignedOut'];

    const patientOptions = (Array.isArray(patients) ? patients : []).map(p =>
        '<option value="' + safeId(p.id) + '">' + escapeHTML((p.name || '') + ' (#' + p.id + ')') + '</option>'
    ).join('');

    content.innerHTML = `
        <h2>${tr('Pathology', '??? ??????')}</h2>
        <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(150px,1fr));gap:12px;margin-bottom:16px">
            <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e65100">${specimens.length}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Total Specimens', '????? ??????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#fce4ec"><h3 style="margin:0;color:#c62828">${cnt('Received')}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Received', '?????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1565c0">${cnt('Processing')} + cnt('Grossing')</h3><p style="margin:4px 0 0;font-size:12px">${tr('In Process', '??? ??????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${cnt('SignedOut')}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Signed Out', '?? ??????')}</p></div>
        </div>
        <div style="display:grid;grid-template-columns:1fr 2fr;gap:16px">
            <div class="card" style="padding:20px">
                <h4 style="margin:0 0 12px">${tr('Accession Specimen', '?????? ?????')}</h4>
                <div class="form-group"><label>${tr('Patient', '?????')}</label>
                    <select class="form-input" id="pathPatient"><option value="">${tr('Select...', '????...')}</option>${patientOptions}</select></div>
                <div class="form-group"><label>${tr('Specimen Type', '??? ??????')}</label>
                    <select class="form-input" id="pathType">
                        <option value="biopsy">${tr('Biopsy', '????')}</option>
                        <option value="cytology">${tr('Cytology', '????')}</option>
                        <option value="surgical">${tr('Surgical', '????')}</option>
                        <option value="autopsy">${tr('Autopsy', '????')}</option>
                        <option value="frozen">${tr('Frozen Section', '???? ?????')}</option>
                    </select></div>
                <div class="form-group"><label>${tr('Site', '????')}</label><input class="form-input" id="pathSite">
            </div>
            <div class="form-group"><label>${tr('Clinical Details', '???????? ?????????')}</label><textarea class="form-input" id="pathDetails" rows="2"></textarea></div>
            <div class="form-group"><label>${tr('Priority', '????????')}</label>

```



```

        <select class="form-input" id="pathPriority">
            <option value="routine">${tr('Routine', '????')}</option>
            <option value="urgent">${tr('Urgent', '????')}</option>
            <option value="stat">${tr('STAT', '????')}</option>
        </select></div>
        <button class="btn btn-primary w-full" onclick="pathRegisterSpecimen()">? ${tr('Accession', '???????')}
    }</button>

    <p style="font-size:11px;color:#888;margin-top:8px">${tr('Accession number is generated by the server.
    ', '??? ????? ???? ??????? ?? ??????.')}</p>
</div>
<div class="card" style="padding:20px">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px">
        <h4 style="margin:0">${tr('Specimen List', '????? ???????')}</h4>
        <input class="form-input" id="pathSearch" placeholder="${tr('Search...', '???...')}" style="width:20
0px" oninput="pathFilterTable()">
    </div>
    <div id="pathTable"></div>
    <div id="pathDetailPane" style="margin-top:16px"></div>
</div>
</div>`;

window._pathData = specimens;
const stateBadge = (st) => {
    const colors = { Received: '#fce4ec', Grossing: '#fff3e0', Processing: '#e3f2fd', Reported: '#ede7f6', Sig
nedOut: '#e8f5e9' };
    return rawHtml('<span style="padding:2px 8px;border-radius:4px;font-size:11px;background:' + (colors[st] |
| '#eee') + '">' + escapeHTML(st || 'Received') + '</span>');
};
const pt = document.getElementById('pathTable');
if (pt) {
    createTable(pt, 'pathTbl',
        [tr('Accession', '????????'), tr('Patient', '?????'), tr('Type', '?????'), tr('Site', '?????'), tr('P
riority', '????????'), tr('State', '?????'), tr('Flags', '????????'), tr('Actions', '????????')],
        specimens.map(s => {
            const flags = (s.malignancy_flag ? '?' + tr('Malignant', '????') : '') + (s.critical_flag ? ' ?? ' +
tr('Critical', '????') : '');
            return {
                cells: [s.accession_number || '', s.patient_name || ('#' + s.patient_id), s.specimen_type || '', s.s
ite || '',
                    rawHtml('<span style="padding:2px 8px;border-radius:4px;font-size:11px;background:' + (s.priority
=== 'stat' ? '#fce4ec' : s.priority === 'urgent' ? '#fff3e0' : '#e8f5e9') + '">' + escapeHTML(s.priority || 'r
outine') + '</span>'),
                    stateBadge(s.state),
                    rawHtml('<span style="font-size:11px">' + escapeHTML(flags) + '</span>'),
                    rawHtml('<button class="btn btn-sm" onclick="viewPathSpecimen(' + parseInt(s.id, 10) + ')">' + t
r('Open', '???') + '</button>')],
                id: s.id
            };
        })
    );
}

// ---- Register: posts FK patient_id to the safe E15 route (no free-text identity) ----
window.pathRegisterSpecimen = async () => {

```

```

const pid = parseInt(document.getElementById('pathPatient').value, 10);
if (!Number.isInteger(pid)) { showToast(tr('Select a patient', '???? ?????'), 'error'); return; }
try {
  await API.post('/api/pathology/specimens', {
    patient_id: pid,
    specimen_type: document.getElementById('pathType').value,
    site: document.getElementById('pathSite').value,
    clinical_details: document.getElementById('pathDetails').value,
    priority: document.getElementById('pathPriority').value
  });
  showToast(tr('Specimen accessioned!', '?? ?????? ??????!'));
  navigateTo(currentPage);
} catch (e) { showToast(tr('Error', '???'), 'error'); }
};

// ---- Detail pane: blocks, slides, report, state machine, sign-out, addendum ----
window.viewPathSpecimen = async (id) => {
  const sid = parseInt(id, 10);
  if (!Number.isInteger(sid)) return;
  const data = await API.get('/api/pathology/specimens/' + sid).catch(() => null);
  const pane = document.getElementById('pathDetailPane');
  if (!data || !pane) { if (pane) pane.innerHTML = ''; return; }
  const sp = data.specimen, rep = data.report || {};
  const immutable = sp.state === 'SignedOut';
  const nextStates = STATES.slice(STATES.indexOf(sp.state) + 1);
  const stateBtns = immutable ? '' : nextStates.map(ns =>
    '<button class="btn btn-sm" style="margin:2px" onclick="pathAdvanceState(' + sid + ', \'' + jsStr(ns) + '
\\')">?? ' + escapeHTML(ns) + '</button>').join('');
  const blocksHtml = (data.blocks || []).map(b =>
    '<li>' + escapeHTML(b.block_no) + ' (' + escapeHTML(b.embedding_type || '') + ') ' +
    (immutable ? '' : '<button class="btn btn-sm" onclick="pathAddSlide(' + parseInt(b.id, 10) + ')">+ ' + t
r('Slide', '????') + '</button>') +
    '<ul>' + (data.slides || []).filter(sl => sl.block_id === b.id).map(sl => '<li>' + escapeHTML(sl.slide_n
o) + ' ? ' + escapeHTML(sl.stain_type || '') + '</li>').join('') + '</ul></li>'
  ).join('');
  pane.innerHTML = `
<div class="card" style="padding:16px;border:2px solid #1565c0">
  <h4 style="margin:0 0 8px">${escapeHTML(sp.accession_number)} ? ${stateBadge(sp.state).html}</h4>
  <p style="font-size:12px;margin:0 0 8px">${tr('Type', '????')}: ${escapeHTML(sp.specimen_type || '')}
| ${tr('Site', '????')}: ${escapeHTML(sp.site || '')}</p>
  <div style="margin-bottom:8px"><strong>${tr('Workflow', '??? ?????')}</strong> ${stateBtns || '<em>'
+ tr('Signed out (immutable)', '?? ???????? (??? ???? ???????') + '</em>'}</div>
  <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px">
    <div>
      <strong>${tr('Blocks / Slides', '??????? / ???????')}</strong>
      ${immutable ? '' : '<button class="btn btn-sm" style="margin-inline-start:8px" onclick="pathAddBlo
ck(' + sid + ')">+ ' + tr('Block', '????') + '</button>'}
      <ul style="font-size:12px">${blocksHtml || '<li><em>' + tr('None', '?? ????') + '</em></li>'}</ul>
    </div>
    <div>
      <strong>${tr('Report', '??????')}</strong>
      <div class="form-group"><label>${tr('Gross', '????? ???????')}</label><textarea class="form-input"
id="rGross" rows="2" ${immutable ? 'disabled' : ''}>${escapeHTML(rep.gross_text || '')}</textarea></div>
      <div class="form-group"><label>${tr('Microscopic', '????? ???????')}</label><textarea class="form-

```

```

input" id="rMicro" rows="2" ${immutable ? 'disabled' : ''}>${escapeHTML(rep.micro_text || '')}</textarea></div>
>
    <div class="form-group"><label>${tr('Diagnosis', '???????')}</label><input class="form-input" id="
rDx" value="${escapeHTML(rep.diagnosis || '')}" ${immutable ? 'disabled' : ''}></div>
    <div class="form-group"><label>${tr('SNOMED codes (comma)', '???? SNOMED (?????)')}</label><input
class="form-input" id="rSnomed" value="${escapeHTML((rep.snomed_codes || []).join(','))}" ${immutable ? 'disab
led' : ''}></div>
    <p style="font-size:11px">${rep.malignancy_flag ? '?' + tr('Malignant', '????') : ''} ${rep.criti
cal_flag ? '?' + tr('Critical', '???') : ''}</p>
    ${immutable
    ? '<div class="form-group"><label>' + tr('Addendum', '????') + '</label><textarea class="form-in
put" id="rAddendum" rows="2"></textarea></div><button class="btn btn-sm" onclick="pathAddendum(' + sid + ')">?
' + tr('Add Addendum', '????? ?????') + '</button>'
    : '<button class="btn btn-primary btn-sm" onclick="pathSaveReport(' + sid + ')">? ' + tr('Save R
eport', '??? ???????') + '</button> <button class="btn btn-sm" style="background:#2e7d32;color:#fff" onclick="
pathSignOut(' + sid + ')">?? ' + tr('Sign Out', '??????') + '</button>'}
    </div>
  </div>
</div>;
};

window.pathAdvanceState = async (id, st) => {
  try { await API.put('/api/pathology/specimens/' + parseInt(id, 10) + '/state', { state: st }); showToast('
?'); navigateTo(currentPage); }
  catch (e) { showToast(e?.message || tr('Invalid transition', '?????? ??? ?????'), 'error'); }
};
window.pathAddBlock = async (id) => {
  const no = prompt(tr('Block number', '??? ??????')); if (no === null) return;
  try { await API.post('/api/pathology/specimens/' + parseInt(id, 10) + '/blocks', { block_no: no }); window
.viewPathSpecimen(id); }
  catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.pathAddSlide = async (blockId) => {
  const no = prompt(tr('Slide number', '??? ???????')); if (no === null) return;
  const stain = prompt(tr('Stain (H&E/IHC...)', '??????')) || 'H&E';
  try { await API.post('/api/pathology/blocks/' + parseInt(blockId, 10) + '/slides', { slide_no: no, stain_t
ype: stain }); navigateTo(currentPage); }
  catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.pathSaveReport = async (id) => {
  const snomed = (document.getElementById('rSnomed').value || '').split(',').map(x => x.trim()).filter(Boole
an);
  try {
    await API.put('/api/pathology/specimens/' + parseInt(id, 10) + '/report', {
      gross_text: document.getElementById('rGross').value,
      micro_text: document.getElementById('rMicro').value,
      diagnosis: document.getElementById('rDx').value,
      snomed_codes: snomed
    });
    showToast(tr('Report saved', '?? ??? ???????')); window.viewPathSpecimen(id);
  } catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.pathSignOut = async (id) => {
  if (!confirm(tr('Sign out is final and locks the report. Continue?', '???????? ?????? ?????? ????????. ??????

```

```

?')) return;
    try { await API.post('/api/pathology/specimens/' + parseInt(id, 10) + '/signout', {}); showToast(tr('Signe
d out', '?? ??????')); navigateTo(currentPage); }
    catch (e) { showToast(e.message || tr('Cannot sign out', '???? ??????'), 'error'); }
};
window.pathAddendum = async (id) => {
    const text = document.getElementById('rAddendum')?.value;
    if (!text || !text.trim()) { showToast(tr('Addendum text required', '?? ??????'), 'error'); return; }
    try { await API.post('/api/pathology/specimens/' + parseInt(id, 10) + '/addendum', { text }); showToast(tr
('Addendum added', '??? ??????')); window.viewPathSpecimen(id); }
    catch (e) { showToast(tr('Error', '???'), 'error'); }
};
window.pathFilterTable = () => { const t = (document.getElementById('pathSearch')?.value || '').toLowerCase(
); document.querySelectorAll('#pathTbl tbody tr').forEach(r => r.style.display = r.textContent.toLowerCase().i
ncludes(t) ? '' : 'none'); };
}

// ===== SOCIAL WORK =====
async function renderSocialWork(el) {

    const cases = await API.get('/api/social-work/cases').catch(() => []);
    content.innerHTML = `
        <h2>${tr('Social Work', '?????')}</h2>
        <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(180px,1fr));gap:12px;margin-bottom:1
6px">
            <div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0">${cases
.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Total Cases', '?????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0">${cases
.filter?.(c => c.status === 'open')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Open', '?
????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0">${cases
.filter?.(c => c.status === 'resolved')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px">${tr('Resol
ved', '?? ????')}</p></div>
        </div>
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px">
            <div class="card" style="padding:20px">
                <h4 style="margin:0 0 12px">${tr('New Case', '????')}</h4>
                <div class="form-group"><label>${tr('Patient', '????')}</label><input class="form-input" id="swPatie
nt"></div>
                <div class="form-group"><label>${tr('Case Type', '???')}</label>
                <select class="form-input" id="swType"><option value="financial">${tr('Financial Assistance', '?????
????')}</option><option value="abuse">${tr('Abuse/Neglect', '????/????')}</option><option value="discharge_
planning">${tr('Discharge Planning', '????')}</option><option value="counseling">${tr('Counseling', '?
????')}</option><option value="legal">${tr('Legal Issues', '????')}</option><option value="housing">${
tr('Housing', '????')}</option></select></div>
                <div class="form-group"><label>${tr('Priority', '????')}</label>
                <select class="form-input" id="swPriority"><option value="low">${tr('Low', '????')}</option><option
value="medium" selected>${tr('Medium', '????')}</option><option value="high">${tr('High', '????')}</option>
<option value="urgent">${tr('Urgent', '????')}</option></select></div>
                <div class="form-group"><label>${tr('Description', '????')}</label><textarea class="form-input" id="s
wDesc" rows="3"></textarea></div>
                <div class="form-group"><label>${tr('Assigned To', '????')}</label><input class="form-input" i

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d="swAssigned"></div>
    <button class="btn btn-primary w-full" onclick="saveSWCase()">? ${tr('Create Case', '????? ???')}</button>
</div>
<div class="card" style="padding:20px">
    <h4 style="margin:0 0 12px">${tr('Cases', '???????')}</h4>
    <div id="swList"></div>
</div>
</div>`;

const sl = document.getElementById('swList');
if (sl) { sl.innerHTML = cases.length ? cases.map(c => '<div style="padding:10px;margin:6px 0;background:#f8f9fa;border-radius:8px;border-right:4px solid ' + (c.status === 'resolved' ? '#4caf50' : c.priority === 'urgent' ? '#cc0000' : '#ff9800') + '"><strong>' + escapeHTML(c.patient_name || '') + '</strong> - ' + escapeHTML(c.case_type || '') + '<br><span style="font-size:12px;color:#666">' + escapeHTML(c.status || '') + ' | ' + escapeHTML(c.priority || '') + (c.assigned_to ? ' | ' + escapeHTML(c.assigned_to) : '') + '</span></div>').join('') : '<p style="color:#999;text-align:center">' + tr('No cases', '?? ???? ?????') + '</p>'; }

window.saveSWCase = async () => {
    try {
        await API.post('/api/social-work/cases', { patient_name: document.getElementById('swPatient')?.value, case_type: document.getElementById('swType')?.value, priority: document.getElementById('swPriority')?.value, description: document.getElementById('swDesc')?.value, assigned_to: document.getElementById('swAssigned')?.value, status: 'open' });
        showToast(tr('Case created', '?? ????? ??????'));
        renderSocialWork(content);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}

// ===== MORTUARY =====
async function renderMortuary(el) {

    const cases = await API.get('/api/mortuary/cases').catch(() => []);
    content.innerHTML = `
        <h2>${tr('Mortuary', '???????')}</h2>
        <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(180px,1fr));gap:12px;margin-bottom:16px">
            <div class="card" style="padding:16px;text-align:center;background:#f3e5f5"><h3 style="margin:0">${cases.length || 0}</h3><p style="margin:4px 0 0;font-size:13px;color:#666">${tr('Total Cases', '?????? ???????')}</p></div>
            <div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0">${cases.filter?.(c => c.status === 'pending')?.length || 0}</h3><p style="margin:4px 0 0;font-size:13px;color:#666">${tr('Pending Release', '??????? ???????')}</p></div>
        </div>
        <div style="display:grid;grid-template-columns:1fr 1fr;gap:16px">
            <div class="card" style="padding:20px">
                <h4 style="margin:0 0 12px">${tr('Register Case', '????? ???')}</h4>
                <div class="form-group"><label>${tr('Deceased Name', '??? ???????')}</label><input class="form-input" id="mortName"></div>
                <div class="form-group"><label>${tr('National ID', '??????')}</label><input class="form-input" id="mortID"></div>
                <div class="form-group"><label>${tr('Date/Time of Death', '????? ???? ??????')}</label><input type="date"

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tetime-local" class="form-input" id="mortDate"></div>
    <div class="form-group"><label>${tr('Cause of Death', '??? ?????')}</label><input class="form-input"
id="mortCause"></div>
    <div class="form-group"><label>${tr('Attending Doctor', '????? ?????')}</label><input class="form-i
nput" id="mortDoctor"></div>
    <div class="form-group"><label>${tr('Next of Kin', '???? ?????')}</label><input class="form-input" i
d="mortKin"></div>
    <div class="form-group"><label>${tr('Contact Phone', '???? ?????')}</label><input class="form-input"
id="mortPhone"></div>
    <div class="form-group"><label>${tr('Storage Location', '???? ?????')}</label><input class="form-inp
ut" id="mortLoc" placeholder="e.g. Unit-3"></div>
    <button class="btn btn-primary w-full" onclick="saveMortuaryCase()">? ${tr('Register', '????')}</butt
on>
</div>
<div class="card" style="padding:20px">
    <h4 style="margin:0 0 12px">${tr('Cases', '????')}</h4>
    <div id="mortList"></div>
</div>
</div>`;

const ml = document.getElementById('mortList');
if (ml) { ml.innerHTML = cases.length ? cases.map(c => '<div style="padding:10px;margin:6px 0;background:#f8
f9fa;border-radius:8px;border-right:4px solid ' + (c.status === 'released' ? '#4caf50' : '#ff9800') + '"><stro
ng>' + escapeHTML(c.name || '') + '</strong><br><span style="font-size:12px;color:#666">' + escapeHTML(c.cause
_of_death || '') + ' | ' + escapeHTML(c.status || '') + '</span></div>').join('') : '<p style="color:#999;text
-align:center">' + tr('No cases', '?? ???? ?????') + '</p>'; }

window.saveMortuaryCase = async () => {
    try {
        await API.post('/api/mortuary/cases', { name: document.getElementById('mortName')?.value, national_id: d
ocument.getElementById('mortID')?.value, death_datetime: document.getElementById('mortDate')?.value, cause_of_
death: document.getElementById('mortCause')?.value, doctor: document.getElementById('mortDoctor')?.value, next
_of_kin: document.getElementById('mortKin')?.value, contact_phone: document.getElementById('mortPhone')?.value
, storage_location: document.getElementById('mortLoc')?.value, status: 'pending' });
        showToast(tr('Case registered', '?? ????? ?????'));
        renderMortuary(content);
    } catch (e) { showToast(tr('Error', '???'), 'error'); }
};

}

// ===== CME (Continuing Medical Education) =====
let cmeTab = 'activities';
async function renderCME(el) {
    const content = el;

    const events = await API.get('/api/cme/events').catch(() => []);
    const totalHours = events.reduce((s, e) => s + parseFloat(e.cme_hours || 0), 0);
    const upcoming = events.filter(e => e.status === 'upcoming').length;

    content.innerHTML = `
    <h2>${tr('CME - Continuing Medical Education', '??????? ????? ?????')}</h2>
    <div style="display:grid;grid-template-columns:repeat(auto-fit,minmax(160px,1fr));gap:12px;margin-bottom:1
6px">

```

```

<div class="card" style="padding:16px;text-align:center;background:#e3f2fd"><h3 style="margin:0;color:#1565c0">${events.length}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Total Events', '????? ?????????')}</p></div>

<div class="card" style="padding:16px;text-align:center;background:#e8f5e9"><h3 style="margin:0;color:#2e7d32">${totalHours.toFixed(1)}</h3><p style="margin:4px 0 0;font-size:12px">${tr('CME Hours', '????? CME')}</p></div>

<div class="card" style="padding:16px;text-align:center;background:#fff3e0"><h3 style="margin:0;color:#e65100">${upcoming}</h3><p style="margin:4px 0 0;font-size:12px">${tr('Upcoming', '????')}</p></div>
</div>
<div style="display:grid;grid-template-columns:1fr 2fr;gap:16px">
  <div class="card" style="padding:20px">
    <h4 style="margin:0 0 12px">${tr('Add Event', '????? ?????')}</h4>
    <div class="form-group"><label>${tr('Title', '???????')}</label><input class="form-input" id="cmeTitle"></div>
    <div class="form-group"><label>${tr('Speaker', '???????')}</label><input class="form-input" id="cmeSpeaker"></div>
    <div class="form-group"><label>${tr('Date', '???????')}</label><input type="date" class="form-input" id="cmeDate"></div>
    <div class="form-group"><label>${tr('CME Hours', '????? CME')}</label><input type="number" class="form-input" id="cmeHours" min="0.5" step="0.5" value="1"></div>
    <div class="form-group"><label>${tr('Category', '????')}</label>
    <select class="form-input" id="cmeCat"><option value="lecture">${tr('Lecture', '????')}</option><option value="workshop">${tr('Workshop', '????')}</option><option value="conference">${tr('Conference', '????')}</option><option value="online">${tr('Online', '????')}</option></select></div>
    <div class="form-group"><label>${tr('Department', '????')}</label><input class="form-input" id="cmeDept"></div>
    <button class="btn btn-primary w-full" onclick="saveCmeEvent()">? ${tr('Save', '???')}</button>
  </div>
  <div class="card" style="padding:20px">
    <div style="display:flex;justify-content:space-between;margin-bottom:12px">
      <h4 style="margin:0">${tr('Events', '????????')}</h4>
      <button class="btn btn-sm" onclick="exportToCSV(window._cmeData||[],'cme')" style="background:#e0f7fa;color:#00838f">? ${tr('Export', '????')}</button>
    </div>
    <div id="cmeTable"></div>
  </div>
</div>`;

window._cmeData = events;
const et = document.getElementById('cmeTable');
if (et) {
  createTable(et, 'cmeTbl',
    [tr('Title', '???????'), tr('Speaker', '???????'), tr('Date', '???????'), tr('Hours', '????'), tr('Category', '????'), tr('Status', '????')],
    events.map(e => ({ cells: [e.title, e.speaker || '', e.event_date ? new Date(e.event_date).toLocaleDateString('ar-SA') : '', e.cme_hours || 0, e.category || '', statusBadge(e.status)], id: e.id })));
};
}
window.saveCmeEvent = async () => {
  try { await API.post('/api/cme/events', { title: document.getElementById('cmeTitle').value, speaker: document.getElementById('cmeSpeaker').value, event_date: document.getElementById('cmeDate').value, cme_hours: document.getElementById('cmeHours').value, category: document.getElementById('cmeCat').value, department: document.getElementById('cmeDept').value }); showToast(tr('Event added!', '??? ????? ?????????!')); navigateTo(currentPage); } catch (e) { showToast(tr('Error', '???'), 'error'); }
}

```

```

};

}

window.addCME = async function () {
    await API.post('/api/cme/activities', { title: document.getElementById('cmeTitle').value, category: document
        .getElementById('cmeCat').value, credit_hours: document.getElementById('cmeHours').value, activity_date: docum
        ent.getElementById('cmeDate').value, location: document.getElementById('cmeLoc').value });
    showToast(tr('Activity added', '??? ??????')); cmeTab = 'activities'; navigateTo(40);
};

// ===== COSMETIC / PLASTIC SURGERY =====
let cosTab = 'procedures';
async function renderCosmeticSurgery(el) {
    const [procedures, cases, consents, followups, patients] = await Promise.all([
        API.get('/api/cosmetic/procedures').catch(() => []),
        API.get('/api/cosmetic/cases').catch(() => []),
        API.get('/api/cosmetic/consents').catch(() => []),
        API.get('/api/cosmetic/followups').catch(() => []),
        API.get('/api/patients').catch(() => [])
    ]);
    const scheduled = cases.filter(c => c.status === 'Scheduled').length;
    const completed = cases.filter(c => c.status === 'Completed').length;
    const revenue = cases.reduce((s, c) => s + Number(c.total_cost || 0), 0);
    const catIcons = { Face: '?', Body: '?', 'Non-Surgical': '?', Laser: '?', Hair: '?' };
    el.innerHTML = `
        <div class="page-title">? ${tr('Cosmetic & Plastic Surgery', '????? ??????? ??????? ?????????')}</div>
        <div class="stats-grid" style="grid-template-columns:repeat(5,1fr)">
            <div class="stat-card" style="--stat-color:#ec4899"><span class="stat-icon">?</span><div class="stat-lab
            el">${tr('Procedures', '????????')}</div><div class="stat-value">${procedures.length}</div></div>
            <div class="stat-card" style="--stat-color:#f59e0b"><span class="stat-icon">?</span><div class="stat-lab
            el">${tr('Scheduled', '?????')}</div><div class="stat-value">${scheduled}</div></div>
            <div class="stat-card" style="--stat-color:#4ade80"><span class="stat-icon">?</span><div class="stat-lab
            el">${tr('Completed', '?????')}</div><div class="stat-value">${completed}</div></div>
            <div class="stat-card" style="--stat-color:#3b82f6"><span class="stat-icon">?</span><div class="stat-lab
            el">${tr('Consents', '????????')}</div><div class="stat-value">${consents.length}</div></div>
            <div class="stat-card" style="--stat-color:#a78bfa"><span class="stat-icon">?</span><div class="stat-lab
            el">${tr('Revenue', '????????')}</div><div class="stat-value">${revenue.toLocaleString()}</div></div>
        </div>
        <div class="tab-bar">
            <button class="tab-btn ${cosTab === 'procedures' ? 'active' : ''}" onclick="cosTab='procedures';navigate
            To(41)">? ${tr('Procedures', '????????')}</button>
            <button class="tab-btn ${cosTab === 'cases' ? 'active' : ''}" onclick="cosTab='cases';navigateTo(41)">?
            ${tr('Cases', '????????')}</button>
            <button class="tab-btn ${cosTab === 'newcase' ? 'active' : ''}" onclick="cosTab='newcase';navigateTo(41)
            ">? ${tr('New Case', '???? ?????')}</button>
            <button class="tab-btn ${cosTab === 'consents' ? 'active' : ''}" onclick="cosTab='consents';navigateTo(4
            1)">? ${tr('Consents', '????????')}</button>
            <button class="tab-btn ${cosTab === 'newconsent' ? 'active' : ''}" onclick="cosTab='newconsent';navigate
            To(41)">? ${tr('New Consent', '????? ?????')}</button>
            <button class="tab-btn ${cosTab === 'followups' ? 'active' : ''}" onclick="cosTab='followups';navigateTo
            (41)">? ${tr('Follow-ups', '????????')}</button>
        </div>
        <div id="cosContent"></div>`;
    const mc = document.getElementById('cosContent');

```



```

if (cosTab === 'procedures') {
  // Group by category
  const cats = {};
  procedures.forEach(p => { if (!cats[p.category]) cats[p.category] = []; cats[p.category].push(p); });
  mc.innerHTML = Object.entries(cats).map(([cat, procs]) => `
    <div class="card mb-16">
      <div class="card-title">${catIcons[cat] || '?'} ${escapeHTML(cat)}</div>
      <div style="display:grid;grid-template-columns:repeat(3,1fr);gap:12px">
        ${procs.map(p => `
          <div style="border:1px solid var(--border-color,#e5e7eb);border-radius:12px;padding:14px;background
d:var(--card-bg)">
            <div style="font-weight:700;font-size:15px;margin-bottom:6px">${escapeHTML(isArabic ? p.name_ar
: p.name_en)}</div>
            <div style="font-size:12px;color:var(--text-muted);margin-bottom:8px">${escapeHTML(p.description
?.substring(0, 80) || '')}</div>
            <div style="display:grid;grid-template-columns:repeat(2,1fr);gap:4px;font-size:11px">
              <span>?? ${escapeHTML(p.estimated_duration)} ${tr('min', '?')}</span>
              <span>? ${escapeHTML(p.anesthesia_type)}</span>
              <span>? ${Number(p.average_cost).toLocaleString()} SAR</span>
              <span>? ${escapeHTML(p.recovery_days)} ${tr('days', '???')}</span>
            </div>
            <div style="margin-top:8px;font-size:11px;color:#ef4444">?? ${escapeHTML(p.risks?.substring(0, 6
0) || '')}...</div>
          </div>
        `).join('')}
      </div>
    `).join('')}
  </div>
} else if (cosTab === 'cases') {
  mc.innerHTML = `<div class="card">${cases.length ? makeTable(
    [tr('Patient', '??????'), tr('Procedure', '???????'), tr('Surgeon', '??????'), tr('Date', '???????'), tr
('Cost', '???????'), tr('Payment', '?????'), tr('Status', '???????'), tr('Actions', '???????')],
    cases.map(c => ({
      cells: [c.patient_name, c.procedure_name, c.surgeon, c.surgery_date, Number(c.total_cost).toLocaleStri
ng() + ' SAR',
      c.payment_status === 'Paid' ? rawHtml('<span class="badge badge-success">' + tr('Paid', '????') + '</
span>') : rawHtml('<span class="badge badge-danger">' + tr('Pending', '????') + '</span>'),
      statusBadge(c.status),
      c.status === 'Scheduled' ? rawHtml('<button class="btn btn-sm btn-success" onclick="completeCosCase(${
parseInt(c.id, 10)}"> ${tr('Complete', '????')}</button>') : ''
    ]
  })))
  ) : `<div class="empty-state"><span style="font-size:48px">?</span><p>${tr('No cases yet', '?? ???? ?????
???)}</p></div>`</div>`;
} else if (cosTab === 'newcase') {
  mc.innerHTML = `<div class="card"><h3>? ${tr('Schedule New Cosmetic Case', '????? ???? ????? ?????')}</h3>
<div class="form-grid">
  <div><label>${tr('Patient', '??????')}</label><select id="cosPatient" class="form-input">${patients.map(
p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.file_nu
mber)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}</select></div>
  <div><label>${tr('Procedure', '???????')}</label><select id="cosProc" class="form-input" onchange="updat
eCosFields()">${procedures.map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(isArabic ? p.name
_ar : p.name_en)}" data-cost="${escapeHTML(p.average_cost)}" data-anes="${escapeHTML(p.anesthesia_type)}" data

```

```

-dur="${escapeHTML(p.estimated_duration)}">${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')
}</select></div>
    <div><label>${tr('Date', '??????')}</label><input id="cosSurgDate" type="date" class="form-input"></div>
    <div><label>${tr('Time', '????')}</label><input id="cosSurgTime" type="time" class="form-input"></div>
    <div><label>${tr('Anesthesia', '??????')}</label><select id="cosAnes" class="form-input"><option>Local<
/option><option>General</option><option>Sedation</option><option>None</option></select></div>
    <div><label>${tr('Operating Room', '??? ????')}</label><input id="cosOR" class="form-input" placeho
lder="${tr('OR-1', '?.-1')}"></div>
    <div><label>${tr('Cost (SAR)', '????')}</label><input id="cosCost" type="number" class="form-input"><
/div>
    <div style="grid-column:1/-1"><label>${tr('Pre-Op Notes', '????? ? ? ? ?')}</label><textarea id
="cosPreNotes" class="form-input" rows="2"></textarea></div>
</div>
<button class="btn btn-primary" onclick="saveCosCase()" style="margin-top:10px;width:100%;height:44px">? $
{tr('Schedule Case', '???? ????')}</button></div>`;
// Auto-fill fields from selected procedure
setTimeout(() => {
    const sel = document.getElementById('cosProc');
    if (sel && sel.options.length) {
        const opt = sel.options[sel.selectedIndex];
        document.getElementById('cosCost').value = opt.dataset.cost || '';
        document.getElementById('cosAnes').value = opt.dataset.anes || 'Local';
    }
}, 100);
} else if (cosTab === 'consents') {
    mc.innerHTML = `<div class="card">${consents.length ? makeTable(
        [tr('Patient', '????'), tr('Procedure', '????'), tr('Type', '????'), tr('Surgeon', '????'), tr('
Date', '????'), tr('?', '????'), tr('?', '????'), tr('?', '??? ?'), tr('Status', '????'), tr('Actions'
, '????')],
        consents.map(c => ({
            cells: [c.patient_name, c.procedure_name, c.consent_type, c.surgeon, c.consent_date,
                c.is_photography_consent ? '?' : '?', c.is_anesthesia_consent ? '?' : '?', c.is_blood_transfusion_cons
ent ? '?' : '?',
                statusBadge(c.status),
                rawHtml(`<button class="btn btn-sm" onclick="printCosConsent(${parseInt(c.id, 10)})">?? ${tr('Print',
'????')}</button>`)
            ]
        }))
    ) : `<div class="empty-state"><span style="font-size:48px">?</span><p>${tr('No consents', '?? ???? ?')}</p>`</div>`;
} else if (cosTab === 'newconsent') {
    mc.innerHTML = `<div class="card"><h3>?? ${tr('New Consent Form', '???? ???? ?')}</h3>
<div class="form-grid">
    <div><label>${tr('Patient', '????')}</label><select id="conPatient" class="form-input">${patients.map(
p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(p.name_ar || p.name_en)}">${escapeHTML(p.file_nu
mber)} - ${escapeHTML(isArabic ? p.name_ar : p.name_en)}</option>`).join('')}</select></div>
    <div><label>${tr('Procedure', '????')}</label><select id="conProc" class="form-input" onchange="fillC
onsentRisks()">${procedures.map(p => `<option value="${safeId(p.id)}" data-name="${escapeHTML(isArabic ? p.nam
e_ar : p.name_en)}" data-risks="${escapeHTML(p.risks)}" data-desc="${escapeHTML(p.description)}">${escapeHTML(
isArabic ? p.name_ar : p.name_en)}</option>`).join('')}</select></div>
    <div><label>${tr('Consent Type', '??? ?')}</label><select id="conType" class="form-input"><option
value="Surgery">${tr('Surgery Consent', '???? ?')}</option><option value="Non-Surgical">${tr('Non-Surgica
l', '??? ?')}</option><option value="Anesthesia">${tr('Anesthesia', '????')}</option></select></div>

```

```

<div><label>${tr('Witness', '?????')}</label><input id="conWitness" class="form-input"></div>
<div style="grid-column:1/-1"><label>? ${tr('Risks Explained', '?????? ?????')}</label><textarea id="conRisks" class="form-input" rows="3"></textarea></div>
<div style="grid-column:1/-1"><label>? ${tr('Alternatives Explained', '?????? ?????')}</label><textarea id="conAlts" class="form-input" rows="2" placeholder="${tr('Non-surgical options, different techniques...', '?????? ?? ?????? ?????? ??????...')}"></textarea></div>
<div style="grid-column:1/-1"><label>? ${tr('Expected Results', '?????? ?????')}</label><textarea id="conResults" class="form-input" rows="2"></textarea></div>
<div style="grid-column:1/-1"><label>? ${tr('Limitations', '????? ?????')}</label><textarea id="conLimits" class="form-input" rows="2" placeholder="${tr('Results may vary, revision may be needed...', '?????? ?? ????? ?? ????? ?????? ??????...')}"></textarea></div>
</div>
<div style="margin:16px 0;padding:16px;background:var(--hover);border-radius:12px">
<h4 style="margin-bottom:12px">${tr('Additional Consents', '?????? ?????')}</h4>
<div style="display:grid;grid-template-columns:repeat(3,1fr);gap:12px">
<label style="display:flex;align-items:center;gap:8px;cursor:pointer;font-size:14px"><input type="checkbox" id="conPhoto"> ? ${tr('Photography Consent', '?????? ?? ?????')}</label>
<label style="display:flex;align-items:center;gap:8px;cursor:pointer;font-size:14px"><input type="checkbox" id="conAnesC"> ? ${tr('Anesthesia Consent', '?????? ?? ?????')}</label>
<label style="display:flex;align-items:center;gap:8px;cursor:pointer;font-size:14px"><input type="checkbox" id="conBlood"> ? ${tr('Blood Transfusion', '?? ????')}</label>
</div>
</div>
<button class="btn btn-primary" onclick="saveCosConsent()" style="width:100%;height:48px;font-size:16px">? ${tr('Sign & Save Consent', '???? ?? ?????')}</button></div>`;
setTimeout(() => fillConsentRisks(), 100);
} else if (cosTab === 'followups') {
mc.innerHTML = `<div class="card">${followups.length ? makeTable(
[tr('Patient', '?????'), tr('Date', '??????'), tr('Days Post-Op', '???? ?? ?????'), tr('Healing', '??????'), tr('Pain', '?????'), tr('Swelling', '?????'), tr('Satisfaction', '?????'), tr('Next', '??????')],
followups.map(f => ({
cells: [f.patient_name, f.followup_date, f.days_post_op + ' ' + tr('days', '???'),
f.healing_status === 'Excellent' ? '?' + tr('Excellent', '????') : f.healing_status === 'Good' ? '?' + tr('Good', '???') : '?' + tr('Poor', '????'),
rawHtml(`<span style="color:${f.pain_level} >= 7 ? '#ef4444' : f.pain_level >= 4 ? '#f59e0b' : '#22c55e'`)>${escapeHTML(f.pain_level)}/10</span>`),
f.swelling, '?'.repeat(Math.min(f.patient_satisfaction || 0, 5)), f.next_followup || '-'
])
}))
): `<div class="empty-state"><span style="font-size:48px">?</span><p>${tr('No follow-ups', '?? ???? ??????')}</p></div>`</div>`;
}
}

// Cosmetic Surgery Helper Functions
window.updateCosFields = function () {
const sel = document.getElementById('cosProc');
if (!sel) return;
const opt = sel.options[sel.selectedIndex];
document.getElementById('cosCost').value = opt.dataset.cost || '';
document.getElementById('cosAnes').value = opt.dataset.anes || 'Local';
};
window.fillConsentRisks = function () {
const sel = document.getElementById('conProc');

```

```

    if (!sel) return;
    const opt = sel.options[sel.selectedIndex];
    const risksEl = document.getElementById('conRisks');
    if (risksEl) risksEl.value = opt.dataset.risks || '';
};

window.saveCosCase = async function () {
    const patSel = document.getElementById('cosPatient');
    const procSel = document.getElementById('cosProc');
    await API.post('/api/cosmetic/cases', {
        patient_id: patSel.value, patient_name: patSel.options[patSel.selectedIndex].dataset.name,
        procedure_id: procSel.value, procedure_name: procSel.options[procSel.selectedIndex].dataset.name,
        surgery_date: document.getElementById('cosSurgDate').value, surgery_time: document.getElementById('cosSurg
Time').value,
        anesthesia_type: document.getElementById('cosAnes').value, operating_room: document.getElementById('cosOR'
).value,
        total_cost: document.getElementById('cosCost').value, pre_op_notes: document.getElementById('cosPreNotes')
.value
    });
    showToast(tr('Case scheduled!', '??? ??????!!')); cosTab = 'cases'; navigateTo(41);
};

window.completeCosCase = async function (id) {
    await API.put('/api/cosmetic/cases/' + id, { status: 'Completed' });
    showToast(tr('Case completed', '??? ??????!!')); navigateTo(41);
};

window.saveCosConsent = async function () {
    const patSel = document.getElementById('conPatient');
    const procSel = document.getElementById('conProc');
    await API.post('/api/cosmetic/consents', {
        patient_id: patSel.value, patient_name: patSel.options[patSel.selectedIndex].dataset.name,
        procedure_name: procSel.options[procSel.selectedIndex].dataset.name,
        consent_type: document.getElementById('conType').value,
        risks_explained: document.getElementById('conRisks').value,
        alternatives_explained: document.getElementById('conAlts').value,
        expected_results: document.getElementById('conResults').value,
        limitations: document.getElementById('conLimits').value,
        is_photography_consent: document.getElementById('conPhoto').checked,
        is_anesthesia_consent: document.getElementById('conAnesC').checked,
        is_blood_transfusion_consent: document.getElementById('conBlood').checked,
        witness_name: document.getElementById('conWitness').value
    });
    showToast(tr('Consent signed!', '?? ??????!!')); cosTab = 'consents'; navigateTo(41);
};

window.printCosConsent = async function (id) {
    const consents = await API.get('/api/cosmetic/consents');
    const c = consents.find(x => x.id === id);
    if (!c) return;
    const w = window.open('', '_blank', 'width=800,height=1000');
    w.document.write(`<!DOCTYPE html><html dir="rtl"><head><title>????? ?????? - Consent Form</title>
<style>body{font-family:'Segoe UI',Tahoma,sans-serif;padding:30px;color:#333;direction:rtl}
.header{text-align:center;border-bottom:3px double #333;padding:20px 0;margin-bottom:20px}
.header h1{margin:0;font-size:22px;color:#1a365d} .header h2{margin:5px 0;font-size:16px;color:#666}
.section{margin:20px 0;padding:15px;border:1px solid #ddd;border-radius:8px}
.section h3{color:#1a365d;border-bottom:1px solid #eee;padding-bottom:8px;margin-top:0}
.field{margin:10px 0;line-height:1.8} .field label{font-weight:700;color:#555}

```

```

.sig-area{display:grid;grid-template-columns:repeat(3,1fr);gap:20px;margin-top:30px}
.sig-box{text-align:center;border-top:2px solid #333;padding-top:10px}
.checkbox{margin:8px 0;font-size:14px}
@media print{body{padding:20px}}</style></head><body>
<div class="header">
  <h1>????? ?????? ??????? ??? ?????? ???????</h1>
  <h2>Cosmetic Procedure Consent Form</h2>
  <p style="margin:5px 0;color:#888">jumanaMedical - ?????? ??????</p>
</div>
<div class="section">
  <h3>? ?????? ?????? / Patient Information</h3>
  <div class="field"><label>??? ?????? / Patient Name:</label> ${escapeHTML(c.patient_name)}</div>
  <div class="field"><label>??????? / Procedure:</label> ${escapeHTML(c.procedure_name)}</div>
  <div class="field"><label>??????? / Surgeon:</label> ${escapeHTML(c.surgeon)}</div>
  <div class="field"><label>??????? / Date:</label> ${escapeHTML(c.consent_date)} &nbsp; <label>????? / Time
:</label> ${escapeHTML(c.consent_time)}</div>
</div>
<div class="section">
  <h3>?? ??????? ?????????? ?????????? / Risks & Complications</h3>
  <p>${escapeHTML(c.risks_explained || 'N/A')}</p>
</div>
<div class="section">
  <h3>? ??????? ??????? / Available Alternatives</h3>
  <p>${escapeHTML(c.alternatives_explained || 'N/A')}</p>
</div>
<div class="section">
  <h3>? ??????? ?????????? / Expected Results</h3>
  <p>${escapeHTML(c.expected_results || 'N/A')}</p>
</div>
<div class="section">
  <h3>? ?????? ?????????? / Limitations</h3>
  <p>${escapeHTML(c.limitations || 'N/A')}</p>
</div>
<div class="section">
  <h3>? ??????? ??????? / Additional Consents</h3>
  <div class="checkbox">${c.is_photography_consent ? '?' : ''} ??????? ??? ??????? / Photography Consent</
div>
  <div class="checkbox">${c.is_anesthesia_consent ? '?' : ''} ??????? ??? ??????? / Anesthesia Consent</di
v>
  <div class="checkbox">${c.is_blood_transfusion_consent ? '?' : ''} ??????? ??? ??? ??? / Blood Transfus
ion Consent</div>
</div>
<div style="margin:25px 0;padding:15px;background:#f8f9fa;border-radius:8px;font-size:13px">
  <strong>????? / Declaration:</strong><br>
  ??? ??? ?????? ?????? ?????? ?? ?????? ?????? ??????? ??????? ??????? ??? ??? ??????? ?????????? ??????
?? ??????? ??????? ???. ?????? ??? ?????? ??????? ?????? ??????.<br><br>
  <em>I, the undersigned, declare that I have fully understood the nature of the cosmetic procedure describe
d above, and the risks, complications, and alternatives have been explained to me. I consent to the procedure
of my own free will.</em>
</div>
<div class="sig-area">
  <div class="sig-box"><strong>????? ??????<br>Patient Signature</strong></div>
  <div class="sig-box"><strong>????? ??????<br>Surgeon: ${escapeHTML(c.surgeon)}</strong></div>
  <div class="sig-box"><strong>????? ??????<br>Witness: ${escapeHTML(c.witness_name || '')}</strong></div>

```

```

</div>
</body></html>`);
setTimeout(() => { w.print(); }, 500);
};

// =====
// ===== CARDIOLOGY DEPARTMENT MODULE (G01) =====
// =====

window.e1RenderCardiology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.2;min-width:300px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('ECG Records', '????? ????? (ECG)}')</h4>
        <button class="btn btn-sm btn-info mb-12" onclick="e1SimulateECG(${safeId(pid)})">? ${tr('Simulate New
ECG', '????? ????? ??? ?????')}</button>
        <div id="e1EcgList">${tr('Loading...', '???? ???????...')}</div>
      </div>
      <div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
        <h4 style="margin:0 0 12px;color:var(--primary)">%tr%Procedure & Echo Reports%?????? ??????? ???????%
</h4>
        <div class="form-group mb-8">
          <label>${tr('Procedure Type', '??? ???????')}</label>
          <select class="form-input" id="e1CardioProcType">
            <option value="Echocardiography">${tr('Echocardiography', '???? - Echocardiography')}</option>
            <option value="Catheterization">${tr('Catheterization', '????? - Catheterization')}</option>
            <option value="Cardiac Stress Test">${tr('Cardiac Stress Test', '??? ??? - Stress Test')}</option>
          </select>
        </div>
        <div class="form-group mb-8">
          <label>${tr('Findings', '????????')}</label>
          <textarea class="form-input form-textarea" id="e1CardioFindings" rows="3" placeholder="${tr('Describ
e clinical findings...', '???? ?????? ?????? ???????...')}" style="min-height:70px"></textarea>
        </div>
        <div class="form-group mb-8">
          <label>${tr('Recommendations', '????????')}</label>
          <textarea class="form-input form-textarea" id="e1CardioRecs" rows="2" placeholder="${tr('Clinical re
commendations...', '????????? ???????...')}" style="min-height:50px"></textarea>
        </div>
        <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddCardiologyProcedure(${safeId(pid)})"
>? ${tr('Save Report', '??? ???????')}</button>
        <div id="e1CardioProcsList">${tr('Loading...', '???? ???????...')}</div>
      </div>
    </div>
  `;

  window.e1LoadEcgList(pid);
  window.e1LoadProcsList(pid);
};

```

```

window.e1LoadEcgList = async (pid) => {
  const container = document.getElementById('e1EcgList');
  if (!container) return;
  try {
    const ecgs = await API.get('/api/cardiology/ecg/patient/' + pid);
    if (!ecgs.length) {
      container.innerHTML = `

```

window.e1LoadProcsList = async (pid) => {
  const container = document.getElementById('e1CardioProcsList');
  if (!container) return;
  try {
    const procs = await API.get('/api/cardiology/procedures/patient/' + pid);
    if (!procs.length) {
      container.innerHTML = `
```


```

```

window.e1AddCardiologyProcedure = async (pid) => {
  const type = document.getElementById('e1CardioProcType')?.value;
  const findings = document.getElementById('e1CardioFindings')?.value;
  const recs = document.getElementById('e1CardioRecs')?.value;
  if (!findings || !findings.trim()) {
    showToast(tr('Enter findings', '???? ????? ???????'), 'error');
    return;
  }
  try {
    await API.post('/api/cardiology/procedures', {
      patient_id: pid,
      procedure_type: type,
      findings: findings,
      recommendations: recs
    });
    showToast(tr('Report saved!', '?? ??? ??????? ??????!'));
    window.e1LoadProcsList(pid);
    document.getElementById('e1CardioFindings').value = '';
    document.getElementById('e1CardioRecs').value = '';
  } catch (err) {
    showToast(tr('Error saving report', '??? ?? ??? ???????'), 'error');
  }
};

window.e1SimulateECG = async (pid) => {
  try {
    showToast(tr('Simulating ECG recording...', '???? ????? ????? ??????...'));
    const leads = ['I', 'II', 'III', 'aVR', 'aVL', 'aVF', 'V1', 'V2', 'V3', 'V4', 'V5', 'V6'];
    const leads_data = {};
    leads.forEach(l => {
      leads_data[l] = Array.from({ length: 500 }, (_, i) => window.getECGValue(i / 250, l));
    });
    const heart_rate = Math.floor(Math.random() * (95 - 65 + 1)) + 65;
    const interpretation = heart_rate > 85
      ? tr('Sinus tachycardia, otherwise normal', '????? ?????? ?????? ??????? ?????? ?????')
      : tr('Normal sinus rhythm', '????? ??? ?????? (Normal Sinus Rhythm)');
    await API.post('/api/cardiology/ecg', {
      patient_id: pid,
      leads_data,
      heart_rate,
      interpretation
    });
    showToast(tr('ECG recorded successfully!', '?? ??? ?????? ?????? ??????! ?'));
    window.e1LoadEcgList(pid);
  } catch (err) {
    showToast(tr('Error simulating ECG', '??? ?? ??????? ???????'), 'error');
  }
};

window.e1ViewECG = async (id) => {
  try {
    const ecg = await API.get('/api/cardiology/ecg/' + id);
    if (!ecg) return;
  }
};

```



```

const modal = document.createElement('div');
modal.className = 'modal-overlay';
modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.6);z-index:9999;display:flex;align-items:center;justify-content:center';
modal.innerHTML = `
  <div style="background:#fff;border-radius:16px;padding:20px;width:1050px;max-width:95vw;max-height:95vh;overflow:auto;direction:rtl;box-shadow:0 10px 25px rgba(0,0,0,0.2)">
    <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:12px;border-bottom:1px solid #eee;padding-bottom:8px">
      <h3 style="margin:0;color:var(--primary)">? ${tr('12-Lead ECG Waveform Viewer', '???? ?? ????? ??? ? ? ? 12 ??????')}</h3>
      <button class="btn btn-secondary btn-sm" onclick="this.closest('.modal-overlay').remove()">?</button>
    </div>
    <div style="display:flex;gap:12px;margin-bottom:10px;font-size:12px;background:#f3f4f6;padding:8px 12px;border-radius:8px">
      <div><strong>${tr('Heart Rate:', '???? ????? ?????')}</strong> ${ecg.heart_rate} bpm</div>
      <div style="margin-right:15px"><strong>${tr('Interpretation:', '?????? ??????')}</strong> ${ecg.interpretation}</div>
      <div style="margin-right:15px"><strong>${tr('Date/Time:', '?????? ??????')}</strong> ${new Date(ecg.created_at).toLocaleString('ar-SA')}</div>
      <div style="margin-right:auto;color:#ef4444;font-weight:bold">? Calibration: 25 mm/s, 10 mm/mV</div>
    </div>
    <div style="display:flex;justify-content:center;background:#fff;padding:4px;border:1px solid #ddd;border-radius:8px;overflow:auto">
      <canvas id="ecgCanvas" width="1000" height="600" style="background:#fff;display:block"></canvas>
    </div>
  </div>
`;
document.body.appendChild(modal);
setTimeout(() => { window.draw12LeadECG('ecgCanvas', ecg.leads_data); }, 100);
modal.onclick = (e) => { if (e.target === modal) modal.remove(); };
} catch (err) {
  showToast(tr('Error loading ECG details', '?? ? ????? ?????? ??????'), 'error');
}
};

```

```

window.getECGValue = (t, leadName) => {
  const hr = 72;
  const period = 60 / hr;
  const phase = (t % period) / period;
  let val = 0;
  if (phase > 0.06 && phase < 0.14) {
    val += 0.12 * Math.sin(Math.PI * (phase - 0.06) / 0.08);
  }
  if (phase > 0.18 && phase < 0.20) {
    val -= 0.12 * Math.sin(Math.PI * (phase - 0.18) / 0.02);
  }
  if (phase > 0.20 && phase < 0.23) {
    val += 1.1 * Math.sin(Math.PI * (phase - 0.20) / 0.03);
  }
  if (phase > 0.23 && phase < 0.26) {
    val -= 0.22 * Math.sin(Math.PI * (phase - 0.23) / 0.03);
  }
}

```

```

    if (phase > 0.38 && phase < 0.53) {
      val += 0.26 * Math.sin(Math.PI * (phase - 0.38) / 0.15);
    }
    let multiplier = 1.0;
    if (leadName === 'aVR') multiplier = -0.9;
    else if (leadName === 'V1' || leadName === 'V2') multiplier = -0.5;
    else if (leadName === 'V5' || leadName === 'V6') multiplier = 1.1;
    else if (leadName === 'III') multiplier = 0.4;
    else if (leadName === 'aVL') multiplier = 0.3;
    const noise = (Math.random() - 0.5) * 0.015;
    return val * multiplier + noise;
  };

window.draw12LeadECG = (canvasId, leadsData) => {
  const canvas = document.getElementById(canvasId);
  if (!canvas) return;
  const ctx = canvas.getContext('2d');
  const w = canvas.width;
  const h = canvas.height;
  ctx.fillStyle = 'ffffff';
  ctx.fillRect(0, 0, w, h);
  ctx.strokeStyle = '#ffe4e6';
  ctx.lineWidth = 0.5;
  for (let x = 0; x < w; x += 10) {
    ctx.beginPath(); ctx.moveTo(x, 0); ctx.lineTo(x, h); ctx.stroke();
  }
  for (let y = 0; y < h; y += 10) {
    ctx.beginPath(); ctx.moveTo(0, y); ctx.lineTo(w, y); ctx.stroke();
  }
  ctx.strokeStyle = '#fca5a5';
  ctx.lineWidth = 1.0;
  for (let x = 0; x < w; x += 50) {
    ctx.beginPath(); ctx.moveTo(x, 0); ctx.lineTo(x, h); ctx.stroke();
  }
  for (let y = 0; y < h; y += 50) {
    ctx.beginPath(); ctx.moveTo(0, y); ctx.lineTo(w, y); ctx.stroke();
  }
  const colWidth = w / 4;
  const rowHeight = h / 3;
  const leadsLayout = [
    ['I', 'aVR', 'V1', 'V4'],
    ['II', 'aVL', 'V2', 'V5'],
    ['III', 'aVF', 'V3', 'V6']
  ];
  ctx.font = 'bold 12px sans-serif';
  ctx.fillStyle = '#1e293b';
  ctx.strokeStyle = '#cbd5e1';
  ctx.lineWidth = 1.5;
  for (let c = 1; c < 4; c++) {
    ctx.beginPath(); ctx.moveTo(c * colWidth, 0); ctx.lineTo(c * colWidth, h); ctx.stroke();
  }
  for (let r = 1; r < 3; r++) {
    ctx.beginPath(); ctx.moveTo(0, r * rowHeight); ctx.lineTo(w, r * rowHeight); ctx.stroke();
  }

```

```

for (let r = 0; r < 3; r++) {
  for (let c = 0; c < 4; c++) {
    const leadName = leadsLayout[r][c];
    const xOffset = c * colWidth;
    const yOffset = r * rowHeight;
    ctx.fillStyle = '#1e293b';
    ctx.fillText(leadName, xOffset + 10, yOffset + 20);
    ctx.strokeStyle = '#dc2626';
    ctx.lineWidth = 1.5;
    ctx.beginPath();
    const centerY = yOffset + rowHeight / 2;
    for (let x = 0; x < colWidth; x++) {
      const t = (xOffset + x) / 250;
      const val = window.getECGValue(t, leadName);
      const y = centerY - (val * 100);
      if (x === 0) {
        ctx.moveTo(xOffset + x, y);
      } else {
        ctx.lineTo(xOffset + x, y);
      }
    }
    ctx.stroke();
  }
}
};

// =====
// ===== GASTROENTEROLOGY DEPARTMENT MODULE (G02) =====
// =====

window.e1RenderGastro = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
<div style="display:flex;gap:16px;flex-wrap:wrap">
  <div style="flex:1.2;min-width:300px">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Endoscopy Reports', '????? ???????')}</h4>
    <div class="form-group mb-8">
      <label>${tr('Endoscopy Type', '??? ???????')}</label>
      <select class="form-input" id="e1GastroEndoType">
        <option value="Gastrosocopy">${tr('Gastrosocopy', '????? ???? - Gastrosocopy')}</option>
        <option value="Colonoscopy">${tr('Colonoscopy', '????? ????? - Colonoscopy')}</option>
        <option value="ERCP">${tr('ERCP', '????? ??????? ??????? - ERCP')}</option>
      </select>
    </div>
    <div class="form-group mb-8">
      <label>${tr('Indications', '????? ?????')}</label>
      <input class="form-input" id="e1GastroEndoIndications" placeholder="${tr('e.g. Dyspepsia, bleeding..', '?????: ??? ???? ????...')}">
    </div>
    <div class="form-group mb-8">
      <label>${tr('Findings', '??????')}</label>
      <textarea class="form-input form-textarea" id="e1GastroEndoFindings" rows="3" placeholder="${tr('Des

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cribe endoscopy findings...', '???? ????? ??? ???????...')}" style="min-height:70px"></textarea>
</div>
<div class="form-group mb-8">
  <label>${tr('Complications', '????????')}</label>
  <input class="form-input" id="e1GastroEndoComplications" placeholder="${tr('None, or specify...', '?
? ????? ?? ???...')}">
</div>
<div class="form-group mb-8">
  <label>${tr('Recommendations', '????????')}</label>
  <textarea class="form-input form-textarea" id="e1GastroEndoRecs" rows="2" placeholder="${tr('Clinica
l recommendations...', '???????? ???????...')}" style="min-height:50px"></textarea>
</div>
<button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddEndoscopyReport(${safeId(pid)})">? $
${tr('Save Endoscopy Report', '??? ????? ???????')}</button>
<div id="e1GastroEndoList">${tr('Loading...', '???? ???????...')}</div>
</div>
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">

  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Biopsy Tracking', '???? ????? ???????')}</h4>
  <div class="form-group mb-8">
    <label>${tr('Specimen Source', '???? ??????')}</label>
    <input class="form-input" id="e1GastroBiopsySource" placeholder="${tr('e.g. Gastric Antrum, Duodenum
...', '?????: ??? ???? ??? ??? ??? ???...')}">
  </div>
  <div class="form-group mb-8">
    <label>${tr('Clinical Notes', '?????? ??????')}</label>
    <textarea class="form-input form-textarea" id="e1GastroBiopsyNotes" rows="2" placeholder="${tr('Clin
ical context...', '????? ??????? ???????...')}" style="min-height:50px"></textarea>
  </div>
  <button class="btn btn-warning btn-sm mb-12 w-full" onclick="e1AddBiopsyRequest(${safeId(pid)})">? ${t
r('Request Biopsy', '??? ????? ???')}</button>
  <div id="e1GastroBiopsiesList">${tr('Loading...', '???? ???????...')}</div>
</div>
</div>
`;

window.e1LoadEndoscopiesList(pid);
window.e1LoadBiopsiesList(pid);
};

window.e1LoadEndoscopiesList = async (pid) => {
  const container = document.getElementById('e1GastroEndoList');
  if (!container) return;
  try {
    const reports = await API.get('/api/gastro/endoscopy/patient/' + pid);
    if (!reports.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No endoscopy reports foun
d', '?? ??? ???? ???? ????')}</div>`;
      return;
    }
  }
  container.innerHTML = reports.map(r => `
    <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px
solid #8b5cf6;font-size:12px">
      <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">

```

```

        <span>? ${escapeHTML(r.endoscopy_type)}</span>
        <span style="font-weight:400;color:var(--text-dim)">${new Date(r.created_at).toLocaleDateString('ar-
SA')}</span>
    </div>
    <div style="margin-top:4px"><strong>${tr('Indications:', '???????')}</strong> ${escapeHTML(r.indicati
ons || '-')}</div>
    <div style="margin-top:2px"><strong>${tr('Findings:', '???????')}</strong> ${escapeHTML(r.findings)}<
/div>
    ${r.complications ? `<div style="margin-top:2px;color:#dc2626"><strong>${tr('Complications:', '???????
??')}</strong> ${escapeHTML(r.complications)}</div>` : ''}
    ${r.recommendations ? `<div style="margin-top:2px"><strong>${tr('Recs:', '???????')}</strong> ${esca
peHTML(r.recommendations)}</div>` : ''}
    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name || ''
)}</div>
</div>
`).join('');
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ?????')}</div>`;
}
};

window.e1loadBiopsiesList = async (pid) => {
    const container = document.getElementById('e1GastroBiopsiesList');
    if (!container) return;
    try {
        const biopsies = await API.get('/api/gastro/biopsy/patient/' + pid);
        if (!biopsies.length) {
            container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No biopsy samples found',
'?? ???? ???? ???? ????')}</div>`;
            return;
        }
        container.innerHTML = biopsies.map(b => {
            const isPending = b.status === 'Pending';
            const statusBadge = isPending
                ? `<span class="badge badge-warning">? ${tr('Pending', '????????')}</span>`
                : `<span class="badge badge-success">? ${tr('Resulted', '????')}</span>`;

            const simulateBtn = isPending
                ? `<button class="btn btn-sm btn-secondary mt-8" onclick="e1SimulateBiopsyResult(${safeId(b.id)}, ${sa
feId(pid)})">? ${tr('Simulate Result', '???? ???? ????')}</button>`
                : '';

            return `
                <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4
px solid ${isPending ? '#f59e0b' : '#10b981'};font-size:12px">
                    <div style="display:flex;justify-content:space-between;align-items:center">
                        <strong>? ${escapeHTML(b.specimen_source)}</strong>
                        ${statusBadge}
                    </div>
                    ${b.clinical_notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '????
???')}</strong> ${escapeHTML(b.clinical_notes)}</div>` : ''}
                    ${b.result_findings ? `<div style="margin-top:6px;padding:6px;background:#f0fdf4;border:1px solid #b
bf7d0;border-radius:6px;color:#16a34a"><strong>${tr('Pathology Result:', '???? ?????????')}</strong> ${esca
peHTML(b.result_findings)}</div>` : ''}
            `;
        });
    } catch (err) {
        container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ?????')}</div>`;
    }
};

```

```

        <div style="font-size:10px;color:var(--text-dim);margin-top:4px">?? ${new Date(b.created_at).toLocaleString('ar-SA')}}</div>
        ${simulateBtn}
    </div>
    `;
    }).join('');
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
}
};

window.e1AddEndoscopyReport = async (pid) => {
    const type = document.getElementById('e1GastroEndoType')?.value;
    const indications = document.getElementById('e1GastroEndoIndications')?.value;
    const findings = document.getElementById('e1GastroEndoFindings')?.value;
    const complications = document.getElementById('e1GastroEndoComplications')?.value;
    const recs = document.getElementById('e1GastroEndoRecs')?.value;

    if (!findings || !findings.trim()) {
        showToast(tr('Enter findings', '???? ????? ???????'), 'error');
        return;
    }
    try {
        await API.post('/api/gastro/endoscopy', {
            patient_id: pid,
            endoscopy_type: type,
            indications: indications,
            findings: findings,
            complications: complications,
            recommendations: recs
        });
        showToast(tr('Endoscopy report saved!', '?? ??? ?????? ??????? ?????!'));
        window.e1LoadEndoscopiesList(pid);
        document.getElementById('e1GastroEndoIndications').value = '';
        document.getElementById('e1GastroEndoFindings').value = '';
        document.getElementById('e1GastroEndoComplications').value = '';
        document.getElementById('e1GastroEndoRecs').value = '';
    } catch (err) {
        showToast(tr('Error saving report', '??? ?? ??? ?????? ???????'), 'error');
    }
};

window.e1AddBiopsyRequest = async (pid) => {
    const source = document.getElementById('e1GastroBiopsySource')?.value;
    const notes = document.getElementById('e1GastroBiopsyNotes')?.value;

    if (!source || !source.trim()) {
        showToast(tr('Enter specimen source', '???? ?????? ??? ????'), 'error');
        return;
    }
    try {
        await API.post('/api/gastro/biopsy', {
            patient_id: pid,
            specimen_source: source,

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        clinical_notes: notes
    });
    showToast(tr('Biopsy requested successfully!', '?? ??? ?????? ??????!'));
    window.e1LoadBiopsiesList(pid);
    document.getElementById('e1GastroBiopsySource').value = '';
    document.getElementById('e1GastroBiopsyNotes').value = '';
} catch (err) {
    showToast(tr('Error requesting biopsy', '??? ?? ??? ??????'), 'error');
}
};

window.e1SimulateBiopsyResult = async (id, pid) => {
    try {
        const result_findings = tr(
            'Chronic superficial gastritis. No dysplasia or malignancy detected. Helicobacter pylori: Positive (+).'
        ,
            '?????? ???? ???? ?? ??????. ?? ????? ?????? ???????? ?? ??? ??????. ?????? ??????: ?????? (+).'
        );
        await API.put(`/api/gastro/biopsy/${id}/result`, { result_findings });
        showToast(tr('Biopsy result recorded!', '?? ?????? ?????? ?????? ?????? ?!'));
        window.e1LoadBiopsiesList(pid);
    } catch (err) {
        showToast(tr('Error recording result', '??? ?? ?????? ????????'), 'error');
    }
};

// =====
// ===== ENDOCRINOLOGY & DIABETES MODULE (G03) =====
// =====

window.e1RenderEndocrine = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <div style="flex:1.2;min-width:300px">
                <h4 style="margin:0 0 12px;color:var(--primary)"> ${tr('Blood Glucose Tracker', '?????? ???? ???? ?
                ? ???')}</h4>
                <div class="flex gap-8 mb-8">
                    <div class="form-group" style="flex:1">
                        <label>${tr('Glucose Value (mg/dL)', '????? ?????? (mg/dL)')}</label>
                        <input class="form-input" type="number" id="e1EndoGlucoseVal" placeholder="e.g. 120" min="20" max=
                    "600">
                    </div>
                    <div class="form-group" style="flex:1">
                        <label>${tr('Measurement Type', '??? ??????')}</label>
                        <select class="form-input" id="e1EndoGlucoseType">
                            <option value="Fasting">${tr('Fasting', '???? - Fasting')}</option>
                            <option value="Postprandial">${tr('Postprandial', '??? ????? - Postprandial')}</option>
                            <option value="Random">${tr('Random', '?????? - Random')}</option>
                        </select>
                    </div>
                </div>
            </div>
        </div>
    `

```

```

<div class="form-group mb-8">
  <label>${tr('Notes', '???????')}</label>
  <input class="form-input" id="e1EndoGlucoseNotes" placeholder="${tr('e.g. Before lunch, feeling dizz
y...', '?????: ??? ?????? ??? ??????...')}">
</div>
  <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddGlucoseLog(${safeId(pid)})">? ${tr('
Save Reading', '??? ??????')}</button>
  <div id="e1EndoGlucoseList">${tr('Loading...', '??? ??????...')}</div>
</div>
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Active Insulin Regimens', '???? ????????? ??
????')}</h4>
  <div class="form-group mb-8">
    <label>${tr('Insulin Type', '??? ?????????')}</label>
    <select class="form-input" id="e1EndoInsulinType">
      <option value="Rapid-acting (NovoRapid)">Rapid-acting (NovoRapid)</option>
      <option value="Short-acting (Actrapid)">Short-acting (Actrapid)</option>
      <option value="Intermediate-acting (Insulatard)">Intermediate-acting (Insulatard)</option>
      <option value="Long-acting (Lantus)">Long-acting (Lantus)</option>
      <option value="Premixed (Mixtard 30)">Premixed (Mixtard 30)</option>
    </select>
  </div>
  <div class="form-group mb-8">
    <label>${tr('Dosage / Instructions', '????? ?????????')}</label>
    <input class="form-input" id="e1EndoInsulinDosage" placeholder="${tr('e.g. 12 units before breakfast
', '?????: 12 ??? ???? ??????')}">
  </div>
  <button class="btn btn-warning btn-sm mb-12 w-full" onclick="e1AddInsulinRegimen(${safeId(pid)})">? ${
tr('Prescribe Insulin', '??? ???? ??????')}</button>
  <div id="e1EndoInsulinList">${tr('Loading...', '??? ??????...')}</div>
</div>
</div>
`;

window.e1LoadGlucoseList(pid);
window.e1LoadInsulinList(pid);
};

window.e1LoadGlucoseList = async (pid) => {
  const container = document.getElementById('e1EndoGlucoseList');
  if (!container) return;
  try {
    const logs = await API.get('/api/endocrine/glucose/patient/' + pid);
    if (!logs.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No glucose logs found', '
?? ???? ?????? ??? ??????')}</div>`;
      return;
    }
  }
  container.innerHTML = `
  <div style="max-height: 250px; overflow-y: auto; padding-right: 4px">
    ${logs.map(l => {
      let color = '#10b981';
      const val = parseFloat(l.glucose_value);

```



```

    if (l.log_type === 'Fasting' && (val < 70 || val > 110)) color = '#ef4444';
    else if (l.log_type !== 'Fasting' && (val < 70 || val > 140)) color = '#ef4444';
    return `
      <div style="padding:8px;margin:4px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid ${color};font-size:12px;display:flex;justify-content:space-between;align-items:center">
        <div>
          <strong style="color:${color};font-size:13px">${val} mg/dL</strong>
          <span class="badge" style="margin:0 4px">${escapeHTML(l.log_type)}</span>
          ${l.notes ? `<span style="color:var(--text-dim)">${escapeHTML(l.notes)}</span>` : ''}
        </div>
        <div style="font-size:10px;color:var(--text-dim)">${new Date(l.created_at).toLocaleTimeString('ar-SA')}</div>
      </div>
    `;
  }).join('')
</div>
`;
} catch (err) {
  container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
}
};

window.e1LoadInsulinList = async (pid) => {
  const container = document.getElementById('e1EndoInsulinList');
  if (!container) return;
  try {
    const regimens = await API.get('/api/endocrine/insulin/patient/' + pid);
    if (!regimens.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No active insulin regimens found', '?? ???? ???? ???? ????')}</div>`;
      return;
    }
    container.innerHTML = regimens.map(r => `
      <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid #f59e0b;font-size:12px;display:flex;justify-content:space-between;align-items:center">
        <div>
          <div style="font-weight:700;color:var(--primary)">? ${escapeHTML(r.insulin_type)}</div>
          <div style="margin-top:2px"><strong>${tr('Dose:', '?????')}</strong> ${escapeHTML(r.dosage)}</div>
          <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name || '')}</div>
        </div>
        <button class="btn btn-sm btn-secondary" onclick="e1DeactivateInsulin(${safeId(r.id)}, ${safeId(pid)})">? ${tr('Stop', '????')}</button>
      </div>
    `).join('');
  } catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
  }
};

window.e1AddGlucoseLog = async (pid) => {
  const val = document.getElementById('e1EndoGlucoseVal')?.value;
  const type = document.getElementById('e1EndoGlucoseType')?.value;
  const notes = document.getElementById('e1EndoGlucoseNotes')?.value;

```

```

if (!val || !val.trim()) {
  showToast(tr('Enter glucose value', '???? ????? ????? ?????'), 'error');
  return;
}
try {
  await API.post('/api/endocrine/glucose', {
    patient_id: pid,
    glucose_value: parseFloat(val),
    log_type: type,
    notes: notes
  });
  showToast(tr('Glucose log saved!', '?? ??? ????? ????? ?????!'));
  window.e1LoadGlucoseList(pid);
  document.getElementById('e1EndoGlucoseVal').value = '';
  document.getElementById('e1EndoGlucoseNotes').value = '';
} catch (err) {
  showToast(tr('Error saving reading', '??? ?? ??? ???????'), 'error');
}
};

window.e1AddInsulinRegimen = async (pid) => {
  const type = document.getElementById('e1EndoInsulinType')?.value;
  const dosage = document.getElementById('e1EndoInsulinDosage')?.value;
  if (!dosage || !dosage.trim()) {
    showToast(tr('Enter dosage instructions', '???? ????? ?????????????'), 'error');
    return;
  }
  try {
    await API.post('/api/endocrine/insulin', {
      patient_id: pid,
      insulin_type: type,
      dosage: dosage
    });
    showToast(tr('Insulin regimen prescribed!', '?? ??? ??? ????????? ?????!'));
    window.e1LoadInsulinList(pid);
    document.getElementById('e1EndoInsulinDosage').value = '';
  } catch (err) {
    showToast(tr('Error prescribing insulin', '??? ?? ??? ??????'), 'error');
  }
};

window.e1DeactivateInsulin = async (id, pid) => {
  if (!confirm(tr('Are you sure you want to stop this insulin regimen?', '?? ??? ????? ?? ????? ??? ?????????
  ????'))) return;
  try {
    await API.put(`/api/endocrine/insulin/${id}/deactivate`, {});
    showToast(tr('Insulin regimen stopped!', '?? ????? ??? ?????????!'));
    window.e1LoadInsulinList(pid);
  } catch (err) {
    showToast(tr('Error stopping regimen', '??? ?? ????? ??????'), 'error');
  }
};

// =====

```

```
// ===== NEPHROLOGY & DIALYSIS MODULE (G04) =====
// =====

window.e1RenderNephrology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.2;min-width:300px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Dialysis Session Recorder', '????? ?????? ????'
?????')}</h4>
        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Pre-Weight (kg)', '????? ??? ?????? (???)')}</label>
            <input class="form-input" type="number" step="0.1" id="e1NephroWeightPre" placeholder="e.g. 74.5">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Post-Weight (kg)', '????? ??? ?????? (???)')}</label>
            <input class="form-input" type="number" step="0.1" id="e1NephroWeightPost" placeholder="e.g. 71.2"
>
          </div>
        </div>
        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Blood Flow Rate (mL/min)', '????? ?????? (??/?????)')}</label>
            <input class="form-input" type="number" id="e1NephroBloodFlow" placeholder="e.g. 300" min="50" max
="500">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('UF Volume (Liters)', '??? ?????? ????????? (???)')}</label>
            <input class="form-input" type="number" step="0.1" id="e1NephroUFVolume" placeholder="e.g. 3.2">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Duration (Hours)', '??? ?????? (?????)')}</label>
            <input class="form-input" type="number" step="0.5" id="e1NephroDuration" placeholder="e.g. 4.0" mi
n="1" max="8">
          </div>
        </div>
        <div class="form-group mb-8">
          <label>${tr('Clinical Notes', '???????? ??????')}</label>
          <textarea class="form-input form-textarea" id="e1NephroNotes" rows="2" placeholder="${tr('Describe s
ession details...', '???? ?????? ??????...')}" style="min-height:50px"></textarea>
        </div>
        <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddDialysisSession(${safeId(pid)})">? $
${tr('Save Dialysis Record', '??? ??? ??????')}</button>
      </div>
      <div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Session History', '??? ?????? ??????')}</h4>
        <div id="e1NephroDialysisList">${tr('Loading...', '???? ??????...')}</div>
      </div>
    </div>
  `
}

```

```

`;

window.e1LoadDialysisList(pid);
};

window.e1LoadDialysisList = async (pid) => {
  const container = document.getElementById('e1NephroDialysisList');
  if (!container) return;
  try {
    const sessions = await API.get('/api/nephrology/dialysis/patient/' + pid);
    if (!sessions.length) {
      container.innerHTML = `
```

```

        ultrafiltration_volume: uf_volume ? parseFloat(uf_volume) : null,
        duration_hours: duration ? parseFloat(duration) : null,
        notes: notes
    });
    showToast(tr('Dialysis session record saved!', '?? ??? ??? ????? ?????? ??????!'));
    window.e1LoadDialysisList(pid);
    document.getElementById('e1NephroWeightPre').value = '';
    document.getElementById('e1NephroWeightPost').value = '';
    document.getElementById('e1NephroBloodFlow').value = '';
    document.getElementById('e1NephroUFVolume').value = '';
    document.getElementById('e1NephroDuration').value = '';
    document.getElementById('e1NephroNotes').value = '';
} catch (err) {
    showToast(tr('Error saving dialysis record', '??? ?? ??? ??? ??????'), 'error');
}
};

// =====
// ===== PULMONOLOGY MODULE (G05) =====
// =====

window.e1RenderPulmonology = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <div style="flex:1.2;min-width:300px">
                <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Pulmonary Function Test (PFT) Entry', '????
                ??? ????? ????? (PFT)')}</h4>
                <div class="flex gap-8 mb-8">
                    <div class="form-group" style="flex:1">
                        <label>${tr('FEV1 (L)', '??? ?????? ?????? FEV1 (???)')}</label>
                        <input class="form-input" type="number" step="0.01" id="e1PulmoFEV1" placeholder="e.g. 3.2" oninput
                    t="e1PulmoCalcRatio()">
                    </div>
                    <div class="form-group" style="flex:1">
                        <label>${tr('FVC (L)', '????? ?????? ?????? FVC (???)')}</label>
                        <input class="form-input" type="number" step="0.01" id="e1PulmoFVC" placeholder="e.g. 4.0" oninput
                    ="e1PulmoCalcRatio()">
                    </div>
                    <div class="form-group" style="flex:1.2">
                        <label>${tr('FEV1/FVC Ratio (%)', '???? FEV1/FVC (%)')}</label>
                        <input class="form-input" type="number" step="0.1" id="e1PulmoRatio" placeholder="Auto-calculated"
                    readonly style="background:var(--hover,#f8f9fa)">
                    </div>
                </div>
                <div class="flex gap-8 mb-8">
                    <div class="form-group" style="flex:1">
                        <label>${tr('PEF (L/min)', '???? ?????? ?????? PEF (???/?????)')}</label>
                        <input class="form-input" type="number" step="1" id="e1PulmoPEF" placeholder="e.g. 450">
                    </div>
                    <div class="form-group" style="flex:1.5">
                        <label>${tr('Interpretation', '??????? ??????')}</label>

```

```

        <select class="form-input" id="e1PulmoInterpretation">
            <option value="Normal">${tr('Normal', '???? - Normal')}

```

```

    else if (r.interpretation === 'Restrictive') badgeColor = '#3b82f6';
    else if (r.interpretation === 'Mixed') badgeColor = '#ef4444';

    return `
      <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4
px solid ${badgeColor};font-size:12px">
        <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between;align-it
ems:center">
          <span>? ${new Date(r.test_date).toLocaleDateString('ar-SA')}</span>
          <span class="badge" style="background:${badgeColor};color:#fff">${escapeHTML(r.interpretation)}</s
pan>
        </div>
        <div style="margin-top:4px;display:grid;grid-template-columns:1fr 1fr;gap:4px">
          <div><strong>${tr('FEV1:', '??? ?????? FEV1:')}</strong> ${r.fev1 || '-'} L</div>
          <div><strong>${tr('FVC:', '????? ???????? FVC:')}</strong> ${r.fvc || '-'} L</div>
          <div><strong>${tr('FEV1/FVC:', '???????')}</strong> ${r.fev1_fvc_ratio || '-'}%</div>
          <div><strong>${tr('PEF:', '????? ????????')}</strong> ${r.pef || '-'} L/m</div>
        </div>
        ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '???????')}</
strong> ${escapeHTML(r.notes)}</div>` : ''}
        <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
      </div>
    `;
  }).join('');
} catch (err) {
  container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
}
};

window.e1AddPFT = async (pid) => {
  const fev1 = document.getElementById('e1PulmoFEV1')?.value;
  const fvc = document.getElementById('e1PulmoFVC')?.value;
  const ratio = document.getElementById('e1PulmoRatio')?.value;
  const pef = document.getElementById('e1PulmoPEF')?.value;
  const interpretation = document.getElementById('e1PulmoInterpretation')?.value;
  const notes = document.getElementById('e1PulmoNotes')?.value;
  if (!fev1 || !fvc) {
    showToast(tr('Enter FEV1 and FVC values', '????? ?????? ??? FEV1 ? FVC'), 'error');
    return;
  }
  try {
    await API.post('/api/pulmonology/pft', {
      patient_id: pid,
      fev1: parseFloat(fev1),
      fvc: parseFloat(fvc),
      fev1_fvc_ratio: ratio ? parseFloat(ratio) : null,
      pef: pef ? parseFloat(pef) : null,
      interpretation: interpretation,
      notes: notes
    });
    showToast(tr('PFT record saved successfully!', '?? ??? ??? ?????? ?????? ??????!'));
    window.e1LoadPFTList(pid);
    document.getElementById('e1PulmoFEV1').value = '';
  }
}

```

```

document.getElementById('e1PulmoFVC').value = '';
document.getElementById('e1PulmoRatio').value = '';
document.getElementById('e1PulmoPEF').value = '';
document.getElementById('e1PulmoNotes').value = '';
} catch (err) {
  showToast(tr('Error saving PFT record', '??? ?? ??? ??? ????? ?????'), 'error');
}
};

// =====
// ===== RHEUMATOLOGY MODULE (G06) =====
// =====

window.e1RenderRheumatology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.2;min-width:300px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Joint Assessment & DAS28 Calculator', '????
        ?????? ?????? DAS28')}</h4>
        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Tender Joints (0-28)', '??? ?????? ?????? (0-28)')}</label>
            <input class="form-input" type="number" id="e1RheumTJC" min="0" max="28" placeholder="TJC28" oninp
ut="e1RheumCalcDAS28()">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Swollen Joints (0-28)', '??? ?????? ?????? (0-28)')}</label>
            <input class="form-input" type="number" id="e1RheumSJC" min="0" max="28" placeholder="SJC28" oninp
ut="e1RheumCalcDAS28()">
          </div>
        </div>
        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Patient Global Health VAS (0-100)', '????? ?????? VAS ?? ?????? (0-100)')}</label>
            <input class="form-input" type="number" id="e1RheumVAS" min="0" max="100" placeholder="Pain VAS" o
ninput="e1RheumCalcDAS28()">
          </div>
          <div class="form-group" style="flex:1.2">
            <label>${tr('Calculated DAS28 Score', '????? ?????? ?????? ?????? DAS28')}</label>
            <input class="form-input" type="text" id="e1RheumDAS28" placeholder="Auto-calculated" readonly sty
le="background:var(--hover,#f8f9fa);font-weight:700">
          </div>
        </div>
        <div class="form-group mb-8">
          <label>${tr('Clinical Notes', '??????? ??????')}</label>
          <textarea class="form-input form-textarea" id="e1RheumNotes" rows="2" placeholder="${tr('Describe jo
int involvement, morning stiffness...', '???? ?????? ?????? ?????? ?????? ?????? ??????...')}" style="min-height:50
px"></textarea>
        </div>
        <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddJointAssessment(${safeId(pid)})">? ${
tr('Save Joint Assessment', '??? ?????? ??????')}</button>

```



```

    </div>
    <div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Assessment History', '??? ??????? ???????')}
    </h4>
        <div id="e1RheumJointList">${tr('Loading...', '???? ???????...')}</div>
    </div>
</div>
`;

window.e1LoadJointList(pid);
};

window.e1RheumCalcDAS28 = () => {
    const tjc = parseInt(document.getElementById('e1RheumTJC')?.value);
    const sjc = parseInt(document.getElementById('e1RheumSJC')?.value);
    const vas = parseInt(document.getElementById('e1RheumVAS')?.value);
    const scoreField = document.getElementById('e1RheumDAS28');

    if (scoreField) {
        if (!isNaN(tjc) && !isNaN(sjc) && !isNaN(vas)) {
            const score = (0.56 * Math.sqrt(tjc) + 0.28 * Math.sqrt(sjc) + 0.014 * vas);
            let status = '';
            if (score < 2.6) status = ` (${tr('Remission', '???? ??????')})`;
            else if (score < 3.2) status = ` (${tr('Low Activity', '???? ??????')})`;
            else if (score <= 5.1) status = ` (${tr('Moderate Activity', '???? ??????')})`;
            else status = ` (${tr('High Activity', '???? ??????')})`;

            scoreField.value = score.toFixed(2) + status;
        } else {
            scoreField.value = '';
        }
    }
};

window.e1LoadJointList = async (pid) => {
    const container = document.getElementById('e1RheumJointList');
    if (!container) return;
    try {
        const records = await API.get('/api/rheumatology/joints/patient/' + pid);
        if (!records.length) {
            container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No joint assessments found', '?? ???? ??????? ?????? ??????')}</div>`;
            return;
        }
        container.innerHTML = records.map(r => {
            const score = parseFloat(r.das28_score || 0);
            let border = '4px solid #10b981';
            if (score >= 5.1) border = '4px solid #ef4444';
            else if (score > 3.2) border = '4px solid #f59e0b';

            return `
                <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:
                ${border};font-size:12px">

```

```

<div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">
  <span>? ${new Date(r.assessment_date).toLocaleDateString('ar-SA')}</span>
  <span style="font-weight:700;color:var(--accent,#0ea5e9)">DAS28: ${score.toFixed(2)}</span>
</div>
<div style="margin-top:4px;display:grid;grid-template-columns:1fr 1fr;gap:4px">
  <div><strong>${tr('Tender (TJC):', '???????')}</strong> ${r.tender_joint_count}</div>
  <div><strong>${tr('Swollen (SJC):', '???????')}</strong> ${r.swollen_joint_count}</div>
  <div><strong>${tr('Pain VAS:', '????? ?????')}</strong> ${r.vas_pain || '-'}</div>
</div>
${r.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '???????')}</strong> ${escapeHTML(r.notes)}</div>` : ''}
  <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name || '')}</div>
</div>
`;
}).join('');
} catch (err) {
  container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ?????')}</div>`;
}
};

window.e1AddJointAssessment = async (pid) => {
  const tjc = document.getElementById('e1RheumTJC')?.value;
  const sjc = document.getElementById('e1RheumSJC')?.value;
  const vas = document.getElementById('e1RheumVAS')?.value;
  const notes = document.getElementById('e1RheumNotes')?.value;
  if (!tjc || !sjc || !vas) {
    showToast(tr('Enter TJC, SJC and VAS values', '???? ????? ??? ?????? ?????? ?????? ?????? VAS'), 'error');
    return;
  }
  const score = (0.56 * Math.sqrt(parseInt(tjc)) + 0.28 * Math.sqrt(parseInt(sjc)) + 0.014 * parseInt(vas));
  try {
    const resp = await API.post('/api/rheumatology/joints', {
      patient_id: pid,
      tender_joint_count: parseInt(tjc),
      swollen_joint_count: parseInt(sjc),
      vas_pain: parseInt(vas),
      das28_score: parseFloat(score.toFixed(2)),
      notes: notes
    });
    if (resp && resp.error) {
      showToast(tr('Rejected: ', '?????: ') + resp.error, 'error');
      return;
    }
    showToast(tr('Joint assessment saved successfully!', '?? ??? ????? ?????? ?????!'));
    window.e1LoadJointList(pid);
    document.getElementById('e1RheumTJC').value = '';
    document.getElementById('e1RheumSJC').value = '';
    document.getElementById('e1RheumVAS').value = '';
    document.getElementById('e1RheumDAS28').value = '';
    document.getElementById('e1RheumNotes').value = '';
  } catch (err) {
    showToast(tr('Error saving assessment', '??? ?? ??? ??????'), 'error');
  }
}

```

```

    }
};

// =====
// ===== NEUROLOGY MODULE (G07) =====
// =====

window.e1RenderNeurology = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <div style="flex:1.2;min-width:300px">
                <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Neurological Assessment & GCS Calculator', '?
                ?????? ?????? ?????? ?????? ?????? ?????????')}</h4>

                <!-- GCS Section -->
                <h5 style="margin:0 0 8px;color:var(--accent,#0ea5e9)">? ${tr('Glasgow Coma Scale (GCS)', '????? ?????
                ? ????????? (GCS)')}</h5>
                <div class="flex gap-8 mb-8">
                    <div class="form-group" style="flex:1">
                        <label>${tr('Eye Opening (E: 1-4)', '??????? ????? (E: 1-4)')}</label>
                        <select class="form-input" id="e1NeuroGCSEye" onchange="e1NeuroCalcGCS()">
                            <option value="">-- E --</option>
                            <option value="4">4 - ${tr('Spontaneous', '?????')}</option>
                            <option value="3">3 - ${tr('To sound / speech', '????? / ?????')}</option>
                            <option value="2">2 - ${tr('To pressure / pain', '????? / ?????')}</option>
                            <option value="1">1 - ${tr('None', '?? ???? ?????')}</option>
                        </select>
                    </div>
                    <div class="form-group" style="flex:1">
                        <label>${tr('Verbal Response (V: 1-5)', '????????? ???????? (V: 1-5)')}</label>
                        <select class="form-input" id="e1NeuroGCSVerbal" onchange="e1NeuroCalcGCS()">
                            <option value="">-- V --</option>
                            <option value="5">5 - ${tr('Oriented', '????? ?????')}</option>
                            <option value="4">4 - ${tr('Confused', '????')}</option>
                            <option value="3">3 - ${tr('Inappropriate words', '????? ??? ?????')}</option>
                            <option value="2">2 - ${tr('Incomprehensible sounds', '????? ?????')}</option>
                            <option value="1">1 - ${tr('None', '?? ???? ?????')}</option>
                        </select>
                    </div>
                </div>

                <div class="flex gap-8 mb-8">
                    <div class="form-group" style="flex:1">
                        <label>${tr('Motor Response (M: 1-6)', '????????? ???????? (M: 1-6)')}</label>
                        <select class="form-input" id="e1NeuroGCSMotor" onchange="e1NeuroCalcGCS()">
                            <option value="">-- M --</option>
                            <option value="6">6 - ${tr('Obeys commands', '????? ?????')}</option>
                            <option value="5">5 - ${tr('Localises pain', '????? ?????')}</option>
                            <option value="4">4 - ${tr('Normal flexion / Withdraws', '??? ????? ??? ?????')}</option>
                            <option value="3">3 - ${tr('Abnormal flexion / Decorticate', '????? ??? ?????')}</option>
                            <option value="2">2 - ${tr('Extension / Decerebrate', '??? ??? ????? ?????')}</option>
                    </div>
                </div>
            </div>
        </div>
    `
}

```

```

        <option value="1">1 - ${tr('None', '?? ??? ????')}</option>
    </select>
</div>
<div class="form-group" style="flex:1">
    <label>${tr('Total GCS Score (3-15)', '????? ???? GCS (3-15)')}</label>
    <input class="form-input" type="text" id="e1NeuroGCSTotal" placeholder="Auto-calculated" readonly
style="background:var(--hover,#f8f9fa);font-weight:700">
</div>
</div>

<!-- NIHSS & Reflexes -->
<h5 style="margin:12px 0 8px;color:var(--accent,#0ea5e9)">?? ${tr('Stroke & Reflexes', '???? ???? ?
????? ?????')}</h5>
<div class="flex gap-8 mb-8">
    <div class="form-group" style="flex:1">
        <label>${tr('NIH Stroke Scale (0-42)', '???? ???? ????? NIHSS (0-42)')}</label>
        <input class="form-input" type="number" id="e1NeuroNIHSS" min="0" max="42" placeholder="0-42">
    </div>
    <div class="form-group" style="flex:1">
        <label>${tr('Deep Tendon Reflexes', '????? ??????')}</label>
        <select class="form-input" id="e1NeuroReflexes">
            <option value="Normal">${tr('Normal (2+)', '???? (2+)')}</option>
            <option value="Hyperreflexia">${tr('Hyperreflexia (3+ / 4+)', '???? ???? (3+ / 4+)')}</option>
            <option value="Hyporeflexia">${tr('Hyporeflexia (1+)', '???? ???? (1+)')}</option>
            <option value="Absent">${tr('Absent (0)', '???? ????? (0)')}</option>
        </select>
    </div>
</div>

<div class="form-group mb-8">
    <label>${tr('Clinical Notes / Findings', '????? ???? / ???? ????')}</label>
    <textarea class="form-input form-textarea" id="e1NeuroNotes" rows="2" placeholder="${tr('Describe cr
anial nerves, motor/sensory deficits...', '???? ????? ?????? ?????? ???? ???? ? ????...')}" style="mi
n-height:50px"></textarea>
</div>
<button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddNeurologyAssessment(${safeId(pid)})"
>? ${tr('Save Neurology Assessment', '?? ??????')}</button>
</div>
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Assessment History', '?? ??????')}
</h4>
    <div id="e1NeuroList">${tr('Loading...', '???? ??????')}</div>
</div>
</div>
`;

window.e1LoadNeuroList(pid);
};

window.e1NeuroCalcGCS = () => {
    const eye = parseInt(document.getElementById('e1NeuroGCSEye')?.value);
    const verbal = parseInt(document.getElementById('e1NeuroGCSVerbal')?.value);
    const motor = parseInt(document.getElementById('e1NeuroGCSMotor')?.value);

```

```

const scoreField = document.getElementById('e1NeuroGCSTotal');

if (scoreField) {
  if (!isNaN(eye) && !isNaN(verbal) && !isNaN(motor)) {
    const score = eye + verbal + motor;
    let status = '';
    if (score <= 8) status = ` (${tr('Severe / Coma', '????? / ????? ?????')} `;
    else if (score <= 12) status = ` (${tr('Moderate', '???? ?????')} `;
    else status = ` (${tr('Mild', '???? ?????')} `;

    scoreField.value = `${score} / 15${status}`;
  } else {
    scoreField.value = '';
  }
}
};

window.e1LoadNeuroList = async (pid) => {
  const container = document.getElementById('e1NeuroList');
  if (!container) return;
  try {
    const records = await API.get('/api/neurology/assessments/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No neurological assessments found', '?? ??? ???? ?? ?? ??')}</div>`;
      return;
    }
    container.innerHTML = records.map(r => {
      // A NULL total = incomplete/unassessed GCS ? must render as unmeasured ('?'),
      // never as 0/15-Severe, and unmeasured components must not display as worst-score 1.
      const gcs = (r.gcs_total_score === null || r.gcs_total_score === undefined) ? null : r.gcs_total_score;
      let border = '4px solid #10b981';
      if (gcs === null) border = '4px solid #94a3b8';
      else if (gcs <= 8) border = '4px solid #ef4444';
      else if (gcs <= 12) border = '4px solid #f59e0b';
      const comp = (v) => (v === null || v === undefined) ? '?' : v;

      return `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:
        ${border};font-size:12px">
          <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">
            <span>? ${new Date(r.assessment_date).toLocaleDateString('ar-SA')}</span>
            <span style="font-weight:700;color:var(--accent,#0ea5e9)">GCS: ${gcs === null ? '?' : gcs + '/15'}
          </span>
          </div>
          <div style="margin-top:4px;display:grid;grid-template-columns:1fr 1fr;gap:4px">
            <div><strong>${tr('GCS (E,V,M):', '???? ???? ?????')}</strong> ${comp(r.gcs_eye)},${comp(r.gcs_verbal)},${comp(r.gcs_motor)}</div>
            <div><strong>${tr('NIHSS Score:', '???? ????? ?????')}</strong> ${r.nihss_score !== null ? r.nihss_score : '-'}</div>
            <div style="grid-column: span 2"><strong>${tr('Reflexes:', '???????? ?????')}</strong> ${r.reflexes_status || 'Normal'}</div>
          </div>
          ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '???? ????

```

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?:'})</strong> ${escapeHTML(r.notes)}</div>` : ''}
    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
''})</div>
    </div>
    `;
    }).join('');
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
}
};

window.e1AddNeurologyAssessment = async (pid) => {
    const eye = document.getElementById('e1NeuroGCSEye').value;
    const verbal = document.getElementById('e1NeuroGCSVerbal').value;
    const motor = document.getElementById('e1NeuroGCSMotor').value;
    const nihss = document.getElementById('e1NeuroNIHSS').value;
    const reflexes = document.getElementById('e1NeuroReflexes').value;
    const notes = document.getElementById('e1NeuroNotes').value;
    if (!eye || !verbal || !motor) {
        showToast(tr('Please select all GCS components', '???? ?????? ????? ?????? ?????? ?????????'), 'error')
    };
    return;
}
try {
    const resp = await API.post('/api/neurology/assessments', {
        patient_id: pid,
        gcs_eye: parseInt(eye),
        gcs_verbal: parseInt(verbal),
        gcs_motor: parseInt(motor),
        nihss_score: nihss ? parseInt(nihss) : null,
        reflexes_status: reflexes,
        notes: notes
    });
    if (resp && resp.error) {
        showToast(tr('Rejected: ', '?????: ') + resp.error, 'error');
        return;
    }
    showToast(tr('Neurological assessment saved successfully!', '?? ??? ??????? ?????? ??????!'));
    window.e1LoadNeuroList(pid);
    document.getElementById('e1NeuroGCSEye').value = '';
    document.getElementById('e1NeuroGCSVerbal').value = '';
    document.getElementById('e1NeuroGCSMotor').value = '';
    document.getElementById('e1NeuroGCSTotal').value = '';
    document.getElementById('e1NeuroNIHSS').value = '';
    document.getElementById('e1NeuroNotes').value = '';
} catch (err) {
    showToast(tr('Error saving assessment', '??? ?? ??? ??????? ???????'), 'error');
}
};

// =====
// ===== NEPHROLOGY MODULE (G02) =====
// =====

```

```

window.e1RenderNephrology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.2;min-width:300px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Nephrology & Dialysis Session Log', '?? ?????
?????? ???? ?????? ??????')}</h4>

        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Dialysis Type', '??? ?????? ??????')}</label>
            <select class="form-input" id="e1NephroType">
              <option value="Hemodialysis">${tr('Hemodialysis (HD)', '???? ???? (HD)')}</option>
              <option value="Peritoneal Dialysis">${tr('Peritoneal Dialysis (PD)', '???? ??????? (PD)')}</opti
on>
            </select>
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Duration (Hours)', '??? ?????? (????)')}</label>
            <input class="form-input" type="number" id="e1NephroDuration" step="0.5" min="0" placeholder="e.g.
4.0">
          </div>
        </div>

        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Ultrafiltration Target (Liters)', '?????? ??? ??????? (???)')}</label>
            <input class="form-input" type="number" id="e1NephroUFTarget" step="0.1" min="0" placeholder="e.g.
2.5">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Blood Flow Rate (mL/min)', '???? ???? ???? (??/????)')}</label>
            <input class="form-input" type="number" id="e1NephroBloodFlow" min="0" placeholder="e.g. 300">
          </div>
        </div>

        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Dialysate Flow Rate (mL/min)', '???? ???? ???? ?????')}</label>
            <input class="form-input" type="number" id="e1NephroDialysateFlow" min="0" placeholder="e.g. 500">
          </div>
          <div class="form-group" style="flex:1">
            <label>${tr('Weight Pre-Dialysis (kg)', '???? ??? ?????? (???)')}</label>
            <input class="form-input" type="number" id="e1NephroWeightPre" step="0.1" min="0" placeholder="e.g
. 75.2" oninput="e1NephroCalcWeightLoss()">
          </div>
        </div>

        <div class="flex gap-8 mb-8">
          <div class="form-group" style="flex:1">
            <label>${tr('Weight Post-Dialysis (kg)', '????? ??? ?????? (???)')}</label>
            <input class="form-input" type="number" id="e1NephroWeightPost" step="0.1" min="0" placeholder="e.

```

```

g. 72.8" oninput="e1NephroCalcWeightLoss()">
  </div>
  <div class="form-group" style="flex:1">
    <label>${tr('Fluid Removed / Loss (kg)', '?????? ???? ??? ?????? (???)')}</label>
    <input class="form-input" type="text" id="e1NephroWeightLoss" placeholder="Auto-calculated" readon
ly style="background:var(--hover,#f8f9fa);font-weight:700">
  </div>
</div>

<div class="form-group mb-8">
  <label>${tr('Clinical Notes / Access Status', '?????? ???? / ??? ???? ?????')}</label>
  <textarea class="form-input form-textarea" id="e1NephroNotes" rows="2" placeholder="${tr('Describe A
V fistula status, vital signs during session...', '??? ???? ????? ????????? ????????? ????????? ?????
?????..')}" style="min-height:50px"></textarea>
</div>
<button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddNephrologySession(${safeId(pid)})">?
${tr('Save Dialysis Session', '??? ?????')}</button>
</div>
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Dialysis Session History', '??? ????? ??????
????????')}</h4>
  <div id="e1NephroList">${tr('Loading...', '??? ??????..')}</div>
</div>
</div>
`;

window.e1LoadNephroList(pid);
};

window.e1NephroCalcWeightLoss = () => {
  const pre = parseFloat(document.getElementById('e1NephroWeightPre')?.value);
  const post = parseFloat(document.getElementById('e1NephroWeightPost')?.value);
  const lossField = document.getElementById('e1NephroWeightLoss');

  if (lossField) {
    if (!isNaN(pre) && !isNaN(post)) {
      const diff = pre - post;
      lossField.value = `${diff.toFixed(2)} kg`;
    } else {
      lossField.value = '';
    }
  }
};

window.e1LoadNephroList = async (pid) => {
  const container = document.getElementById('e1NephroList');
  if (!container) return;
  try {
    const records = await API.get('/api/nephrology/dialysis/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No dialysis sessions foun
d', '?? ???? ????? ???? ?????')}</div>`;
      return;
    }
  }
};

```



```

    }
    container.innerHTML = records.map(r => {
      const pre = parseFloat(r.pre_weight_kg || 0);
      const post = parseFloat(r.post_weight_kg || 0);
      const loss = pre && post ? (pre - post).toFixed(2) : '-';

      return `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:
4px solid var(--accent,#0ea5e9);font-size:12px">
          <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">
            <span>? ${new Date(r.session_date).toLocaleDateString('ar-SA')}</span>
            <span style="font-weight:700;color:var(--accent,#0ea5e9)">${escapeHTML(r.dialysis_type)}</span>
          </div>
          <div style="margin-top:4px;display:grid;grid-template-columns:1fr 1fr;gap:4px">
            <div><strong>${tr('Duration:', '?????:')}</strong> ${r.duration_hours || '-'} hrs</div>
            <div><strong>${tr('UF Target:', '??? ??????:')}</strong> ${r.ultrafiltration_target_liters || '-'}
} L</div>
            <div><strong>${tr('Blood Flow:', '???? ?????:')}</strong> ${r.blood_flow_rate_ml_min || '-'} mL/min
</div>
            <div><strong>${tr('Weight Loss:', '????? ??????:')}</strong> ${loss} kg</div>
          </div>
          ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '????????:')}</
strong> ${escapeHTML(r.notes)}</div>` : ''}
          <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
        </div>
      `;
    }).join('');
  } catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
  }
};

```

```

window.e1AddNephrologySession = async (pid) => {
  const type = document.getElementById('e1NephroType')?.value;
  const duration = document.getElementById('e1NephroDuration')?.value;
  const ufTarget = document.getElementById('e1NephroUFTarget')?.value;
  const bloodFlow = document.getElementById('e1NephroBloodFlow')?.value;
  const dialysateFlow = document.getElementById('e1NephroDialysateFlow')?.value;
  const weightPre = document.getElementById('e1NephroWeightPre')?.value;
  const weightPost = document.getElementById('e1NephroWeightPost')?.value;
  const notes = document.getElementById('e1NephroNotes')?.value;

  if (!type) {
    showToast(tr('Please select dialysis type', '???? ?????? ??? ?????? ??????'), 'error');
    return;
  }

  try {
    await API.post('/api/nephrology/dialysis', {
      patient_id: pid,
      dialysis_type: type,
      duration_hours: duration ? parseFloat(duration) : null,
      ultrafiltration_target_liters: ufTarget ? parseFloat(ufTarget) : null,

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```

        blood_flow_rate_ml_min: bloodFlow ? parseInt(bloodFlow) : null,
        dialysate_flow_rate_ml_min: dialysateFlow ? parseInt(dialysateFlow) : null,
        pre_weight_kg: weightPre ? parseFloat(weightPre) : null,
        post_weight_kg: weightPost ? parseFloat(weightPost) : null,
        notes: notes
    });
    showToast(tr('Dialysis session saved successfully!', '?? ?? ?????? ?????!'));
    window.e1LoadNephroList(pid);
    document.getElementById('e1NephroDuration').value = '';
    document.getElementById('e1NephroUFTarget').value = '';
    document.getElementById('e1NephroBloodFlow').value = '';
    document.getElementById('e1NephroDialysateFlow').value = '';
    document.getElementById('e1NephroWeightPre').value = '';
    document.getElementById('e1NephroWeightPost').value = '';
    document.getElementById('e1NephroWeightLoss').value = '';
    document.getElementById('e1NephroNotes').value = '';
} catch (err) {
    showToast(tr('Error saving dialysis session', '??? ?? ?? ??????'), 'error');
}
};

// =====
// ===== OPHTHALMOLOGY MODULE (G14) =====
// =====

window.e1RenderOphthalmology = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <div style="flex:1.4;min-width:320px">
                <h4 style="margin:0 0 12px;color:var(--primary)">?? ${tr('Ophthalmology Examination', '??? ?? ?????? ?
                ??????')}</h4>

                <!-- Visual Acuity Section -->
                <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
                ttom:12px">
                    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Visual Acuity (Snellen)',
                    '??? ?????? (???????)')}</legend>
                    <div class="flex gap-8 mb-8">
                        <div class="form-group" style="flex:1">
                            <label>?? OD (${tr('Right Eye Uncorrected', '????? ?????? ???? ??????')})</label>
                            <select class="form-input" id="e1EyeVaOdUncorrected">
                                <option value="20/20">20/20</option>
                                <option value="20/25">20/25</option>
                                <option value="20/30">20/30</option>
                                <option value="20/40">20/40</option>
                                <option value="20/50">20/50</option>
                                <option value="20/70">20/70</option>
                                <option value="20/100">20/100</option>
                                <option value="20/200">20/200</option>
                                <option value="CF">${tr('CF (Counting Fingers)', '?? ???????')}</option>
                                <option value="HM">${tr('HM (Hand Motion)', '???? ?????')}</option>

```

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        <option value="LP">${tr('LP (Light Perception)', '????? ?????')}</option>
        <option value="NLP">${tr('NLP (No Light)', '?? ???? ????? ?????')}</option>
    </select>
</div>
<div class="form-group" style="flex:1">
    <label>?? OS (${tr('Left Eye Uncorrected', '????? ?????? ???? ?????')}</label>
    <select class="form-input" id="e1EyeVa0sUncorrected">
        <option value="20/20">20/20</option>
        <option value="20/25">20/25</option>
        <option value="20/30">20/30</option>
        <option value="20/40">20/40</option>
        <option value="20/50">20/50</option>
        <option value="20/70">20/70</option>
        <option value="20/100">20/100</option>
        <option value="20/200">20/200</option>
        <option value="CF">CF</option>
        <option value="HM">HM</option>
        <option value="LP">LP</option>
        <option value="NLP">NLP</option>
    </select>
</div>
</div>
<div class="flex gap-8">
    <div class="form-group" style="flex:1">
        <label>? OD (${tr('Right Eye Corrected', '????? ?????? ?????????')}</label>
        <select class="form-input" id="e1EyeVa0dCorrected">
            <option value="20/20">20/20</option>
            <option value="20/25">20/25</option>
            <option value="20/30">20/30</option>
            <option value="20/40">20/40</option>
            <option value="20/50">20/50</option>
            <option value="20/70">20/70</option>
            <option value="20/100">20/100</option>
            <option value="20/200">20/200</option>
            <option value="CF">CF</option>
            <option value="HM">HM</option>
            <option value="LP">LP</option>
            <option value="NLP">NLP</option>
        </select>
    </div>
    <div class="form-group" style="flex:1">
        <label>? OS (${tr('Left Eye Corrected', '????? ?????? ?????????')}</label>
        <select class="form-input" id="e1EyeVa0sCorrected">
            <option value="20/20">20/20</option>
            <option value="20/25">20/25</option>
            <option value="20/30">20/30</option>
            <option value="20/40">20/40</option>
            <option value="20/50">20/50</option>
            <option value="20/70">20/70</option>
            <option value="20/100">20/100</option>
            <option value="20/200">20/200</option>
            <option value="CF">CF</option>
            <option value="HM">HM</option>
            <option value="LP">LP</option>
        </select>
    </div>

```

```
        <option value="NLP">NLP</option>
    </select>
</div>
</div>
</fieldset>

<!-- Intraocular Pressure (IOP) Section -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Intraocular Pressure (IOP)', '??? ?????')}</legend>
    <div class="flex gap-8 mb-8">
        <div class="form-group" style="flex:1">
            <label>OD (${tr('Right Eye - mmHg', '???? ????? - ?? ???')})</label>
            <input class="form-input" type="number" id="e1EyeIop0d" step="0.5" placeholder="e.g. 15.0" oninput="e1EyeCheckIopAlert()">
        </div>
        <div class="form-group" style="flex:1">
            <label>OS (${tr('Left Eye - mmHg', '???? ????? - ?? ???')})</label>
            <input class="form-input" type="number" id="e1EyeIop0s" step="0.5" placeholder="e.g. 16.0" oninput="e1EyeCheckIopAlert()">
        </div>
    </div>
    <div class="form-group">
        <label>${tr('Measurement Method', '???? ?????')}</label>
        <select class="form-input" id="e1EyeIopMethod">
            <option value="Goldmann">${tr('Goldmann Applanation', '???? ?????? ?????')}</option>
            <option value="Tonopen">${tr('Tonopen', '????')}</option>
            <option value="Airpuff">${tr('Non-Contact (Airpuff)', '?? ????? (??? ???)')}</option>
        </select>
    </div>
    <div id="e1EyeIopWarning" style="margin-top:6px;font-size:12px;color:red;display:none;font-weight:700">
        ?? ${tr('Warning: High Intraocular Pressure detected (> 21 mmHg)!', '????: ?? ??? ??? ??? ????? (> 21 ?? ?????)!')}
    </div>
</fieldset>

<!-- Refraction Section -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Refraction', '??? ?????? ? (????)')}</legend>
    <div class="flex gap-8 mb-8">
        <div class="form-group" style="flex:1">
            <label>OD - ${tr('Sphere', '????')}</label>
            <input class="form-input" type="number" id="e1EyeSphere0d" step="0.25" placeholder="-1.50">
        </div>
        <div class="form-group" style="flex:1">
            <label>OD - ${tr('Cylinder', '????????')}</label>
            <input class="form-input" type="number" id="e1EyeCylinder0d" step="0.25" placeholder="-0.50">
        </div>
        <div class="form-group" style="flex:1">
            <label>OD - ${tr('Axis', '????')}</label>
        </div>
    </div>
</fieldset>
```

```

        <input class="form-input" type="number" id="e1EyeAxis0d" min="0" max="180" placeholder="90">
    </div>
</div>
<div class="flex gap-8">
    <div class="form-group" style="flex:1">
        <label>OS - ${tr('Sphere', '????')}</label>
        <input class="form-input" type="number" id="e1EyeSphere0s" step="0.25" placeholder="-1.25">
    </div>
    <div class="form-group" style="flex:1">
        <label>OS - ${tr('Cylinder', '????????')}</label>
        <input class="form-input" type="number" id="e1EyeCylinder0s" step="0.25" placeholder="-0.75">
    </div>
    <div class="form-group" style="flex:1">
        <label>OS - ${tr('Axis', '????')}</label>
        <input class="form-input" type="number" id="e1EyeAxis0s" min="0" max="180" placeholder="85">
    </div>
</div>
</fieldset>

<!-- Findings & Notes -->
<div class="form-group mb-8">
    <label>${tr('Slit Lamp Examination', '??? ?????? ????? (???????? ??????)')}</label>
    <textarea class="form-input form-textarea" id="e1EyeSlitLamp" rows="2" placeholder="${tr('Cornea clear, anterior chamber deep and quiet...', '?????? ?????? ?????? ????????? ?????? ??????...')}" style="min-height:45px"></textarea>
</div>

<div class="form-group mb-8">
    <label>${tr('Fundoscopy / Retina Exam', '??? ??? ????? ?????????')}</label>
    <textarea class="form-input form-textarea" id="e1EyeFundoscopy" rows="2" placeholder="${tr('Optic disc pink with sharp margins, cup-to-disc ratio 0.3...', '????? ?????? ??? ???? ???? ???? ???? ???? 0.3...')}" style="min-height:45px"></textarea>
</div>

<div class="form-group mb-8">
    <label>${tr('Clinical Notes', '??????? ??????')}</label>
    <input class="form-input" id="e1EyeNotes" placeholder="${tr('e.g. Advised patient to follow up in 6 months', '?????: ??? ?????? ?????????? ??? 6 ?????')}">
</div>

    <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddEyeExam(${safeId(pid)})">? ${tr('Save Eye Examination', '??? ??? ?????')}</button>
</div>
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Eye Examination History', '??? ?????? ????? ? ??????')}</h4>
    <div id="e1EyeExamsList">${tr('Loading...', '???? ?????????...')}</div>
</div>
</div>
`;

window.e1LoadEyeExamsList(pid);
};

```

```

window.e1EyeCheckIopAlert = () => {
  const od = parseFloat(document.getElementById('e1EyeIop0d')?.value || 0);
  const os = parseFloat(document.getElementById('e1EyeIop0s')?.value || 0);
  const warning = document.getElementById('e1EyeIopWarning');

  if (warning) {
    if (od > 21 || os > 21) {
      warning.style.display = 'block';
    } else {
      warning.style.display = 'none';
    }
  }
};

window.e1LoadEyeExamsList = async (pid) => {
  const container = document.getElementById('e1EyeExamsList');
  if (!container) return;
  try {
    const exams = await API.get('/api/ophthalmology/exams/patient/' + pid);
    if (!exams.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No eye exams found', '??
      ??? ???? ???? ????')}</div>`;
      return;
    }
    container.innerHTML = exams.map(e => {
      const iop0d = parseFloat(e.od_iop || 0);
      const iop0s = parseFloat(e.os_iop || 0);
      const isIopHigh = iop0d > 21 || iop0s > 21;
      const iopStyle = isIopHigh ? 'color:red;font-weight:700' : 'color:var(--text-color)';

      return `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4
        px solid var(--primary);font-size:12px">
          <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">
            <span>? ${new Date(e.exam_date).toLocaleDateString('ar-SA')}</span>
            <span>?? VA: OD ${escapeHTML(e.od_va_uncorrected || '-')} | OS ${escapeHTML(e.os_va_uncorrected ||
            '-')}</span>
          </div>
          <div style="margin-top:6px;display:grid;grid-template-columns:1fr 1fr;gap:4px">
            <div style="${iopStyle}"><strong>${tr('IOP (OD/OS):', '??? ?????:')}</strong> ${e.od_iop || '-'} /
            ${e.os_iop || '-'} mmHg</div>
            <div><strong>${tr('Method:', '???????:')}</strong> ${escapeHTML(e.iop_method || '-')}</div>
            <div style="grid-column:span 2"><strong>${tr('Refraction (OD):', '????? ??????:')}</strong> S:${e
            .od_sphere || '0'} C:${e.od_cylinder || '0'} A:${e.od_axis || '0'}°</div>
            <div style="grid-column:span 2"><strong>${tr('Refraction (OS):', '????? ??????:')}</strong> S:${e
            .os_sphere || '0'} C:${e.os_cylinder || '0'} A:${e.os_axis || '0'}°</div>
          </div>
          ${e.slit_lamp_exam ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Slit Lamp:', '
          ?????? ??????:')}</strong> ${escapeHTML(e.slit_lamp_exam)}</div>` : ''}
          ${e.fundoscopy_exam ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Fundoscopy:',
          '??? ??????:')}</strong> ${escapeHTML(e.fundoscopy_exam)}</div>` : ''}
          ${e.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '???????:')}</
          strong> ${escapeHTML(e.notes)}</div>` : ''}
      `;
    });
  } catch {
    // Error handling
  }
};

```

```

        <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(e.doctor_name ||
    '')}</div>
    </div>
    `;
    }).join('');
    } catch (err) {
        container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ?????')}</div>`;
    }
};

window.e1AddEyeExam = async (pid) => {
    const odVaUn = document.getElementById('e1EyeVa0Uncorrected')?.value;
    const osVaUn = document.getElementById('e1EyeVa0Uncorrected')?.value;
    const odVaCor = document.getElementById('e1EyeVa0Corrected')?.value;
    const osVaCor = document.getElementById('e1EyeVa0Corrected')?.value;

    const odIop = document.getElementById('e1EyeIop0d')?.value;
    const osIop = document.getElementById('e1EyeIop0s')?.value;
    const iopMethod = document.getElementById('e1EyeIopMethod')?.value;

    const odSph = document.getElementById('e1EyeSphere0d')?.value;
    const osSph = document.getElementById('e1EyeSphere0s')?.value;
    const odCyl = document.getElementById('e1EyeCylinder0d')?.value;
    const osCyl = document.getElementById('e1EyeCylinder0s')?.value;
    const odAx = document.getElementById('e1EyeAxis0d')?.value;
    const osAx = document.getElementById('e1EyeAxis0s')?.value;

    const slit = document.getElementById('e1EyeSlitLamp')?.value;
    const fundus = document.getElementById('e1EyeFundoscopy')?.value;
    const notes = document.getElementById('e1EyeNotes')?.value;

    try {
        await API.post('/api/ophthalmology/exams', {
            patient_id: pid,
            od_va_uncorrected: odVaUn,
            os_va_uncorrected: osVaUn,
            od_va_corrected: odVaCor,
            os_va_corrected: osVaCor,
            od_iop: odIop,
            os_iop: osIop,
            iop_method: iopMethod,
            od_sphere: odSph,
            os_sphere: osSph,
            od_cylinder: odCyl,
            os_cylinder: osCyl,
            od_axis: odAx,
            os_axis: osAx,
            slit_lamp_exam: slit,
            fundoscopy_exam: fundus,
            notes: notes
        });

        showToast(tr('Eye examination saved successfully!', '?? ??? ?? ???? ?????!'));
        window.e1LoadEyeExamsList(pid);
    }
};

```

```

// Reset fields
document.getElementById('e1EyeIop0d').value = '';
document.getElementById('e1EyeIop0s').value = '';
document.getElementById('e1EyeSphere0d').value = '';
document.getElementById('e1EyeSphere0s').value = '';
document.getElementById('e1EyeCylinder0d').value = '';
document.getElementById('e1EyeCylinder0s').value = '';
document.getElementById('e1EyeAxis0d').value = '';
document.getElementById('e1EyeAxis0s').value = '';
document.getElementById('e1EyeSlitLamp').value = '';
document.getElementById('e1EyeFundoscopy').value = '';
document.getElementById('e1EyeNotes').value = '';
} catch (err) {
  showToast(tr('Error saving eye examination', '??? ?? ??? ??? ?????'), 'error');
}
};

// =====
// ===== ENT (OTOLARYNGOLOGY) MODULE (G13) =====
// =====

window.e1RenderEnt = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.4;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Otolaryngology (ENT) Examination', '??? ?????
        ?????? ?????????')}</h4>

        <!-- Audiogram Section -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Pure Tone Audiometry (AC
          Thresholds in dB)', '????? ?????? (???????? ????????? ?????????)')}</legend>

          <div style="margin-bottom:8px;font-weight:700;color:var(--primary);font-size:12px">? ${tr('Right Ear
          (OD / AD) - Air Conduction', '????? ?????? - ????????? ?????????')}</div>
          <div class="flex gap-4 mb-8" style="flex-wrap:wrap">
            <div class="form-group" style="flex:1;min-width:60px">
              <label>250Hz</label>
              <input class="form-input" type="number" id="e1EntAcR250" min="0" max="120" step="5" placeholder=
              "e.g. 15" oninput="e1EntCalcPta()">
            </div>
            <div class="form-group" style="flex:1;min-width:60px">
              <label>500Hz</label>
              <input class="form-input" type="number" id="e1EntAcR500" min="0" max="120" step="5" placeholder=
              "15" oninput="e1EntCalcPta()">
            </div>
            <div class="form-group" style="flex:1;min-width:60px">
              <label>1kHz</label>
              <input class="form-input" type="number" id="e1EntAcR1000" min="0" max="120" step="5" placeholder

```



```

="20" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>2kHz</label>
        <input class="form-input" type="number" id="e1EntAcR2000" min="0" max="120" step="5" placeholder=
="20" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>4kHz</label>
        <input class="form-input" type="number" id="e1EntAcR4000" min="0" max="120" step="5" placeholder=
="25" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>8kHz</label>
        <input class="form-input" type="number" id="e1EntAcR8000" min="0" max="120" step="5" placeholder=
="25">
    </div>
</div>

<div style="margin-bottom:8px;font-weight:700;color:var(--primary);font-size:12px">? ${tr('Left Ear
(OS / AS) - Air Conduction', '????? ?????? - ??????? ???????')}</div>
<div class="flex gap-4 mb-8" style="flex-wrap:wrap">
    <div class="form-group" style="flex:1;min-width:60px">
        <label>250Hz</label>
        <input class="form-input" type="number" id="e1EntAcL250" min="0" max="120" step="5" placeholder=
"e.g. 10" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>500Hz</label>
        <input class="form-input" type="number" id="e1EntAcL500" min="0" max="120" step="5" placeholder=
"10" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>1kHz</label>
        <input class="form-input" type="number" id="e1EntAcL1000" min="0" max="120" step="5" placeholder=
="15" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>2kHz</label>
        <input class="form-input" type="number" id="e1EntAcL2000" min="0" max="120" step="5" placeholder=
="15" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>4kHz</label>
        <input class="form-input" type="number" id="e1EntAcL4000" min="0" max="120" step="5" placeholder=
="20" oninput="e1EntCalcPta()">
    </div>
    <div class="form-group" style="flex:1;min-width:60px">
        <label>8kHz</label>
        <input class="form-input" type="number" id="e1EntAcL8000" min="0" max="120" step="5" placeholder=
="20">
    </div>
</div>

```

```

<!-- PTA Average Indicators -->
<div style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;display:flex;justify-content:space-between;align-items:center;font-size:12px">
  <div>
    <strong>${tr('Right PTA (0.5-4kHz):', '???? ?????:')}</strong> <span id="e1EntPtaR">-</span> dB

    <span id="e1EntGradeR" class="badge" style="margin-left:6px;display:none"></span>
  </div>
  <div>
    <strong>${tr('Left PTA (0.5-4kHz):', '???? ?????:')}</strong> <span id="e1EntPtaL">-</span> dB
    <span id="e1EntGradeL" class="badge" style="margin-left:6px;display:none"></span>
  </div>
</div>
</fieldset>

<!-- Tympanometry & Otoscopy Section -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
  <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Otoscopy & Tympanometry', '????? ?????? ?????? ??? ??????')}</legend>
  <div class="flex gap-8 mb-8">
    <div class="form-group" style="flex:1">
      <label>${tr('Tympanometry Right', '????? ?????? ??????')}</label>
      <select class="form-input" id="e1EntTympR">
        <option value="Type A">${tr('Type A (Normal)', '??? ? (?????)')}</option>
        <option value="Type B">${tr('Type B (Fluid/Perf)', '??? ? (?????/????)')}</option>
        <option value="Type C">${tr('Type C (Negative Press)', '??? ? (??? ?????)')}</option>
      </select>
    </div>
    <div class="form-group" style="flex:1">
      <label>${tr('Tympanometry Left', '????? ?????? ??????')}</label>
      <select class="form-input" id="e1EntTympL">
        <option value="Type A">${tr('Type A (Normal)', '??? ? (?????)')}</option>
        <option value="Type B">${tr('Type B (Fluid/Perf)', '??? ? (?????/????)')}</option>
        <option value="Type C">${tr('Type C (Negative Press)', '??? ? (??? ?????)')}</option>
      </select>
    </div>
  </div>
  <div class="flex gap-8">
    <div class="form-group" style="flex:1">
      <label>${tr('Otoscopy Right', '????? ?????? ??????')}</label>
      <textarea class="form-input form-textarea" id="e1EntOtoscopyR" rows="2" placeholder="${tr('Tympanic membrane intact, translucent...', '???? ?????? ????? ??????...')}" style="min-height:45px"></textarea>
    </div>
    <div class="form-group" style="flex:1">
      <label>${tr('Otoscopy Left', '????? ?????? ??????')}</label>
      <textarea class="form-input form-textarea" id="e1EntOtoscopyL" rows="2" placeholder="${tr('Tympanic membrane intact, translucent...', '???? ?????? ????? ??????...')}" style="min-height:45px"></textarea>
    </div>
  </div>
</fieldset>

<!-- Speech Audiometry & Interpretation -->
<div class="flex gap-8 mb-8">

```

```

<div class="form-group" style="flex:1">
  <label>${tr('Speech Reception Threshold (SRT) - R/L (dB)', '???? ??? ?????? - ???/???? (?????)')}
}</label>

  <div class="flex gap-4">
    <input class="form-input" type="number" id="e1EntSrtR" placeholder="R">
    <input class="form-input" type="number" id="e1EntSrtL" placeholder="L">
  </div>
</div>

<div class="form-group" style="flex:1">
  <label>${tr('Speech Discrimination (SD) - R/L (%)', '????? ?????? - ???/???? (%)')}</label>
  <div class="flex gap-4">
    <input class="form-input" type="number" id="e1EntSdR" min="0" max="100" placeholder="R%">
    <input class="form-input" type="number" id="e1EntSdL" min="0" max="100" placeholder="L%">
  </div>
</div>
</div>

<div class="form-group mb-8">
  <label>${tr('Clinical Interpretation', '??????? ???????? ????????')}</label>
  <textarea class="form-input form-textarea" id="e1EntInterpretation" rows="2" placeholder="${tr('Normal hearing bilaterally...', '??? ????? ?? ??? ????????...')}" style="min-height:45px"></textarea>
</div>

<div class="form-group mb-8">
  <label>${tr('Clinical Notes', '??????? ????????')}</label>
  <input class="form-input" id="e1EntNotes" placeholder="${tr('e.g. Advised ear drops for cerumen', '?
????: ??? ?????? ??? ?????? ??????')}">
</div>

  <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddEntExam(${safeId(pid)})">? ${tr('Save ENT Examination', '??? ??? ?????? ?????? ????????')}</button>
</div>

<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">

  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('ENT History Log', '??? ???????? ????????')}</h
4>

  <div id="e1EntHistoryList">${tr('Loading...', '???? ????????...')}</div>
</div>
</div>
`;

window.e1LoadEntHistoryList(pid);
};

window.e1EntGetImpairmentGrade = (pta) => {
  if (pta < 20) return { label: tr('Normal', '?????'), color: 'badge-success' };
  if (pta >= 20 && pta < 35) return { label: tr('Mild', '?????'), color: 'badge-info' };
  if (pta >= 35 && pta < 50) return { label: tr('Moderate', '?????'), color: 'badge-warning' };
  if (pta >= 50 && pta < 65) return { label: tr('Mod. Severe', '??? ????????'), color: 'badge-warning' };
  if (pta >= 65 && pta < 80) return { label: tr('Severe', '?????'), color: 'badge-danger' };
  return { label: tr('Profound', '????? ??????'), color: 'badge-danger' };
};

window.e1EntCalcPta = () => {

```

```

const getVal = (id) => parseFloat(document.getElementById(id)?.value || NaN);

const r500 = getVal('e1EntAcR500');
const r1000 = getVal('e1EntAcR1000');
const r2000 = getVal('e1EntAcR2000');
const r4000 = getVal('e1EntAcR4000');

const l500 = getVal('e1EntAcL500');
const l1000 = getVal('e1EntAcL1000');
const l2000 = getVal('e1EntAcL2000');
const l4000 = getVal('e1EntAcL4000');

const calc = (vals) => {
  const valid = vals.filter(v => !isNaN(v));
  if (!valid.length) return null;
  return (valid.reduce((sum, v) => sum + v, 0) / valid.length).toFixed(1);
};

const ptaR = calc([r500, r1000, r2000, r4000]);
const ptaL = calc([l500, l1000, l2000, l4000]);

const ptaRText = document.getElementById('e1EntPtaR');
const ptaLText = document.getElementById('e1EntPtaL');
const badgeR = document.getElementById('e1EntGradeR');
const badgeL = document.getElementById('e1EntGradeL');

if (ptaRText && ptaR !== null) {
  ptaRText.innerText = ptaR;
  const grade = window.e1EntGetImpairmentGrade(parseFloat(ptaR));
  if (badgeR) {
    badgeR.style.display = 'inline-block';
    badgeR.innerText = grade.label;
    badgeR.className = 'badge ' + grade.color;
  }
} else if (ptaRText) {
  ptaRText.innerText = '-';
  if (badgeR) badgeR.style.display = 'none';
}

if (ptaLText && ptaL !== null) {
  ptaLText.innerText = ptaL;
  const grade = window.e1EntGetImpairmentGrade(parseFloat(ptaL));
  if (badgeL) {
    badgeL.style.display = 'inline-block';
    badgeL.innerText = grade.label;
    badgeL.className = 'badge ' + grade.color;
  }
} else if (ptaLText) {
  ptaLText.innerText = '-';
  if (badgeL) badgeL.style.display = 'none';
}
};

window.e1LoadEntHistoryList = async (pid) => {

```

```

const container = document.getElementById('e1EntHistoryList');
if (!container) return;
try {
  const records = await API.get('/api/ent/audiograms/patient/' + pid);
  if (!records.length) {
    container.innerHTML = `
```

```

        ${r.interpretation ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Interpretation
:', '???????')}</strong> ${escapeHTML(r.interpretation)}</div>` : ''}
        ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Notes:', '???????')}</
strong> ${escapeHTML(r.notes)}</div>` : ''}
        <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
    </div>
    `;
    }).join('');
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ???????')}</div>`;
}
};

```

```

window.e1AddEntExam = async (pid) => {
    const getVal = (id) => document.getElementById(id)?.value;

    try {
        await API.post('/api/ent/audiograms', {
            patient_id: pid,
            right_ac_250: getVal('e1EntAcR250'),
            right_ac_500: getVal('e1EntAcR500'),
            right_ac_1000: getVal('e1EntAcR1000'),
            right_ac_2000: getVal('e1EntAcR2000'),
            right_ac_4000: getVal('e1EntAcR4000'),
            right_ac_8000: getVal('e1EntAcR8000'),

            left_ac_250: getVal('e1EntAcL250'),
            left_ac_500: getVal('e1EntAcL500'),
            left_ac_1000: getVal('e1EntAcL1000'),
            left_ac_2000: getVal('e1EntAcL2000'),
            left_ac_4000: getVal('e1EntAcL4000'),
            left_ac_8000: getVal('e1EntAcL8000'),

            tympanometry_right: getVal('e1EntTympR'),
            tympanometry_left: getVal('e1EntTympL'),
            otoscopy_right: getVal('e1EntOtoscopyR'),
            otoscopy_left: getVal('e1EntOtoscopyL'),

            right_srt: getVal('e1EntSrtR'),
            left_srt: getVal('e1EntSrtL'),
            right_sd_score: getVal('e1EntSdR'),
            left_sd_score: getVal('e1EntSdL'),

            interpretation: getVal('e1EntInterpretation'),
            notes: getVal('e1EntNotes')
        });

        showToast(tr('ENT examination saved successfully!', '?? ??? ??? ????? ?????? ??????? ??????!'));
        window.e1LoadEntHistoryList(pid);

        // Reset fields
        const fields = [
            'e1EntAcR250', 'e1EntAcR500', 'e1EntAcR1000', 'e1EntAcR2000', 'e1EntAcR4000', 'e1EntAcR8000',

```

```

    'e1EntAcL250', 'e1EntAcL500', 'e1EntAcL1000', 'e1EntAcL2000', 'e1EntAcL4000', 'e1EntAcL8000',
    'e1EntOtoscopyR', 'e1EntOtoscopyL', 'e1EntSrtR', 'e1EntSrtL', 'e1EntSdR', 'e1EntSdL',
    'e1EntInterpretation', 'e1EntNotes'
  ];
  fields.forEach(f => {
    const el = document.getElementById(f);
    if (el) el.value = '';
  });

  const ptaRText = document.getElementById('e1EntPtaR');
  const ptaLText = document.getElementById('e1EntPtaL');
  if (ptaRText) ptaRText.innerText = '-';
  if (ptaLText) ptaLText.innerText = '-';
  const badgeR = document.getElementById('e1EntGradeR');
  const badgeL = document.getElementById('e1EntGradeL');
  if (badgeR) badgeR.style.display = 'none';
  if (badgeL) badgeL.style.display = 'none';

} catch (err) {
  showToast(tr('Error saving ENT examination', '??? ?? ??? ??? ????? ?????? ?????????'), 'error');
}
};

// =====
// ===== PLASTIC & BURNS MODULE (G12) =====
// =====

window.e1RenderPlasticBurns = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.4;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Plastic & Burns Assessment', '????? ?????? ??
        ??????')}</h4>

        <!-- Weight & Rule of Nines Section -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Rule of Nines & Fluid Cal
          culator', '???? ?????? ?????? ?????????? ?????????')}</legend>

          <div class="form-group mb-8">
            <label><strong>${tr('Patient Weight (kg) *', '??? ?????? (???) *')}</strong></label>
            <input class="form-input" type="number" id="e1BurnWeight" step="0.1" placeholder="e.g. 70" oninput
            ="e1BurnCalcFluid()">
          </div>

          <div style="margin-bottom:8px;font-weight:700;color:var(--primary);font-size:12px">${tr('Burn Area P
          ercentages (%)', '??? ?????????? ?????????? ??? ?????????? (%)')}</div>

          <div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(120px, 1fr));gap:8px;margin-
          bottom:12px">

```

```

<div class="form-group">
  <label>${tr('Head & Neck (9%)', '????? ??????? (9%)')}</label>
  <input class="form-input" type="number" id="e1BurnHead" min="0" max="9" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Torso Front (18%)', '????? ??????? (18%)')}</label>
  <input class="form-input" type="number" id="e1BurnTorsoFront" min="0" max="18" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Torso Back (18%)', '????? ??????? (18%)')}</label>
  <input class="form-input" type="number" id="e1BurnTorsoBack" min="0" max="18" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Left Arm (9%)', '????? ??????? (9%)')}</label>
  <input class="form-input" type="number" id="e1BurnLeftArm" min="0" max="9" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Right Arm (9%)', '????? ??????? (9%)')}</label>
  <input class="form-input" type="number" id="e1BurnRightArm" min="0" max="9" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Left Leg (18%)', '????? ??????? (18%)')}</label>
  <input class="form-input" type="number" id="e1BurnLeftLeg" min="0" max="18" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Right Leg (18%)', '????? ??????? (18%)')}</label>
  <input class="form-input" type="number" id="e1BurnRightLeg" min="0" max="18" step="0.5" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
<div class="form-group">
  <label>${tr('Perineum (1%)', '????? ??????? (1%)')}</label>
  <input class="form-input" type="number" id="e1BurnPerineum" min="0" max="1" step="0.1" placeholder="0" oninput="e1BurnCalcFluid()">
</div>
</div>

<!-- Calculated Output -->
<div style="background:var(--hover,#f8f9fa);padding:12px;border-radius:8px;font-size:13px;border-left:4px solid var(--primary)">
  <div class="mb-4">
    <strong>${tr('Total Burned Surface Area (TBSA):', '????? ??????? (TBSA:')}</strong> <span id="e1BurnTbsa">0</span>%
  </div>
  <div class="mb-4">
    <strong>${tr('Parkland Fluid Required (24 Hours):', '???? ??????? ??????? (???? 24 ?????:')}</strong> <span id="e1BurnFluidTotal">-</span> mL
  </div>
  <div style="display:flex;gap:12px;font-size:12px;color:var(--text-dim)">

```



```


> <strong>${tr('First 8 Hours:', '??? 8 ?????:')}}</strong> <span id="e1BurnFluid8h"></span>
> mL</div>



> <strong>${tr('Next 16 Hours:', '??? 16 ???? ??????:')}}</strong> <span id="e1BurnFluid16h
"></span> mL</div>
</div>
</div>
</fieldset>

<!-- Clinical Notes & Save -->
<div class="form-group mb-8">
  <label>${tr('Clinical Notes & Assessment Details', '??????? ??????? ??????? ???????')}}</label>
  <textarea class="form-input form-textarea" id="e1BurnNotes" rows="2" placeholder="${tr('Describe bur
n depth (2nd degree, 3rd degree), locations, dressing applied...', '?? ??? ?????? ?????? ?????? ?? ??????? ?
?????? ??? ?????? ???????...')}}" style="min-height:45px"></textarea>
</div>

<button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddBurnAssessment(${safeId(pid)})">? ${
tr('Save Burn Assessment', '??? ????? ??????')}}</button>

<!-- Clinical Photos Section -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-to
p:12px">
  <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Clinical Photo Registry (
Metadata Only)', '??? ????? ??????? (??????? ???????')}}</legend>
  <div class="flex gap-4 mb-4">
    <div class="form-group" style="flex:1">
      <label>${tr('Body Region', '???? ?????')}}</label>
      <input class="form-input" id="e1PhotoRegion" placeholder="e.g. Left forearm">
    </div>
    <div class="form-group" style="flex:1">
      <label>${tr('Photo Date', '???? ?????')}}</label>
      <input class="form-input" type="date" id="e1PhotoDate">
    </div>
  </div>
  <div class="form-group mb-8">
    <label>${tr('Description / Reference', '???? ?? ?????? ?????? ??????')}}</label>
    <input class="form-input" id="e1PhotoDesc" placeholder="e.g. Pre-debridement photo ref #PB-984">
  </div>
  <div class="form-group mb-8" style="display:flex;align-items:center;gap:8px">
    <input type="checkbox" id="e1PhotoConfidential" checked>
    <label for="e1PhotoConfidential">? ${tr('Mark as Highly Confidential', '???? ?????? ??? ??????')}}
</label>
  </div>
  <button class="btn btn-secondary btn-sm w-full" onclick="e1AddClinicalPhoto(${safeId(pid)})">? ${tr(
'Register Photo Metadata', '???? ?????? ??????')}}</button>
</fieldset>
</div>

<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Burn & Photo History', '??? ?????? ?????? ???
?????')}}</h4>
  <div id="e1BurnHistoryList" class="mb-12">${tr('Loading...', '???? ???????...')}}</div>


```

```

        <h4 style="margin:12px 0 12px;color:var(--primary)">? ${tr('Registered Clinical Photos', '????? ??????
?? ???????')}</h4>
        <div id="e1PhotoHistoryList">${tr('Loading...', '???? ???????...')}</div>
    </div>
</div>
`;

// Set default date for photo
const photoDateEl = document.getElementById('e1PhotoDate');
if (photoDateEl) photoDateEl.value = new Date().toISOString().slice(0, 10);

window.e1LoadBurnHistory(pid);
};

window.e1BurnCalcFluid = () => {
    const getVal = (id) => parseFloat(document.getElementById(id)?.value || 0);

    const weight = parseFloat(document.getElementById('e1BurnWeight')?.value || 0);

    const head = getVal('e1BurnHead');
    const front = getVal('e1BurnTorsoFront');
    const back = getVal('e1BurnTorsoBack');
    const lArm = getVal('e1BurnLeftArm');
    const rArm = getVal('e1BurnRightArm');
    const lLeg = getVal('e1BurnLeftLeg');
    const rLeg = getVal('e1BurnRightLeg');
    const peri = getVal('e1BurnPerineum');

    const tbsa = head + front + back + lArm + rArm + lLeg + rLeg + peri;

    const tbsaEl = document.getElementById('e1BurnTbsa');
    if (tbsaEl) tbsaEl.innerText = tbsa.toFixed(1);

    const fluidTotalEl = document.getElementById('e1BurnFluidTotal');
    const fluid8hEl = document.getElementById('e1BurnFluid8h');
    const fluid16hEl = document.getElementById('e1BurnFluid16h');

    if (weight > 0 && tbsa > 0) {
        // Parkland Formula: 4mL * weight_kg * TBSA%
        const totalFluid = 4 * weight * tbsa;
        const first8h = totalFluid / 2;
        const next16h = totalFluid / 2;

        if (fluidTotalEl) fluidTotalEl.innerText = Math.round(totalFluid).toLocaleString();
        if (fluid8hEl) fluid8hEl.innerText = Math.round(first8h).toLocaleString();
        if (fluid16hEl) fluid16hEl.innerText = Math.round(next16h).toLocaleString();
    } else {
        if (fluidTotalEl) fluidTotalEl.innerText = '-';
        if (fluid8hEl) fluid8hEl.innerText = '-';
        if (fluid16hEl) fluid16hEl.innerText = '-';
    }
};

window.e1LoadBurnHistory = async (pid) => {

```

```

const burnContainer = document.getElementById('e1BurnHistoryList');
const photoContainer = document.getElementById('e1PhotoHistoryList');

if (burnContainer) {
  try {
    const records = await API.get('/api/plastic-burns/assessments/patient/' + pid);
    if (!records.length) {
      burnContainer.innerHTML = `
```

```

<div style="padding:8px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:
4px solid #10b981;font-size:12px">
  <div style="font-weight:700;display:flex;justify-content:space-between">
    <span>? ${escapeHTML(p.body_region)}</span>
    <span>? ${new Date(p.photo_date).toLocaleDateString('ar-SA')}</span>
  </div>
  <div style="margin-top:4px;color:var(--text-dim)">${escapeHTML(p.description || '')}</div>
  <div style="margin-top:4px;display:flex;justify-content:space-between;align-items:center">
    <span style="font-size:10px;color:var(--text-dim)">???? ${escapeHTML(p.doctor_name || '')}</span>
  >
  ${p.is_confidential ? `<span class="badge badge-danger" style="font-size:9px;padding:2px 6px">?
${tr('Confidential', '???? ?????')}</span>` : ''}
  </div>
</div>
`).join('');
}
} catch (err) {
  photoContainer.innerHTML = `<div style="color:red">${tr('Error loading', '??? ?? ?????')}</div>`;
}
};

window.e1AddBurnAssessment = async (pid) => {
  const getVal = (id) => document.getElementById(id)?.value || '0';
  const weight = document.getElementById('e1BurnWeight')?.value;

  if (!weight || parseFloat(weight) <= 0) {
    showToast(tr('Please enter a valid patient weight', '???? ????? ??? ????? ?????'), 'error');
    return;
  }

  // Calculate total TBSA and fluids
  const head = parseFloat(getVal('e1BurnHead'));
  const front = parseFloat(getVal('e1BurnTorsoFront'));
  const back = parseFloat(getVal('e1BurnTorsoBack'));
  const lArm = parseFloat(getVal('e1BurnLeftArm'));
  const rArm = parseFloat(getVal('e1BurnRightArm'));
  const lLeg = parseFloat(getVal('e1BurnLeftLeg'));
  const rLeg = parseFloat(getVal('e1BurnRightLeg'));
  const peri = parseFloat(getVal('e1BurnPerineum'));

  const tbsa = head + front + back + lArm + rArm + lLeg + rLeg + peri;
  if (tbsa <= 0) {
    showToast(tr('TBSA must be greater than 0%', '??? ?? ????? ?????? ????? ?? 0%'), 'error');
    return;
  }

  const totalFluid = 4 * parseFloat(weight) * tbsa;
  const first8h = totalFluid / 2;
  const next16h = totalFluid / 2;

  try {
    await API.post('/api/plastic-burns/assessments', {
      patient_id: pid,

```

```

        weight_kg: weight,
        head_percent: head,
        torso_front_percent: front,
        torso_back_percent: back,
        left_arm_percent: lArm,
        right_arm_percent: rArm,
        left_leg_percent: lLeg,
        right_leg_percent: rLeg,
        perineum_percent: peri,
        tbsa_percent: tbsa,
        parkland_fluid_ml: totalFluid,
        fluid_first_8h_ml: first8h,
        fluid_next_16h_ml: next16h,
        clinical_notes: document.getElementById('e1BurnNotes')?.value || ''
    });

    showToast(tr('Burn assessment saved successfully!', '?? ??? ????? ?????? ??????!'));
    window.e1LoadBurnHistory(pid);

    // Reset inputs
    document.getElementById('e1BurnWeight').value = '';
    document.getElementById('e1BurnNotes').value = '';
    const areas = ['e1BurnHead', 'e1BurnTorsoFront', 'e1BurnTorsoBack', 'e1BurnLeftArm', 'e1BurnRightArm', 'e1BurnLeftLeg', 'e1BurnRightLeg', 'e1BurnPerineum'];
    areas.forEach(a => {
        const el = document.getElementById(a);
        if (el) el.value = '';
    });
    window.e1BurnCalcFluid();

} catch (err) {
    showToast(tr('Error saving burn assessment', '??? ?? ??? ????? ??????'), 'error');
}
};

window.e1AddClinicalPhoto = async (pid) => {
    const bodyRegion = document.getElementById('e1PhotoRegion')?.value;
    if (!bodyRegion) {
        showToast(tr('Please specify the body region', '???? ?????? ?????? ?????? ??????'), 'error');
        return;
    }

    try {
        await API.post('/api/plastic-burns/photos', {
            patient_id: pid,
            body_region: bodyRegion,
            photo_date: document.getElementById('e1PhotoDate')?.value,
            description: document.getElementById('e1PhotoDesc')?.value || '',
            is_confidential: document.getElementById('e1PhotoConfidential')?.checked
        });

        showToast(tr('Clinical photo registered successfully!', '?? ?????? ?????? ?????? ?????????? ??????!'));
        window.e1LoadBurnHistory(pid);
    }

```

```

// Reset
document.getElementById('e1PhotoRegion').value = '';
document.getElementById('e1PhotoDesc').value = '';
document.getElementById('e1PhotoConfidential').checked = true;
document.getElementById('e1PhotoDate').value = new Date().toISOString().slice(0, 10);

} catch (err) {
  showToast(tr('Error registering clinical photo', '??? ?? ????? ?????? ?????? ??????'), 'error');
}
};

// =====
// ===== INTENSIVE CARE MODULE (G17) =====
// =====

window.e1RenderIntensiveCare = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('ICU Clinical Severity Scores', '????? ?????
        ?????? ??????')}</h4>

        <!-- APACHE II Score Section -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('APACHE II Score Calculato
          r', '???? ???? APACHE II')}</legend>

          <div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(200px, 1fr));gap:8px;margin-
          bottom:12px">
            <div class="form-group">
              <label>${tr('Temperature (°C)', '???? ?????? (°?')}</label>
              <select class="form-input" id="e1IcuTemp" onchange="e1IcuCalcApache()">
                <option value="0">36°C - 38.4°C (0)</option>
                <option value="1">38.5°C - 38.9°C (+1)</option>
                <option value="3">39.0°C - 40.9°C (+3)</option>
                <option value="4">?41°C (+4)</option>
                <option value="1">34°C - 35.9°C (+1)</option>
                <option value="2">32°C - 33.9°C (+2)</option>
                <option value="3">30°C - 31.9°C (+3)</option>
                <option value="4">?29.9°C (+4)</option>
              </select>
            </div>

            <div class="form-group">
              <label>${tr('Mean Arterial Pressure (MAP)', '????? ??? ???? ????????? (MAP')}</label>
              <select class="form-input" id="e1IcuMap" onchange="e1IcuCalcApache()">
                <option value="0">70 - 109 mmHg (0)</option>
                <option value="2">110 - 129 mmHg (+2)</option>
                <option value="3">130 - 159 mmHg (+3)</option>
                <option value="4">?160 mmHg (+4)</option>
              </select>
            </div>
          </div>
        </fieldset>
      </div>
    </div>
  `
}

```

```
    <option value="2">50 - 69 mmHg (+2)</option>
    <option value="4">?49 mmHg (+4)</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>${tr('Heart Rate (bpm)', '????? ????? (????/?????)')}</label>
  <select class="form-input" id="e1IcuHr" onchange="e1IcuCalcApache()">
    <option value="0">70 - 109 bpm (0)</option>
    <option value="2">110 - 139 bpm (+2)</option>
    <option value="3">140 - 179 bpm (+3)</option>
    <option value="4">?180 bpm (+4)</option>
    <option value="2">55 - 69 bpm (+2)</option>
    <option value="3">40 - 54 bpm (+3)</option>
    <option value="4">?39 bpm (+4)</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>${tr('Respiratory Rate (bpm)', '???? ?????? (???/?????)')}</label>
  <select class="form-input" id="e1IcuRr" onchange="e1IcuCalcApache()">
    <option value="0">12 - 24 bpm (0)</option>
    <option value="1">25 - 34 bpm (+1)</option>
    <option value="3">35 - 49 bpm (+3)</option>
    <option value="4">?50 bpm (+4)</option>
    <option value="1">10 - 11 bpm (+1)</option>
    <option value="2">6 - 9 bpm (+2)</option>
    <option value="4">?5 bpm (+4)</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>${tr('Oxygenation (PaO2 mmHg)', '????? ????? (PaO2)')}</label>
  <select class="form-input" id="e1IcuPao2" onchange="e1IcuCalcApache()">
    <option value="0">?70 mmHg (0)</option>
    <option value="1">61 - 69 mmHg (+1)</option>
    <option value="3">55 - 60 mmHg (+3)</option>
    <option value="4">&lt;55 mmHg (+4)</option>
  </select>
</div>
```

```
<div class="form-group">
  <label>${tr('Arterial pH', '????? ???? (pH)')}</label>
  <select class="form-input" id="e1IcuPh" onchange="e1IcuCalcApache()">
    <option value="0">7.33 - 7.49 (0)</option>
    <option value="1">7.50 - 7.59 (+1)</option>
    <option value="3">7.60 - 7.69 (+3)</option>
    <option value="4">?7.70 (+4)</option>
    <option value="2">7.25 - 7.32 (+2)</option>
    <option value="3">7.15 - 7.24 (+3)</option>
    <option value="4">&lt;7.15 (+4)</option>
  </select>
</div>
```

```

<div class="form-group">
  <label>${tr('Serum Sodium (Na+ mEq/L)', '?????? ???? (Sodium)')}</label>
  <select class="form-input" id="e1IcuNa" onchange="e1IcuCalcApache()">
    <option value="0">130 - 149 mEq/L (0)</option>
    <option value="1">150 - 154 mEq/L (+1)</option>
    <option value="2">155 - 159 mEq/L (+2)</option>
    <option value="4">?160 mEq/L (+4)</option>
    <option value="2">120 - 129 mEq/L (+2)</option>
    <option value="3">111 - 119 mEq/L (+3)</option>
    <option value="4">?110 mEq/L (+4)</option>
  </select>
</div>

<div class="form-group">
  <label>${tr('Serum Potassium (K+ mEq/L)', '???????? ???? (Potassium)')}</label>
  <select class="form-input" id="e1IcuK" onchange="e1IcuCalcApache()">
    <option value="0">3.5 - 5.4 mEq/L (0)</option>
    <option value="1">5.5 - 5.9 mEq/L (+1)</option>
    <option value="3">6.0 - 6.9 mEq/L (+3)</option>
    <option value="4">?7.0 mEq/L (+4)</option>
    <option value="1">3.0 - 3.4 mEq/L (+1)</option>
    <option value="2">2.5 - 2.9 mEq/L (+2)</option>
    <option value="4">&lt;2.5 mEq/L (+4)</option>
  </select>
</div>

<div class="form-group">
  <label>${tr('Serum Creatinine (mg/dL)', '????????? ???? (Creatinine)')}</label>
  <select class="form-input" id="e1IcuCreatinine" onchange="e1IcuCalcApache()">
    <option value="0">0.6 - 1.4 mg/dL (0)</option>
    <option value="2">1.5 - 1.9 mg/dL (+2)</option>
    <option value="3">2.0 - 3.4 mg/dL (+3)</option>
    <option value="4">?3.5 mg/dL (+4)</option>
    <option value="2">&lt;0.6 mg/dL (+2)</option>
  </select>
</div>

<div class="form-group">
  <label>${tr('Hematocrit (Hct %)', '??? ?????? ?????? (Hct)')}</label>
  <select class="form-input" id="e1IcuHct" onchange="e1IcuCalcApache()">
    <option value="0">30% - 45.9% (0)</option>
    <option value="1">46% - 49.9% (+1)</option>
    <option value="2">50% - 59.9% (+2)</option>
    <option value="4">?60% (+4)</option>
    <option value="2">20% - 29.9% (+2)</option>
    <option value="4">&lt;20% (+4)</option>
  </select>
</div>

<div class="form-group">
  <label>${tr('White Blood Cells (WBC x103)', '????? ???? ??????? (WBC)')}</label>
  <select class="form-input" id="e1IcuWbc" onchange="e1IcuCalcApache()">
    <option value="0">3.0 - 14.9 x103/mcL (0)</option>
    <option value="1">15.0 - 19.9 x103/mcL (+1)</option>

```



```

        <option value="2">20.0 - 39.9 x103/mCL (+2)</option>
        <option value="4">240.0 x103/mCL (+4)</option>
        <option value="2">1.0 - 2.9 x103/mCL (+2)</option>
        <option value="4">&lt;1.0 x103/mCL (+4)</option>
    </select>
</div>

```

```

<div class="form-group">
    <label>${tr('Glasgow Coma Scale (GCS)', '????? ?????? ?????? (GCS)')}</label>
    <select class="form-input" id="e1IcuGcs" onchange="e1IcuCalcApache()">
        <option value="0">GCS 15 (0)</option>
        <option value="1">GCS 14 (+1)</option>
        <option value="2">GCS 13 (+2)</option>
        <option value="3">GCS 12 (+3)</option>
        <option value="4">GCS 11 (+4)</option>
        <option value="5">GCS 10 (+5)</option>
        <option value="6">GCS 9 (+6)</option>
        <option value="7">GCS 8 (+7)</option>
        <option value="8">GCS 7 (+8)</option>
        <option value="9">GCS 6 (+9)</option>
        <option value="10">GCS 5 (+10)</option>
        <option value="11">GCS 4 (+11)</option>
        <option value="12">GCS 3 (+12)</option>
    </select>
</div>

```

```

<div class="form-group">
    <label>${tr('Age Points', '???? ?????')}</label>
    <select class="form-input" id="e1IcuAge" onchange="e1IcuCalcApache()">
        <option value="0">244 years (0)</option>
        <option value="2">45 - 54 years (+2)</option>
        <option value="3">55 - 64 years (+3)</option>
        <option value="5">65 - 74 years (+5)</option>
        <option value="6">75 years (+6)</option>
    </select>
</div>

```

```

<div class="form-group">
    <label>${tr('Chronic Health History', '???????? ?????? ??????')}</label>
    <select class="form-input" id="e1IcuChronic" onchange="e1IcuCalcApache()">
        <option value="0">None / No severe organ insufficiency (0)</option>
        <option value="5">Non-operative or Emergency post-op (+5)</option>
        <option value="2">Elective post-operative (+2)</option>
    </select>
</div>
</div>

```

```

<div style="background:var(--hover,#f8f9fa);padding:12px;border-radius:8px;font-size:13px;border-left:4px solid var(--primary);margin-top:8px">
    <strong>${tr('APACHE II Score:', '???? APACHE II ??????????:')}</strong> <span id="e1ApacheScoreText" style="font-weight:700;font-size:15px">0</span>
    <span id="e1ApacheMortalityText" class="badge badge-warning" style="margin-left:12px;font-size:11px">Mortality: ~4%</span>
</div>

```

```
</fieldset>
```

```
<!-- SOFA Score Section -->
```

```
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
```

```
<legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('SOFA Score Calculator', '
???? ???? SOFA')}</legend>
```

```
<div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(200px, 1fr));gap:8px;margin-bottom:12px">
```

```
<div class="form-group">
```

```
<label>${tr('Respiration (PaO2/FiO2 ratio)', '????? (???? PaO2/FiO2)')}</label>
```

```
<select class="form-input" id="e1SofaResp" onchange="e1SofaCalc()">
```

```
<option value="0">?400 (0)</option>
```

```
<option value="1">&lt;400 (1)</option>
```

```
<option value="2">&lt;300 (2)</option>
```

```
<option value="3">&lt;200 with mechanical ventilation (3)</option>
```

```
<option value="4">&lt;100 with mechanical ventilation (4)</option>
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>${tr('Coagulation (Platelets x103/mcL)', '????? (?????? ??????)')}</label>
```

```
<select class="form-input" id="e1SofaCoag" onchange="e1SofaCalc()">
```

```
<option value="0">?150 (0)</option>
```

```
<option value="1">&lt;150 (1)</option>
```

```
<option value="2">&lt;100 (2)</option>
```

```
<option value="3">&lt;50 (3)</option>
```

```
<option value="4">&lt;20 (4)</option>
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>${tr('Liver (Bilirubin mg/dL)', '???? (?????????)')}</label>
```

```
<select class="form-input" id="e1SofaLiver" onchange="e1SofaCalc()">
```

```
<option value="0">&lt;1.2 (0)</option>
```

```
<option value="1">1.2 - 1.9 (1)</option>
```

```
<option value="2">2.0 - 5.9 (2)</option>
```

```
<option value="3">6.0 - 11.9 (3)</option>
```

```
<option value="4">?12.0 (4)</option>
```

```
</select>
```

```
</div>
```

```
<div class="form-group">
```

```
<label>${tr('Cardiovascular (Hypotension)', '???? ??????? (??? ???/? ??????)')}</label>
```

```
<select class="form-input" id="e1SofaCardio" onchange="e1SofaCalc()">
```

```
<option value="0">No hypotension (0)</option>
```

```
<option value="1">MAP &lt;70 mmHg (1)</option>
```

```
<option value="2">Dopamine ?5 or Dobutamine any dose (2)</option>
```

```
<option value="3">Dopamine >5 OR Epinephrine ?0.1 OR Norepinephrine ?0.1 (3)</option>
```

```
<option value="4">Dopamine >15 OR Epinephrine >0.1 OR Norepinephrine >0.1 (4)</option>
```

```
</select>
```

```
</div>
```

```

<div class="form-group">
  <label>${tr('CNS (Glasgow Coma Scale)', '????? ???? (??? GCS)')}</label>
  <select class="form-input" id="e1SofaCns" onchange="e1SofaCalc()">
    <option value="0">GCS 15 (0)</option>
    <option value="1">GCS 13 - 14 (1)</option>
    <option value="2">GCS 10 - 12 (2)</option>
    <option value="3">GCS 6 - 9 (3)</option>
    <option value="4">GCS <6 (4)</option>
  </select>
</div>

<div class="form-group">
  <label>${tr('Renal (Creatinine mg/dL)', '???? (????????/????)')}</label>
  <select class="form-input" id="e1SofaRenal" onchange="e1SofaCalc()">
    <option value="0"><1.2 (0)</option>
    <option value="1">1.2 - 1.9 (1)</option>
    <option value="2">2.0 - 3.4 (2)</option>
    <option value="3">3.5 - 4.9 or Urine <500 mL/day (3)</option>
    <option value="4">≥5.0 or Urine <200 mL/day (4)</option>
  </select>
</div>
</div>

<div style="background:var(--hover,#f8f9fa);padding:12px;border-radius:8px;font-size:13px;border-left:4px solid #10b981;margin-top:8px">
  <strong>${tr('SOFA Score:', '??? SOFA ??????')}</strong> <span id="e1SofaScoreText" style="font-weight:700;font-size:15px">0</span>
</div>
</fieldset>

<!-- Clinical Notes & Save -->
<div class="form-group mb-8">
  <label>${tr('Clinical Notes / Critical Care Plan', '????? ???? ???? ????')}</label>
  <textarea class="form-input form-textarea" id="e1IcuNotes" rows="2" placeholder="${tr('Describe patient status, ventilation settings, vasopressor titration...', '?? ???? ???? ???? ???? ???? ???? ???? ???? ???? ???? ????...')}" style="min-height:45px"></textarea>
</div>

  <button class="btn btn-primary btn-sm mb-12 w-full" onclick="e1AddIcuAssessment(${safeId(pid)})">? ${tr('Save ICU Assessment', '??? ???? ???? ????')}</button>
</div>

<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('ICU Assessment History', '??? ?????? ?????? ??????')}</h4>
  <div id="e1IcuHistoryList">${tr('Loading...', '???? ??????...')}</div>
</div>
</div>
`;

window.e1LoadIcuHistory(pid);
};

```

```

window.e1IcuCalcApache = () => {
  const getVal = (id) => parseInt(document.getElementById(id)?.value || 0);

  const score = getVal('e1IcuTemp') +
    getVal('e1IcuMap') +
    getVal('e1IcuHr') +
    getVal('e1IcuRr') +
    getVal('e1IcuPao2') +
    getVal('e1IcuPh') +
    getVal('e1IcuNa') +
    getVal('e1IcuK') +
    getVal('e1IcuCreatinine') +
    getVal('e1IcuHct') +
    getVal('e1IcuWbc') +
    getVal('e1IcuGcs') +
    getVal('e1IcuAge') +
    getVal('e1IcuChronic');

  const scoreEl = document.getElementById('e1ApacheScoreText');
  if (scoreEl) scoreEl.innerText = score;

  let mortality = '~4%';
  if (score >= 5 && score <= 9) mortality = '~8%';
  else if (score >= 10 && score <= 14) mortality = '~15%';
  else if (score >= 15 && score <= 19) mortality = '~25%';
  else if (score >= 20 && score <= 24) mortality = '~40%';
  else if (score >= 25 && score <= 29) mortality = '~55%';
  else if (score >= 30 && score <= 34) mortality = '~73%';
  else if (score >= 35) mortality = '~85%';

  const mortalityEl = document.getElementById('e1ApacheMortalityText');
  if (mortalityEl) {
    mortalityEl.innerText = `${tr('Predicted Mortality:', '???? ?????? ??????:')} ${mortality}`;
    if (score >= 25) {
      mortalityEl.className = 'badge badge-danger';
    } else if (score >= 15) {
      mortalityEl.className = 'badge badge-warning';
    } else {
      mortalityEl.className = 'badge badge-success';
    }
  }
};

window.e1SofaCalc = () => {
  const getVal = (id) => parseInt(document.getElementById(id)?.value || 0);

  const score = getVal('e1SofaResp') +
    getVal('e1SofaCoag') +
    getVal('e1SofaLiver') +
    getVal('e1SofaCardio') +
    getVal('e1SofaCns') +
    getVal('e1SofaRenal');

  const scoreEl = document.getElementById('e1SofaScoreText');

```

```

    if (scoreEl) scoreEl.innerText = score;
  };

window.e1LoadIcuHistory = async (pid) => {
  const container = document.getElementById('e1IcuHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/icu/assessments/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No ICU assessments found'
, '?? ???? ?????? ????? ????? ?????')}</div>`;
    } else {
      container.innerHTML = records.map(r => `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4
px solid var(--primary);font-size:12px">
          <div style="font-weight:700;color:var(--primary);display:flex;justify-content:space-between">
            <span>? ${new Date(r.assessment_date).toLocaleDateString('ar-SA')}</span>
          </div>
          <div style="margin-top:6px;display:flex;gap:12px">
            <div><strong>APACHE II:</strong> <span class="badge ${r.apache_ii_score >= 25 ? 'badge-danger' : r
.apache_ii_score >= 15 ? 'badge-warning' : 'badge-success'}" style="font-size:11px">${r.apache_ii_score}</span>
          </div>
            <div><strong>SOFA:</strong> <span class="badge badge-success" style="font-size:11px;background:#10
b981">${r.sofa_score}</span></div>
          </div>
          ${r.clinical_notes ? `<div style="margin-top:6px;color:var(--text-dim);line-height:1.3"><strong>${tr
('Notes:', '???????')}</strong> ${escapeHTML(r.clinical_notes)}</div>` : ''}
          <div style="font-size:10px;color:var(--text-dim);margin-top:4px">??? ${escapeHTML(r.doctor_name ||
'')}</div>
        </div>
      `).join('');
    }
  } catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
  }
};

```

```

window.e1AddIcuAssessment = async (pid) => {
  const getVal = (id) => parseInt(document.getElementById(id)?.value || 0);

  const apache_ii_score = getVal('e1IcuTemp') +
    getVal('e1IcuMap') +
    getVal('e1IcuHr') +
    getVal('e1IcuRr') +
    getVal('e1IcuPao2') +
    getVal('e1IcuPh') +
    getVal('e1IcuNa') +
    getVal('e1IcuK') +
    getVal('e1IcuCreatinine') +
    getVal('e1IcuHct') +
    getVal('e1IcuWbc') +
    getVal('e1IcuGcs') +
    getVal('e1IcuAge') +

```

```

        getVal('e1IcuChronic'));

const sofa_score = getVal('e1SofaResp') +
    getVal('e1SofaCoag') +
    getVal('e1SofaLiver') +
    getVal('e1SofaCardio') +
    getVal('e1SofaCns') +
    getVal('e1SofaRenal');

try {
    await API.post('/api/icu/assessments', {
        patient_id: pid,
        apache_temp: getVal('e1IcuTemp'),
        apache_map: getVal('e1IcuMap'),
        apache_hr: getVal('e1IcuHr'),
        apache_rr: getVal('e1IcuRr'),
        apache_pao2: getVal('e1IcuPao2'),
        apache_ph: getVal('e1IcuPh'),
        apache_na: getVal('e1IcuNa'),
        apache_k: getVal('e1IcuK'),
        apache_creatinine: getVal('e1IcuCreatinine'),
        apache_hct: getVal('e1IcuHct'),
        apache_wbc: getVal('e1IcuWbc'),
        apache_gcs: getVal('e1IcuGcs'),
        apache_age_points: getVal('e1IcuAge'),
        apache_chronic_points: getVal('e1IcuChronic'),
        apache_ii_score,
        sofa_pao2_fio2: getVal('e1SofaResp'),
        sofa_platelets: getVal('e1SofaCoag'),
        sofa_bilirubin: getVal('e1SofaLiver'),
        sofa_map_vasopressor: getVal('e1SofaCardio'),
        sofa_gcs: getVal('e1SofaCns'),
        sofa_creatinine: getVal('e1SofaRenal'),
        sofa_score,
        clinical_notes: document.getElementById('e1IcuNotes')?.value || ''
    });

    showToast(tr('ICU assessment saved successfully!', '?? ?? ????? ??????? ??????? ??????!'));
    window.e1LoadIcuHistory(pid);

    // Reset inputs
    document.getElementById('e1IcuNotes').value = '';
    const selects = ['e1IcuTemp', 'e1IcuMap', 'e1IcuHr', 'e1IcuRr', 'e1IcuPao2', 'e1IcuPh', 'e1IcuNa', 'e1IcuK',
        'e1IcuCreatinine', 'e1IcuHct', 'e1IcuWbc', 'e1IcuGcs', 'e1IcuAge', 'e1IcuChronic', 'e1SofaResp', 'e1SofaCoag', 'e1SofaLiver', 'e1SofaCardio', 'e1SofaCns', 'e1SofaRenal'];
    selects.forEach(s => {
        const el = document.getElementById(s);
        if (el) el.selectedIndex = 0;
    });
    window.e1IcuCalcApache();
    window.e1SofaCalc();

} catch (err) {
    showToast(tr('Error saving ICU assessment', '??? ?? ?? ????? ??????? ???????'), 'error');
}

```

```

    }
};

// =====
// ===== ORTHOPEDICS MODULE (G10) =====
// =====

window.e1RenderOrthopedics = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <!-- Left Column: Forms -->
            <div style="flex:1.5;min-width:320px">
                <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Orthopedic Clinical Tools', '???? ? ? ????
                ?????')}</h4>

                <!-- Joint Range of Motion (ROM) Assessment -->
                <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
                ttom:12px">
                    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Joint Range of Motion (RO
                    M)', '???? ???? ??????? (ROM)')}</legend>

                    <div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(180px, 1fr));gap:8px;margin-
                    bottom:12px">
                        <div class="form-group">
                            <label>${tr('Joint', '????')}</label>
                            <select class="form-input" id="e1RomJoint" onchange="e1RomUpdateMovements()">
                                <option value="Knee">${tr('Knee', '????')}</option>
                                <option value="Hip">${tr('Hip', '????')}</option>
                                <option value="Shoulder">${tr('Shoulder', '????')}</option>
                                <option value="Elbow">${tr('Elbow', '????')}</option>
                                <option value="Ankle">${tr('Ankle', '????')}</option>
                                <option value="Wrist">${tr('Wrist', '????')}</option>
                            </select>
                        </div>

                        <div class="form-group">
                            <label>${tr('Side', '????')}</label>
                            <select class="form-input" id="e1RomSide">
                                <option value="Right">${tr('Right', '????')}</option>
                                <option value="Left">${tr('Left', '????')}</option>
                                <option value="Bilateral">${tr('Bilateral', '????')}</option>
                            </select>
                        </div>

                        <div class="form-group">
                            <label>${tr('Movement', '????')}</label>
                            <select class="form-input" id="e1RomMovement" onchange="e1RomCheckNormal()"></select>
                        </div>

                        <div class="form-group">
                            <label>${tr('Angle (Degrees)', '???? (????')}</label>

```

```

        <input type="number" class="form-input" id="e1RomAngle" value="120" min="0" max="360" oninput="e
1RomCheckNormal()">
    </div>
</div>

<div id="e1RomFeedback" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-s
ize:13px;border-left:4px solid #10b981;margin-bottom:8px">
    <strong>${tr('Normal Range:', '????? ?????:')}</strong> <span id="e1RomNormalText">0-135°</span
>
    <span id="e1RomStatusBadge" class="badge badge-success" style="margin-left:12px;font-size:11px">${
tr('Normal', '????')}</span>
</div>

<button class="btn btn-primary btn-sm w-full" onclick="e1AddRomAssessment(${safeId(pid)})">? ${tr('S
ave ROM Assessment', '??? ????? ???? ?????')}</button>
</fieldset>

<!-- Orthopedic Implant Registry -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Orthopedic Implant Tracki
ng', '???? ?????? ??????? ?????????')}</legend>

    <div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(180px, 1fr));gap:8px;margin-
bottom:12px">
        <div class="form-group">
            <label>${tr('Implant Type', '??? ?????/????')}</label>
            <select class="form-input" id="e1ImplantType">
                <option value="Total Knee Joint">${tr('Total Knee Joint', '???? ???? ????')}</option>
                <option value="Total Hip Joint">${tr('Total Hip Joint', '???? ????')}</option>
                <option value="Plate">${tr('Plate / Orthopedic Plate', '???? ????')}</option>
                <option value="Screw">${tr('Bone Screw', '???? ????')}</option>
                <option value="Rod">${tr('Intramedullary Rod / Nail', '???? ????')}</option>
                <option value="Pin / Wire">${tr('K-Wire / Pin', '???? / ??? ?????')}</option>
                <option value="Other">${tr('Other Implant', '???? ????')}</option>
            </select>
        </div>

        <div class="form-group">
            <label>${tr('Manufacturer', '????? ?????')}</label>
            <input type="text" class="form-input" id="e1ImplantMfg" placeholder="e.g. Zimmer Biomet, Stryker
">
        </div>

        <div class="form-group">
            <label>${tr('Model Name', '??? ?????')}</label>
            <input type="text" class="form-input" id="e1ImplantModel" placeholder="e.g. Persona Knee System"
>
        </div>

        <div class="form-group">
            <label>${tr('Serial Number (S/N)', '???? ?????')}</label>
            <input type="text" class="form-input" id="e1ImplantSn" placeholder="e.g. SN-98483120">
        </div>

```



```

<div class="form-group">
  <label>${tr('Size / Dimensions', '????? / ?????')}</label>
  <input type="text" class="form-input" id="e1ImplantSize" placeholder="e.g. Size 6, L: 120mm">
</div>

<div class="form-group">
  <label>${tr('Batch / Lot Number', '?? ?????')}</label>
  <input type="text" class="form-input" id="e1ImplantLot" placeholder="e.g. LOT-2026-X">
</div>
</div>

<div class="form-group mb-8">
  <label>${tr('Surgical & Placement Notes', '????? ?????? ????? ?????')}</label>
  <textarea class="form-input form-textarea" id="e1ImplantNotes" rows="2" placeholder="${tr('Describe implant placement, stability, or any intraoperative findings...', '?? ??? ?????? ?? ?????? ?? ?? ??? ????? ?????? ??????...')}" style="min-height:45px"></textarea>
</div>

  <button class="btn btn-primary btn-sm w-full" onclick="e1AddImplantRegistry(${safeId(pid)})">? ${tr(
'Register Implant', '???? ?????? ?????')}</button>
</fieldset>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">

  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Orthopedic Patient History', '???? ??????
????')}</h4>

  <div style="margin-bottom:16px">
    <h5 style="margin:0 0 8px;font-weight:700;color:var(--text)">? ${tr('Joint ROM History', '??? ??? ?
????')}</h5>
    <div id="e1RomHistoryList" style="max-height:220px;overflow-y:auto;border:1px solid var(--border-col
or,#e5e7eb);border-radius:8px;padding:8px">
      ${tr('Loading...', '???? ??????...')}
    </div>
  </div>

  <div>
    <h5 style="margin:0 0 8px;font-weight:700;color:var(--text)">? ${tr('Registered Implants', '??? ???
??? ?????')}</h5>
    <div id="e1ImplantHistoryList" style="max-height:220px;overflow-y:auto;border:1px solid var(--border
-color,#e5e7eb);border-radius:8px;padding:8px">
      ${tr('Loading...', '???? ??????...')}
    </div>
  </div>
</div>
</div>
`;

window.e1RomUpdateMovements();
window.e1LoadOrthopedicHistory(pid);
};

```

```

// ROM Normals Database
const ROM_NORMALS = {
  Knee: {
    Flexion: { normal: '0-135°', min: 130 },
    Extension: { normal: '0°', min: 0 }
  },
  Hip: {
    Flexion: { normal: '0-120°', min: 110 },
    Extension: { normal: '0-20°', min: 15 },
    Abduction: { normal: '0-45°', min: 40 },
    Adduction: { normal: '0-30°', min: 25 }
  },
  Shoulder: {
    Flexion: { normal: '0-180°', min: 160 },
    Extension: { normal: '0-60°', min: 50 },
    Abduction: { normal: '0-180°', min: 160 }
  },
  Elbow: {
    Flexion: { normal: '0-150°', min: 140 },
    Extension: { normal: '0°', min: 0 }
  },
  Ankle: {
    Dorsiflexion: { normal: '0-20°', min: 15 },
    Plantarflexion: { normal: '0-50°', min: 40 }
  },
  Wrist: {
    Flexion: { normal: '0-80°', min: 70 },
    Extension: { normal: '0-70°', min: 60 }
  }
};

window.e1RomUpdateMovements = () => {
  const joint = document.getElementById('e1RomJoint').value;
  const movementSel = document.getElementById('e1RomMovement');
  if (!movementSel) return;

  const movements = Object.keys(ROM_NORMALS[joint] || {});
  movementSel.innerHTML = movements.map(m => `<option value="${m}">${tr(m, m)}</option>`).join('');

  window.e1RomCheckNormal();
};

window.e1RomCheckNormal = () => {
  const joint = document.getElementById('e1RomJoint').value;
  const movement = document.getElementById('e1RomMovement').value;
  const angle = parseInt(document.getElementById('e1RomAngle').value || 0);

  const rule = ROM_NORMALS[joint]?.[movement];
  if (!rule) return;

  const feedbackEl = document.getElementById('e1RomFeedback');
  const normalTextEl = document.getElementById('e1RomNormalText');
  const badgeEl = document.getElementById('e1RomStatusBadge');

```

```

if (normalTextEl) normalTextEl.innerText = rule.normal;

const isRestricted = angle < rule.min;

if (badgeEl) {
  if (isRestricted) {
    badgeEl.className = 'badge badge-danger';
    badgeEl.innerText = tr('Restricted ROM', '???? ???? ?????');
    if (feedbackEl) feedbackEl.style.borderLeftColor = 'red';
  } else {
    badgeEl.className = 'badge badge-success';
    badgeEl.innerText = tr('Normal', '?????');
    if (feedbackEl) feedbackEl.style.borderLeftColor = '#10b981';
  }
}
};

window.e1LoadOrthopedicHistory = async (pid) => {
  const romContainer = document.getElementById('e1RomHistoryList');
  const implantContainer = document.getElementById('e1ImplantHistoryList');

  try {
    // 1. Load ROM History
    if (romContainer) {
      const romRecords = await API.get('/api/orthopedics/rom/patient/' + pid);
      if (!romRecords.length) {
        romContainer.innerHTML = `<div style="color:var(--text-dim);font-size:12px">${tr('No ROM records found', '?? ???? ????? ????? ?????')}</div>`;
      } else {
        romContainer.innerHTML = romRecords.map(r => `
          <div style="padding:6px;margin:4px 0;border-radius:6px;background:var(--hover,#f8f9fa);font-size:11px;border-right:3px solid ${r.is_restricted ? 'red' : '#10b981'}">
            <strong>${tr(r.joint_name, r.joint_name)} (${tr(r.lateral_side, r.lateral_side)})</strong> - ${tr(r.movement_type, r.movement_type)}
            <div style="margin-top:2px;display:flex;justify-content:space-between">
              <span>${r.angle_degrees}°</span>
              <span class="badge ${r.is_restricted ? 'badge-danger' : 'badge-success'}" style="font-size:9px">
                ${r.is_restricted ? tr('Restricted', '????') : tr('Normal', '????')}</span>
            </div>
            <div style="font-size:9px;color:var(--text-dim);margin-top:2px">? ${new Date(r.assessment_date).toLocaleDateString('ar-SA')} | ??? ${escapeHTML(r.doctor_name || '')}</div>
          </div>
        `).join('');
      }
    }

    // 2. Load Implant History
    if (implantContainer) {
      const implantRecords = await API.get('/api/orthopedics/implants/patient/' + pid);
      if (!implantRecords.length) {
        implantContainer.innerHTML = `<div style="color:var(--text-dim);font-size:12px">${tr('No registered implants found', '?? ???? ????? ????? ?????')}</div>`;
      } else {

```

```

    implantContainer.innerHTML = implantRecords.map(r => `
      <div style="padding:6px;margin:4px 0;border-radius:6px;background:var(--hover,#f8f9fa);font-size:11px;border-right:3px solid var(--primary)">
        <strong>${escapeHTML(r.Implant_type)}</strong>
        <div style="margin-top:2px;color:var(--text-dim)">
          Mfg: ${escapeHTML(r.manufacturer)} | Model: ${escapeHTML(r.model_name)}<br>
          S/N: <code>${escapeHTML(r.serial_number)}</code> | Size: ${escapeHTML(r.size_dimension)}
        </div>
        ${r.clinical_notes ? `<div style="margin-top:2px;color:var(--text-dim);font-style:italic"> "${escapeHTML(r.clinical_notes)}" </div>` : ''}
        <div style="font-size:9px;color:var(--text-dim);margin-top:2px">? ${new Date(r.Implant_date).toLocaleDateString('ar-SA')} | ???? ${escapeHTML(r.doctor_name || '')}</div>
      </div>
    `).join('');
  }
}
} catch (err) {
  if (romContainer) romContainer.innerHTML = `<div style="color:red;font-size:12px">Error</div>`;
  if (implantContainer) implantContainer.innerHTML = `<div style="color:red;font-size:12px">Error</div>`;
}
};

```

```

window.e1AddRomAssessment = async (pid) => {
  const joint_name = document.getElementById('e1RomJoint').value;
  const lateral_side = document.getElementById('e1RomSide').value;
  const movement_type = document.getElementById('e1RomMovement').value;
  const angle_degrees = parseInt(document.getElementById('e1RomAngle').value || 0);

  const rule = ROM_NORMALS[joint_name]?.[movement_type];
  const is_restricted = rule ? angle_degrees < rule.min : false;

  try {
    await API.post('/api/orthopedics/rom', {
      patient_id: pid,
      joint_name,
      lateral_side,
      movement_type,
      angle_degrees,
      is_restricted
    });

    showToast(tr('ROM assessment saved successfully!', '?? ?? ???? ???? ?????? ?????!'));
    window.e1LoadOrthopedicHistory(pid);
  } catch (err) {
    showToast(tr('Error saving ROM assessment', '??? ?? ??? ?????? ????? ??????'), 'error');
  }
};

```

```

window.e1AddImplantRegistry = async (pid) => {
  const implant_type = document.getElementById('e1ImplantType').value;
  const manufacturer = document.getElementById('e1ImplantMfg').value.trim();
  const model_name = document.getElementById('e1ImplantModel').value.trim();
  const serial_number = document.getElementById('e1ImplantSn').value.trim();
  const size_dimension = document.getElementById('e1ImplantSize').value.trim();

```

```

const batch_lot_number = document.getElementById('e1ImplantLot').value.trim();
const clinical_notes = document.getElementById('e1ImplantNotes').value.trim();

if (!manufacturer || !serial_number) {
    return showToast(tr('Manufacturer and Serial Number are required', '????? ?????? ?????? ??????? ??????'), 'error');
}

try {
    await API.post('/api/orthopedics/implants', {
        patient_id: pid,
        implant_type,
        manufacturer,
        model_name,
        serial_number,
        size_dimension,
        batch_lot_number,
        clinical_notes
    });

    showToast(tr('Implant registered successfully!', '?? ????? ?????? ?????? ?????!'));
    window.e1LoadOrthopedicHistory(pid);

    // Reset
    document.getElementById('e1ImplantMfg').value = '';
    document.getElementById('e1ImplantModel').value = '';
    document.getElementById('e1ImplantSn').value = '';
    document.getElementById('e1ImplantSize').value = '';
    document.getElementById('e1ImplantLot').value = '';
    document.getElementById('e1ImplantNotes').value = '';
} catch (err) {
    showToast(tr('Error registering implant', '??? ?? ?????? ?????? ??????'), 'error');
}

};

// =====
// ===== OPHTHALMOLOGY MODULE (G14) =====
// =====

window.e1RenderOphthalmology = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
<div style="display:flex;gap:16px;flex-wrap:wrap">
    <!-- Left Column: Forms -->
    <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">?? ${tr('Ophthalmology Examination', '??? ?? ?????? ?
?????')}</h4>

        <!-- Visual Acuity & Refraction -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">
            <legend style="padding:0 8px;font-weight:700;color:var(--primary)">?? ${tr('Visual Acuity & Refracti

```

```
on', '??? ?????? ??? ??????')}</legend>

<div style="display:grid;grid-template-columns:1fr 1fr;gap:16px;margin-bottom:12px">
  <!-- Right Eye (OD) -->
  <div style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;border-right:4px solid
var(--primary)">
    <h5 style="margin:0 0 8px;font-weight:700;color:var(--primary)">?? ${tr('Right Eye (OD)', '????
????? (OD)')}</h5>
    <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px">
      <div class="form-group">
        <label>${tr('Uncorrected VA', '???? ???? ?????')}</label>
        <select class="form-input" id="e10phVa0dUnread">
          <option value="20/20">20/20</option>
          <option value="20/25">20/25</option>
          <option value="20/30">20/30</option>
          <option value="20/40">20/40</option>
          <option value="20/50">20/50</option>
          <option value="20/70">20/70</option>
          <option value="20/100">20/100</option>
          <option value="20/200">20/200</option>
          <option value="CF">${tr('CF (Count Fingers)', '?? ??????')}</option>
          <option value="HM">${tr('HM (Hand Motion)', '???? ?????')}</option>
          <option value="LP">${tr('LP (Light Perception)', '????? ?????')}</option>
          <option value="NLP">${tr('NLP (No Light)', '??? ????? ?????')}</option>
        </select>
      </div>
      <div class="form-group">
        <label>${tr('Corrected VA', '???? ?????????')}</label>
        <select class="form-input" id="e10phVa0dCorrected">
          <option value="20/20">20/20</option>
          <option value="20/25">20/25</option>
          <option value="20/30">20/30</option>
          <option value="20/40">20/40</option>
          <option value="20/50">20/50</option>
          <option value="20/70">20/70</option>
          <option value="20/100">20/100</option>
          <option value="20/200">20/200</option>
          <option value="CF">CF</option>
          <option value="HM">HM</option>
          <option value="LP">LP</option>
          <option value="NLP">NLP</option>
        </select>
      </div>
    </div>
  <div style="display:grid;grid-template-columns:repeat(4, 1fr);gap:6px;margin-top:8px">
    <div class="form-group">
      <label style="font-size:10px">SPH</label>
      <input type="number" step="0.25" class="form-input" id="e10phSph0d" value="0.00" style="padd
ing:4px">
    </div>
    <div class="form-group">
      <label style="font-size:10px">CYL</label>
      <input type="number" step="0.25" class="form-input" id="e10phCyl0d" value="0.00" style="padd
ing:4px">
    </div>
  </div>
</div>
```

```

    </div>
    <div class="form-group">
      <label style="font-size:10px">AXIS</label>
      <input type="number" class="form-input" id="e10phAxis0d" value="0" min="0" max="180" style="padding:4px">
    </div>
    <div class="form-group">
      <label style="font-size:10px">ADD</label>
      <input type="number" step="0.25" class="form-input" id="e10phAdd0d" value="0.00" style="padding:4px">
    </div>
  </div>
</div>

<!-- Left Eye (OS) -->
<div style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;border-right:4px solid #10b981">
  <h5 style="margin:0 0 8px;font-weight:700;color:#10b981">?? ${tr('Left Eye (OS)', '????? ?????? (OS)')}</h5>
  <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px">
    <div class="form-group">
      <label>${tr('Uncorrected VA', '????? ???? ?????')}</label>
      <select class="form-input" id="e10phVa0sUnread">
        <option value="20/20">20/20</option>
        <option value="20/25">20/25</option>
        <option value="20/30">20/30</option>
        <option value="20/40">20/40</option>
        <option value="20/50">20/50</option>
        <option value="20/70">20/70</option>
        <option value="20/100">20/100</option>
        <option value="20/200">20/200</option>
        <option value="CF">${tr('CF (Count Fingers)', '?? ???????')}</option>
        <option value="HM">${tr('HM (Hand Motion)', '???? ?????')}</option>
        <option value="LP">${tr('LP (Light Perception)', '????? ?????')}</option>
        <option value="NLP">${tr('NLP (No Light)', '??? ?????? ?????')}</option>
      </select>
    </div>
    <div class="form-group">
      <label>${tr('Corrected VA', '????? ?????????')}</label>
      <select class="form-input" id="e10phVa0sCorrected">
        <option value="20/20">20/20</option>
        <option value="20/25">20/25</option>
        <option value="20/30">20/30</option>
        <option value="20/40">20/40</option>
        <option value="20/50">20/50</option>
        <option value="20/70">20/70</option>
        <option value="20/100">20/100</option>
        <option value="20/200">20/200</option>
        <option value="CF">CF</option>
        <option value="HM">HM</option>
        <option value="LP">LP</option>
        <option value="NLP">NLP</option>
      </select>
    </div>
  </div>

```

```

    </div>
    <div style="display:grid;grid-template-columns:repeat(4, 1fr);gap:6px;margin-top:8px">
      <div class="form-group">
        <label style="font-size:10px">SPH</label>
        <input type="number" step="0.25" class="form-input" id="e10phSph0s" value="0.00" style="padding:4px">
      </div>
      <div class="form-group">
        <label style="font-size:10px">CYL</label>
        <input type="number" step="0.25" class="form-input" id="e10phCyl0s" value="0.00" style="padding:4px">
      </div>
      <div class="form-group">
        <label style="font-size:10px">AXIS</label>
        <input type="number" class="form-input" id="e10phAxis0s" value="0" min="0" max="180" style="padding:4px">
      </div>
      <div class="form-group">
        <label style="font-size:10px">ADD</label>
        <input type="number" step="0.25" class="form-input" id="e10phAdd0s" value="0.00" style="padding:4px">
      </div>
    </div>
  </div>
</div>
</div>
</div>
</div>
</fieldset>

<!-- Intraocular Pressure (IOP) -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
  <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Intraocular Pressure (IOP)', '??? ????? (IOP)')}</legend>

  <div style="display:grid;grid-template-columns:repeat(auto-fill, minmax(140px, 1fr));gap:8px;margin-bottom:12px">
    <div class="form-group">
      <label>${tr('IOP OD (mmHg)', '??? ????? ??????')}</label>
      <input type="number" class="form-input" id="e10phIop0d" value="15" min="0" max="80" oninput="e10phCheckIop()">
    </div>
    <div class="form-group">
      <label>${tr('IOP OS (mmHg)', '??? ????? ??????')}</label>
      <input type="number" class="form-input" id="e10phIop0s" value="15" min="0" max="80" oninput="e10phCheckIop()">
    </div>
    <div class="form-group">
      <label>${tr('Measurement Method', '????? ??????')}</label>
      <select class="form-input" id="e10phIopMethod">
        <option value="GAT">Goldmann Applanation Tonometer (GAT)</option>
        <option value="Air-puff">${tr('Air-puff Tonometer', '???? ???? ??????')}</option>
        <option value="Tonopen">Tonopen</option>
        <option value="iCare">iCare Tonometer</option>
      </select>
    </div>
  </div>
</fieldset>

```



```

</div>

<div id="e1IopFeedback" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-size:13px;border-left:4px solid #10b981;margin-bottom:8px">
    <strong>${tr('Normal Range:', '????? ??????:')}</strong> 10-21 mmHg
    <span id="e1IopStatusBadge" class="badge badge-success" style="margin-left:12px;font-size:11px">${tr('Normal', '????')}</span>
</div>
</fieldset>

<!-- Slit Lamp & Fundoscopy -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Clinical Examinations', '??????? ???????? ?????')}</legend>

    <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px">
        <div class="form-group">
            <label>${tr('Slit Lamp Examination', '??? ?????? ?????')}</label>
            <textarea class="form-input form-textarea" id="e10phSlitExam" rows="2" placeholder="${tr('Cornea , anterior chamber, iris, lens...', '??????? ?????? ?????????? ????????? ??????...')}" style="min-height:45px">
</textarea>
        </div>
        <div class="form-group">
            <label>${tr('Fundus Examination (Fundoscopy)', '??? ??? ?????? ?????????')}</label>
            <textarea class="form-input form-textarea" id="e10phFundusExam" rows="2" placeholder="${tr('Optic disc, macula, vessels, retina...', '????? ????????? ?????? ?????????? ????????? ?????????...')}" style="min-height:45px">
</textarea>
        </div>
    </div>
</fieldset>

<div class="form-group mb-8">
    <label>${tr('Clinical Notes / Diagnosis', '??????? ?????? ?????????')}</label>
    <textarea class="form-input form-textarea" id="e10phNotes" rows="2" placeholder="${tr('Any additional findings or diagnostic notes...', '?? ?????? ?????? ?? ?????? ?????????...')}" style="min-height:45px">
</textarea>
</div>

<button class="btn btn-primary btn-sm w-full mb-12" onclick="e1AddEyeExam(${safeId(pid)})">? ${tr('Save Eye Exam', '??? ??? ?????')}</button>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Eye Exams History', '??? ?????? ?????')}</h4>
    <div id="e10phHistoryList">${tr('Loading...', '???? ?????????...')}</div>
</div>
</div>
`;

window.e1Load0phHistory(pid);
};

```

```

window.e10phCheckIop = () => {
  const iop0d = parseInt(document.getElementById('e10phIop0d').value || 0);
  const iop0s = parseInt(document.getElementById('e10phIop0s').value || 0);

  const feedbackEl = document.getElementById('e1IopFeedback');
  const badgeEl = document.getElementById('e1IopStatusBadge');

  const isHigh = iop0d > 21 || iop0s > 21;

  if (badgeEl) {
    if (isHigh) {
      badgeEl.className = 'badge badge-danger';
      badgeEl.innerText = tr('High IOP / Glaucoma Risk', '??? ??? ????? / ??? ??????????');
      if (feedbackEl) feedbackEl.style.borderLeftColor = 'red';
    } else {
      badgeEl.className = 'badge badge-success';
      badgeEl.innerText = tr('Normal', '?????');
      if (feedbackEl) feedbackEl.style.borderLeftColor = '#10b981';
    }
  }
};

window.e1LoadOphHistory = async (pid) => {
  const container = document.getElementById('e10phHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/ophthalmology/exams/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `
```

```

px">${r.os_iop} mmHg</span>
    </div>
</div>
    ${r.slit_lamp_exam ? `<div style="margin-top:4px;color:var(--text-dim)"><strong>${tr('Slit Lamp:', '
?????? ?????:')}</strong> ${escapeHTML(r.slit_lamp_exam)}</div>` : ''}
    ${r.fundoscopy_exam ? `<div style="margin-top:2px;color:var(--text-dim)"><strong>${tr('Fundus:', '??
? ??????:')}</strong> ${escapeHTML(r.fundoscopy_exam)}</div>` : ''}
    ${r.notes ? `<div style="margin-top:2px;color:var(--text-dim);font-style:italic">"${escapeHTML(r.notes)}"</div>` : ''}
    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
    </div>
    `).join('');
}
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '?? ? ???? ????')}</div>`;
}
};

window.e1AddEyeExam = async (pid) => {
    const getVal = (id) => parseFloat(document.getElementById(id)?.value || 0);
    const getStr = (id) => document.getElementById(id)?.value || '';

    try {
        await API.post('/api/ophthalmology/exams', {
            patient_id: pid,
            od_va_uncorrected: getStr('e10phVa0dUnread'),
            os_va_uncorrected: getStr('e10phVa0sUnread'),
            od_va_corrected: getStr('e10phVa0dCorrected'),
            os_va_corrected: getStr('e10phVa0sCorrected'),
            od_iop: parseInt(document.getElementById('e10phIop0d')?.value || 15),
            os_iop: parseInt(document.getElementById('e10phIop0s')?.value || 15),
            iop_method: getStr('e10phIopMethod'),
            od_sphere: getVal('e10phSph0d'),
            os_sphere: getVal('e10phSph0s'),
            od_cylinder: getVal('e10phCyl0d'),
            os_cylinder: getVal('e10phCyl0s'),
            od_axis: parseInt(document.getElementById('e10phAxis0d')?.value || 0),
            os_axis: parseInt(document.getElementById('e10phAxis0s')?.value || 0),
            od_add: getVal('e10phAdd0d'),
            os_add: getVal('e10phAdd0s'),
            slit_lamp_exam: getStr('e10phSlitExam'),
            fundoscopy_exam: getStr('e10phFundusExam'),
            notes: getStr('e10phNotes')
        });

        showToast(tr('Eye exam saved successfully!', '?? ??? ??? ????? ?????!'));
        window.e1LoadOphHistory(pid);

        // Reset fields
        document.getElementById('e10phSlitExam').value = '';
        document.getElementById('e10phFundusExam').value = '';
        document.getElementById('e10phNotes').value = '';
        document.getElementById('e10phSph0d').value = '0.00';
    }
}

```

```

document.getElementById('e10phSph0s').value = '0.00';
document.getElementById('e10phCyl0d').value = '0.00';
document.getElementById('e10phCyl0s').value = '0.00';
document.getElementById('e10phAxis0d').value = '0';
document.getElementById('e10phAxis0s').value = '0';
document.getElementById('e10phAdd0d').value = '0.00';
document.getElementById('e10phAdd0s').value = '0.00';
document.getElementById('e10phIop0d').value = '15';
document.getElementById('e10phIop0s').value = '15';
window.e10phCheckIop();
} catch (err) {
  showToast(tr('Error saving eye exam', '??? ?? ??? ??? ?????'), 'error');
}
};

// =====
// ===== GENERAL SURGERY MODULE (G09) =====
// =====

window.e1RenderGeneralSurgery = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Surgical Management', '????? ????????? ?????????
?')}</h4>

        <!-- WHO Surgical Safety Checklist -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('WHO Surgical Safety Check
list', '????? ?????? ?? ????? ????????? ?????????')}</legend>

          <div class="form-group mb-8">
            <label>${tr('Procedure Name', '??? ??????? ?????????')}</label>
            <input type="text" class="form-input" id="e1SurgProcName" placeholder="e.g. Laparoscopic Cholecyst
ectomy">
          </div>

          <div style="display:grid;grid-template-columns:1fr;gap:8px;margin-bottom:8px">
            <label style="display:flex;align-items:center;gap:8px;font-size:13px">
              <input type="checkbox" id="e1SurgSignIn">
              <span><strong>${tr('Sign In (Before Anesthesia)', '????? ?????? (??? ?????????')}</strong>: ${tr(
'Patient identity, site, consent, and safety checks confirmed.', '?? ?????? ???? ?????? ????????? ?????????
?? ??????')}</span>
            </label>
            <label style="display:flex;align-items:center;gap:8px;font-size:13px">
              <input type="checkbox" id="e1SurgTimeOut">
              <span><strong>${tr('Time Out (Before Incision)', '????? ?????? (??? ???? ?????????')}</strong>: ${
tr('All team members introduced; procedure and site re-confirmed.', '?? ?????? ???? ?????? ?????? ?????? ??????
????????')}</span>

```

```

        </label>
        <label style="display:flex;align-items:center;gap:8px;font-size:13px">
            <input type="checkbox" id="e1SurgSignOut">
            <span><strong>${tr('Sign Out (Before Patient Leaves OR)', '???? ????? (?? ????? ????? ????)')}</strong>: ${tr('Instrument count, specimen labeling, and equipment checks done.', '?? ?? ?????? ?????? ?? ?????? ???????? ??? ??????')}</span>
        </label>
    </div>

    <div class="form-group mb-8">
        <label>${tr('Surgical Notes', '?????? ?????')}</label>
        <textarea class="form-input form-textarea" id="e1SurgNotes" rows="2" style="min-height:45px"></textarea>
    </div>

    <button class="btn btn-primary btn-sm w-full" onclick="e1AddSurgicalChecklist(${safeId(pid)})">? ${tr('Save Checklist', '??? ????? ?????')}</button>
</fieldset>

<!-- Intraoperative Time Logger -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">?? ${tr('Intraoperative Time Logger', '???? ????? ???????? ????????')}</legend>

    <div class="form-group mb-8">
        <label>${tr('Active Procedure', '?????? ?????')}</label>
        <input type="text" class="form-input" id="e1TimeLogProcName" placeholder="e.g. Appendectomy">
    </div>

    <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-bottom:12px">
        <div class="form-group">
            <label>${tr('Anesthesia Start', '??? ??????')}</label>
            <div style="display:flex;gap:4px">
                <input type="text" class="form-input" id="e1AnesStart" readonly placeholder="Not Set" style="font-size:11px">
                <button class="btn btn-secondary btn-sm" onclick="e1SetTime('e1AnesStart')">??</button>
            </div>
        </div>
        <div class="form-group">
            <label>${tr('Surgical Incision', '???? ??????')}</label>
            <div style="display:flex;gap:4px">
                <input type="text" class="form-input" id="e1IncisionTime" readonly placeholder="Not Set" style="font-size:11px">
                <button class="btn btn-secondary btn-sm" onclick="e1SetTime('e1IncisionTime')">??</button>
            </div>
        </div>
        <div class="form-group">
            <label>${tr('Wound Closure', '???? ?????')}</label>
            <div style="display:flex;gap:4px">
                <input type="text" class="form-input" id="e1ClosureTime" readonly placeholder="Not Set" style="font-size:11px">
                <button class="btn btn-secondary btn-sm" onclick="e1SetTime('e1ClosureTime')">??</button>
            </div>
        </div>
    </div>

```

```

    </div>
    <div class="form-group">
      <label>${tr('Anesthesia End', '???? ?????')}</label>
      <div style="display:flex;gap:4px">
        <input type="text" class="form-input" id="e1AnesEnd" readonly placeholder="Not Set" style="font-size:11px">
        <button class="btn btn-secondary btn-sm" onclick="e1SetTime('e1AnesEnd')">??</button>
      </div>
    </div>
  </div>

  <button class="btn btn-primary btn-sm w-full" onclick="e1AddSurgicalTimeLog(${safeId(pid)})">? ${tr(
'Save Time Log', '??? ??? ?????')}</button>
</fieldset>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Surgical History', '??? ?????? ?????')}</h4>
  <div id="e1SurgHistoryList">${tr('Loading...', '??? ??????...')}</div>
</div>
</div>
`;

window.e1LoadSurgHistory(pid);
};

window.e1SetTime = (id) => {
  const el = document.getElementById(id);
  if (el) {
    el.value = new Date().toISOString();
  }
};

window.e1LoadSurgHistory = async (pid) => {
  const container = document.getElementById('e1SurgHistoryList');
  if (!container) return;

  try {
    const [checklists, timelogs] = await Promise.all([
      API.get('/api/surgery/checklists/patient/' + pid),
      API.get('/api/surgery/timelogs/patient/' + pid)
    ]);

    let html = '';

    if (checklists.length || timelogs.length) {
      if (checklists.length) {
        html += `<h5 style="margin:8px 0 4px;color:var(--primary)">? ${tr('Safety Checklists', '????? ?????? ?
????? ??????')}</h5>`;
        html += checklists.map(c => `
          <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right

```

```

:4px solid var(--primary);font-size:12px">
    <div style="font-weight:700;color:var(--primary)">${escapeHTML(c.procedure_name)}</div>
    <div style="margin:4px 0">
        <span class="badge ${c.sign_in_confirmed ? 'badge-success' : 'badge-danger'}">Sign In</span>
        <span class="badge ${c.time_out_confirmed ? 'badge-success' : 'badge-danger'}">Time Out</span>
        <span class="badge ${c.sign_out_confirmed ? 'badge-success' : 'badge-danger'}">Sign Out</span>
    </div>
    ${c.notes ? `<div style="color:var(--text-dim);font-style:italic">${escapeHTML(c.notes)}</div>`
: ''}

    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">? ${new Date(c.surgery_date).toLocaleDateString('ar-SA')} | ??? ${escapeHTML(c.doctor_name || '')}</div>
    </div>
    `).join('');
}

if (timelogs.length) {
    html += `<h5 style="margin:12px 0 4px;color:var(--primary)">?? ${tr('Intraoperative Time Logs', '?????
????? ?????????')}</h5>`;
    html += timelogs.map(t => {
        const formatTime = (isoStr) => isoStr ? new Date(isoStr).toLocaleTimeString('ar-SA', { hour: '2-digit', minute: '2-digit', second: '2-digit' }) : '-';
        return `
            <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid #10b981;font-size:12px">
                <div style="font-weight:700;color:#10b981">${escapeHTML(t.procedure_name)}</div>
                <div style="margin-top:4px;display:grid;grid-template-columns:1fr 1fr;gap:4px;font-size:11px">
                    <div>? Anes Start: ${formatTime(t.anesthesia_start_time)}</div>
                    <div>? Incision: ${formatTime(t.incision_time)}</div>
                    <div>? Closure: ${formatTime(t.closure_time)}</div>
                    <div>? Anes End: ${formatTime(t.anesthesia_end_time)}</div>
                </div>
                <div style="font-size:10px;color:var(--text-dim);margin-top:4px">? ${new Date(t.created_at).toLocaleDateString('ar-SA')} | ??? ${escapeHTML(t.doctor_name || '')}</div>
            </div>
            `;
    }).join('');
}
} else {
    html = `<div style="color:var(--text-dim);font-size:13px">${tr('No surgical records found', '?? ???? ??
?? ?????? ??????')}</div>`;
}

container.innerHTML = html;
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
}
};

window.e1AddSurgicalChecklist = async (pid) => {
    const procName = document.getElementById('e1SurgProcName').value;
    if (!procName) {
        showToast(tr('Procedure name is required', '??? ?????? ?????? ??????'), 'error');
        return;
    }
}

```

```

try {
  await API.post('/api/surgery/checklists', {
    patient_id: pid,
    procedure_name: procName,
    sign_in_confirmed: document.getElementById('e1SurgSignIn').checked,
    time_out_confirmed: document.getElementById('e1SurgTimeOut').checked,
    sign_out_confirmed: document.getElementById('e1SurgSignOut').checked,
    notes: document.getElementById('e1SurgNotes').value
  });

  showToast(tr('Surgical checklist saved successfully!', '?? ??? ????? ????? ?????? ?????!'));
  window.e1LoadSurgHistory(pid);

  document.getElementById('e1SurgProcName').value = '';
  document.getElementById('e1SurgSignIn').checked = false;
  document.getElementById('e1SurgTimeOut').checked = false;
  document.getElementById('e1SurgSignOut').checked = false;
  document.getElementById('e1SurgNotes').value = '';
} catch (err) {
  showToast(tr('Error saving checklist', '??? ?? ??? ????? ????? ??????'), 'error');
}
};

window.e1AddSurgicalTimeLog = async (pid) => {
  const procName = document.getElementById('e1TimeLogProcName').value;
  if (!procName) {
    showToast(tr('Procedure name is required', '??? ?????? ?????? ?????'), 'error');
    return;
  }

  try {
    await API.post('/api/surgery/timelogs', {
      patient_id: pid,
      procedure_name: procName,
      anesthesia_start_time: document.getElementById('e1AnesStart').value || null,
      incision_time: document.getElementById('e1IncisionTime').value || null,
      closure_time: document.getElementById('e1ClosureTime').value || null,
      anesthesia_end_time: document.getElementById('e1AnesEnd').value || null
    });

    showToast(tr('Surgical time log saved successfully!', '?? ??? ??? ?????? ?????? ?????!'));
    window.e1LoadSurgHistory(pid);

    document.getElementById('e1TimeLogProcName').value = '';
    document.getElementById('e1AnesStart').value = '';
    document.getElementById('e1IncisionTime').value = '';
    document.getElementById('e1ClosureTime').value = '';
    document.getElementById('e1AnesEnd').value = '';
  } catch (err) {
    showToast(tr('Error saving time log', '??? ?? ??? ??? ?????? ??????'), 'error');
  }
};

```



```

// =====
// ===== UROLOGY MODULE (G15) =====
// =====

window.e1RenderUrology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Urodynamic Studies & Diagnostics', '????? ?
????? ???? ????')}</h4>

        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Urodynamic Parameters',
'??????? ????')}</legend>

          <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-bottom:12px">
            <div class="form-group">
              <label>${tr('Max Flow Rate (Qmax - mL/s)', '??? ????')}</label>
              <input type="number" step="0.1" class="form-input" id="e1UroFlowRate" value="18" oninput="e1UroC
heckParams()">
            </div>
            <div class="form-group">
              <label>${tr('Voided Volume (mL)', '????')}</label>
              <input type="number" class="form-input" id="e1UroVolume" value="250">
            </div>
            <div class="form-group">
              <label>${tr('Post-Void Residual (PVR - mL)', '???')}</label>
              <input type="number" class="form-input" id="e1UroPvr" value="20" oninput="e1UroCheckParams()">
            </div>
            <div class="form-group">
              <label>${tr('Detrusor Pressure (pdet - cmH2O)', '???')}</label>
              <input type="number" class="form-input" id="e1UroPressure" value="30">
            </div>
          </div>

          <!-- Dynamic Alerts & Indicators -->
          <div id="e1UroFlowAlert" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-s
ize:13px;border-left:4px solid #10b981;margin-bottom:8px;display:flex;justify-content:space-between;align-ite
ms:center">
            <span><strong>Qmax Status:</strong> <span id="e1UroFlowStatusText">${tr('Normal (>15 mL/s)', '???
?')}</span></span>
            <span id="e1UroFlowBadge" class="badge badge-success">${tr('Normal', '????')}</span>
          </div>

          <div id="e1UroPvrAlert" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-s
ize:13px;border-left:4px solid #10b981;margin-bottom:8px;display:flex;justify-content:space-between;align-ite
ms:center">
            <span><strong>PVR Status:</strong> <span id="e1UroPvrStatusText">${tr('Normal (<50 mL)', '????')}

```

```

</span></span>
    <span id="e1UroPvrBadge" class="badge badge-success">${tr('Normal', '????')}</span>
  </div>
</fieldset>

  <div class="form-group mb-8">
    <label>${tr('Clinical Interpretation', '?????? ???? ????')}</label>
    <textarea class="form-input form-textarea" id="e1UroInterpretation" rows="2" placeholder="${tr('Any
diagnostic findings, e.g., Bladder Outlet Obstruction (B00)...', '?? ???? ???? ???? ???? ???? ????...
')}" style="min-height:45px"></textarea>
  </div>

  <button class="btn btn-primary btn-sm w-full mb-12" onclick="e1AddUroStudy(${safeId(pid)})">? ${tr('Sa
ve Urodynamic Study', '?? ???? ???? ????')}</button>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Urodynamic History', '?? ???? ????')}
  </h4>
  <div id="e1UroHistoryList">${tr('Loading...', '???')}</div>
</div>
</div>
`;

window.e1LoadUroHistory(pid);
window.e1UroCheckParams();
};

window.e1UroCheckParams = () => {
  const flowRate = parseFloat(document.getElementById('e1UroFlowRate')?.value || 0);
  const pvr = parseFloat(document.getElementById('e1UroPvr')?.value || 0);

  const flowAlert = document.getElementById('e1UroFlowAlert');
  const flowText = document.getElementById('e1UroFlowStatusText');
  const flowBadge = document.getElementById('e1UroFlowBadge');

  const pvrAlert = document.getElementById('e1UroPvrAlert');
  const pvrText = document.getElementById('e1UroPvrStatusText');
  const pvrBadge = document.getElementById('e1UroPvrBadge');

  if (flowAlert && flowText && flowBadge) {
    if (flowRate < 15) {
      flowAlert.style.borderLeftColor = 'red';
      flowText.innerHTML = tr('Low Flow Rate (<15 mL/s) - Suspicion of Obstruction / Weak Bladder', '???? ????
???? - ????? ? ? ????? ? ? ? ? ?????');
      flowBadge.className = 'badge badge-danger';
      flowBadge.innerHTML = tr('Abnormal', '???');
    } else {
      flowAlert.style.borderLeftColor = '#10b981';
      flowText.innerHTML = tr('Normal (>15 mL/s)', '????');
      flowBadge.className = 'badge badge-success';
      flowBadge.innerHTML = tr('Normal', '????');
    }
  }
}

```

```

    }
}

if (pvrAlert && pvrText && pvrBadge) {
  if (pvr > 50) {
    pvrAlert.style.borderLeftColor = 'red';
    pvrText.innerText = tr('High Post-Void Residual (>50 mL) - Urinary Retention Risk', '??? ????? ????? - ?
    ?? ?????? ??????');
    pvrBadge.className = 'badge badge-danger';
    pvrBadge.innerText = tr('Retention Warning', '???? ??????');
  } else {
    pvrAlert.style.borderLeftColor = '#10b981';
    pvrText.innerText = tr('Normal (<50 mL)', '?????');
    pvrBadge.className = 'badge badge-success';
    pvrBadge.innerText = tr('Normal', '?????');
  }
}
};

window.e1LoadUroHistory = async (pid) => {
  const container = document.getElementById('e1UroHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/urology/urodynamics/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `
```

```

    } catch (err) {
        container.innerHTML = `
```

```

<div class="form-group">
  <label>${tr('Bypass Pump Time (mins)', '??? ????? ?????? (????????)')}</label>
  <input type="number" class="form-input" id="e1CpbPumpTime" value="90">
</div>
<div class="form-group">
  <label>${tr('Cross-Clamp Time (mins)', '??? ????? ?????? ??????')}</label>
  <input type="number" class="form-input" id="e1CpbClampTime" value="60">
</div>
<div class="form-group">
  <label>${tr('Mean Flow Rate (L/min/m²)', '????? ???? ???? ?????')}</label>
  <input type="number" step="0.01" class="form-input" id="e1CpbFlowRate" value="2.4">
</div>
<div class="form-group">
  <label>${tr('Minimum Temp (°C)', '??? ???? ????? ?????? ??????')}</label>
  <input type="number" step="0.1" class="form-input" id="e1CpbTemp" value="32.0">
</div>
</div>

<div class="form-group mb-8">
  <label>${tr('Perfusion & Operative Notes', '?????? ??????? ???????')}</label>
  <textarea class="form-input form-textarea" id="e1CpbNotes" rows="2" placeholder="e.g., Cold blood cardioplegia administered..." style="min-height:45px"></textarea>
</div>

  <button class="btn btn-primary btn-sm w-full" onclick="e1AddCpbLog(${safeId(pid)})">? ${tr('Save CPB Log', '??? ??? ????? ?????')}</button>
</fieldset>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px"
">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Bypass History', '??? ????? ??????')}</h4>
  <div id="e1CpbHistoryList">${tr('Loading...', '???? ???????...')}</div>
</div>
</div>
`;

window.e1LoadCpbHistory(pid);
};

window.e1LoadCpbHistory = async (pid) => {
  const container = document.getElementById('e1CpbHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/surgery/cpb/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No cardiopulmonary bypass logs found', '?? ????? ?????? ??? ??? ??????')}</div>`;
    } else {
      container.innerHTML = records.map(r => `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4

```

```

px solid var(--primary);font-size:12px">
    <div style="font-weight:700;color:var(--primary)">? ${new Date(r.bypass_date).toLocaleDateString('ar
-SA')}</div>
    <div style="margin-top:6px;display:grid;grid-template-columns:1fr 1fr;gap:8px;font-size:11px">
        <div>
            <strong>Pump Time:</strong> ${r.pump_time} mins<br>
            <strong>Clamp Time:</strong> ${r.cross_clamp_time} mins
        </div>
        <div>
            <strong>Flow Rate:</strong> ${r.flow_rate} L/min/m²<br>
            <strong>Min Temp:</strong> ${r.min_temp} °C
        </div>
    </div>
    ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim);font-style:italic">"${escapeHTML(r.notes)}"</div>` : ''}
    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
</div>
`).join('');
}
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
}
};

```

```

window.e1AddCpbLog = async (pid) => {
    const getVal = (id) => parseFloat(document.getElementById(id)?.value || 0);
    const getStr = (id) => document.getElementById(id)?.value || '';

    try {
        await API.post('/api/surgery/cpb', {
            patient_id: pid,
            pump_time: getVal('e1CpbPumpTime'),
            cross_clamp_time: getVal('e1CpbClampTime'),
            flow_rate: getVal('e1CpbFlowRate'),
            min_temp: getVal('e1CpbTemp'),
            notes: getStr('e1CpbNotes')
        });

        showToast(tr('CPB log saved successfully!', '?? ??? ??? ????? ?????? !'));
        window.e1LoadCpbHistory(pid);

        document.getElementById('e1CpbPumpTime').value = '90';
        document.getElementById('e1CpbClampTime').value = '60';
        document.getElementById('e1CpbFlowRate').value = '2.4';
        document.getElementById('e1CpbTemp').value = '32.0';
        document.getElementById('e1CpbNotes').value = '';
    } catch (err) {
        showToast(tr('Error saving CPB log', '??? ?? ??? ??? ????? ??????'), 'error');
    }
};

```

```
// =====
```

```
// ===== ANESTHESIA & PAIN MODULE (G19) =====
// =====

window.e1RenderAnesthesiaPain = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Anesthesia & Acute Pain Management', '??????
        ?????? ?????? ??????')}</h4>

        <!-- Visual Analog Scale (VAS) Pain Score -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Visual Analog Scale (VAS)
          Pain Score', '????? ??? ?????? ?????????')}</legend>

          <div style="display:flex;align-items:center;gap:16px;margin-bottom:12px">
            <input type="range" min="0" max="10" value="4" class="form-input" id="e1PainScoreVas" style="flex:
            1;height:10px" oninput="e1PainCheckVas()">
            <div style="display:flex;flex-direction:column;align-items:center;min-width:70px">
              <span id="e1PainEmoji" style="font-size:32px">?</span>
              <span id="e1PainText" style="font-size:13px;font-weight:700">Moderate</span>
              <span id="e1PainVal" style="font-size:16px;font-weight:900;color:var(--primary)">4 / 10</span>
            </div>
          </div>
        </fieldset>

        <!-- Patient-Controlled Analgesia (PCA) Pump -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Patient-Controlled Analge
          sia (PCA) Pump Log', '??? ?????? ?????? ?????? ?????????? ?????? ??????')}</legend>

          <label style="display:flex;align-items:center;gap:8px;font-size:13px;margin-bottom:8px">
            <input type="checkbox" id="e1PcaUsed" onchange="e1PainTogglePca()">
            <span><strong>${tr('PCA Pump Active', '???? PCA ?????')}</strong></span>
          </label>

          <div id="e1PcaFields" style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-bottom:12px;
          opacity:0.5:pointer-events:none">
            <div class="form-group">
              <label>${tr('Total Demands', '?????? ??????? ?? ??????')}</label>
              <input type="number" class="form-input" id="e1PcaDemands" value="0">
            </div>
            <div class="form-group">
              <label>${tr('Total Deliveries', '?????? ??????? ??????? ???????')}</label>
              <input type="number" class="form-input" id="e1PcaDeliveries" value="0">
            </div>
          </div>
        </fieldset>
      </div>
    </div>
  `
}
```

```

<div class="form-group mb-8">
  <label>${tr('Anesthetic Notes & Pain Plan', '??????? ?????? ??? ???? ?????')}</label>
  <textarea class="form-input form-textarea" id="e1PainNotes" rows="2" placeholder="e.g. Epidural infusion running smoothly..." style="min-height:45px"></textarea>
</div>

  <button class="btn btn-primary btn-sm w-full" onclick="e1AddPainAssessment(${safeId(pid)})">? ${tr('Save Pain Assessment', '??? ????? ?????')}</button>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Pain History', '??? ?????? ????????? ????? ??? ?????')}</h4>
  <div id="e1PainHistoryList">${tr('Loading...', '???? ?????????...')}</div>
</div>
</div>
`;

window.e1LoadPainHistory(pid);
window.e1PainCheckVas();
};

window.e1PainCheckVas = () => {
  const vas = parseInt(document.getElementById('e1PainScoreVas')?.value || 0);
  const emojiEl = document.getElementById('e1PainEmoji');
  const textEl = document.getElementById('e1PainText');
  const valEl = document.getElementById('e1PainVal');

  if (!emojiEl || !textEl || !valEl) return;

  valEl.innerText = `${vas} / 10`;

  if (vas <= 2) {
    emojiEl.innerText = '?';
    textEl.innerText = tr('Mild / ????', 'Mild / ?????');
    textEl.style.color = '#10b981';
  } else if (vas <= 5) {
    emojiEl.innerText = '?';
    textEl.innerText = tr('Moderate / ?????', 'Moderate / ?????');
    textEl.style.color = '#f59e0b';
  } else if (vas <= 8) {
    emojiEl.innerText = '?';
    textEl.innerText = tr('Severe / ????', 'Severe / ?????');
    textEl.style.color = '#ef4444';
  } else {
    emojiEl.innerText = '?';
    textEl.innerText = tr('Worst Pain / ??? ?? ?????', 'Worst Pain / ??? ?? ?????');
    textEl.style.color = '#b91c1c';
  }
};

```



```

window.e1PainTogglePca = () => {
  const active = document.getElementById('e1PcaUsed')?.checked;
  const fields = document.getElementById('e1PcaFields');
  if (fields) {
    fields.style.opacity = active ? '1' : '0.5';
    fields.style.pointerEvents = active ? 'auto' : 'none';
  }
};

window.e1LoadPainHistory = async (pid) => {
  const container = document.getElementById('e1PainHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/anesthesia/pain/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No pain assessments found', '?? ???? ?????? ?? ????')}</div>`;
    } else {
      container.innerHTML = records.map(r => `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid var(--primary);font-size:12px">
          <div style="font-weight:700;color:var(--primary)">? ${new Date(r.assessment_time).toLocaleString('ar-SA', { dateStyle: 'short', timeStyle: 'short' })}</div>
          <div style="margin-top:6px;display:flex;justify-content:space-between;align-items:center">
            <span><strong>Pain Score (VAS):</strong> <span class="badge" style="background:${parseInt(r.pain_score_vas) <= 2 ? '#10b981' : parseInt(r.pain_score_vas) <= 5 ? '#f59e0b' : '#ef4444'};color:white">${r.pain_score_vas} / 10</span></span>
          </div>
          ${r.pca_pump_used ? `
            <div style="margin-top:4px;font-size:11px;background:#e0f2fe;padding:4px 8px;border-radius:4px">
              <strong>PCA Pump Active:</strong> Demands: ${r.pca_demands} | Deliveries: ${r.pca_deliveries}
            </div>
          ` : ''}
          ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim);font-style:italic">"${escapeHTML(r.notes)}"</div>` : ''}
          <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name || '')}</div>
        </div>
      `).join('');
    }
  } catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
  }
};

window.e1AddPainAssessment = async (pid) => {
  try {
    await API.post('/api/anesthesia/pain', {
      patient_id: pid,
      pain_score_vas: parseInt(document.getElementById('e1PainScoreVas').value),
      pca_pump_used: document.getElementById('e1PcaUsed').checked,
      pca_demands: parseInt(document.getElementById('e1PcaDemands').value || 0),
      pca_deliveries: parseInt(document.getElementById('e1PcaDeliveries').value || 0),
    });
  }
};

```

```

    notes: document.getElementById('e1PainNotes').value
  });

  showToast(tr('Pain assessment saved successfully!', '?? ??? ????? ????? ?????!'));
  window.e1LoadPainHistory(pid);

  document.getElementById('e1PainScoreVas').value = 4;
  document.getElementById('e1PcaUsed').checked = false;
  document.getElementById('e1PcaDemands').value = 0;
  document.getElementById('e1PcaDeliveries').value = 0;
  document.getElementById('e1PainNotes').value = '';
  window.e1PainCheckVas();
  window.e1PainTogglePca();
} catch (err) {
  showToast(tr('Error saving pain assessment', '??? ?? ??? ????? ?????'), 'error');
}
};

// =====
// ===== NEONATAL & PEDIATRICS MODULE (G20) =====
// =====

window.e1RenderPediatrics = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Neonatal & Pediatric Care', '?? ??????? ?????
? ???????')}</h4>

        <!-- APGAR Score Calculator -->
        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('APGAR Score Calculator',
'????? ?????? ?????? ?????? (????? ??????')}</legend>

          <div style="display:grid;grid-template-columns:1fr;gap:8px;margin-bottom:12px;font-size:12px">
            <div style="display:flex;justify-content:space-between;align-items:center">
              <span><strong>Appearance (Skin Color):</strong></span>
              <select id="e1ApgarApp" class="form-input" style="width:160px;padding:2px" onchange="e1CalcApgar
()">
                <option value="0">0 - Blue/Pale</option>
                <option value="1">1 - Body pink, extremities blue</option>
                <option value="2" selected>2 - Completely pink</option>
              </select>
            </div>
            <div style="display:flex;justify-content:space-between;align-items:center">
              <span><strong>Pulse (Heart Rate):</strong></span>
              <select id="e1ApgarPulse" class="form-input" style="width:160px;padding:2px" onchange="e1CalcApg
ar()">

```

```

        <option value="0">0 - Absent</option>
        <option value="1">1 - < 100 bpm</option>
        <option value="2" selected>2 - ? 100 bpm</option>
    </select>
</div>
<div style="display:flex;justify-content:space-between;align-items:center">
    <span><strong>Grimace (Reflex Irritability):</strong></span>
    <select id="e1ApgarGrim" class="form-input" style="width:160px;padding:2px" onchange="e1CalcApgar
r())">
        <option value="0">0 - Floppy (No response)</option>
        <option value="1">1 - Grimace on stimulation</option>
        <option value="2" selected>2 - Cry, sneeze, or cough</option>
    </select>
</div>
<div style="display:flex;justify-content:space-between;align-items:center">
    <span><strong>Activity (Muscle Tone):</strong></span>
    <select id="e1ApgarAct" class="form-input" style="width:160px;padding:2px" onchange="e1CalcApgar
())">
        <option value="0">0 - Absent (Flaccid)</option>
        <option value="1">1 - Some flexion of arms/legs</option>
        <option value="2" selected>2 - Active motion</option>
    </select>
</div>
<div style="display:flex;justify-content:space-between;align-items:center">
    <span><strong>Respiration (Effort):</strong></span>
    <select id="e1ApgarResp" class="form-input" style="width:160px;padding:2px" onchange="e1CalcApgar
r())">
        <option value="0">0 - Absent</option>
        <option value="1">1 - Slow, irregular, weak cry</option>
        <option value="2" selected>2 - Vigorous cry</option>
    </select>
</div>
</div>

<!-- Total APGAR Output -->
<div id="e1ApgarResultPanel" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-size:13px;border-left:4px solid #10b981;display:flex;justify-content:space-between;align-items:center">
    <span><strong>APGAR Score:</strong> <span id="e1ApgarInterpretation">${tr('Normal (7-10)', '????')}</span></span>
    <span id="e1ApgarScoreVal" class="badge badge-success" style="font-size:14px;padding:4px 10px">10 / 10</span>
</div>
</fieldset>

<!-- Growth Parameters -->
<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bottom:12px">
    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Growth & Physical Parameters', '????? ???? ?????? ?????? ?????')}</legend>

    <div style="display:grid;grid-template-columns:1fr 1fr 1fr;gap:8px">
        <div class="form-group">
            <label>${tr('Weight (kg)', '???? (???)')}</label>
            <input type="number" step="0.01" class="form-input" id="e1PedWeight" placeholder="e.g. 3.20">

```

```
</div>
<div class="form-group">
  <label>${tr('Height (cm)', '???? (??)')}</label>
  <input type="number" step="0.1" class="form-input" id="e1PedHeight" placeholder="e.g. 50.5">
</div>
<div class="form-group">
  <label>${tr('Head Circ. (cm)', '???? ????? (??)')}</label>
  <input type="number" step="0.1" class="form-input" id="e1PedHead" placeholder="e.g. 34.0">
</div>
</div>
</fieldset>
```

```
  <button class="btn btn-primary btn-sm w-full mb-12" onclick="e1AddPedRecord(${safeId(pid)})">? ${tr('Save Pediatric Record', '??? ??? ?????')}</button>
</div>
```

```
<!-- Right Column: History Logs & Growth Chart -->
<div style="flex:1.5;min-width:320px;padding-left:16px">
  <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('WHO Growth Percentile Chart', '?????? ?????
?????? ?????? ????? ??????')}</h4>
```

```
  <div style="padding:8px;background:#ffffff;border:1px solid var(--outline);border-radius:8px;text-align:center;margin-bottom:12px">
    <div style="font-weight:bold;font-size:11px;margin-bottom:4px;display:flex;justify-content:space-between;align-items:center">
```

```
      <span>? ${tr('Growth Curve Visualization', '????? ?????? ??? ?????')}</span>
      <select class="form-input" id="e1PedChartType" style="padding:1px 4px;font-size:9px;width:auto;margin:0" onchange="window.e1UpdateGrowthChart(${safeId(pid)})">
        <option value="weight">${tr('Weight for Age', '????? ?????? ?????')}</option>
        <option value="height">${tr('Height for Age', '????? ?????? ?????')}</option>
        <option value="head">${tr('Head Circumference', '???? ?????? ?????? ?????')}</option>
      </select>
    </div>
```

```
    <svg id="e1PedGrowthChart" viewBox="0 0 300 150" style="width:100%;height:auto;background:#fafafa;border-radius:4px">
      <line x1="30" y1="130" x2="290" y2="130" stroke="#ddd" stroke-width="1"/>
      <line x1="30" y1="10" x2="30" y2="130" stroke="#ddd" stroke-width="1"/>
      <path id="e1PedPercentile3" d="M 30,120 Q 100,105 180,95 T 280,85" fill="none" stroke="#fca5a5" stroke-dasharray="2,2" stroke-width="1.5"/>
      <path id="e1PedPercentile50" d="M 30,110 Q 100,85 180,75 T 280,60" fill="none" stroke="#22c55e" stroke-width="1.5"/>
      <path id="e1PedPercentile97" d="M 30,95 Q 100,65 180,50 T 280,30" fill="none" stroke="#fca5a5" stroke-dasharray="2,2" stroke-width="1.5"/>
      <text x="160" y="145" font-size="8" text-anchor="middle">${tr('Age (Months)', '????? (???????)')}</text>
      <text id="e1PedYLabel" x="10" y="70" font-size="8" text-anchor="middle" transform="rotate(-90 10 70)">${tr('Weight (kg)', '????? (???)')}</text>
      <g id="e1PedGrowthPoints"></g>
    </svg>
  </div>
```

```
  <h4 style="margin:12px 0 12px;color:var(--primary)">? ${tr('Pediatric History', '??? ?????????? ??????????
? ?????????')}</h4>
  <div id="e1PedHistoryList">${tr('Loading...', '???? ?????????...')}</div>
```

```

    </div>
  </div>
`;

window.e1LoadPedHistory(pid);
window.e1CalcApgar();
};

window.e1CalcApgar = () => {
  const getSel = (id) => parseInt(document.getElementById(id)?.value || 0);
  const total = getSel('e1ApgarApp') + getSel('e1ApgarPulse') + getSel('e1ApgarGrim') + getSel('e1ApgarAct') +
    getSel('e1ApgarResp');

  const panel = document.getElementById('e1ApgarResultPanel');
  const interp = document.getElementById('e1ApgarInterpretation');
  const scoreVal = document.getElementById('e1ApgarScoreVal');

  if (!panel || !interp || !scoreVal) return;

  scoreVal.innerText = `${total} / 10`;

  if (total >= 7) {
    panel.style.borderLeftColor = '#10b981';
    interp.innerText = tr('Normal (7-10)', '????? - ??? ???? ????');
    scoreVal.className = 'badge badge-success';
  } else if (total >= 4) {
    panel.style.borderLeftColor = '#f59e0b';
    interp.innerText = tr('Moderately Depressed (4-6)', '????? - ?? ???? ???? ????');
    scoreVal.className = 'badge badge-warning';
  } else {
    panel.style.borderLeftColor = 'red';
    interp.innerText = tr('Severely Depressed (0-3)', '????? - ???? ???? ???? ????');
    scoreVal.className = 'badge badge-danger';
  }
};

window.e1LoadPedHistory = async (pid) => {
  const container = document.getElementById('e1PedHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/pediatrics/growth/patient/' + pid);
    window.e1PedHistoryCache = records || [];

    if (!records.length) {
      container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No pediatric growth records found', '?? ???? ???? ????')}</div>`;
    } else {
      container.innerHTML = records.map(r => `
        <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4px solid var(--primary);font-size:12px">
          <div style="font-weight:700;color:var(--primary)">? ${new Date(r.record_date).toLocaleDateString('ar-SA')}</div>
          <div style="margin-top:6px;display:grid;grid-template-columns:1fr 1fr;gap:8px;font-size:11px">

```

```

        <div>
            <strong>Weight:</strong> ${r.weight_kg ? `${r.weight_kg} kg` : '-'}<br>
            <strong>Height:</strong> ${r.height_cm ? `${r.height_cm} cm` : '-'}
        </div>
        <div>
            <strong>Head Circ:</strong> ${r.head_circ_cm ? `${r.head_circ_cm} cm` : '-'}<br>
            <strong>APGAR:</strong> ${r.apgar_1min !== null ? `1m: ${r.apgar_1min} | 5m: ${r.apgar_5min}` :
        '-'}

        </div>
    </div>
    <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
    '')}</div>
    </div>
    `).join('');
}

// Update the growth curves
if (typeof window.e1UpdateGrowthChart === 'function') {
    window.e1UpdateGrowthChart(pid);
}
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
}
};
window.e1UpdateGrowthChart = (pid) => {
    const chartTypeSelect = document.getElementById('e1PedChartType');
    const chart = document.getElementById('e1PedGrowthChart');
    if (!chart) return;

    const chartType = chartTypeSelect ? chartTypeSelect.value : 'weight';
    const records = window.e1PedHistoryCache || [];

    // 1. Update paths and Y-axis label
    const p3 = document.getElementById('e1PedPercentile3');
    const p50 = document.getElementById('e1PedPercentile50');
    const p97 = document.getElementById('e1PedPercentile97');
    const yLabel = document.getElementById('e1PedYLabel');

    if (chartType === 'weight') {
        if (yLabel) yLabel.textContent = tr('Weight (kg)', '???? (??)');
        if (p3) p3.setAttribute('d', 'M 30,120 Q 100,105 180,95 T 280,85');
        if (p50) p50.setAttribute('d', 'M 30,110 Q 100,85 180,75 T 280,60');
        if (p97) p97.setAttribute('d', 'M 30,95 Q 100,65 180,50 T 280,30');
    } else if (chartType === 'height') {
        if (yLabel) yLabel.textContent = tr('Height (cm)', '???? (??)');
        if (p3) p3.setAttribute('d', 'M 30,125 Q 100,105 180,80 T 280,60');
        if (p50) p50.setAttribute('d', 'M 30,115 Q 100,90 180,65 T 280,40');
        if (p97) p97.setAttribute('d', 'M 30,105 Q 100,75 180,50 T 280,20');
    } else if (chartType === 'head') {
        if (yLabel) yLabel.textContent = tr('Head Circ (cm)', '???? ????? (??)');
        if (p3) p3.setAttribute('d', 'M 30,120 Q 100,95 180,85 T 280,75');
        if (p50) p50.setAttribute('d', 'M 30,110 Q 100,80 180,65 T 280,50');
        if (p97) p97.setAttribute('d', 'M 30,95 Q 100,60 180,45 T 280,30');
    }
}

```

```

// 2. Fetch patient birth_date to compute age in months
const patient = window.currentPatientsList ? window.currentPatientsList.find(x => x.id === pid) : null;
const dobStr = patient ? (patient.birth_date || patient.date_of_birth) : null;
const birth = dobStr ? new Date(dobStr) : new Date();

// 3. Map historical records to coordinates
let pointsStr = '';
let circlesHtml = '';

const sorted = [...records]
  .filter(r => r.record_date)
  .sort((a, b) => new Date(a.record_date) - new Date(b.record_date));

sorted.forEach(r => {
  const recordDate = new Date(r.record_date);
  const diffMs = recordDate.getTime() - birth.getTime();
  // age in months
  const ageMonths = Math.max(0, parseFloat((diffMs / (1000 * 60 * 60 * 24 * 30.4375)).toFixed(1)));

  // x coordinate: age range 0 to 36 months maps to SVG width x = 30 to 280 (delta = 250)
  const x = 30 + Math.min(Math.max(ageMonths, 0), 36) / 36 * 250;

  let val = 0;
  let minVal = 0;
  let maxVal = 18;

  if (chartType === 'weight') {
    val = parseFloat(r.weight_kg) || 0;
    minVal = 0;
    maxVal = 18;
    if (!r.weight_kg) return;
  } else if (chartType === 'height') {
    val = parseFloat(r.height_cm) || 40;
    minVal = 40;
    maxVal = 105;
    if (!r.height_cm) return;
  } else if (chartType === 'head') {
    val = parseFloat(r.head_circ_cm) || 30;
    minVal = 30;
    maxVal = 55;
    if (!r.head_circ_cm) return;
  }
}

const range = maxVal - minVal;
// y coordinate: value range minVal to maxVal maps to SVG height y = 130 to 10 (delta = 120)
const y = 130 - Math.min(Math.max(val - minVal, 0), range) / range * 120;

pointsStr += `${x},${y} `;
circlesHtml += `<circle cx="${x}" cy="${y}" r="3.5" fill="#ef4444" stroke="#ffffff" stroke-width="1"><title>Age: ${ageMonths}m, Value: ${val}</title></circle>`;
});

const lineHtml = pointsStr ? `<polyline points="${pointsStr.trim()}" fill="none" stroke="#ef4444" stroke-wid

```

```

th="2" />` : '';
const ptsContainer = document.getElementById('e1PedGrowthPoints');
if (ptsContainer) {
  ptsContainer.innerHTML = lineHtml + circlesHtml;
}
};

window.e1AddPedRecord = async (pid) => {
  const getVal = (id) => document.getElementById(id)?.value || '';

  try {
    const apgarScore = parseInt(document.getElementById('e1ApgarApp').value) +
      parseInt(document.getElementById('e1ApgarPulse').value) +
      parseInt(document.getElementById('e1ApgarGrim').value) +
      parseInt(document.getElementById('e1ApgarAct').value) +
      parseInt(document.getElementById('e1ApgarResp').value);

    await Promise.all([
      API.post('/api/pediatrics/growth', {
        patient_id: pid,
        apgar_1min: apgarScore,
        apgar_5min: apgarScore,
        weight_kg: getVal('e1PedWeight'),
        height_cm: getVal('e1PedHeight'),
        head_circ_cm: getVal('e1PedHead')
      }),
      API.post('/api/pediatrics/apgar', {
        patient_id: pid,
        mother_id: null,
        apgar_1min: apgarScore,
        apgar_5min: apgarScore,
        apgar_10min: null,
        details: {
          appearance: parseInt(document.getElementById('e1ApgarApp').value),
          pulse: parseInt(document.getElementById('e1ApgarPulse').value),
          grimace: parseInt(document.getElementById('e1ApgarGrim').value),
          activity: parseInt(document.getElementById('e1ApgarAct').value),
          respiration: parseInt(document.getElementById('e1ApgarResp').value)
        },
        notes: 'Logged via growth parameters screen.'
      })
    ]);
  });

  showToast(tr('Pediatric & APGAR record saved successfully!', '?? ??? ??? ????? ?????? ????? ?????!'));
  window.e1LoadPedHistory(pid);

  document.getElementById('e1PedWeight').value = '';
  document.getElementById('e1PedHeight').value = '';
  document.getElementById('e1PedHead').value = '';
  document.getElementById('e1ApgarApp').value = '2';
  document.getElementById('e1ApgarPulse').value = '2';
  document.getElementById('e1ApgarGrim').value = '2';
  document.getElementById('e1ApgarAct').value = '2';
  document.getElementById('e1ApgarResp').value = '2';

```



```

        window.e1CalcApgar();
    } catch (err) {
        showToast(tr('Error saving pediatric record', '??? ?? ??? ??? ?????'), 'error');
    }
};

// =====
// ===== OBGYN MODULE (G21) =====
// =====

window.e1RenderObgyn = async (pid) => {
    const body = document.getElementById('e1TabBody');
    if (!body) return;

    body.innerHTML = `
        <div style="display:flex;gap:16px;flex-wrap:wrap">
            <!-- Left Column: Forms -->
            <div style="flex:1.5;min-width:320px">
                <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Obstetrics & Gynecology (OBGYN)', '?? ?????
                ???????')}</h4>

                <!-- GPAL obstetrics history -->
                <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
                ttom:12px">
                    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Obstetric History (GPAL)'
                    , '?????? ??????? (GPAL)')}</legend>
                    <div style="display:grid;grid-template-columns:1fr 1fr 1fr 1fr;gap:8px">
                        <div class="form-group">
                            <label>${tr('Gravida (G)', '??? ??? ????')}</label>
                            <input type="number" class="form-input" id="e1obgynGravida" value="1" min="0">
                        </div>
                        <div class="form-group">
                            <label>${tr('Para (P)', '??? ??? ????')}</label>
                            <input type="number" class="form-input" id="e1obgynPara" value="0" min="0">
                        </div>
                        <div class="form-group">
                            <label>${tr('Abortions (A)', '??? ??? ????')}</label>
                            <input type="number" class="form-input" id="e1obgynAbortions" value="0" min="0">
                        </div>
                        <div class="form-group">
                            <label>${tr('Living (L)', '????? ????')}</label>
                            <input type="number" class="form-input" id="e1obgynLiving" value="0" min="0">
                        </div>
                    </div>
                </fieldset>

                <!-- Pregnancy Calculator -->
                <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
                ttom:12px">
                    <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Gestational Age & EDD Cal
                    culator', '???? ? ? ? ? ? ? ? ? ? ?')}</legend>

                    <div class="form-group mb-8">
                        <label>${tr('Last Menstrual Period (LMP) Date', '???? ? ? ? ?')}</label>

```

```

        <input type="date" class="form-input" id="e10bgynLmp" onchange="e10bgynCalcPregnancy()">
    </div>

    <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-top:12px">
        <div style="background:var(--hover,#f8f9fa);padding:8px;border-radius:6px">
            <div style="font-size:11px;color:var(--text-dim)">${tr('Estimated Date of Delivery (EDD)', '????
            ?????? ??????')}</div>
            <div id="e10bgynEdd" style="font-size:14px;font-weight:700;color:var(--primary)">-</div>
        </div>
        <div style="background:var(--hover,#f8f9fa);padding:8px;border-radius:6px">
            <div style="font-size:11px;color:var(--text-dim)">${tr('Gestational Age', '??? ????? ??????')}</
div>
            <div id="e10bgynWeeks" style="font-size:14px;font-weight:700;color:var(--primary)">-</div>
        </div>
    </div>
</fieldset>

<div class="form-group mb-8">
    <label>${tr('Clinical Notes & Antenatal Plan', '????????? ?????????? ??? ???? ????')}</label>
    <textarea class="form-input form-textarea" id="e10bgynNotes" rows="2" placeholder="e.g., Normal feta
l heart rate detected on Doppler..." style="min-height:45px"></textarea>
</div>

    <button class="btn btn-primary btn-sm w-full" onclick="e1Add0bgynPregnancy(${safeId(pid)})">? ${tr('Sa
ve Pregnancy Record', '??? ??? ?????')}</button>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">

    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Pregnancy Records', '???????? ??????')}</h4>
    <div id="e10bgynHistoryList">${tr('Loading...', '??? ?????????...')}</div>
</div>
</div>
`;

window.e1Load0bgynHistory(pid);
};

window.e10bgynCalcPregnancy = () => {
    const lmpStr = document.getElementById('e10bgynLmp')?.value;
    const eddEl = document.getElementById('e10bgynEdd');
    const weeksEl = document.getElementById('e10bgynWeeks');

    if (!lmpStr || !eddEl || !weeksEl) return;

    const lmp = new Date(lmpStr);
    if (isNaN(lmp.getTime())) return;

    // Naegele's rule: LMP + 9 months + 7 days
    const edd = new Date(lmp);
    edd.setMonth(edd.getMonth() + 9);
    edd.setDate(edd.getDate() + 7);

```

```

eddel.innerText = edd.toLocaleDateString('ar-SA');

// Calculate Gestational Weeks: (Current Date - LMP) / 7 days
const diffTime = Math.abs(new Date() - lmp);
const diffDays = Math.ceil(diffTime / (1000 * 60 * 60 * 24));
const weeks = Math.floor(diffDays / 7);
const remainingDays = diffDays % 7;

weeksEl.innerText = `${weeks} ${tr('weeks', '????')} ${remainingDays} ${tr('days', '??')}`;
};

window.e1LoadObgynHistory = async (pid) => {
  const container = document.getElementById('e10bgynHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/obgyn/pregnancies/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `
```

```

if (lmpStr) {
  const lmp = new Date(lmpStr);
  const edd = new Date(lmp);
  edd.setMonth(edd.getMonth() + 9);
  edd.setDate(edd.getDate() + 7);
  eddDate = edd.toISOString().slice(0, 10);

  const diffTime = Math.abs(new Date() - lmp);
  const diffDays = Math.ceil(diffTime / (1000 * 60 * 60 * 24));
  gestWeeks = Math.floor(diffDays / 7);
}

try {
  await API.post('/api/obgyn/pregnancies', {
    patient_id: pid,
    gravida: getInt('e10bgynGravida'),
    para: getInt('e10bgynPara'),
    abortions: getInt('e10bgynAbortions'),
    living: getInt('e10bgynLiving'),
    lmp_date: lmpStr || null,
    edd_date: eddDate,
    gestational_weeks: gestWeeks,
    notes: getVal('e10bgynNotes')
  });

  showToast(tr('Pregnancy record saved successfully!', '?? ??? ??? ????? ?????!'));
  window.e1LoadObgynHistory(pid);

  document.getElementById('e10bgynGravida').value = '1';
  document.getElementById('e10bgynPara').value = '0';
  document.getElementById('e10bgynAbortions').value = '0';
  document.getElementById('e10bgynLiving').value = '0';
  document.getElementById('e10bgynLmp').value = '';
  document.getElementById('e10bgynNotes').value = '';
  document.getElementById('e10bgynEdd').innerText = '-';
  document.getElementById('e10bgynWeeks').innerText = '-';
} catch (err) {
  showToast(tr('Error saving pregnancy record', '??? ?? ??? ??? ?????'), 'error');
}
};

// =====
// ===== PSYCHIATRY MODULE (G24) =====
// =====

window.e1RenderPsychiatry = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">

```

```
<h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Psychiatric Evaluation & Mental Status', '???
???? ?????? ????? ??? ???????')}</h4>

<fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
ttom:12px">

  <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Mental Status Examination
(MSE)', '??? ?????? ???????')}</legend>

  <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-bottom:12px;font-size:12px">
    <div class="form-group">
      <label>${tr('Appearance & Grooming', '????? ?????????')}</label>
      <select id="e1MseAppearance" class="form-input" style="padding:2px">
        <option value="Neat / Clean">${tr('Neat / Clean', '???? ?????')}</option>
        <option value="Disheveled / Unkempt">${tr('Disheveled / Unkempt', '???? ?????? ??? ?????')}</op
tion>

        <option value="Undernourished">${tr('Undernourished', '???? ????? ??? ?????????')}</option>
      </select>
    </div>
    <div class="form-group">
      <label>${tr('Behavior & Psychomotor', '????? ????????? ?????????')}</label>
      <select id="e1MseBehavior" class="form-input" style="padding:2px">
        <option value="Calm & Cooperative">${tr('Calm & Cooperative', '???? ?????????')}</option>
        <option value="Agitated / Restless">${tr('Agitated / Restless', '????? / ??? ?????????')}</option
>

        <option value="Withdrawn / Catatonic">${tr('Withdrawn / Catatonic', '????? / ???? ?????')}</opt
ion>

      </select>
    </div>
    <div class="form-group">
      <label>${tr('Speech', '??????')}</label>
      <select id="e1MseSpeech" class="form-input" style="padding:2px">
        <option value="Normal Rate & Volume">${tr('Normal Rate & Volume', '???? ?????? ??? ?????????')}</o
ption>

        <option value="Pressured / Rapid">${tr('Pressured / Rapid', '???? ?????? ?????????')}</option>
        <option value="Slow / Monotonous">${tr('Slow / Monotonous', '???? ??????')}</option>
      </select>
    </div>
    <div class="form-group">
      <label>${tr('Mood & Affect', '?????? ?????????')}</label>
      <select id="e1MseMood" class="form-input" style="padding:2px">
        <option value="Euthymic (Normal)">Euthymic (Normal)</option>
        <option value="Depressed / Dysphoric">Depressed / Dysphoric</option>
        <option value="Manic / Euphoric">Manic / Euphoric</option>
        <option value="Anxious / Apprehensive">Anxious / Apprehensive</option>
      </select>
    </div>
    <div class="form-group">
      <label>${tr('Thought Process', '???? ?????????')}</label>
      <select id="e1MseThoughtProcess" class="form-input" style="padding:2px">
        <option value="Linear & Goal-directed">${tr('Linear & Goal-directed', '?????? ??????')}</option
>

        <option value="Flight of Ideas">${tr('Flight of Ideas', '????? ?????????')}</option>
        <option value="Tangential / Loose">${tr('Tangential / Loose', '????? ?????? ???? ?????????')}</op
tion>
```

```

        </select>
    </div>
    <div class="form-group">
        <label>${tr('Thought Content', '????? ??????')}</label>
        <select id="e1MseThoughtContent" class="form-input" style="padding:2px">
            <option value="No delusions or suicidal ideation">${tr('Normal / No suicidal ideation', '?????
/ ?? ????? ??????')}</option>
            <option value="Delusions present">${tr('Delusions present', '????? / ?????? ?????')}</optio
n>
            <option value="Suicidal ideation present">${tr('Suicidal ideation present', '????? ?????? ??
??')}</option>
        </select>
    </div>
    <div class="form-group">
        <label>${tr('Insight', '????????? ??????')}</label>
        <select id="e1MseInsight" class="form-input" style="padding:2px">
            <option value="Good (Full awareness)">Good (Full awareness)</option>
            <option value="Partial / Limited">Partial / Limited</option>
            <option value="Poor (No awareness)">Poor (No awareness)</option>
        </select>
    </div>
    <div class="form-group">
        <label>${tr('Judgment', '??????? ??? ?????')}</label>
        <select id="e1MseJudgment" class="form-input" style="padding:2px">
            <option value="Intact / Good">Intact / Good</option>
            <option value="Impaired / Poor">Impaired / Poor</option>
        </select>
    </div>
</div>
</fieldset>

<div class="form-group mb-8">
    <label>${tr('Diagnostic Summary & Management Plan', '?????? ?????????? ??? ?????? ?????? ?????????')}<
/label>
    <textarea class="form-input form-textarea" id="e1PsychSummary" rows="2" placeholder="Describe clinic
al symptoms, medication adjustments..." style="min-height:45px"></textarea>
</div>

    <button class="btn btn-primary btn-sm w-full" onclick="e1AddPsychEvaluation(${safeId(pid)})">? ${tr('S
ave Evaluation', '??? ?????? ?????')}</button>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px
">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Evaluation History', '??? ?????????? ?????? ??
????')}</h4>
    <div id="e1PsychHistoryList">${tr('Loading...', '???? ?????????...')}</div>
</div>
</div>
`;

window.e1LoadPsychHistory(pid);
};

```

```

window.e1LoadPsychHistory = async (pid) => {
  const container = document.getElementById('e1PsychHistoryList');
  if (!container) return;

  try {
    const records = await API.get('/api/psychiatry/evaluations/patient/' + pid);
    if (!records.length) {
      container.innerHTML = `

```

window.e1AddPsychEvaluation = async (pid) => {
  const getVal = (id) => document.getElementById(id)?.value || '';

  try {
    await API.post('/api/psychiatry/evaluations', {
      patient_id: pid,
      mse_appearance: getVal('e1MseAppearance'),
      mse_behavior: getVal('e1MseBehavior'),
      mse_speech: getVal('e1MseSpeech'),
      mse_mood: getVal('e1MseMood'),
      mse_affect: getVal('e1MseMood'), // simplified
      mse_thought_process: getVal('e1MseThoughtProcess'),
      mse_thought_content: getVal('e1MseThoughtContent'),
      mse_perception: 'Normal',
      mse_cognition: 'Intact',
      mse_insight: getVal('e1MseInsight'),
      mse_judgment: getVal('e1MseJudgment'),
      diagnostic_summary: getVal('e1PsychSummary')
    });
  }
};

```


```

```

showToast(tr('Psychiatric evaluation saved successfully!', '?? ??? ?????? ?????? ?????!'));
window.e1LoadPsychHistory(pid);

document.getElementById('e1MseAppearance').selectedIndex = 0;
document.getElementById('e1MseBehavior').selectedIndex = 0;
document.getElementById('e1MseSpeech').selectedIndex = 0;
document.getElementById('e1MseMood').selectedIndex = 0;
document.getElementById('e1MseThoughtProcess').selectedIndex = 0;
document.getElementById('e1MseThoughtContent').selectedIndex = 0;
document.getElementById('e1MseInsight').selectedIndex = 0;
document.getElementById('e1MseJudgment').selectedIndex = 0;
document.getElementById('e1PsychSummary').value = '';
} catch (err) {
  showToast(tr('Error saving psychiatric evaluation', '??? ?? ?? ?????? ??????'), 'error');
}
};

// =====
// ===== DERMATOLOGY MODULE (G25) =====
// =====

window.e1RenderDermatology = async (pid) => {
  const body = document.getElementById('e1TabBody');
  if (!body) return;

  body.innerHTML = `
    <div style="display:flex;gap:16px;flex-wrap:wrap">
      <!-- Left Column: Forms -->
      <div style="flex:1.5;min-width:320px">
        <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Dermatology & Skin Lesion Log', '?? ?????? ?
        ?????? ?????? ??????')}</h4>

        <fieldset style="border:1px solid var(--border-color,#e5e7eb);border-radius:8px;padding:12px;margin-bo
        ttom:12px">
          <legend style="padding:0 8px;font-weight:700;color:var(--primary)">? ${tr('Skin Lesion Examination',
          '??? ?????? ?????? ??????')}</legend>

          <div style="display:grid;grid-template-columns:1fr 1fr;gap:12px;margin-bottom:12px">
            <div class="form-group">
              <label>${tr('Anatomical Site / Location', '????? ?????????? ?? ?????')}</label>
              <input type="text" class="form-input" id="e1DermSite" placeholder="e.g. Left forearm">
            </div>
            <div class="form-group">
              <label>${tr('Lesion Type', '??? ?????? ?????????')}</label>
              <select id="e1DermType" class="form-input" style="padding:2px">
                <option value="Macule">${tr('Macule', '???? ?????')}</option>
                <option value="Papule">${tr('Papule', '????? ?????')}</option>
                <option value="Nodule">${tr('Nodule', '????? ?????')}</option>
                <option value="Vesicle">${tr('Vesicle', '?????? ?????')}</option>
                <option value="Plaque">${tr('Plaque', '????? ?????')}</option>
              </select>
            </div>
            <div class="form-group">

```



```

        <label>${tr('Color', '????')}</label>
        <input type="text" class="form-input" id="e1DermColor" placeholder="e.g. Erythematous (Red)">
    </div>
    <div class="form-group">
        <label>${tr('Size (mm)', '???? (?????)')}</label>
        <input type="number" step="0.1" class="form-input" id="e1DermSize" value="5.0">
    </div>
    <div class="form-group">
        <label>${tr('Distribution / Pattern', '???? ???? ????')}</label>
        <input type="text" class="form-input" id="e1DermDist" placeholder="e.g. Asymmetrical / Isolated"
>

    </div>
    <div class="form-group" style="display:flex;align-items:center;margin-top:20px">
        <label style="display:flex;align-items:center;gap:8px;cursor:pointer">
            <input type="checkbox" id="e1DermBiopsy">
            <span><strong>${tr('Skin Biopsy Taken', '?? ??? ????')}</strong></span>
        </label>
    </div>
</div>

<div class="form-group mb-8">
    <label>${tr('Morphology Description & Treatment Plan', '???? ???? ????')}</label>
    <textarea class="form-input form-textarea" id="e1DermNotes" rows="2" placeholder="e.g., Margins are well-defined, mild scaling present..." style="min-height:45px"></textarea>
</div>

    <button class="btn btn-primary btn-sm w-full" onclick="e1AddDermLesion(${safeId(pid)})">? ${tr('Save Lesion Record', '??? ???')}</button>
</fieldset>
</div>

<!-- Right Column: History Logs -->
<div style="flex:1;min-width:280px;border-right:1px solid var(--border-color,#e5e7eb);padding-right:16px"
">
    <h4 style="margin:0 0 12px;color:var(--primary)">? ${tr('Skin Lesion History', '???')}</h4>
    <div id="e1DermHistoryList">${tr('Loading...', '????')}</div>
</div>
</div>
`;

window.e1LoadDermHistory(pid);
};

window.e1LoadDermHistory = async (pid) => {
    const container = document.getElementById('e1DermHistoryList');
    if (!container) return;

    try {
        const records = await API.get('/api/dermatology/lesions/patient/' + pid);
        if (!records.length) {
            container.innerHTML = `<div style="color:var(--text-dim);font-size:13px">${tr('No skin lesions logged', '???')}</div>`;
        } else {

```

```

        container.innerHTML = records.map(r => `
            <div style="padding:10px;margin:6px 0;border-radius:8px;background:var(--hover,#f8f9fa);border-right:4
px solid var(--primary);font-size:12px">
                <div style="font-weight:700;color:var(--primary)">? ${new Date(r.exam_date).toLocaleDateString('ar-S
A')}</div>
                <div style="margin-top:6px;font-size:11px">
                    <strong>Site:</strong> ${escapeHTML(r.body_site)}<br>
                    <strong>Type:</strong> ${escapeHTML(r.lesion_type)} (${r.size_mm} mm)<br>
                    <strong>Color:</strong> ${escapeHTML(r.color)} | <strong>Pattern:</strong> ${escapeHTML(r.distribu
tion)}<br>
                    ${r.biopsy_taken ? `<span class="badge badge-warning" style="margin-top:4px;display:inline-block">
${tr('Biopsy Performed', '?? ?? ????')}</span>` : ''}
                </div>
                ${r.notes ? `<div style="margin-top:4px;color:var(--text-dim);font-style:italic">"${escapeHTML(r.not
es)}"</div>` : ''}
                <div style="font-size:10px;color:var(--text-dim);margin-top:4px">???? ${escapeHTML(r.doctor_name ||
'')}</div>
            </div>
        `).join('');
    }
} catch (err) {
    container.innerHTML = `<div style="color:red">${tr('Error loading history', '??? ?? ????? ?????')}</div>`;
}
};

```

```

window.e1AddDermLesion = async (pid) => {
    const getVal = (id) => document.getElementById(id)?.value || '';

    try {
        await API.post('/api/dermatology/lesions', {
            patient_id: pid,
            body_site: getVal('e1DermSite'),
            lesion_type: getVal('e1DermType'),
            color: getVal('e1DermColor'),
            size_mm: parseFloat(document.getElementById('e1DermSize')?.value || 0),
            distribution: getVal('e1DermDist'),
            biopsy_taken: document.getElementById('e1DermBiopsy').checked,
            notes: getVal('e1DermNotes')
        });

        showToast(tr('Skin lesion record saved successfully!', '?? ??? ??? ????? ??????? ??????!'));
        window.e1LoadDermHistory(pid);

        document.getElementById('e1DermSite').value = '';
        document.getElementById('e1DermType').selectedIndex = 0;
        document.getElementById('e1DermColor').value = '';
        document.getElementById('e1DermSize').value = '5.0';
        document.getElementById('e1DermDist').value = '';
        document.getElementById('e1DermBiopsy').checked = false;
        document.getElementById('e1DermNotes').value = '';
    } catch (err) {
        showToast(tr('Error saving skin lesion record', '??? ?? ??? ??? ????? ???????'), 'error');
    }
};

```

```
// =====
// DYNAMIC SPECIALTY TEMPLATES & CALCULATORS
// =====
window.renderSpecialtyTemplate = (specialty) => {
  const container = document.getElementById('drSpecialtyTemplateContainer');
  if (!container) return;
  if (!specialty) {
    container.innerHTML = '';
    return;
  }

  let html = `
```



```
        </div>
        <div>
            <label>${tr('Reflexes', '????????')}</label>
            <select class="form-input" id="apgarReflex" style="padding:2px;font-size:10px" onchange="window.calculateAPGAR()">
                <option value="0">0 - None</option>
                <option value="1">1 - Grimace</option>
                <option value="2" selected>2 - Cry/Sneeze</option>
            </select>
        </div>
    </div>
</div>

<div style="display:flex;flex-direction:column;gap:10px">
    <div style="padding:8px;background:#ffffff;border:1px solid var(--outline);border-radius:8px;text-align:center">
        <div style="font-weight:bold;font-size:11px;margin-bottom:4px;display:flex;justify-content:space-between;align-items:center">
            <span>? WHO Growth Percentile</span>
            <select class="form-input" id="pedChartType" style="padding:1px 4px;font-size:9px;width:auto;margin:0" onchange="window.updatePediatricGrowthChart()">
                <option value="weight">${tr('Weight for Age', '???? ????')}</option>
                <option value="height">${tr('Height for Age', '???? ????')}</option>
                <option value="head">${tr('Head Circumference', '???? ????')}</option>
            </select>
        </div>
        <svg id="pedGrowthChart" viewBox="0 0 300 150" style="width:100%;height:auto;background:#fafafa;border-radius:4px">
            <line x1="30" y1="130" x2="290" y2="130" stroke="#ddd" stroke-width="1"/>
            <line x1="30" y1="10" x2="30" y2="130" stroke="#ddd" stroke-width="1"/>
            <path id="pedPercentile3" d="M 30,120 Q 100,105 180,95 T 280,85" fill="none" stroke="#fca5a5" stroke-dasharray="2,2" stroke-width="1.5"/>
            <path id="pedPercentile50" d="M 30,110 Q 100,85 180,75 T 280,60" fill="none" stroke="#22c55e" stroke-width="1.5"/>
            <path id="pedPercentile97" d="M 30,95 Q 100,65 180,50 T 280,30" fill="none" stroke="#fca5a5" stroke-dasharray="2,2" stroke-width="1.5"/>
            <text x="160" y="145" font-size="8" text-anchor="middle">${tr('Age (Months)', '???? (????)')}</text>
            <text id="pedYLabel" x="10" y="70" font-size="8" text-anchor="middle" transform="rotate(-90 10 70)">${tr('Weight (kg)', '???? (???)')}</text>
            <circle id="pedGrowthPoint" cx="100" cy="85" r="4.5" fill="#ef4444" stroke="ffffff" stroke-width="1.5">
                <animate attributeName="r" values="3.5;5.5;3.5" dur="1.5s" repeatCount="indefinite"/>
            </circle>
        </svg>
    </div>

    <div style="padding:8px;background:#ffffff;border:1px solid var(--outline);border-radius:8px">
        <div style="font-weight:bold;font-size:11px;margin-bottom:4px;display:flex;justify-content:space-between;align-items:center">
            <span>? ${tr('National Vaccine Schedule', '?? ????')}</span>
            <span id="vaccineAlertBadge" class="badge badge-warning" style="font-size:9px">${tr('2 Pending', '2 ???')}</span>
        </div>
    </div>
</div>
```

```

</div>
<div style="max-height:100px;overflow-y:auto;font-size:10px" id="pedVaccineList">
  <div style="display:flex;justify-content:space-between;padding:4px;border-bottom:1px solid #eee"
>
    <span>BCG (Tuberculosis) - Birth</span>
    <span style="color:#22c55e;font-weight:bold">? ${tr('Given', '?? ?????')}</span>
  </div>
  <div style="display:flex;justify-content:space-between;padding:4px;border-bottom:1px solid #eee"
>
    <span>HepB (Hepatitis B) - Birth</span>
    <span style="color:#22c55e;font-weight:bold">? ${tr('Given', '?? ?????')}</span>
  </div>
  <div style="display:flex;justify-content:space-between;align-items:center;padding:4px;border-bottom:1px solid #eee" id="vaccine_dtap">
    <span>Hexavalent (DTaP/IPV/Hib/HepB) - 2m</span>
    <button class="btn btn-xs btn-primary" onclick="window.administerVaccine('dtap')">? ${tr('Give', '????')}</button>
  </div>
  <div style="display:flex;justify-content:space-between;align-items:center;padding:4px;border-bottom:1px solid #eee" id="vaccine_pcv">
    <span>PCV (Pneumococcal) - 2m</span>
    <button class="btn btn-xs btn-primary" onclick="window.administerVaccine('pcv')">? ${tr('Give', '????')}</button>
  </div>
</div>
</div>
</div>
`
} else if (specLower.includes('obgyn') || specLower.includes('obstet') || specLower.includes('gynec')) {
  html += `
    <div style="display:flex;gap:8px" class="mb-8">
      <div class="form-group" style="flex:1">
        <label>${tr('Gestational Age (wks)', '?? ???? ?????')}</label>
        <input type="number" class="form-input dr-spec-input" name="gestational_age" data-label="${tr('Gestational Age', '?? ????')}" placeholder="weeks">
      </div>
      <div class="form-group" style="flex:1">
        <label>${tr('Fetal Heart Rate (bpm)', '?? ???? (??)')}</label>
        <input type="number" class="form-input dr-spec-input" name="fetal_hr" data-label="${tr('Fetal HR', '?? ?????')}" placeholder="bpm">
      </div>
    </div>
    <div style="display:flex;gap:4px" class="mb-8">
      <div style="flex:1"><label style="font-size:10px">Gravida (G)</label><input type="number" class="form-input dr-spec-input" name="gravida" data-label="Gravida (G)" placeholder="G"></div>
      <div style="flex:1"><label style="font-size:10px">Para (P)</label><input type="number" class="form-input dr-spec-input" name="para" data-label="Para (P)" placeholder="P"></div>
      <div style="flex:1"><label style="font-size:10px">Abortus (A)</label><input type="number" class="form-input dr-spec-input" name="abortus" data-label="Abortus (A)" placeholder="A"></div>
    </div>
    <div class="form-group mb-8">
      <label>${tr('Last Menstrual Period (LMP)', '?? ????')}</label>
      <input type="date" class="form-input" id="drLmpDate" onchange="window.calculateEDD()">
    </div>
  `
}

```

```

</div>
<div class="form-group mb-8">
  <label>${tr('Expected Delivery Date (EDD)', '????? ?????? ??????')}</label>
  <input type="text" class="form-input dr-spec-input" id="drEddOutput" name="edd" data-label="${tr('EDD', '????? ?????? ??????')}" readonly>
</div>
`;
} else if (specLower.includes('emerg') || specLower.includes('er')) {
  html += `
    <div class="form-group mb-8">
      <label>${tr('Triage ESI Level', '????? ????? ESI')}</label>
      <select class="form-input dr-spec-input" name="esi_level" data-label="${tr('ESI Level', '????? ?????')}">
        <option>ESI 1 - Resuscitation</option>
        <option>ESI 2 - Emergent</option>
        <option>ESI 3 - Urgent</option>
        <option>ESI 4 - Less Urgent</option>
        <option>ESI 5 - Non-Urgent</option>
      </select>
    </div>
    <div style="padding:8px;background:var(--bg-secondary,#f8f9fa);border:1px solid var(--border-color,#e5e7eb);border-radius:8px;margin-bottom:8px;font-size:11px">
      <div style="font-weight:600;margin-bottom:4px">? ${tr('GCS Calculator', '????? ????? ?????')}</div>
      <div style="display:grid;grid-template-columns:repeat(3,1fr);gap:4px">
        <div>
          <label>${tr('Eyes', '????')}</label>
          <select class="form-input" id="gcsEyes" style="padding:2px;font-size:10px" onchange="window.calculateGCS()">
            <option value="4">4 - Spontaneous</option>
            <option value="3">3 - To Speech</option>
            <option value="2">2 - To Pain</option>
            <option value="1">1 - None</option>
          </select>
        </div>
        <div>
          <label>${tr('Verbal', '????')}</label>
          <select class="form-input" id="gcsVerbal" style="padding:2px;font-size:10px" onchange="window.calculateGCS()">
            <option value="5">5 - Oriented</option>
            <option value="4">4 - Confused</option>
            <option value="3">3 - Inappropriate</option>
            <option value="2">2 - Incomprehensible</option>
            <option value="1">1 - None</option>
          </select>
        </div>
        <div>
          <label>${tr('Motor', '????')}</label>
          <select class="form-input" id="gcsMotor" style="padding:2px;font-size:10px" onchange="window.calculateGCS()">
            <option value="6">6 - Obeys commands</option>
            <option value="5">5 - Localizes pain</option>
            <option value="4">4 - Withdraws</option>
            <option value="3">3 - Flexion (decorticate)</option>
            <option value="2">2 - Extension (decerebrate)</option>
          </select>
        </div>
      </div>
    </div>
  `;
}

```

```

        <option value="1">1 - None</option>
    </select>
</div>
</div>
<div class="form-group mb-8">
    <label>${tr('GCS Total Score', '????? ?????? ?????')}</label>
    <input type="text" class="form-input dr-spec-input" id="drGcsOutput" name="gcs_score" data-label="GCS
Score" value="15" readonly>
</div>
`;
} else if (specLower.includes('icu') || specLower.includes('critical')) {
    html += `
        <div class="form-group mb-8">
            <label>${tr('SOFA Score (0-24)', '????? SOFA ?????')}</label>
            <input type="number" min="0" max="24" class="form-input dr-spec-input" name="sofa_score" data-label="S
OFA Score" placeholder="e.g. 5">
        </div>
        <div class="form-group mb-8">
            <label>${tr('Ventilator Mode', '??? ???? ?????? ???????')}</label>
            <select class="form-input dr-spec-input" name="vent_mode" data-label="${tr('Vent Mode', '??? ??????')}>
                <option>None (Room Air)</option>
                <option>Nasal Cannula</option>
                <option>CPAP / BiPAP</option>
                <option>Volume Control (VC)</option>
                <option>Pressure Control (PC)</option>
                <option>SIMV</option>
            </select>
        </div>
    `;
} else {
    html += `
        <div class="form-group mb-8">
            <label>S - ${tr('Subjective', '?????? ?????????? ?????????')}</label>
            <textarea class="form-input form-textarea dr-spec-input" name="soap_s" data-label="SOAP - Subjective"
rows="1" style="min-height:40px"></textarea>
        </div>
        <div class="form-group mb-8">
            <label>O - ${tr('Objective', '????? ?????????? ?????????')}</label>
            <textarea class="form-input form-textarea dr-spec-input" name="soap_o" data-label="SOAP - Objective" r
ows="1" style="min-height:40px"></textarea>
        </div>
        <div class="form-group mb-8">
            <label>A - ${tr('Assessment', '????????? ?????????? ?????')}</label>
            <textarea class="form-input form-textarea dr-spec-input" name="soap_a" data-label="SOAP - Assessment"
rows="1" style="min-height:40px"></textarea>
        </div>
        <div class="form-group mb-8">
            <label>P - ${tr('Plan', '????? ?????????? ?????????')}</label>
            <textarea class="form-input form-textarea dr-spec-input" name="soap_p" data-label="SOAP - Plan" rows="
1" style="min-height:40px"></textarea>
        </div>
    `;
}

```



```

}

html += `</div>`;
container.innerHTML = html;
if (specLower.includes('pediatr') || specLower.includes('neonat')) {
  setTimeout(() => { if (typeof window.updatePediatricGrowthChart === 'function') window.updatePediatricGrowthChart(); }, 50);
}
};

window.calculateGCS = () => {
  const e = parseInt(document.getElementById('gcsEyes')?.value || '4');
  const v = parseInt(document.getElementById('gcsVerbal')?.value || '5');
  const m = parseInt(document.getElementById('gcsMotor')?.value || '6');
  const total = e + v + m;
  const output = document.getElementById('drGcsOutput');
  if (output) output.value = `${total} / 15`;
};

window.calculateAPGAR = () => {
  const hr = parseInt(document.getElementById('apgarHR')?.value || '2');
  const reflex = parseInt(document.getElementById('apgarReflex')?.value || '2');
  const total = hr + reflex + 6;
  const input = document.getElementById('drApgarInput');
  if (input) input.value = total;
};

window.calculateEDD = () => {
  const lmpVal = document.getElementById('drLmpDate')?.value;
  if (!lmpVal) return;
  const lmp = new Date(lmpVal);
  const edd = new Date(lmp);
  edd.setDate(lmp.getDate() + 7);
  edd.setMonth(lmp.getMonth() + 9);
  const output = document.getElementById('drEddOutput');
  if (output) {
    output.value = edd.toISOString().split('T')[0];
  }
};

window.ICD10_DICTIONARY = [
  { code: 'I10', en: 'Essential (primary) hypertension', ar: '????? ???? ??????' },
  { code: 'E11.9', en: 'Type 2 diabetes mellitus without complications', ar: '??? ?????? ????? ??????' },
  { code: 'J06.9', en: 'Acute upper respiratory infection, unspecified', ar: '????? ???? ?? ?????? ?????? ???' },
  { code: 'K59.0', en: 'Constipation', ar: '???????' },
  { code: 'M54.5', en: 'Low back pain', ar: '??? ???? ??????' },
  { code: 'N39.0', en: 'Urinary tract infection, site not specified', ar: '????? ?????? ??????' },
  { code: 'H66.9', en: 'Otitis media, unspecified', ar: '????? ?????? ??????' },
  { code: 'J45.909', en: 'Unspecified asthma, uncomplicated', ar: '????? ??? ??????' },
  { code: 'Z00.00', en: 'Encounter for general adult medical examination', ar: '??? ??? ???' },
  { code: 'O80', en: 'Encounter for full-term uncomplicated delivery', ar: '????? ?????? ??? ??????' },
  { code: 'P07.3', en: 'Preterm infant', ar: '??? ??????' },
  { code: 'R07.9', en: 'Chest pain, unspecified', ar: '??? ?? ??????' }
];

```

```

];

window.handleDiagAutocomplete = (input) => {
  const q = input.value.trim().toLowerCase();
  const box = document.getElementById('drDiagSuggestions');
  if (!box) return;
  if (!q) {
    box.style.display = 'none';
    return;
  }
  const filtered = window.ICD10_DICTIONARY.filter(item =>
    item.code.toLowerCase().includes(q) ||
    item.en.toLowerCase().includes(q) ||
    item.ar.includes(q)
  );
  if (filtered.length === 0) {
    box.style.display = 'none';
    return;
  }
  box.style.display = 'block';
  box.innerHTML = filtered.map(item => `
    <div class="autocomplete-item" style="padding:8px 12px;cursor:pointer;border-bottom:1px solid var(--border-color,#f3f4f6);font-size:12px;" onclick="window.selectDiagSuggestion('${item.code}', '${window.jsStr(isArabic ? item.ar : item.en)}')">
      <strong style="color:var(--primary,#0ea5e9)">${escapeHTML(item.code)}</strong> - ${escapeHTML(isArabic ?
item.ar : item.en)}
    </div>
  `).join('');
};

window.selectDiagSuggestion = (code, diagText) => {
  const diagInput = document.getElementById('drDiag');
  const icdInput = document.getElementById('drIcd');
  if (diagInput) diagInput.value = diagText;
  if (icdInput) icdInput.value = code;
  const box = document.getElementById('drDiagSuggestions');
  if (box) box.style.display = 'none';
};

// Dismiss autocomplete box on outside click
document.addEventListener('click', (e) => {
  const box = document.getElementById('drDiagSuggestions');
  if (box && !e.target.closest('#drDiag')) {
    box.style.display = 'none';
  }
});

window.updatePediatricGrowthChart = () => {
  const ageInput = document.getElementById('drPedAge');
  const weightInput = document.getElementById('drPedWeight');
  const heightInput = document.getElementById('drPedHeight');
  const headInput = document.getElementById('drPedHeadCirc');
  const chartTypeSelect = document.getElementById('pedChartType');
  const point = document.getElementById('pedGrowthPoint');

```

```

if (!ageInput || !point) return;

const age = parseFloat(ageInput.value) || 0;
const chartType = chartTypeSelect ? chartTypeSelect.value : 'weight';

// Update UI Elements
const p3 = document.getElementById('pedPercentile3');
const p50 = document.getElementById('pedPercentile50');
const p97 = document.getElementById('pedPercentile97');
const yLabel = document.getElementById('pedYLabel');

const x = 30 + Math.min(Math.max(age, 0), 36) / 36 * 250;
let y = 130;

if (chartType === 'weight') {
  if (yLabel) yLabel.textContent = tr('Weight (kg)', '????? (???)');
  if (p3) p3.setAttribute('d', 'M 30,120 Q 100,105 180,95 T 280,85');
  if (p50) p50.setAttribute('d', 'M 30,110 Q 100,85 180,75 T 280,60');
  if (p97) p97.setAttribute('d', 'M 30,95 Q 100,65 180,50 T 280,30');

  const weight = weightInput ? parseFloat(weightInput.value) || 0 : 0;
  y = 130 - Math.min(Math.max(weight, 0), 18) / 18 * 120;

} else if (chartType === 'height') {
  if (yLabel) yLabel.textContent = tr('Height (cm)', '????? (??)');
  if (p3) p3.setAttribute('d', 'M 30,125 Q 100,105 180,80 T 280,60');
  if (p50) p50.setAttribute('d', 'M 30,115 Q 100,90 180,65 T 280,40');
  if (p97) p97.setAttribute('d', 'M 30,105 Q 100,75 180,50 T 280,20');

  const height = heightInput ? parseFloat(heightInput.value) || 0 : 0;
  y = 130 - Math.min(Math.max(height - 40, 0), 65) / 65 * 120;

} else if (chartType === 'head') {
  if (yLabel) yLabel.textContent = tr('Head Circ (cm)', '???? ???? (??)');
  if (p3) p3.setAttribute('d', 'M 30,120 Q 100,95 180,85 T 280,75');
  if (p50) p50.setAttribute('d', 'M 30,110 Q 100,80 180,65 T 280,50');
  if (p97) p97.setAttribute('d', 'M 30,95 Q 100,60 180,45 T 280,30');

  const head = headInput ? parseFloat(headInput.value) || 0 : 0;
  y = 130 - Math.min(Math.max(head - 30, 0), 25) / 25 * 120;
}

point.setAttribute('cx', String(x));
point.setAttribute('cy', String(y));
};

window.administerVaccine = (vaccineId) => {
  const row = document.getElementById('vaccine_' + vaccineId);
  if (row) {
    let name = vaccineId === 'dtap' ? 'Hexavalent' : 'PCV';
    row.innerHTML = `<span>${name}</span><span style="color:#22c55e;font-weight:bold">? ${tr('Given', '??
?????')}</span>`;
    showToast(tr('Vaccine administered successfully', '?? ????? ?????? ?????? ??????'));
  }
}

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const badge = document.getElementById('vaccineAlertBadge');
if (badge) {
  const current = badge.textContent.includes('2') ? '1 Pending' : tr('Completed', '?????');
  badge.textContent = current;
  if (current === tr('Completed', '?????')) {
    badge.className = 'badge badge-success';
  }
}
};

window.showOBGYNUltrasoundForm = async (pregId, patientId) => {
  let rows = '';
  try {
    const entries = await API.get('/api/obgyn/ultrasounds/' + pregId);
    rows = entries.map(e => `
      <tr>
        <td>${escapeHTML(String(e.scan_date || '').replace('T', ' ').slice(0, 10))}</td>
        <td>BPD: ${e.bpd}mm, HC: ${e.hc}mm, AC: ${e.ac}mm, FL: ${e.fl}mm</td>
        <td>${e.efw}g (${escapeHTML(e.efw_percentile || '-')}</td>
        <td>AFI: ${e.amniotic_fluid_index}cm</td>
        <td>${escapeHTML(e.placenta_location)} (Gr: ${escapeHTML(e.placenta_grade)})</td>
        <td>${e.fetal_heart_rate} bpm</td>
      </tr>
    `).join('');
  } catch (e) { }

  const html = `
    <div style="max-height:200px;overflow-y:auto;margin-bottom:16px;border:1px solid var(--outline);border-radius:8px">
      <table class="data-table">
        <thead>
          <tr>
            <th>${tr('Date', '???????')}</th>
            <th>${tr('Biometry', '???????')}</th>
            <th>${tr('EFW', '????? ?????')}</th>
            <th>AFI</th>
            <th>${tr('Placenta', '???????')}</th>
            <th>FHR</th>
          </tr>
        </thead>
        <tbody>${rows}</tbody>
      </table>
    </div>
    ` : `<p style="color:var(--text-muted);margin-bottom:16px">${tr('No previous ultrasound scans recorded', '
    ?? ???? ???? ????? ?????')}</p>`;

  <h4 style="margin-bottom:8px">${tr('New Obstetric Ultrasound Scan', '??? ????? ?????')}</h4>
  <div class="form-grid" style="gap:8px">
    <div class="form-group"><label>BPD (Biparietal Diameter - mm)</label><input type="number" id="usBpd" class="form-control" placeholder="e.g. 45"></div>
    <div class="form-group"><label>HC (Head Circumference - mm)</label><input type="number" id="usHc" class="

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"form-control" placeholder="e.g. 175"></div>
    <div class="form-group"><label>AC (Abdominal Circumference - mm)</label><input type="number" id="usAc" c
lass="form-control" placeholder="e.g. 160"></div>
    <div class="form-group"><label>FL (Femur Length - mm)</label><input type="number" id="usFl" class="form-
control" placeholder="e.g. 35"></div>
    <div class="form-group"><label>EFW (Est. Fetal Weight - grams)</label><input type="number" id="usEfw" cl
ass="form-control" placeholder="e.g. 450"></div>
    <div class="form-group"><label>AFI (Amniotic Fluid Index - cm)</label><input type="number" id="usAfi" cl
ass="form-control" placeholder="e.g. 12" step="0.1"></div>
    <div class="form-group"><label>Placenta Location</label><select id="usPlacentaLoc" class="form-control">
<option>Anterior</option><option>Posterior</option><option>Fundal</option><option>Previa</option></select></di
v>
    <div class="form-group"><label>Placenta Grade</label><select id="usPlacentaGrade" class="form-control"><
option>Grade 0</option><option>Grade I</option><option>Grade II</option><option>Grade III</option></select></d
iv>
    <div class="form-group"><label>Fetal Heart Rate (bpm)</label><input type="number" id="usFhr" class="form
-control" placeholder="e.g. 140"></div>
    <div class="form-group"><label>Number of Fetuses</label><input type="number" id="usFetuses" class="form-
control" value="1"></div>
    <div class="form-group"><label>Cervical Length (mm)</label><input type="number" id="usCervix" class="for
m-control" placeholder="e.g. 35"></div>
    <div class="form-group"><label>Fetal Gender</label><select id="usGender" class="form-control"><option>No
t determined</option><option>Male</option><option>Female</option></select></div>
</div>
<div class="form-group" style="margin-top:8px"><label>Findings</label><textarea id="usFindings" class="for
m-control" rows="2" placeholder="e.g. Normal fetal anatomy..."></textarea></div>
<div class="form-group"><label>Impression</label><textarea id="usImpression" class="form-control" rows="2"
placeholder="e.g. Single active fetus matching GA..."></textarea></div>
<button class="btn btn-primary" onclick="window.saveOBGYNUltrasound('${pregId}', '${patientId}')" style="margi
n-top:8px">? ${tr('Save Scan', '??? ??? ???????')}</button>
`;
    showModal(tr('Obstetric Ultrasound / Fetal Biometry', '???? ??????? ?????????? ??????? ???????'), html);
};

```

```

window.saveOBGYNUltrasound = async (pregId, patientId) => {
  try {
    await API.post('/api/obgyn/ultrasounds', {
      pregnancy_id: pregId,
      patient_id: patientId,
      scan_type: 'Routine Obstetric',
      bpd: parseFloat(document.getElementById('usBpd').value) || 0,
      hc: parseFloat(document.getElementById('usHc').value) || 0,
      ac: parseFloat(document.getElementById('usAc').value) || 0,
      fl: parseFloat(document.getElementById('usFl').value) || 0,
      efw: parseFloat(document.getElementById('usEfw').value) || 0,
      amniotic_fluid_index: parseFloat(document.getElementById('usAfi').value) || 0,
      placenta_location: document.getElementById('usPlacentaLoc').value,
      placenta_grade: document.getElementById('usPlacentaGrade').value,
      fetal_heart_rate: parseInt(document.getElementById('usFhr').value) || 0,
      number_of_fetuses: parseInt(document.getElementById('usFetuses').value) || 1,
      cervical_length: parseFloat(document.getElementById('usCervix').value) || 0,
      fetal_gender: document.getElementById('usGender').value,
      findings: document.getElementById('usFindings').value,
      impression: document.getElementById('usImpression').value
    })
  }
}

```

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    });
    showToast(tr('Ultrasound scan saved successfully!', '?? ??? ??? ??????? ?????!'));
    document.querySelector('.modal-overlay')?.remove();
    navigateTo(currentPage);
  } catch (e) {
    showToast(tr('Error saving scan', '??? ?? ??? ???????'), 'error');
  }
};

window.simulateBarcodeScan = async (barcodeText, orderId) => {
  const statusEl = document.getElementById('lblScanStatus');
  if (statusEl) {
    statusEl.textContent = 'SCANNED_IN_ANALYZER';
    statusEl.style.background = '#22c55e';
    statusEl.style.color = 'fff';
  }
  showToast(tr('Barcode Scanned: ' + barcodeText, '?? ??? ?????????: ' + barcodeText), 'success');

  try {
    await API.put(`/api/lab/orders/${orderId}`, { status: 'In Progress' });
    showToast(tr('Analyzer updated: Sample state is now In Progress', '?? ????? ??????: ??? ?????? ??? ? ?
    ?????'), 'success');
    setTimeout(() => {
      document.querySelector('.modal-overlay')?.remove();
      navigateTo(currentPage);
    }, 1500);
  } catch (e) { }
};

window.printThermalLabelContent = (labelId) => {
  const labelHtml = document.getElementById('thermalLabelContainer')?.innerHTML || '';
  const printWin = window.open('', '_blank', 'width=350,height=300');
  printWin.document.write(`
    <!DOCTYPE html>
    <html>
    <head>
      <title>Thermal Label Print</title>
      <style>
        body { margin:0; padding:10px; display:flex; justify-content:center; align-items:center; background:
    #fff; }
        #thermalLabelContainer { box-shadow: none !important; border: 1px solid #000 !important; }
      </style>
    </head>
    <body>
      \${labelHtml}
      <script>setTimeout(() => { window.print(); window.close(); }, 300);</script>
    </body>
    </html>
  `);
  printWin.document.close();
};

window.__dicomState = { contrast: 1, brightness: 1, scale: 1, panX: 0, panY: 0 };

```

```

window.adjustDicom = () => {
  const contrast = parseFloat(document.getElementById('dicomContrast')?.value || '1');
  const brightness = parseFloat(document.getElementById('dicomBrightness')?.value || '1');
  window.__dicomState.contrast = contrast;
  window.__dicomState.brightness = brightness;

  const img = document.getElementById('dicomImage');
  if (img) {
    img.style.setProperty('--d-contrast', String(contrast));
    img.style.setProperty('--d-brightness', String(brightness));
  }

  const meta = document.getElementById('dicomMeta');
  if (meta) {
    meta.textContent = `WL: ${Math.round(brightness * 40)} / WW: ${Math.round(contrast * 400)}`;
  }
};

window.zoomDicom = (delta) => {
  window.__dicomState.scale = Math.min(Math.max(window.__dicomState.scale + delta, 0.5), 3.0);
  const img = document.getElementById('dicomImage');
  if (img) {
    img.style.setProperty('--d-scale', String(window.__dicomState.scale));
  }
};

window.panDicom = (dx, dy) => {
  window.__dicomState.panX += dx;
  window.__dicomState.panY += dy;
  const img = document.getElementById('dicomImage');
  if (img) {
    img.style.setProperty('--d-pan-x', window.__dicomState.panX + 'px');
    img.style.setProperty('--d-pan-y', window.__dicomState.panY + 'px');
  }
};

window.resetDicom = () => {
  window.__dicomState = { contrast: 1, brightness: 1, scale: 1, panX: 0, panY: 0 };
  const contrastInput = document.getElementById('dicomContrast');
  const brightnessInput = document.getElementById('dicomBrightness');
  if (contrastInput) contrastInput.value = '1';
  if (brightnessInput) brightnessInput.value = '1';

  const img = document.getElementById('dicomImage');
  if (img) {
    img.style.setProperty('--d-contrast', '1');
    img.style.setProperty('--d-brightness', '1');
    img.style.setProperty('--d-scale', '1');
    img.style.setProperty('--d-pan-x', '0px');
    img.style.setProperty('--d-pan-y', '0px');
  }
  const meta = document.getElementById('dicomMeta');
  if (meta) meta.textContent = 'WL: 40 / WW: 400';
};

```

```

window.applyRadReportTemplate = (val) => {
  const findings = document.getElementById('rrFindings');
  const impression = document.getElementById('rrImpression');
  if (!findings || !impression) return;

  if (val === 'BI-RADS') {
    findings.value = 'BREAST DENSITY: Scattered fibroglandular tissues.\n\nFINDINGS: No dominant mass, architectural distortion, or suspicious microcalcifications in either breast.\n\nAXILLARY LYMPH NODES: Normal in appearance.';
    impression.value = 'No mammographic evidence of malignancy.';
    const birads = document.getElementById('rrBirads');
    if (birads) birads.value = '1';
  } else {
    findings.value = '';
    impression.value = '';
    const birads = document.getElementById('rrBirads');
    if (birads) birads.value = '';
  }
};

// =====
// ===== PHASE B & C & D & E UI MODULES =====
// =====

// ??? PHASE B: SAUDI COMPLIANCE UI ?????????????????????????????????????????????????????????????
window.renderNphiesRemittance = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const summary = await API.get('/api/nphies/remittance/summary').catch(() => ({}));
    const list = await API.get('/api/nphies/remittance').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('NPHIES Remittance Advice (RA)', '??????? ?????? ?????? ??????')}</h3>
        <div class="grid grid-cols-4 gap-12 mb-16">
          <div class="card p-12 bg-primary-container text-center">
            <div class="text-xs text-on-surface-variant">${tr('Total Paid', '??????? ???????? ????????')}</div>
            <div class="text-xl font-bold text-primary">SAR ${parseFloat(summary.total_paid||0).toFixed(2)}</div>
          </div>
          <div class="card p-12 bg-secondary-container text-center">
            <div class="text-xs text-on-surface-variant">${tr('Total Adjustments', '????????')}</div>
            <div class="text-xl font-bold text-secondary">SAR ${parseFloat(summary.total_adjusted||0).toFixed(2)}</div>
          </div>
          <div class="card p-12 bg-error-container text-center">
            <div class="text-xs text-on-surface-variant">${tr('Total Denied', '????????')}</div>
            <div class="text-xl font-bold text-error">SAR ${parseFloat(summary.total_denied||0).toFixed(2)}</div>
          </div>
          <div class="card p-12 bg-surface-variant text-center">
            <div class="text-xs text-on-surface-variant">${tr('Unposted Approved', '????? ??? ?????')}</div>

```



```

        <div class="text-xl font-bold">${summary.unposted_approved || 0}</div>
    </div>
</div>

<h4 class="font-bold mb-8">? ${tr('Remittance Transactions', '?? ?????? ???????')}</h4>
<div class="overflow-x-auto">
    <table class="w-full text-sm">
        <thead>
            <tr class="bg-surface-variant">
                <th class="p-8 text-right">${tr('Claim ID', '?? ???????')}</th>
                <th class="p-8 text-right">${tr('Payer', '???? ???????')}</th>
                <th class="p-8 text-right">${tr('Payment Date', '???? ?????')}</th>
                <th class="p-8 text-right">${tr('Reference', '?? ??????')}</th>
                <th class="p-8 text-right">${tr('Paid Amount', '?????')}</th>
                <th class="p-8">${tr('Status', '?????')}</th>
                <th class="p-8">${tr('Actions', '????????')}</th>
            </tr>
        </thead>
        <tbody>
            ${list.map(r => `
                <tr class="border-b">
                    <td class="p-8 font-medium">${r.claim_number || r.claim_id}</td>
                    <td class="p-8">${escapeHTML(r.patient_name || 'Insurance Payer')}</td>
                    <td class="p-8">${r.payment_date ? new Date(r.payment_date).toLocaleDateString('ar-SA') : '-
'}</td>

                    <td class="p-8 text-xs font-mono">${escapeHTML(r.payment_reference || '')}</td>
                    <td class="p-8 font-bold text-primary">SAR ${parseFloat(r.payment_amount).toFixed(2)}</td>
                    <td class="p-8">${r.adjudication_status === 'approved' ? '<span class="badge bg-secondary-co
ntainer">Approved</span>' : '<span class="badge bg-error-container">Denied</span>'}</td>
                    <td class="p-8">
                        ${r.posted_to_gl ? '<span class="text-xs text-secondary">? Posted</span>' : '<button class
="btn btn-sm btn-primary" onclick="postRemittanceToAR(${r.id}, '${containerId}')">? Post to AR</button>`}
                    </td>
                </tr>
            `).join('') || `<tr><td colspan="7" class="text-center p-8 text-on-surface-variant">${tr('No rec
ords found', '?? ???? ????? ?????')}</td></tr>`}
        </tbody>
    </table>
</div>
</div>
`;
} catch (err) { el.innerHTML = `<div class="card p-20 text-error">Error: ${err.message}</div>`; }
};

window.postRemittanceToAR = async function(id, containerId) {
    try {
        await API.post(`/api/nphies/remittance/${id}/post-to-ar`);
        showToast(tr('Remittance posted to Accounts Receivable', '?? ????? ??????? ???? ????????? ?????????'));
        renderNphiesRemittance(containerId);
    } catch (e) { showToast(e.message || 'Error', 'error'); }
};

window.renderZatcaCreditNote = async function(containerId, invoiceId) {
    const el = document.getElementById(containerId);

```

```

if (!el) return;
try {
  const cnList = await API.get('/api/zatca/credit-notes').catch(() => []);
  const chainInfo = await API.get('/api/zatca/invoice-chain').catch(() => ({}));
  el.innerHTML = `
    <div class="card p-20 mb-16">
      <h3 class="card-title text-primary">? ${tr('ZATCA Credit Notes & Chaining', '????????? ?????? ?????? ?
????? ????? ??????')}</h3>
      <div class="mb-16 p-12 rounded ${chainInfo.chain_valid ? 'bg-secondary-container/20 text-secondary' :
'bg-error-container/20 text-error'}">
        <strong>? ${tr('Invoice Chain Status:', '???? ????? ????? ???????:')}</strong>
        ${chainInfo.chain_valid ? tr('Valid (All invoice hashes cryptographically chained)', '???? (???? ??
?? ????????? ??????? ??????)') : tr('Chain Break Detected', '???? ?????? ?? ????? ??????')}
      </div>

      <div class="grid grid-cols-2 gap-16 mb-16">
        <div class="card p-16">
          <h4 class="font-bold mb-8">? ${tr('Generate Credit Note', '???? ????? ???/?????')}</h4>
          <div class="space-y-8">
            <div>
              <label class="label">${tr('Invoice reference ID', '?? ????????? ??????')}</label>
              <input id="cnInvoiceId" class="input w-full" type="number" placeholder="e.g. 1" value="${invoi
ceId||''}">
            </div>
            <div>
              <label class="label">${tr('Reason for credit', '?? ?????')}</label>
              <select id="cnReasonCode" class="input w-full">
                <option value="CANCEL">???? ?????</option>
                <option value="RETURN">???? ?????/????</option>
                <option value="DISCOUNT">?? ?????</option>
                <option value="ERROR">???? ?? ?????</option>
              </select>
            </div>
            <div>
              <label class="label">${tr('Reason description', '?? ????? ??????')}</label>
              <input id="cnReasonDesc" class="input w-full" placeholder="e.g. Patient discharged early">
            </div>
            <button class="btn btn-primary mt-8" onclick="submitCreditNote('${containerId}')">? Generate Not
e</button>
          </div>
        </div>

        <div class="card p-16">
          <h4 class="font-bold mb-8">? ${tr('Recent Credit Notes', '????????? ??????')}</h4>
          <div class="overflow-y-auto max-h-200 space-y-8">
            ${cnList.map(c => `
              <div class="p-8 border-b text-xs flex justify-between items-center">
                <div>
                  <strong>${escapeHTML(c.credit_note_number)}</strong> (Orig: #${c.original_invoice_id})<br>
                  <span class="text-on-surface-variant">${escapeHTML(c.credit_reason)}</span>
                </div>
                <div>
                  <span class="font-bold">SAR ${parseFloat(c.total_with_vat).toFixed(2)}</span>
                  ${c.submission_status === 'Submitted_Mock' || c.submission_status === 'RECORDED' ? `<span

```

```

class="badge bg-secondary-container ml-4"> Cleared</span>` : `<button class="btn btn-xs btn-outline ml-4" oncl
lick="submitCnToZatca(${c.id}, '${containerId}')">Submit</button>`
    </div>
  </div>
  `).join('') || `<p class="text-on-surface-variant">${tr('No credit notes','?? ???? ?????? ?????
')}}</p>`
    </div>
  </div>
</div>
</div>
</div>
`;
} catch (err) { el.innerHTML = `<div class="card p-20 text-error">Error: ${err.message}</div>`; }
};

window.submitCreditNote = async function(containerId) {
  const invoice_id = document.getElementById('cnInvoiceId')?.value;
  const credit_reason = document.getElementById('cnReasonCode')?.value;
  const credit_reason_description = document.getElementById('cnReasonDesc')?.value;
  if (!invoice_id) { showToast('Invoice ID required', 'error'); return; }
  try {
    await API.post('/api/zatca/credit-note', { invoice_id, credit_reason, credit_reason_description });
    showToast(tr('Credit note created successfully','?? ????? ??????? ?????? ??????'));
    renderZatcaCreditNote(containerId);
  } catch (e) { showToast(e.message||'Error', 'error'); }
};

window.submitCnToZatca = async function(id, containerId) {
  try {
    await API.post(`/api/zatca/credit-note/${id}/submit`);
    showToast(tr('Submitted & cleared by ZATCA sandbox','?? ??????? ??????? ??? ???? ?????? ?????????'));
    renderZatcaCreditNote(containerId);
  } catch (e) { showToast(e.message||'Error', 'error'); }
};

window.renderHrSaudi = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const listCred = await API.get('/api/hr/credentialing').catch(() => []);
    const nitaqat = await API.get('/api/hr/nitaqat').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">?? ${tr('HR Saudi Localization (GOSI, WPS, Nitaqat, Credentials)',
'???? ??????? ????????? ????????? ?????????')}</h3>

        <div class="grid grid-cols-3 gap-16 mb-16">
          <div class="card p-16">
            <h4 class="font-bold text-secondary mb-8">? ${tr('Nitaqat & Saudization','???? ?????? ??? ????')
'></h4>
            <div class="mb-12">
              <label class="label">Required Saudization %</label>
              <input id="nitaqatReq" class="input w-full mb-8" type="number" value="25">
              <button class="btn btn-sm btn-primary" onclick="calculateNitaqat('${containerId}')">Recalculate
Band</button>

```

```

    </div>
    ${nitaqat.length ? `
      <div class="p-12 bg-primary-container text-center rounded">
        <div class="text-xs">Current Saudization</div>
        <div class="text-xl font-bold">${parseFloat(nitaqat[0].saudization_pct).toFixed(2)}%</div>
        <div class="text-xs mt-4">Nitaqat Band: <strong>${escapeHTML(nitaqat[0].nitaqat_band)}</strong>
    </div>

    </div>
    ` : '<p class="text-xs text-on-surface-variant">No snapshot calculated yet</p>'}
  </div>

  <div class="card p-16">
    <h4 class="font-bold text-secondary mb-8">?? ${tr('GOSI Monthly Contribution','????????? ?????????? ?
    ?????????? GOSI'))}</h4>
    <div class="mb-12">
      <label class="label">Select Month</label>
      <input id="gosiMonth" class="input w-full mb-8" type="date" value="${new Date().toISOString().sl
    ice(0,7)}-01">
      <button class="btn btn-sm btn-primary" onclick="calculateGosi('${containerId}')">Calculate GOSI<
    /button>
    </div>
    <div id="gosiCalcResult" class="text-xs"></div>
  </div>

  <div class="card p-16">
    <h4 class="font-bold text-secondary mb-8">?? ${tr('WPS (Wage Protection SIF)','????? ?????? ??????? ?
    ?????? SIF'))}</h4>
    <div class="mb-12">
      <label class="label">WPS SIF Month</label>
      <input id="wpsMonth" class="input w-full mb-8" type="date" value="${new Date().toISOString().sli
    ce(0,7)}-01">
      <button class="btn btn-sm btn-primary" onclick="generateWps('${containerId}')">Generate SIF File
    </button>
    </div>
    <div id="wpsResult" class="text-xs"></div>
  </div>

  <div class="card p-16">
    <h4 class="font-bold mb-8">? ${tr('Physician Credentialing & Medical Licenses','??????? ??????? ???????
    ?????????????')}</h4>
    <div class="grid grid-cols-4 gap-8 mb-12">
      <input id="credEmpId" class="input w-full" type="number" placeholder="Employee ID">
      <input id="credNumber" class="input w-full" placeholder="License Number">
      <input id="credType" class="input w-full" placeholder="Type (e.g. SCFHS License)">
      <input id="credExpiry" class="input w-full" type="date">
    </div>
    <button class="btn btn-primary btn-sm mb-12" onclick="saveCredential('${containerId}')">Add Credenti
    al</button>

    <div class="overflow-x-auto">
      <table class="w-full text-xs">
        <thead>
          <tr class="bg-surface-variant">

```

```

        <th class="p-8 text-right">Physician</th>
        <th class="p-8 text-right">License</th>
        <th class="p-8 text-right">Expiry Date</th>
        <th class="p-8">Verification Status</th>
        <th class="p-8">Action</th>
    </tr>
</thead>
<tbody>
    ${listCred.map(c => `
        <tr class="border-b">
            <td class="p-8">${escapeHTML(c.employee_name)}</td>
            <td class="p-8">${escapeHTML(c.credential_type)} (#${escapeHTML(c.credential_number)})</td>
            <td class="p-8 font-bold text-error">${c.expiry_date ? new Date(c.expiry_date).toLocaleDateString('ar-SA') : ''}</td>
            <td class="p-8"><span class="badge ${c.verification_status==='verified'? 'bg-secondary-container': 'bg-error-container'}">${c.verification_status}</span></td>
            <td class="p-8">
                ${c.verification_status !== 'verified' ? `<button class="btn btn-xs btn-outline" onclick="verifyCred(${c.id}, '${containerId}')">Verify</button>` : ''}
            </td>
        </tr>
    `).join('')
</tbody>
</table>
</div>
</div>
</div>
`;
    } catch (err) { el.innerHTML = `<div class="card p-20 text-error">Error: ${err.message}</div>`; }
};

window.calculateNitaqat = async function(containerId) {
    const req_pct = document.getElementById('nitaqatReq')?.value;
    try {
        await API.post('/api/hr/nitaqat/calculate', { required_pct: req_pct });
        showToast(tr('Nitaqat snapshot generated', '?? ????? ????? ?????'));
        renderHrSaudi(containerId);
    } catch (e) { showToast(e.message, 'error'); }
};

window.calculateGosi = async function(containerId) {
    const month_year = document.getElementById('gosiMonth')?.value;
    if (!month_year) return;
    try {
        const res = await API.post('/api/hr/gosi/calculate', { month_year });
        document.getElementById('gosiCalcResult').innerHTML = `
            <div class="mt-8 bg-surface-variant/40 p-8 rounded">
                <strong>? Calculated!</strong><br>
                Employees: ${res.employee_count}<br>
                Total Contribution: SAR ${parseFloat(res.total_contributions).toFixed(2)}
            </div>
        `;
    } catch (e) { showToast(e.message, 'error'); }
};

```

```

};

window.generateWps = async function(containerId) {
  const payroll_month = document.getElementById('wpsMonth')?.value;
  if (!payroll_month) return;
  try {
    const res = await API.post('/api/hr/wps/generate', { payroll_month });
    document.getElementById('wpsResult').innerHTML = `
      <div class="mt-8 bg-surface-variant/40 p-8 rounded font-mono text-xs">
        <strong>? File SIF Ref: ${res.wps_file.file_reference}</strong><br>
        <pre class="overflow-x-auto max-w-full">${escapeHTML(res.preview.join('\n'))}</pre>
      </div>
    `;
  } catch (e) { showToast(e.message, 'error'); }
};

window.saveCredential = async function(containerId) {
  const employee_id = document.getElementById('credEmpId')?.value;
  const credential_number = document.getElementById('credNumber')?.value;
  const credential_type = document.getElementById('credType')?.value;
  const expiry_date = document.getElementById('credExpiry')?.value;
  if (!employee_id || !credential_number) return;
  try {
    await API.post('/api/hr/credentialing', { employee_id, credential_number, credential_type, expiry_date });
    showToast(tr('Credential added', '?? ????? ??????? ?????'));
    renderHrSaudi(containerId);
  } catch (e) { showToast(e.message, 'error'); }
};

window.verifyCred = async function(id, containerId) {
  try {
    await API.put(`/api/hr/credentialing/${id}/verify`, { verification_status: 'verified' });
    showToast(tr('Credential verified', '?? ????? ??????? ?????'));
    renderHrSaudi(containerId);
  } catch (e) { showToast(e.message, 'error'); }
};

// ??? PHASE C: CLINICAL QUALITY UI ?????????????????????????????????????????????????????????????
window.renderControlledSubstances = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const csList = await API.get('/api/pharmacy/controlled-substances').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('Controlled Substances Safe Ledger', '??? ?????? ????????? ??
        ??????? ???????')}</h3>

        <div class="grid grid-cols-2 gap-16 mb-16">
          <div class="card p-16">
            <h4 class="font-bold text-secondary mb-8">?? ${tr('Double-Witness Dispensing', '??? ????? ??????
            ??? (???? ?????)}</h4>
            <div class="space-y-8">

```

```

<div>
  <label class="label">Select Controlled Drug</label>
  <select id="dispenseCsId" class="input w-full">
    ${csList.map(c => `<option value="${c.id}">${escapeHTML(c.drug_name)} (Bal: ${c.closing_balance} ${c.unit})</option>`).join('')}
  </select>
</div>
<div>
  <label class="label">Patient ID</label>
  <input id="dispensePatId" class="input w-full" type="number">
</div>
<div>
  <label class="label">Dispensing Quantity</label>
  <input id="dispenseQty" class="input w-full" type="number" value="1">
</div>
<div class="p-8 bg-surface-variant/40 rounded">
  <span class="text-xs text-secondary-container">? Double-Witness Authentication</span>
  <div class="grid grid-cols-2 gap-8 mt-4">
    <input id="dispenseWitnessName" class="input w-full text-xs" placeholder="Witness Nurse Name"
  >
    <input id="dispenseWitnessId" class="input w-full text-xs" type="number" placeholder="Witness User ID">
  </div>
</div>
<button class="btn btn-primary mt-8" onclick="dispenseControlledDrug('${containerId}')">? Double-Sign & Dispense</button>
</div>
</div>

<div class="card p-16">
  <h4 class="font-bold text-secondary mb-8">? ${tr('Safe Reconciliation Audit','????? ?????? ?????? ??????')}</h4>
  <div class="space-y-8">
    <input id="reconDrugName" class="input w-full" placeholder="Drug Name">
    <input id="reconDrugCode" class="input w-full" placeholder="Drug Formulary Code">
    <input id="reconClosing" class="input w-full" type="number" placeholder="Physical Count">
    <button class="btn btn-outline mt-8" onclick="reconcileControlledDrug('${containerId}')">Audit safe count</button>
  </div>
</div>
</div>

<h4 class="font-bold mb-8">? ${tr('Current Controlled Stock Ledger','???? ????? ?????? ?????? ??????')}</h4>
<div class="overflow-x-auto">
  <table class="w-full text-xs">
    <thead>
      <tr class="bg-surface-variant">
        <th class="p-8 text-right">Substance</th>
        <th class="p-8 text-right">Schedule</th>
        <th class="p-8 text-right">Physical Count</th>
        <th class="p-8 text-right">Discrepancy</th>
        <th class="p-8 text-right">Witnessed By</th>
      </tr>
    </thead>
  </table>
</div>

```

```

</thead>
<tbody>
  ${csList.map(c => `
    <tr class="border-b">
      <td class="p-8 font-medium">${escapeHTML(c.drug_name)}</td>
      <td class="p-8"><span class="badge bg-error-container">Class ${c.schedule_class}</span></td>
      <td class="p-8 font-bold">${c.closing_balance} ${c.unit}</td>
      <td class="p-8 ${parseFloat(c.discrepancy) !== 0 ? 'text-error font-bold':''}>${c.discrepancy}<
/td>

      <td class="p-8 text-xs">${escapeHTML(c.witnessed_by||'')}</td>
    </tr>
  `).join('')} || `<tr><td colspan="5" class="text-center p-8 text-on-surface-variant">${tr('No controlled stock recorded today','?? ??? ???? ???? ????')}</td></tr>`
  </tbody>
</table>
</div>
</div>
`;
} catch (err) { el.innerHTML = `<div class="card p-20 text-error">Error: ${err.message}</div>`; }
};

window.dispenseControlledDrug = async function(containerId) {
  const cs_id = document.getElementById('dispenseCsId')?.value;
  const quantity = document.getElementById('dispenseQty')?.value;
  const patient_id = document.getElementById('dispensePatId')?.value;
  const witness2_name = document.getElementById('dispenseWitnessName')?.value;
  const witness2_id = document.getElementById('dispenseWitnessId')?.value;
  if (!cs_id || !quantity || !witness2_name) { showToast('Controlled drug, quantity, and witness required', 'error'); return; }
  try {
    await API.post('/api/pharmacy/controlled-substances/dispense', { cs_id, quantity, patient_id, witness2_name, witness2_id });
    showToast(tr('Controlled Drug Dispensed & Logged','?? ??? ?????? ?????? ?????? ?????? ??????'));
    renderControlledSubstances(containerId);
  } catch (e) { showToast(e.message, 'error'); }
};

window.reconcileControlledDrug = async function(containerId) {
  const drug_name = document.getElementById('reconDrugName')?.value;
  const drug_code = document.getElementById('reconDrugCode')?.value;
  const closing_balance = document.getElementById('reconClosing')?.value;
  if (!drug_name || !drug_code || !closing_balance) return;
  try {
    await API.post('/api/pharmacy/controlled-substances/reconcile', { drug_name, drug_code, closing_balance });
  }
  ;
  showToast(tr('Safe audit logged','?? ????? ?????? ????????? ?????????'));
  renderControlledSubstances(containerId);
} catch (e) { showToast(e.message, 'error'); }
};

window.renderMedicationReconciliation = async function(containerId, patientId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {

```



```

const list = patientId ? await API.get('/api/clinical/medication-reconciliation/' + patientId).catch(() =>
[]) : [];
el.innerHTML = `
<div class="card p-20 mb-16">
  <h3 class="card-title text-primary">? ${tr('Medication Reconciliation','????? ?????? ??????')}</h3>
  ${patientId ? `
    <div class="grid grid-cols-2 gap-16 mb-16">
      <div class="card p-16">
        <h4 class="font-bold mb-8">? Record Med Rec</h4>
        <div class="space-y-8">
          <div>
            <label class="label">Type</label>
            <select id="mrType" class="input w-full">
              <option value="Admission">Admission (????)</option>
              <option value="Discharge">Discharge (????)</option>
              <option value="Transfer">Transfer (???)</option>
            </select>
          </div>
          <div>
            <label class="label">Home Medications List (JSON format)</label>
            <textarea id="mrHome" class="input w-full" rows="3" placeholder='[{"name":"Aspirin","dose":"
100mg"}]`></textarea>
          </div>
          <div class="flex items-center space-x-12 mt-8">
            <label class="flex items-center"><input id="mrAllergy" type="checkbox" class="mr-4"> Allergy
Verified</label>
            <label class="flex items-center ml-12"><input id="mrHighAlert" type="checkbox" class="mr-4">
High Alert Checked</label>
          </div>
          <button class="btn btn-primary mt-8 w-full" onclick="saveMedRecon(${patientId}, '${containerId
}')">Save Med Reconciliation</button>
        </div>
      </div>

      <div class="card p-16">
        <h4 class="font-bold mb-8">? Reconciliation History</h4>
        <div class="overflow-y-auto max-h-250 space-y-8">
          ${list.map(l => `
            <div class="p-12 border rounded bg-surface-variant/20 text-xs">
              <div class="flex justify-between font-bold">
                <span>${l.reconciliation_type}</span>
                <span>${new Date(l.performed_at).toLocaleDateString('ar-SA')}</span>
              </div>
              <div class="text-on-surface-variant mt-4">By: ${escapeHTML(l.performed_by_name)}</div>
              <div class="mt-4">Home Meds: ${JSON.stringify(l.home_medications)}</div>
            </div>
          `).join('')} || <p class="text-on-surface-variant">No history found</p>`
        </div>
      </div>
    </div>
  ` : `<p class="text-on-surface-variant text-center">${tr('Please select a patient to view Medication R
econciliation','???? ?????? ????? ?????? ???????? ????????')}</p>`
  </div>
`;

```

```

    } catch (err) { el.innerHTML = `<div class="card p-20 text-error">Error: ${err.message}</div>`; }
  };

window.saveMedRecon = async function(patientId, containerId) {
  const reconciliation_type = document.getElementById('mrType')?.value;
  const homeText = document.getElementById('mrHome')?.value || '[]';
  const allergy_verified = document.getElementById('mrAllergy')?.checked;
  const high_alert_checked = document.getElementById('mrHighAlert')?.checked;

  let home_medications = [];
  try { home_medications = JSON.parse(homeText); } catch (_) { showToast('Invalid home meds JSON', 'error'); return; }

  try {
    await API.post('/api/clinical/medication-reconciliation', {
      patient_id: patientId, reconciliation_type, home_medications, allergy_verified, high_alert_checked
    });
    showToast(tr('Medication reconciliation saved', '?? ??? ?????? ???????'));
    renderMedicationReconciliation(containerId, patientId);
  } catch (e) { showToast(e.message, 'error'); }
};

window.renderLabMicrobiology = async function(containerId, patientId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const list = patientId ? await API.get('/api/lab/microbiology/' + patientId).catch(() => []) : [];
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('Lab Microbiology & Culture Sensitivity', '??? ??????? ??????
        ?? ???????')}</h3>
        ${patientId ? `
          <div class="grid grid-cols-2 gap-16">
            <div class="card p-16">
              <h4 class="font-bold mb-8">? Record Culture Result</h4>
              <div class="space-y-8">
                <input id="micSpecimen" class="input w-full" placeholder="Specimen Type (e.g. Blood)">
                <input id="micOrganism" class="input w-full" placeholder="Organism Identified (e.g. E. coli)">
                <textarea id="micSensitivity" class="input w-full" rows="3" placeholder='Sensitivities [{"anti
                biotic": "Amoxicillin", "mic": "<=2", "interpretation": "S"}]'}></textarea>
                <div class="flex items-center mt-8">
                  <label class="flex items-center text-error font-bold">
                    <input id="micCritical" type="checkbox" class="mr-4"> Critical Pathogen / Urgent Alert
                  </label>
                </div>
                <button class="btn btn-primary w-full mt-8" onclick="saveMicroResult(${patientId}, '${containe
                rId}')">Submit Culture Report</button>
              </div>
            </div>
          </div>

          <div class="card p-16">
            <h4 class="font-bold mb-8">? Result History</h4>
            <div class="overflow-y-auto max-h-250 space-y-8">
              ${list.map(l => `

```



```

<div>
  <label class="label">ICD-10 Code Lookup</label>
  <input id="probIcd10" class="input w-full" placeholder="e.g. J45.9" onkeyup="lookupIcd10Drop
down(this.value)">
  <div id="icd10Dropdown" class="bg-surface border rounded max-h-120 overflow-y-auto hidden ab
solute z-10 w-250"></div>
</div>
<div>
  <label class="label">Problem Name</label>
  <input id="probName" class="input w-full" placeholder="Asthma / Diabetes...">
</div>
<div class="flex items-center mt-8">
  <label class="flex items-center">
    <input id="probPdx" type="checkbox" class="mr-4"> Principal Diagnosis (Pdx)
  </label>
</div>
<button class="btn btn-primary w-full mt-8" onclick="addPatientProblem(${patientId}, '${contai
nerId}')">Add to List</button>
</div>
</div>

<div class="card p-16">
  <h4 class="font-bold mb-8">? Active & Resolved Problems</h4>
  <div class="space-y-8">
    ${list.map(p => `
      <div class="p-8 border-b text-xs flex justify-between items-center">
        <div>
          <strong>${escapeHTML(p.problem_name)}</strong>
          ${p.icd10_code ? `<span class="badge bg-primary-container ml-4">${escapeHTML(p.icd10_cod
e)}</span>` : ''}
          ${p.principal_diagnosis ? `<span class="badge bg-error-container ml-4">Pdx</span>` : ''}
        </div>
        <span class="badge ${p.status==='Active'? 'bg-secondary-container': 'bg-surface-variant'}">${
p.status}</span>
      </div>
    `).join('')} || 'No active problems</p>'}
  </div>
</div>
</div>
` : `<p class="text-on-surface-variant text-center">${tr('Select patient to view Problem List','??? ?
??? ???? ????? ?????????')}</p>`}
</div>
`;
} catch(e) { el.innerHTML = `<div class="card p-20 text-error">Error: ${e.message}</div>`; }
};

```

```

window.lookupIcd10Dropdown = async function(val) {
  const dd = document.getElementById('icd10Dropdown');
  if (!dd) return;
  if (val.length < 2) { dd.classList.add('hidden'); return; }
  try {
    const res = await API.get('/api/clinical/icd10?query=' + encodeURIComponent(val)).catch(() => []);
    if (!res.length) { dd.classList.add('hidden'); return; }
    dd.classList.remove('hidden');
  }

```

```

    dd.innerHTML = res.map(i => `
      <div class="p-8 hover:bg-surface-variant cursor-pointer text-xs" onclick="selectIcd10Code('${i.code}', '
    ${escapeHTML(i.description_en)}')">
        <strong>${i.code}</strong> - ${escapeHTML(i.description_en)}
      </div>
    `).join('');
  } catch(_) {}
};

```

```

window.selectIcd10Code = function(code, desc) {
  const icdEl = document.getElementById('probIcd10');
  const nameEl = document.getElementById('probName');
  if (icdEl) icdEl.value = code;
  if (nameEl) nameEl.value = desc;
  const dd = document.getElementById('icd10Dropdown');
  if (dd) dd.classList.add('hidden');
};

```

```

window.addPatientProblem = async function(patientId, containerId) {
  const icd10_code = document.getElementById('probIcd10')?.value;
  const problem_name = document.getElementById('probName')?.value;
  const principal_diagnosis = document.getElementById('probPDx')?.checked;
  if (!problem_name) return;
  try {
    await API.post('/api/clinical/problem-list', { patient_id: patientId, icd10_code, problem_name, principal_
diagnosis });
    showToast(tr('Problem added to list','??? ????? ??????? ??????? ??????'));
    renderProblemListIcd10(containerId, patientId);
  } catch (e) { showToast(e.message, 'error'); }
};

```

// ??? PHASE D: FINANCE & OPERATIONS UI ?????????????????????????????????????????????????????????????

```

window.renderAccountsPayableReceivable = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const apList = await API.get('/api/finance/ap').catch(() => []);
    const arList = await API.get('/api/finance/ar').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('Subledger Accounts (AP & AR)','????????? ??????? ? ?????? ??
        ????? ?????????')}</h3>

        <div class="grid grid-cols-2 gap-16">
          <div class="card p-16">
            <h4 class="font-bold text-error mb-8">? Accounts Payable (AP - ????????)</h4>
            <div class="overflow-y-auto max-h-250 space-y-8 text-xs">
              ${apList.map(a => `
                <div class="p-8 border rounded flex justify-between items-center bg-surface-variant/20">
                  <div>
                    <strong>${escapeHTML(a.vendor_name)}</strong> (Inv: #${escapeHTML(a.invoice_number)})<br>
                    Due: ${a.due_date ? new Date(a.due_date).toLocaleDateString('ar-SA') : ''}
                  </div>

```

```

        <div class="text-right">
            <span class="font-bold block">SAR ${parseFloat(a.balance_due||a.total_amount).toFixed(2)}<
    /span>

            ${a.payment_status === 'Paid' ? '<span class="text-secondary font-bold">? Paid</span>' : `
    <button class="btn btn-xs btn-error mt-4" onclick="payApInvoice(${a.id}, ${a.total_amount}, '${containerId}')"
    >Pay</button>`}

        </div>
    </div>
    `).join('') || '<p class="text-on-surface-variant">No AP invoices outstanding</p>'}
    </div>
</div>

<div class="card p-16">
    <h4 class="font-bold text-secondary mb-8">? Accounts Receivable (AR - ?????? ????????)</h4>
    <div class="overflow-y-auto max-h-250 space-y-8 text-xs">
        ${arList.map(a => `
            <div class="p-8 border rounded flex justify-between items-center bg-surface-variant/20">
                <div>
                    <strong>${escapeHTML(a.patient_name || a.payer_name)}</strong> (Inv: #${escapeHTML(a.invoi
ce_number)})<br>
                    Payer Type: ${a.payer_type}
                </div>
                <div class="text-right">
                    <span class="font-bold block">SAR ${parseFloat(a.balance_due||a.total_amount).toFixed(2)}<
    /span>

                    ${a.collection_status === 'Collected' ? '<span class="text-secondary font-bold">? Collecte
d</span>' : `<button class="btn btn-xs btn-primary mt-4" onclick="collectArInvoice(${a.id}, ${a.total_amount},
    '${containerId}')">Collect</button>`}
                </div>
            </div>
        `).join('') || '<p class="text-on-surface-variant">No AR balances outstanding</p>'}
    </div>
</div>
</div>
    `;
    } catch (e) { el.innerHTML = `<div class="card p-20 text-error">Error: ${e.message}</div>`; }
};

window.payApInvoice = async function(id, total, containerId) {
    try {
        await API.post(`/api/finance/ap/${id}/pay`, { payment_amount: total });
        showToast(tr('AP invoice paid and journal ledger posted','?? ???? ??????? ?????? ?????? ?????? ??????'));
        renderAccountsPayableReceivable(containerId);
    } catch (e) { showToast(e.message, 'error'); }
};

window.collectArInvoice = async function(id, total, containerId) {
    try {
        await API.post(`/api/finance/ar/${id}/collect`, { collection_amount: total });
        showToast(tr('AR balance collected','?? ???? ???? ?????? ??????'));
        renderAccountsPayableReceivable(containerId);
    } catch (e) { showToast(e.message, 'error'); }
};

```

```

window.renderVendors = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const list = await API.get('/api/vendors').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('Approved Vendor Register','?? ?????? ????????? ?????????
?')}</h3>

        <div class="grid grid-cols-2 gap-16 mb-16">
          <div class="card p-16">
            <h4 class="font-bold mb-8">? Add New Vendor</h4>
            <div class="space-y-8">
              <input id="vendName" class="input w-full" placeholder="Vendor Name (Arabic)">
              <input id="vendVat" class="input w-full" placeholder="VAT Number (3xxxxxxxxxxxxx)">
              <input id="vendCR" class="input w-full" placeholder="Commercial Register (CR)">
              <input id="vendIBAN" class="input w-full" placeholder="IBAN (SAxxxxxxxxxxxxxxxxxxxxx)">
              <button class="btn btn-primary mt-8 w-full" onclick="saveVendor('${containerId}')">Add Approved
Vendor</button>
            </div>
          </div>

          <div class="card p-16">
            <h4 class="font-bold mb-8">? Vendor Directory</h4>
            <div class="overflow-y-auto max-h-200 space-y-8 text-xs">
              ${list.map(v => `
                <div class="p-8 border-b">
                  <strong>${escapeHTML(v.vendor_name_ar)}</strong> (Code: ${escapeHTML(v.vendor_code)})<br>
                  VAT: ${escapeHTML(v.vat_number)} | CR: ${escapeHTML(v.commercial_register)}
                </div>
              `).join('')} || '<p class="text-on-surface-variant">No vendors registered</p>'
            </div>
          </div>
        </div>
      </div>
    `;
  } catch (e) { el.innerHTML = `<div class="card p-20 text-error">Error: ${e.message}</div>`; }
};

window.saveVendor = async function(containerId) {
  const vendor_name_ar = document.getElementById('vendName')?.value;
  const vat_number = document.getElementById('vendVat')?.value;
  const commercial_register = document.getElementById('vendCR')?.value;
  const iban = document.getElementById('vendIBAN')?.value;
  if (!vendor_name_ar) return;
  try {
    await API.post('/api/vendors', { vendor_name_ar, vat_number, commercial_register, iban });
    showToast(tr('Vendor registered','?? ?????? ?????? ??????'));
    renderVendors(containerId);
  } catch (e) { showToast(e.message, 'error'); }
};

```

```

window.renderFinancialReportsSnapshots = async function(containerId) {
  const el = document.getElementById(containerId);
  if (!el) return;
  try {
    const list = await API.get('/api/finance/reports/snapshots').catch(() => []);
    el.innerHTML = `
      <div class="card p-20 mb-16">
        <h3 class="card-title text-primary">? ${tr('Financial Statement Snapshots (Profit & Loss / Balance She
et)', '????? ?????? ?????? ?????? ??????')}</h3>

        <div class="grid grid-cols-2 gap-16">
          <div class="card p-16">
            <h4 class="font-bold mb-8">?? ${tr('Generate New Statement Snapshot', '????? ?????? ??????')}</h4>
            <div class="space-y-8">
              <div>
                <label class="label">Report Type</label>
                <select id="fsType" class="input w-full">
                  <option value="PL">Profit & Loss (??????? ????????)</option>
                  <option value="BS">Balance Sheet (????????? ????????)</option>
                </select>
              </div>
              <div>
                <label class="label">Start Date</label>
                <input id="fsStart" class="input w-full" type="date" value="${new Date().toISOString().slice(0,1
0)}-01-01">
              </div>
              <div>
                <label class="label">End Date</label>
                <input id="fsEnd" class="input w-full" type="date" value="${new Date().toISOString().slice(0,1
0)}">
              </div>
              <button class="btn btn-primary w-full mt-8" onclick="generateFinancialReport('${containerId}')">
Generate Statement</button>
            </div>
          </div>

          <div class="card p-16">
            <h4 class="font-bold mb-8">? ${tr('Generated Report Archive', '????? ?????????? ???????')}</h4>
            <div class="overflow-y-auto max-h-250 space-y-8 text-xs">
              ${list.map(s => `
                <div class="p-12 border rounded bg-surface-variant/20">
                  <div class="flex justify-between font-bold">
                    <span>${s.report_type === 'PL' ? 'Profit & Loss' : 'Balance Sheet'}</span>
                    <span>${new Date(s.generated_at).toLocaleDateString('ar-SA')}</span>
                  </div>
                  <div class="mt-4">Period: ${new Date(s.report_period_start).toLocaleDateString('ar-SA')} to
${new Date(s.report_period_end).toLocaleDateString('ar-SA')}</div>
                  <div class="mt-8 grid grid-cols-3 gap-4 text-center">
                    <div class="bg-primary-container p-4 rounded"><div class="text-xs">Revenue</div><strong>SA
R ${parseFloat(s.total_revenue).toFixed(0)}</strong></div>
                    <div class="bg-error-container/10 p-4 rounded"><div class="text-xs">Expenses</div><strong>
SAR ${parseFloat(s.total_expenses).toFixed(0)}</strong></div>
                    <div class="bg-secondary-container p-4 rounded"><div class="text-xs">Net Income</div><stro
ng>SAR ${parseFloat(s.net_income).toFixed(0)}</strong></div>
              `)}
            </div>
          </div>
        </div>
      `
  }
}

```



```

        </div>
    </div>
    `).join('') || '<p class="text-on-surface-variant">No report snapshots yet</p>'}
</div>
</div>
</div>
</div>
`
} catch (e) { el.innerHTML = `<div class="card p-20 text-error">Error: ${e.message}</div>`; }
};

window.generateFinancialReport = async function(containerId) {
    const report_type = document.getElementById('fsType')?.value;
    const period_start = document.getElementById('fsStart')?.value;
    const period_end = document.getElementById('fsEnd')?.value;
    if (!period_start || !period_end) return;
    try {
        await API.post('/api/finance/reports/generate', { report_type, period_start, period_end });
        showToast(tr('Financial report snapshot compiled', '?? ????? ????? ?????? ?????? ??????'));
        renderFinancialReportsSnapshots(containerId);
    } catch (e) { showToast(e.message, 'error'); }
};

// ??? PHASE E: INTEGRATION & AI UI ?????????????????????????????????????????????
window.renderFhirResourceStore = async function(containerId) {
    const el = document.getElementById(containerId);
    if (!el) return;
    el.innerHTML = `
        <div class="card p-20 mb-16">
            <h3 class="card-title text-primary">? ${tr('FHIR R4 Resource Store Inspector', '?????? ??????? ?????? ???
???? FHIR R4')}</h3>
            <div class="grid grid-cols-3 gap-8 mb-12">
                <input id="fhirType" class="input w-full" placeholder="Resource Type (e.g. Patient)" value="Patient">
                <input id="fhirId" class="input w-full" placeholder="Resource Logical ID" value="dummy-patient">
                <button class="btn btn-primary" onclick="inspectFhirResource()">Inspect Resource JSON</button>
            </div>
            <div id="fhirInspectResult" class="bg-surface-variant/30 p-12 rounded font-mono text-xs overflow-x-auto
max-h-200">
                JSON Output will appear here...
            </div>
        </div>
    `;
};

window.inspectFhirResource = async function() {
    const t = document.getElementById('fhirType')?.value;
    const id = document.getElementById('fhirId')?.value;
    if (!t || !id) return;
    try {
        const res = await API.get(`/api/fhir/${t}/${id}`);
        document.getElementById('fhirInspectResult').innerHTML = `<pre>${escapeHTML(JSON.stringify(res, null, 2))}
</pre>`;
    } catch (e) {

```

```

document.getElementById('fhirInspectResult').innerHTML = `

```

```

<div class="grid grid-cols-2 gap-16">
  <div class="card p-16">
    <h4 class="font-bold text-secondary mb-8">?? ${tr('Voice Dictation to SOAP Note','????? ?????? ???
?? ?????? SOAP')}</h4>
    <div class="space-y-8">
      <textarea id="aiRawAudioText" class="input w-full" rows="4" placeholder="Type or paste simulated p
atient spoken notes here... e.g. Patient complains of chest pain since 6 AM, pulse is 88, ECG completed."></te
xtarea>
      <button id="btnVoiceFinalize" class="btn btn-primary w-full" onclick="simulateVoiceFinalize(${pati
entId})">? Process Dictation (AI Structure)</button>
    </div>
    <div id="aiVoiceResult" class="mt-12 bg-surface-variant/40 p-12 rounded text-xs hidden"></div>
  </div>

  <div class="card p-16">
    <h4 class="font-bold text-secondary mb-8">?? ${tr('Clinical Decision Hooks CDS','????? ??? ?????? ??
?????')}</h4>
    <div class="space-y-8">
      <select id="cdsContext" class="input w-full">
        <option value="Differential">Differential Diagnosis</option>
        <option value="DoseCheck">Pediatric Weight Dosing Alert</option>
      </select>
      <button class="btn btn-outline w-full" onclick="triggerCdsHooks(${patientId})">? Run CDS Hooks Che
ck</button>
    </div>
    <div id="aiCdsResult" class="mt-12 bg-surface-variant/40 p-12 rounded text-xs hidden"></div>
  </div>
</div>
`;
};

window.simulateVoiceFinalize = async function(patientId) {
  const transcript_raw = document.getElementById('aiRawAudioText')?.value;
  if (!transcript_raw) return;
  try {
    // Start session
    const startRes = await API.post('/api/ai/voice-dictation/start', { patient_id: patientId, session_type: 'C
linical Note' });
    const sessionId = startRes.session.id;
    // Finalize
    const res = await API.post(`/api/ai/voice-dictation/${sessionId}/finalize`, { transcript_raw });
    const div = document.getElementById('aiVoiceResult');
    div.classList.remove('hidden');
    div.innerHTML = `
      <h5 class="font-bold text-secondary mb-4">Structured SOAP Note Output (AI mapped):</h5>
      <pre class="bg-surface p-8 rounded font-mono overflow-x-auto max-h-150">${escapeHTML(res.session.final_t
ext)}</pre>
      <div class="text-xs text-on-surface-variant mt-4">Confidence Score: ${res.session.confidence_score * 100
}%</div>
    `;
    showToast(tr('Voice note structured successfully','?? ????? ?????? ?????? ??????'));
  } catch (e) { showToast(e.message, 'error'); }
};

```

```

window.triggerCdsHooks = async function(patientId) {
  if (!patientId) { showToast('Select a patient first', 'error'); return; }
  const context_type = document.getElementById('cdsContext')?.value;
  try {
    const res = await API.post('/api/ai/cds-hooks', { patient_id: patientId, context_type, input_data: { symptoms: ['chest pain', 'shortness of breath'] } });
    const div = document.getElementById('aiCdsResult');
    div.classList.remove('hidden');
    div.innerHTML = `
      <div class="p-8 border-l-4 border-error bg-error-container/10">
        <strong>${escapeHTML(res.cards[0].summary)}</strong><br>
        <p class="mt-4">${escapeHTML(res.cards[0].detail)}</p>
        <span class="text-xs text-on-surface-variant block mt-4">Source: ${escapeHTML(res.cards[0].source.label)}</span>
      </div>
    `;
  } catch (e) { showToast(e.message, 'error'); }
};

// ===== INTERACTIVE CLINICAL ASSESSMENTS MODALS =====
window.openBradenModal = function() {
  const patients = window.nursingPatients || [];
  if (!patients.length) { showToast(tr('No patient records loaded', '?? ???? ????? ???? ?????'), 'error'); return; }

  let patientOptions = patients.map(p => `<option value="${p.id}">${escapeHTML(p.file_number)} - ${escapeHTML(isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</option>`).join('');

  const modal = document.createElement('div');
  modal.id = 'bradenModal';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-items:center;justify-content:center;padding:16px;overflow-y:auto';

  modal.innerHTML = `
    <div class="card glass-card-premium" style="width:100%;max-width:550px;max-height:90vh;overflow-y:auto;padding:24px;border:1px solid rgba(255,255,255,0.2)">
      <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px;border-bottom:1px solid var(--border-color);padding-bottom:8px">
        <h3 style="margin:0;color:var(--primary)"> ${tr('Braden Scale Assessment', '????? ????? ????? ????? ?????')}</h3>
        <button class="btn btn-sm btn-secondary" onclick="document.getElementById('bradenModal').remove()">?</button>
      </div>

      <div class="form-group mb-12">
        <label>? ${tr('Select Patient', '???? ?????')}</label>
        <select id="bradenPatientId" class="form-input">${patientOptions}</select>
      </div>

      <div style="font-size:12px;margin-bottom:16px">
        <div class="form-group mb-8">
          <label>?? ${tr('Sensory Perception', '??????? ?????')}</label>
          <select id="bradenSensory" class="form-input" onchange="calcBradenTotal()">

```

```

        <option value="1">1 - Completely Limited (????? ??????)</option>
        <option value="2">2 - Very Limited (????? ??????)</option>
        <option value="3">3 - Slightly Limited (????? ??????)</option>
        <option value="4" selected>4 - No Impairment (?? ???? ???)</option>
    </select>
</div>
<div class="form-group mb-8">
    <label>? ${tr('Moisture', '???????')}</label>
    <select id="bradenMoisture" class="form-input" onchange="calcBradenTotal()">
        <option value="1">1 - Constantly Moist (??? ????????)</option>
        <option value="2">2 - Very Moist (??? ??????)</option>
        <option value="3">3 - Occasionally Moist (??? ????????)</option>
        <option value="4" selected>4 - Rarely Moist (?????? ?? ???? ??????)</option>
    </select>
</div>
<div class="form-group mb-8">
    <label>? ${tr('Activity', '?????? ??????')}</label>
    <select id="bradenActivity" class="form-input" onchange="calcBradenTotal()">
        <option value="1">1 - Bedfast (???? ????????)</option>
        <option value="2">2 - Chairfast (???? ????????)</option>
        <option value="3">3 - Walks Occasionally (???? ????????)</option>
        <option value="4" selected>4 - Walks Frequently (???? ???? ??????)</option>
    </select>
</div>
<div class="form-group mb-8">
    <label>? ${tr('Mobility', '?????? ?? ????')}</label>
    <select id="bradenMobility" class="form-input" onchange="calcBradenTotal()">
        <option value="1">1 - Completely Immobile (???? ?????? ??????)</option>
        <option value="2">2 - Very Limited (????? ??????)</option>
        <option value="3">3 - Slightly Limited (????? ??????)</option>
        <option value="4" selected>4 - No Limitation (?? ???? ??????)</option>
    </select>
</div>
<div class="form-group mb-8">
    <label>? ${tr('Nutrition', '???????')}</label>
    <select id="bradenNutrition" class="form-input" onchange="calcBradenTotal()">
        <option value="1">1 - Very Poor (???? ??????)</option>
        <option value="2">2 - Probably Inadequate (??? ?????? ?? ?????)</option>
        <option value="3">3 - Adequate (?????)</option>
        <option value="4" selected>4 - Excellent (?????)</option>
    </select>
</div>
<div class="form-group mb-8">
    <label>? ${tr('Friction & Shear', '???????? ?????')}</label>
    <select id="bradenFriction" class="form-input" onchange="calcBradenTotal()">
        <option value="1">1 - Problem (????? ??????)</option>
        <option value="2">2 - Potential Problem (????? ??????)</option>
        <option value="3" selected>3 - No Apparent Problem (?? ???? ?????? ??????)</option>
    </select>
</div>
</div>

```

```

<div id="bradenResultPanel" style="background:var(--hover,#f8f9fa);padding:12px;border-radius:8px;font-size:14px;border-left:4px solid #10b981;margin-bottom:16px;display:flex;justify-content:space-between;align-items:

```

```

ms:center">
    <span><strong>${tr('Total Score', '????? ?????????')}</strong> <span id="bradenInterpretation">${tr(
'No Risk', '?? ???? ???')}</span></span>
    <span id="bradenScoreVal" class="badge badge-success" style="font-size:16px;padding:4px 10px">23 / 23<
/span>
</div>

<div style="display:flex;gap:12px">
    <button class="btn btn-secondary" style="flex:1" onclick="document.getElementById('bradenModal').remov
e()">${tr('Cancel', '????')}</button>
    <button class="btn btn-primary" style="flex:2" onclick="saveBradenAssessment()">${tr('Save Assessment'
, '??? ?????')}</button>
</div>
</div>
`;
document.body.appendChild(modal);
window.calcBradenTotal();
};

window.calcBradenTotal = () => {
    const getSel = id => parseInt(document.getElementById(id)?.value || 0);
    const total = getSel('bradenSensory') + getSel('bradenMoisture') + getSel('bradenActivity') + getSel('braden
Mobility') + getSel('bradenNutrition') + getSel('bradenFriction');

    const panel = document.getElementById('bradenResultPanel');
    const interp = document.getElementById('bradenInterpretation');
    const scoreVal = document.getElementById('bradenScoreVal');
    if (!panel || !interp || !scoreVal) return;

    scoreVal.innerText = `${total} / 23`;

    if (total <= 9) {
        panel.style.borderLeftColor = 'red';
        interp.innerText = tr('Severe Risk (9)', '??? ???? ???? (???? ????');
        scoreVal.className = 'badge badge-danger';
    } else if (total <= 12) {
        panel.style.borderLeftColor = '#ef4444';
        interp.innerText = tr('High Risk (10-12)', '??? ?????');
        scoreVal.className = 'badge badge-danger';
    } else if (total <= 14) {
        panel.style.borderLeftColor = '#f59e0b';
        interp.innerText = tr('Moderate Risk (13-14)', '??? ?????');
        scoreVal.className = 'badge badge-warning';
    } else if (total <= 18) {
        panel.style.borderLeftColor = '#3b82f6';
        interp.innerText = tr('Mild Risk (15-18)', '??? ?????');
        scoreVal.className = 'badge badge-info';
    } else {
        panel.style.borderLeftColor = '#10b981';
        interp.innerText = tr('No Risk (19-23)', '?? ???? ???');
        scoreVal.className = 'badge badge-success';
    }
};

```

```

window.saveBradenAssessment = async () => {
  const patientId = document.getElementById('bradenPatientId').value;
  const getSel = id => parseInt(document.getElementById(id)?.value || 0);
  const total = getSel('bradenSensory') + getSel('bradenMoisture') + getSel('bradenActivity') + getSel('braden
Mobility') + getSel('bradenNutrition') + getSel('bradenFriction');

  let riskLevel = 'No Risk';
  if (total <= 9) riskLevel = 'Severe Risk';
  else if (total <= 12) riskLevel = 'High Risk';
  else if (total <= 14) riskLevel = 'Moderate Risk';
  else if (total <= 18) riskLevel = 'Mild Risk';

  try {
    await API.post('/api/nursing/risk-assessment', {
      patient_id: patientId,
      assessment_type: 'Braden Scale',
      total_score: total,
      risk_level: riskLevel,
      details: {
        sensory: getSel('bradenSensory'),
        moisture: getSel('bradenMoisture'),
        activity: getSel('bradenActivity'),
        mobility: getSel('bradenMobility'),
        nutrition: getSel('bradenNutrition'),
        friction: getSel('bradenFriction')
      }
    });
    showToast(tr('Braden Scale assessment saved!', '?? ??? ????? ????? ????? ?????!'));
    document.getElementById('bradenModal').remove();
    await navigateTo(11);
  } catch (e) {
    showToast(e.message || tr('Error saving assessment', '??? ?? ??? ???????'), 'error');
  }
};

window.openMorseModal = function() {
  const patients = window.nursingPatients || [];
  if (!patients.length) { showToast(tr('No patient records loaded', '?? ??? ???? ???? ????'), 'error'); ret
urn; }

  let patientOptions = patients.map(p => `<option value="${p.id}">${escapeHTML(p.file_number)} - ${escapeHTML(
isArabic ? (p.name_ar || p.name_en) : (p.name_en || p.name_ar))}</option>`).join('');

  const modal = document.createElement('div');
  modal.id = 'morseModal';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center;padding:16px;overflow-y:auto';

  modal.innerHTML = `
    <div class="card glass-card-premium" style="width:100%;max-width:550px;max-height:90vh;overflow-y:auto;pad
ding:24px;border:1px solid rgba(255,255,255,0.2)">
      <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px;border-bott
om:1px solid var(--border-color);padding-bottom:8px">
        <h3 style="margin:0;color:var(--primary)">? ${tr('Morse Fall Risk Assessment', '???? ???? ????? (??

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??? ?????')}]</h3>
    <button class="btn btn-sm btn-secondary" onclick="document.getElementById('morseModal').remove()">?</b
utton>
</div>

<div class="form-group mb-12">
    <label>? ${tr('Select Patient', '???? ?????')}</label>
    <select id="morsePatientId" class="form-input">${patientOptions}</select>
</div>

<div style="font-size:12px;margin-bottom:16px">
    <div class="form-group mb-8">
        <label>? ${tr('History of Falling (Within 3 Months)', '????? ?????? ??? 3 ???')}</label>
        <select id="morseHistory" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>No (0) (??)</option>
            <option value="25">Yes (25) (???)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label>? ${tr('Secondary Diagnosis', '???? ?????? ??????')}</label>
        <select id="morseSecondary" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>No (0) (??)</option>
            <option value="15">Yes (15) (???)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label>? ${tr('Ambulatory Aid', '????????? ?? ?????? / ????????? ?????????')}</label>
        <select id="morseAid" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>None / Bed Rest / Nurse Assist (0) (?? ????? / ?????? ?????? / ?????? ???
            <option value="15">Crutches / Cane / Walker (15) (????? / ??? / ?????)</option>
            <option value="30">Furniture / Clinging to objects (30) (????? ??? ?????)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label>? ${tr('IV / Heparin Lock', '????????? ????????? / ??? ?????????')}</label>
        <select id="morseIV" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>No (0) (??)</option>
            <option value="20">Yes (20) (???)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label>? ${tr('Gait / Transferring', '?????? ????????? ??? ?????????')}</label>
        <select id="morseGait" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>Normal / Bedrest / Immobile (0) (????? / ??? ??????)</option>
            <option value="10">Weak (10) (???? ??????)</option>
            <option value="20">Impaired (20) (???? ??????)</option>
        </select>
    </div>
    <div class="form-group mb-8">
        <label>? ${tr('Mental Status', '?????? ????????? ?????? ?????? ??????')}</label>
        <select id="morseMental" class="form-input" onchange="calcMorseTotal()">
            <option value="0" selected>Oriented to own ability (0) (???? ????????? ??????)</option>
            <option value="15">Overestimates or forgets limitations (15) (????? ?? ?????? ?????? ?? ????? ??????)<

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```

/option>
    </select>
</div>
</div>

<div id="morseResultPanel" style="background:var(--hover,#f8f9fa);padding:12px;border-radius:8px;font-size:14px;border-left:4px solid #10b981;margin-bottom:16px;display:flex;justify-content:space-between;align-items:center">
    <span><strong>${tr('Total Score', '????? ?????????')}</strong> <span id="morseInterpretation">${tr('Low Risk', '????? ??????')}</span></span>
    <span id="morseScoreVal" class="badge badge-success" style="font-size:16px;padding:4px 10px">0 / 125</span>
</div>

<div style="display:flex;gap:12px">
    <button class="btn btn-secondary" style="flex:1" onclick="document.getElementById('morseModal').remove()">${tr('Cancel', '????')}</button>
    <button class="btn btn-primary" style="flex:2" onclick="saveMorseAssessment()">${tr('Save Assessment', '??? ?????')}</button>
</div>
</div>
`;
document.body.appendChild(modal);
window.calcMorseTotal();
};

window.calcMorseTotal = () => {
    const getSel = id => parseInt(document.getElementById(id)?.value || 0);
    const total = getSel('morseHistory') + getSel('morseSecondary') + getSel('morseAid') + getSel('morseIv') + getSel('morseGait') + getSel('morseMental');

    const panel = document.getElementById('morseResultPanel');
    const interp = document.getElementById('morseInterpretation');
    const scoreVal = document.getElementById('morseScoreVal');
    if (!panel || !interp || !scoreVal) return;

    scoreVal.innerText = `${total} / 125`;

    if (total >= 45) {
        panel.style.borderColor = 'red';
        interp.innerText = tr('High Fall Risk (≥45)', '????? ???? ????? - ???? ????? ????????? ???? ??????');
        scoreVal.className = 'badge badge-danger';
    } else if (total >= 25) {
        panel.style.borderColor = '#f59e0b';
        interp.innerText = tr('Moderate Fall Risk (25-44)', '????? ???? ????? - ???? ????? ?????? ??????');
        scoreVal.className = 'badge badge-warning';
    } else {
        panel.style.borderColor = '#10b981';
        interp.innerText = tr('Low Fall Risk (0-24)', '????? ???? ?????');
        scoreVal.className = 'badge badge-success';
    }
};

window.saveMorseAssessment = async () => {

```

```

const patientId = document.getElementById('morsePatientId').value;
const getSel = id => parseInt(document.getElementById(id)?.value || 0);
const total = getSel('morseHistory') + getSel('morseSecondary') + getSel('morseAid') + getSel('morseIv') + g
etSel('morseGait') + getSel('morseMental');

let riskLevel = 'Low Risk';
if (total >= 45) riskLevel = 'High Risk';
else if (total >= 25) riskLevel = 'Moderate Risk';

try {
  await API.post('/api/nursing/risk-assessment', {
    patient_id: patientId,
    assessment_type: 'Morse Fall Risk',
    total_score: total,
    risk_level: riskLevel,
    details: {
      history: getSel('morseHistory'),
      secondary: getSel('morseSecondary'),
      aid: getSel('morseAid'),
      iv: getSel('morseIv'),
      gait: getSel('morseGait'),
      mental: getSel('morseMental')
    }
  });
  showToast(tr('Morse Fall Risk assessment saved!', '?? ??? ?????? ??? ???? ???? ???? ???? ????!'));
  document.getElementById('morseModal').remove();
  await navigateTo(11);
} catch (e) {
  showToast(e.message || tr('Error saving assessment', '??? ?? ??? ??????'), 'error');
}
};

window.openSurgicalCountModal = function(surgeryId) {
  if (!surgeryId) { showToast(tr('Select a surgery session first', '???? ?????? ?????? ??????'), 'error'); retur
n; }

  const modal = document.createElement('div');
  modal.id = 'surgCountModal';
  modal.style.cssText = 'position:fixed;inset:0;background:rgba(0,0,0,0.5);z-index:9999;display:flex;align-ite
ms:center;justify-content:center;padding:16px';

  modal.innerHTML = `
    <div class="card glass-card-premium" style="width:100%;max-width:500px;padding:24px;border:1px solid rgba(
255,255,255,0.2)">
      <div style="display:flex;justify-content:space-between;align-items:center;margin-bottom:16px;border-bott
om:1px solid var(--border-color);padding-bottom:8px">
        <h3 style="margin:0;color:var(--primary)">? ${tr('Surgical Count Verification Sheet', '???? ?????? ???
? ?????? ?????? ??????')}</h3>
        <button class="btn btn-sm btn-secondary" onclick="document.getElementById('surgCountModal').remove()">
?</button>
      </div>

      <div class="form-group mb-12">
        <label>? ${tr('Surgery ID', '??? ??????')}</label>

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```

        <input type="text" class="form-input" value="{surgeryId}" readonly disabled>
    </div>

    <div style="font-size:12px;margin-bottom:16px">
        <div style="display:grid;grid-template-columns:2fr 1fr 1fr;gap:8px;align-items:center;font-weight:bold
;margin-bottom:8px">
            <div>${tr('Item', '???? / ?????')}</div>
            <div>${tr('Initial Count', '???? ?????')}</div>
            <div>${tr('Final Count', '???? ?????')}</div>
        </div>

        <div style="display:grid;grid-template-columns:2fr 1fr 1fr;gap:8px;align-items:center;margin-bottom:8p
x">
            <div>? ${tr('Sponges (???? ?????)', '???? ?????')}</div>
            <input type="number" id="countSpongeInit" class="form-input" value="10" min="0" onchange="checkCount
Match()">
            <input type="number" id="countSpongeFinal" class="form-input" value="10" min="0" onchange="checkCoun
tMatch()">
        </div>

        <div style="display:grid;grid-template-columns:2fr 1fr 1fr;gap:8px;align-items:center;margin-bottom:8p
x">
            <div>? ${tr('Needles (???? ?????)', '????')}</div>
            <input type="number" id="countNeedleInit" class="form-input" value="5" min="0" onchange="checkCountM
atch()">
            <input type="number" id="countNeedleFinal" class="form-input" value="5" min="0" onchange="checkCount
Match()">
        </div>

        <div style="display:grid;grid-template-columns:2fr 1fr 1fr;gap:8px;align-items:center;margin-bottom:12
px">
            <div>? ${tr('Instruments (????? ?????)', '????? ?????')}</div>
            <input type="number" id="countInstInit" class="form-input" value="24" min="0" onchange="checkCountMa
tch()">
            <input type="number" id="countInstFinal" class="form-input" value="24" min="0" onchange="checkCountM
atch()">
        </div>

        <div style="display:grid;grid-template-columns:1fr 1fr;gap:8px;margin-bottom:12px">
            <div class="form-group">
                <label>${tr('Scrub Nurse (M1)', '???? ?????')}</label>
                <input type="text" id="countWitness1" class="form-input" value="Nurse Fatima" placeholder="Witness
1">
            </div>
            <div class="form-group">
                <label>${tr('Circulating Nurse (M2)', '????? ?????')}</label>
                <input type="text" id="countWitness2" class="form-input" value="Nurse Ahmad" placeholder="Witness
2">
            </div>
        </div>

        <div id="countMatchPanel" style="background:var(--hover,#f8f9fa);padding:10px;border-radius:8px;font-siz
e:13px;border-left:4px solid #10b981;margin-bottom:16px;display:flex;justify-content:space-between;align-items

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:center">
    <span id="countMatchStatus"><strong>${tr('Verification Status', '???? ???????')}</strong> <span id="
countInterpretation" style="color:#10b981;font-weight:bold">${tr('Perfect Match ?', '?????? ?????? ?')}</span
></span>
    </div>

    <div style="display:flex;gap:12px">
        <button class="btn btn-secondary" style="flex:1" onclick="document.getElementById('surgCountModal').re
move()">${tr('Cancel', '????')}</button>
        <button class="btn btn-primary" style="flex:2" onclick="saveSurgicalCountSheet(${surgeryId})">${tr('Sa
ve & Verify counts', '??? ?????? ?????')}</button>
    </div>
</div>
`;
document.body.appendChild(modal);
window.checkCountMatch();
};

window.checkCountMatch = () => {
    const getVal = id => parseInt(document.getElementById(id)?.value || 0);
    const spongeOk = getVal('countSpongeInit') === getVal('countSpongeFinal');
    const needleOk = getVal('countNeedleInit') === getVal('countNeedleFinal');
    const instOk = getVal('countInstInit') === getVal('countInstFinal');

    const panel = document.getElementById('countMatchPanel');
    const interp = document.getElementById('countInterpretation');
    if (!panel || !interp) return;

    if (spongeOk && needleOk && instOk) {
        panel.style.borderLeftColor = '#10b981';
        interp.innerText = tr('Perfect Match ?', '?????? ?????? ?');
        interp.style.color = '#10b981';
    } else {
        panel.style.borderLeftColor = 'red';
        interp.innerText = tr('DISCREPANCY DETECTED ?', '???? ?????? ?? ?????! ?');
        interp.style.color = 'red';
    }
};

window.saveSurgicalCountSheet = async (surgeryId) => {
    const getVal = id => parseInt(document.getElementById(id)?.value || 0);
    const spongeOk = getVal('countSpongeInit') === getVal('countSpongeFinal');
    const needleOk = getVal('countNeedleInit') === getVal('countNeedleFinal');
    const instOk = getVal('countInstInit') === getVal('countInstFinal');

    const sponge_init = getVal('countSpongeInit');
    const sponge_final = getVal('countSpongeFinal');
    const needle_init = getVal('countNeedleInit');
    const needle_final = getVal('countNeedleFinal');
    const instrument_init = getVal('countInstInit');
    const instrument_final = getVal('countInstFinal');
    const witness_1 = document.getElementById('countWitness1')?.value || '';
    const witness_2 = document.getElementById('countWitness2')?.value || '';

```

```

try {
  await API.post('/api/surgery/count-sheet', {
    surgery_id: surgeryId,
    sponge_init, sponge_final,
    needle_init, needle_final,
    instrument_init, instrument_final,
    witness_1, witness_2
  });

  showToast(tr('Surgical count sheet saved!', '?? ??? ??? ???? ???? ???? ???? ???? ???? !'));

  // Automatically set Counts Verified option in UI if matched
  const match = spongeOk && needleOk && instOk;
  const select = document.getElementById('op_counts');
  if (select) {
    select.value = match ? '1' : '0';
  }

  document.getElementById('surgCountModal').remove();
} catch (e) {
  showToast(e.message || tr('Error saving count sheet', '??? ? ???? ????'), 'error');
}
};

```